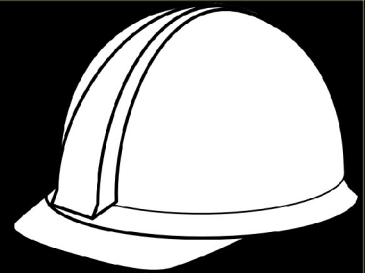


PE

Practice Problems

Civil Engineering



Shahriar Jahanian, PhD



Practice Problems

Civil Engineering

PE Exam

Practice Problems for PE Civil Engineering Exam

First Edition

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Preface

Beginning January 2015, the National Council of Examiners for Engineering and Surveying (NCEES) changed the Civil Engineering Specification for Civil PE exam. Now Civil PE are offered in the following Disciplines;

Transportation
Construction
Structure
Geotechnical Engineering
Water Resource and Environmental

Each discipline is divided to two areas of breadth and Depth.
The breadth part is common for all disciplines and includes the following specifications.

- a) Project Planning
- b) Means and Methods
- c) Soil Mechanics
- d) Structural Mechanics
- e) Hydraulics and Hydrology
- f) Geometrics
- g) Materials
- h) Site Development

In this book we have presented more than 1000 problems to prepare you for the breadth part of exam. Problems in each specification have been separated. This way you may concentrate on your area of strength if you want to. 4 sample tests similar to breadth PE is also presented.

I hope you find this book helpful for passing the PE exam.

Shawn (Shahriar) Jahanian, Ph.D, PE
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Part 1 : Project Planning

84 Problems

- 1) Which of the following joints join concrete that has been paved at different times?
- (A) Expansion joints
 - (B) Construction joints
 - (C) Control joints
 - (D) Contraction joints

The Answers is B**Construction Joints**

When joining concrete that has been paved at different times, construction joints are used. There are two different types of construction joints, transverse and longitudinal. Transverse are used at the end of a paving segment where as longitudinal join lanes paved at different times

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2) You are mixing a batch of concrete and decide to increase the water-to-cement ratio. What will be true of the new mix?

- | | |
|---------------------------|------------------------------|
| (A) Decreased Strength | (B) More Durable |
| (C) Increased air content | (D) Increased aggregate size |

The Answer is A

As the water to cement ratio increases, the concrete will decrease in strength and become less durable. Just changing the water to cement ratio however will not affect air content or aggregate size.

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3) A new roadway construction requires the sub base soil to have a dry density of 125 pcf and optimum moisture content (OMC) of 12.5%. A smooth drum roller will be used to compact the soil in 4-inch-thick lifts while the width is 32 ft. The soil has been tested in place and the results show moisture content of 6%. The water must be added to the stationing length of 100ft to obtain the required moisture content for compaction. How many gallons per yard must be added to meet the requirements?

- (A) 3.42 gal/yd²
- (B) 5.63 gal/yd²
- (C) 2.34 gal/yd²
- (D) 2.93 gal/yd²

The Answers is D

Gallons of Water =

$$\begin{aligned}
 &= \text{Desired dry density (pcf)} \times \frac{\text{Compacted cubic feet of soil}}{8.33 \text{ lb/gal}} \\
 &\quad \times \frac{\text{Goal water content\%} - \text{Existing water content \%}}{100} \\
 &= 125 \text{ pcf} \times \frac{32\text{ft} \times 100\text{ft}(\text{sta.}) \times (4\text{in}/12 \text{ in/ft})}{8.33 \text{ lb/gal}} \times \frac{12.5\% - 6\%}{100} = 1040.42 \text{ gal /sta.} \\
 \text{Gallonsyd}^2 &= \frac{1040.42 \text{ gal/sta.}}{(32\text{ft} \times 100\text{ft/sta})/9 \text{ ft}^2 \text{ /yd}^2} = 2.93 \text{ gal /yd}^2
 \end{aligned}$$

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4) A contractor is constructing a conventional two lane highway using unreinforced Portland Cement Concrete (PCC). Dowels and tie bars have not been used along the joints. The slabs measure 15.5 feet in width and 16 feet in length. When the temperature is 100°F , the longitudinal joints have a width of 4 inches. Given a coefficient of thermal expansion of 0.000006 and a new temperature of 65°F what is the maximum separation between two adjacent slabs?

- (A) 0.11 in (B) 0.22 in (C) 0.04 in (D) 0.078 in

The Answer is D

When the temperature decreases the longitudinal joint width will increase. Mechanics of materials shows that:

$$\Delta L = \alpha_t \times L \times (\Delta T) = 0.000006 \frac{1}{^\circ\text{F}} \times 15.5 \times (100^\circ\text{F} - 65^\circ\text{F}) \times 2 \text{ slabs} = 0.00651\text{ft} \\ = 0.078"$$

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5) What is the approximate density of the pavement mixture?

The following are the contents of an asphalt mix containing coarse and fine aggregate as well as asphalt. The amounts in the mix are as follows:

Ingredient	Fine Aggregate	Coarse Aggregate
Specific Gravity	2.25	4.05
% (by weight)	36	64

The asphalt has a specific gravity of 1.2. The pavement mixture has 6.3% asphalt by weight. The aggregate will absorb the asphalt 1.1% by weight. A sample of the mixture taken and compacted and a briquette is formed that 6-in diameter and 3 in height. The weight of the sample is 6 lbs.

- (A) 115 lb/ft³
- (B) 64 lb/ft³
- (C) 122 lb/ft³
- (D) 61 lb/ft³

The Answers is C

Volume of the sample =

$$\frac{\pi D^2}{4} \times h = \pi r^2 \times h = \frac{\pi(6''/12)^2}{4} \times (3''/12) = 0.049\text{ft}^3$$

Sample density

$$\rho = \frac{\text{Weight}}{\text{Volume}} = \frac{W}{V} = \frac{6\text{lb}}{0.049\text{ft}^3} = 122 \text{ pcf}$$

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6) What is the total weight of the asphalt?

The following are the contents of an asphalt mix containing coarse and fine aggregate as well as asphalt. The amounts in the mix are as follows:

Ingredient	Fine Aggregate	Coarse Aggregate
Specific Gravity	2.25	4.05
% (by weight)	36	64

The asphalt has a specific gravity of 1.2. The pavement mixture has 6.3% asphalt by weight. The aggregate will absorb the asphalt 1.1% by weight. A sample of the mixture taken and compacted and a briquette is formed that 6-in diameter and 3 in height. The weight of the sample is 6 lbs.

- (A) 7.7 lb
- (B) 8.00lb
- (C) 3.23 lb
- (D) 3.05lb

The Answer is A

The pavement mixture is approximately 6.3% which includes both the absorbed and binding portions.

$$\text{Asphalt weight} = (6.3\%) \times 122 \frac{\text{lb}}{\text{ft}^3} \times 1 \text{ ft}^3 = 7.7 \text{ lb}$$

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7) When compared to low water-cement ratio concrete, a high water-to cement ratio concrete is

- I) Weaker
- II) More durable
- III) Less permeable
- IV) Less drying shrinkage

- (A) I
- (B) I & III
- (C) II & IV
- (D) I, II, III & IV

The Answers is A

- I- TRUE statement- As water-cement ratio increases, strength decreases
- II- NOT TRUE – High w/c ratio is less permeable
- III- NOT TRUE - High w/c ratio is more permeable
- IV- NOT TRUE – as II and III are not true

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8) Three standard 6" $\times$ 12" cylinders were tested for their compressive and tensile strength. A beam test was also performed on a 20" $\times$ 6" $\times$ 6" beam using third point loading set up. The following table shows the results of the tests.

Batch No.	Splitting Tensile Strength Failure Load (lb)	Beam Test Failure Load (lb)	Compressive Strength Failure Load (lb)
1	52,500	2,780	123,000
2	49,600	3,320	132,200
3	51,000	3,100	127,560

What is the maximum compressive strength of batch # 1?

- (A) 4350 psi
- (B) 4676 psi
- (C) 1088 psi
- (D) 1128 psi

The Answer is A

Compressive strength, f'_c

$$f'_c = \frac{P}{A} = \frac{123,000}{\pi \times 3^2} = 4350.24 \text{ psi}$$

Where:

A = cross-sectional area of the cylinder

P = maximum failure load

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9) Three standard 6" $\times$ 12" cylinders were tested for their compressive and tensile strength. A beam test was also performed on a 20" $\times$ 6" $\times$ 6" beam using third point loading set up. The following table shows the results of the tests.

Batch No.	Splitting Tensile Strength Failure Load (lb)	Beam Test Failure Load (lb)	Compressive Strength Failure Load (lb)
1	52,500	2,780	123,000
2	49,600	3,320	132,200
3	51,000	3,100	127,560

What is the splitting tensile strength of batch #3?

- (A) 225 psi
- (B) 451 psi
- (C) 438 psi
- (D) 219 psi

The Answer is B

Splitting tensile strength, f_{ct}

$$f_{ct} = \frac{2P}{\pi DL} = \frac{2 \times 51,000}{\pi \times 6" \times 12"} = 451 \text{ psi}$$

Where:

P = maximum failure load

D = diameter

L = length

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10) Three standard 6" $\times$ 12" cylinders were tested for their compressive and tensile strength. A beam test was also performed on a 20" $\times$ 6" $\times$ 6" beam using third point loading set up. The following table shows the results of the tests.

Batch No.	Splitting Tensile Strength Failure Load (lb)	Beam Test Failure Load (lb)	Compressive Strength Failure Load (lb)
1	52,500	2,780	123,000
2	49,600	3,320	132,200
3	51,000	3,100	127,560

What is the modulus of rupture for batch #2?

- (A) 461 psi (B) 2766 psi (C) 386 psi (D) 495 psi

The Answer is A

Modulus of rupture (beam test), f_r

$$f_r = \frac{MC}{I} = \frac{\{(3320 \text{ lb/2}) \times 10''\}6''/2}{6'' \times 6^3/12} = 461 \text{ psi}$$

Where:

M = maximum bending moment

c = distance from the neutral axis to the outer surface of the beam

$I = bh^3/12$ (for a rectangle section $b \times h$)

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11) Three standard 6" $\times$ 12" cylinders were tested for their compressive and tensile strength. A beam test was also performed on a 20" $\times$ 6" $\times$ 6" beam using third point loading set up. The following table shows the results of the tests.

Batch No.	Splitting Tensile Strength Failure Load (lb)	Beam Test Failure Load (lb)	Compressive Strength Failure Load (lb)
1	52,500	2,780	123,000
2	49,600	3,320	132,200
3	51,000	3,100	127,560

For batch #2, what is the ratio of modulus of rupture to the compressive strength?

- (A) 6.3%
- (B) 9.9%
- (C) 4.8%
- (D) 9.6%

The Answer is B

$$f'_c = \frac{P}{A} = \frac{132,200}{\pi \times 3^2} = 4675.62 \text{ psi}$$
$$\text{Ratio} = \frac{461 \text{ psi}}{4676 \text{ psi}} = 9.9 \%$$

*See Fundamentals of Building Construction: Materials and Methods Wiley

12) Three standard 6" $\times$ 12" cylinders were tested for their compressive and tensile strength. A beam test was also performed on a 20" $\times$ 6" $\times$ 6" beam using third point loading set up. The following table shows the results of the tests.

Batch No.	Splitting Tensile Strength Failure Load (lb)	Beam Test Failure Load (lb)	Compressive Strength Failure Load (lb)
1	52,500	2,780	123,000
2	49,600	3,320	132,200
3	51,000	3,100	127,560

What is the ratio of the tensile strength to the compressive strength for batch # 3?

- (A) 20% (B) 16% (C) 14% (D) 10%

The Answer is D

$$f'_c = \frac{P}{A} = \frac{127,560}{\pi \times 6^2/4} = 4512 \text{ psi}$$

$$\text{Ratio} = \frac{451 \text{ psi}}{4512 \text{ psi}} = 10 \%$$

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13) A town is looking for a construction company to manufacture 5,200 catch basin frames. The construction company can either manufacture these frames with hand tools or using an automated system. It will cost \$5500.00 for the tools with \$10.75 manufacturing cost per frame. The automated system will have an initial cost of \$175,000.00 with a \$4.00 manufacturing cost per frame. What is the break-even point of the two frame manufacturing methods

- | | |
|---------------|---------------|
| (A) 1.6 year | (B) 3.0 years |
| (C) 2.6 years | (D) 4.8 yeas |

The Answers is D

Savings annually = $(5,200 \times \$10.75) - (5,200 \times \$4.0) = \$35,100$

Investment annually = $\$175,500 - \$5500 = \$170,000$

Payback period $170,000/35,100 = 4.8$ years

* See Basics of Engineering Economy, Pearson Prentice Hall

14) Contractors have been hired to construct a masonry wall that measures 10 feet in height and 27 feet in length. The individual bricks will measure $13/4" \times 3 1/4" \times 7 1/2"$ (height, thickness, length). The mortar around the bricks will have a thickness of $1/2"$. How many bricks will be needed to construct the wall?

- | | |
|----------------|------------------|
| (A) 960 bricks | (B) 2,160 bricks |
| (C) 900 bricks | (D) 1350 bricks |

The Answers is B

Each brick is:

$$1\frac{3}{4} \text{ (height)} \times 7\frac{1}{2} \text{ (length)} \times 3\frac{1}{4} \text{ (thick)}$$

Mortar dimensions around brick ($1/2$ " on all sides, only care about height and length as we are only concerned about wall face)

$$\text{Height} = 1\frac{3}{4} \text{ "} + 1/2 \text{ "} = 2\frac{1}{4} \text{ "} = 2.25 \text{ "}$$

$$\text{Length} = 7\frac{1}{2} \text{ "} + 1/2 \text{ "} = 8 \text{ "}$$

$$\text{Face of brick area} = 2.25 \times 8 = 18 \text{ in}^2$$

$$\text{Wall surface area} = 10 \text{ ft} \times 27 \text{ ft} = 270 \text{ ft}^2$$

$$\text{Bricks needed for wall} = \frac{270 \text{ ft}^2}{18 \text{ in}^2 \times \left(\frac{\text{ft}^2}{144 \text{ in}^2}\right)} = 2,160 \text{ bricks}$$

*See Fundamentals of Building Construction: Materials and Methods Wiley

15) A construction company has just bought a new excavator for \$45,000. The salvage value of the excavator after seven years is \$12,000. Using the straight line method, what is the annual depreciation of the excavator

- | | |
|--------------|-------------|
| (A) \$12,000 | (B) \$1,714 |
| (C) \$4,714 | (D) \$6,429 |

The Answer is C

$$\text{Depreciation} = \frac{\$45,000 - \$12,000}{7} = \$4,714$$

*See Basics of Engineering Economy, Pearson Prentice Hall

16) A construction company has just bought a new excavator for \$45,000. The salvage value of the excavator after seven years is \$12,000. Using the straight line depreciation method, what is the book value of the excavator after four years?

- (A) \$30,858 (B) \$26,144
(C) \$12,000 (D) \$45,000

The Answer is B

$$\text{Book Value} = \$45,000 - 4 \times \$4,714 = \$26,144$$

*Taken from Basics of Engineering Economy, Pearson Prentice Hall

17) A construction company has just bought a new excavator for \$45,000. The salvage value of the excavator after seven years is \$12,000. Using the sum of the year's digits method, what is the depreciation of the excavator for the fourth year?

- (A) \$4,714 (B) \$3,536
(C) \$3,143 (D) \$12,000

The Answer is A

$$\text{SOYD} = \frac{n(n+1)}{2} = \frac{7 \times 8}{2} = 28$$

$$\text{Depreciation} = (\$45,000 - \$12,000) (4/28) = \$4,714$$

*Taken from Basics of Engineering Economy, Pearson Prentice Hall

18) Using the sum of the year's digits method, what is the book value of the excavator at the end of the third year?

- (A) \$12,000 (B) \$19,071
(C) \$9,643 (D) \$25,928

The Answer is B

$$\text{Book Value} = \$45,000 - (7/28 \times \$33,000 + 6/28 \times \$33,000 + 5/28 \times \$33,000 + 4/28 \times \$33,000) = \$19,071$$

*Taken from Basics of Engineering Economy, Pearson Prentice Hall

19) Which of the following is not true regarding the cost performance index (CPI)?

- (A) $CPI > \$1$ means money is being spent inefficiently
- (B) Shows how efficiently money is being spent
- (C) $CPI = \text{Earned Value} / \text{Actual cost}$
- (D) $CPI = \$1$, means the project is on track

The Answer is A

If the CPI is greater than \$1 this means that the money is being spent efficiently. It means that if you spend \$1 on the project you are getting more than \$1's worth in return. This is an efficient use of money.

*See Basics of Engineering Economy, Pearson Prentice Hall

20) Use the following table shows the activities and duration for the installation of a new water line. The project is to be fast tracked thus activities can overlap. Activity "E" is to be lagged 3 days after activity "D" is completed.

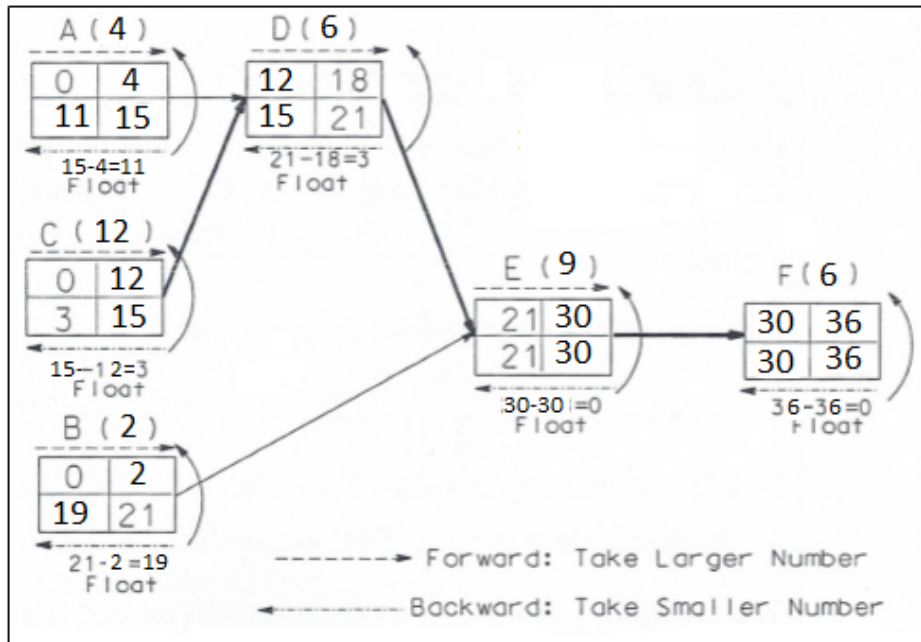
Activity	Symbol	Duration (days)	Predecessors
Install Straw Waddles for Erosion	A	4	-
Clearing the Site	B	2	-
Dig trench	C	12	-
Place Water Line Segments	D	6	A&C
Backfill	E	9	B,D
Compact	F	6	E

What is the CPM diagram for the table of activities?

The Answers is D

Two types of methods to draw out the critical path method (CPM) diagram are the activity on arrow and activity on node.

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21) Use the following table shows the activities and duration for the installation of a new water line. The project is to be fast tracked thus activities can overlap. Activity "E" is to be lagged 3 days after activity "D" is completed.

Activity	Symbol	Duration (days)	Predecessors
Install Straw Waddles for Erosion	A	4	-
Clearing the Site	B	2	-
Dig trench	C	12	-
Place Water Line Segments	D	6	A&C
Backfill	E	9	B,D
Compact	F	6	E

What is the critical path for the schedule?

- (A) CBEF
- (B) ADEF
- (C) FEDA
- (D) CDEF

The Answer is D

The critical path is the longest path through the CPM diagram. The critical path will also be the minimum amount of time needed to complete the project. The critical path is CDEF.

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22) Use the following table shows the activities and duration for the installation of a new water line. The project is to be fast tracked thus activities can overlap. Activity "E" is to be lagged 3 days after activity "D" is completed.

Activity	Symbol	Duration (days)	Predecessors
Install Straw Waddles for Erosion	A	4	-
Clearing the Site	B	2	-
Dig trench	C	12	-
Place Water Line Segments	D	6	A&C
Backfill	E	9	B,D
Compact	F	6	E

What is the critical path for the schedule?

- (A) CBEF
- (B) ADEF
- (C) FEDA
- (D) CDEF

The Answer is D

The critical path is the longest path through the CPM diagram. The critical path will also be the minimum amount of time needed to complete the project. The critical path is CDEF.

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23) Use the following table shows the activities and duration for the installation of a new water line. The project is to be fast tracked thus activities can overlap. Activity "E" is to be lagged 3 days after activity "D" is completed.

Activity	Symbol	Duration (days)	Predecessors
Install Straw Waddles for Erosion	A	4	-
Clearing the Site	B	2	-
Dig trench	C	12	-
Place Water Line Segments	D	6	A&C
Backfill	E	9	B,D
Compact	F	6	E

What is the float for activity A?

- (A) 3 days
- (B) 19 days
- (C) 11 days
- (D) Zero

The Answer is C

$$\text{Total Float} = \text{LF} - \text{EF} = \text{LS} - \text{ES} = 15 - 4 = 11$$

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24) Use the following table shows the activities and duration for the installation of a new water line. The project is to be fast tracked thus activities can overlap. Activity "E" is to be lagged 3 days after activity "D" is completed.

Activity	Symbol	Duration (days)	Predecessors
Install Straw Waddles for Erosion	A	4	-
Clearing the Site	B	2	-
Dig trench	C	12	-
Place Water Line Segments	D	6	A&C
Backfill	E	9	B,D
Compact	F	6	E

What would not be a reason to use time-cost trade off analysis?

- (A) Finish a project early to meet a deadline
- (B) Free up equipment or personnel for another project
- (C) Make budget cuts to a project
- (D) Recover from previous delays in the project

The Answer is C

The sole reason to use time-cost trade off analysis would not be to make budget cuts because the purpose of the analysis is to decrease the original project duration, not cost. Money will usually be necessary when trying to shorten the project schedule thus this is not a good budget cutting procedure.

*See Basics of Engineering Economy, Pearson Prentice Hall

25) Which of the following is not true regarding crashing?

- (A) Technique for making cost/time trade offs by compressing time
- (B) Decreases the duration of critical path activities
- (C) Changes the scope of some critical path activities
- (D) Speeding up activities by decreasing money spent on them

The Answer is D

Activities can always be sped up by increasing the amount of money spent on them. Therefore D is not correct.

*See Fundamentals of Building Construction: Materials and Methods, Wiley

26) A construction company is buying a new hand compactor for \$1200 and has an annual maintenance of \$65. After 7 years its salvage value is \$420. The interest rate is 6%. What is the equivalent uniform annual cost?

- | | |
|-----------|-----------|
| (A) \$164 | (B) \$133 |
| (C) \$68 | (D) \$229 |

The Answer is D

$$\begin{aligned} \text{EUAC} &= \$1200(A/P)_{n=7}^{i=6\%} + \$65 - \$420(A/F)_{n=7}^{i=6\%} \\ &= \$1200(0.179) + \$65 - \$420(0.12) = \$229.4 \end{aligned}$$

$$\text{Recall that } A/P = \frac{i(1+i)^n}{(1+i)^n - 1} \text{ and } A/F = \frac{i}{(1+i)^n - 1}$$

*See Basics of Engineering Economy, Pearson Prentice Hall

27) A large corporation wants to get its headquarters built and running quickly in time for its peak season in a year. Cost is not a large consideration as they have a fairly large budget for the new building. However they do want it constructed quickly, which type of contract/procurement method should they use for the project?

- | | |
|-------------------------|------------------------------|
| (A) Unit price | (B) Lump sum |
| (C) Cost plus fixed fee | (D) Cost plus percentage fee |

The Answer is C

Since the owner wants the project completed quickly and efficiently the should choose a cost plus fixed fee. In this method, the contractor is paid for the material, labor and overhead in addition to a fixed fee. This encourages the contractor to be as efficient as possible because the faster the project is completed the more profit they pocket. It is not wise to choose a cost plus percentage because the contractor has no incentive to complete timely or cost effectively as they make more profit the more expensive the project is.

*See Fundamentals of Building Construction: Materials and Methods, Wiley

28) A construction company is weighing their options for replacing a cooling system. They are deciding between two options that will both have a life span of 12 years and are similar in performance. Given an annual interest rate of 8%, determine the benefit-cost ratio of the better option?

Item	Option I	Option II
Initial Cost	\$8500	\$15000
Salvage Value	\$250	\$1300
Annual Savings	\$3000	\$5400

- (A) 0.54 (B) 3.98 (C) 0.34 (D) 2.81

The Answers is D**Option I:**

$$\text{Benefit} = B = \$3000 (P/A)_{12}^{8\%} = \$3000 (7.5361) = \$22,608.3$$

$$\text{Cost} = C = \$8500 - \$250 \left(\frac{P}{F} \right)_{12}^{8\%} = \$8500 - \$250(0.3971) = \$8,400.73$$

$$\left(\frac{B}{C} \right)_{\text{Sys1}} = \frac{\$22,608.3}{\$8,400.73} = 2.69$$

Option II:

$$\text{Benefit} = B = \$5,400 (P/A)_{12}^{8\%} = \$5,400 (7.5361) = \$40,696.6$$

$$\text{Cost} = C = \$15,000 - \$1,300 \left(\frac{P}{F} \right)_{12}^{8\%} = \$15,000 - \$1,300(0.3971) = \$14,483.77$$

$$\left(\frac{B}{C} \right)_{\text{Sys2}} = \frac{\$40,696.6}{\$14,483.77} = 2.81$$

Using incremental analysis

$$\frac{B_{\text{Sys2}} - B_{\text{Sys1}}}{C_{\text{Sys2}} - C_{\text{Sys1}}} = \frac{\$40,696.6 - \$22,608.3}{\$14,483.77 - \$8,400.73} = 2.97$$

Since the ratio is greater than one, system "2" is better than system "1"

*See Basics of Engineering Economy, Pearson Prentice Hall

29). The following are the batch weights for one cubic yard of a concrete mix. Following the Standard Test Method for Concrete Yield (ASTM C138), a $\frac{1}{2}$ cubic foot container is filled with concrete and has a total weight of 103.2 pounds. The empty weight of the container is 21.5 pounds. What is the yield in ft^3 of the concrete?

Water - 243-lbs/cu yd
Cement – 540-lbs/cu yd
Sand – 1233 lbs/cu yd
Coarse aggregate - 1924 lbs/cu yd
5% air entrainment

- (A) 25.3 ft^3
- (B) 24.1 ft^3
- (C) 81.7 ft^3
- (D) 48.2 ft^3

The Answers is B

318.41-70, ACI 318-08 Building Code Requirements for Structural Concrete, 2008, American Concrete Institute, Farmington Hills, MI, www.concrete.org.

Step 1: Total Weight of Concrete Mix (lbs/cu yd)

Total (lbs/cu yd) = 243+540+ 1233+ 1924 = 3,940 lbs/cu yd. (Remember air does not add any weight)

Step 2: Weight of ½ cubic foot sample

Sample weight (lbs) = 103.2 – 21.5 = 81.7 lbs

Step 3: Density of Sample

Density = 81.7lbs/(½ft³) = 163.4 lbs/ft

Step 4: Yield

Yield (ft³) = M/D = 3,940 lbs/(163.4lbs/ft) = 24.1 ft³

30) The total volume of concrete poured on site is 5 CY. The air content is tested to be 4%. What is the total volume of solids in the concrete?

- (A) 0.4 CY
- (B) 5.0 CY
- (C) 0.2 CY
- (D) 4.8 CY

The Answer is D

318.41-70, ACI 318-08 Building Code Requirements for Structural Concrete, 2008, American Concrete Institute, Farmington Hills, MI, www.concrete.org.

Step 1: Volume of Air

Air content (%) = Volume of Air (CY)/Total Volume of Concrete (CY)

0.04 = Volume of Air/ 5CY

Volume of Air = 0.2 CY

Step 2: Volume of Solids

Total Volume = Volume of Solids + Volume of Air

5 CY = Volume of Solids + 0.2 CY

Volume of Solids = 4.8 CY

31) A sample of soil was oven dried using the procedure outlined in ASTM D 2216-98 to determine the water content of soil. The following data points were collected. What is the water content (%) of the soil?

Mass of container and wet specimen (M_{cws}) = 30.2g

Mass of container and oven dry specimen (M_{cds}) = 28.4g

Mass of container (M_c) = 2g

(A) 14.4%

(B) 6.8%

(C) 7.6%

(D) 6.0%

The Answers is B

Use ASTM D2216-98 Standard test Method for Laboratory Determination of Water Content of Soil and Rock by Mass

Step 1: Equation for Water Content

$$W = [(M_{cws} - M_{cds}) / (M_{cs} - M_c)] \times 100 = M_w / M_s \times 100$$

Step 2: Mass Water

$$M_w = M_{cws} - M_{cds} = 30.2 - 28.4 \text{ g} = 1.8 \text{ g}$$

Step 3: Mass Solids

$$M_d = M_{cds} - M_c = 28.4 - 2 = 26.4 \text{ g}$$

Step 4: Water Content

$$W = 1.8 \text{ g} / 26.4 \text{ g} \times 100 = 6.8\%$$

32) An experiment was done using ASTM C127-88 to determine the specific gravity of bituminous material. Given the following data points determine the bulk specific gravity of the sample of bituminous material.

Mass of oven-dry test sample in air (A) = 1243.0g

Mass of saturated-surface-dry test sample in air (B) = 1248.2g

Mass of saturated test sample in water (C) = 885.1

(A)3.4

(B)2.4

(C)3.5

(D)2.5

The Answer is A

Use ASTM C 127-88 Standard test Method for Specific Gravity and Absorption of Coarse Aggregate.

Step 1: Bulk Specific Gravity Equation

$$\text{Bulk Specific Gravity} = A/(B-C)$$

Where: Mass of oven-dry test sample in air (A)

Mass of saturated-surface-dry test sample in air (B)
saturated test sample in water (C)

Mass of

Step 2: Calculate

$$\text{Bulk Specific Gravity} = 1243.0/(1248.2-885.1) = 1243.0/363.1 = 3.4$$

33) An experiment was done using ASTM C127-88 to determine the specific gravity of coarse aggregate. Given the following data points determine the absorption of the sample of coarse aggregate.

Mass of oven-dry test sample in air (A) = 2536.1g

Mass of saturated-surface-dry test sample in air (B) = 2551.2g

Mass of saturated test sample in water (C) = 1846.3g

(A)0.4%

(B)0.3%

(C)3.0%

(D)0.6%

The Answer is D

Use ASTM C 127-88 Standard test Method for Specific Gravity and Absorption of Coarse Aggregate.

Step 1: Absorption Equation

$$\% \text{ Absorption} = (B-A)/A \times 100$$

Where: Mass of oven-dry test sample in air (A)

Mass of saturated-surface-dry test sample in air (B)

Step 2: Calculate

$$\text{Absorption} = (2551.2-2536.1) / 2546.1 \times 100 = 0.6\%$$

34) For an existing concrete water reservoir the contractor has tested the existing compressive strength of concrete in 50 different location. The average of these tests shows the value of 4.1 KSI and the standard deviation shows the value of 0.5 KSI. If the contractor wants to take the risk of 5% for the rehabilitation of the structure, find the target strength of the structure that should be considered for the new design? (Use normal distribution function.)

- (A) 3.43 KSI
- (B) 4.77 KSI
- (C) 4.1 KSI
- (D) 4.6 KSI

The Answers is A

ACI 318-08, 5.3.2.1

The required compressive strength for the $f'_c < 5$ KSI is given by the following formula:

$$f'_{cr} = f'_c + 1.34S_s$$

S_s = to the standard deviation of the samples and 1.34 according to the normal distribution function represents the 90% success in the samples and 10% failure. That formula is used for design the concrete mixes and so, the safety margin in concrete mixes will be 1.34 S_s . Inversely for the existing structures the formula should be written as:

$$f'_{cr} = f'_c - 1.34S_s$$

This will give the safety factor with the 90% probability of success for the existing concrete, because for the existing concrete the 1.34 S_s will make the safety margin and the existing average should be decreased prorate to the probability of failure.

So:

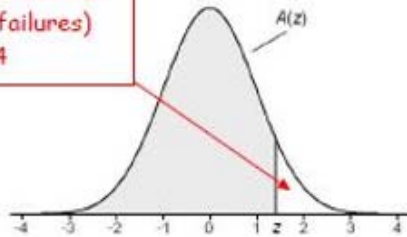
$$f'_{cr} = 4.1 - 1.34 * 0.5 = 3.43 \text{ KSI}$$

Note: in the normal distribution function, for $Z=1.34$, the probability of success is given as 0.9099. (See next page.)

TABLE A.1

Cumulative Standardized Normal Distribution

Example
For 95% success
(ie 5% failures)
 $Z = 1.64$



$A(z)$ is the integral of the standardized normal distribution from $-\infty$ to z (in other words, the area under the curve to the left of z). It gives the probability of a normal random variable not being more than z standard deviations above its mean. Values of z of particular importance:

z	$A(z)$	
1.645	0.9500	Lower limit of right 5% tail
1.960	0.9750	Lower limit of right 2.5% tail
2.326	0.9900	Lower limit of right 1% tail
2.576	0.9950	Lower limit of right 0.5% tail
3.090	0.9990	Lower limit of right 0.1% tail
3.291	0.9995	Lower limit of right 0.05% tail

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998

35) Which of the following item identified in a bid take-off would normally not be bid on the basis of weight?

(A) Shotcrete (gunite)

(B) Riprap

(C) Reinforcing Steel

(D) Steel Plates

The Answers is A

Shotcrete or guniting are measured per square foot or yard. All other items are measured by weights.

36) The Building Cost Index (BCI) for structural iron work developed by which formula?

(A) $BCI = 68.36 \text{ hours of skilled worker at 20-city average of bricklayers + carpenters + structural ironworkers rate} + 25 \text{ cwt of standard steel shapes at the mill price prior to 1996.}$

(B) $BCI = 200 \text{ hours of common labor at 20-city average of common labor rates} + 25 \text{ cwt of standard steel structural shape prior to 1996}$

(C) $\text{Fabricated 20-city price from 1996} + 1.128 \text{ tons of Portland cement} + 1088 \text{ board-ft of 2x4 lumber}$

(D) Both A & C

The Answer is D

The Engineering News-records BCI is developed with the explanation of both A & C.
The explanation of “B” is about the Construction Cost Index (CCI).

37) A precast concrete wall with the thickness of 8", height of 8' and width of 6'. A hauler truck should carry these precast walls. The capacity of the hauler is 20 US tons. How many pieces of wall can be transported by the hauler?

(A) 10

(B) 8

(C) 12

(D) 3

The Answer is B

Concrete density is given equal to 150 pcf. So the weight of each piece will be equal to:

$$W = 150 * (8/12 * 8 * 6) = 4800 \text{ lbs}$$

1 US Ton = 2000 lbs, each wall = $4800/2000 = 2.4$ ton,

Therefore; $20/2.4 = 8.33$, so the truck can take 8 walls.

- 38) Which one of these explanations shows the concept of the Gantt chart?
- (A) It is an activity chart (activity on node (AON) or arrow (AOA) network). The arrows may represent the activities or predecessor-successor relationship and the node may represent the instant time indicating or the activities for AON and AOA respectively.
 - (B) It is a bar chart often used to track the scheduling of activities, budget, resources, and equipment usage. The precedence relationships are difficult to display.
 - (C) It is a program evaluation and program technique, uses the probabilistic approach for each task.
 - (D) It formally identifies tasks which must be completed on time for whole project to be completed on time and identifies tasks which can be delayed.

The Answers is B

This question shows the meaning and concept for the three important project scheduling methods.

Choice “A” represents the CPA or Critical Path Analysis methods. This method uses two techniques for scheduling including AOA and AON which has described in “A”.

Choice “B” represents Gantt chart which is the answer. Also called as time relationship between tasks and it is a bar chart.

Choice “C” is the explanation for the PERT method of scheduling. PERT needs a reach history and data for each tasks to make it feasible for the use of probabilistic functions. The PERT chart is very similar to CPA, however in CPA a single deterministic values are used instead of probabilistic value.

Choice “D” represents Critical Path Analysis method (CPA). A network diagram including nodes and arrows with deterministic values represents the CPA charts.

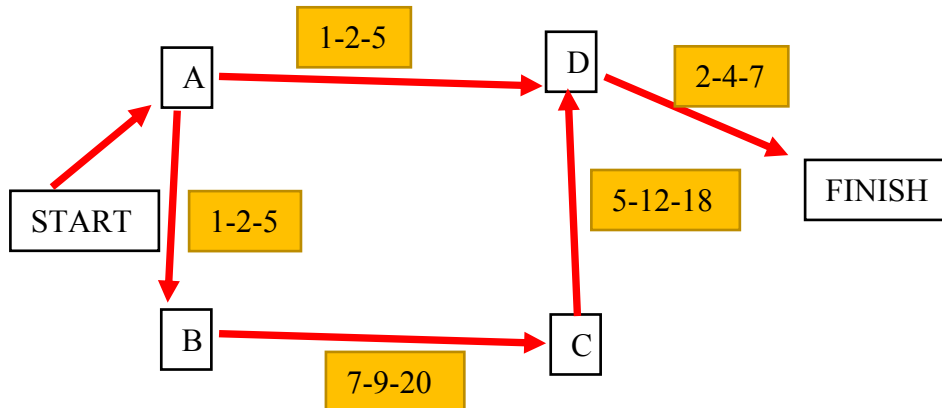
39) PERT activities are listed in the following table. It shows the characteristic completion time. If the project starts on day 15, what is the probability that the project will be completed on or before day 42 if the expected time for the critical path is given equal to $\mu_{\text{total}}=43.833$ and variance of the path is given equal to $\sigma^2_{\text{Total}}=10.52$?

Activity	Predecessors	Successors	a	m	b
START	-	A	0	0	0
A	START	B,D	1	2	5
B	A	C	7	9	20
C	B	D	5	12	18
D	A,C	FINISH	2	4	7
FINISH	D	-	0	0	0

- (A) 10.5% (B) 71.2% (C) 28.8% (D) 43.8%

The Answer is C

PERT Activities for this project is:



PERT uses a probabilistic approach that allows three time estimates for each activity:

Optimistic time a: Time an activity if everything goes perfectly.

Most likely time m: Most realistic time estimate to complete the activity.

Pessimistic time b: Time an activity if everything goes wrong.

To find the answer use the following steps:

Step 1: Find the critical path, expected values, and variances for the critical path, however for this question the expected values and variances and consequently standard deviation are available as follows:

There are 2 paths:

Path 1: Start-A-B-C-D-Finish = 15-27-50

Path 2: Start-A-D-Finish = 3-6-12

The A-B-C-D path is critical path, the following data are available:

Variance = Standard deviation²

$$\text{Standard Deviation} = \sigma = \frac{(b-a)}{6}$$

$$\text{Expected time} = \mu = \frac{(a + 4m + b)}{6}$$

$$\mu_{\text{Total}} = \sum \text{Expected time for each task} = 43.83$$

$$\sigma^2_{\text{Total}} = \sum \text{Variances} = 10.52$$

$$\sigma_{\text{Total}} = \sqrt{\sum \text{Variances}} = 3.24$$

Step 2: Find the Z value as the indicator of normal distribution function to find the probability for that particular target:

Expected day for the project completion is given equal to 42:

$$z = \frac{t - \mu_{\text{Total}}}{\sigma} = \frac{42 - 43.833}{3.244} = 0.565$$

Using the following table for the normal distribution function:

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
1.0	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214

For the $T > 42$, $P = 0.712$, so, $T < = 1 - 0.712 = 0.288 * 100 = 28.8\%$

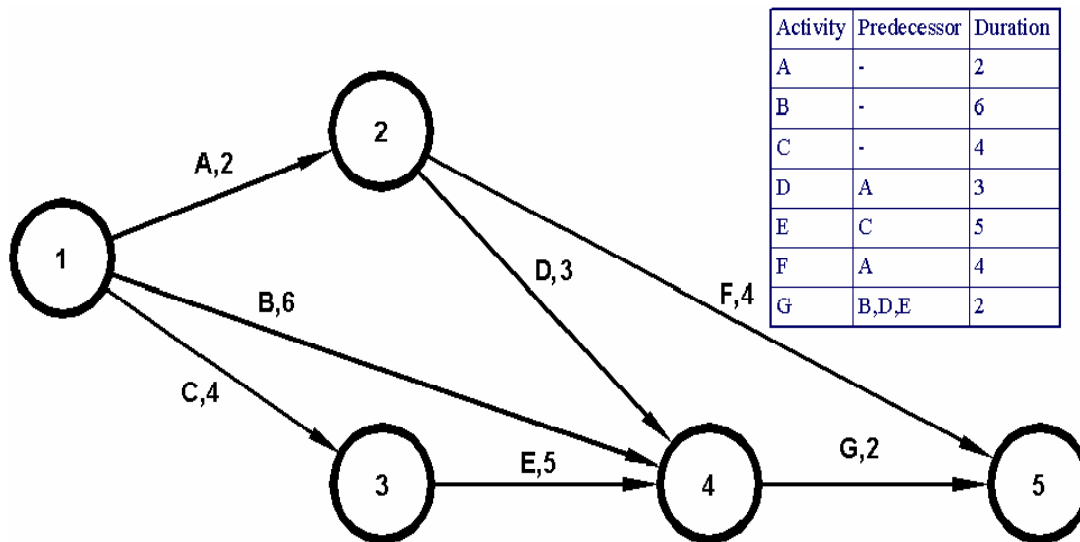
40) A CPM diagram for a project shows five activities according to the following diagram. The minimum days required to finish the project is mostly near:

(A) 26

(B) 11

(C) 6

(D) 15



The Answers is B

There are 4 critical paths as free floats available:

Path 1: $1-2-5 = 6$

Path 2: $1-2-4-5 = 7$

Path 3: $1-4-5 = 8$

Path 4: $1-3-4-5 = 11$

So, the longest one shows the minimum duration for the project.

41) The hourly labor cost is a \$120 per worker for 8 hours in a workday. How many days required to complete the budget of \$8000 job if 4 workers are assigned to the job.

(A) 5

(B) 2

(C) 7

(D) 8

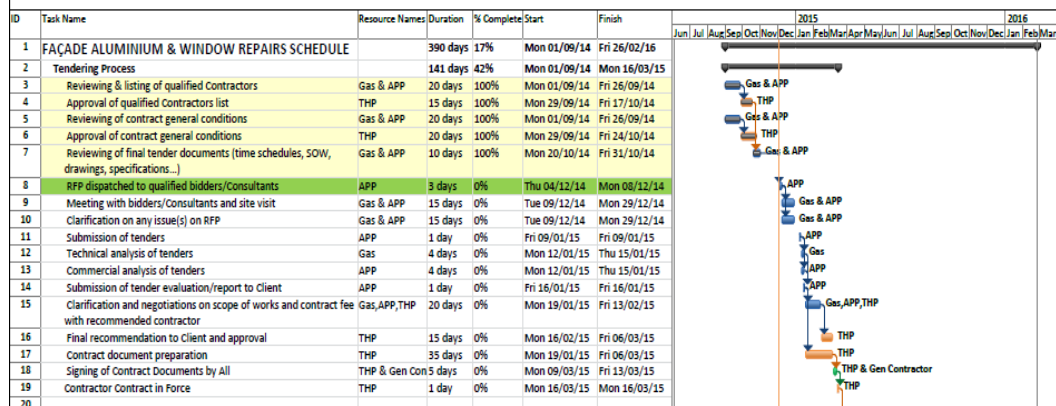
The Answers is B

Each labor should be paid daily for: $120 \times 8 = \$960$

The compensation for 4 workers is $= 960 \times 4 = \$3840$

For the budget of \$8000 the number of days is given equal to $= 8000/3840 = 2.083$ days

42) The following Gantt chart is given for a project. What type of Gantt chart is illustrated?



(A) Milestone and Timeline Gantt

(B) Gantt with dependencies

(C) Baseline Gantt

(D) All choices together

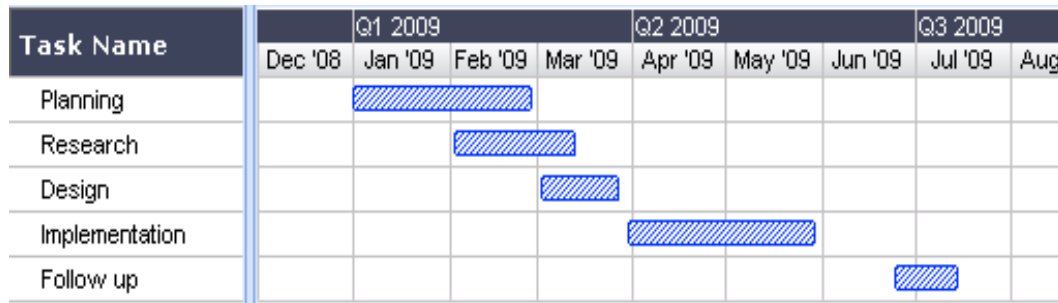
The Answer is D

A Gantt chart, commonly used in project management, is one of the most popular and useful ways of showing activities (tasks or events) displayed against time. On the left of the chart is a list of the activities and along the top is a suitable time scale. Each activity is represented by a bar; the position and length of the bar reflects the start date, duration and end date of the activity. This allows you to see at a glance:

- What the various activities are.
- When each activity begins and ends (dependencies).
- How long each activity is scheduled to last (time line).
- Where activities overlap with other activities, and by how much
- The start and end date of the whole project (milestones and base line).

So, this chart shows altogether, the correct answer is “D”.

43) A Gantt chart is given in the following figure. What type of Gantt chart is illustrated in the Figure?



(A) Milestone Gantt

(B) Gantt with dependencies

(C) Timeline Gantt

(D) Baseline Gantt

The Answers is C

This Gantt chart shows tasks against time. On the left of the chart is a list of the activities and each activity is represented by a bar which reflects the start date, duration and end date of the activity. So this is timeline Gantt chart and it cannot shows the dependencies and clear idea about the base line and milestones.

44) A dump-hauler has a purchase price of \$109,000. Freight for delivery is \$1000. Tires are an additional 21,000 with the estimated life time of 4500 hours. The hauler expected to operate 1500 hours annually and for 11 years. Maintenance fees for the hauler is estimated at \$18000. What is the before-tax estimated hourly cost of operation excluding operator labor cost?

(A) 35

(B) 41

(C) 24

(D) 15

The Answers is C

The best way to estimate the hourly cost is to find all expenditures and cost for a year, then the hourly cost can be estimated prorated:

The total hauler cost = $109000 + 1000 = \$110,000$

The hauler price per year: $110,000 / 11 = \$10,000$ per year

Tires will work 4500 hr and every year 1500 years of operation is expected so:
 $4500 / 1500 = 3$ years is the life time for the tires

Therefore, tire costs for a year is become: $21000 / 3 = \$7000$

Total annual expenditures = $\$10,000 + \$7,000 + \$18,000 = \$35,000$

Hourly rate = $35000 / 1500 \text{ hr/year} = 23.3$ say \$24.00

45) A project is described by the following precedence table. The project manager wants to decrease the normal project time by 4 days. Most nearly, how much will it cost to reduce the project completion time by three days?

Activity	Predecessors	Normal time (days)	Crash time	Normal cost daily	Crash cost daily
A	-	8	6	50	100
B	A	2	1	80	140
C	A	6	4	80	100
D	B	2	1	100	150
E	C	6	3	90	200
F	E	3	1	80	160
G	D,F	4	2	120	300

(A) 200

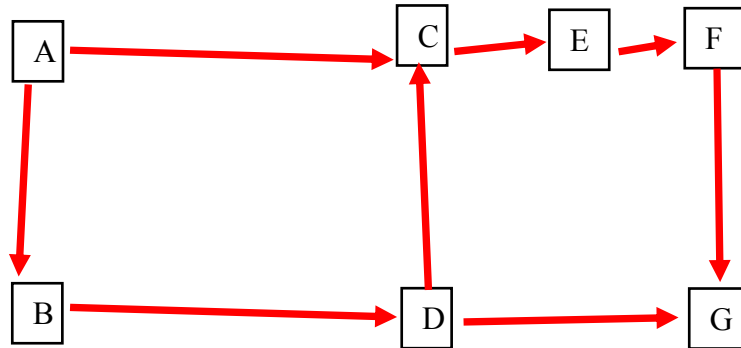
(B) 120

(C) 180

(D) 140

The Answers is D

According to the table the CPM chart represents the following free floats:



Free floats are:

Path 1: A-B-D-G= 16

Path 2: A-B-D-E-F-G=25

Path 3: A-C-E-F-G= 27 critical path

In order to reduce the overall project duration by 4 days, the most inexpensive operation is to allocate additional resources (crash) to activities C, D, and A.

For 4 days we can consider C for 2 days and A for the other 2 days. So

The critical path will be equal to $27 - 4 = 23$ days.

The additional costs will be equal to:

For C: $100 - 80 = \$20$ daily, $2 * 20 = \$40$ for two days

For A: $100 - 50 = \$50$ daily, $2 * 50 = \$100$ for two days.

The total additional cost due to the project crash = $100 + 40 = \$140$

46) A design firm uses unbalanced bidding to bid a four phase design project. The unbalanced bid results in phase billings of 35%, 40%, 15%, and 10% respectively. The billing roughly evenly placed along the design period. The increase in profit from this practice is mostly near:

- | | |
|-----------|----------|
| (A) 0% | (B) 5% |
| (C) 0.02% | (D) 1.2% |

The Answer is A

The unbalanced bidding is used by contractors and design firms to improve cash flow by distorting cost in certain periods. While the overall bid remains competitive by virtue of the total cost, the payments are structured to suit the firm's needs. It cannot increase the gross profit and it is just the breakdown of the profit. So no increase in profit will be expected which means 0%.

47) The amount of sales for contract has given equal to \$1,000,000 including \$100,000 for tax and \$70,000 for the insurance. At the end of the project contractor has owned \$50,000 of equipment and tools (assets) in addition \$400,000 assets that he does. Contractor paid \$650,000 for the man power, equipment, and materials for this project. Find the gross profit, operation profit, net profit, and return on assets.

Choice	Gross profit	Operational profit	Net profit	Return on assert
A	18%	40%	83%	35%
B	35%	83%	18%	40%
C	40%	18%	35%	83%
D	83%	35%	40%	18%

The Answers is B

Net income= $1000000 - 650000 - 100000 - 70000 = \180000

Total Assets = $\$50000 + 400000 = \450000

Return on assets=net income/total assets = $180000/450000 = 0.4 * 100 = 40\%$

Operation profit margin ratio = earnings before interest and taxes/sales= $(1000000 - 100000 - 70000)/1000000 = 0.83 * 100 = 83\%$

Net profit margin ratio=net income/sales = $180000/1000000 = 0.18 * 100 = 18\%$

Gross profit margin ratio = gross profit/sales= $(1000000 - 650000)/1000000 = 0.35 * 100 = 35\%$

48) You are doing a sidewalk project and your concrete crew is one foreman at \$50/hr and 2 laborers \$25/hr, they work 8 hours a day. The crew can lay down 46 yd³/day and must do 4238 ft³ for the job. What is the cost of the crew for this job?

- (A)\$2730
- (B)\$4238
- (C)\$1000
- (D)\$2047

The Answer is A

Answer "A" is correct:

Convert ft^3 to $\text{yd}^3 \rightarrow 4238 \text{ ft}^3 / 27 (\text{ft}^3/\text{yd}^3) = 157 \text{ yd}^3$

- Number of hours to complete project

$-157 \text{ yd}^3 / (46 \text{ yd}^3/\text{day}) = 3.4 \text{ days}$

$-3.4 \text{ days} * 8 \text{ hr/day} = 27.3 \text{ hours}$

- Cost of crew/hour = $\$50 + 2 * \$25 = \$100/\text{hr}$

- Total crew cost = $27.3 \text{ hours} * \$100/\text{hr} = \2730

49) The construction crew is made up of 1 foreman paid \$60/hour and 4 laborers at \$32/hour each. They are working on excavating a trench for future pipe laying. Since it is summer, they are working 10 hours per day however 2 hours is overtime meaning each crewmember gets paid time and a half. They are able to excavate 15 cubic yards per hour and need to excavate a total of 500 cubic yards. What is the cost of the crew for this job?

- (A)\$6203
- (B)\$9391
- (C)\$6824
- (D)\$4230

The Answer is C

Answer "C" is correct:

- Number of hours to complete project:

$$-500\text{yd}^3 / (15\text{yd}^3/\text{hour}) = 33.3 \text{ hours}$$

– Thus working 10-hour days this can be completed in 3 days and 3.3 non-overtime hours

- Cost of crew/normal hour = $\$60 + 4 * \$32 = \$188/\text{hr}$
- Cost of crew/overtime = $\$90 + 4 * \$48 = \$282/\text{hour}$

$$\text{Total Cost} = (8\text{hour} * 3 + 3.3\text{hours}) * 188/\text{hour} + (2\text{hours} * 3) * \$282/\text{hour} = \$6824.4$$

50) A university building is about to be built in Florida similar to another one on campus built in 1995 that cost \$4,000,000. The client wants to know how much it would cost to hire the necessary labor to build the project. The labor index for 1995 was 734.2 whereas the current labor index is 1129.3. Use the formula $C_2 = C_1 (I_2/I_1)$?

- (A) 4 million
- (B) 2.6 million
- (C) 6.2 million
- (D) 3.4 million

The Answer is C

Answer "C" is correct:

- Cost = 4.0 million * (1129.3/734.2) = 6.15 million

51) A university building is about to be built in Florida similar to another one on built in Europe 1995 that cost 5,000,000 Euros. The client wants to know how much it would cost to hire the necessary labor to build the project. The labor index for 1995 was 734.2 whereas the current labor index is 1129.3. The current exchange rate is 1 euro to \$1.5. Use the formula $C_2 = C_1 (I_2/I_1)$.

- (A) 7.69 million
- (B) 11.5 million
- (C) 6.15 million
- (D) 3.25 million

The Answer is B

Answer "B" is correct:

Cost = 5.0 million Euros * (1129.3/734.2) = 7.69 million Euros

7.69 million Euros * \$1.5/Euro = 11.5 million dollars

52) What is not a direct cost of construction?

- (A) Labor
- (B) Materials
- (C) Profit
- (D) Equipment
- (E) Subcontractors

The Answer is C

Answer “A”, “B”, “D” and “E” are not correct: Labor, materials equipment and subcontractors are all direct costs because they directly contribute to the act of constructing the project.

Answer “C” is correct: The profit is not a direct construction cost. It is a cost that the contractor charges the client as a compensation for performing the work.

- 53) A project was constructed in City B and cost in \$287,403. A new project is being built in city D. Given the following table and the formula $C_2 = C_1 (I_2/I_1)$, what is the cost of the proposed project.

Location	Index
City A	1,025
City B	1,170
City C	1,260
City D	1,105
City E	1,240

- (A)\$287,403
(B)\$271,436
(C)\$304,309
(D)\$276,349

The Answer is B

Answer "B" is correct: Using the table:

(Cost of project B/Index B) = (Cost of project D/Index D)

Cost of project D = $1,105/1170 * \$287,403 = \$271,436$

54) A 200 gallons per minute capacity pump costs \$35,000. The cost capacity factor is 0.8. What is a cost of a similar pump that has a capacity of 350-gallon per minute? Use the formula $C_2 = C_1 (Q_2/Q_1)^x$?

- (A)\$54,764
- (B)\$22,368
- (C)\$61,250
- (D)\$20,000

The Answer is A

Answer "A" is correct:

Remember that $C_2 = C_1 (Q_2/Q_1)^x = \$35,000 * (350/200)^{0.8} = \$54,764$

55) What is true about a definitive estimate?

- A) Prepared from completed plans and specifications
- B) Forecasts the project costs by combining the conceptual and detailed information
- C) Prepared early before design completion
- D) Estimate of the direct and indirect costs

The Answer is B

Answer “B” is correct: A definitive estimate is a forecast of the project cost within the allowable limits using a combination of conceptual and detailed information that comes from partial contracts and other procurement awards.

56) A client is working on building a 2000 SF restaurant for which they want to estimate the cost. This restaurant is very similar to a 4500 SF restaurant down the road that was built in 1993 and cost 2.4 million. The cost capacity factor is 0.7. The current BCI is 5532% and the BCI for 1993 was 2994%. What is the estimated cost of the new restaurant using the equation using the formula $C_2 = C_1 (I_b / I_a) (Q_2 / Q_1)^x$?

- (A) 4.4 million
- (B) 5.4 million
- (C) 4.2 million
- (D) 7.8 million

The Answer is D

Answer “D” is correct: Recall that $C_2 = C_1 (I_b / I_a) (Q_2 / Q_1)^x$

$$C_2 = 2.4 \text{ million } (5532 / 2994) (4500/2000)^{0.7}$$

$$C_2 = 7.8 \text{ million}$$

57) A lift station is being built on a concrete pad that is 4" thick and 20' x 15'. An additional 10% of concrete material is needed to cover waste and unforeseen circumstances. What volume of concrete is necessary for this slab?

- (A) 3.7 yd³
- (B) 4.07 yd³
- (C) 44.4 yd³
- (D) 48.8 yd³

The Answer is B

Answer "B" is correct:

$$\text{Volume} = (1/3' \times 20' \times 15') / 27(\text{ft}^3/\text{yd}^3) = 3.7 \text{ yd}^3$$

$$\text{Volume with Safety} = 3.7 + 3.7 * 10\% = 4.07 \text{ yd}^3$$

58) What is not included as a fixed material in the building cost index?

- (A) Portland Cement
- (B) General Labor
- (C) Structural Steel
- (D) Lumber

The Answer is B

Answer “B” is correct: In the building cost index, the fixed cost is skilled labor not general labor. In order to construct a building the BCI takes into account the skilled labor required including bricklayers, carpenters and ironworkers, as they are specific to building construction

59) In the preliminary phase in the construction of a school, the owner knows they want to have a 100,000 square foot building with enough parking to service their staff and parents. The budget for the project is 2 million for the building and parking. From previous projects they have basic estimates on file that it costs \$4500 per parking spot and \$360 to construct 90 square feet of a school building. How many parking spots will they be able to afford?

- (A)36
- (B)35
- (C)5
- (D)79

The Answer is B

Answer "B" is correct:

$$100,000 \text{ ft}^2 \text{ building} * (\$360/90\text{ft}^2) = 400,000$$

They have 2,000,000-400,000 left or 1,600,000 for parking spots.

$$\text{Thus } \$1,600,000 / (\$4,500/\text{spot}) = 35.6 \rightarrow \text{rounding down } 35$$

60) A 10 million gallon per day waste water treatment plant was constructed in 1986 and cost 20 million. Estimate the cost of a 17 million gallon per day wastewater treatment plant constructed in 2003 if the cost capacity factor is 0.4 and the $BCI_{1986} = 2301$ and the $BCI_{2003} = 3894$. Use the formula $C_2 = C_1 (Q_2/Q_1)^x$?

- A) 24.7 M
- B) 34.0M
- C) 16.2 M
- D) 11.8 M

The Answer is C

Answer "C" is correct:

$$C_{2003} = C_{1986} * (Q_{2003}/Q_{1986})^x = 20M * (17M \text{ GPD}/10M \text{ GPD})^{0.4} = 16.2 \text{ M}$$

61) Your company is making an estimate for a 6.5m high concrete retail store that is 2000 m². In your records there is building from 7 years ago that is 9m high, 3250 m² and cost 3.5 million dollars. 7 years ago the cost index was 115. If the cost index is currently 160, what is the cost estimate per cubic meter using the equation $C_2 = C_1 (I_2/I_1)$?

- A)\$166.3/m³
- B)\$119.65/m³
- C)\$179.49/m³
- D)\$249.04/m³

The Answer is A

Answer "A" is correct:

$$\text{Cost/m}^3_{-7 \text{ years}} = 3,500,000 / (9\text{m} * 3250 \text{ m}^2) = \$119.65/\text{m}^3$$

$$\text{Cost/m}^3_{\text{current}} = \$119.55 * (160/115) = \$166.3/\text{m}^3$$

62) A 150ft x 250ft room must be painted on every side. The walls are 8 feet high and need two coats of paint. From previous experience, your company has found that one gallon of paint covers 250 ft^2 . How many gallons of paint are required for this room?

- A)22 gal
- B)64 gal
- C)23 gal
- D)43 gal

The Answer is D

Answer "D" is correct:

$$\text{Area for painting} = 2(150' \times 8') + 2(250' \times 8') = 6400\text{ft}^2$$

$$\text{Paint}_{1 \text{ coat}} = 6400\text{ft}^2 / (300\text{ft}^2/\text{gal}) = 21.3$$

$$\text{Paint}_{2 \text{ coats}} = 2 \times 21.3 = 42.6 \text{ gallons}$$

63) You are a contractor that owns a dump truck that you plan on using to transport dirt from one location to another. You know how much dirt you want to haul, the production rates for the excavator and how much it costs to operate. How would you calculate the cost of the equipment for this specific job?

- A) $\text{Quantity} * \text{Production Rate} * \text{Equipment Rate}$
- B) $\text{Quantity} / (\text{Equipment Rate} * \text{Production Rate})$
- C) $(\text{Production Rate} / \text{Equipment Rate}) * \text{Quantity}$
- d) $\text{Equipment rate} * (\text{Quantity} / \text{Production Rate})$

The Answer is D

Answer "D" is correct:

In order to determine the cost of the equipment we must first determine the time in which it would take to complete the project – this information is found using the Quantity/Production rate. By multiplying the time by the equipment rate (\$/hour) you can determine the cost of the equipment for the specific job.

64) Which of the following not a cost included in the Unit price of the RS means?

- A) Materials
- B) Subcontractor
- C) Labor
- D) Equipment
- E) Overhead
- F) Profit

The Answer is B

Answer "B" is correct:

The cost to hire subcontractors varies significantly depending on the area, type of project, competition, relationship and situation. For this reason, the cost to hire a subcontractor is not included in the RS means as it is too variable. Instead the subcontractor cost is typically found by calling subcontractors in the area, describing the project's specific needs and getting a quote from them.

- 65) You are a contractor and have made a bid sheet for a construction project as follows. Excavation for this project is to occur in the winter meaning it will take more time for the crew to complete this task. You have calculated that this extra time will increase the labor cost by 27%. How much are you charging the client for the excavation work given the new situation?

RECAP SHEET FOR UNIT PRICE BID										
Project: <u>Holloman Taxiways and Aprons</u>				Bid Date: <u>August 9, 20--</u>				Estimator: <u>GAS</u>		
Item No.	Bid Item	Unit	Estimated Quantity	Labor Cost	Equipment Cost	Material Cost	Subcontract Cost	Direct Cost	Bid	
									Unit	Total
1	Clearing	l.s.	job	\$5,139	\$11,097	-	-	\$16,236	\$20,132.64	\$20,133
2	Demolition	l.s.	job	\$5,726	\$7,383	-	-	\$13,109	\$16,255.16	\$16,255
3	Excavation	c.y.	127,000	\$75,636	\$175,194	-	-	\$250,830	\$311,029.20	\$311,029
4	Base Course	ton	79,500	\$352,670	\$651,995	\$159,479	-	\$1,164,144	\$1,443,538.56	\$1,443,539
5	Concrete Pavement, 9"	s.y.	90,000	\$1,001,795	\$202,708	\$791,063	-	\$1,995,566	\$2,474,501.84	\$2,474,502
6	Concrete Pavement, 11"	s.y.	115,400	\$1,572,819	\$318,252	\$1,241,967	-	\$3,133,038	\$3,884,967.12	\$3,884,967
7	Asphalt Concrete Surface	ton	150	\$1,035	\$1,304	\$2,208	-	\$4,547	\$5,638.28	\$5,638
8	Concrete Pipe, 12"	l.f.	1,000	\$5,183	\$5,439	\$15,477	-	\$26,099	\$32,362.76	\$32,363
9	Concrete Pipe, 36"	l.f.	300	\$2,937	\$3,756	\$6,809	-	\$13,502	\$16,742.48	\$16,742
10	Inlet	ea.	2	\$111	\$164	\$801	-	\$1,076	\$1,334.24	\$1,334
11	Fiber Duct, 4-way	l.f.	600	\$3,278	\$10,367	\$5,126	-	\$18,771	\$23,276.04	\$23,276
12	Fiber Duct, 8-way	l.f.	1,200	\$9,035	\$20,576	\$16,095	-	\$45,706	\$56,675.44	\$56,675
13	Electrical Manhole	ea.	6	-	-	-	\$9,315	\$9,315	\$11,550.60	\$11,551
14	Underground Cable	l.f.	34,000	-	-	-	\$255,957	\$255,957	\$317,386.68	\$317,387
15	Taxiway Lights	ea.	120	-	-	-	\$45,710	\$45,710	\$56,680.40	\$56,680
16	Apron Lights	ea.	70	-	-	-	\$26,313	\$26,313	\$32,628.12	\$32,628
17	Taxiway marking	l.s.	job	\$3,560	\$675	\$4,683	-	\$8,918	\$11,058.32	\$11,058
18	Fence	l.f.	26,000	\$28,506	\$7,977	\$129,477	-	\$165,960	\$205,790.40	\$205,790
Totals				\$3,067,430	\$1,416,887	\$2,373,185	\$337,295	\$7,194,797		\$8,921,548
Factor = $\frac{\$8,921,548}{\$7,194,797} = 1.240$								Job Overhead	\$272,091	
									\$7,466,888	
								Markup, 15%	\$1,120,033	
									\$8,586,921	
								Bond	\$71,815	
									\$8,658,736	
								Sales Tax	\$259,762	
								Total Project Bid	\$8,918,498	

- A) \$271,252
 B) \$318,553
 C) \$336,352
 D) \$395,007

The Answer is C

Answer "C" is correct:

The new labor cost for the excavation = $\$75,636 + \$75,636 * 27\% = \$96,057.72$ The cost of the equipment stay the same, thus the direct cost of the excavation is = $\$96,057.72 + \$175,194 = \$271,251.72$.

The charge to the client is $\$271,251.72 * 1.24 = \$336,352.13$

- 66) You are a contractor and have made a bid sheet for a construction project as follows. You want to use the same subcontractor that is doing the apron lights for another project as they will give you \$50 off the cost of each light. In this other project 124 lights are to be installed. What will be the total direct cost for the lights?

RECAP SHEET FOR UNIT PRICE BID										
Project: <u>Holloman Taxiways and Aprons</u>				Bid Date: <u>August 9, 20--</u>				Estimator: <u>CAS</u>		
Item No.	Bid Item	Unit	Estimated Quantity	Labor Cost	Equipment Cost	Material Cost	Subcontract Cost	Direct Cost	Bid	
									Unit	Total
1	Clearing	l.s.	job	\$5,139	\$11,097	-	-	\$16,236	\$20,132.64	\$20,133
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3	Excavation	c.y.	127,000	\$75,636	\$175,194	-	-	\$250,830	\$311,029.20	\$311,029
4	Base Course	ton	79,500	\$352,670	\$651,995	\$159,479	-	\$1,164,144	\$1,443,538.56	\$1,443,539
5	Concrete Pavement, 9"	s.y.	90,000	\$1,001,795	\$202,708	\$791,063	-	\$1,995,566	\$2,474,501.84	\$2,474,502
6	Concrete Pavement, 11"	s.y.	115,400	\$1,572,819	\$318,252	\$1,241,967	-	\$3,133,038	\$3,884,967.12	\$3,884,967
7	Asphalt Concrete Surface	ton	150	\$1,035	\$1,304	\$2,208	-	\$4,547	\$5,638.28	\$5,638
8	Concrete Pipe, 12"	l.f.	1,000	\$5,183	\$5,439	\$15,477	-	\$26,099	\$32,362.76	\$32,363
9	Concrete Pipe, 36"	l.f.	300	\$2,937	\$3,756	\$6,809	-	\$13,502	\$16,742.48	\$16,742
10	Inlet	ea.	2	\$111	\$164	\$801	-	\$1,076	\$1,334.24	\$1,334
11	Fiber Duct, 4-way	l.f.	600	\$3,278	\$10,367	\$5,126	-	\$18,771	\$23,276.04	\$23,276
12	Fiber Duct, 8-way	l.f.	1,200	\$9,035	\$20,576	\$16,095	-	\$45,706	\$56,675.44	\$56,675
13	Electrical Manhole	ea.	6	-	-	-	\$9,315	\$9,315	\$11,550.60	\$11,551
14	Underground Cable	l.f.	34,000	-	-	-	\$255,957	\$255,957	\$317,386.68	\$317,387
15	Taxiway Lights	ea.	120	-	-	-	\$45,710	\$45,710	\$56,680.40	\$56,680
16	Apron Lights	ea.	70	-	-	-	\$26,313	\$26,313	\$32,628.12	\$32,628
17	Taxiway marking	l.s.	job	\$3,560	\$675	\$4,683	-	\$8,918	\$11,058.32	\$11,058
18	Fence	l.f.	26,000	\$28,506	\$7,977	\$129,477	-	\$165,960	\$205,790.40	\$205,790
Totals				\$3,067,430	\$1,416,887	\$2,373,185	\$337,295	\$7,194,797	\$8,921,548	
Factor = <u>\$8,921,548</u> = 1.240 \$7,194,797				Job Overhead				\$272,091		
				Markup, 15%				\$7,466,888		
								\$1,120,033		
				Bond				\$8,586,921		
								\$71,815		
				Sales Tax				\$8,658,736		
Total Project Bid				\$259,762						
				\$8,918,498						

- A) \$40,412
 B) \$50,110
 C) \$46,612
 D) \$57,798

The Answer is A

Answer "A" is correct:

The apron lights have a unit cost of $\$26,313/70 = \$375.90/\text{light}$

The future cost per light on the next project is $\$375.90 - \$50.00 = \$325.90/\text{light}$

Therefore the cost for 124 lights is $\$325.90 * 124 = \$40,411.6$

67) What is not a cost associated with the cost of labor?

- A) Social Security
- B) Fringe Benefits
- C) Wage Premiums
- D) Insurance
- E) None of the above

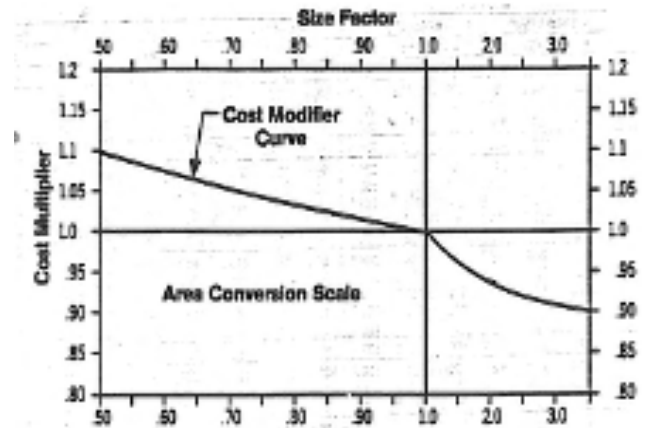
The Answer is E

Answer “E” is correct:

All of the following are associated with the cost of labor and are a costs that a company must pay when employing their workers.

- 68) You are thinking about making a nightclub that is 15,000 SF. Using the project size modification table (NCEES p172) and cost modifier curve, what is the predicted cost per SF?

Building Type	Median Cost per S.F.	Typical Size Gross S.F.
Apartments, Low Rise	\$ 54.05	21,000
Apartments, Mid Rise	68.25	50,000
Apartments, High Rise	78.30	310,000
Auditoriums	90.35	25,000
Auto Sales	55.90	20,000
Banks	121.00	4,200
Churches	81.60	9,000
Clubs, Country	81.40	6,500
Clubs, Social	79.15	10,000
Clubs, YMCA	81.60	28,300



- A) \$79.15/SF
 B) \$75.98/SF
 C) \$87.07/SF
 D) \$118.7/SF

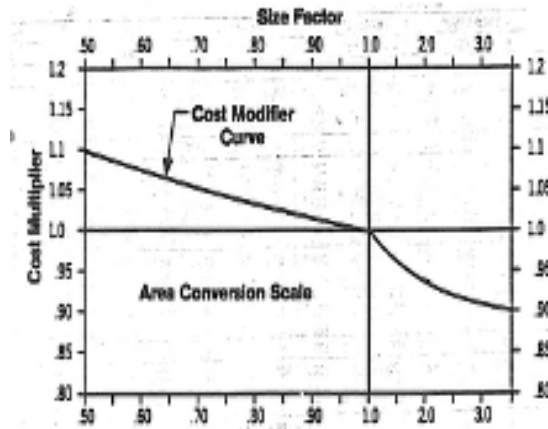
The Answer is B

Answer "B" is correct:

The proposed area of the club is 15,000 and the typical area shown by the table is 10,000. Therefore the size factor is $15,000/10,000$ or 1.5. Drawing a vertical line down from 1.5 to intersect cost modifier curve. Then a horizontal line from this point is drawn to the cost modifier, which is 0.96. Looking back at the table we can find a median cost of \$79.15 which can be adjusted by the cost modifier to 79.15×0.96 or \$75.98/SF

- 69) You are thinking about making a bank that is 8,400 SF. Using the project size modification table (NCEES p172) and cost modifier curve, what is the predicted cost per SF?

Building Type	Median Cost per S.F.	Typical Size Gross S.F.
Apartments, Low Rise	\$ 54.05	21,000
Apartments, Mid Rise	68.25	50,000
Apartments, High Rise	78.30	310,000
Auditoriums	90.35	25,000
Auto Sales	55.90	20,000
Banks	121.00	4,200
Churches	81.60	9,000
Clubs, Country	81.40	6,500
Clubs, Social	79.15	10,000
Clubs, YMCA	81.60	28,300



- A) \$121.00/SF
 B) \$75.98/SF
 C) \$135.52/SF
 D) \$118.70/SF

The Answer is B

Answer "B" is correct:

The proposed area of the office is 8,400 and the typical area shown by the table is 4,200. Therefore the size factor is $8,400/4,200$ or 2. Drawing a vertical line down from 2 to intersect cost modifier curve. Then a horizontal line from this point is drawn to the cost modifier, which is 0.925. Looking back at the table we can find a median cost of \$121.00 which can be adjusted by the cost modifier to $\$121 \times 0.925$ or \$118.70/SF

- 70) Given the following excerpt from the RS means what is the production efficiency of a crew working 7 days a week, 9 hours a day for 3 weeks?

General Requirements

R011

Overhead & Miscellaneous Data

R01100-110 Overtime

One way to improve the completion date of a project or eliminate negative float from a schedule is to compress activity duration times. This can be achieved by increasing the crew size or working overtime with the proposed crew.

cost chart based on a five, six, or seven day week with an eight through twelve hour day. Payroll percentage increases for time and one half and double time are shown for the various working days.

To determine the costs of working overtime to compress activity duration times, consider the following examples. Below is an overtime efficiency and

Days per Week	Hours per Day	Production Efficiency					Payroll Cost Factors	
		1 Week	2 Weeks	3 Weeks	4 Weeks	Average 4 Weeks	@ 1-1/2 Times	@ 2 Times
5	8	100%	100%	100%	100%	100 %	100 %	100 %
	9	100	100	95	90	96.25	105.6	111.1
	10	100	95	90	85	91.25	110.0	120.0
	11	95	90	75	65	81.25	113.6	127.3
	12	90	85	70	60	76.25	116.7	133.3
6	8	100	100	95	90	96.25	108.3	116.7
	9	100	95	90	85	92.50	113.0	125.9
	10	95	90	85	80	87.50	116.7	133.3
	11	95	85	70	65	78.75	119.7	139.4
	12	90	80	65	60	73.75	122.2	144.4
7	8	100	95	85	75	88.75	114.3	128.6
	9	95	90	80	70	83.75	118.3	136.5
	10	90	85	75	65	78.75	121.4	142.9
	11	85	80	65	60	72.50	124.0	148.1
	12	85	75	60	55	68.75	126.2	152.4

1 GENERAL REQUIREMENTS

- A)95%
B)90%
C)85%
D)80%

The Answer is D

Answer “D” is correct:

At the given days per week, hours per day and weeks worked the production efficiency of the crew would be 80%

71) What is not a type of preliminary or conceptual estimate?

- A) Cost Indices
- B) Bid Estimate
- C) Cost Capacity factor
- D) Parameter Cost

The Answer is B

Answer "B" is correct:

A bid estimate is a type of detailed estimate whereas the cost indices, bid estimate, and cost capacity factor are part of preliminary estimates.

72) You are replacing 12, 18" wide 2 use beams and 10, 24" wide 1-use beams. How much does labor cost for this project?

3 CONCRETE

03100 Concrete Forms & Accessories										
03110 Structural C.I.P. Forms			CREW	DAILY OUTPUT	LABOR HOURS	UNIT	2006 BARE COSTS			
							MAT.	LABOR	EQUIP.	TOTAL INCL O&P
300	0010	EXPANSION JOINT See division 03150-250								
405	0010	FORMS IN PLACE, BEAMS AND GIRDERS								
	0020	See also Elevated Slabs, Division 03110-240								
	0500	Exterior spandrel, job-built plywood, 12" wide, 1 use								
	0550	2 use								
	0600	3 use								
	0650	4 use								
	1000	18" wide, 1 use								
	1050	2 use								
	1100	3 use								
	1150	4 use								
	1500	24" wide, 1 use								
	1550	2 use								
	1600	3 use								
	1650	4 use								
	2000	Interior beam, job-built plywood, 12" wide, 1 use								
	2050	2 use								
	2100	3 use								
	2150	4 use								
	2500	24" wide, 1 use								
	2550	2 use								
	2600	3 use								
	2650	4 use								

- A)\$1
35.10
B)\$72
.60
C)\$75
.00
D)\$1
35.50

The Answer is A

Answer "A" is correct:

18", 2 use = $\$6.05 * 12 = \72.6

– 24", 1 use = $\$6.25 * 10 = \62.5

Total = $\$72.6 + \$62.5 = \$135.1$

12 –
10

73) As a project manager, you have calculated the CPI for your project to be 1.1, and you already know that you have spent \$85,000 on the completed work items. How much of the project budget had been allocated to these work items?

(A) \$85,000

(B) \$93,500

(C) \$71,000

(D) \$80,000

Answer is (C).

The cost performance index (CPI) is defined as follows:

$$\text{CPI} = \frac{\text{Budegeted Cost of Work Performed (BCWP)}}{\text{Actual Cost of Work Performed (ACWP)}}$$

Then, the BCWP can be calculated to be,

$$1.1 = \frac{\text{BCWP}}{\$85,000} \quad \rightarrow \quad \text{BCWP} = \$93,500$$

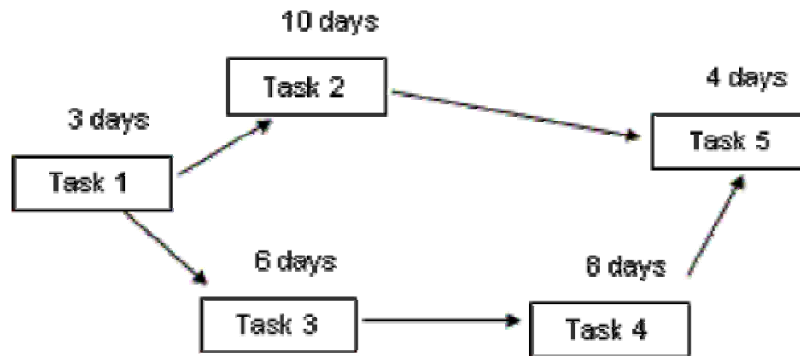
Short Cut

A $\text{CPI} > 1$ is always an indicator that you are performing under budget. So, what you have spent is less than what has been allocated in the budget. The only answer which is larger than \$85,000 is \$93,500.

74) In the activity network below, what is the total float of Task 5?

- (A) 0
- (C) 2

- (B) 1
- (D) 3



Answer is (A).

The complete solution to this problem requires calculating the ES, LS, EF, LF, and the float (as follows).

	ES	LS	EF	LF	Float
1	0	0	3	3	0
2	3	7	13	17	4
3	3	3	9	9	0
4	9	9	17	17	0
5	17	17	21	21	0

Therefore, task 5 has a total float of 0.

Short Cut

There is no need to calculate anything in this problem. Task 5 is the finishing task of this project and therefore, must be on the critical path. All tasks that are on critical path have always zero floats. Therefore, (A) is the correct answer.

75) A contractor has three active construction projects: A, B, and C. All three projects have expected durations of 36 months. However, project A has a duration standard deviation of 2 months, project B has a duration standard deviation of 3 months, and project C has a duration standard deviation of 4 months. Using the PERT technique, which project has a higher probability to be completed within ± 1 standard deviation of its expected duration?

- (A) A
- (C) C

- (B) B
- (D) All the same.

Answer is (D).

The Z score corresponding to +1 standard deviation of the expected duration for project A is:

$$z = \frac{(36 + 2) - 36}{2} = +1$$

The Z score corresponding to -1 standard deviation of the expected duration for project A is:

$$z = \frac{(36 - 2) - 36}{2} = -1$$

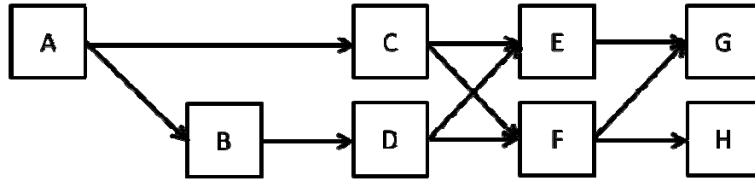
Therefore, for project A, we need to find $P(-1 \leq Z \leq +1)$. From the Z-table, this probability is 31.74%.

Similarly, for projects B and C, the z score corresponding to +1 and -1 standard deviation of the expected duration is calculated as +1 and -1, respectively. So, for projects B and C, this probability is also 31.74%.

Short Cut

There is no need to calculate anything in this problem. The probability that a measurement is within +/-1 standard deviation of its mean (expected value) is not a function of the expected value itself neither is it a function of the standard deviation value, because the Z score normalizes the ratio.

76) Given the diagram below and activity durations, what is the early finish time of activity E?



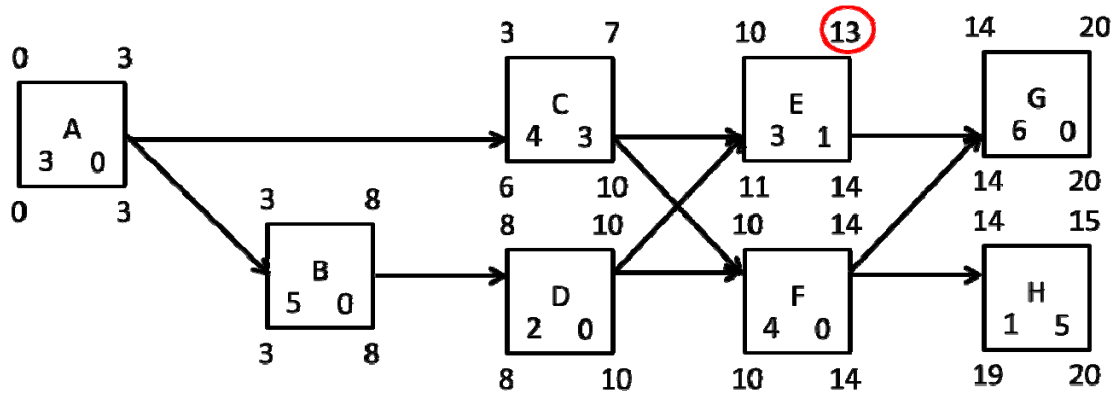
Activity	Duration
A	3
B	5
C	4
D	2
E	3
F	4
G	6
H	1

(A) 7
(C) 13

(B) 10
(D) 14

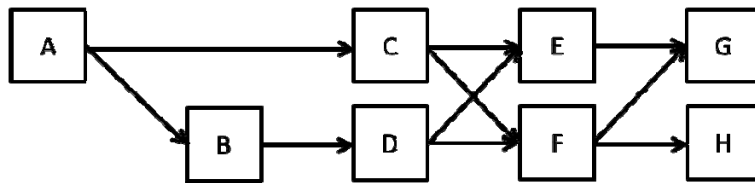
Answer is (C).

Step 1: Using forward and backward passes, calculate the ES, EF, LS, and LF of all activities, as shown below:



Step 2: Early finish time of activity E is marked on this network.

77) Given the diagram and activity durations, what is the total project duration?



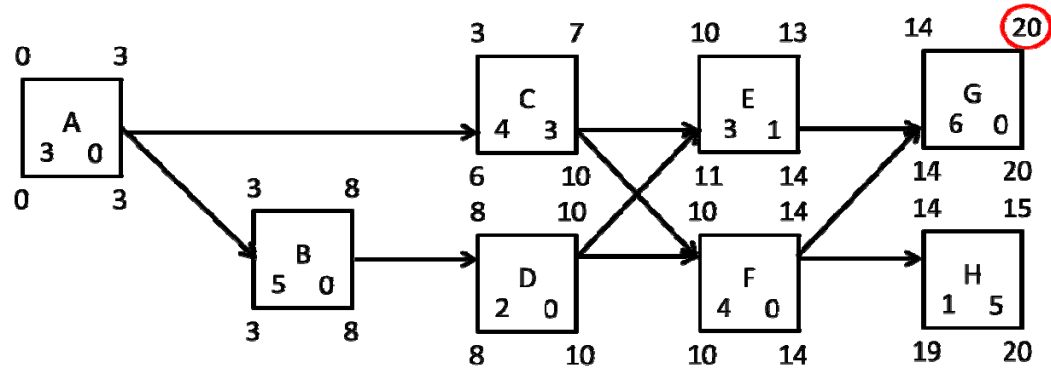
Activity	Duration
A	3
B	5
C	4
D	2
E	3
F	4
G	6
H	1

(A) 15
(C) 25

(B) 20
(D) 30

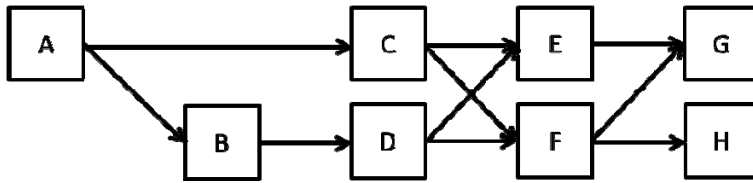
Answer is (B).

Step 1: Using forward and backward passes, calculate the ES, EF, LS, and LF of all activities, as shown below:



Step 2: Total project duration is the maximum EF value as marked on this network.

79) Given the diagram and activity durations, what is the critical path?



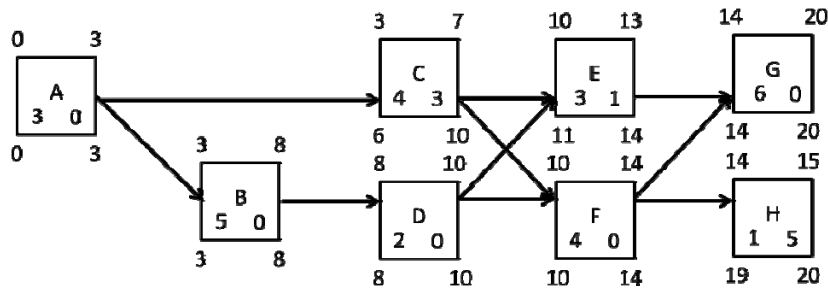
Activity	Duration
A	3
B	5
C	4
D	2
E	3
F	4
G	6
H	1

- (A) A-B-D-E-G
(C) A-B-D-F-H

- (B) A-B-D-F-G
(D) A-C-E-G

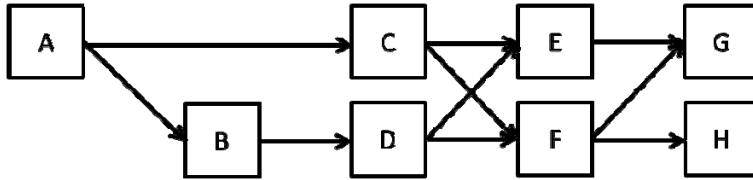
Answer is (B).

Step 1: Using forward and backward passes, calculate the ES, EF, LS, and LF of all activities, as shown below:



Step 2: Mark all activities that have zero float. These activities constitute the critical path.

80) Given the diagram and activity durations, what is the total and free float of activity C?



Activity	Duration
A	3
B	5
C	4
D	2
E	3
F	4
G	6
H	1

(A) 3, 3

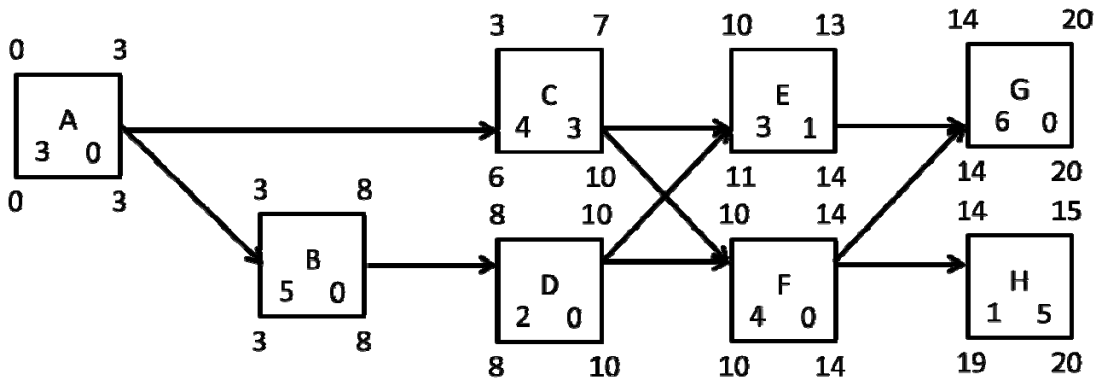
(B) 3, 4

(C) 4, 3

(D) 4, 4

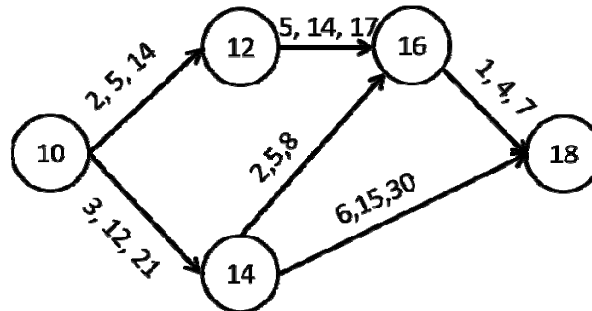
Answer is (A).

Step 1: Using forward and backward passes, calculate the ES, EF, LS, and LF of all activities, as shown below:



Step 2: The total float of activity C is $TF = LS - ES = 6 - 3 = 3$. The free float of activity is the maximum days it can be delayed without delaying any subsequent activity which is: $10 - 7 = 3$

81) Given the arrow diagram and activity durations, what is the probability that the project is completed by the end of day 23?



- (A) 10
(C) 14

- (B) 12
(D) 16

Answer is (D).

Step 1: For activity 10-12, we have:

$$t_e = \frac{a + 4m + b}{6} = \frac{2 + 4 \times 5 + 14}{6} = 6$$

$$\sigma_e = \frac{b - a}{6} = \frac{14 - 2}{6} = 2$$

$$v = \sigma_e^2 = 2^2 = 4$$

The same formulas can be applied to all other activities. Results are shown in the table below:

Activity	t_e	σ_e	v
10, 12	6	2	4
10, 14	12	3	9
12, 16	13	2	4
14, 16	5	1	1
16, 18	4	1	1
14, 18	16	4	16

Step 2: Doing the forward and backward pass calculations, the critical path passes through two activities: 10-14 and 14-18, with a total expected duration of $12+16=28$ days. The variance is $9 + 16 = 25$. Therefore,

$$\sigma_E = \sqrt{25} = 5$$

Step 3: The Z score corresponding to the probability of completing this project by the end of day 23 is:

$$Z = \frac{T - T_E}{\sigma_E} = \frac{23 - 28}{5} = -1.0$$

Based on the Z score table, the probability for $Z = -1.0$ is around 16%.

82) A crawler tractor with an expected life of 5 years has an initial cost of \$35,000 and at the end of each year it has a salvage value of \$5,000. What is the depreciation at year 2?

- (A) \$4,000
- (B) \$6,000
- (C) \$8,000
- (D) \$10,000

Answer is (C).

Step 1: Depreciation in each year (n) can be calculated as follows:

$$D_n = \frac{\text{Expected life} - \text{Year digit} + 1}{\text{Sum of years' digits}} \times \text{Amount to be depreciated}$$

$$\text{Sum of years' digits} = 1 + 2 + 3 + 4 + 5 = 15$$

$$D_n = \frac{5 - 2 + 1}{15} \times (35,000 - 5,000) = 8,000$$

83) An excavator cost \$40,000 and has a 4-year life time. Assuming that the machine will operates 2,000 hours during the year, and the repair cost for the second year is \$2.40 per hour, what is the lifetime repair cost?

- (A) \$24,000
- (B) \$28,000
- (C) \$30,000
- (D) \$32,000

Answer is (A).

Step 1: The Hourly repair cost is calculated as follows:

$$\text{Hourly repair cost} = \frac{\text{Year digit}}{\text{Sum of years' digits}} \times \frac{\text{Lifetime repair cost}}{\text{Hours operated}}$$

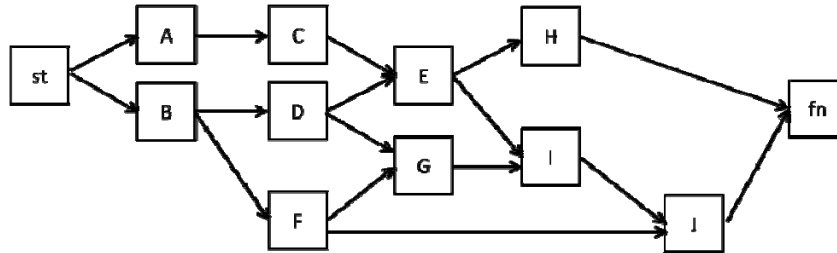
Therefore, the lifetime repair cost is:

$$\text{Lifetime repair cost} = \frac{\text{Hourly repair cost} \times \text{Sum of years' digits} \times \text{Hours operated}}{\text{Year digit}}$$

$$\text{Sum of years' digits} = 1 + 2 + 3 + 4 = 10$$

$$\text{Lifetime repair cost} = \frac{2.4 \times 10 \times 2000}{2} = \$24,000$$

84) Consider the network below for a construction project. In this project, some activities have known durations, and some has optimistic, most likely, and pessimistic durations. All of the durations are provided below. What is the expected duration of activities A and D?



Activities with Unknown Duration			
Activity	Optimistic	Most Likely	Pessimistic
A	8	10	12
C	3	5	7
D	10	12	14
G	13	15	17
H	8	10	12

Activities with Unknown Duration	
Activity	Duration
B	15
E	14
F	8
I	6
J	9

(A) 8, 10
(C) 5, 10

(B) 10, 12
(D) 10, 15

Answer is (B).

Step 1: For activities with optimistic, most likely, and pessimistic durations, the PERT formulations can be used to calculate the activity duration.

$$t_e = \frac{a + 4m + b}{6}$$

$$\sigma_e = \frac{b - a}{6}$$

$$v = \sigma_e^2$$

Step 2: The results are shown in the following table:

Activity	a	m	b	t_e	v
A	8	10	12	10	0.44
C	3	5	7	5	0.44
D	10	12	14	12	0.44
G	13	15	17	15	0.44
H	8	10	12	10	0.44

Therefore, for activities A and D duration values are 10 and 12, respectively.

Part 2: Means and Methods

55 Problems

1) A wall is 8'-4" high and 30'-2" long. $\frac{3}{4}$ " plywood sheathing is to be placed on the formwork. The sheathing is 3' wide vertically and 5' horizontally. What is the total amount of sheets of sheathing required for the entire wall?

- A.24
- B.8
- C.48
- D.16

The Answer is C

P347-16. ACI 347-04 Guide to Formwork for Concrete, 2004, American Concrete Institute, Farmington Hills, MI, www.concrete.org (in ACI SP-4, 7th edition appendix).

Sheets vertically: $8'-4\frac{1}{2}"/3' \text{ sheet} = 2.77' = 3'$, Use 3 sheets

Sheets horizontal: $30'-2\frac{1}{2}"/5' = 7.5'$, Use 8 sheets

Number of sheets per side, $3 \times 8 = 24$

Remember that the wall needs to have sheathing on both sides, thus double the number of sheets for the entire wall, 48 sheets

2) Given the following design for a wall, determine the maximum wall pressure of the design:

Cement Type: Type II without Retarders

Concrete Density = 140 pcf

Wall Height (h) = 13'-4"

Placement Rate (R) = 5ft/hr

Concrete Temperature (T) = 80F

A.712

B.854

C.562

D.704

The Answer is A

P 347-7. ACI 347-04 Guide to Formwork for Concrete, 2004, American Concrete Institute, Farmington Hills, MI, www.concrete.org (in ACI SP-4, 7th edition appendix).

Step 1: Find Equation for Maximum Wall pressure that produces the least P

$$P_{\max} = C_w C_c [150 + 9000R/T]$$

Step 2: Determine C_w and C_c

Table 2.1—Unit weight coefficient C_w

Inch-pound version		SI version	
Unit weight of concrete	C_w	Density of concrete	C_w
Less than 140 lb/ft ³	$C_w = 0.5[1 + (w/145 \text{ lb/ft}^3)]$ but not less than 0.80	Less than 2240 kg/m ³	$C_w = 0.5[1 + (w/2320 \text{ kg/m}^3)]$ but not less than 0.80
140 to 150 lb/ft ³	1.0	2240 to 2400 kg/m ³	1.0
More than 150 lb/ft ³	$C_w = w/145 \text{ lb/ft}^3$	More than 2400 kg/m ³	$C_w = w/2320 \text{ kg/m}^3$

Table 2.2—Chemistry coefficient C_c

Cement type or blend	C_c
Types I, II, and III without retarders*	1.0
Types I, II, and III with a retarder	1.2
Other types or blends containing less than 70% slag or 40% fly ash without retarders*	1.2
Other types or blends containing less than 70% slag or 40% fly ash with a retarder*	1.4
Blends containing more than 70% slag or 40% fly ash	1.4

*Retarders include any admixture, such as a retarder, retarding water reducer, retarding midrange water-reducing admixture, or high-range water-reducing admixture (superplasticizer), that delays setting of concrete.

Step 3: Calculate

$$P = (1)(1) * [150 + 9000R/T]$$

$$= 150 + 9000(5)/80 = 712 \text{ psf}$$

Intuitive way: Compare to Table of Maximum Pressures

**TABLE 5-4: MAXIMUM LATERAL PRESSURE
FOR DESIGN OF WALL FORMS**

Applies only for normal weight concrete made with Type I cement, no admixtures or pozzolans, slump no more than 4 in., and vibration depth limited to 4 ft or less.

Rate of placement, R, ft per hr	p, maximum lateral pressure, psf, for temperature indicated					
	90F	80F	70F	60F	50F	40F
1	600 psf minimum governs					
2						
3						
4			664	750	870	1050
5	650	712	793	900	1050	1275
6	750	825	921	1050	1230	1500
7	850	938	1050	1200	1410	1725
8	881	973	1090	1246	1466	1795
9	912	1008	1130	1293	1522	1865
10	943	1043	1170	1340	1578	1935

NOTE: Do not use design pressures in excess of 150 x height of fresh concrete in forms.

3) Determine the maximum lateral pressure on a 15 ft high formwork of freshly placed concrete having a unit weight of 135 lb/ft^3 .

- A. 1323 lb/ft^2
- B. 1000 lb/ft^2
- C. 2025 lb/ft^2
- D. 9 lb/ft^2

The Answer is C

P 347-7. ACI 347-04 Guide to Formwork for Concrete, 2004, American Concrete Institute, Farmington Hills, MI, www.concrete.org (in ACI SP-4, 7th edition appendix).

Maximum lateral pressure is determined using:

$$p = wh$$

$$p = (135\text{lb/ft}^3)(15\text{ft}) = 2025\text{lb/ft}^2$$

- 4) Determine the maximum lateral pressure given the following design conditions

Wall Height = 10'-6"

Placement Rate = 6 ft/hour

Concrete Temperature = 75F

$C_w = 1$

$C_c = 1$

A. 870 lb/ft²

B. 722 lb/ft²

C. 720 lb/ft²

D. 957 lb/ft²

The Answer is A

P 347-7. ACI 347-04 Guide to Formwork for Concrete, 2004, American Concrete Institute, Farmington Hills, MI, www.concrete.org (in ACI SP-4, 7th edition appendix).

that produces the least pressure

$$P_{\max} = C_w C_c [150 + 9000R/T]$$

Step 2: Calculate

$$P = (1)(1) * [150 + 9000(6)/75] = 870$$

- 5) Determine the maximum lateral pressure given the following design conditions

Wall Height = 3'

Placement Rate = 8 ft/hour

Concrete Temperature = 80F

$C_w = 1$

$C_c = 1$

A. 972.5 lb/ft²

B. 450 lb/ft²

C. 2000 lb/ft²

D. 1050 lb/ft²

The Answer is B

P 347-7. ACI 347-04 Guide to Formwork for Concrete, 2004, American Concrete Institute, Farmington Hills, MI, www.concrete.org (in ACI SP-4, 7th edition appendix).

Step 1: Find Equation for Maximum Wall pressure that yields the lowest P

Compare results of 3 equations-

1. Since R is 7 -10 fph: $P_{\max} = C_w C_c [150 + 43,400/T + 2,800R/T]$

2. $P = 150h$

3. $P = 2000\text{psf}$

Step 2: Calculate

1. $P = (1)(1) * [150 + 43,400/80 + 2800(8)/80] = 972.5 \text{ psf}$

2. $P = 150 * 3' = 450 \text{ psf}$

3. 2000psf

Step 3: Determine Lowest

Lowest comes from 150h, therefore answer is 450 psf

- 6) Determine the maximum lateral pressure given the following design conditions

Wall Height = 12'-6"

Placement Rate = 12 ft/hour

Concrete Temperature = 75F

$C_w = 1$

$C_c = 1$

A. 2000 lb/ft²

B. 1590 lb/ft²

C. 1875 lb/ft²

D. 1065 lb/ft²

The Answer is C

P 347-7. ACI 347-04 Guide to Formwork for Concrete, 2004, American Concrete Institute, Farmington Hills, MI, www.concrete.org (in ACI SP-4, 7th edition appendix).

If the placement of concrete exceeds a rate of 10ft/hour the lateral pressure is equal to 150h.

Therefore, $150(12.5) = 1875\text{psf}$

7) Which of the following is not a dead load?

- (A) Structure weight
- (B) Walls
- (C) Rood
- (D) People

The Answer is D

Dead Loads	Variable Loads
Fixed in magnitude and position Examples: Loads that will not be moved in the future	Varies in magnitude and position during the construction process Examples: Temporary storage of construction materials

*From Fundamentals of Building Construction: Materials and Methods Wiley

People are not a dead load because we are always moving and the load does not remain constant over time.

8. A contractor is building a concrete to separate a house from a nearby golf course. The wall is to measure 3 ft $\times$ 10 ft $\times$ 45 ft (width $\times$ height $\times$ length). When mixing the concrete, they will use Type I cement without any added admixtures. The concrete is designed to have a 28- day compressive strength of 4000 psi and a 5.5-inch slump. The concrete will be pumped into the form at a rate of 15 cubic yards per hour and will have a temperature of 80F. What is the maximum pressure at the base of the form?

- (A) 524 psf
- (B) 600 psf
- (C) 488psf
- (D) 1500psf

The Answer is B

Using p347-7 ACI 347-04 Guide to Formwork for Concrete, 2004, American Concrete Institute, Farmington Hills, MI, (in ACI SP-4, 7th edition appendix).

The rate of concrete placement:

$$R = \frac{15.0 \text{ cy/hr}}{45\text{ft} \times 3\text{ft}(1 \text{ cy} / 27\text{cu ft})} = 3\text{ft/h}$$

a- Using Equations:

If rate of placement $< 7 \text{ ft (2 m)}$ per hour:

$$C_c = (150 + 9,000 R/T)$$

Making sure that $\rightarrow 600\text{psf} \leq C_c \leq 2000 \text{ psf}$

$$C_c \leq 150 \text{ h}$$

R = rate of concrete placement, ft/h ; and

T = the temperature of concrete in the forms, °F (°C)

$$C_c = \left(150 + 9,000 \frac{R}{T}\right) = \left(150 + 9,000 \times \frac{3}{80}\right) = 487.5 \text{ psf}$$

Since $487.5\text{psf} \leq 600\text{psf}$, then the maximum pressure is 600psf

*See Fundamentals of Building Construction: Materials and Methods Wiley

9. Using ASCE 37-02, what is the minimum concentrated load for the wheel of a powered vehicle?

- (A) 250 lb
- (B) 8.90 kN
- (C) 55 lb
- (D) 2.22 Kn

The Answer is B

Action	Minimum Load ^a lb(kN)	Area load Application in.× in. (mm × mm)
Each person	250 (1.11)	12 × 12(300 × 300) ^b
Wheel of manually powered vehicle	50 (2.22)	Load divided by tire pressure ^c
Wheel of powered equipment	2000 (8.90)	Load divided by tire pressure ^c

ASCE 37-02 Design Loads on Structures During Construction, 2002, American Society of Civil Engineers, Reston, VA, www.asce.org.

10) A construction crew will use the slipform method to create a concrete structure that is: 3.25 ft at the base, 60 inch high and 4.5 inch at the top. The concrete mix will have Type I cement without any admixtures. The concrete is designed to have a 4000 psi 28-day compressive strength and a 4.5-inch slump. The concrete will be at a temperature (T) of 70° F. The concrete has a unit weight of 142 pcf and it will be placed at rate (R) of 7 ft/hr requiring additional vibration. What is the maximum lateral pressure C_C at the base of the structure?

- (A) 700 psf (B) 850 psf
(C) 750 psf (D) 192 psf

The Answer is C

p27. ACI 347-04 Guide to Formwork for Concrete, 2004, American Concrete Institute, Farmington Hills, MI, www.concrete.org (in ACI SP-4, 7th edition appendix).

During a slipform concreting the concrete lateral pressure is:

$$C_C = c + 6,000 R / T [\text{U.S.}]$$

Where:

$c = 150$ psf for concrete that requires additional vibration

$R =$ the rate of concrete placement, ft/h (m/h);

$C_C =$ lateral pressure, psf (kPa);

and

$T =$ the temperature of concrete in the forms, °F(°C).

$$C_C = c + 6,000 R/T = 150 \text{ psf} + 6,000 \times 7\text{ft/h}/70 = 750 \text{ psf}$$

*Taken from Basics of Engineering Economy, Pearson Prentice Hall

11. A roof is to be sloped 8 inches per foot, what is the reduction in gravity loads for personnel and equipment on this roof?

- (A) 0.90
- (B) 0.80
- (C) 0.60
- (D) 0.50

The Answer is B

p15 ASCE 37-02 Design Loads on Structures During Construction, 2002, American Society of Civil Engineers, Reston, VA, www.asce.org.

The reduction factor $R = 1.2 - 0.05F$

Where F is defined as the slope of the roof (inches per foot)

Therefore $R = 1.2 - 0.05(8) = 0.8$

12. A reinforced concrete wall 2 ft $\times$ 20 ft $\times$ 60 ft (thickness $\times$ height $\times$ length) will be constructed using Type I cement without any admixture added. The concrete has a 28- day compressive strength of 3500 psi and a 4-inch slump. The concrete will be delivered at a rate of 12 cubic yards per hour and the concrete has a temperature of 650 F. The maximum pressure at the base of the form is most nearly:

- (A) 524 psf
- (B) 600 psf
- (C) 645 psf
- (D) 945 psf

The Answer is B

The rate of concrete placement:

$$R = \frac{12.0 \text{ cy/hr}}{60\text{ft} \times 2\text{ft}(1 \text{ cy} / 27\text{cu ft})} = 2.70\text{ft/h}$$

a- Using Equations:

For walls:

Rate of placement less than 7 ft (2 m) per hour, therefore:

$$C_C = (150 + 9,000 R/T)$$

$$C_{C \text{ SI}} = 7.2 + \frac{785 R_{\text{SI}}}{T_{\text{SI}} + 17.8}$$

With the following limitations $\rightarrow 600\text{psf} \leq C_C \leq 2000 \text{ psf}$

$$C_C \leq 150 \text{ h}$$

$R (R_{\text{SI}})$ = rate of concrete placement, ft/h ; and

$T (T_{\text{SI}})$ = the temperature of concrete in the forms, °F (°C)

$$C_C = (150 + 9,000 R/T) = (150 + 9,000 \times 2.70/65) = 523.85 \text{ psf}$$

Check limitations:

$$\text{i) } 600\text{psf} \leq 523.85\text{psf} \leq 2000\text{psf} \quad \therefore \text{NG} \quad \text{USE } 600\text{psi}$$

$$\text{ii) } 523.85 \text{ psf} \leq 150 \text{ h} \leq 150 \times 20 \text{ ft} = 3000 \text{ psf} \quad \therefore \text{OK}$$

b- Using Table E-5:

$R \rightarrow$ rate of concrete placement, ft/h = 2.70 ft/h

$T \rightarrow$ the temperature of concrete in the forms °F = 65° F

Concrete $\rightarrow$ type I cement, 4 – inch slump and a unit weight of 150 pcf

$\therefore$ Table E – 5 can be used, @ 3.0 ft/h and 65° F $\Rightarrow$ 600psi

13- Per ASCE 37-02, the minimum concentrated personnel load for each person including the equipment is:

- (A) 150 lb
- (B) 175 lb
- (C) 200 lb
- (D) 250 lb

The Answer is D

Action	Minimum Load ^a lb(kN)	Area of load Application in. × in. (mm × mm)
Each person	250 (1.11)	12 × 12 (300 × 300) ^b
Wheel of manually powered vehicle	50 (2.22)	Load divided by tire pressure ^c
Wheel of powered equipment	2000 (8.90)	Load divided by tire pressure ^c

SOURCE: Design Loads on Structures During Construction, ASCE37-02, reprinted with permission

a. Use actual loads when they are larger than the tabulated here.

b. Need not less than 18 in. (457 mm) c. to c.

c. For hard rubber tires, distribute load over an area lin. (25 mm) by the width of the tire.

14. A reduction of gravity construction loads for personnel and equipment is permitted per ASCE 37-02. What would be the reduction for a sloped roof 6 inches per foot?

- (A) 0.90
- (B) 0.70
- (C) 0.60
- (D) 0.50

The Answer is A**§4.8.3.3 Personnel and Equipment Loads on Sloping Roofs**

A reduction in gravity construction loads for personnel and equipment on a roof is also permitted based upon the slope of the roof. The reduction factor

R is:

$$R = 1.2 - 0.05F$$

Where F is the slope of the roof expressed in inches per foot (in SI system $F = 0.12 \times$ slope of the roof expressed in percentage points). R need not exceed 1.0 and shall not be less than 0.6. This reduction may be combined by multiplication with the reduction based on area, but the reduced load shall not be less than 60% of the basic unreduced load.

15. The major difference between the fixed material loads (FML) and variable materials loads (VML) in construction industry is:

- (A) In fact there is no difference between the two construction loads
- (B) VML is the load from materials that varies in magnitude during the construction process while FML is the load from materials that do not vary in magnitude during construction
- (C) FML is always large than VML
- (D) VML is larger than FML

The Answer is B

Table E - 1 Fixed and Variable Material load

Fixed Material Loads (FML)	Variable Material Loads (VML)
Fixed in magnitude Examples: Loads that are placed In their final end use position.	Varies in magnitude during The construction process Examples: Stockpiling of any material (scaffold, forms, rebar)

The FML is the load from materials that is fixed in magnitude. The VML is the load from materials that varies in magnitude during the construction process. If the local magnitude of a material load varies during the construction process, then that load must be considered a VML.

16. A reinforced concrete wall 1 ft x 12 ft x 60 ft (thickness x height x length) will be constructed using Type I cement without any admixture added. This normal weight concrete has a 3000 psi 28-day compressive strength and a 4-inch slump. The concrete will be pumped from the base (to prevent segregation) of the wall at a rate of 6 cubic yards per hour and the concrete has a temperature of 70°F. The concrete has a unit weight of 145 pcf.

The maximum pressure at the base of the form is most nearly:

- (A) 2175 psf
- (B) 2275 psf
- (C) 2475 psf
- (D) 2685 psf

The Answer is A

§4.7.1.3 of ASCE37-02 is applicable since concrete is pumped from the base to prevent concrete segregation

$$C_C = W \times h \times 1.25 \quad [\text{U.S.}]$$

$$C_{C\text{ SI}} = 23.5 h_{\text{SI}} \times 1.25 \quad [\text{SI}]$$

$$C_C = W \times h \times 1.25 = 145 \text{ pcf} \times 12 \times 1.25 = 2,175 \text{ psf}$$

Note: Maximum and minimum values given for other pressure formulas do not apply to Equations E-5.

17. A reinforced concrete barrier to be constructed in a median of a freeway using slip form construction method. The concrete barrier is Type 60G with the following dimensions: 2 ft at the base, 57 inch high and 6 inch at the top.

The barrier will be constructed using Type I cement without any admixture added. This normal weight concrete has a 3000 psi 28-day compressive strength and a 4-inch slump. The concrete will be placed in an 10-inch lift with slight vibration and the concrete has a temperature of 65° F. The concrete has a unit weight of 150 pcf and it will be placed at rate of 6 ft/hr.

The maximum pressure at the base of the slip form is most nearly:

- | | |
|-------------|-------------|
| (A) 450 psf | (B) 530 psf |
| (C) 620 psf | (D) 655 psf |

The Answer is D

For a slip form concreting operation, the lateral pressure of fresh concrete to be used in designing the forms, bracing, and wales shall be calculated as:

$$C_C = c + 6,000 R / T \quad [\text{U.S.}]$$

$$C_{C\text{ SI}} = C_{\text{SI}} + \frac{524 R_{\text{SI}}}{T_{\text{SI}} + 17.8} \quad [\text{SI}]$$

Where:

$C(C_{\text{SI}})$ = 100 psf(4.79 kPa) for concrete placed in 6 to 10-in. (150 to 250- mm)
lift with slight vibration or no revibration
= 150 psf (7.19 kPa) for concrete that requires additional vibration,

such as gastight or containment structures;

C_C ($C_{C\text{ SI}}$) = lateral pressure, psf (kPa);

R (R_{SI}) = the rate of concrete placement, ft/h (m/h); and

T (T_{SI}) = the temperature of concrete in the forms, °F(°C).

$$C_C = c + 6,000 R/T = 100 \text{ psf} + 6,000 \times 6\text{ft/h}/65 = 653.85 \text{ psf}$$

18. Which of the following is not a non-destructive weld testing method?

- (A) X-ray weld inspection
- (B) Acoustic Emission
- (C) Eddy current test
- (D) Guided bend tests

The Answer is D

The guided bend test is a destructive testing method which involves the demolition of a completed weld to determine its characteristics. The other choices (A-C) are non-destructive methods that do not involve breaking existing welds.

19. Which of the following is not an example of a quality assurance activity?

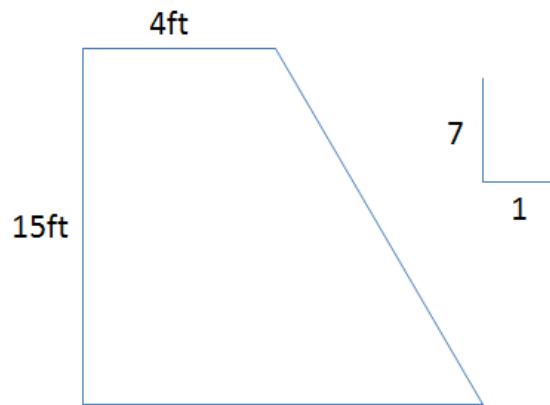
- (A) Reviewing and approving shop drawings
- (B) Evaluate the contractor's schedule for the project
- (C) Observe field sampling or testing that occurs on site
- (D) All of the above

The Answer is D

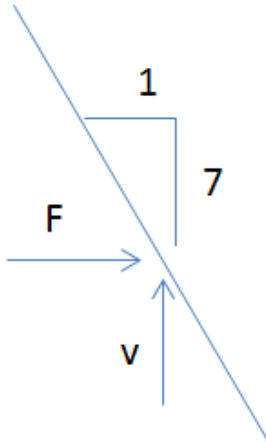
The answers A-C are all types of quality assurance methods.

20- A retaining wall shown in the figure below is to be constructed for the project. The wall is to be 15-ft high and sloped on one side at a ratio of 1:7. The concrete pressure in the form is to have a maximum pressure of 2,000 psf. It will take less than an hour to pour the 135 pcf concrete.

What is the uplift force the form will be subjected to?



- (A) 0 lb/ft
- (B) 3750 lb/ft
- (C) 2143 lb/ft
- (D) 467 lb/ft

The Answer is C

The form's vertical force (uplift) is:

$$V = \frac{F}{7} = \frac{\left(\frac{1}{2}\right)(2000)(15)}{7} = 2,143 \text{ lb/ft}$$

21- A contractor is working on a deep trench in which they are installing a water line. Instead of using the typical vertical planks, the contractor is looking to use plywood instead. Which of the following is true regarding this situation?

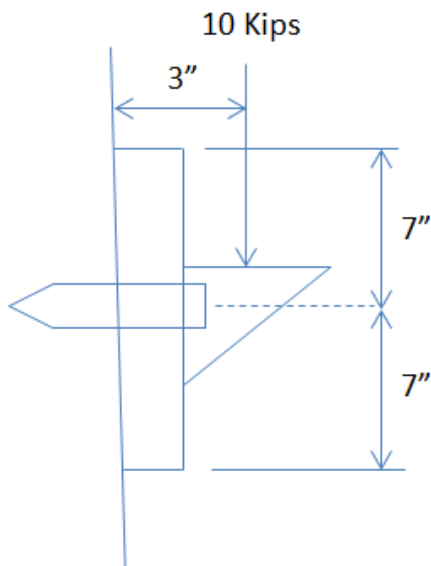
- (A) Any plywood is able to be substituted to only specific types of planks
- (B) Plywood can only be substituted if the plywood is a 1" thick
- (C) Plywood should not substituted for vertical planks
- (D) Plywood can be used if 3 layers are used

The Answer is C

Subpart P – Excavations, OSHA Occupational Safety and Health Standards for the Construction Industry, 29 CFR Part 1926 (US federal version), US Department of Labor, Washington, DC.

The vertical planks are chosen using tables in Appendix C (CFR 1926, Subpart P, Appendix C). However, plywood cannot be used instead of the uprights.

22- The following diagram is of a bracket screwed into a concrete wall through an imbedded insert. What is the tension in the bolt?



- (A) 4.29 kips (B) 23.3 kips (C) 2.14 kips (D) 32.7 kips

The Answer is A

Solve for tension (T) by summing the moment at the bottom of the bracket:

$$\sum M = 0$$

$$(10 \text{ kips})(3 \text{ in.}) - T(7 \text{ in.}) = 0$$

$$T = \frac{(10 \text{ kips})(3 \text{ in})}{7} \text{ in.} = 4.29 \text{ kips}$$

23- Which of the following is not an example of a construction load as determined by the SEI/ASCE 37-02 Standards?

- (A) Dead load from temporary structures
- (B) Erection and fitting forces
- (C) Personnel and equipment loads
- (D) Thermal loads

The Answer is D

p. 7, ASCE 37-02 Design Loads on Structures During Construction, 2002, American Society of Civil Engineers, Reston, VA, www.asce.org.

Thermal loads are indicated in SEI/ASACE 37-02 Standards to be an environmental not a construction load.

24- Which of the following terms is defined as “a factor that accounts for the degree of hazard to human life and damage to property”

- (A) Building Factor
- (B) Importance Factor
- (C) Design Factor
- (D) Environmental Factor

The Answer is B

p. 25, §6.1 Importance Factor, ASCE 37-02 Design Loads on Structures During Construction, 2002, American Society of Civil Engineers, Reston, VA, www.asce.org.

The definition is that of the importance factor.

25- A project is to take 1.5 years to complete. The design wind speed for the construction load calculations equals the basic wind speed x the factor _____?

- (A) 0.75
- (B) 0.8
- (C) 0.85
- (D) 0.9

The Answer is C

§6.2.1 Design Velocity, p. 25, ASCE 37-02 Design Loads on Structures During Construction, 2002, American Society of Civil Engineers, Reston, VA, www.asce.org.

The factor for determining the design wind speed is determined using the following table:

Construction Period	Factor
Less than 6 weeks	0.75
6 weeks to 1 year	0.80
1 to 2 years	0.85
2 to 5 years	0.90

26- A building is to be designed so that the overturning movement created by lateral forces does not exceed $\frac{2}{3}$ multiplied by the _____ stabilizing moment. Fill in the blank:

- (A) Dead load
- (B) Live Load
- (C) Temporary Load
- (D) None of the above

The Answer is A

§2.3.3 Overturning and Sliding, p. 25, ASCE 37-02 Design Loads on Structures During Construction, 2002, American Society of Civil Engineers, Reston, VA, www.asce.org.

The paragraph on overturning and sliding says that the overturning moment “does not exceed two thirds of the dead load stabilizing moment unless the building structure is anchored to resist the excess moment.”

27- A project in the Midwest is to have a construction time of two years. Utilizing the ASCE 7-95, the snow loads during the construction period are to be calculated. In order to determine this load, what is the factor that is used to modify the ground snow load?

- (A) 0.75
- (B) 0.8
- (C) 0.85
- (D) 1.0

The Answer is B

§6.4.1 Ground Snow Loads, p. 25, ASCE 37-02 Design Loads on Structures During Construction, 2002, American Society of Civil Engineers, Reston, VA, www.asce.org.

Construction Period	Factor
5 years or less	0.80
More than 5 years	1.00

- 28) Which one of the following reasons is not the cause for the formwork failure?
- A) Inadequate bracings, temperature, rate, and vibrations
 - B) Improper stripping, unstable soil under mudsills, and shore removal
 - C) Poor quality detailing and insufficient nailing prorate to the construction weights
 - D) Heavy weight concrete

The Answer is D

Choice A, B, and C are the cause of many large accidents. But the heavy weight concrete is not the cause. In fact the formwork shall be designed based on the construction weights with the proper detailing. So, either heavy or light weight construction loads shall have a well-designed formwork, otherwise it will fail.

29) For a retaining wall with 9' height find the lateral pressure on the wall forms.

A) 150 psf

B) 1350 psf

C) 1500 psf

D) 1000 psf

The Answer is B

According to the ASCE 37-02 the lateral pressure from wet concrete is to be taken as the hydrostatic pressure:

$$C_c = w * h$$

30) Which load combination according to the ASCE 37-02 is not correct?

- A) 1.2 Dead Load+1.6 Variable load
- B) 1.4 Dead Load+1.2 Variable Load
- C) 1.2 Dead Load+1.4 Variable Load
- D) 1.2 Dead Load+1.4 Variable Load +1.6 Personnel and equipment Loads

The Answer is A

ASCE 37-02

According to the code A is not correct.

- 31) Horizontal construction loads means:
- A) Earthquake and wind loads
 - B) 2% of dead load, or 50lb per person at the level of the platform
 - C) 0.2times the fully loaded weight or 1 times the fully loaded weight of the wheeled vehicle transporting material
 - D) B&C are correct

The Answers is D**ASCE 37-02**

According to the code B& C are mandatory for the lateral design of the construction structures, but the A may or may not be considered.

32) What is the design live load for the construction of post tension slab in a 10 stories building.

- A) 20 psf
- B) 25 psf
- C) 50 psf
- D) 75 psf

The Answer is C

According to the load classification, 20psf is used for the sparsely populated with personnel with hand tools, 25 psf is used for the light weight constructions, 50psf is for staging materials for the average construction, and 75 psf is used for the heavy construction with motorized buggies and heavy equipment.

For the pre-stressed concrete slabs, it is expected for staging materials and equipment including jacks and it can be classified as average construction. So the 50 psf would provide enough safety confidence for this structure.

- 33) The following formwork should support the 22' by 20' slab. Find the live load from personnel and equipment acting temporary during construction on this formwork if it is classified as heavy duty construction.



- A) LL=75 psf
B) LL= 73 psf
C) LL= 37.5 psf
D) LL=70 psf

The Answer is B**ASCE 37-02**

Area of the formwork = $22 \times 20 = 440 \text{ sf} > 400 \text{ sf}$

So, the following reduction may be used for the actual live load on the form work:

$$C_p = L_o \left(0.25 + \frac{15}{\sqrt{A_I}} \right)$$

For the heavy duty construction $L_o = 75 \text{ psf}$

A_I = the influence area greater than $400 \text{ sf} = 440 \text{ sf}$

So, $C_p = 0.96 \times 75 = 72.4 \text{ psf}$

- 34) The Figure below shows a scaffolding for a bridge as a heavy construction. If the deck thickness of the bridge is given equal to 4', and the span length between scaffolding columns is given equal to 4', find the axial force in those columns for the dead and personnel and equipment combination. (No variable material on the scaffolding will be stored.)



- A) $P = 2400$ lbs B) $P = 13440$ lbs C) $P = 9600$ lbs D) $P = 1200$ lbs

The Answer is B

According to the load classification, the 75 psf live load is used for the heavy construction with motorized buggies and heavy equipment. There is no variable material on scaffolding so only the live load of the personnel and their equipment shall be considered.

The area on each column is less than 400sf so no reduction of live loads shall be considered.

Concrete density = 150 pcf

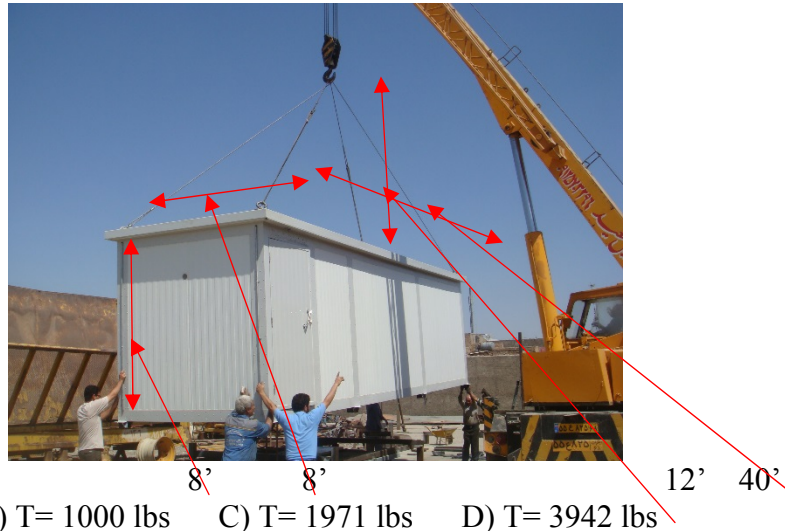
So, Dead load = $4 \times 150 = 600$ lb/sf

Live load = 75 lb/sf

$U = 1.2D + 1.6L = 1.2 \times 600 + 1.6 \times 75 = 840$ lb/sf

For the 4'x4' scaffolding grid the axial force will be equal to: $840 \times 4 \times 4 = 13440$ lbs

35) A symmetric prefabricated SCADA building (8ft wide x 40 ft long x 8 ft height) is lifted by a crane utilizing four cable as shown in the Figure below (at four corners). The cables are connected to the vertical hoisting cable at the point which is 15 ft directly above the center of gravity of the building. If the weight of the prefab building is given equal to 2 US tons, what is the most nearly the tension in the cables?



- A) $T = 4000$ lbs B) $T = 1000$ lbs C) $T = 1971$ lbs D) $T = 3942$ lbs

The Answers is C

According to the vectors' law:

$$T = \sqrt{F_x^2 + F_y^2 + F_z^2} = \text{Tension in the cable}$$

Equation of equilibrium:

$$\sum F_z = 0; \quad 4F_z = 2 * 2000 = 4000 \text{ lbs}$$

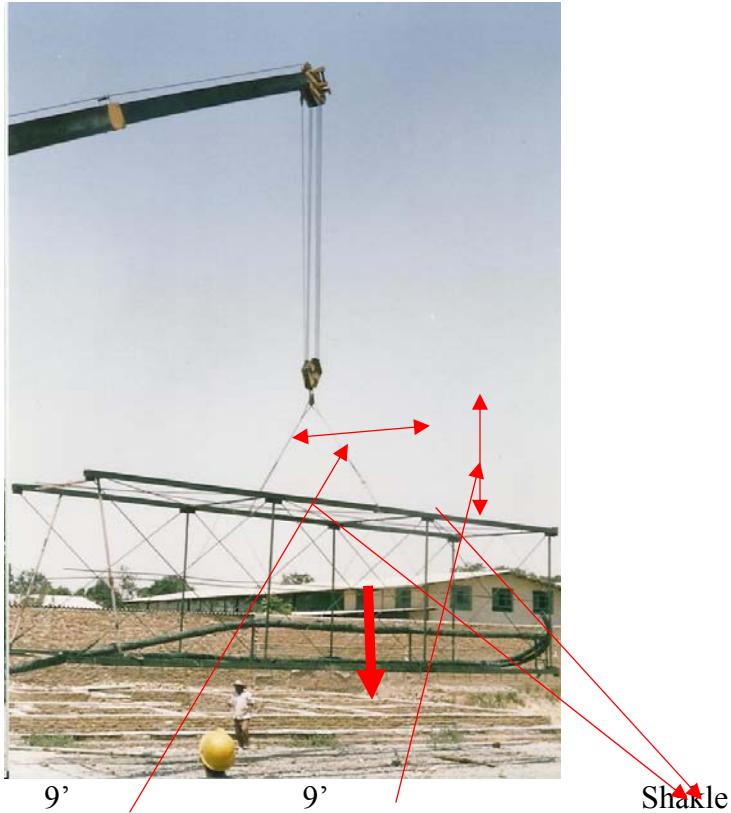
So, $F_z = 1000 \text{ lbs}$

Dimensions of the container represents the ratios for the force vector, so for the 40x8x8 feet container, the equivalent vector for each cable can be represented as:

$$T = \sqrt{T^2(20^2 + 4^2 + 12^2)} = 23.66T$$

$$T/F_z = 23.66T / 12T = 1.971 \text{ so, } T = 1.971 * F_z = 1.971 * 1000 = 1971 \text{ Lbs}$$

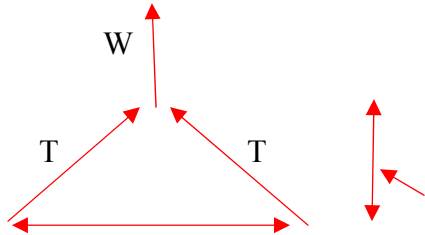
- 36) A prefabricated jacket shown below weighs 4000 lb. During erection it is lifted using 2 cables as shown. Find the working loads on shackle if we need the safety factor of 2?



- A) $T=1788$ lbs B) $T= 4000$ lbs C) $T= 2000$ lbs D) $T= 3578$ lbs

The Answer is D

Each cable should carry the half of the load, but the lining cable will increase the force inside the cable. So according to the vectors' law:



$T = \frac{W}{2} * \cos \phi$ So the measure of the angle between T and the vertical axis is 26.5 degree and so $T = \frac{4000}{2} * \cos 26.5 = 1788.85 \text{ lb}$

- 37) If the weight of the white car is 1.5 US tons, for lifting up by a crane like the left figure, find the minimum safe resistive moment for the crane to prevent of an accident shown in the right figure if the required lever arm is 30'. If the weight of crane is given equal to 10 tons, then find the required distance between cranes jacks on the ground



30'

- A) $M_R < 90,000 \text{ lb-ft}$ B) $M_R = 90,000 \text{ lb-ft}$
C) $M_R \geq 90,000 \text{ lb-ft}$ D) None of the answers

The Answer is C

The resistive moment should be equal to or bigger than the overturning moment.

$$M_{\text{overturning}} = \text{Force} * \text{Lever arm} = 1.5 * 2000 \text{ lbs} * 30' = 90,000 \text{ lb-ft}$$

So, to have the safe lifting the resistive moment (M_R) should be bigger than the overturning moment: $M_R \geq 90,000 \text{ lb-ft}$

- 38) A crane with the capacity of 100 tons when the crane boom stands at 80 degree and with the 90' length of boom. This crane should carry a truss and install it at the distance of 150' (below figure). What would be maximum weight of truss?



150'

The truss position

- A) 5 tons
B) 8 tons
C) 10 tons
D) 20 tons

The Answer is C

According to the crane specification the safe resistive moment of crane can found as:

Length of boom = 90' stands at 80 degree, so the horizontal length of the boom will be:

$$L_x = 90' * \cos 80 = 15.62'$$

$$\text{Resistive moment} = L_x \cdot \text{Capacity} = 15.62 * 100 = 1562 \text{ ton-ft}$$

$$\text{Overturning moment for carrying the truss} = 150' * \text{weight of truss} = 1562 \text{ ton ft}$$

$$\text{So the maximum weight of truss} = 10.4 \text{ ton}$$

39) Find the construction load for the fresh concrete on steel beams to make the composite flooring system (below picture) if the concrete thickness is 4" and the building is the considered as average construction. Beams spacing is given equal to 10' (below picture).



4" fresh concrete



Beam Spacing = 10'

- A) 700 lb/ft B) 140 lb/ft
C) 100 lb/ft D) 1400 lb/ft

The Answers is D**ASCE 37-02**

Dead load of concrete = $150 * 4/12 = 50$ psf

Live load for the average construction = 50 psf

Load combination = $1.2 D + 1.6 L = 1.2 * 50 + 1.6 * 50 = 140$ psf

Construction load on the beams = $140 * 10 = 1400$ lb/ft

40) Which sequence shows the correct priority of the ability to influence construction cost over time. (From highest to the lowest.)

- A) 1-Conceptual and feasibility, 2-design engineering, 3-Procurement and construction.
- B) 1-design engineering, 2-Conceptual and feasibility, 3-Procurement and construction.
- C) 1-Procurement and construction, 2-Conceptual and feasibility, 3-design engineering.
- D) 1-Procurement and construction, 2-design engineering, 3-Conceptual and feasibility.

The Answers is A

According to the experiences, the conceptual design, planning and feasibility studies has main influence on the construction costs. After it engineering and design process is in the second place and procurement and construction process is in the third.

41) A bulldozer has used at the construction site. The LCY/hr (loose cubic yard per hour) production of this bulldozer is given equal to 1000 for the 300 ft dozing distance. If the job efficiency is 0.83 find the actual production:

- A) 830 B) 1200 C) 1000 D) 500

The Answer is A

Actual production = LCY/hr . job efficiency factor, So:

$$1000 * 0.83 = 830 \text{ cubic yard}$$

42) For the question number 2, determine the required days for using one bulldozer if the excavation site has the dimension of 500' x 100' x 30'. The realistic time of operation is 55 minute in an hour and the operation is continue for 8 hours per day.

- A) 8 days B) 12 days C) 10 days D) 5 days

The Answer is C

Total excavation volume = $500 \times 100 \times 30 = 1,500,000 \text{ cf} = 55,556 \text{ CY}$

From question 2, the actual production is 830 LCY/hr, Modification factor for the effect of operation time per hour = $55/60 \times 830 = 760.8 \text{ LCY/hr}$

Required hours = $55,556/760.8 = 73 \text{ hours}$, Number of days = 9.1 days say 10 days.

43) An auger works on the pile foundations as shown in the following picture. If the pile diameter is given equal to 4' and pile depth is 60'. If price of auger including the freight expenses is \$1,000,000. The expected cost including 20% profit for each cubic yard is determined as \$100 per cubic yard. How many piles should this auger make to compensate the original price from the profit?



- A) 1200 pile B) 2000 piles C) 1786 piles D) 2500 piles

The Answer is C

Volume of each pile = $A = \pi r^2 * h = 3.14 * 2^2 * 60 = 753.6 \text{ cf} = 27.9 \text{ say } 28 \text{ CY}$

Cost of each pile = $28 * 100 = \$2800$, Profit = 20%, So Profit = $\$2800 * 0.2 = \560

Number piles = $\$1,000,000 / \$560 = 1785.7 = 1786 \text{ piles}$

44) For the following haul road crown what is the maximum possible slope?



- A) 6%
- B) 12%
- C) 3%
- D) 1%

The Answer is c

The ideal crown slope is 3%. Lower slopes may allow water to pool on the road and more than 3% cause the uneven tire wear.

45) A truck should work on a stream bank. To minimize the disturbance on the ground the pressure exerted by tires should be less than:

- A) 25 PSI
- C) 12 PSI

- B) 6 PSI
- D) 1 PSI

The Answer is B

To minimize the disturbance the pressure exerted by the tire should be between 5-6 PSI.

46. The contractor “Bob Builders” has subcontracted part of construction project to “Toilets 4 You”, a plumbing subcontractor. Right before Toilets 4 You was about to start work the local seller of the PVC piping they needed sold out. In order to get the piping, Toilets 4 You needs to get it shipped from out of town costing more money and delaying the schedule. How must the subcontractor update their contractor with Bob Builders?
- a. They must write up a new contract and have it signed by Bob Builders and Toilets 4 You.
 - b. Make an addendum to the contract and have it signed by Bob Builders and Toilets 4 you
 - c. Bob Builder must put in a Request for Information (RFI) for Toilets 4 You in order to get more information about pricing and scheduling
 - d. Nothing, Toilets 4 You does not have to change anything in the contract.

Answer is B

Answer "A" is not correct: While writing an entirely new contract is possible it is time consuming for both parties and is not a likely situation.

Answer "B" is correct: This is best option, just making an addendum to the contract is time efficient and allows both parties to agree upon the new price and scheduling. This can also be referred to as a change order.

Answer "C" is not correct: The purpose of an RFI is usually for a contractor to get more information about a detail in the drawing set. An RFI would not be used to ask the subcontractor for information about goods or services.

Answer "D" is not correct: If there is a change in price or schedule, the contract must be changed.

47. You are contractor and have signed a cost plus incentive fee contract to construct a new building. The goal price was \$500,000 with a target profit of 10%. However, the project came to \$400,000 instead. The incentive given was a 90/10 split. What is the total cost of the contract?
- a. \$550,000
 - b. \$460,000
 - c. \$450,000
 - d. \$500,000

Answer is B

Answer "A", "C" and "D" are not correct: These do not use the proper method to calculate the total cost of contract.

Answer "B" is correct: The target cost is \$500,000. The profit on the target cost is \$50,000. The actual finished cost was \$400,000, which is a difference in \$100,000 between the target cost and the actual cost. The contract awards the contractor a split of 90/10 of the under budget amount. Thus to calculate you would use the actual cost (\$400,000) + profit margin (\$50,000) + 0.1*under budget amount ($0.1 * \$100,000$) = \$460,000

48. You are a contractor negotiating a contract for a project whose design is not finalized. The owner has told you that he wants you to start work anyway even though the design has incomplete specifications. What type of contract should you ask for in these talks with the owner?
- a. Lump sum
 - b. Time and Materials
 - c. Cost plus incentive fee
 - d. Lump sum plus fee

Answer is B

Answer "A" is not correct: This is not ideal because you have no idea the final cost of the project because you don't have the full design yet.

Answer "B" is correct: Since you don't know the entire scope of the project time and materials is the least risky option. That way you are bearing none of the risks of the unknown costs of the project

Answer "C" is not correct: This is still a risk because if the target cost is exceeded the contractor ends up losing profits.

Answer "D" is not correct: Same as lump sum in risk, even though you get paid a certain fee, it might not cover the unknown project cost.

49. You are the owner of a construction project and the language of the contract says that the “Contractor agrees to design, build, purchase the land and finance the project”. What type of contract did you enter into?
- a. Turnkey
 - b. Design Build
 - c. Design Bid Build
 - d. Joint Venture

Answer is A

Answer "A" is correct: This is the definition of a turnkey project

Answer "B" is not correct: In design build, the contractor does not have to purchase the land and finance the project

Answer "C" is not correct: In design bid build, separate entities are in charge of designing and building.

Answer "D" is not correct: This is not a joint venture because two contractors are not going in on the project together

50. You as the owner of a construction project enter into contracts with the architect/engineer, each trade contractor, as well as a management company that will coordinate construction, costs and scheduling. What type of contract are you entered into?
- a. Joint Venture
 - b. Construction Management
 - c. Turnkey
 - d. Design Build

Answer is B

Answer "A" is not correct: This does not involve two contractors cooperating over the construction of the project.

Answer "B" is correct: This is the definition of a construction management project

Answer "C" is not correct: This is not a turnkey project because it does not involve one party, designing, building, financing the project.

Answer "D" is not correct: This is not a design build because the architect/engineer team is separate from the construction group.

51. You as the owner of a construction project enter into contracts that stipulate that if the estimated quantities of work vary by more than 15%, the price will be adjusted. What type of contract is this?
- a. Cost plus Fee
 - b. Cost plus incentive
 - c. Lump sum
 - d. Unit Price

Answer is D

Answer "A" "B" and "C" are not correct: None of these types of contracts have prices that are based directly on exact quantities of materials.

Answer "D" is correct: This is the definition of a unit price project; this is the only type of contract stipulate changes in the contract price based on a percentage change in quantities. Only in a unit price contract are exact quantities kept track of.

52. Which type of project would you not want to use a unit price contract for?
- a. Road
 - b. Bridge
 - c. Refinery
 - d. Excavation

Answer is B

Answer "A" B" and "D" are not correct: The road, bridge, and excavation type projects are all simple in terms of the number of different types of materials and it is also easy to gauge quantities of supplies necessary. Thus it makes it ideal to do a unit price contract with these types of projects.

Answer "C" is correct: A refinery is a complicated project that involves many multiple parts, equipment, services and goods. Thus it would not make sense to price out everything by quantities and items.

53. What is not an advantage of having a pre-bid meeting with all the key players in the project and potential contractors?
- a. It allows for clarification of plans
 - b. Architect can touch on key points in the documents
 - c. The owner and architect can gauge contractor interest.
 - d. Rewards contractors in attendance by giving them answers to questions that the other absent contractors don't get to hear.

Answer is D

Answer "A" B" and "C" are not correct: These are all advantages of the pre-bid meeting. It's a good way for contractors to clear up things they don't understand and to get focused on the main points of the plan.

Answer "D" is correct: Technically, the architects and owners are supposed to make sure that all bidders receive the same answers as others. If they provide an answer that is not found on the documents they must release a follow-up addendum open to all the bidders. Contractors that attend the meeting are not supposed to be rewarded with additional information.

54. An owner is going to send out a sealed document that requests a contractor's firm price to complete a project. What is this document called?
- a. RFI
 - b. Only IFB
 - c. Only RFQ
 - d. A or C
 - e. B or C

Answer is E

Answer “A” is not correct: The RFI is a request for information and is usually done to request information about how to complete the work. Thus D is also wrong.

Answer “B” and “C” are not correct: Because these two are synonymous they can be combined into answer E but they are not correct on their own.

Answer “E” is correct: The invitation for bid (IFB) and request for quotation (RFQ) is generally the same thing. They are an invitation to the contractor or vender to submit a price that they will do the project for.

55. You are an owner working on a long term construction project. To compensate for potential price increases for materials, the contractor has included an additional amount in his fixed price bid. If these cost increases never occurred, you have a clause in the contract that protects you from paying extra. What is this clause called?

- a. Escalation
- b. Contingency
- c. Changing conditions
- d. Allocation of risk

Answer A is Correct

Answer "A" is correct: The escalation clause allows the owner to avoid paying money for services they did not receive. The escalation clause uses published indexes as their base point for change in labor and material prices.

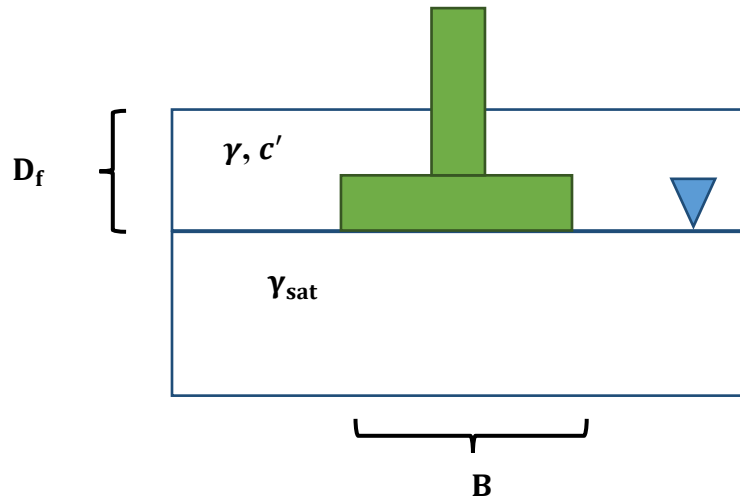
Answer "B" is not correct: Contingency is specifically designed to compensate for potential error not for potential cost increases.

Answer "C" and "D" are not correct: These are unrelated to the variable price for materials.

Part 3 : Soil Mechanics

136 Problems

1) A continuous foundation is given in the following diagram. Determine the distance of D_f . Given equation: $q_u = c'N_c + qN_q + \frac{1}{2}\gamma B N_\gamma = 1500 \text{ kN/m}^2$.
 $\gamma = 16 \text{ kN/m}^3$, $\gamma_{\text{sat}} = 18 \text{ kN/m}^3$, $c' = 20 \text{ kN/m}^2$, $B = 6 \text{ m}$, Friction angle = 20°



A) 10.35m

B) 13.5m

C) 13.0m

D) 12.5m

The Answer is A

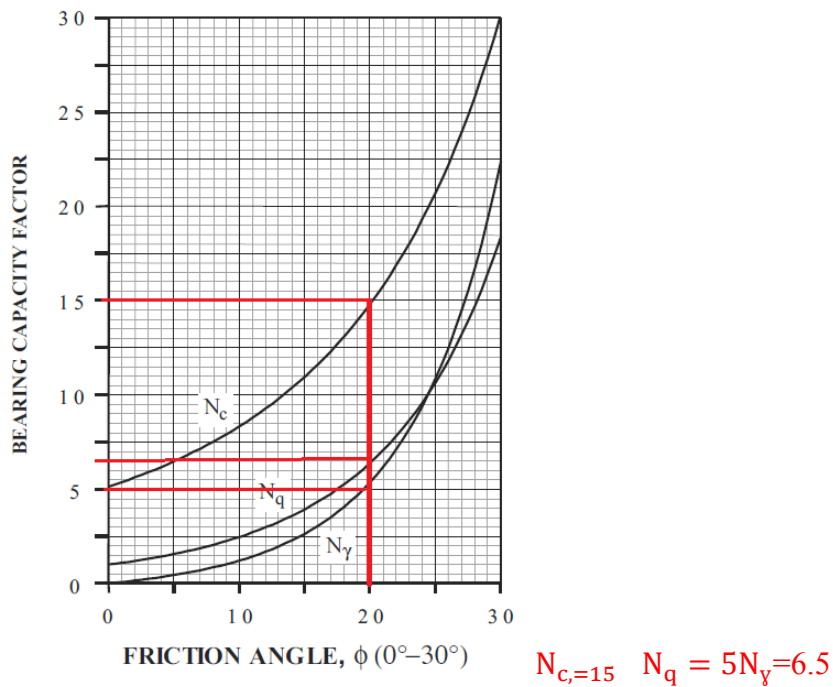
Step 1: From this problem, the equation of bearing capacity is given as,

$$q_u = c'N_c + qN_q + \frac{1}{2}YBN_r$$

Vertical effective stress at the level of the foundation can be given,

$$q = \gamma D_f = (16 \text{ kN/m}^3)D_f = 16D_f$$

Substitute all known values into the equation of bearing capacity, we can get the equation shown below



$$q_u = c'N_c + qN_q + \frac{1}{2}YBN_r$$

$$= (20 \text{ kN/m}^2)(15) + (16D_f)(5)$$

$$+ (0.5)(18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3)(6 \text{ m})(6.5) = 1500 \text{ kN/m}^2$$

Solve this equation,

$$D_f = 10.35 \text{ m} \sim 10.35 \text{ m}$$

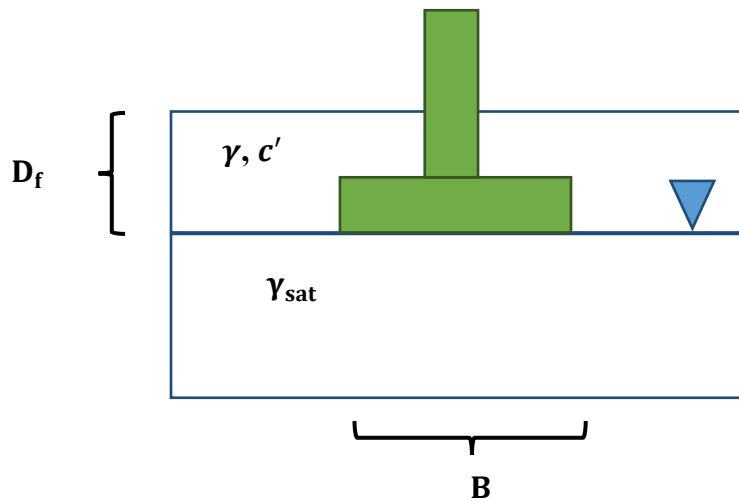
2. A continuous foundation is given in the following diagram. Determine the value of B. Given equation: $q_u = c'N_c + qN_q + \frac{1}{2}\gamma B N_\gamma = 2300 \text{ kN/m}^2$. $\gamma = 16 \text{ kN/m}^3$, $\gamma_{\text{sat}} = 18 \text{ kN/m}^3$, $c' = 20 \text{ kN/m}^2$, Friction angle = 30°
 $D_f = 4 \text{ m}$.

A) 4.6m

B) 3.85m

C) 5.2m

D) 6.2m



The Answer is B

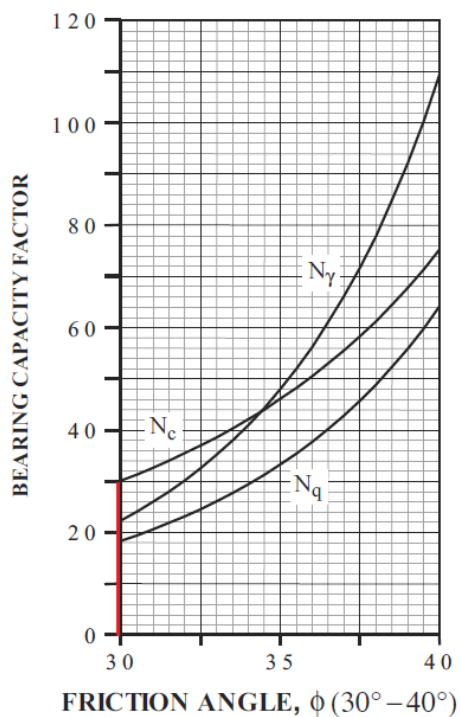
Step 1: From this problem, the equation of bearing capacity is given as,

$$q_u = c'N_c + qN_q + \frac{1}{2}\gamma BN_r$$

Vertical effective stress at the level of the foundation can be given,

$$q = \gamma D_f = (16 \text{ kN/m}^3)(4\text{m}) = 64\text{kN/m}^2$$

Substitute all known values into the equation of bearing capacity, we can get the equation shown below



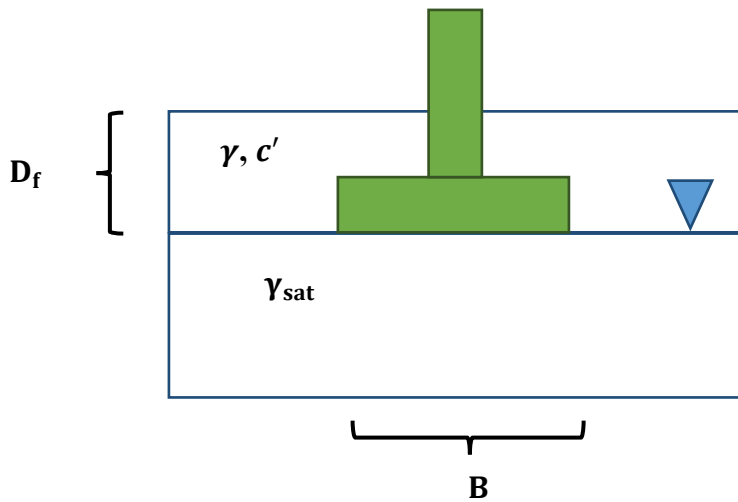
$$N_c = 30, \quad N_r = 22, \quad N_q = 18$$

$$\begin{aligned}q_u &= c'N_c + qN_q + \frac{1}{2}\gamma BN_r \\&= (20\text{kN/m}^2)(30) + (64\text{kN/m}^2)(18) \\&\quad + (0.5)(18\text{kN/m}^3 - 9.8\text{kN/m}^3)B(22) = 2100\text{kN/m}^2\end{aligned}$$

Solve this equation,

$$B = 3.85\text{m} \sim 3.85\text{m}$$

3. A continuous square foundation is given in the following diagram. Determine the area of this square foundation. Given equation: $q_u = 1.3c'N_c + qN_q + 0.5\gamma B N_\gamma = 5500 \text{ kN/m}^2$. $\gamma = 16 \text{ kN/m}^3$, $\gamma_{\text{sat}} = 18 \text{ kN/m}^3$, $c' = 15 \text{ kN/m}^2$, $D_f = 4 \text{ m}$. Friction angle $= 37.5^\circ$
- A) 40 m^2 B) 48 m^2 C) 27 m^2 D) 36 m^2



The Answer is C

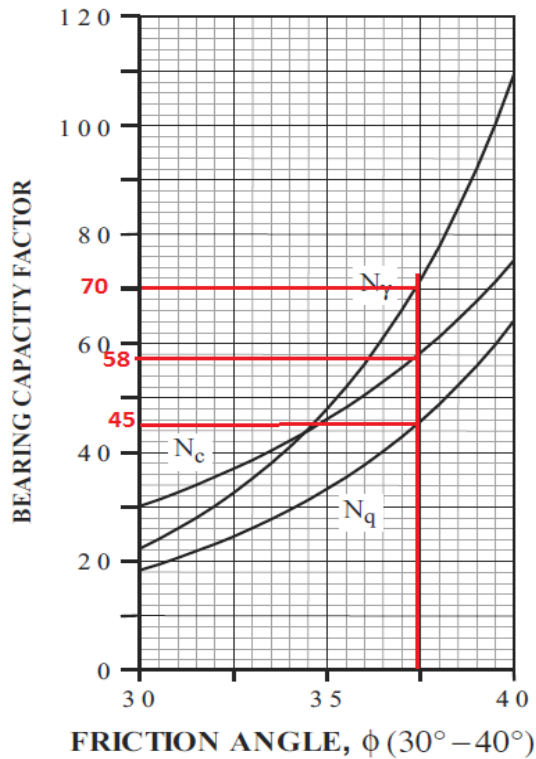
Step 1: From this problem, the equation of bearing capacity is given as,

$$q_u = 1.3c'N_c + qN_q + 0.4\gamma B N_\gamma$$

Vertical effective stress at the level of the foundation can be given,

$$q = \gamma D_f = (16 \text{ kN/m}^3)(4\text{m}) = 64\text{kN/m}^2$$

Substitute all known values into the equation of bearing capacity, we can get the equation shown below



$$N_c = 58, \quad N_\gamma = 70, \quad N_q = 45$$

$$q_u = 1.3c'N_c + qN_q + 0.4\gamma B N_\gamma$$

$$= (1.3)(15\text{kN/m}^2)(58) + (64\text{kN/m}^2)(45)$$

$$+ (0.5)(18\text{kN/m}^3 - 9.8\text{kN/m}^3)B(70) = 5500\text{kN/m}^2$$

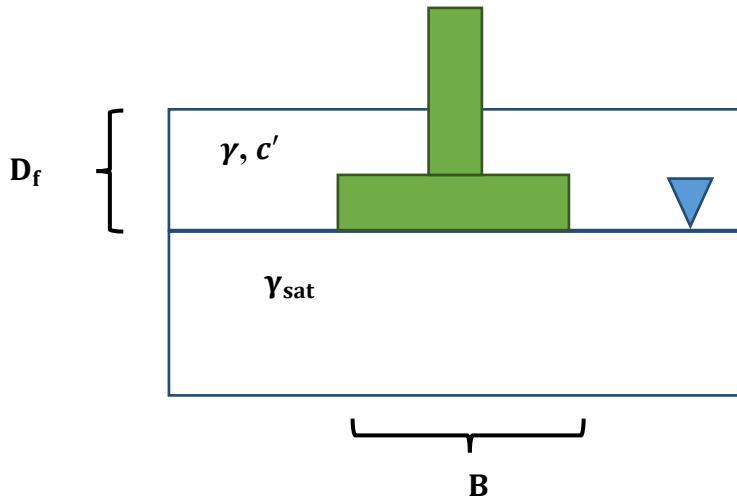
Solve this equation,

$$B = 5.2\text{m}$$

The area of this foundation can be calculated by,

$$A = B^2 = (5.2\text{m})^2 = 27\text{m}^2 \sim 27\text{m}^2$$

4. A continuous foundation is given in the following diagram. Determine the value of γ_{sat} . Given equation: $q_u = c'N_c + qN_q + \frac{1}{2}\gamma B N_\gamma = 350 \text{ kN/m}^2$.
 $\gamma = 16 \text{ kN/m}^3$, $B = 6 \text{ m}$, $c' = 20 \text{ kN/m}^2$, $D_f = 4 \text{ m}$.
Friction angle = 10.0°



- A) 19 kN/m^3 B) 18 kN/m^3 C) 16 kN/m^3 D) 16.5 kN/m^3

The Answer is D

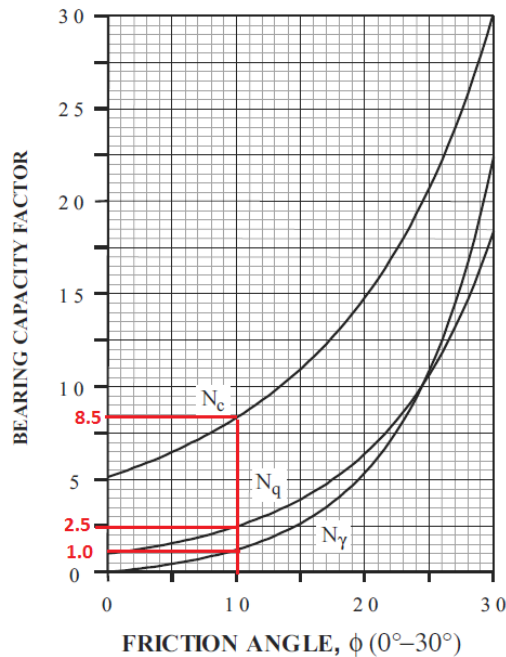
Step 1: From this problem, the equation of bearing capacity is given as,

$$q_u = c'N_c + qN_q + \frac{1}{2}\gamma B N_\gamma$$

Vertical effective stress at the level of the foundation can be given,

$$q = \gamma D_f = (16 \text{ kN/m}^3)(4\text{m}) = 64\text{kN/m}^2$$

Substitute all known values into the equation of bearing capacity, we can get the equation shown below

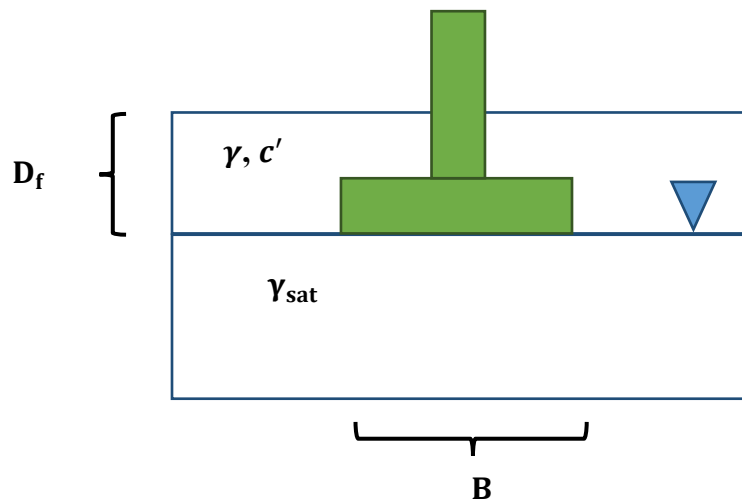


$$\begin{aligned} q_u &= c'N_c + qN_q + \frac{1}{2}\gamma B N_\gamma \\ &= (20\text{kN/m}^2)(8.5) + (64\text{kN/m}^2)(2.5) \\ &\quad + (0.5)(\gamma_{\text{sat}} - 9.8\text{kN/m}^3)(6\text{m})(1) = 350\text{kN/m}^2 \end{aligned}$$

Solve this equation,

$$\gamma_{\text{sat}} = 16.5\text{kN/m}^3 \sim 16.5\text{kN/m}^3$$

5) A continuous foundation is given in the following diagram. Determine the value of γ . Given equation: $q_u = c'N_c + qN_q + \frac{1}{2}\gamma B N_\gamma = 700 \text{ kN/m}^2$. $\gamma_{\text{sat}} = 18 \text{ kN/m}^3$, $B = 6 \text{ m}$, $c' = 20 \text{ kN/m}^2$, $D_f = 4 \text{ m}$. Friction angle = 20.0°

A) 19 kN/m^3 B) 18 kN/m^3 C) 10.6 kN/m^3 D) 17 kN/m^3

The Answer is C

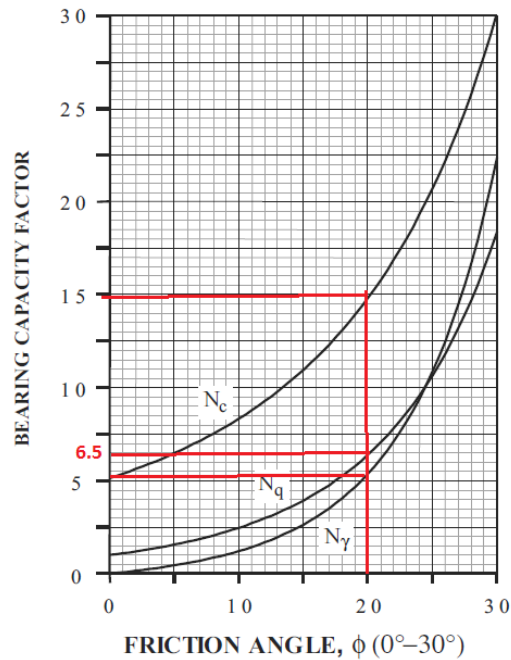
Step 1: From this problem, the equation of bearing capacity is given as,

$$q_u = c'N_c + qN_q + \frac{1}{2} \gamma B N_\gamma$$

Vertical effective stress at the level of the foundation can be given,

$$q = \gamma D_f = \gamma(4\text{m}) = 4\gamma$$

Substitute all known values into the equation of bearing capacity, we can get the equation shown below



$$\begin{aligned} q_u &= c'N_c + qN_q + \frac{1}{2} \gamma B N_\gamma \\ &= \left(\frac{20\text{kN}}{\text{m}^2}\right)(15) + (4\gamma)(6.5) \\ &\quad + (0.5)(18\text{kN/m}^3 - 9.8\text{kN/m}^3)(6\text{m})(5.0) = 700\text{kN/m}^2 \end{aligned}$$

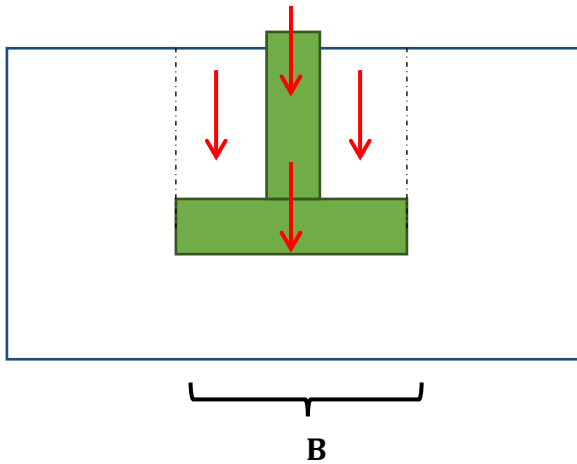
Solve this equation,

$$\gamma = 10.6\text{kN/m}^3 \sim 10.6\text{kN/m}^3$$

6) A circular foundation is given in the following diagram. It is known that this foundation subjects three loads as dead and live load, W_{D+L} , self-weight load, W_F , and soil load, W_s . Assume that the size of this foundation is given as 4m.

$W_{D+L} = 300\text{kN}$, $W_F = 100\text{kN}$, $W_s = 450\text{kN}$. Determine the gross allowable bearing capacity of this foundation.

- A) 68kN/m^2 B) 72kN/m^2 C) 54kN/m^2 D) 63kN/m^2



The Answer is A

Step 1: From this problem, the equation of gross allowable bearing capacity of this foundation is given as,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A}$$

Step 2: The area of this foundation can be calculated based on the information of size, B, as,

$$A = \frac{\pi B^2}{4} = \frac{\pi(4m)^2}{4} = 12.6m^2$$

Step 3: Solve the gross allowable bearing capacity of this foundation,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A} = \frac{300kN + 100kN + 450kN}{12.6m^2} = 67.5kN/m^2$$

~68kN/m²

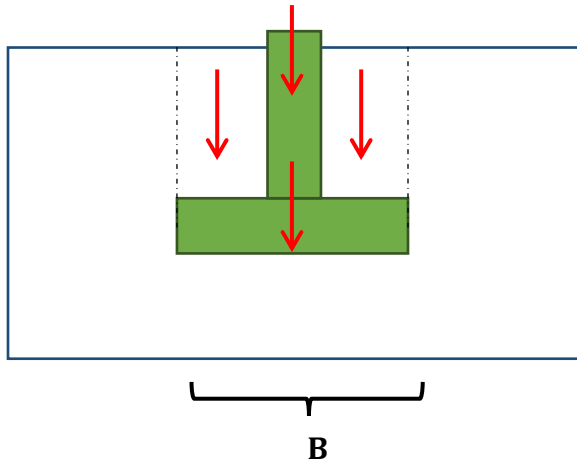
7) A square foundation is given in the following diagram. It is known that this foundation subjects three loads as dead and live load, W_{D+L} , self-weight load, W_F , and soil load, W_s . Assume that the maximum gross allowable bearing capacity of this foundation is 70 kN/m^2 . $W_{D+L} = 300 \text{ kN}$, $W_F = 100 \text{ kN}$, $W_s = 450 \text{ kN}$. Determine the size of this foundation.

A) 4.0m

B) 3.5m

C) 3.0m

D) 4.5m



The Answer is B

Step 1: From this problem, the equation of gross allowable bearing capacity of this foundation is given as,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A}$$

Step 2: The area of this foundation can be calculated based on the information of size, B, as,

$$A = B^2$$

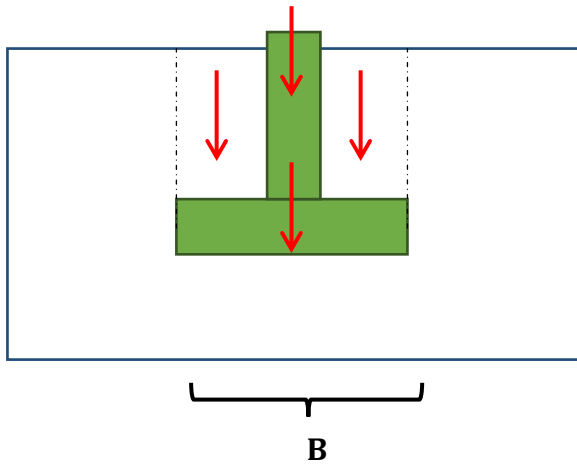
Step 3: Solve the gross allowable bearing capacity of this foundation,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A} = \frac{300\text{kN} + 100\text{kN} + 450\text{kN}}{B^2} = 70\text{kN/m}^2$$

Solve this equation we got,

$$B = 3.5\text{m} \sim 3.5\text{m}$$

8) A triangle foundation is given in the following diagram. The length of each side of this triangle is the same. It is known that this foundation subjects three loads as dead and live load, W_{D+L} , self-weight load, W_F , and soil load, W_s . Assume that the maximum gross allowable bearing capacity of this foundation is 70 kN/m^2 . $W_{D+L} = 300 \text{ kN}$, $W_F = 100 \text{ kN}$, $W_s = 450 \text{ kN}$. Determine the size of this foundation.



A) 4.0m

B) 3.5m

C) 5.3m

D) 4.5m

The Answer is C

Step 1: From this problem, the equation of gross allowable bearing capacity of this foundation is given as,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A}$$

Step 2: The area of this foundation can be calculated based on the information of size, B, as,

$$A = \frac{\sqrt{3}}{4} B^2$$

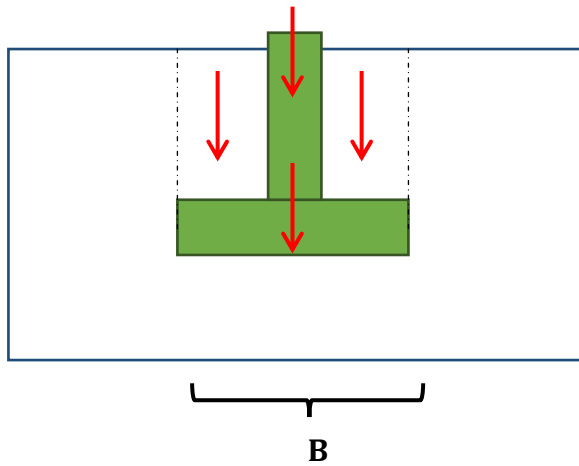
Step 3: Solve the gross allowable bearing capacity of this foundation,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A} = \frac{300\text{kN} + 100\text{kN} + 450\text{kN}}{\frac{\sqrt{3}}{4} B^2} = 70\text{kN/m}^2$$

Solve this equation we got,

$$B = 5.3\text{m} \sim 5.3\text{m}$$

9) A square foundation is given in the following diagram. It is known that this foundation subjects three loads as dead and live load, W_{D+L} , self-weight load, W_F , and soil load, W_s . Assume that the maximum gross allowable bearing capacity of this foundation is 70 kN/m^2 . $W_{D+L} = 300 \text{ kN}$, $W_s = 450 \text{ kN}$. Determine the maximum self-weight of this foundation if size of it is 4m.



A) 370kN

B) 520kN

C) 450kN

D) 320kN

The Answer is C

Step 1: From this problem, the equation of gross allowable bearing capacity of this foundation is given as,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A}$$

Step 2: The area of this foundation can be calculated based on the information of size, B, as,

$$A = B^2 = (4\text{m})^2 = 16\text{m}^2$$

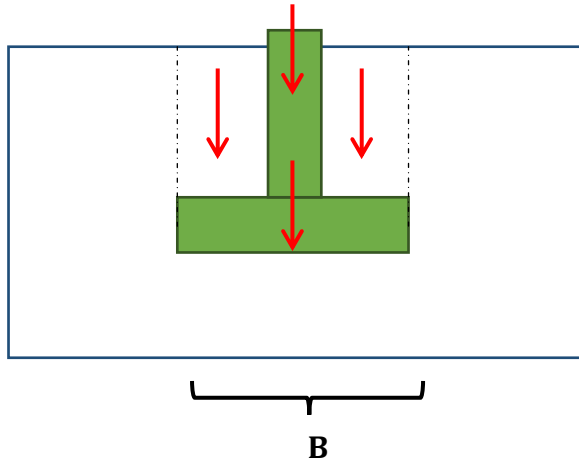
Step 3: Solve the gross allowable bearing capacity of this foundation,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A} = \frac{300\text{kN} + W_F + 450\text{kN}}{16\text{m}^2} = 70\text{kN/m}^2$$

Solve this equation we got,

$$W_F = 370\text{kN} \sim 370\text{kN}$$

10) A circular foundation is given in the following diagram. It is known that this foundation subjects three loads as dead and live load, W_{D+L} , self-weight load, W_F , and soil load, W_s . Assume that the maximum gross allowable bearing capacity of this foundation is 70kN/m^2 . $W_{D+L} = 300\text{kN}$, $W_s = 450\text{kN}$. Determine the maximum self-weight of this foundation if size of it is 4m.



A) 150kN

B) 130kN

C) 170kN

D) 110kN

The Answer is B

Step 1: From this problem, the equation of gross allowable bearing capacity of this foundation is given as,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A}$$

Step 2: The area of this foundation can be calculated based on the information of size, B, as,

$$A = \frac{\pi B^2}{4} = \frac{\pi(4\text{m})^2}{4} = 12.6\text{m}^2$$

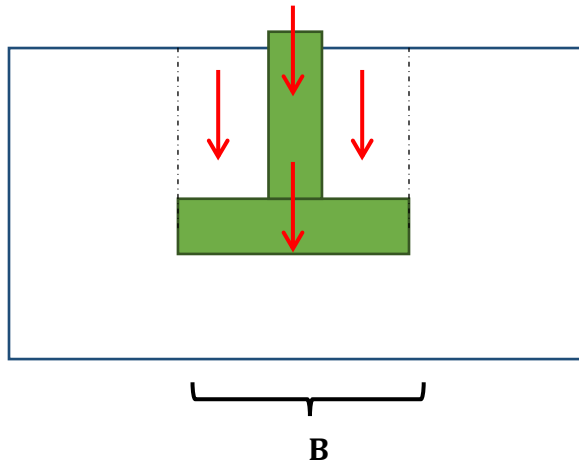
Step 3: Solve the gross allowable bearing capacity of this foundation,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A} = \frac{300\text{kN} + W_F + 450\text{kN}}{12.6\text{m}^2} = 70\text{kN/m}^2$$

Solve this equation we got,

$$W_F = 130\text{kN} \sim 130\text{kN}$$

11) A square foundation is given in the following diagram. It is known that this foundation subjects three loads as dead and live load, $\mathbf{W_{D+L}}$, self-weight load, $\mathbf{W_F}$, and soil load, $\mathbf{W_s}$. Assume that the maximum gross allowable bearing capacity of this foundation is 70kN/m^2 . Dead load is 200kN , $\mathbf{W_s = 450kN}$. $\mathbf{W_F = 200kN}$. Determine the maximum live load this foundation can carry if size of it is 4m .



- A) 250kN B) 230kN C) 270kN D) 210kN

The Answer is C

Step 1: From this problem, the equation of gross allowable bearing capacity of this foundation is given as,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A}$$

Step 2: The area of this foundation can be calculated based on the information of size, B, as,

$$A = B^2 = (4m)^2 = 16m^2$$

Step 3: Solve the gross allowable bearing capacity of this foundation,

$$q_{all} = \frac{W_{D+L} + W_F + W_s}{A} = \frac{W_{D+L} + 200kN + 450kN}{16m^2} = 70kN/m^2$$

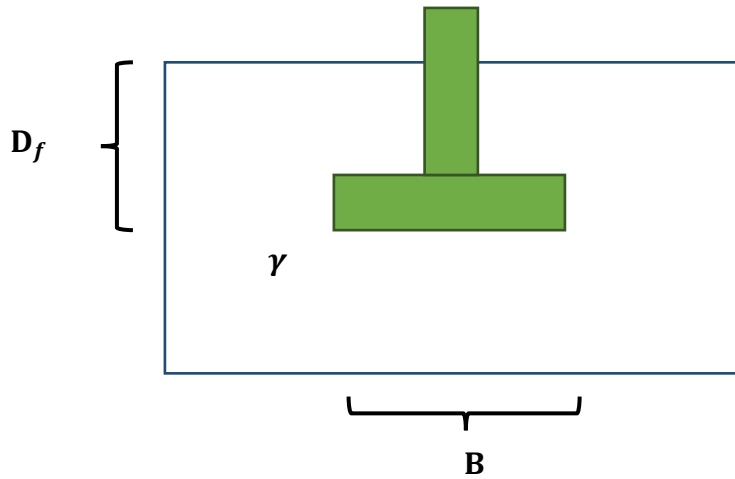
Solve this equation we got,

$$W_{D+L} = 470kN$$

Since we know the value of dead load, we can calculate the maximum live load as,

$$W_L = W_{D+L} - W_D = 470kN - 200kN = 270kN \sim 270kN$$

12) A continuous foundation is given in the following. It is known that the ultimate load, $q_u = 1500 \text{ kN/m}^2$. If the unit weight of soil is assumed to be 16 kN/m^3 , $D_f = 4 \text{ m}$, factor of safety is 3. Determine the net allowable bearing capacity of this foundation.



- A) 454 kN/m^2
C) 433 kN/m^2

- B) 479 kN/m^2
D) 542 kN/m^2

The Answer is B

Step 1: Vertical effective stress at the level of the foundation can be calculated by,

$$q = \gamma D_f = (16 \text{ kN/m}^3)(4 \text{ m}) = 64 \text{ kN/m}^2$$

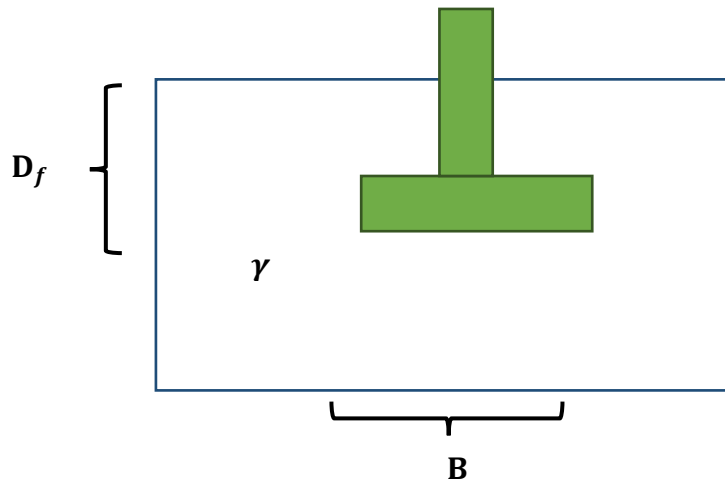
Step 2: Net ultimate load can be calculated by subtracting vertical effective stress from the ultimate load as,

$$q_{u(\text{net})} = q_u - q = 1500 \text{ kN/m}^2 - 64 \text{ kN/m}^2 = 1436 \text{ kN/m}^2$$

Step 3: Net allowable bearing load can be calculated by,

$$q_{\text{all}(\text{net})} = \frac{q_{u(\text{net})}}{F_s} = \frac{1436 \text{ kN/m}^2}{3} = 479 \text{ kN/m}^2 \sim 479 \text{ kN/m}^2$$

13) A continuous foundation is given in the following. It is known that the ultimate load, $q_u = 1500 \text{ kN/m}^2$ and net allowable bearing load is 450 kN/m^2 . If the unit weight of soil is assumed to be 16 kN/m^3 , factor of safety is 3. Determine the value of D_f .



A) 7.6m

B) 8.3m

C) 9.4m

D) 10.2m

The Answer is C

Step 1: Vertical effective stress at the level of the foundation can be calculated by,

$$q = \gamma D_f = (16 \text{ kN/m}^3) D_f = 16 D_f$$

Step 2: Net ultimate load can be calculated by subtracting vertical effective stress from the ultimate load as,

$$q_{u(\text{net})} = q_u - q = 1500 \text{ kN/m}^2 - 16 D_f$$

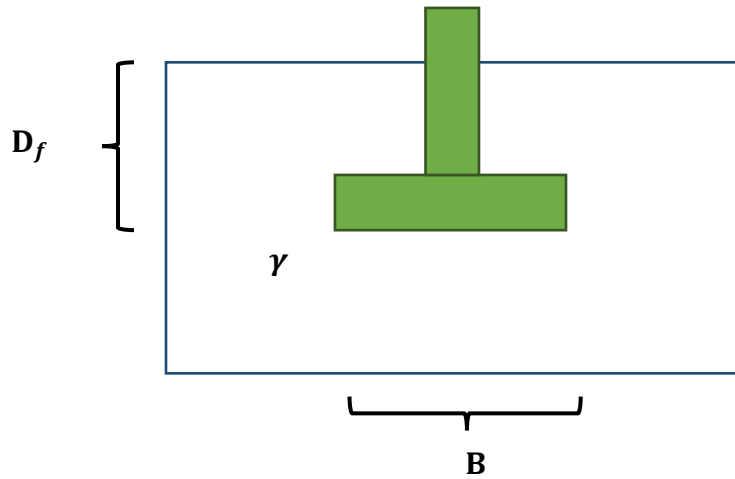
Step 3: Net allowable bearing load can be calculated by,

$$q_{\text{all}(\text{net})} = \frac{q_{u(\text{net})}}{F_s} = \frac{1500 \text{ kN/m}^2 - 16 D_f}{3} = 450 \text{ kN/m}^2$$

Solve the equations above we got,

$$D_f = 9.4 \text{ m} \sim 9.4 \text{ m}$$

14) A continuous foundation is given in the following. It is known that the ultimate load, $q_u = 1500 \text{ kN/m}^2$ and net allowable bearing load is 450 kN/m^2 . If the unit weight of soil is assumed to be 16 kN/m^3 , $D_f = 6 \text{ m}$. Determine the factor of safety.



- A) 2 B) 3 C) 2.5 D) 3.5

The Answer is B

Step 1: Vertical effective stress at the level of the foundation can be calculated by,

$$q = \gamma D_f = (16 \text{ kN/m}^3)(6 \text{ m}) = 96 \text{ kN/m}^2$$

Step 2: Net ultimate load can be calculated by subtracting vertical effective stress from the ultimate load as,

$$q_{u(\text{net})} = q_u - q = 1500 \text{ kN/m}^2 - 96 \text{ kN/m}^2 = 1404 \text{ kN/m}^2$$

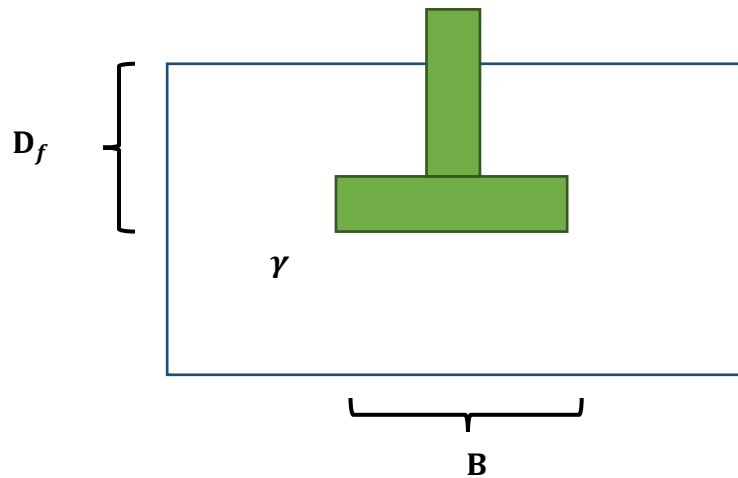
Step 3: Net allowable bearing load can be calculated by,

$$q_{\text{all}(\text{net})} = \frac{q_{u(\text{net})}}{F_s} = \frac{1404 \text{ kN/m}^2}{F_s} = 450 \text{ kN/m}^2$$

Solve the equations above we got,

$$F_s = 3.12 \sim 3$$

15) A continuous foundation is given in the following. It is known that the ultimate load, $q_u = 1500 \text{ kN/m}^2$ and net allowable bearing load is 450 kN/m^2 . If the factor of safety is 3.1, $D_f = 6 \text{ m}$. Determine the unit weight of this soil.



A) 17.5 kN/m^3
C) 15.2 kN/m^3

B) 16.4 kN/m^3
D) 18.6 kN/m^3

The Answer is A

Step 1: Vertical effective stress at the level of the foundation can be calculated by,

$$q = \gamma D_f = \gamma(6\text{m}) = 6\gamma$$

Step 2: Net ultimate load can be calculated by subtracting vertical effective stress from the ultimate load as,

$$q_{u(\text{net})} = q_u - q = 1500\text{kN/m}^2 - 6\gamma$$

Step 3: Net allowable bearing load can be calculated by,

$$q_{\text{all}(\text{net})} = \frac{q_{u(\text{net})}}{F_s} = \frac{1500\text{kN/m}^2 - 6\gamma}{3.1} = 450\text{kN/m}^2$$

Solve the equations above we got,

$$\gamma = 17.5\text{kN/m}^3 \sim 17.5\text{kN/m}^3$$

16. A standard penetration test (SPT) was conducted on a ground to determine the bearing capacity of soil. It was measured that the number of blows needed for the sampler to penetrate each 6-inch depth is 12, 14, 16, and 18 for the 1st, 2nd, 3rd, and 4th 6-inch depth increment. What is the SPT resistance (N-value)?

A) 12

B) 14

C) 16

D) 30

The Answer is D

Step 1:

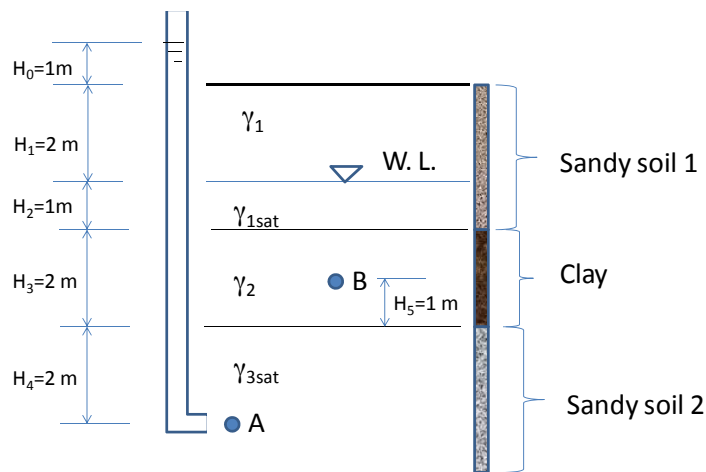
In a SPT, the sum of the number of blows required for the second and third 6-inch penetration is termed the "standard penetration resistance" or the "N-value".

Step 2:

From the given information, it is known that the number of blows for the 2nd and 3rd penetration of 6-inch depth is 14 and 16, respectively. Therefore, the N-value is

$$14 + 16 = 30$$

17) As shown in the figure below, a clay layer exists between two sandy soil layers. The underground water level is 1 m below the ground. The second sandy soil layer contains confined water. At a point A, which is 7 m below the ground (in the second sandy soil layer), the head of water pressure is 1 m above the ground. It is known that the unit weight of soil above the water level is 16.5 kN/m^3 , the saturated unit weight of the first layer of sandy soil below the water level is 19.2 kN/m^3 , the saturated unit weight of the second layer of sandy soil is 20.2 kN/m^3 , the saturated unit weight of the clay soil is 18.4 kN/m^3 . Determine the pore water pressure at point B, which is in the middle of the clay layer.



A) 0 kPa

B) 23 kPa

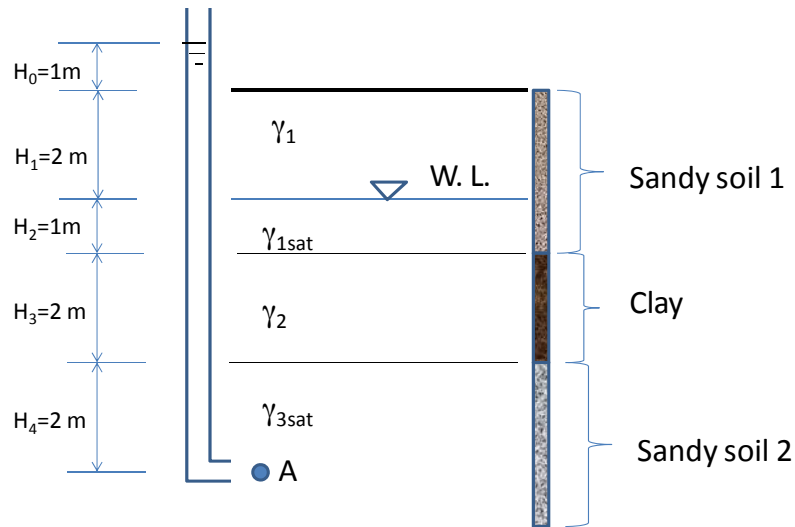
C) 35 kPa

D) 51 kPa

The Answer is D

Since the second sandy soil layer contains confined water, and it can be seen that the head of water pressure at point A exceeds the groundwater level, the clay layer is an aquiclude layer, which is impermeable. Therefore, the pore water pressure at point B is zero.

18) As shown in the figure below, a clay layer exists between two sandy soil layers. The underground water level is 1 m below the ground. The second sandy soil layer contains confined water. At a point A, which is 7 m below the ground (in the second sandy soil layer), the head of water pressure is 1 m above the ground. It is known that the unit weight of soil above the water level is 16.5 kN/m^3 , the saturated unit weight of the first layer of sandy soil below the water level is 19.2 kN/m^3 , the saturated unit weight of the second layer of sandy soil is 20.2 kN/m^3 , the saturated unit weight of the clay soil is 18.4 kN/m^3 . Determine the effective vertical soil pressure at point A.



A) 23 kPa

B) 35 kPa

C) 51 kPa

D) 88 kPa

The Answer is C

Effective stress σ' is equal to the total stress σ minus pore water pressure

$$\sigma' = \sigma - u.$$

Step 1:

The total stress σ is calculated by $\sigma = \gamma_1 H_1 + \gamma_{1sat} H_2 + \gamma_2 H_3 + \gamma_{3sat} H_4 =$
 $(16.5 \text{ kN/m}^3)(2 \text{ m}) + (19.2 \text{ kN/m}^3)(1 \text{ m}) + (18.4 \text{ kN/m}^3)(2 \text{ m}) +$
 $(20.2 \text{ kN/m}^3)(2 \text{ m}) = 129.4 \text{ kPa}$

Step 2:

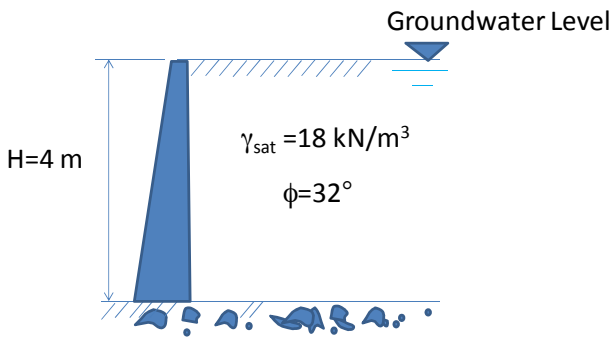
The pore water pressure u is calculated by $u = \gamma_w H = (9.8 \text{ kN/m}^3)(1 \text{ m} +$
 $2 \text{ m} + 1 \text{ m} + 2 \text{ m} + 2 \text{ m}) = 78.4 \text{ kPa}$

Step 3:

The effective stress σ' is

$$\sigma' = \sigma - u = 129.4 \text{ kPa} - 78.4 \text{ kPa} = 51 \text{ kPa}$$

19) Determine the active lateral earth pressure on the frictionless wall, as shown in the figure below, at the base of the wall. The soil behind of the wall is clean sand with a saturated unit weight of 18.0 kN/m^3 and angle of internal friction of 32° . Groundwater level is at the surface of the ground.



A) 5.2 kPa

B) 10.1 kPa

C) 16.3 kPa

D) 18.5 kPa

The Answer is B

Step 1:

Based on the Rankine theory, the Rankine active earth pressure coefficient

$$K_A = \tan^2 \left(45^\circ - \frac{\phi}{2} \right) = \tan^2 \left(45^\circ - \frac{32^\circ}{2} \right) = 0.307.$$

Step 2:

The vertical effective stress at the base of the wall, σ'_v , is

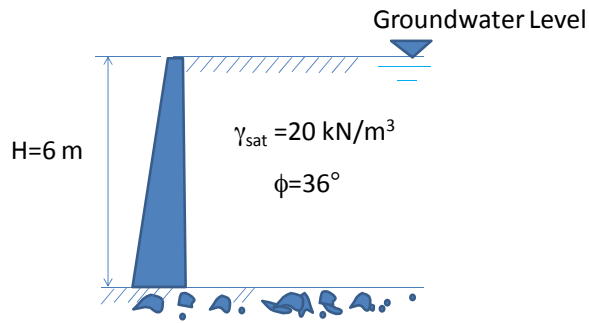
$$\sigma'_v = \sigma_v - u = (\gamma_{\text{sat}} - \gamma_w)H = \left(18 \frac{\text{kN}}{\text{m}^3} - 9.8 \frac{\text{kN}}{\text{m}^3} \right) (4 \text{ m}) = 32.8 \text{ kPa}$$

Step 3:

The active lateral earth pressure at the base of the wall, σ_a , is

$$\sigma_a = K_A \sigma'_v = 0.307 \times 32.8 \text{ kPa} = 10.1 \text{ kPa}$$

20) Determine the lateral force (per unit length of wall) on the frictionless retaining wall, as shown in the figure below. The soil behind of the wall is clean sand with a saturated unit weight of 20.0 kN/m^3 and angle of internal friction of 36° . Groundwater level is at the surface of the ground.



- A) 124.1 kN/m B) 202.1 kN/m
C) 224.1 kN/m D) 420.1 kN/m

The Answer is C

Step 1:

Based on the Rankine theory, the Rankine active earth pressure coefficient

$$K_A = \tan^2 \left(45^\circ - \frac{\phi}{2} \right) = \tan^2 \left(45^\circ - \frac{36^\circ}{2} \right) = 0.260.$$

Step 2:

At the surface of the ground, the vertical effective stress and the pore water pressure are both zero, $\sigma'_v = u = 0$.

At the base of the wall,

$$\sigma'_v = \sigma_v - u = (\gamma_{\text{sat}} - \gamma_w)H = \left(20 \frac{\text{kN}}{\text{m}^3} - 9.8 \frac{\text{kN}}{\text{m}^3} \right) (6 \text{ m}) = 61.2 \text{ kPa}$$

$$u = \gamma_w H = \left(9.8 \frac{\text{kN}}{\text{m}^3} \right) (6 \text{ m}) = 58.8 \text{ kPa}$$

Step 3:

The active lateral earth pressure at the base of the wall, σ_a , is

$$\sigma_a = K_A \sigma'_v = 0.260 \times 61.2 \text{ kPa} = 15.9 \text{ kPa}$$

Step 4:

Based on the distributions of lateral earth pressure and pore water pressure,



15.9 kPa

Lateral earth pressure



58.8 kPa

Hydrostatic pressure

the lateral force (per unit length of wall) on the wall, $\mathbf{P}_a$, is

$$\begin{aligned}\mathbf{P}_a &= \mathbf{P}_s + \mathbf{P}_w = \frac{1}{2}\sigma_a H + \frac{1}{2}uH = \frac{1}{2}(\sigma_a + u)H \\ &= \frac{1}{2}(15.9 \text{ kPa} + 58.8 \text{ kPa})(6 \text{ m}) = 224.1 \text{ kN/m}\end{aligned}$$

where $\mathbf{P}_s$ is the lateral force due to soil solids and $\mathbf{P}_w$ is the lateral force due to the pore water.

21) A 10-inch diameter steel pile is driven 60 ft into insensitve clay, which has an undrained shear strength $s_u = 1300 \text{ lbf/ft}^2$. The groundwater table is at the ground surface. Assume the entire pile length is effective, the base resistance $f_b = 9s_u$ and the adhesive stress $f_s = 0.5s_u$, determine the allowable bearing capacity of the pile for a factor of safety $FS_q = 3$.

A) 20 kips

B) 33 kips

C) 60 kips

D) 72 kips

The Answer is B

Step 1:

The ultimate bearing capacity of the pile due to friction, Q_f , is the product of the adhesive stress, f_s , and the surface area of the shaft.

$$f_s = 0.5s_u = 0.5 \left(1300 \frac{\text{lbf}}{\text{ft}^2} \right) = 650 \frac{\text{lbf}}{\text{ft}^2}$$

$$Q_f = f_s A = f_s \pi D L = \left(650 \frac{\text{lbf}}{\text{ft}^2} \right) \pi \left(\frac{10 \text{ inches}}{12 \frac{\text{inches}}{\text{ft}}} \right) (60 \text{ ft}) \cong 102102 \text{ lbf}$$

$$= \frac{102102 \text{ lbf}}{1000 \frac{\text{lbf}}{\text{kip}}} \cong 102.1 \text{ kips}$$

Step 2:

The ultimate bearing capacity of the pile due to end bearing, Q_p , is the product of the base resistance, f_b , and the base area.

$$f_b = 9s_u = 9 \left(1300 \frac{\text{lbf}}{\text{ft}^2} \right) = 11700 \frac{\text{lbf}}{\text{ft}^2}$$

$$Q_p = f_b \frac{\pi D^2}{4} = \left(11700 \frac{\text{lbf}}{\text{ft}^2} \right) \frac{\pi \left(\frac{10 \text{ inches}}{12 \frac{\text{inches}}{\text{ft}}} \right)^2}{4} \cong 6381 \text{ lbf} = \frac{6381 \text{ lbf}}{1000 \frac{\text{lbf}}{\text{kip}}}$$

$$\cong 6.4 \text{ kips}$$

Step 3:

The allowable bearing capacity is

$$Q_a = \frac{Q_f + Q_p}{FS_q} = \frac{102.1 \text{ kips} + 6.4 \text{ kips}}{3} \cong 36.2 \text{ kips}$$

22) For an over consolidated clay of 3 m thickness, the following data are given: average effective pressure $\sigma_0 = 100 \text{ kPa}$; initial void ratio $e_0 = 1.1$; average increase of pressure in the clay layer $\Delta\sigma = 30 \text{ kPa}$; compression index $C_c = 0.34$; recompression index $C_R = 0.12$; past maximum consolidation stress $p_c = 150 \text{ kPa}$. Determine the primary consolidation settlement of the clay.

- A) 0.020 m B) 0.040 m C) 0.060 m D) 0.080 m

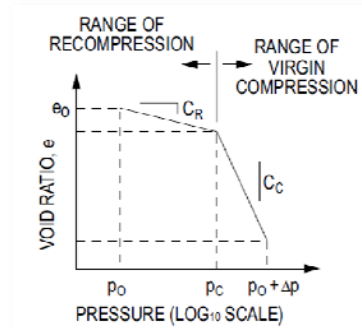
The Answer is A

Step 1:

Since the clay is over consolidated, the final effective pressure needs to be compared with the past maximum consolidation stress to determine which compression index should be used. Given the initial average effective pressure $\sigma_0 = 100 \text{ kPa}$ and pressure increase $\Delta\sigma = 30 \text{ kPa}$, the final effective pressure is $\sigma_1 = \sigma_0 + \Delta\sigma = 100 \text{ kPa} + 30 \text{ kPa} = 130 \text{ kPa} < p_c = 150 \text{ kPa}$.

Therefore, the clay is in recompression under the increased pressure.

Recompression index $C_R = 0.12$ should be used.



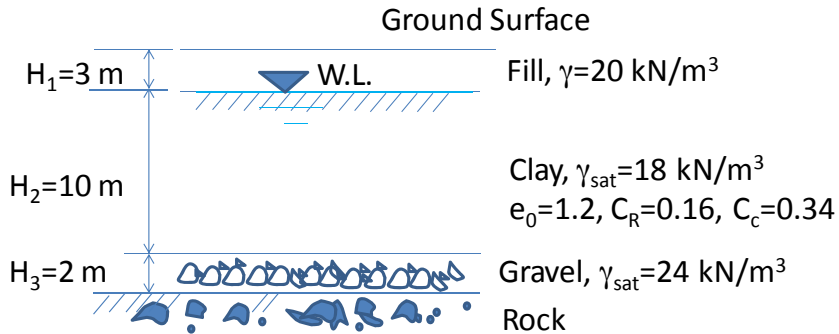
Step 2:

The primary consolidation settlement of the clay is

$$S = H \epsilon_v = H \frac{\Delta e}{1 + e_0} = H \frac{C_R \log_{10} \left(\frac{\sigma_1}{\sigma_0} \right)}{1 + e_0} = (3 \text{ m}) \frac{(0.12) \log_{10} \left(\frac{130 \text{ kPa}}{100 \text{ kPa}} \right)}{1 + 1.1}$$

$$\cong 0.020 \text{ m}$$

23) As shown in the figure below, a clay layer of thickness 10 m is above a gravel layer. The groundwater level is at the surface of the clay layer. The clay is over consolidated and the past maximum consolidation stress is $p_c = 81 \text{ kPa}$. If a 3 m thick fill layer is placed above the clay layer, determine the settlement of the fill layer due to the primary consolidation of the clay layer.



A) 0.4 m

B) 1.2 m

C) 2.2 m

D) 3.4 m

The Answer is A

Step 1:

Before the fill layer is placed, the vertical stress in the clay is due to the self weight of the soil, which increases linearly with depth. Since the clay is saturated, the effective vertical stress at center of the clay layer is

$$\sigma_0 = \frac{1}{2} H_2 (\gamma_{\text{sat}} - \gamma_w) = \frac{1}{2} (10 \text{ m}) \left(18 \frac{\text{kN}}{\text{m}^3} - 9.8 \frac{\text{kN}}{\text{m}^3} \right) = 41 \text{ kPa}$$

Step 2:

After the fill layer is placed, the effective vertical stress at center of the clay layer is

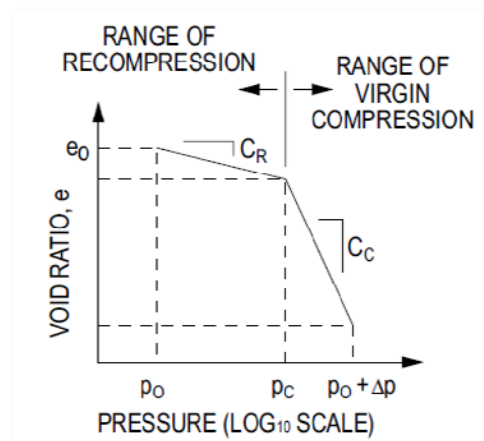
$$\sigma_1 = \sigma_0 + H_1 \gamma_{\text{fill}} = 41 \text{ kPa} + (3 \text{ m}) \left(20 \frac{\text{kN}}{\text{m}^3} \right) = 101 \text{ kPa}$$

Step 3:

Since the clay is over consolidated, before the effective vertical stress reaches the past maximum consolidation stress, $p_c = 81 \text{ kPa}$, the settlement of the fill layer is

Continuing:

$$\begin{aligned} S_1 &= H_2 \epsilon_{v1} = H_2 \frac{\Delta e_1}{1 + e_0} \\ &= H_2 \frac{C_R \log_{10} \left(\frac{\sigma_p}{\sigma_0} \right)}{1 + e_0} \\ &= (10 \text{ m}) \frac{(0.16) \log_{10} \left(\frac{81 \text{ kPa}}{41 \text{ kPa}} \right)}{1 + 1.2} \\ &= 0.215 \text{ m} \end{aligned}$$



Step 4:

After the effective vertical stress passes the past maximum consolidation stress and reaches the final effective vertical stress, the settlement of the fill layer is

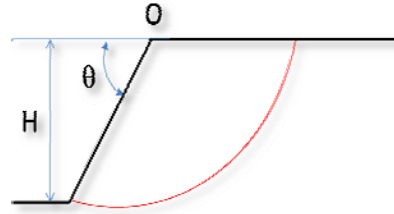
$$\begin{aligned} S_2 &= H_2 \epsilon_{v2} = H_2 \frac{\Delta e_2}{1 + e_0} = H_2 \frac{C_c \log_{10} \left(\frac{\sigma_1}{\sigma_p} \right)}{1 + e_0} \\ &= (10 \text{ m}) \frac{(0.34) \log_{10} \left(\frac{101 \text{ kPa}}{81 \text{ kPa}} \right)}{1 + 1.2} \\ &= 0.148 \text{ m} \end{aligned}$$

Step 5:

Therefore, the total settlement of the fill layer is

$$S = S_1 + S_2 = 0.215 \text{ m} + 0.148 \text{ m} = 0.363 \text{ m}$$

24) An embankment constructed with a clay soil has a height $H = 10 \text{ m}$, and a slope angle $\theta = 20^\circ$. The soil has a cohesion $c = 60 \text{ kPa}$, unit weight $\gamma = 20 \text{ kN/m}^3$, and friction angle $\phi = 0^\circ$. A potential failure plane is an arc with its center at the edge of the embankment, O, as shown in the figure. Determine the factor of safety against slope rotational instability, FS, for this plane.



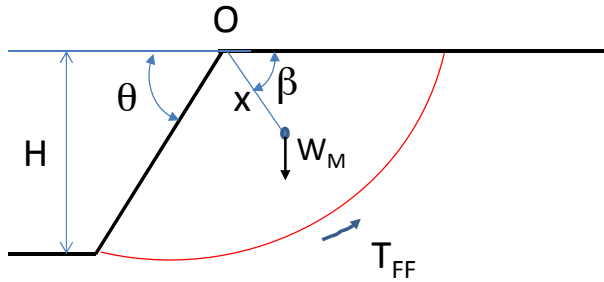
- A) 1.0 B) 1.5 C) 2.0 D) 2.5

The Answer is D

Step 1:

The radius of the potential failure plane is

$$R = \frac{H}{\sin(\theta)} = \frac{(10 \text{ m})}{\sin(20^\circ)} \cong 29.24 \text{ m}$$



Step 2:

Let β be the angle that bisects the top slope angle, then $2\beta = 180^\circ - \theta = 180^\circ - 20^\circ = 160^\circ$. The weight of the sliding soil block is

$$W_M = \gamma A = \gamma \pi R^2 \frac{2\beta}{360^\circ} = \gamma \pi R^2 \frac{160^\circ}{360^\circ} = \left(20 \frac{\text{kN}}{\text{m}^3}\right) \pi (29.24 \text{ m})^2 \frac{160^\circ}{360^\circ} \cong 23872.3 \text{ kN/m}$$

Step 3:

The centroid of a circular sector is on the line that bisects the sector at the distance

$$\text{from the center } x = \frac{2R \sin(\beta)}{3 \beta} = \frac{2(29.24 \text{ m}) \sin(80^\circ)}{3 \frac{80^\circ}{180^\circ} \pi} \cong 13.75 \text{ m}$$

Step 4:

Use the arc center O as the point of rotation, the mobilized moment is

$$M_{\text{MOB}} = W_M x \cos(\beta) = \left(23872.3 \frac{\text{kN}}{\text{m}}\right) (13.75 \text{ m}) \cos(80^\circ) = 56999 \text{ kN} \cdot \text{m/m}$$

Step 5:

Since the soil has a zero friction angle, the resistance moment along the potential failure plane is

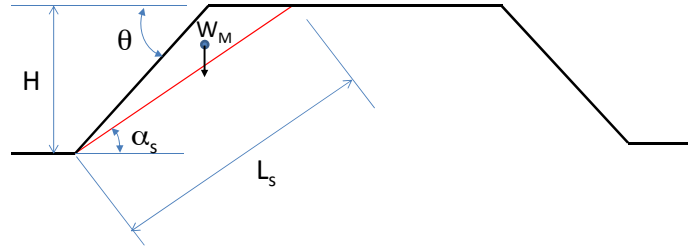
$$\begin{aligned} M_{FF} &= T_{FF}R = cL_sR = c(2\beta R)R \\ &= (60 \text{ kPa}) \left(\frac{2 \times 80^\circ}{180^\circ} \pi (29.24 \text{ m}) \right) (29.24 \text{ m}) \\ &= 143253 \text{ kN} \cdot \text{m/m} \end{aligned}$$

Step 6:

The factor of safety against slope instability, FS is

$$FS = \frac{M_{FF}}{M_{MOB}} = \frac{143253 \text{ kN} \cdot \text{m/m}}{56999 \text{ kN} \cdot \text{m/m}} \cong 2.5$$

25) An embankment constructed with a sandy clay soil has a height $H = 10 \text{ m}$, and a slope angle $\theta = 30^\circ$. The soil has a cohesion $c = 10 \text{ kPa}$, unit weight $\gamma = 19 \text{ kN/m}^3$, and friction angle $\phi = 20^\circ$. The factor of safety against slope instability, FS, for a plane that has an angle α_s from the horizontal surface, is 3. Determine the angle α_s .



- A) 10° B) 15° C) 20° D) 30°

The Answer is A

Step 1:

The top width of the sliding wedge is

$$w = \frac{H}{\tan(\alpha_s)} - \frac{H}{\tan(\theta)} = \frac{(10 \text{ m})}{\tan(\alpha_s)} - \frac{(10 \text{ m})}{\tan(30^\circ)}$$

Step 2:

The weight of the sliding soil wedge is

$$\begin{aligned} W_M &= \gamma A = \gamma \frac{1}{2} wH = \left(19 \frac{\text{kN}}{\text{m}^3}\right) (0.5) \left(\frac{(10 \text{ m})}{\tan(\alpha_s)} - \frac{(10 \text{ m})}{\tan(30^\circ)}\right) (10 \text{ m}) \\ &= \left(950 \frac{\text{kN}}{\text{m}}\right) \left(\frac{1}{\tan(\alpha_s)} - \frac{1}{\tan(30^\circ)}\right) \end{aligned}$$

Step 3:

Since the length of the slip surface is $L_s = H/\sin(\alpha_s)$, the shear resistance along the slip plane is

$$\begin{aligned} T_{FF} &= cL_s + W_M \cos(\alpha_s) \tan(\phi) = cH/\sin(\alpha_s) + W_M \cos(\alpha_s) \tan(\phi) \\ &= \frac{(10 \text{ kPa})(10 \text{ m})}{\sin(\alpha_s)} \\ &\quad + \left(\left(950 \frac{\text{kN}}{\text{m}}\right) \left(\frac{1}{\tan(\alpha_s)} - \frac{1}{\tan(30^\circ)}\right)\right) \cos(\alpha_s) \tan(20^\circ) \end{aligned}$$

Step 4:

The mobilized shear force along the slip plane is

$$T_{MOB} = W_M \sin(\alpha_s) = \left(\left(950 \frac{\text{kN}}{\text{m}}\right) \left(\frac{1}{\tan(\alpha_s)} - \frac{1}{\tan(30^\circ)}\right)\right) \sin(\alpha_s)$$

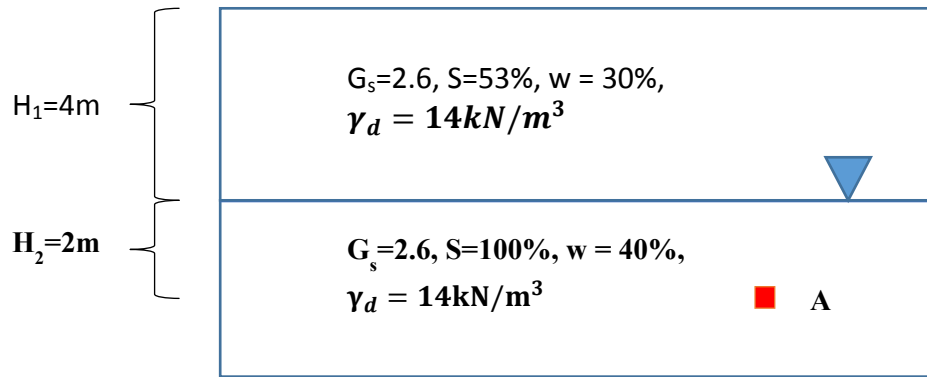
Step 5:

The factor of safety against slope instability, FS is

$$\begin{aligned} \text{FS} &= \frac{T_{\text{FF}}}{T_{\text{MOB}}} \\ &= \frac{\frac{(10 \text{ kPa})(10 \text{ m})}{\sin(\alpha_s)} + \left(\left(950 \frac{\text{kN}}{\text{m}} \right) \left(\frac{1}{\tan(\alpha_s)} - \frac{1}{\tan(30^\circ)} \right) \right) \cos(\alpha_s) \tan(20^\circ)}{\left(\left(950 \frac{\text{kN}}{\text{m}} \right) \left(\frac{1}{\tan(\alpha_s)} - \frac{1}{\tan(30^\circ)} \right) \right) \sin(\alpha_s)} \\ &= 3 \end{aligned}$$

By trial and error, it is obtained that $\alpha_s \cong 10^\circ$

26) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Calculate the effective stress for a soil element at point A.



A) 90kPa

B) 80kPa

C) 85kPa

D) 75kPa

The Answer is C

Step 1: From the diagram, we can see that there are two layers of soil. For each layer, we need to calculate the unit weight. Equation about unit weight is given as,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w$$

The voids ratio is unknown. Therefore, we need to calculate voids ratio of each layer first.

Step 2: Equation of void ratio of soil is shown as,

$$e = \frac{V_v}{V_s}$$

To determine the value of void ratio, we need to find volume of voids, V_v , and volume of solids, V_s . To determine the voids ratio, we can deduce its equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{V_s}{S}} = \frac{\frac{wW}{\gamma_w S}}{\frac{V_s}{S}} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Therefore, for each layer, the voids ratio can be calculated by,

$$e_1 = \gamma_d \frac{w_1}{\gamma_w S_1} = (14 \text{ kN/m}^3) \frac{30\%}{(9.8 \text{ kN/m}^3)(53\%)} = 0.80$$

$$e_2 = \gamma_d \frac{w_2}{\gamma_w S_2} = (14 \text{ kN/m}^3) \frac{40\%}{(9.8 \text{ kN/m}^3)(100\%)} = 0.57$$

Step 3: For the layer above groundwater level, the unit weight is equal to,

$$\gamma_1 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (53\%)(0.8)}{1 + 0.80} \right) (9.8 \text{ kN/m}^3) = 16.5 \text{ kN/m}^3$$

For the layer below groundwater level, the unit weight is equal to,

$$\gamma_2 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (100\%)(0.57)}{1 + 0.57} \right) (9.8 \text{ kN/m}^3) = 19.8 \text{ kN/m}^3$$

Step 4: The total stress at point A can be then calculated as,

$$\sigma = \gamma_1 H_1 + \gamma_2 H_2 = (16.5 \text{ kN/m}^3)(4\text{m}) + (19.8 \text{ kN/m}^3)(2\text{m}) = 105.6 \text{ kPa}$$

The pore water pressure is given as,

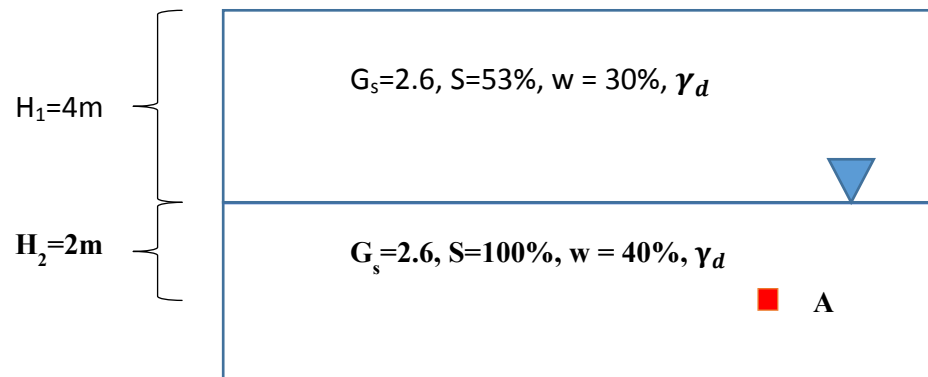
$$u = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(2\text{m}) = 19.6 \text{ kPa}$$

Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = 105.6 \text{ kPa} - 19.6 \text{ kPa} = 86 \text{ kPa}$$

~85kPa

27) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Calculate the dry unit weight of this soil if the effective stress at point A is 85kPa.

A) 12kN/m^3 B) 14kN/m^3 C) 16kN/m^3 D) 18kN/m^3

The Answer is B

Step 1: From the diagram, we can see that there are two layers of soil. For each layer, we need to calculate the unit weight. Equation about unit weight is given as,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w$$

The voids ratio is unknown. Therefore, we need to calculate voids ratio of each layer first.

Step 2: Equation of void ratio of soil can be shown as,

$$e = \frac{V_v}{V_s}$$

To determine the value of void ratio, we need to find volume of voids, V_v , and volume of solids, V_s . To determine the voids ratio, we can deduce its equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{W}{\gamma_w S}} = \frac{W_w}{W} = \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Therefore, for each layer, the voids ratio can be calculated by,

$$e_1 = \gamma_d \frac{w_1}{\gamma_w S_1} = \gamma_d \frac{30\%}{(9.8 \text{ kN/m}^3)(53\%)} = \frac{150\gamma_d}{2597}$$

$$e_2 = \gamma_d \frac{w_2}{\gamma_w S_2} = \gamma_d \frac{40\%}{(9.8 \text{ kN/m}^3)(100\%)} = \frac{2\gamma_d}{49}$$

Step 3: For the layer above groundwater level, the unit weight is equal to,

$$\gamma_1 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (53\%) \left(\frac{2\gamma_d}{49} \right)}{1 + \frac{150\gamma_d}{2597}} \right) (9.8 \text{ kN/m}^3)$$

For the layer below groundwater level, the unit weight is equal to,

$$\gamma_2 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (100\%) \left(\frac{2\gamma_d}{49} \right)}{1 + \frac{2\gamma_d}{49}} \right) (9.8 \text{ kN/m}^3)$$

Step 4: The total stress at point A can be then calculated as,

$$\sigma = \gamma_1 H_1 + \gamma_2 H_2$$

The pore water pressure is given as,

$$u = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(2\text{m}) = 19.6 \text{ kPa}$$

Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = \gamma_1 H_1 + \gamma_2 H_2 - 19.6 \text{ kPa} = 85 \text{ kPa}$$

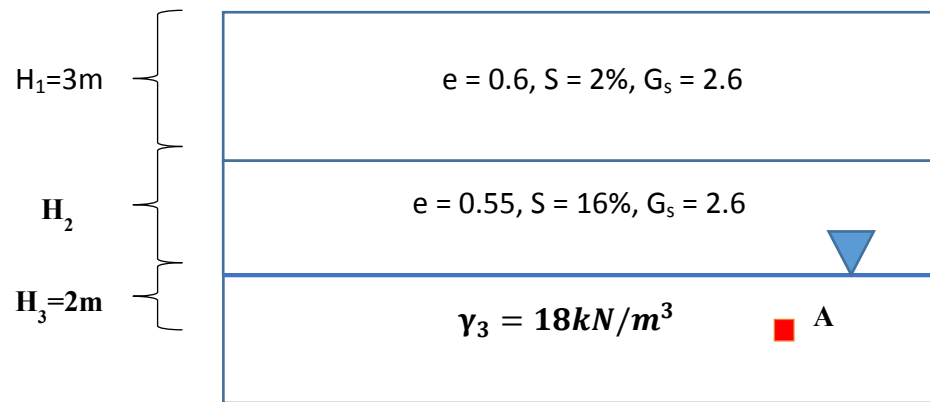
$$= \left(\frac{2.6 + (53\%) \left(\frac{2\gamma_d}{49} \right)}{1 + \frac{150\gamma_d}{2597}} \right) (9.8 \text{ kN/m}^3)(4\text{m})$$

$$+ \left(\frac{2.6 + (100\%) \left(\frac{2\gamma_d}{49} \right)}{1 + \frac{2\gamma_d}{49}} \right) (9.8 \text{ kN/m}^3)(2\text{m}) - 19.6 \text{ kPa}$$

$$\gamma_d = 14 \text{ kN/m}^3$$

$$\sim 14 \text{ kN/m}^3$$

28) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the effective stress at point A is 100kPa, determine the value of H_2 .



A) 2.00m

B) 2.50m

C) 3.00m

D) 3.50m

The Answer is A

Step 1: From the diagram, we can see that there are three layers of soil. For the first layer above groundwater level, the unit weight is equal to,

$$\gamma_1 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (2\%)(0.6)}{1 + 0.6} \right) (9.8 \text{ kN/m}^3) = 16 \text{ kN/m}^3$$

For the second layer above groundwater level, the unit weight is equal to,

$$\gamma_2 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (16\%)(0.55)}{1 + 0.55} \right) (9.8 \text{ kN/m}^3) = 17 \text{ kN/m}^3$$

For the third layer below groundwater level, the unit weight is equal to,

$$\gamma_3 = 18 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\begin{aligned} \sigma &= \gamma_1 H_1 + \gamma_2 H_2 + \gamma_3 H_3 \\ &= (16 \text{ kN/m}^3)(3 \text{ m}) + (17 \text{ kN/m}^3)H_2 + (18 \text{ kN/m}^3)(2 \text{ m}) \\ &= (84 + 17H_2) \text{ kPa} \end{aligned}$$

The pore water pressure is given as,

$$u = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(2 \text{ m}) = 19.6 \text{ kPa}$$

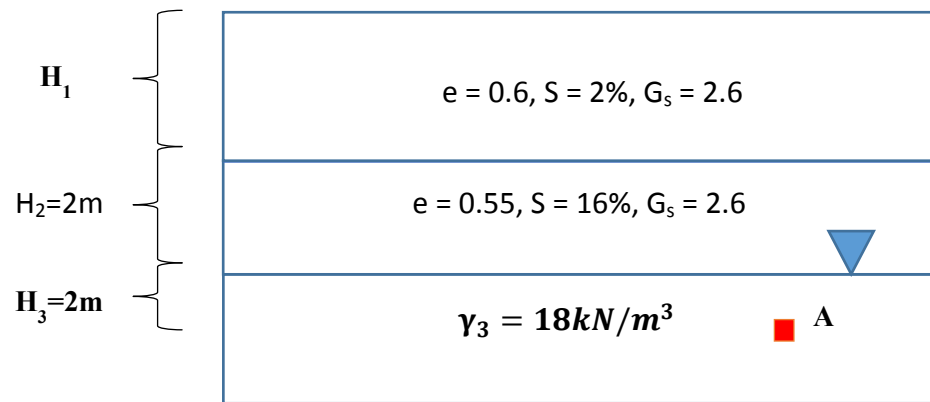
Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = (84 + 17H_2) \text{ kPa} - 19.6 \text{ kPa} = 100 \text{ kPa}$$

Solve the equation above,

$$H_2 = 2 \text{ m} \sim 2 \text{ m}$$

29) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the effective stress at point A is 90kPa, determine the value of H_1 .



A) 2.00m

B) 2.50m

C) 3.00m

D) 3.50m

The Answer is B

Step 1: From the diagram, we can see that there are three layers of soil. For the first layer above groundwater level, the unit weight is equal to,

$$\gamma_1 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (2\%)(0.6)}{1 + 0.6} \right) (9.8 \text{ kN/m}^3) = 16 \text{ kN/m}^3$$

For the second layer above groundwater level, the unit weight is equal to,

$$\gamma_2 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (16\%)(0.55)}{1 + 0.55} \right) (9.8 \text{ kN/m}^3) = 17 \text{ kN/m}^3$$

For the third layer below groundwater level, the unit weight is equal to,

$$\gamma_3 = 18 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\begin{aligned} \sigma &= \gamma_1 H_1 + \gamma_2 H_2 + \gamma_3 H_3 \\ &= (16 \text{ kN/m}^3) H_1 + (17 \text{ kN/m}^3)(2 \text{ m}) \\ &\quad + (18 \text{ kN/m}^3)(2 \text{ m}) = (70 + 16 H_1) \text{ kPa} \end{aligned}$$

The pore water pressure is given as,

$$u = \gamma_w (H_3) = (9.8 \text{ kN/m}^3)(2 \text{ m}) = 19.6 \text{ kPa}$$

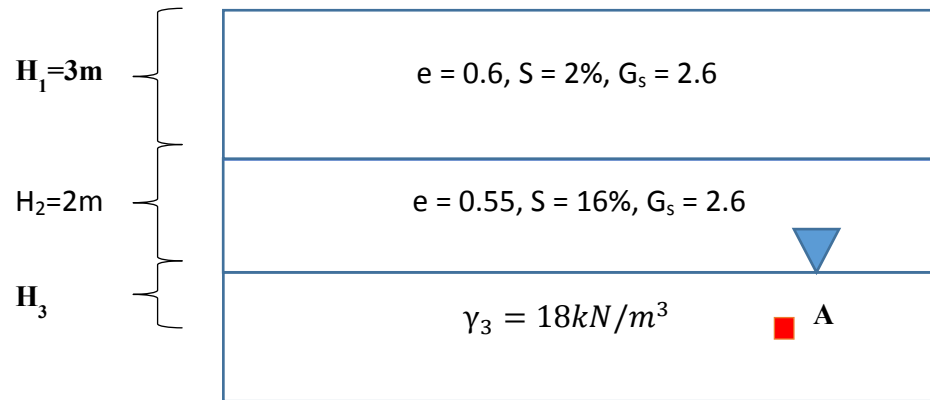
Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = (70 + 16 H_1) \text{ kPa} - 19.6 \text{ kPa} = 90 \text{ kPa}$$

Solve the equation above,

$$H_1 = 2.5 \text{ m} \sim 2.5 \text{ m}$$

30) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the effective stress at point A is 130kPa, determine the value of H_1 .



- A) 3m B) 4m C) 5m D) 6m

The Answer is A

Step 1: From the diagram, we can see that there are three layers of soil. For the first layer above groundwater level, the unit weight is equal to,

$$\gamma_1 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (2\%)(0.6)}{1 + 0.6} \right) (9.8 \text{ kN/m}^3) = 16 \text{ kN/m}^3$$

For the second layer above groundwater level, the unit weight is equal to,

$$\gamma_2 = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (16\%)(0.55)}{1 + 0.55} \right) (9.8 \text{ kN/m}^3) = 17 \text{ kN/m}^3$$

For the third layer below groundwater level, the unit weight is equal to,

$$\gamma_3 = 18 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\begin{aligned} \sigma &= \gamma_1 H_1 + \gamma_2 H_2 + \gamma_3 H_3 \\ &= (16 \text{ kN/m}^3)(3\text{m}) + (17 \text{ kN/m}^3)H_2 + (18 \text{ kN/m}^3)(2\text{m}) \\ &= (84 + 17H_2) \text{ kPa} \end{aligned}$$

The pore water pressure is given as,

$$u = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(2\text{m}) = 19.6 \text{ kPa}$$

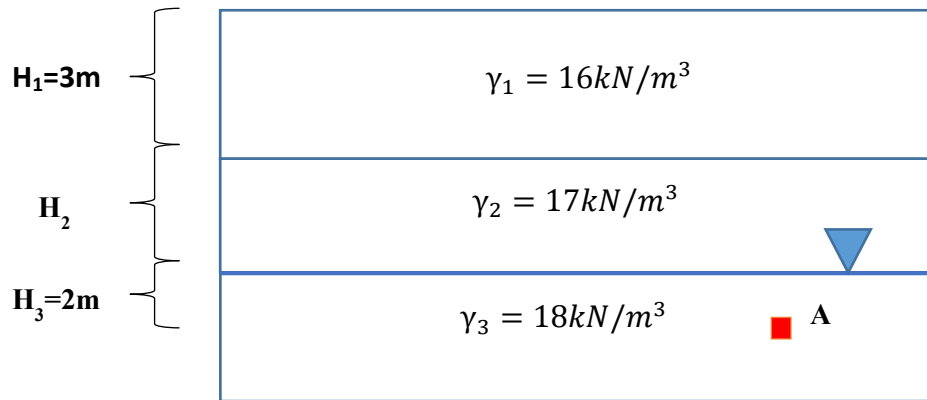
Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = (84 + 17H_2) \text{ kPa} - 19.6 \text{ kPa} = 100 \text{ kPa}$$

Solve the equation above,

$$H_2 = 2\text{m} \sim 2\text{m}$$

31) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the effective stress at point A is 100kPa, determine the value of H_2 .



A) 2.00m

B) 2.50m

C) 3.00m

D) 3.50m

The Answer is A

Step 1: From the diagram, we can see that there are three layers of soil. For the first layer above groundwater level, the unit weight is equal to,

$$\gamma_1 = 16 \text{ kN/m}^3$$

For the second layer above groundwater level, the unit weight is equal to,

$$\gamma_2 = 17 \text{ kN/m}^3$$

For the third layer below groundwater level, the unit weight is equal to,

$$\gamma_3 = 18 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\begin{aligned}\sigma &= \gamma_1 H_1 + \gamma_2 H_2 + \gamma_3 H_3 \\ &= (16 \text{ kN/m}^3)(3 \text{ m}) + (17 \text{ kN/m}^3)H_2 + (18 \text{ kN/m}^3)(2 \text{ m}) \\ &= (84 + 17H_2) \text{ kPa}\end{aligned}$$

The pore water pressure is given as,

$$u = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(2 \text{ m}) = 19.6 \text{ kPa}$$

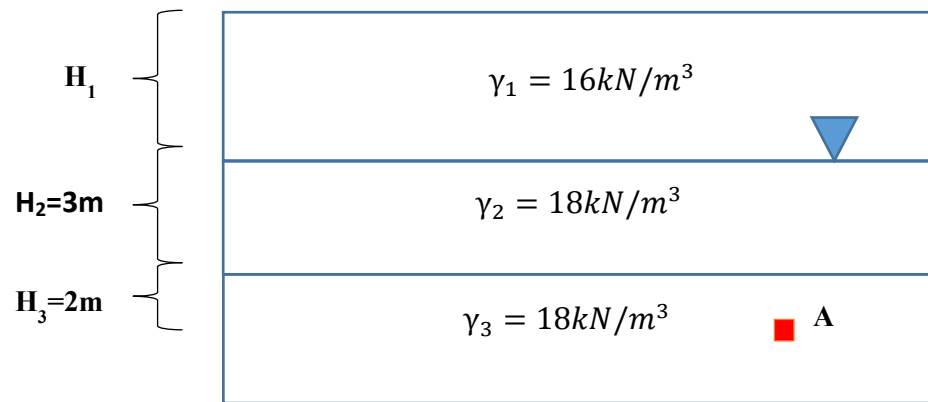
Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = (84 + 17H_2) \text{ kPa} - 19.6 \text{ kPa} = 100 \text{ kPa}$$

Solve the equation above,

$$H_2 = 2 \text{ m} \sim 2 \text{ m}$$

32) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the effective stress at point A is 90kPa, determine the value of H_1 .



A) 2.3m

B) 2.5m

C) 3.5m

D) 3.3m

The Answer is D

Step 1: From the diagram, we can see that there are three layers of soil. For the first layer above groundwater level, the unit weight is equal to,

$$\gamma_1 = 16 \text{ kN/m}^3$$

For the second layer above groundwater level, the unit weight is equal to,

$$\gamma_2 = 17 \text{ kN/m}^3$$

For the third layer below groundwater level, the unit weight is equal to,

$$\gamma_3 = 18 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\begin{aligned}\sigma &= \gamma_1 H_1 + \gamma_2 H_2 + \gamma_3 H_3 \\ &= (16 \text{ kN/m}^3) H_1 + (17 \text{ kN/m}^3)(3 \text{ m}) + (18 \text{ kN/m}^3)(2 \text{ m}) \\ &= (87 + 16 H_1) \text{ kPa}\end{aligned}$$

The pore water pressure is given as,

$$u = \gamma_w (H_2 + H_3) = (9.8 \text{ kN/m}^3)(3 \text{ m} + 2 \text{ m}) = 49 \text{ kPa}$$

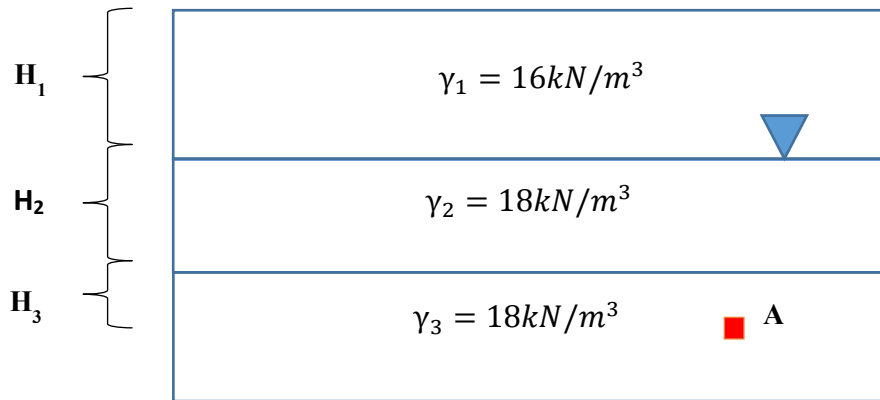
Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = (87 + 16 H_1) \text{ kPa} - 49 \text{ kPa} = 90 \text{ kPa}$$

Solve the equation above,

$$H_2 = 3.25 \text{ m} \sim 3.3 \text{ m}$$

33) An uniform deposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the effective stress at point A is 100kPa, determine the value of H_1 , H_2 , and H_3 if the ratio of them is 2:2:1.



A) 4.2m; 4.2m; 2.1m

B) 3.8m; 3.8m; 1.9m

C) 3.6m; 3.6m; 1.8m

D) 4.0m; 4.0m; 2.0m

The Answer is C

Step 1: From the diagram, we can see that there are three layers of soil. For the first layer above groundwater level, the unit weight is equal to,

$$\gamma_1 = 16 \text{ kN/m}^3$$

For the second layer above groundwater level, the unit weight is equal to,

$$\gamma_2 = 17 \text{ kN/m}^3$$

For the third layer below groundwater level, the unit weight is equal to,

$$\gamma_3 = 18 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\begin{aligned}\sigma &= \gamma_1 H_1 + \gamma_2 H_2 + \gamma_3 H_3 \\ &= (16 \text{ kN/m}^3) 2H_3 + (17 \text{ kN/m}^3) 2H_3 + (18 \text{ kN/m}^3) H_3 \\ &= (84 H_3) \text{ kPa}\end{aligned}$$

The pore water pressure is given as,

$$u = \gamma_w (H_2 + H_3) = (9.8 \text{ kN/m}^3) (2H_3 + H_3) = (29.4 H_3) \text{ kPa}$$

Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = (84 H_3) \text{ kPa} - (29.4 H_3) \text{ kPa} = (54.6 H_3) \text{ kPa} = 100 \text{ kPa}$$

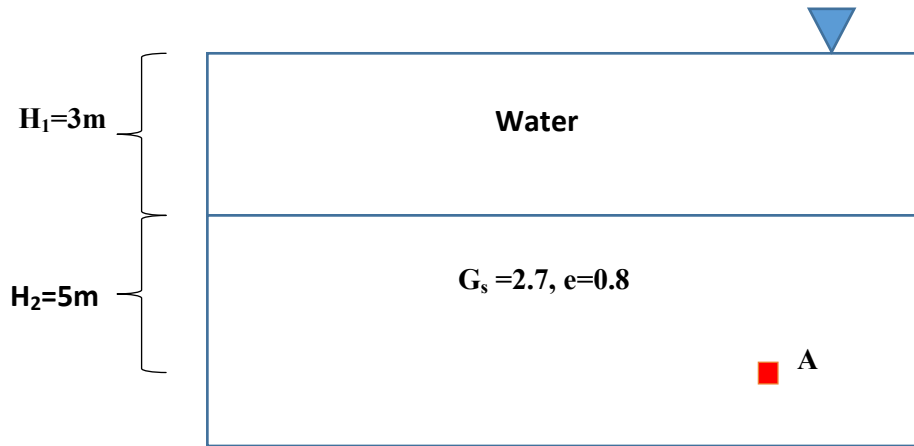
Solve the equation above,

$$H_3 = 1.8 \text{ m}$$

Therefore,

$$H_1 = H_2 = 3.6 \text{ m} \sim 3.6 \text{ m}; 3.6 \text{ m}; 1.8 \text{ m}$$

34) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Determine the effective pressure at point A.



A) 53kPa

B) 46kPa

C) 65kPa

D) 72kPa

The Answer is B

Step 1: From the diagram, we can see that there is only one layer of soil submerged under water. For this layer, the unit weight is equal to,

$$\gamma_{\text{sat}} = \left(\frac{G_s + e}{1 + e} \right) \gamma_w = \left(\frac{2.7 + 0.8}{1 + 0.8} \right) (9.8 \text{ kN/m}^3) = 19 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\sigma = \gamma_1 H_1 + \gamma_{\text{sat}} H_2 = (9.8 \text{ kN/m}^3)(3 \text{ m}) + (19 \text{ kN/m}^3)(5 \text{ m}) = 124.4 \text{ kPa}$$

The pore water pressure is given as,

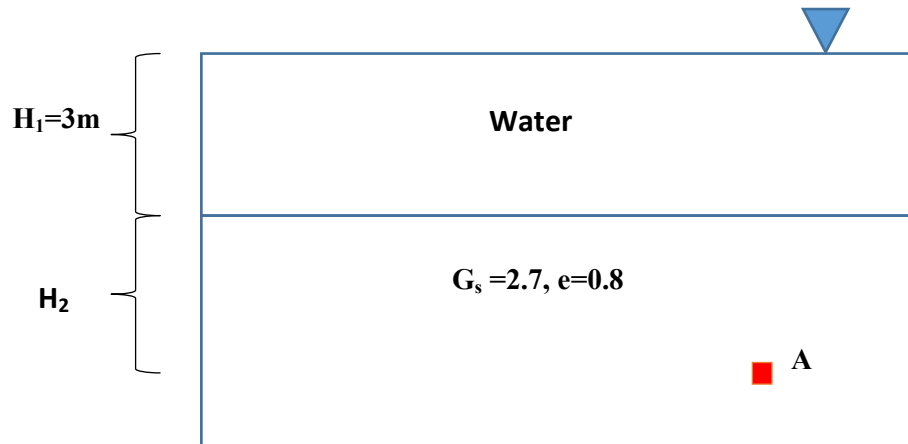
$$u = \gamma_w (H_1 + H_2) = (9.8 \text{ kN/m}^3)(3 \text{ m} + 5 \text{ m}) = 78.4 \text{ kPa}$$

Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = 124.4 \text{ kPa} - 78.4 \text{ kPa} = 46 \text{ kPa}$$

~46kPa

35) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the maximum effective pressure at point A is 50kPa, determine the value of H_2 .



A) 5.8m

B) 4.6m

C) 5.4m

D) 6.3

The Answer is C

Step 1: From the diagram, we can see that there is only one layer of soil submerged under water. For this layer, the unit weight is equal to,

$$\gamma_{\text{sat}} = \left(\frac{G_s + e}{1 + e} \right) \gamma_w = \left(\frac{2.7 + 0.8}{1 + 0.8} \right) (9.8 \text{ kN/m}^3) = 19 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\begin{aligned} \sigma &= \gamma_1 H_1 + \gamma_{\text{sat}} H_2 = (9.8 \text{ kN/m}^3)(3 \text{ m}) + (19 \text{ kN/m}^3) H_2 \\ &= (29.4 + 19 H_2) \text{ kPa} \end{aligned}$$

The pore water pressure is given as,

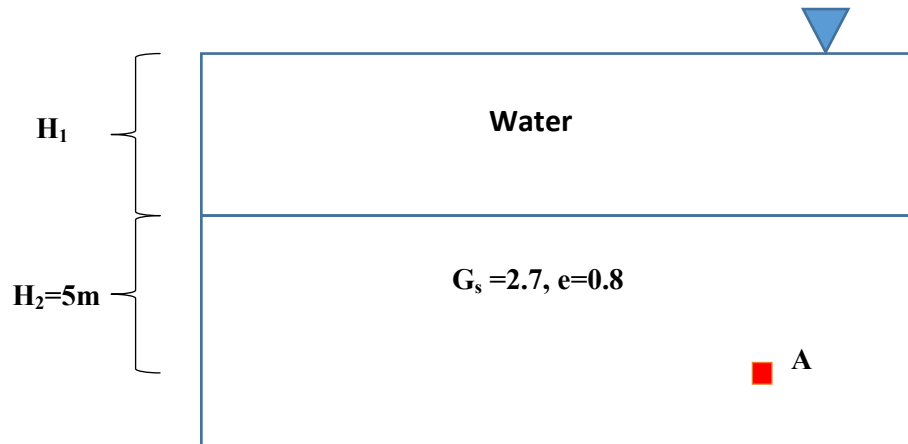
$$u = \gamma_w (H_1 + H_2) = (9.8 \text{ kN/m}^3)(3 \text{ m} + H_2) = (29.4 + 9.8 H_2) \text{ kPa}$$

Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = (29.4 + 19 H_2) \text{ kPa} - (29.4 + 9.8 H_2) \text{ kPa} = 50 \text{ kPa}$$

$$H_2 = 5.4 \text{ m} \sim 5.4 \text{ m}$$

36) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the maximum effective pressure at point A is 46kPa, determine the value of H_1 .



A) 5.8m

B) 4.6m

C) 5.4m

D) Any valu

The Answer is D

Step 1: From the diagram, we can see that there is only one layer of soil submerged under water. For this layer, the unit weight is equal to,

$$\gamma_{\text{sat}} = \left(\frac{G_s + e}{1 + e} \right) \gamma_w = \left(\frac{2.7 + 0.8}{1 + 0.8} \right) (9.8 \text{ kN/m}^3) = 19 \text{ kN/m}^3$$

Step 2: The total stress at point A can be then calculated as,

$$\sigma = \gamma_1 H_1 + \gamma_{\text{sat}} H_2 = (9.8 \text{ kN/m}^3)(3 \text{ m}) + (19 \text{ kN/m}^3)(5 \text{ m}) = 124.4 \text{ kPa}$$

The pore water pressure is given as,

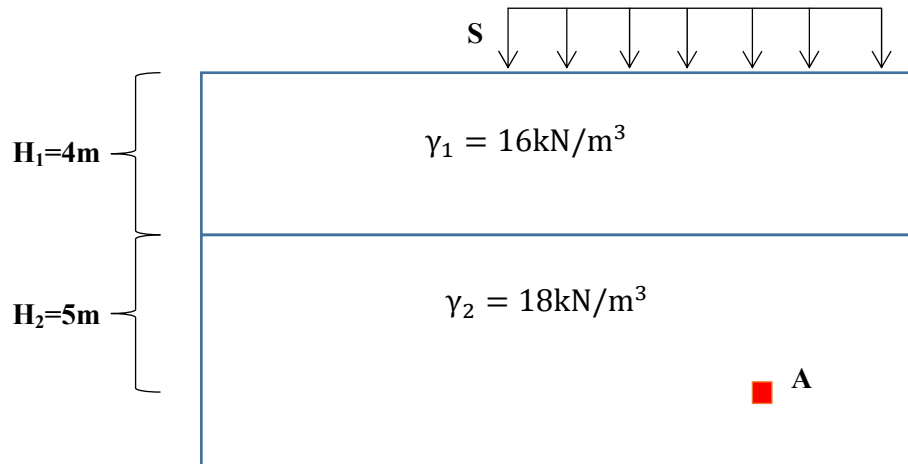
$$u = \gamma_w (H_1 + H_2) = (9.8 \text{ kN/m}^3)(3 \text{ m} + 5 \text{ m}) = 78.4 \text{ kPa}$$

Effective pressure at point A can be finally calculated by,

$$\sigma' = \sigma - u = 124.4 \text{ kPa} - 78.4 \text{ kPa} = 46 \text{ kPa}$$

Therefore, the value of H_1 will not influence the effective pressure at point A. The answer D is correct.

37) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the maximum effective pressure at point A is 200kPa, determine the value of surcharge.



A) 42kPa

B) 46kPa

C) 49kPa

D) 54kPa

The Answer is B

Step 1: From the diagram, we can see that there are two layers of soil. For the first layer, the unit weight is equal to,

$$\gamma_1 = 16 \text{ kN/m}^3$$

For the second layer, the unit weight is equal to,

$$\gamma_2 = 18 \text{ kN/m}^3$$

Step 2: The effective stress at point A can be then calculated as,

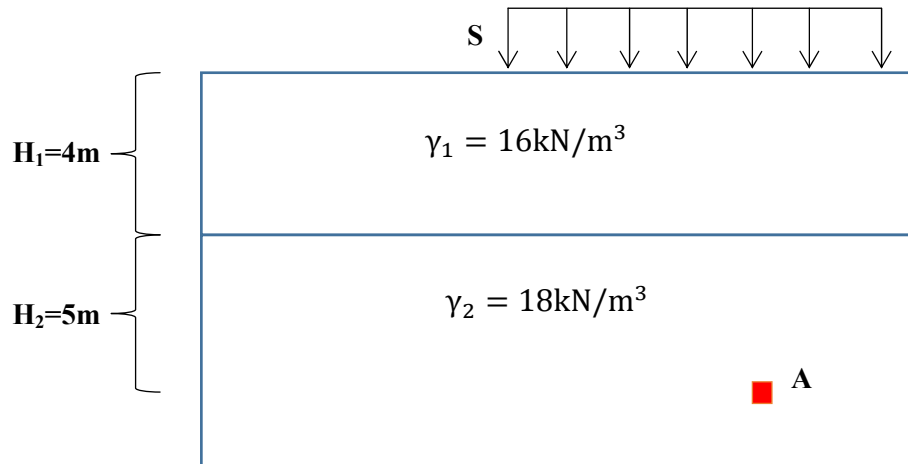
$$\begin{aligned}\sigma &= S + \gamma_1 H_1 + \gamma_2 H_2 = S + (16 \text{ kN/m}^3)(4 \text{ m}) + (18 \text{ kN/m}^3)(5 \text{ m}) \\ &= (S + 154) \text{ kPa}\end{aligned}$$

According to the problem, the effective pressure at point A is equal to 200 kPa, the value of surcharge can be determined by,

$$\sigma = (S + 154) \text{ kPa} = 200 \text{ kPa}$$

$$S = 46 \text{ kPa} \sim 46 \text{ kPa}$$

38) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that value of surcharge is 50kPa. Determine the effective prssure at point A.



A) 100kPa

B) 250kPa

C) 200kPa

D) 150Kp

The Answer is C

Step 1: From the diagram, we can see that there are two layers of soil. For the first layer, the unit weight is equal to,

$$\gamma_1 = 16\text{kN/m}^3$$

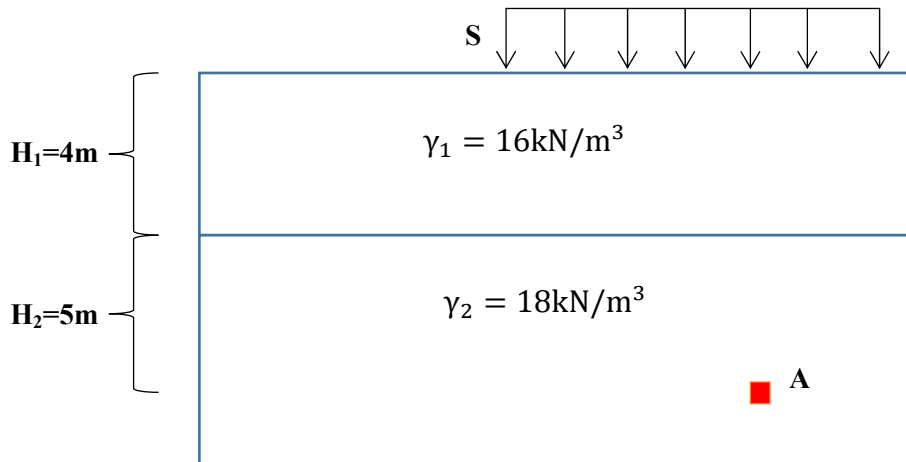
For the second layer, the unit weight is equal to,

$$\gamma_2 = 18\text{kN/m}^3$$

Step 2: The effective stress at point A can be then calculated as,

$$\sigma = S + \gamma_1 H_1 + \gamma_2 H_2 = 50\text{kPa} + (16\text{kN/m}^3)(4\text{m}) + (18\text{kN/m}^3)(5\text{m}) = 204\text{kPa} \sim 200\text{kPa}$$

39) An uniform desposit soil is shown in the following diagram. Information of soil in each layer is given. At first, there is no surcharge at the top of soil. If H_2 is reduced as $0.5H_2$ and in order to keep constant effective pressure at point, surcharge is added at top. Determine the value of this surcharge.



A) 45kPa

B) 50kPa

C) 55kPa

D) 60kPa

The Answer is A

Step 1: From the diagram, we can see that there are two layers of soil. For the first layer, the unit weight is equal to,

$$\gamma_1 = 16 \text{ kN/m}^3$$

For the second layer, the unit weight is equal to,

$$\gamma_2 = 18 \text{ kN/m}^3$$

Step 2: Before changing the value of H_2 , the effective stress at point A can be then calculated as,

$$\sigma_1 = \gamma_1 H_1 + \gamma_2 H_2 = (16 \text{ kN/m}^3)(4 \text{ m}) + (18 \text{ kN/m}^3)(5 \text{ m}) = 154 \text{ kPa}$$

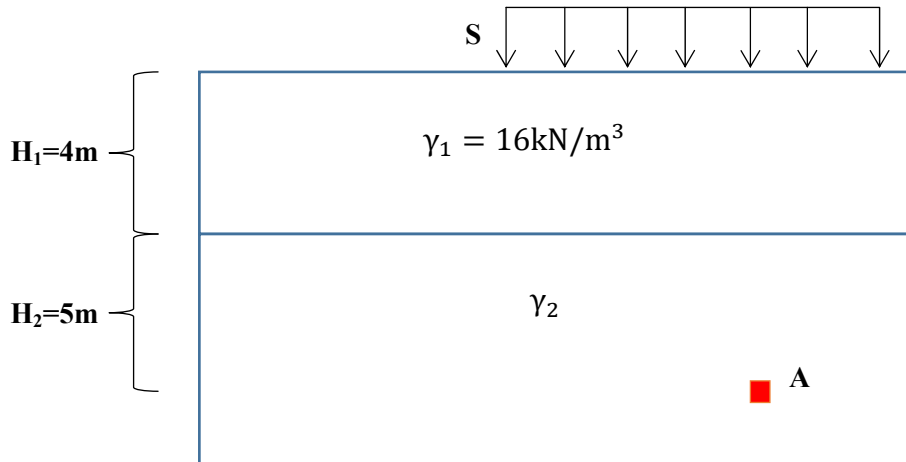
After reducing the value of H_2 , the effective stress at point A can be calculated as,

$$\begin{aligned} \sigma_2 &= S + \gamma_1 H_1 + \frac{1}{2} \gamma_2 H_2 = S + (16 \text{ kN/m}^3)(4 \text{ m}) + (18 \text{ kN/m}^3)\left(\frac{5}{2} \text{ m}\right) \\ &= 154 \text{ kPa} \end{aligned}$$

Solve equation above, $S = 45 \text{ kPa}$

~45kPa

40) An uniform deposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the surcharge is 30kPa acting at the top of soil as shown in the diagram. If the effective pressure at point A is 180kPa, determine the effective density of second layer, γ_2 .



- A) 16.8kN/m^3
C) 17.2kN/m^3

- B) 18.4kN/m^3
D) 19.6kN/m^3

The Answer is C

Step 1: From the diagram, we can see that there are two layers of soil. For the first layer, the unit weight is equal to,

$$\gamma_1 = 16 \text{ kN/m}^3$$

For the second layer, the unit weight is unknown,

$$\gamma_2$$

Step 2: The effective stress at point A can be then calculated as,

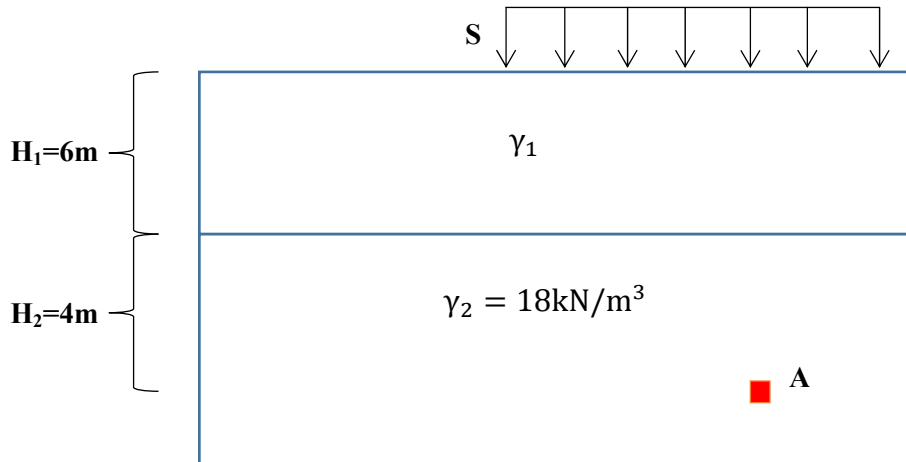
$$\sigma = S + \gamma_1 H_1 + \gamma_2 H_2 = 30 \text{ kPa} + (16 \text{ kN/m}^3)(4 \text{ m}) + \gamma_2(5 \text{ m}) = 180 \text{ kPa}$$

Solve the equation above, we can find the effective density of second layers as,

$$\gamma_2 = 17.2 \text{ kN/m}^3$$

$$\sim 17.2 \text{ kN/m}^3$$

41) An uniform deposit soil is shown in the following diagram. Information of soil in each layer is given. Assume that the surcharge is 45kPa acting at the top of soil as shown in the diagram. If the effective pressure at point A is 200kPa, determine the effective density of first layer, γ_1 .



- A) 15.6kN/m^3
C) 14.4kN/m^3

- B) 12.6kN/m^3
D) 13.8kN/m^3

The Answer is D

Step 1: From the diagram, we can see that there are two layers of soil. For the first layer, the unit weight is unknown,

γ_1

For the second layer, the unit weight is equal to,

$$\gamma_2 = 18 \text{ kN/m}^3$$

Step 2: The effective stress at point A can be then calculated as,

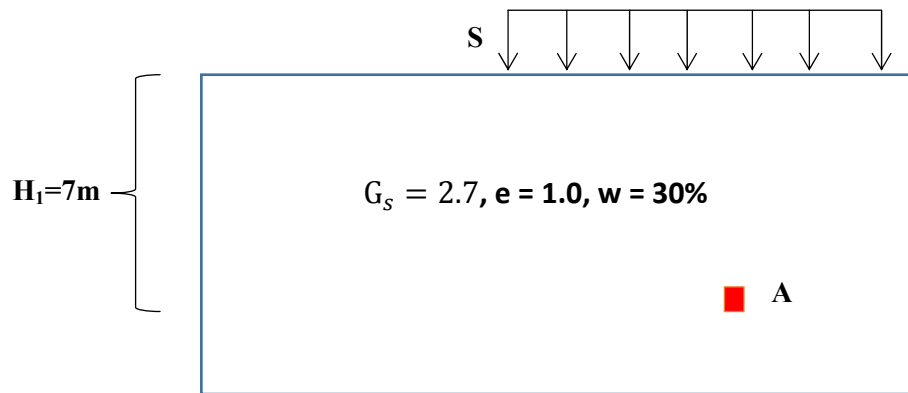
$$\sigma = S + \gamma_1 H_1 + \gamma_2 H_2 = 45 \text{ kPa} + \gamma_1 (6 \text{ m}) + (18 \text{ kN/m}^3)(4 \text{ m}) = 200 \text{ kPa}$$

Solve the equation above, we can find the effective density of second layers as,

$$\gamma_2 = 13.8 \text{ kN/m}^3$$

$$\sim 13.8 \text{ kN/m}^3$$

42) Information of a soil is shown in the following diagram. Assume that the surcharge is 40kPa acting at the top of soil as shown in the diagram. Determine the effective pressure at point A.



A) 160kPa

B) 150kPa

C) 140kPa

D) 130kPa

The Answer is A

Step 1: From the diagram, we can see that there is only one layer of soil. For this layer, the unit weight is equal to,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w$$

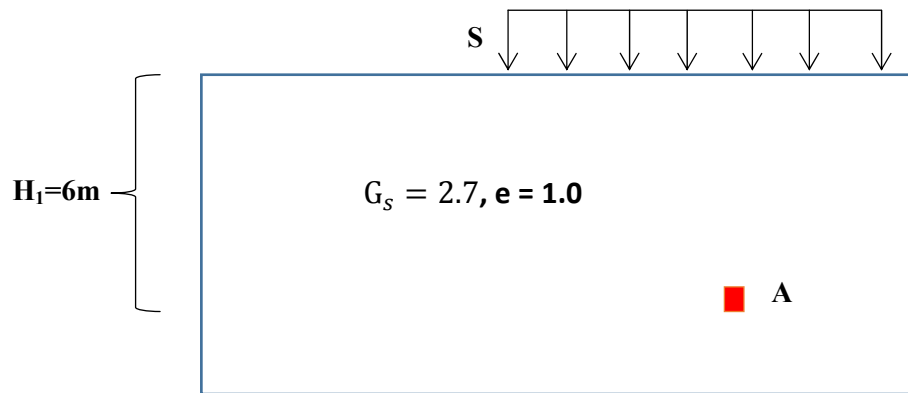
where, $S = \frac{wG_s}{e} = \frac{(30\%)(2.7)}{1.0} = 81\%$, therefore,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.7 + (81\%)(1.0)}{1 + 1.0} \right) (9.8 \text{ kN/m}^3) = 17 \text{ kN/m}^3$$

Step 2: The effective stress at point A can be then calculated as,

$$\sigma = S + \gamma H_1 = 40 \text{ kPa} + (17 \text{ kN/m}^3)(7 \text{ m}) = 159 \text{ kPa} \\ \sim 160 \text{ kPa}$$

43) Information of a soil is shown in the following diagram. Assume that the surcharge is 40kPa acting at the top of soil as shown in the diagram. If the effective pressure at point A is 150kPa. Determine the water content of this soil.



A) 15.6%

B) 14.2%

C) 13.3%

D) 12.1%

The Answer is C

Step 1: From the diagram, we can see that there is only one layer of soil. The effective pressure at point A can be calculated by,

$$\sigma = S + \gamma H_1 = 40 \text{ kPa} + \gamma(6 \text{ m}) = 130 \text{ kPa}$$

Solving this equation, we can know the value of density of this soil as,

$$\gamma = 15 \text{ kN/m}^3$$

Step 2: According to equations of the density of this soil can be described as,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w$$

Therefore,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.7 + S}{1 + 1} \right) (9.8 \text{ kN/m}^3) = 15 \text{ kN/m}^3$$

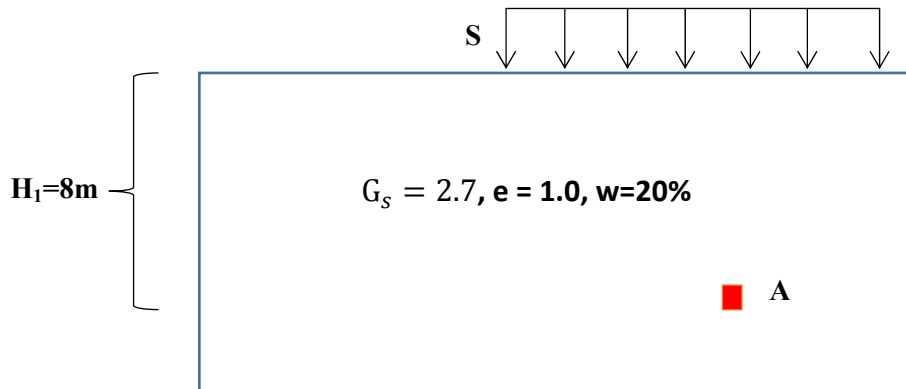
$$S = 36\%$$

According to the following equation water content of soil is,

$$w = \frac{Se}{G_s} = \frac{(36\%)(1.0)}{2.7} = 13.3\%$$

$$\sim 13.3\%$$

44) Information of a soil is shown in the following diagram. Assume that the value of surcharge is unknown, which acting at the top of soil as shown in the diagram. If the effective pressure at point A is 170kPa, determine the value of this surcharge.



A) 39kPa

B) 43kPa

C) 47kPa

D) 52kPa

The Answer is B

Step 1: From the diagram, we can see that there is only one layer of soil. For this layer, the unit weight is equal to,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w$$

where, $S = \frac{wG_s}{e} = \frac{(20\%)(2.7)}{1.0} = 54\%$, therefore,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.7 + (54\%)(1.0)}{1 + 1.0} \right) (9.8 \text{ kN/m}^3) = 15.9 \text{ kN/m}^3$$

Step 2: The effective stress at point A can be then calculated as,

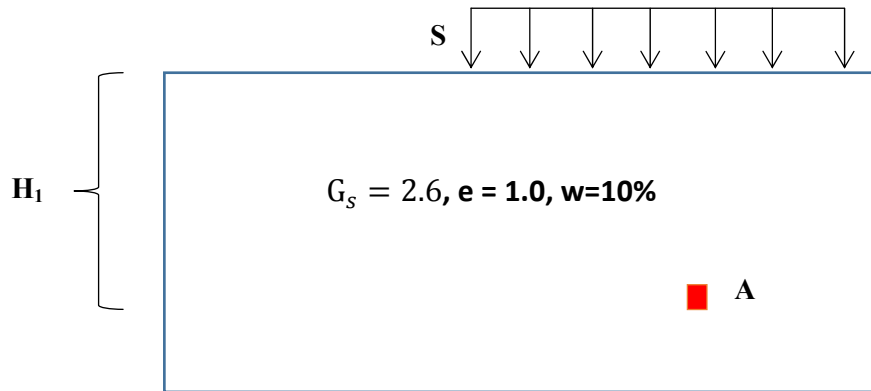
$$\sigma = S + \gamma H_1 = S + (15.9 \text{ kN/m}^3)(8\text{m}) = 170 \text{ kPa}$$

Solving this equation we got,

$$S = 42.8 \text{ kPa}$$

$$\sim 43 \text{ kPa}$$

45) Information of a soil is shown in the following diagram. Assume that the value of surcharge is 25kPa, which acting at the top of soil as shown in the diagram. If the effective pressure at point A is 140kPa, determine the value of H_1 .



A) 8.2m

B) 7.4m

C) 6.6m

D) 5.8m

The Answer is A

Step 1: From the diagram, we can see that there is only one layer of soil. For this layer, the unit weight is equal to,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w$$

where, $S = \frac{wG_s}{e} = \frac{(10\%)(2.6)}{1.0} = 26\%$, therefore,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{2.6 + (26\%)(1.0)}{1 + 1.0} \right) (9.8 \text{ kN/m}^3) = 14 \text{ kN/m}^3$$

Step 2: The effective stress at point A can be then calculated as,

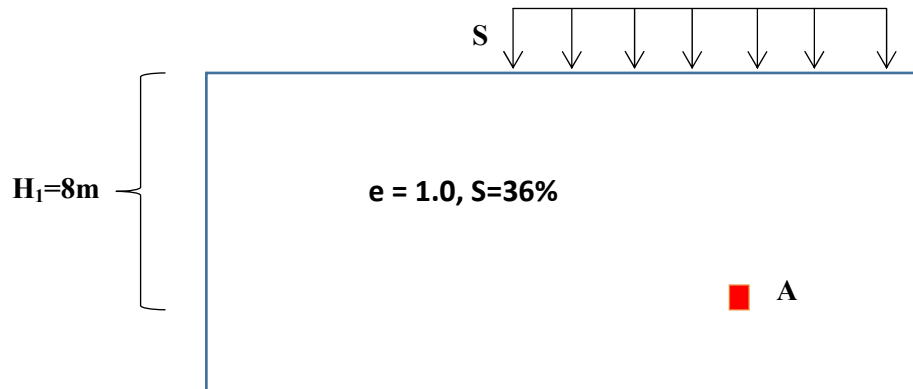
$$\sigma = S + \gamma H_1 = 25 \text{ kPa} + (14 \text{ kN/m}^3) H_1 = 140 \text{ kPa}$$

Solving this equation we got,

$$H_1 = 8.2 \text{ m}$$

~8.2m

46) Information of a soil is shown in the following diagram. Assume that the value of surcharge is 35kPa, which acting at the top of soil as shown in the diagram. If the effective pressure at point A is 160kPa, determine the value of specific gravity of this soil.



A) 2.5

B) 2.6

C) 2.7

D) 2.8

The Answer is D

Step 1: From the diagram, we can see that there is only one layer of soil. The effective pressure at point A can be calculated by,

$$\sigma = S + \gamma H_1 = 35kPa + \gamma(8m) = 160kPa$$

Solving this equation, we can know the value of density of this soil as,

$$\gamma = 15.6kN/m^3$$

Step 2: According to equations of the density of this soil can be described as,

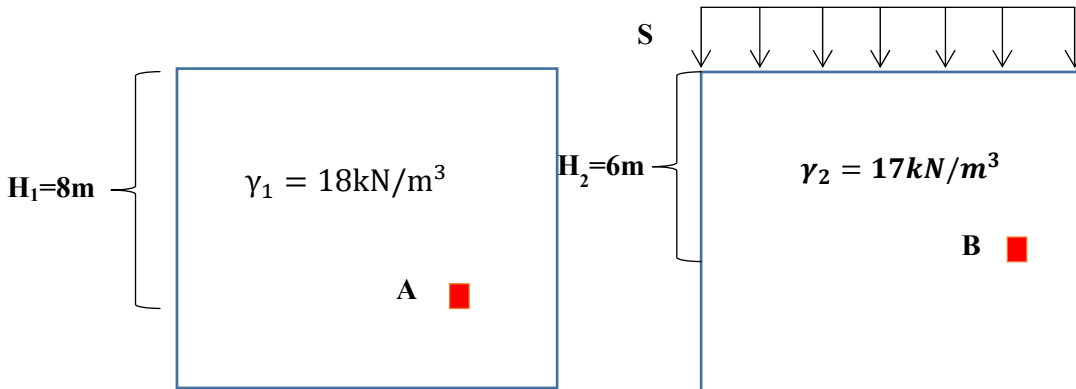
$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w$$

Therefore,

$$\gamma = \left(\frac{G_s + Se}{1 + e} \right) \gamma_w = \left(\frac{G_s + (36\%)(1)}{1 + 1} \right) (9.8kN/m^3) = 15.6kN/m^3$$

$$G_s = 2.8 \sim 2.8$$

47) Assume that there are two soils shown in the following diagram. For the left one, there is no surcharge. For the right one, a surcharge acting on the top of soil but the value is unknown. Assume that effective pressure at point A is the same as that at point B. Determine the value of this surcharge.



A) 15kPa

B) 20kPa

C) 25kPa

D) 30kPa

The Answer is C

Step 1: From the diagram, we can see that there is only one layer of soil on both right and left soil. The effective pressure at point A can be calculated by,

$$\sigma_A = \gamma_1 H_1 = (18 \text{ kN/m}^3)(8 \text{ m}) = 144 \text{ kPa}$$

The effective pressure at point B can be calculated by,

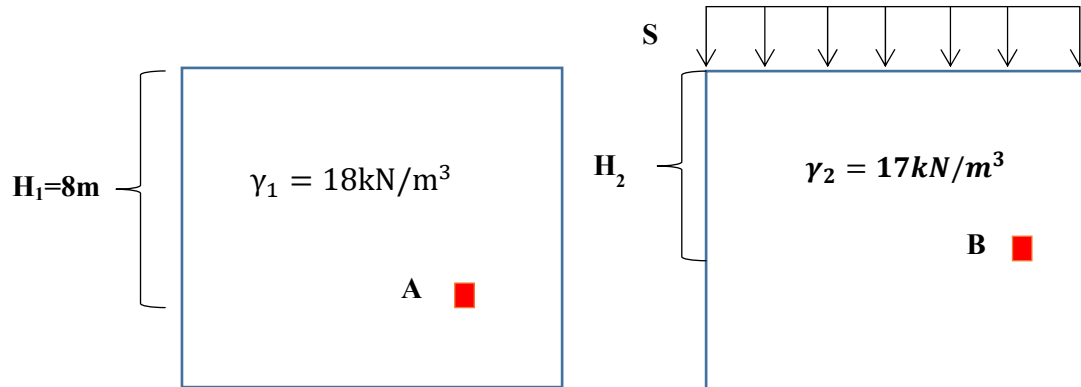
$$\sigma_B = S + \gamma_2 H_2 = S + (17 \text{ kN/m}^3)(7 \text{ m}) = (S + 119) \text{ kPa}$$

Step 2: According to the problem, effective pressure at point A is the same as that at point B, so we got

$$144 \text{ kPa} = (S + 119) \text{ kPa}$$

$$S = 25 \text{ kPa} \sim 25 \text{ kPa}$$

48) Assume that there are two soils shown in the following diagram. For the left one, there is no surcharge. For the right one, a surcharge acting on the top of soil and the value is 30kPa. Assume that effective pressure at point A is the same as that at point B. Determine the value of H_2 .



A) 4.3m

B) 5.8m

C) 6.7m

D) 7.4m

The Answer is C

Step 1: From the diagram, we can see that there is only one layer of soil on both right and left soil. The effective pressure at point A can be calculated by,

$$\sigma_A = \gamma_1 H_1 = (18 \text{ kN/m}^3)(8 \text{ m}) = 144 \text{ kPa}$$

The effective pressure at point B can be calculated by,

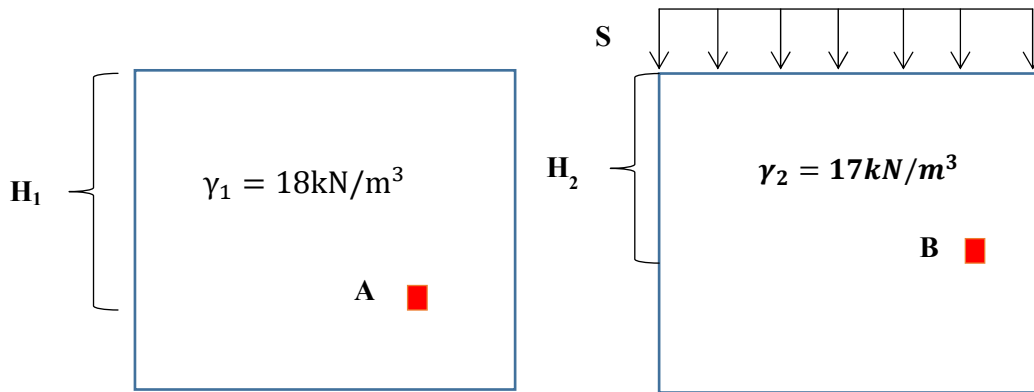
$$\sigma_B = S + \gamma_2 H_2 = 30 \text{ kPa} + (17 \text{ kN/m}^3) H_2 = (30 + 17 H_2) \text{ kPa}$$

Step 2: According to the problem, effective pressure at point A is the same as that at point B, so we got

$$144 \text{ kPa} = (30 + 17 H_2) \text{ kPa}$$

$$H_2 = 6.7 \text{ m} \sim 6.7 \text{ m}$$

49) Assume that there are two soils shown in the following diagram. For the left one, there is no surcharge. For the right one, a surcharge acting on the top of soil and the value is 40kPa. Assume that effective pressure at point A is the same as that at point B. Effective pressure at point A is 90kPa. Determine the ratio between H_1 and H_2 .



- A) 5:3 B) 3:5 C) 4:3 D) 3:4

The Answer is A

Step 1: From the diagram, we can see that there is only one layer of soil on both right and left soil. The effective pressure at point A can be calculated by,

$$\sigma_A = \gamma_1 H_1 = (18 \text{ kN/m}^3) H_1 = (18 H_1) \text{ kPa}$$

The effective pressure at point B can be calculated by,

$$\sigma_B = S + \gamma_2 H_2 = 40 \text{ kPa} + (17 \text{ kN/m}^3) H_2 = (40 + 17 H_2) \text{ kPa}$$

Step 2: According to the problem, effective pressure at point A is the same as that at point B, so we got

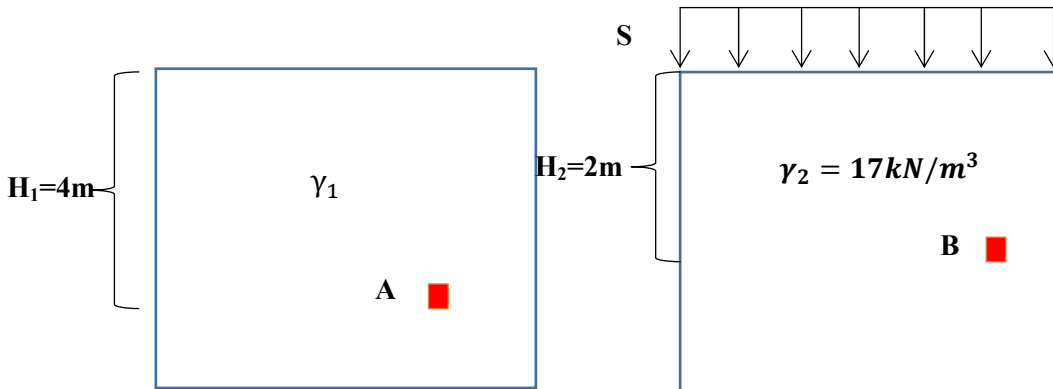
$$(18 H_1) \text{ kPa} = (40 + 17 H_2) \text{ kPa} = 90 \text{ kPa}$$

$$H_1 = 5 \text{ m}$$

$$H_2 = 3 \text{ m}$$

Therefore, the ratio between H_1 and H_2 is 5:3~5:3

50) Assume that there are two soils shown in the following diagram. For the left one, there is no surcharge. For the right one, a surcharge acting on the top of soil and the value is 40kPa. Assume that effective pressure at point A is the same as that at point B. Determine the density of the left soil, γ_1 .



- A) 17kN/m^3 B) 19kN/m^3 C) 21kN/m^3 D) 15kN/m^3

The Answer is B

Step 1: From the diagram, we can see that there is only one layer of soil on both right and left soil. The effective pressure at point A can be calculated by,

$$\sigma_A = \gamma_1 H_1 = \gamma_1 (4m) = (4\gamma_1) kPa$$

The effective pressure at point B can be calculated by,

$$\sigma_B = S + \gamma_2 H_2 = 40 kPa + (17 kN/m^3)(2m) = 74 kPa$$

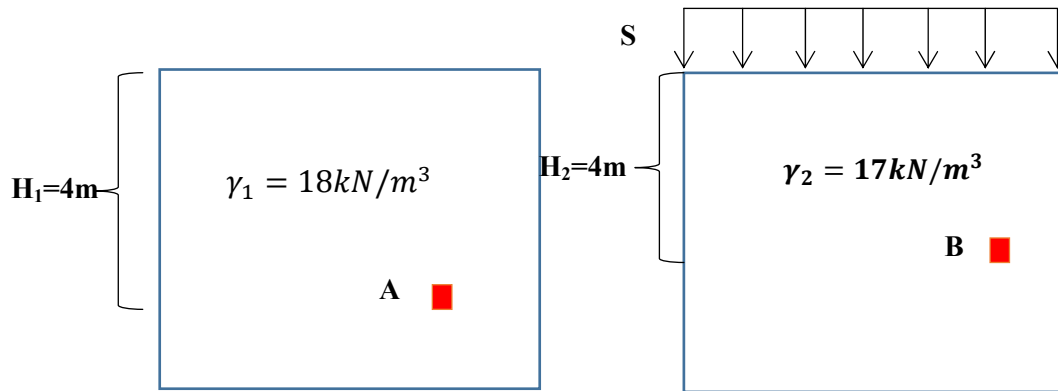
Step 2: According to the problem, effective pressure at point A is the same as that at point B, so we got

$$(4\gamma_1) kPa = 74 kPa$$

$$\gamma_1 = 18.5 kN/m^3$$

$$\sim 19 kN/m^3$$

51) Assume that there are two soils shown in the following diagram. For the left one, there is no surcharge. For the right one, a surcharge acting on the top of soil and the value is 40kPa. Determine the effective pressure ratio between A and B.



- A) 1:1 B) 1:3 C) 2:3 D) 4:3

The Answer is C

Step 1: From the diagram, we can see that there is only one layer of soil on both right and left soil. The effective pressure at point A can be calculated by,

$$\sigma_A = \gamma_1 H_1 = (18 \text{ kN/m}^3)(4 \text{ m}) = 72 \text{ kPa}$$

The effective pressure at point B can be calculated by,

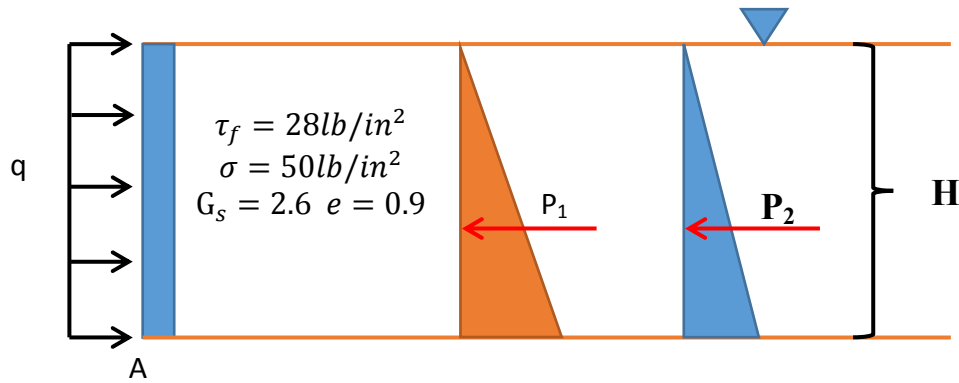
$$\sigma_B = S + \gamma_2 H_2 = 40 \text{ kPa} + (17 \text{ kN/m}^3)(4 \text{ m}) = 108 \text{ kPa}$$

Step 2: Therefore, the effective pressure ratio between A and B is,

$$\sigma_A : \sigma_B = 72 \text{ kPa} : 108 \text{ kPa} = 2 : 3$$

~2:3

52) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that $q=20\text{kN/m}$. Calculate the value of H so that the moment at point A is zero.



A) 4.2m

B) 4.8m

C) 3.8m

D) 3.2m

The Answer is B

Step 1: From the diagram, we can see that there are two horizontal forces acting towards the wall. We need to calculate these forces first. From the problem, the friction angle of this soil can be found by,

$$\phi' = \tan^{-1} \left(\frac{\tau_f}{\sigma} \right) = \tan^{-1} \left(\frac{28 \text{ lb/in}^2}{50 \text{ lb/in}^2} \right) = 30^\circ$$

Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: The saturated unit weight can be calculated by,

$$\gamma_{sat} = \frac{(G_s + e)\gamma_w}{1 + e} = \frac{(2.6 + 0.9)(9.8 \text{ kN/m}^3)}{1 + 0.9} = 18 \text{ kN/m}^3$$

Step 3: The lateral forces due to soil and pore water can be calculated respectively as,

$$P_1 = \frac{1}{2} K_a (\gamma_{sat} - \gamma_w) H^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) H^2 \\ = (1.37 H^2) \text{ kN}$$

$$z_1 = \frac{1}{3} H$$

$$P_2 = \frac{1}{2} \gamma_w H^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) H^2 = (4.9 H^2) \text{ kN}$$

$$z_2 = \frac{1}{3} H$$

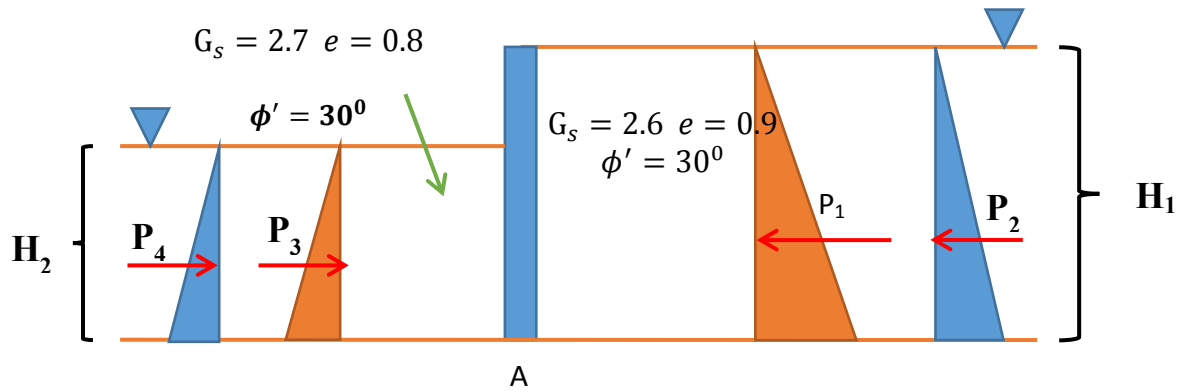
Step 4: In order to make the moment at point A zero we got,

$$\frac{1}{2} q H^2 = P_1 z_1 + P_2 z_2 = (1.37 H^2) \left(\frac{1}{3} H \right) + (4.9 H^2) \frac{1}{3} H = \frac{1}{2} (20 \text{ kN/m}) H^2$$

Solve this equation,

$$H = 4.8m$$

53) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 5m and H_2 is 3m. Calculate the value of resultant force towards this wall.



A) 12kN

B) 14kN

C) 16kN

D) 18

The Answer is A

Step 1: From the diagram, we can see that there are two parts of soil acting towards wall. We need to calculate forces from both parts first. From the problem, Rankine active earth pressure coefficient for right part can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Rankine passive earth pressure coefficient for left part can be calculated by,

$$K_p = \tan^2 \left(45^\circ + \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ + \frac{30^\circ}{2} \right) = 3$$

Step 2: The saturated unit weight can be calculated by,

$$\gamma_1 = \frac{(G_s + e)\gamma_w}{1 + e} = \frac{(2.6 + 0.9)(9.8 \text{ kN/m}^3)}{1 + 0.9} = 18 \text{ kN/m}^3$$

$$\gamma_2 = \frac{(G_s + e)\gamma_w}{1 + e} = \frac{(2.7 + 0.8)(9.8 \text{ kN/m}^3)}{1 + 0.8} = 19 \text{ kN/m}^3$$

Step 3: The lateral forces for right part can be calculated by,

$$P_1 = \frac{1}{2} K_a (\gamma_1 - \gamma_w) H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 34 \text{ kN}$$

$$P_2 = \frac{1}{2} \gamma_w H_1^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 122.5 \text{ kN}$$

The lateral forces for the left part can be calculated by,

$$P_3 = \frac{1}{2} K_p (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} (3) (19 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 124.2 \text{ kN}$$

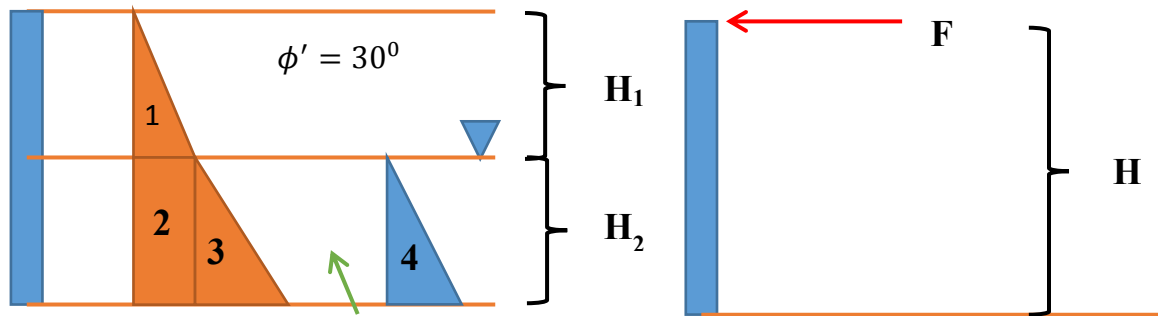
$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

Step 4: The resultant force for this wall can be calculated by,

$$F = P_3 + P_4 - P_1 - P_2 = 124.2 \text{ kN} + 44.1 \text{ kN} - 34 \text{ kN} - 122.5 \text{ kN} = 11.8 \text{ kN}$$

~12kN

54) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 3m. H is 6m. $F = 40\text{kN}$. If moment at point A in the left figure is the same as that in the right figure, calculate the value of H_2 .



- A) 3m B) 4m C) 5m D) 6m

The Answer is A

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: The saturated unit weight can be calculated by,

$$\gamma_2 = \frac{(G_s + e)\gamma_w}{1 + e} = \frac{(2.7 + 0.8)(9.8 \text{ kN/m}^3)}{1 + 0.8} = 19 \text{ kN/m}^3$$

Step 3: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m})^2 = 25.5 \text{ kN}$$

$$z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (3 \text{ m}) + H_2 = 1 + H_2$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m}) H_2 = 17 H_2$$

$$z_2 = \frac{1}{2} H_2 = 0.5 H_2$$

$$P_3 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (19 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) H_2^2$$

$$z_3 = \frac{1}{3} H_2$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (H_2)^2$$

$$z_4 = \frac{1}{3} H_2$$

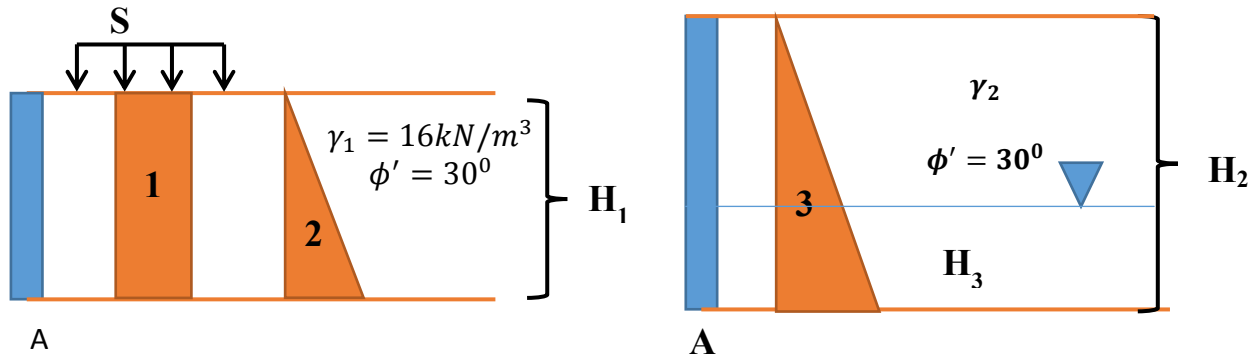
Step 4: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_1 z_1 + P_2 z_2 + P_3 z_3 + P_4 z_4 \\ &= (25.5kN)(1 + H_2) + (17H_2)(0.5H_2) + (1.53H_2^2)\left(\frac{1}{3}H_2\right) \\ &\quad + (4.9H_2^2)\left(\frac{1}{3}H_2\right) = (40kN)(6m) \end{aligned}$$

Solve this equation,

$$H_2 = 3m \sim 3m$$

55) Two figures show two frictionless wall as below. Information about each wall is given. Assume that $H_1=4\text{m}$ and $H_2=5\text{m}$, $H_3=1\text{m}$, $S=20\text{kN/m}$. Assume that moment at point A for left figure and right figure is the same. Determine the density of soil in the right figure, γ_2 .

A) 16kN/m^3 B) 18kN/m^3 C) 20kN/m^3

D) 2

The Answer is A

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = K_a S H_1 = \frac{1}{3} (20 \text{ kN/m}) (4 \text{ m}) = 26.7 \text{ kN}$$

$$z_1 = \frac{1}{2} H_1 = \frac{1}{2} (4 \text{ m}) = 2 \text{ m}$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (16 \text{ kN/m}^3) (4 \text{ m})^2 = 42.7 \text{ kN}$$

$$z_2 = \frac{1}{3} H_1 = \frac{1}{3} (4 \text{ m}) = 1.3 \text{ m}$$

$$P_3 = \frac{1}{2} K_a \gamma_2 H_2^2 - \frac{1}{2} \gamma_w H_3^2 = \frac{1}{2} \left(\frac{1}{3} \right) \gamma_2 (5 \text{ m})^2 - \frac{1}{2} (9.8 \text{ kN/m}^3) (1 \text{ m})^2$$

$$= 4.2 \gamma_2 - 4.9$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (5 \text{ m}) = 1.7 \text{ m}$$

Step 2: The moment at point A is the same. We got,

$$P_1 z_1 + P_2 z_2 = P_3 z_3 = (26.7 \text{ kN}) (2 \text{ m}) + (42.7 \text{ kN}) (1.3 \text{ m})$$

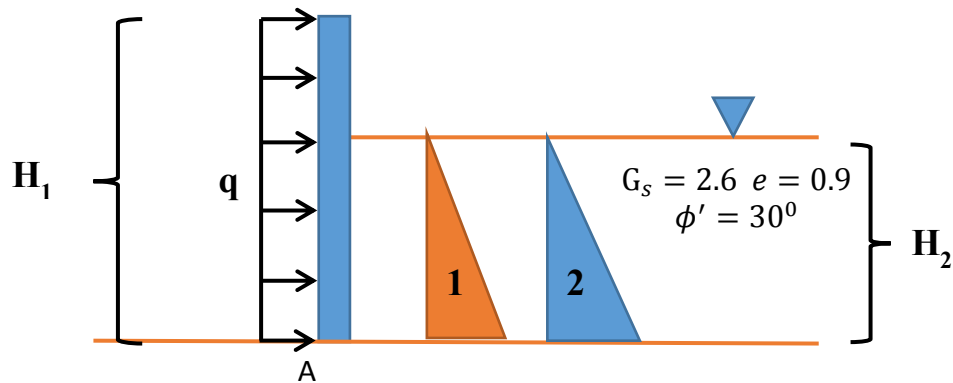
$$= (4.2 \gamma_2 - 4.9) (1.7 \text{ m})$$

Solve this equation,

$$\gamma_2 = 16.4 \text{ kN/m}^3$$

$$\sim 16 \text{ kN/m}^3$$

56) A retaining wall is given in the following diagram. It is known $H_1=5\text{m}$. $q = 10\text{kN/m}$. Determine the value of H_2 when resultant force is equal to zero.



A) 2.8m

B) 3.6m

C) 4.2m

D) 3.8

The Answer is A

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: The saturated unit weight can be calculated by,

$$\gamma = \frac{(G_s + e)\gamma_w}{1 + e} = \frac{(2.6 + 0.9)(9.8 \text{ kN/m}^3)}{1 + 0.9} = 18 \text{ kN/m}^3$$

Step 2: Lateral force for Area 1,

$$\begin{aligned} P_1 &= \frac{1}{2} K_a (\gamma - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (H_2)^2 \\ &= 1.37 (H_2)^2 \end{aligned}$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (H_2)^2 = 4.9 (H_2)^2$$

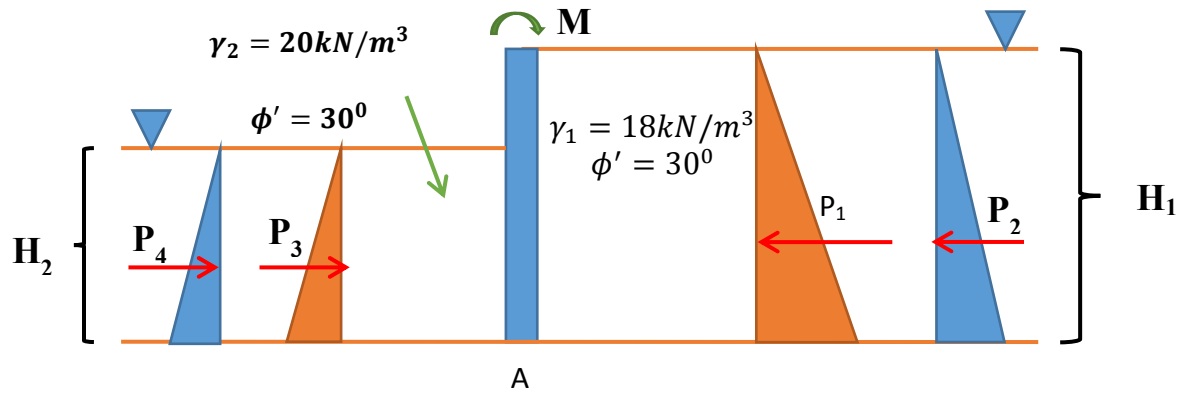
Step 3: The resultant force is equal to zero,

$$F = P_1 + P_2 - qH_1 = 1.37(H_2)^2 + 4.9(H_2)^2 - (10 \text{ kN/m})(5\text{m}) = 0$$

Solve this equation,

$$H_2 = 2.8\text{m} \sim 2.8\text{m}$$

57) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 5m and H_2 is 3m. Calculate the value of moment at point A.



- A) 80kN-m
C) 75kN-m

- B) 70kN-m
D) 65kN-m

The Answer is A

Step 1: From the diagram, we can see that there are two parts of soil acting towards wall. We need to calculate forces from both parts first. From the problem, Rankine active earth pressure coefficient for right part can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Rankine passive earth pressure coefficient for left part can be calculated by,

$$K_p = \tan^2 \left(45^\circ + \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ + \frac{30^\circ}{2} \right) = 3$$

Step 2: The lateral forces for right part can be calculated by,

$$P_1 = \frac{1}{2} K_a (\gamma_1 - \gamma_w) H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 34 \text{ kN}$$

$$P_2 = \frac{1}{2} \gamma_w H_1^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 122.5 \text{ kN}$$

$$z_1 = z_2 = \frac{1}{3} H_1 = \frac{5}{3} \text{ m}$$

The lateral forces for the left part can be calculated by,

$$P_3 = \frac{1}{2} K_p (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} (3) (20 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 137.7 \text{ kN}$$

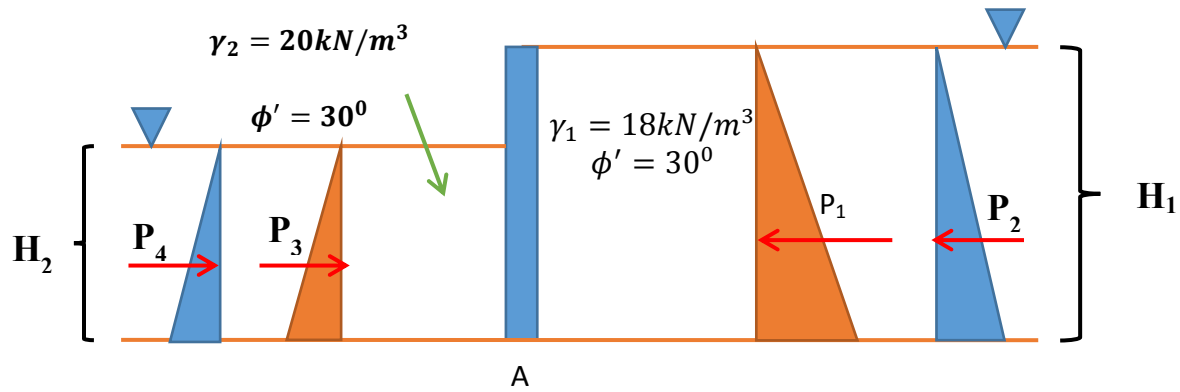
$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

$$z_3 = z_4 = \frac{1}{3} H_2 = 1 \text{ m}$$

Step 3: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_3 z_3 + P_4 z_4 - P_1 z_1 - P_2 z_2 \\ &= (137.7 \text{ kN})(1 \text{ m}) + (44.1 \text{ kN})(1 \text{ m}) - (34 \text{ kN}) \left(\frac{5}{3} \text{ m} \right) \\ &\quad - (122.5 \text{ kN}) \left(\frac{5}{3} \text{ m} \right) = -79 \text{ kN} \cdot \text{m} \\ &\quad \sim 80 \text{ kN} \cdot \text{m} \end{aligned}$$

58) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 5m. Calculate the value of H_2 so that resultant force towards this wall is equal to zero.



A) 2.8m

B) 3.0m

C) 2.6m

D) 2.

The Answer is A

Step 1: From the diagram, we can see that there are two parts of soil acting towards wall. We need to calculate forces from both parts first. From the problem, Rankine active earth pressure coefficient for right part can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Rankine passive earth pressure coefficient for left part can be calculated by,

$$K_p = \tan^2 \left(45^\circ + \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ + \frac{30^\circ}{2} \right) = 3$$

Step 2: The lateral forces for right part can be calculated by,

$$P_1 = \frac{1}{2} K_a (\gamma_1 - \gamma_w) H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 34 \text{ kN}$$

$$P_2 = \frac{1}{2} \gamma_w H_1^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 122.5 \text{ kN}$$

The lateral forces for the left part can be calculated by,

$$P_3 = \frac{1}{2} K_p (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} (3) (20 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) H_2^2 = 15.3 H_2^2$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) H_2^2 = 4.9 H_2^2$$

Step 3: The resultant force for this wall can be calculated by,

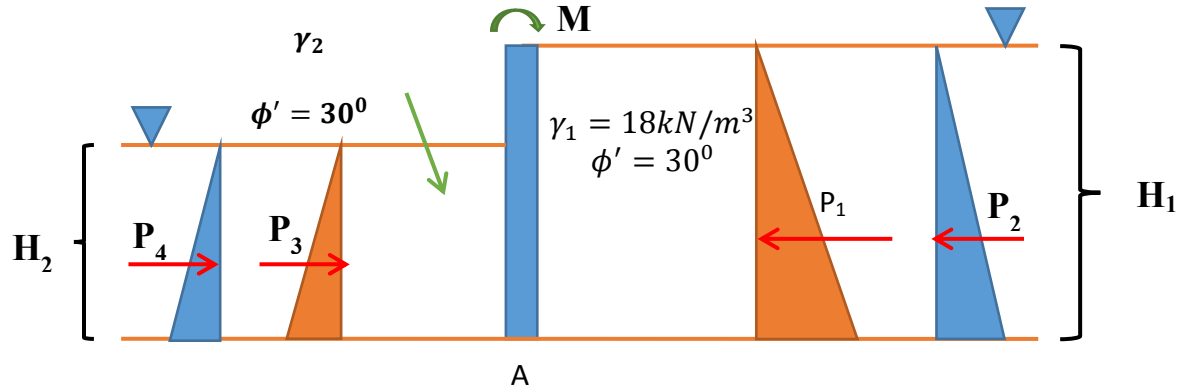
$$F = P_3 + P_4 - P_1 - P_2 = 15.3 H_2^2 + 4.9 H_2^2 - 34 \text{ kN} - 122.5 \text{ kN} = 0$$

Solve this equation,

$$H_2 = 2.8 \text{ m}$$

$$\sim 2.8 \text{ m}$$

59) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 5m and H_2 is 3m. Calculate the value of γ_2 so that moment at point A is equal to zero.



- A) 22 kN/m^3
 C) 24 kN/m^3

- B) 20 kN/m^3
 D) 26 kN/m^3

The Answer is D

Step 1: From the diagram, we can see that there are two parts of soil acting towards wall. We need to calculate forces from both parts first. From the problem, Rankine active earth pressure coefficient for right part can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Rankine passive earth pressure coefficient for left part can be calculated by,

$$K_p = \tan^2 \left(45^\circ + \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ + \frac{30^\circ}{2} \right) = 3$$

Step 2: The lateral forces for right part can be calculated by,

$$P_1 = \frac{1}{2} K_a (\gamma_1 - \gamma_w) H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 34 \text{ kN}$$

$$P_2 = \frac{1}{2} \gamma_w H_1^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 122.5 \text{ kN}$$

$$z_1 = z_2 = \frac{1}{3} H_1 = \frac{5}{3} \text{ m}$$

The lateral forces for the left part can be calculated by,

$$P_3 = \frac{1}{2} K_p (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} (3) (\gamma_2 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

$$z_3 = z_4 = \frac{1}{3} H_2 = 1 \text{ m}$$

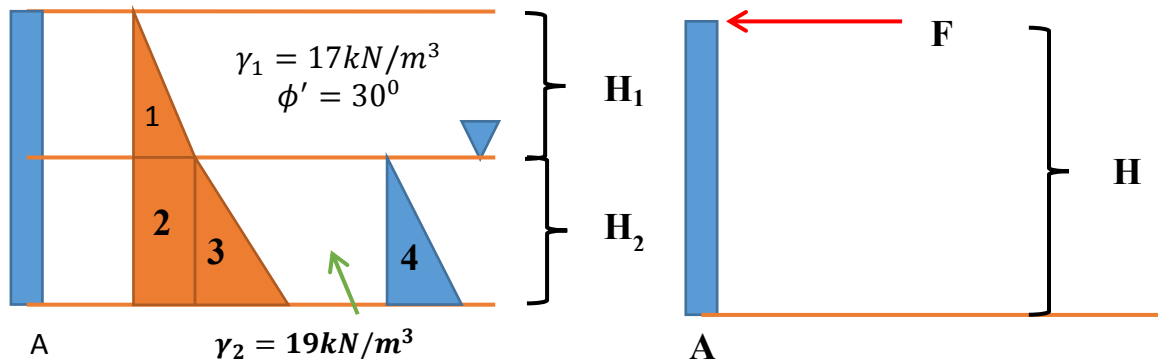
Step 3: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_3 z_3 + P_4 z_4 - P_1 z_1 - P_2 z_2 \\ &= \frac{1}{2} (3) (\gamma_2 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 (1 \text{ m}) + (44.1 \text{ kN}) (1 \text{ m}) \\ &\quad - (34 \text{ kN}) \left(\frac{5}{3} \text{ m} \right) - (122.5 \text{ kN}) \left(\frac{5}{3} \text{ m} \right) = 0 \end{aligned}$$

Solve this equation,

$$\gamma_2 = 25.8 \text{ kN/m}^3 \quad \sim 26 \text{ kN/m}^3$$

60) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 3m and H_2 is 3m. H is 6m. If moment at point A in the left figure is the same as that in the right figure, calculate the value of F .



A) 35kN

B) 45kN

C) 40kN

D) 30kN

The Answer is C

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\Phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m})^2 = 25.5 \text{ kN}$$

$$z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (3 \text{ m}) + (3 \text{ m}) = 4 \text{ m}$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m})^2 = 51 \text{ kN}$$

$$z_2 = \frac{1}{2} H_2 = \frac{1}{2} (3 \text{ m}) = 1.5 \text{ m}$$

$$P_3 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (19 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 13.8 \text{ kN}$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

$$z_4 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

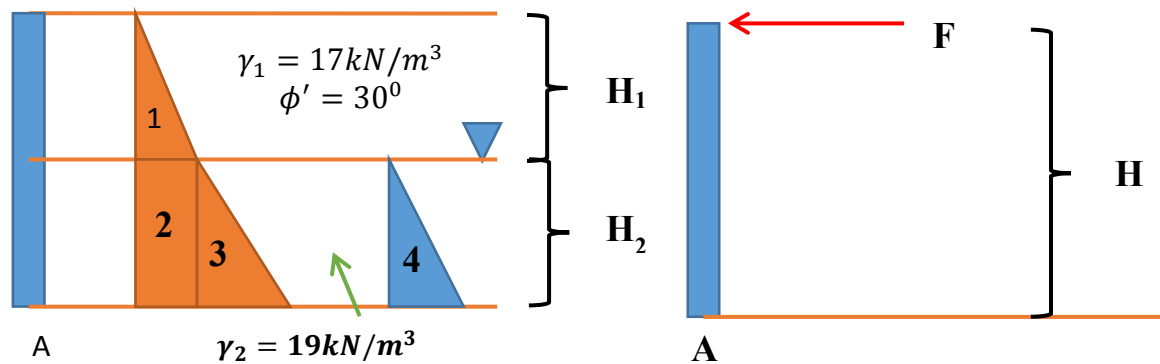
Step 2: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_1 z_1 + P_2 z_2 + P_3 z_3 + P_4 z_4 \\ &= (25.5 \text{ kN})(4 \text{ m}) + (51 \text{ kN})(1.5 \text{ m}) + (13.8 \text{ kN})(1 \text{ m}) \\ &\quad + (44.1 \text{ kN})(1 \text{ m}) = F(6 \text{ m}) \end{aligned}$$

Solve this equation,

$$F = 39.4 \text{ kN} \sim 40 \text{ kN}$$

61) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 3m. H is 6m. $F = 40\text{kN}$. If moment at point A in the left figure is the same as that in the right figure, calculate the value of H_2 .



- A) 3m B) 4m C) 5m D) 6m

The Answer is A

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m})^2 = 25.5 \text{ kN}$$

$$z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (3 \text{ m}) + H_2 = 1 + H_2$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m}) H_2 = 17 H_2$$

$$z_2 = \frac{1}{2} H_2 = 0.5 H_2$$

$$P_3 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (19 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) H_2^2$$

$$z_3 = \frac{1}{3} H_2$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (H_2)^2$$

$$z_4 = \frac{1}{3} H_2$$

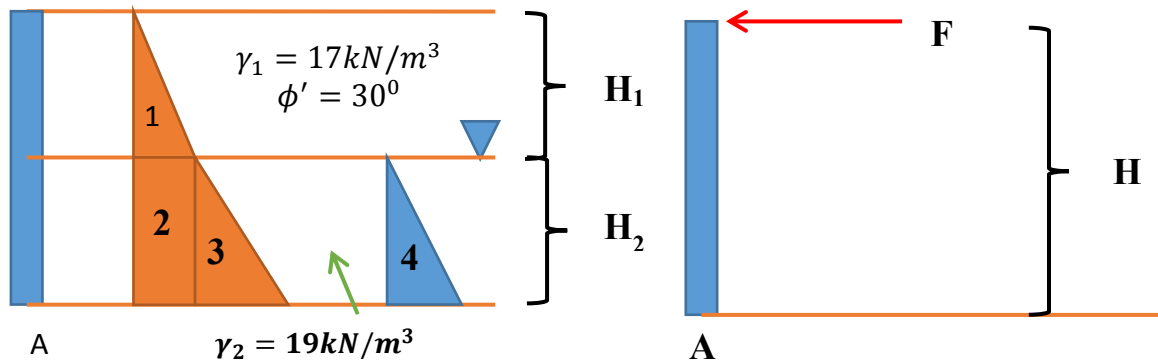
Step 2: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_1 z_1 + P_2 z_2 + P_3 z_3 + P_4 z_4 \\ &= (25.5 \text{ kN})(1 + H_2) + (17 H_2)(0.5 H_2) + (1.53 H_2^2) \left(\frac{1}{3} H_2 \right) \\ &\quad + (4.9 H_2^2) \left(\frac{1}{3} H_2 \right) = (40 \text{ kN})(6 \text{ m}) \end{aligned}$$

Solve this equation,

$$H_2 = 3 \text{ m} \sim 3 \text{ m}$$

62) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_2 is 3m. H is 6m. $F = 40\text{kN}$. If moment at point A in the left figure is the same as that in the right figure, calculate the value of H_1 .



- A) 3m B) 4m C) 5m D) 6m

The Answer is A

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) H_1^2$$

$$z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (H_1) + 3 \text{ m}$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m}) H_1 = 17 H_1$$

$$z_2 = \frac{1}{2} H_2 = \frac{1}{2} (3 \text{ m}) = 1.5 \text{ m}$$

$$P_3 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (19 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 13.8 \text{ kN}$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

$$z_4 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

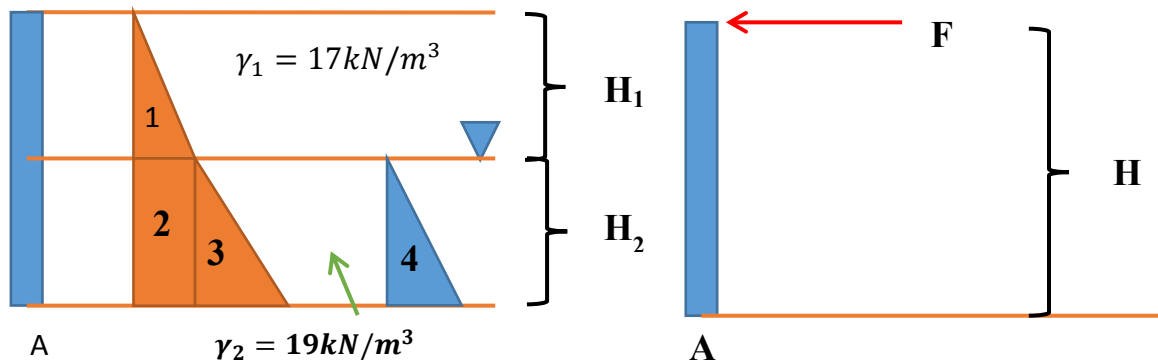
Step 2: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_1 z_1 + P_2 z_2 + P_3 z_3 + P_4 z_4 \\ &= (2.83 H_1^2) \left(\frac{1}{3} (H_1) + 3 \right) + (17 H_1) (1.5 \text{ m}) + (13.8 \text{ kN}) (1 \text{ m}) \\ &\quad + (44.1 \text{ kN}) (1 \text{ m}) = (40 \text{ kN}) (6 \text{ m}) \end{aligned}$$

Solve this equation,

$$H_1 = 3 \text{ m} \sim 3 \text{ m}$$

63) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 3m and H_2 is 3m. H is 6m. $F = 40\text{kN}$. If moment at point A in the left figure is the same as that in the right figure, calculate the value of ϕ' .



- A) 20° B) 35° C) 25° D) 30°

The Answer is D

Step 1: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} K_a (17 \text{ kN/m}^3) (3 \text{ m})^2 = 76.5 K_a$$

$$z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (3 \text{ m}) + 3 \text{ m} = 4 \text{ m}$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = K_a (17 \text{ kN/m}^3) (3 \text{ m})^2 = 153 K_a$$

$$z_2 = \frac{1}{2} H_2 = \frac{1}{2} (3 \text{ m}) = 1.5 \text{ m}$$

$$P_3 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} K_a (19 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 41.4 K_a$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

$$z_4 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

Step 2: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_1 z_1 + P_2 z_2 + P_3 z_3 + P_4 z_4 \\ &= (76.5 K_a)(4 \text{ m}) + (153 K_a)(1.5 \text{ m}) + (41.4 K_a)(1 \text{ m}) \\ &\quad + (44.1 \text{ kN})(1 \text{ m}) = (40 \text{ kN})(6 \text{ m}) \end{aligned}$$

Solve this equation,

$$K_a = \frac{1}{3}$$

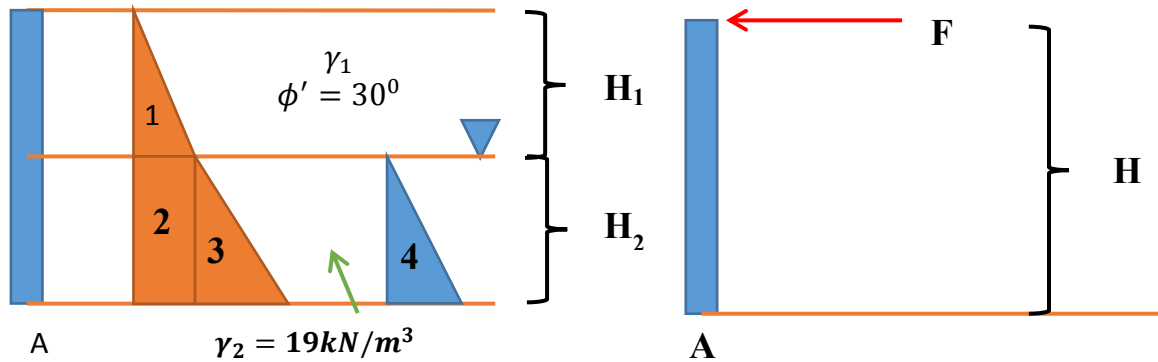
Step 3: Rankine active earth pressure coefficient can be found by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \frac{1}{3}$$

Solve this equation,

$$\phi' = 30^0 \sim 30^0$$

64) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 3m and H_2 is 3m. H is 6m. $F = 40\text{kN}$. If moment at point A in the left figure is the same as that in the right figure, calculate the value of γ_1 .

A) 15kN/m^3 B) 17kN/m^3 C) 19kN/m^3 D) 13kN/m^3

The Answer is B

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) \gamma_1 (3m)^2 = 1.5 \gamma_1$$

$$z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (3m) + 3m = 4m$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) \gamma_1 (3m)^2 = 3 \gamma_1$$

$$z_2 = \frac{1}{2} H_2 = \frac{1}{2} (3m) = 1.5m$$

$$P_3 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (19kN/m^3 - 9.8kN/m^3) (3m)^2 = 13.8kN$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (3m) = 1m$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8kN/m^3) (3m)^2 = 44.1kN$$

$$z_4 = \frac{1}{3} H_2 = \frac{1}{3} (3m) = 1m$$

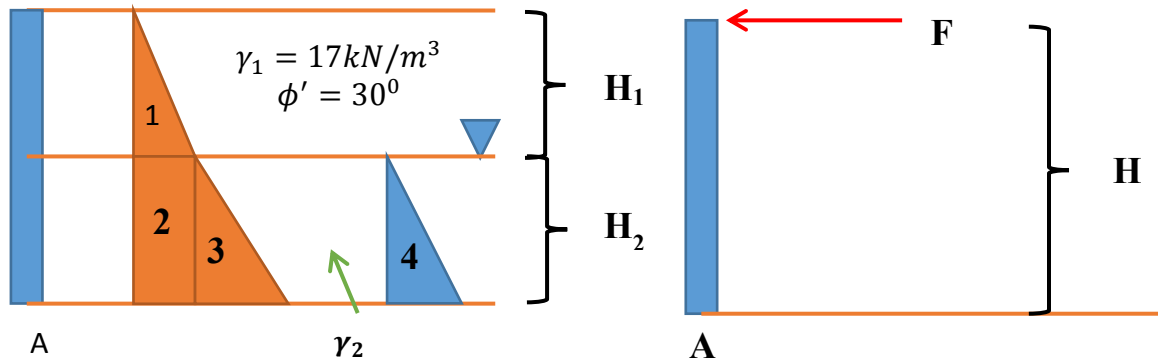
Step 3: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_1 z_1 + P_2 z_2 + P_3 z_3 + P_4 z_4 \\ &= (1.5 \gamma_1)(4m) + (3 \gamma_1)(1.5m) + (13.8kN)(1m) \\ &\quad + (44.1kN)(1m) = (40kN)(6m) \end{aligned}$$

Solve this equation,

$$\gamma_1 = 17kN/m^3 \sim 17kN/m^3$$

65) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 3m and H_2 is 3m. H is 6m. $F = 40\text{kN}$. If moment at point A in the left figure is the same as that in the right figure, calculate the value of γ_2 .

A) 15kN/m^3 B) 17kN/m^3 C) 19kN/m^3 D) 13kN/m^3

The Answer is C

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m})^2 = 25.5 \text{ kN}$$

$$z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (3 \text{ m}) + 3 \text{ m} = 4 \text{ m}$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m})^2 = 51 \text{ kN}$$

$$z_2 = \frac{1}{2} H_2 = \frac{1}{2} (3 \text{ m}) = 1.5 \text{ m}$$

$$P_3 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (\gamma_2 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

$$z_4 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

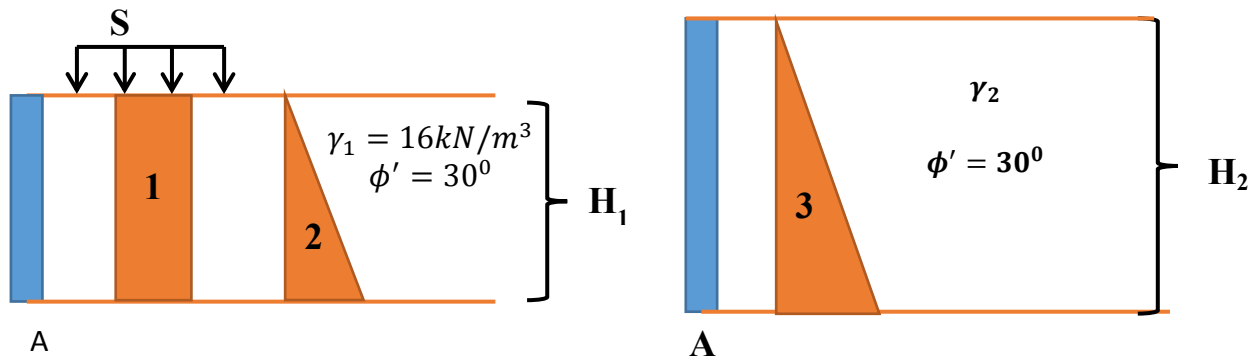
Step 3: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_1 z_1 + P_2 z_2 + P_3 z_3 + P_4 z_4 \\ &= (25.5 \text{ kN})(4 \text{ m}) + (51 \text{ kN})(1.5 \text{ m}) + (1.5(\gamma_2 - 9.8))(1 \text{ m}) \\ &\quad + (44.1 \text{ kN})(1 \text{ m}) = (40 \text{ kN})(6 \text{ m}) \end{aligned}$$

Solve this equation,

$$\gamma_2 = 19 \text{ kN/m}^3 \sim 19 \text{ kN/m}^3$$

66) Two figures show two frictionless wall as below. Information about each wall is given. Assume that $H_1=4\text{m}$ and $H_2=5\text{m}$. $S=20\text{kN/m}$. Assume that moment at point A for left figure and right figure is the same. Determine the density of soil in the right figure, γ_2 .

A) 15kN/m^3 B) 17kN/m^3 C) 19kN/m^3 D) 13kN/m

The Answer is A

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\Phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = K_a S H_1 = \frac{1}{3} (20 \text{ kN/m})(4 \text{ m}) = 26.7 \text{ kN}$$

$$z_1 = \frac{1}{2} H_1 = \frac{1}{2} (4 \text{ m}) = 2 \text{ m}$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (16 \text{ kN/m}^3)(4 \text{ m})^2 = 42.7 \text{ kN}$$

$$z_2 = \frac{1}{3} H_1 = \frac{1}{3} (4 \text{ m}) = 1.3 \text{ m}$$

$$P_3 = \frac{1}{2} K_a \gamma_2 H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) \gamma_2 (5 \text{ m})^2 = 4.2 \gamma_2$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (5 \text{ m}) = 1.7 \text{ m}$$

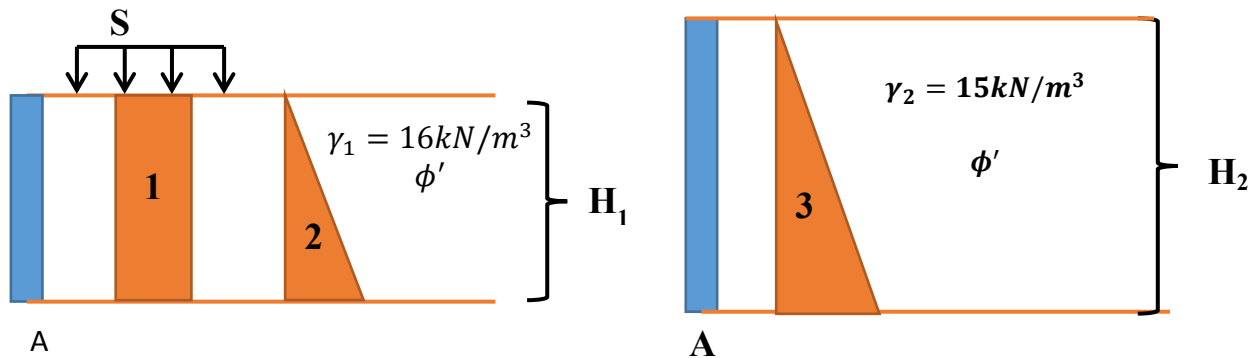
Step 2: The moment at point A is the same. We got,

$$\begin{aligned} P_1 z_1 + P_2 z_2 &= P_3 z_3 = (26.7 \text{ kN})(2 \text{ m}) + (42.7 \text{ kN})(1.3 \text{ m}) \\ &= (4.2 \gamma_2)(1.7 \text{ m}) \end{aligned}$$

Solve this equation,

$$\gamma_2 = 15.3 \text{ kN/m}^3 \sim 15 \text{ kN/m}^3$$

67) Two figures show two frictionless wall as below. Information about each wall is given. Assume that $H_1=4\text{m}$ and $H_2=5\text{m}$. $S=20\text{kN/m}$. Assume that moment at point A for left figure and right figure is the same and the moment is equal to 100kN-m . Determine the friction angle, ϕ' .



- A) 22° B) 44° C) 33° D) 55°

The Answer is B

Step 1: Lateral force for Area 1,

$$P_1 = K_a S H_1 = K_a (20 \text{ kN/m})(4 \text{ m}) = 80 K_a$$

$$z_1 = \frac{1}{2} H_1 = \frac{1}{2} (4 \text{ m}) = 2 \text{ m}$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} K_a (16 \text{ kN/m}^3)(4 \text{ m})^2 = 128 K_a$$

$$z_2 = \frac{1}{3} H_1 = \frac{1}{3} (4 \text{ m}) = 1.3 \text{ m}$$

$$P_3 = \frac{1}{2} K_a \gamma_2 H_2^2 = \frac{1}{2} K_a (15 \text{ kN/m}^3)(5 \text{ m})^2 = 187.5 K_a$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (5 \text{ m}) = 1.7 \text{ m}$$

Step 2: The moment at point A is the same. We got,

$$\begin{aligned} P_1 z_1 + P_2 z_2 &= P_3 z_3 = (80 K_a)(2 \text{ m}) + (128 K_a)(1.3 \text{ m}) \\ &= (187.5 K_a)(1.7 \text{ m}) = 100 \text{ kN} \cdot \text{m} \end{aligned}$$

Solve this equation,

$$K_a = 0.3$$

Step 3: Rankine active earth pressure coefficient can be found by,

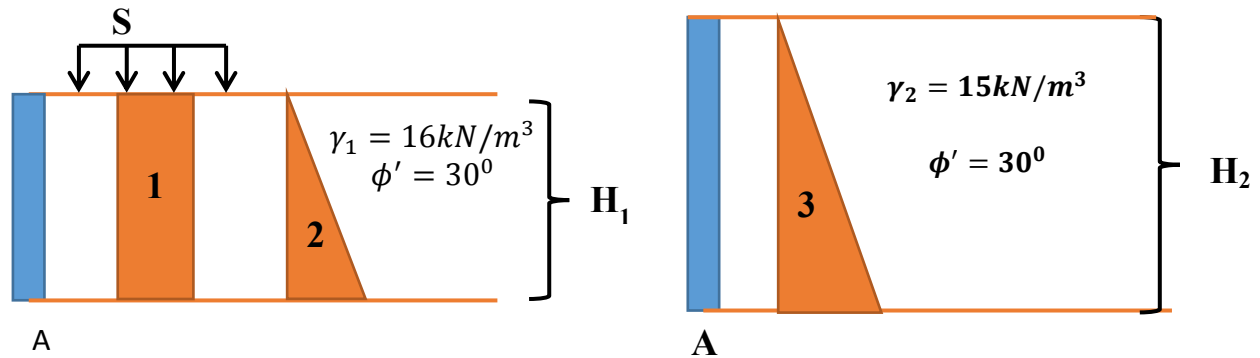
$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = 0.3$$

Solve this equation,

$$\phi' = 33^\circ$$

$$\sim 33^\circ$$

68) Two figures show two frictionless wall as below. Information about each wall is given. Assume that $H_2=5\text{m}$. $S=20\text{kN/m}$. Assume that moment at point A for left figure and right figure is the same. Determine the value of H_1 .



- A) 3m B) 4m C) 5m D) 2m

The Answer is B

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = K_a S H_1 = \frac{1}{3} (20 \text{ kN/m}) H_1 = 6.7 H_1$$

$$z_1 = \frac{1}{2} H_1$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (16 \text{ kN/m}^3) (H_1)^2 = 2.7 H_1^2$$

$$z_2 = \frac{1}{3} H_1$$

$$P_3 = \frac{1}{2} K_a \gamma_2 H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (15 \text{ kN/m}^3) (5 \text{ m})^2 = 62.5 \text{ kN}$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (5 \text{ m}) = 1.7 \text{ m}$$

Step 3: The moment at point A is the same. We got,

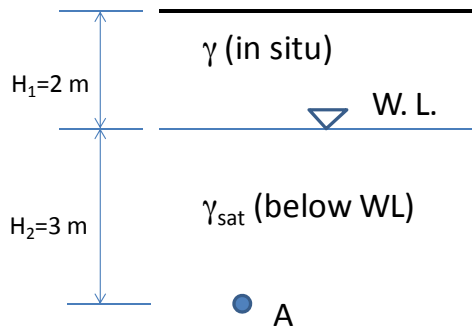
$$P_1 z_1 + P_2 z_2 = P_3 z_3 = (6.7 H_1) \left(\frac{1}{2} H_1 \right) + (2.7 H_1^2) \left(\frac{1}{3} H_1 \right) = (62.5 \text{ kN}) (1.7 \text{ m})$$

Solve this equation,

$$H_1 = 4 \text{ m}$$

~4m

69) A point A in a sandy soil is 5 m below the ground. The underground water level is 2 m below the ground. It is known that the in situ total unit weight of the soil above the water level is 14.6 kN/m^3 , and the saturated unit weight of soil below the water level is 17.8 kN/m^3 . Determine the effective vertical soil pressure at point A.



A) 23.5 kPa

B) 35.2 kPa

C) 53.2 kPa

D) 72.4 kPa

The Answer is C

Step 1:

Effective stress σ' is equal to the total stress σ minus pore water pressure

$$\sigma' = \sigma - u.$$

Step 2:

The total stress σ is calculated by $\sigma = \gamma H_1 + \gamma_{\text{sat}} H_2 = (14.6 \text{ kN/m}^3)(2 \text{ m}) + (17.8 \text{ kN/m}^3)(3 \text{ m}) = 82.6 \text{ kPa}$

Step 3:

The pore water pressure u is calculated by $u = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(3 \text{ m}) = 29.4 \text{ kPa}$

Step 4:

The effective stress σ' is

$$u = \sigma - u = 82.6 \text{ kPa} - 29.4 \text{ kPa} = 53.2 \text{ kPa}$$

70) Deformations of soils are a function of

- A) Effective stresses
- B) Total stresses
- C) Pore water pressure
- D) All of the above

The Answer is A

Based on the principal of effective stress, effective stress is equal to total stress minus pore water pressure. It is the most important principle in soil mechanics. Deformations of soils are a function of effective stresses not total stresses.

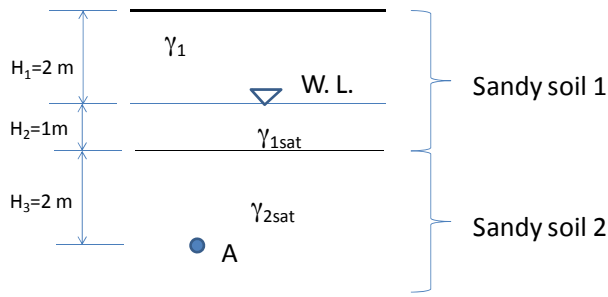
(A) is correct.

(B) is incorrect. Deformations of soils are not a function of total stresses.

(C) is incorrect. Pore water pressure is isotropic and only can cause volumetric changes of soil solids, which is nearly incompressible. Pore water pressure does not cause displacement of soil solids, or deformation of soils.

(D) is incorrect, since options (B) and (C) are incorrect.

71) A point A is 5 m below the ground. The underground water level is 2 m below the ground. There are two deposits of sandy soil at the site. The first deposit of soil has a depth of 3 m, and a total unit weight (above the water level) of 13.2 kN/m^3 , a saturated unit weight of 16.5 kN/m^3 . The second deposit of soil has a saturated unit weight of 17.6 kN/m^3 . Determine the effective vertical soil pressure at point A.



A) 23.5 kPa

B) 48.7 kPa

C) 53.2 kPa

D) 72.4 kPa

The Answer is B

Effective stress σ' is equal to the total stress σ minus pore water pressure

$$\sigma' = \sigma - u.$$

Step 1:

The total stress σ is calculated by $\sigma = \gamma_1 H_1 + \gamma_{1sat} H_2 + \gamma_{2sat} H_3 = (13.2 \text{ kN/m}^3)(2 \text{ m}) + (16.5 \text{ kN/m}^3)(1 \text{ m}) + (17.6 \text{ kN/m}^3)(2 \text{ m}) = 78.1 \text{ kPa}$

Step 2:

The pore water pressure u is calculated by $u = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(1 \text{ m} + 2 \text{ m}) = 29.4 \text{ kPa}$

Step 3:

The effective stress σ' is

$$\sigma' = \sigma - u = 78.1 \text{ kPa} - 29.4 \text{ kPa} = 48.7 \text{ kPa}$$

73) The groundwater level falls at a site due to massive pumping. During such a process, which of the following consequences may occur for the soil above the pumping point?

- A) Effective vertical stress in the soil increases, and ground level rises
- B) Effective vertical stress in the soil decreases, and ground level falls
- C) Effective vertical stress in the soil does not change, and ground level falls
- D) Effective vertical stress in the soil increases, and ground level falls

The Answer is D

Pumping of groundwater leads to water flow through soil, which exerts a frictional drag on the soil particles resulting in head losses. Downward seepage increases the resultant effective stress, while upward seepage decreases the resultant effective stress. For the scenario of the problem, water in the soil above the pumping point seeps downward, which increases the effective vertical stress. With the effective vertical stress increasing, the vertical strain of the soil also increases, which leads to a reduction of the ground level.

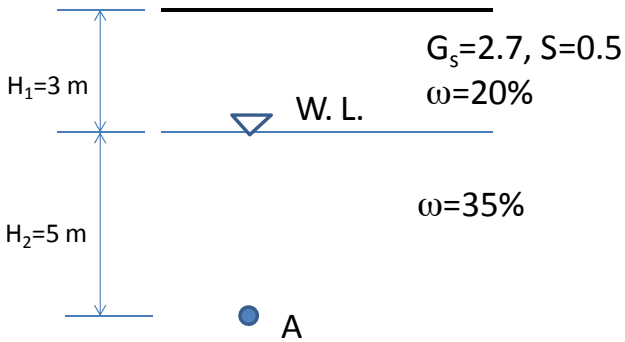
(A) is incorrect. The ground level should fall.

(B) is incorrect. The effective vertical stress in the soil should increase.

(C) is incorrect. The effective vertical stress in the soil should increase.

(D) is correct.

74) A point A in a sandy soil is 8 m below the ground. The underground water level is 3 m below the ground. It is known that the specific gravity of soil solids is 2.7, the water content of soil is 20% above the water level, and 35% below the water level. Determine the effective vertical soil pressure at point A.



A) 23.5 kPa

B) 35.2 kPa

C) 53.2 kPa

D) 88.7 kPa

The Answer is D

Step 1:

It is known that water content $\omega = \frac{W_W}{W_S} \times 100$, specific gravity of soil solids

$G_S = \frac{W_S}{V_S \gamma_W}$, void ratio $e = \frac{V_V}{V_S}$, degree of saturation $S = \frac{V_W}{V_V} \times 100$, and water

volume $V_W = \frac{W_W}{\gamma_W}$. Therefore, $Se = \frac{V_W}{V_V} \times 100 \times \frac{V_V}{V_S} = \frac{V_W}{V_S} \times 100$, and $G_S \omega =$

$\frac{W_S}{V_S \gamma_W} \times \frac{W_W}{W_S} \times 100 = \frac{V_W}{V_S} \times 100$. Therefore, $Se = G_S \omega$. The void ratio of the soil

above the water level is

$$e = \frac{G_S \omega}{S} = \frac{2.7 \times 20\%}{0.5} = 1.08$$

Step 2:

The unit weight of soil above the water level is

$$\gamma_A = \left(\frac{G_S + Se}{1 + e} \right) \gamma_W = \left(\frac{2.7 + 0.5 \times 1.08}{1 + 1.08} \right) \times 9.8 \frac{\text{kN}}{\text{m}^3} \cong 15.27 \frac{\text{kN}}{\text{m}^3}$$

Step 3:

The void ratio of the soil below the water level is

$$e = \frac{G_S \omega}{S} = \frac{2.7 \times 35\%}{1.0} = 0.945$$

The unit weight of soil below the water level is

$$\gamma_B = \left(\frac{G_S + e}{1 + e} \right) \gamma_W = \left(\frac{2.7 + 0.945}{1 + 0.945} \right) \times 9.8 \frac{\text{kN}}{\text{m}^3} \cong 18.37 \frac{\text{kN}}{\text{m}^3}$$

Step 4:

Effective stress σ' is equal to the total stress σ minus pore water pressure

$$\sigma' = \sigma - u.$$

The total stress σ is calculated by $\sigma = \gamma_A H_1 + \gamma_B H_2 = (15.27 \text{ kN/m}^3)(3 \text{ m}) + (18.37 \text{ kN/m}^3)(5 \text{ m}) = 137.66 \text{ kPa}$

Step 5:

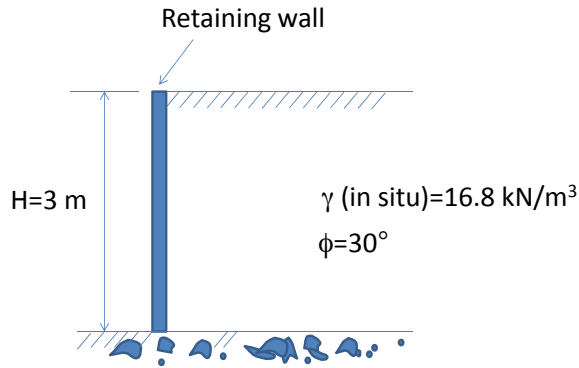
The pore water pressure u is calculated by $u = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(5 \text{ m}) = 49 \text{ kPa}$

Step 6:

The effective stress σ' is

$$u = \sigma - u = 137.66 \text{ kPa} - 49 \text{ kPa} = 88.66 \text{ kPa}$$

75. A 3 m high smooth retaining wall extends from the top of bedrock to the ground surface. The soil behind of the retaining wall is homogeneous and cohesionless, has an in-situ total unit weight of 16.8 kN/m^3 , and angle of internal friction of 30° . Based on the Rankine theory, what is the total active resultant lateral earth force per unit length of retaining wall?



- A) 25.2 kN/m B) 35.2 kN/m
C) 50.4 kN/m D) 75.6 kN/m

The Answer is A

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Step 1:

Based on the Rankine theory, the Rankine active earth pressure coefficient

$$K_A = \tan^2 \left(45^\circ - \frac{\phi}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = 0.333$$

Step 2:

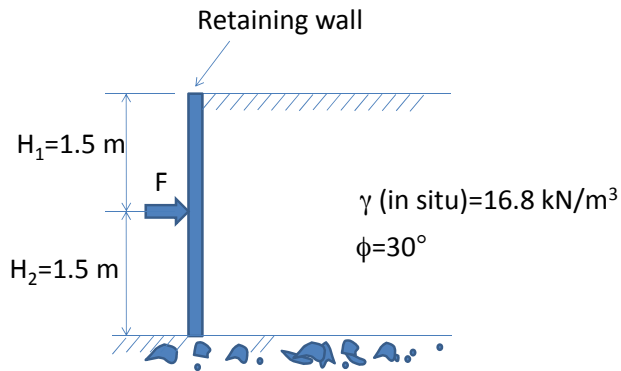
The active lateral earth pressure distribution is linear. The active lateral earth pressure at any depth, h , below the ground surface is calculated by $\sigma_a = K_A \sigma_v = K_A \gamma h$.

Step 3:

The total active lateral earth force per unit length of retaining wall is

$$P_a = \frac{1}{2} K_A \gamma H^2 = \frac{1}{2} (0.333) (16.8 \text{ kN/m}^3) (3 \text{ m})^2 = 25.2 \text{ kN/m}$$

76. A 3 m high smooth retaining wall extends from the top of bedrock to the ground surface. The soil behind of the retaining wall is homogeneous and cohesionless, with an in-situ total unit weight of 16.8 kN/m^3 , and angle of internal friction of 30° . A lateral force F is applied on the opposite side of the wall. Based on the Rankine theory, what is the total passive resultant lateral earth force per unit length of retaining wall?



- A) 25.2 kN/m B) 35.2 kN/m C) 50.4 kN/m D) 226.8 kN/m

The Answer is A

Step 1:

Based on the Rankine theory, the Rankine passive earth pressure coefficient

$$K_p = \tan^2 \left(45^\circ + \frac{\phi}{2} \right) = \tan^2 \left(45^\circ + \frac{30^\circ}{2} \right) = 3$$

Step 2:

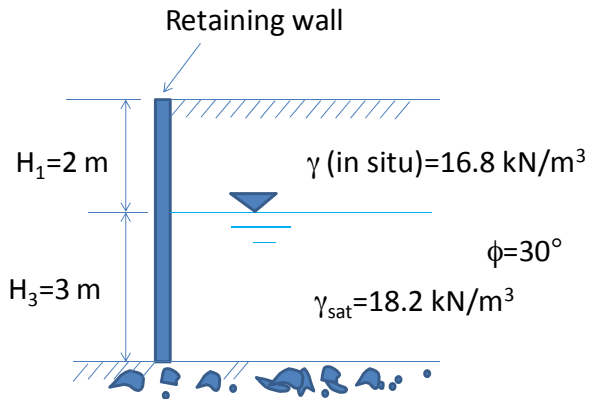
The passive lateral earth pressure distribution is linear. The passive lateral earth pressure at any depth, h , below the ground surface is calculated by $\sigma_p = K_p \sigma_v = K_p \gamma h$.

Step 3:

The total active lateral earth force per unit length of retaining wall is

$$P_p = \frac{1}{2} K_p \gamma H^2 = \frac{1}{2} (3) (16.8 \text{ kN/m}^3) (3 \text{ m})^2 = 226.8 \text{ kN/m}$$

77. A 5 m high smooth retaining wall extends from the top of bedrock to the ground surface. The soil behind of the retaining wall is homogeneous and cohesionless, has an in-situ total unit weight of 16.8 kN/m^3 , and angle of internal friction of 30° . Groundwater level is 2 m below the surface of the soil, as shown in the figure below. The saturated unit weight of the soil is 18.2 kN/m^3 . Based on the Rankine theory, what is the total active resultant lateral earth force per unit length of retaining wall?



- A) 25.2 kN/m B) 35.2 kN/m C) 57.4 kN/m D) 75.6 kN/m

The Answer is C

Step 1:

Based on the Rankine theory, the Rankine active earth pressure coefficient

$$K_A = \tan^2 \left(45^\circ - \frac{\phi}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = 0.333$$

Step 2:

The total vertical stress σ_{vH1} at the groundwater level is calculated by $\sigma_{vH1} = \gamma H_1 = (16.8 \text{ kN/m}^3)(2 \text{ m}) = 33.6 \text{ kPa}$. Since the pore water pressure at the groundwater level is 0, the effective vertical stress $\sigma'_{vH1} = \sigma_{vH1} - u_{H1} = 33.6 \text{ kPa} - 0 = 33.6 \text{ kPa}$.

Step 3:

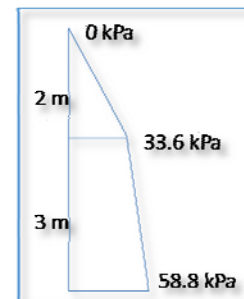
At the bottom level of the retaining wall, the total vertical stress σ_{vH2} is calculated by

$$\sigma_{vH2} = \gamma H_1 + \gamma_{\text{sat}} H_2 = (16.8 \text{ kN/m}^3)(2 \text{ m}) + (18.2 \text{ kN/m}^3)(3 \text{ m}) = 88.2 \text{ kPa}.$$

The pore water pressure u is calculated by $u_{H2} = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(3 \text{ m}) = 29.4 \text{ kPa}$.

The effective vertical stress $\sigma'_{vH2} = \sigma_{vH2} - u_{H2} = 88.2 \text{ kPa} - 29.4 \text{ kPa} = 58.8 \text{ kPa}$.

The distribution of effective vertical stress along depth, σ'_v , is then shown in the plot below:



Step 4:

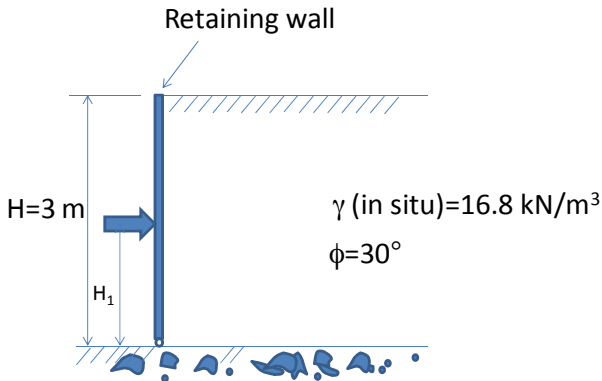
The active lateral earth pressure at any depth, h , below the ground surface is calculated by $\sigma_a = K_A \sigma'_v$.

Step 5:

The total active lateral earth force per unit length of retaining wall is

$$\begin{aligned} P_a &= K_A \left[\frac{1}{2} (0 + \sigma'_{vH1}) H_1 + \frac{1}{2} (\sigma'_{vH1} + \sigma'_{vH2}) H_2 \right] \\ &= (0.333) \left[\frac{1}{2} (0 + 33.6 \text{ kPa})(2 \text{ m}) \right. \\ &\quad \left. + \frac{1}{2} (33.6 \text{ kPa} + 58.8 \text{ kPa})(3 \text{ m}) \right] = 57.4 \text{ kN/m} \end{aligned}$$

78. A 3 m high smooth retaining wall extends from the top of bedrock to the ground surface. The wall is connected to the bedrock by a frictionless hinge. The soil behind of the retaining wall is homogeneous and cohesionless, has an in-situ total unit weight of 16.8 kN/m^3 , and angle of internal friction of 30° . A lateral force is applied at the opposite side of the wall, at a height of $H_1=1.5 \text{ m}$ from the top of bedrock. Based on Rankine theory, what is the minimum force per unit length of retaining wall required to resist the overturning moment?



- A) 16.8 kN/m B) 35.2 kN/m C) 50.4 kN/m D) 75.6 kN/m

The Answer is A

Step 1:

Based on the Rankine theory, the Rankine active earth pressure coefficient

$$K_A = \tan^2 \left(45^\circ - \frac{\phi}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = 0.333$$

Step 2:

The active lateral earth pressure distribution is linear. The active lateral earth pressure at any depth, h , below the ground surface is calculated by

$$\sigma_a = K_A \sigma_v = K_A \gamma h.$$

Step 3:

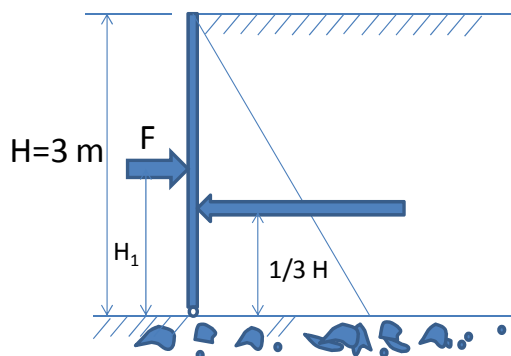
The total active lateral earth force per unit length of retaining wall is

$$P_a = \frac{1}{2} K_A \gamma H^2 = \frac{1}{2} (0.333)(16.8 \text{ kN/m}^3)(3 \text{ m})^2 = 25.2 \text{ kN/m}$$

Based on Page 51 of the NCEES reference manual, the resultant, P_a , acts at a height above the bedrock of

$$\frac{H}{3} = \frac{3 \text{ m}}{3} = 1 \text{ m}.$$

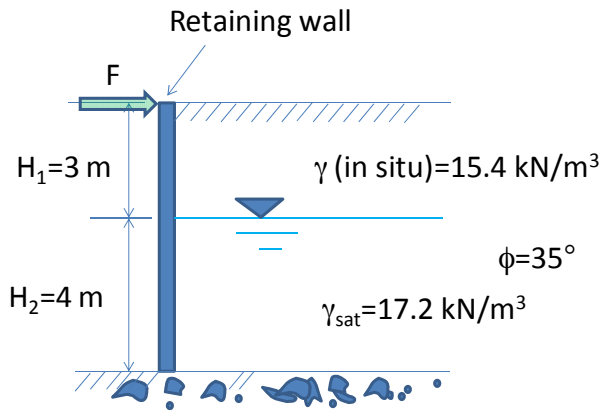
Step 4:



Summing moments on the retaining wall about the hinge gives

$$F = \frac{P_a \frac{H}{3}}{H_1} = \frac{\left(25.2 \frac{\text{kN}}{\text{m}}\right) (1 \text{ m})}{1.5 \text{ m}} = 16.8 \text{ kN/m}$$

79. A 7 m high smooth retaining wall extends from the top of bedrock to the ground surface. The soil behind of the retaining wall is homogeneous and cohesionless, has an in-situ total unit weight of 15.4 kN/m^3 and angle of internal friction of 35° . Groundwater level is 3 m below the surface of the soil, as shown in the figure below. The saturated unit weight of the soil is 17.2 kN/m^3 . A resisting force F is applied on the opposite side of the wall at the top of the wall so that the soil behind the wall reaches the passive earth pressure condition. Based on the Rankine theory, what is the total passive resultant lateral earth force per unit length of retaining wall?



- A) 252 kN/m B) 352 kN/m C) 574 kN/m D) 1156 kN/m

The Answer is D

Step 1:

Based on the Rankine theory, the Rankine passive earth pressure coefficient

$$K_p = \tan^2 \left(45^\circ + \frac{\phi}{2} \right) = \tan^2 \left(45^\circ + \frac{35^\circ}{2} \right) = 3.69.$$

Step 2:

The total vertical stress σ_{vH1} at the groundwater level is calculated by $\sigma_{vH1} = \gamma H_1 = (15.4 \text{ kN/m}^3)(3 \text{ m}) = 46.2 \text{ kPa}$. Since the pore water pressure at the groundwater level is 0, the effective vertical stress $\sigma'_{vH1} = \sigma_{vH1} - u_{H1} = 46.2 \text{ kPa} - 0 = 46.2 \text{ kPa}$.

Step 3:

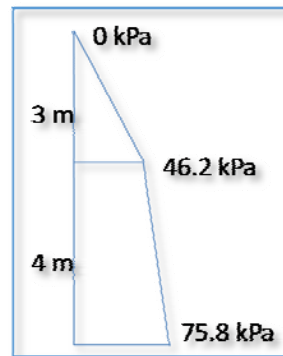
At the bottom level of the retaining wall, the total vertical stress σ_{vH2} is

$$\sigma_{vH2} = \gamma H_1 + \gamma_{\text{sat}} H_2 = (15.4 \text{ kN/m}^3)(3 \text{ m}) + (17.2 \text{ kN/m}^3)(4 \text{ m}) = 115 \text{ kPa}.$$

The pore water pressure u is calculated by $u_{H2} = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(4 \text{ m}) = 39.2 \text{ kPa}$.

The effective vertical stress $\sigma'_{vH2} = \sigma_{vH2} - u_{H2} = 115 \text{ kPa} - 39.2 \text{ kPa} = 75.8 \text{ kPa}$.

The distribution of effective vertical stress along depth, σ'_v , is then shown in the plot:



Step 4:

The passive lateral earth pressure at any depth, h , below the ground surface is calculated by $\sigma_p = K_p \sigma'_v$.

Step 5:

The total passive lateral earth force per unit length of retaining wall is

$$\begin{aligned} P_p &= K_p \left[\frac{1}{2} (0 + \sigma'_{vH1}) H_1 + \frac{1}{2} (\sigma'_{vH1} + \sigma'_{vH2}) H_2 \right] \\ &= (3.69) \left[\frac{1}{2} (0 + 46.2 \text{ kPa})(3 \text{ m}) \right. \\ &\quad \left. + \frac{1}{2} (46.2 \text{ kPa} + 75.8 \text{ kPa})(4 \text{ m}) \right] = 1156 \text{ kN/m} \end{aligned}$$

80. A continuous footing three feet wide is founded 4 ft below the ground surface in a clay soil for which $\gamma = 120 \text{ lbf/ft}^3$, $c = 1300 \text{ lbf/ft}^2$, and $\phi = 10^\circ$. There is no groundwater at the site. Determine the ultimate bearing capacity according to Terzaghi's theory.

- A) $1,336 \frac{\text{lbf}}{\text{ft}^2}$ B) $2,336 \frac{\text{lbf}}{\text{ft}^2}$
C) $12,336 \frac{\text{lbf}}{\text{ft}^2}$ D) $22,336 \frac{\text{lbf}}{\text{ft}^2}$

The Answer is C

Step 1:

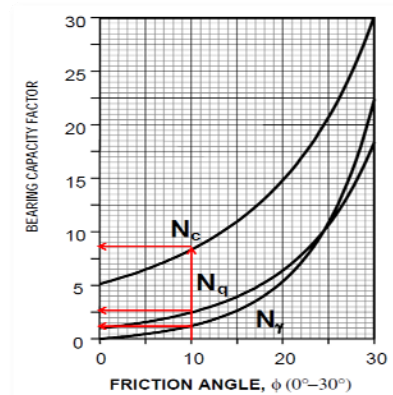
According to Terzaghi's theory, the ultimate bearing capacity, q_{ult} , for a concentrically loaded continuous footing of width B can be expressed as:

$$q_{ult} = cN_c + \gamma_0 H N_q + 0.5 B \gamma N_\gamma$$

in which c is the cohesion of the soil below the base of the footing; $\gamma_0 H$ is the vertical effective stress at the elevation of the footing base; B is the footing width; γ is the unit weight of the soil below the footing base; N_c , N_q , and N_γ are bearing capacity factors, which can be found in the NCEES reference manual.

Step 2:

From the given information, it is known that $\gamma = \gamma_0 = 120 \text{ lbf/ft}^3$, $c = 1300 \text{ lbf/ft}^2$, $H = 4 \text{ ft}$, $B = 3 \text{ ft}$, and $\phi = 10^\circ$. From the figure on Page 138 of the NCEES reference manual, it can be found that $N_c = 8.4$, $N_q = 2.5$, and $N_\gamma = 1.2$.



Step 3:

$$\begin{aligned} \mathbf{q}_{\text{ult}} &= cN_c + \gamma_0 H N_q + 0.5 B \gamma N_\gamma \\ &= \left(1300 \frac{\text{lbf}}{\text{ft}^2} \right) (8.4) + \left(120 \frac{\text{lbf}}{\text{ft}^3} \right) (4 \text{ ft}) (2.5) \\ &\quad + 0.5 (3 \text{ ft}) \left(120 \frac{\text{lbf}}{\text{ft}^3} \right) (1.2) = 12,336 \frac{\text{lbf}}{\text{ft}^2} \end{aligned}$$

81. A shallow foundation is to be constructed below the ground surface in a uniform cohesionless sand. It is found that the bearing capacity ratio for cohesion of soil below the foundation, N_c , is 50. What is the bearing capacity ratio for the vertical effective stress at the elevation of the foundation base, N_q ?

A) **38**

B) **43**

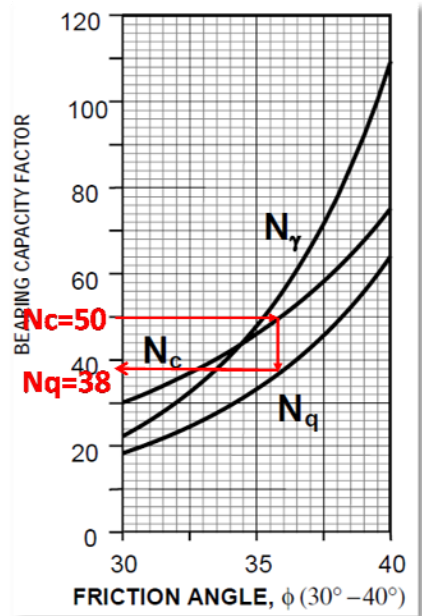
C) **60**

D) **74**

The Answer is A

Step 1:

From the figure on Page 138 of the NCEES reference manual, it can be found that when $N_c = 50$, the friction angle is 36° .



Step2:

For a friction angle of 36° , the bearing capacity ratio for the vertical effective stress at the elevation of the foundation base, N_q , is 38.

82. A continuous footing is founded 5 ft below the ground surface in a clay for which $\gamma = 110 \text{ lbf/ft}^3$, $c = 1200 \text{ lbf/ft}^2$, and $\phi = 0^\circ$. There is no groundwater at the site. If the factor of safety FS is to be at least 3, recommend an allowable bearing capacity according to Terzaghi's theory.

- A) $1,336 \frac{\text{lbf}}{\text{ft}^2}$ B) $2,183 \frac{\text{lbf}}{\text{ft}^2}$
C) $12,336 \frac{\text{lbf}}{\text{ft}^2}$ D) $22,336 \frac{\text{lbf}}{\text{ft}^2}$

The Answer is B

Step 1:

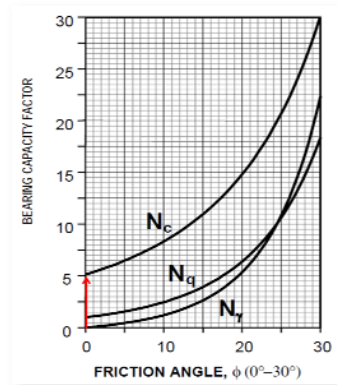
According to Terzaghi's theory, the ultimate bearing capacity, q_{ult} , for a concentrically loaded continuous footing of width B can be expressed as:

$$q_{ult} = cN_c + \gamma_0 H N_q + 0.5 B \gamma N_\gamma$$

in which c is the cohesion of the soil below the base of the footing; $\gamma_0 H$ is the vertical effective stress at the elevation of the footing base; B is the footing width; γ is the unit weight of the soil below the footing base; N_c , N_q , and N_γ are bearing capacity factors, which can be found in the NCEES reference manual.

Step 2:

From the given information, it is known that $\gamma = \gamma_0 = 110 \text{ lbf/ft}^3$, $c = 1200 \text{ lbf/ft}^2$, $H = 5 \text{ ft}$, and $\phi = 0^\circ$. From the figure on Page 138 of the NCEES reference manual, it can be found that $N_c = 5$, $N_q = 1$, and $N_\gamma = 0$.



Step 3:

$$\begin{aligned}
 q_{ult} &= cN_c + \gamma_0 H N_q + 0.5 B \gamma N_\gamma = \left(1200 \frac{\text{lbf}}{\text{ft}^2}\right)(5) + \left(110 \frac{\text{lbf}}{\text{ft}^3}\right)(5 \text{ ft})(1) + 0 \\
 &= 6,550 \frac{\text{lbf}}{\text{ft}^2}
 \end{aligned}$$

Step 4:

For a factor of safety FS=3, the allowable bearing capacity q_a , is

$$q_a = \frac{q_{ult}}{FS} = \frac{6,550 \frac{\text{lbf}}{\text{ft}^2}}{3} = 2183 \frac{\text{lbf}}{\text{ft}^2}$$

83. A continuous footing 3 ft wide is founded 5 ft below the ground surface in a uniform cohesionless sand for which $\gamma = 120 \text{ lbf/ft}^3$, $c = 0$, and $\phi = 35^\circ$. There is no groundwater at the site. If the factor of safety FS is to be at least 3, recommend the allowable bearing force per lineal foot of foundation according to Terzaghi's theory.

- A) **15.2 kips** B) **2.44 kips** C) **12.33 kips** D) **28.44 kips**

The Answer is DStep 1:

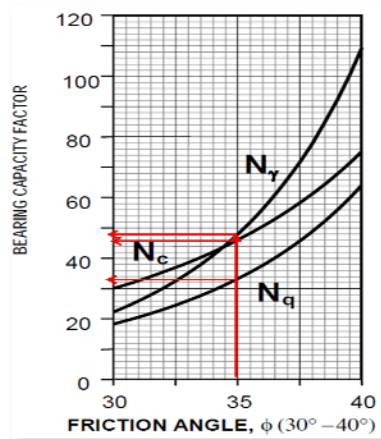
Based on Terzaghi's theory, the ultimate bearing capacity, q_{ult} , for a concentrically loaded continuous footing of width B can be expressed as:

$$q_{ult} = cN_c + \gamma_0 H N_q + 0.5 B \gamma N_\gamma$$

in which c is the cohesion of the soil below the base of the footing; $\gamma_0 H$ is the vertical effective stress at the elevation of the footing base; B is the footing width; γ is the unit weight of the soil below the footing base; N_c , N_q , and N_γ are bearing capacity factors, which can be found in the NCEES reference manual.

Step 2:

From the given information, it is known that $\gamma = \gamma_0 = 120 \text{ lbf/ft}^3$, $c = 0$, $H = 5 \text{ ft}$, $B = 3 \text{ ft}$, and $\phi = 35^\circ$. From the figure on Page 138 of the NCEES reference manual, it can be found that $N_q = 33$, and $N_\gamma = 48$.



Step 3:

$$\begin{aligned}
 q_{ult} &= cN_c + \gamma_0 H N_q + 0.5 B \gamma N_\gamma \\
 &= 0 + \left(120 \frac{\text{lbf}}{\text{ft}^3} \right) (5 \text{ ft}) (33) + 0.5 (3 \text{ ft}) \left(120 \frac{\text{lbf}}{\text{ft}^3} \right) (48) \\
 &= 28,440 \frac{\text{lbf}}{\text{ft}^2}
 \end{aligned}$$

Step 4:

For a factor of safety $FS=3$, the allowable bearing capacity q_a , is

$$q_a = \frac{q_{ult}}{FS} = \frac{28,440 \frac{\text{lbf}}{\text{ft}^2}}{3} = 9480 \frac{\text{lbf}}{\text{ft}^2}$$

Step 5:

For a lineal foot of foundation, the base area is

$$A = B(1 \text{ ft}) = (3 \text{ ft})(1 \text{ ft}) = 3 \text{ ft}^2$$

For a factor of safety $FS=3$, the allowable bearing force per lineal foot of foundation is

$$\begin{aligned}
 Q_{ult} &= q_a A = \left(9480 \frac{\text{lbf}}{\text{ft}^2} \right) (3 \text{ ft}^2) = 28,440 \text{ lbf} = \frac{28,440 \text{ lbf}}{1000 \text{ lbf/kip}} \\
 &= 28.44 \text{ kip}
 \end{aligned}$$

84. A total force of 2000 kN is to be supported by a square footing, which directly rests on a sand ground. The sand is uniform, cohesionless, having a unit weight $\gamma = 17 \text{ kN/m}^3$, and friction angle $\phi = 35^\circ$. If the factor of safety F is 2, determine the minimum width (B) of the footing according to Terzaghi's theory. (Based on Terzaghi's theory, the ultimate bearing capacity, q_{ult} , for a square footing of width B is calculated by $q_{ult} = 1.2cN_c + \gamma_0 H N_q + 0.4B\gamma N_\gamma$, in which c is the cohesion of the soil below the base of the footing; $\gamma_0 H$ is the vertical effective stress at the elevation of the footing base; B is the footing width; γ is the unit weight of the soil below the footing base; N_c , N_q , and N_γ are bearing capacity factors.)

- A) **0.5 m** B) **1.6 m** C) **2.3 m** D) **3.8 m**

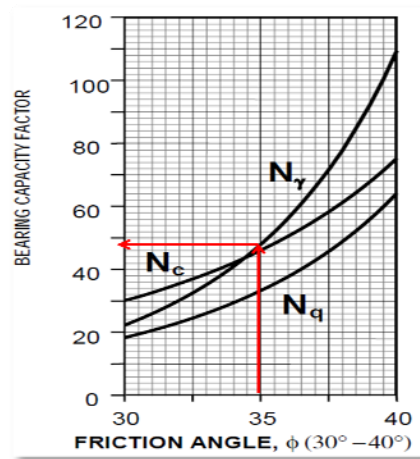
The Answer is C

Step 1:

Since the sand is cohesionless and the footing rests directly on the ground, $c=0$, and $H=0$. Therefore, $q_{ult} = 1.2cN_c + \gamma_0 H N_q + 0.4B\gamma N_\gamma = 0.4B\gamma N_\gamma$.

Step 2:

For $\phi = 35^\circ$, from the figure on Page 138 of the NCEES reference manual, it can be found that $N_\gamma = 48$.



Step 3:

The pressure on the footing should be less than the allowable bearing capacity of the sand.

$$\frac{P}{B^2} \leq \frac{q_{ult}}{F} = \frac{0.4B\gamma N_\gamma}{F}$$

$$B \geq \sqrt[3]{\frac{PF}{0.4\gamma N_\gamma}} = \sqrt[3]{\frac{(2000 \text{ kN})(2)}{0.4(17 \text{ kN/m}^3)(48)}} \cong 2.31 \text{ m}$$

85. Which of the following foundation types are deep foundation?

- I) Spread footings II) Piles III) Wall footings
IV) Mats V) Raft foundation

- A) II B) II and IV C) III and V D) I, II, and V

The Answer is A

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When the term deep foundations is used, it invariably means pile foundations. A pile is a long structural member installed in the ground to transfer loads to soils at some significant depths.

(A) is correct.

(B) is incorrect. Mats are shallow foundations.

(C) is incorrect. Both wall footings and mats are shallow foundations.

(D) is incorrect. Spread footings and raft foundations are both shallow foundations. Raft foundations are the same as mat foundations.

86. Through which is almost all the structural load on a friction pile transferred to the soil?

- A) The bottom end of the pile
- B) Skin friction along the length of the pile
- C) Both A and B
- D) None of the above

The Answer is B

Friction pile is one that transfers almost all the structural load to the soil by skin friction along a substantial length of the pile.

(A) is incorrect. A pile that transfers almost all the structural load to the soil at the bottom end of the pile is named an end bearing or point bearing pile.

(B) is correct.

(C) is incorrect, since A is incorrect.

(D) is incorrect, since B is correct.

- 87.** A strip wall footing of width 4 ft is embedded 3 ft below the ground surface in a clayed sand for which $\gamma = 120 \text{ lbf/ft}^3$, $c = 300 \text{ lbf/ft}^2$, and $\phi = 20^\circ$. The groundwater level 30 ft below the footing base. Use a factor of safety FS_q of 2 and assume the wall footing is sufficiently long, determine the allowable bearing capacity of the wall footing.

- A) $1,325 \frac{\text{lbf}}{\text{ft}^2}$ B) $3,987 \frac{\text{lbf}}{\text{ft}^2}$
C) $2,376 \frac{\text{lbf}}{\text{ft}^2}$ D) $4,146 \frac{\text{lbf}}{\text{ft}^2}$

The Answer is B

Step 1:

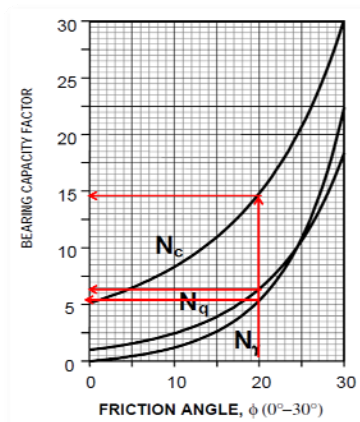
The ultimate bearing capacity, q_{ult} , for a strip footing of width B is:

$$q_{ult} = cN_c + \gamma'D_fN_q + 0.5\gamma'BN_\gamma$$

in which c is the cohesion of the soil; γ' is the effective unit weight of soil; D_f is the depth of footing below ground surface; B is the footing width; N_c , N_q , and N_γ are bearing capacity factors for cohesion, depth, and unit weight, respectively.

Step 2:

From the given information, it is known that $\gamma' = 120 \text{ lbf/ft}^3$, $c = 300 \text{ lbf/ft}^2$, $D_f = 3 \text{ ft}$, and $\phi = 20^\circ$. From the figure on Page 138 of the NCEES reference manual, it can be found that $N_c = 14.5$, $N_q = 6.4$, and $N_\gamma = 5.5$.



Step 3:

$$\begin{aligned}
 q_{ult} &= cN_c + \gamma'D_fN_q + 0.5\gamma'BN_\gamma \\
 &= \left(300 \frac{\text{lbf}}{\text{ft}^2}\right)(14.5) + \left(120 \frac{\text{lbf}}{\text{ft}^3}\right)(3 \text{ ft})(6.4) \\
 &\quad + 0.5\left(120 \frac{\text{lbf}}{\text{ft}^3}\right)(4 \text{ ft})(5.5) = 7,974 \frac{\text{lbf}}{\text{ft}^2}
 \end{aligned}$$

Step 4:

For a factor of safety $FS_q = 2$, the allowable bearing capacity q_a , is

$$q_a = \frac{q_{ult}}{FS_q} = \frac{7,974 \frac{\text{lbf}}{\text{ft}^2}}{2} = 3,987 \frac{\text{lbf}}{\text{ft}^2}$$

88. What type of shallow foundation should be used when the allowable soil pressure is low or where an array of columns and/or walls are so close that individual footings would overlap or nearly touch each other?

- A) Strip footing
- B) Spread footing
- C) Pile
- D) Mat foundation

The Answer is D

when the allowable soil pressure is low or where an array of columns and/or walls are so close that individual footings would overlap or nearly touch each other, a mat or raft foundation is required.

(A) is incorrect. A strip footing, also known as continuous footing, is used for a load-bearing wall, or for a row of columns which are closely spaced. It cannot be used for an array of columns.

(B) is incorrect. A spread footing (or isolated or pad) footing is provided to support an individual column. A spread footing is typically circular, square or rectangular slab of uniform thickness.

(C) is incorrect. Piles are deep foundations, not shallow foundations.

(D) is correct. A mat or raft foundation is a large slab supporting a number of columns and walls under the entire structure or a large part of the structure.

89. A 9-inch diameter concrete pile is driven 50 ft into insensitive clay, which has an in-situ unit weight of **120 lbf/ft³**, and an undrained shear strength **$s_u = 1400 \text{ lbf/ft}^2$** . The groundwater table is at the ground surface. Assume the entire pile length is effective and the adhesive stress **$f_s = 0.5s_u$** , determine the bearing capacity of the pile due to friction.

- A) 60 kips B) 83 kips C) 160 kips D) 500 kips

The Answer is B

Step 1:

The bearing capacity of the pile due to friction is provided by skin friction, Q_f , over the embedded length of the pile, which is the product of the adhesive stress, f_s , and the surface area of the shaft.

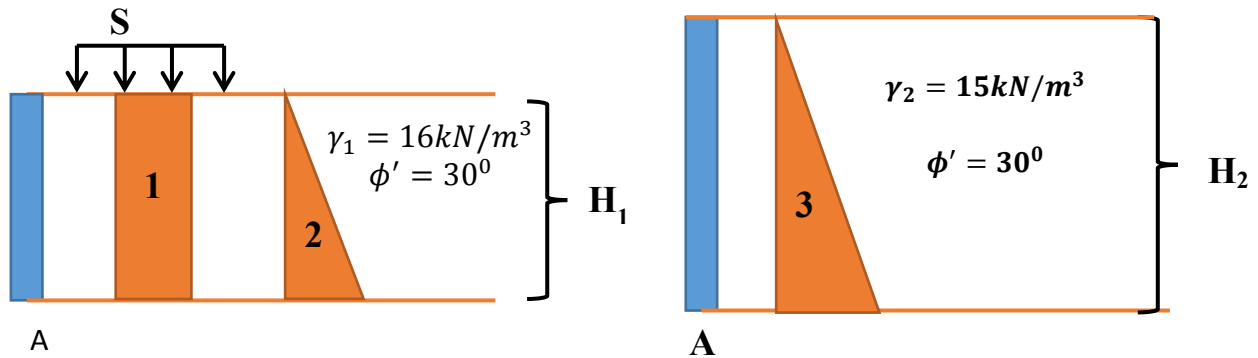
Step 2:

$$f_s = 0.5s_u = 0.5 \left(1400 \frac{\text{lbf}}{\text{ft}^2} \right) = 700 \frac{\text{lbf}}{\text{ft}^2}$$

Step 3:

$$\begin{aligned} Q_f = f_s A = f_s \pi D L &= \left(700 \frac{\text{lbf}}{\text{ft}^2} \right) \pi \left(\frac{9 \text{ inches}}{12 \frac{\text{inches}}{\text{ft}}} \right) (50 \text{ ft}) \cong 82467 \text{ lbf} \\ &= \frac{82467 \text{ lbf}}{1000 \frac{\text{lbf}}{\text{kip}}} \cong 82.5 \text{ kips} \end{aligned}$$

90. Two figures show two frictionless wall as below. Information about each wall is given. Assume that $H_1=4\text{m}$ and $H_2=5\text{m}$. Assume that moment at point A for left figure and right figure is the same. Determine the value of S.



- A) 20kN/m B) 30kN/m C) 25kN/m D) 35kN/m

The Answer is A

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = K_a S H_1 = \frac{1}{3} S (4m) = 1.3S$$

$$z_1 = \frac{1}{2} H_1 = \frac{1}{2} (4m) = 2m$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (16kN/m^3) (4m)^2 = 42.7kN$$

$$z_2 = \frac{1}{3} H_1 = \frac{1}{3} (4m) = 1.3m$$

$$P_3 = \frac{1}{2} K_a \gamma_2 H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (15kN/m^3) (5m)^2 = 62.5kN$$

$$z_3 = \frac{1}{3} H_2 = \frac{1}{3} (5m) = 1.7m$$

Step 3: The moment at point A is the same. We got,

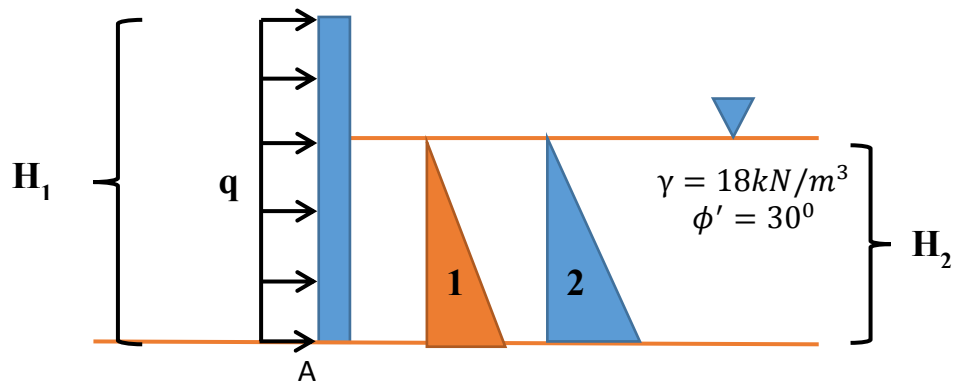
$$P_1 z_1 + P_2 z_2 = P_3 z_3 = (1.3S)(2m) + (42.7kN)(1.3m) = (62.5kN)(1.7m)$$

Solve this equation,

$$S = 19.5kN/m$$

$$\sim 20kN/m$$

91. A retaining wall is given in the following diagram. It is known that $H_1=5\text{m}$ and $H_2=3\text{m}$. Determine the value of q in order to make resultant force of retaining wall zero.



A) 15kN/m

B) 13kN/m

C) 11kN/m

D) 17kN/m

The Answer is C

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a (\gamma - \gamma_w) H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 12.3 \text{ kN}$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

Step 3: The resultant force is zero. We got,

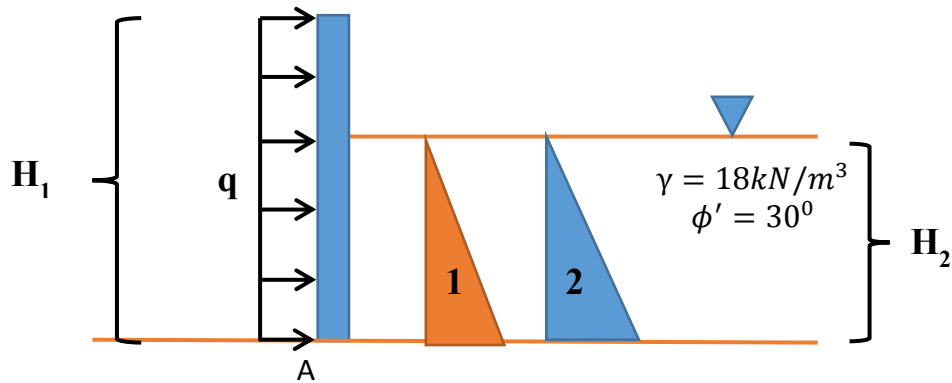
$$P_1 + P_2 = qH_1 = 12.3 \text{ kN} + 44.1 \text{ kN} = q(5 \text{ m})$$

Solve this equation,

$$q = 11.3 \text{ kN/m}$$

$$\sim 11 \text{ kN/m}$$

92. A retaining wall is given in the following diagram. It is known $H_2=3\text{m}$. $q = 10\text{kN/m}$. Determine the value of H_1 in order to make resultant force of retaining wall zero.



- A) 6.3m B) 5.6m C) 5.2m D) 4.8m

The Answer is B

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a (\gamma - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 12.3 \text{ kN}$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

Step 3: The resultant force is zero. We got,

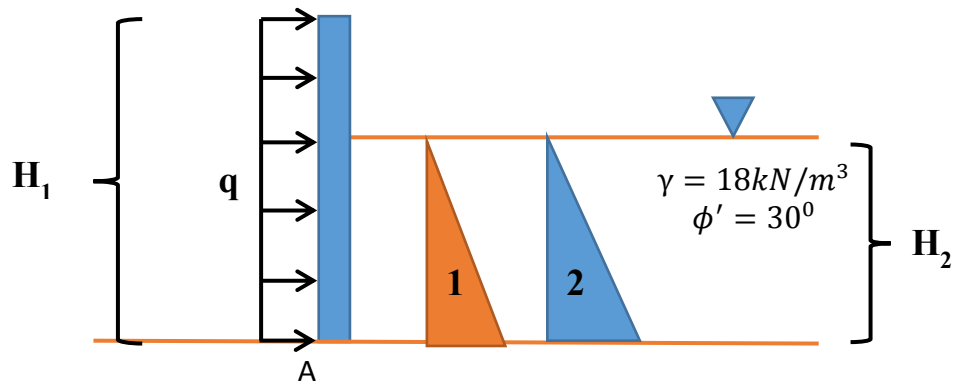
$$P_1 + P_2 = q H_1 = 12.3 \text{ kN} + 44.1 \text{ kN} = (10 \text{ kN/m}) H_1$$

Solve this equation,

$$H_1 = 5.6 \text{ m}$$

$$\sim 5.6 \text{ m}$$

94. A retaining wall is given in the following diagram. It is known $H_1=5\text{m}$ and $H_2=3\text{m}$. $q = 10\text{kN/m}$. Calculate the moment at point A.



- A) 60kN-m B) 50kN-m C) 80kN-m D) 70kN-m

The Answer is D

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a (\gamma - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 12.3 \text{ kN}$$

Lateral force for Area 2,

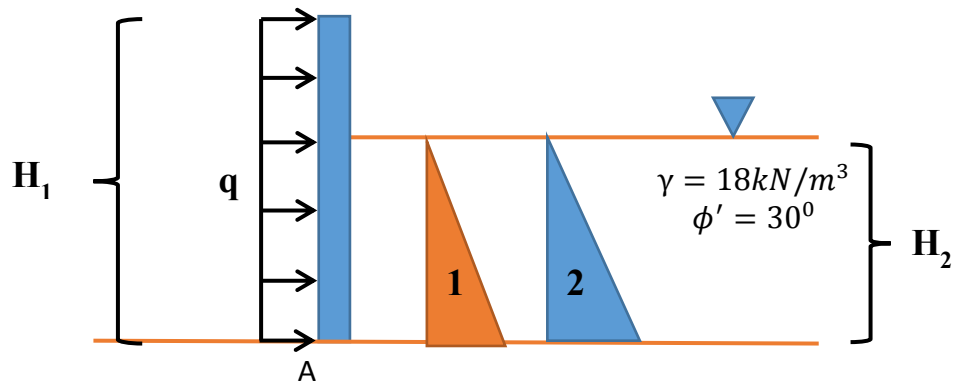
$$P_2 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (3 \text{ m})^2 = 44.1 \text{ kN}$$

$$z_1 = z_2 = \frac{1}{3} H_2 = \frac{1}{3} (3 \text{ m}) = 1 \text{ m}$$

Step 3: The moment at point A can be obtained by,

$$M = \frac{1}{2} q H_1^2 - P_1 z_1 - P_2 z_2 = \frac{1}{2} (10 \text{ kN/m}) (5 \text{ m})^2 - (12.3 \text{ kN}) (1 \text{ m}) - (44.1 \text{ kN}) (1 \text{ m}) = 68.6 \text{ kN} \cdot \text{m} \sim 70 \text{ kN} \cdot \text{m}$$

95. A retaining wall is given in the following diagram. It is known $H_1=5\text{m}$. $q = 10\text{kN/m}$. Determine the value of H_2 when resultant force is equal to zero.



- A) 2.8m B) 3.6m C) 4.2m D) 3.8m

The Answer is A

Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a (\gamma - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (H_2)^2 = 1.37 (H_2)^2$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (H_2)^2 = 4.9 (H_2)^2$$

Step 3: The resultant force is equal to zero,

$$F = P_1 + P_2 - qH_1 = 1.37 (H_2)^2 + 4.9 (H_2)^2 - (10 \text{ kN/m})(5 \text{ m}) = 0$$

Solve this equation,

$$H_2 = 2.8 \text{ m} \sim 2.8 \text{ m}$$

96. A 5m soil is assumed to be in the range of virgin compression. C_c is given as 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 and Δp is equal to 200kN/m^2 . In order to know the initial voids ratio, 100cm^3 initial soil is selected and we find that there is 45cm^3 voids. Determine the settlement of this soil.

A) 32cm

B) 37cm

C) 43cm

D) 48cm

The Answer is B

Step 1: Settlement for soil in the range of virgin compression can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_c \log \frac{p_0 + \Delta p}{p_0} \right]$$

From this problem, we can know everything except for the initial voids ratio. We need to calculate it first.

Step 2: The initial voids ratio can be calculated based on,

$$e = \frac{V_v}{V_s} = \frac{V_v}{V - V_v} = \frac{45 \text{ cm}^3}{100 \text{ cm}^3 - 45 \text{ cm}^3} = 0.8$$

Therefore, we can know the settlement of this soil by,

$$\Delta H = \frac{5 \text{ m}}{1 + 0.8} \left[(0.6) \log \frac{300 \text{ kN/m}^2 + 200 \text{ kN/m}^2}{300 \text{ kN/m}^2} \right] = 37 \text{ cm}$$

~37cm

97. A 3m soil is assumed to be in the range of virgin compression. C_c is given as 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 and Δp is equal to 200kN/m^2 . In order to know the initial voids ratio, 100cm^3 initial soil is selected and we find that there is 75cm^3 soil solids. Determine the settlement of this soil.

A) 34cm

B) 30cm

C) 26cm

D) 38cm

The Answer is B

Step 1: Settlement for soil in the range of virgin compression can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_c \log \frac{p_0 + \Delta p}{p_0} \right]$$

From this problem, we can know everything except for the initial voids ratio. We need to calculate it first.

Step 2: The initial voids ratio can be calculated based on,

$$e = \frac{V_v}{V_s} = \frac{V - V_s}{V_s} = \frac{100\text{cm}^3 - 75\text{cm}^3}{75\text{cm}^3} = 0.33$$

Therefore, we can know the settlement of this soil by,

$$\Delta H = \frac{3\text{m}}{1 + 0.33} \left[(0.6) \log \frac{300\text{kN/m}^2 + 200\text{kN/m}^2}{300\text{kN/m}^2} \right] = 30\text{cm} \sim 30\text{cm}$$

98. A 3m soil is assumed to be in the range of virgin compression. C_c is given as 0.5. If the initial effective pressure p_0 is equal to 300kN/m^2 and Δp is equal to 200kN/m^2 . It is known that it has 30cm settlement. What is the porosity of this soil for the initial state?

A) 0.3

B) 0.2

C) 0.1

D) 0.4

The Answer is C

Step 1: Settlement for soil in the range of virgin compression can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_c \log \frac{p_0 + \Delta p}{p_0} \right]$$

From this problem, we can know everything except for the initial voids ratio. We can calculate it by,

$$\frac{H_0}{1 + e_0} \left[C_c \log \frac{p_0 + \Delta p}{p_0} \right] = \frac{3\text{m}}{1 + e} \left[(0.5) \log \frac{300\text{kN/m}^2 + 200\text{kN/m}^2}{300\text{kN/m}^2} \right] = 30\text{cm}$$

$$e = 0.11$$

Step 2: The initial porosity can be calculated based on,

$$n = \frac{e}{1 + e} = \frac{0.11}{1 + 0.11} = 0.1 \sim 0.1$$

99. A 3m soil is assumed to be in the range of virgin compression. If the initial effective pressure p_0 is equal to 300kN/m^2 and Δp is equal to 200kN/m^2 . It is known that it has 30cm settlement. If the initial porosity is 0.5, then what is the value of C_c ?

A) 0.9

B) 0.8

C) 0.7

D) 0.6

The Answer is A

Step 1: Settlement for soil in the range of virgin compression can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_c \log \frac{p_0 + \Delta p}{p_0} \right]$$

From this problem, we can know everything except for the C_c and voids ratio. We can calculate voids ratio by,

$$e = \frac{n}{1 - n} = \frac{0.5}{1 - 0.5} = 1$$

Step 2: Based on the equation, we have the following as,

$$\begin{aligned} \Delta H &= \frac{H_0}{1 + e_0} \left[C_c \log \frac{p_0 + \Delta p}{p_0} \right] = \frac{3\text{m}}{1 + 1} \left[C_c \log \frac{300\text{kN/m}^2 + 200\text{kN/m}^2}{300\text{kN/m}^2} \right] \\ &= 30\text{cm} \end{aligned}$$

$$C_c = 0.9 \sim 0.9$$

100. A 3m soil is assumed to be in the range of virgin compression. If the initial effective pressure p_0 is equal to 300kN/m^2 . It is known that it has 30cm settlement when adding pressure Δp and C_c is 0.7. If the initial porosity is 0.5, then what is the value of Δp ?

- A) 220kN/m^2 B) 240kN/m^2 C) 260kN/m^2 D) 280kN/m^2

The Answer is D

Step 1: Settlement for soil in the range of virgin compression can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_c \log \frac{p_0 + \Delta p}{p_0} \right]$$

From this problem, we can know everything except for the Δp and voids ratio. We can calculate voids ratio by,

$$e = \frac{n}{1 - n} = \frac{0.5}{1 - 0.5} = 1$$

Step 2: Based on the equation, we have the following as,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_0 + \Delta p}{p_0} \right] = \frac{3\text{m}}{1 + 1} \left[(0.7) \log \frac{300\text{kN/m}^2 + \Delta p}{300\text{kN/m}^2} \right] = 30\text{cm}$$

$$\Delta p = 279\text{kN/m}^2 \sim 280\text{kN/m}^2$$

101. A 5m soil is given. C_R is given as 0.1 and C_c is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 , Δp is equal to 200kN/m^2 , and p_c is 400kN/m^2 , determine the settlement of this soil when the initial porosity of soil is 0.44.

A) 25cm

B) 15cm

C) 20cm

D) 30cm

The Answer is C

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_c \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio. We need to calculate it first.

Step 2: The initial porosity is given as 0.44. Therefore, voids ratio can be calculated by,

$$e = \frac{n}{1 - n} = \frac{0.44}{1 - 0.44} = 0.8$$

Therefore, we can know the settlement of this soil by,

$$\Delta H = \frac{5\text{m}}{1 + 0.8} \left[(0.1) \log \frac{400\text{kN/m}^2}{300\text{kN/m}^2} + (0.6) \log \frac{500\text{kN/m}^2}{400\text{kN/m}^2} \right] = 19.6\text{cm}$$

~20cm

102. A 5m soil is given. C_R is given as 0.1 and C_C is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 , Δp is equal to 200kN/m^2 , and p_c is 400kN/m^2 . In order to know the initial voids ratio, 100cm^3 initial soil is selected and we find that there is 45cm^3 voids. Determine the settlement of this soil.

A) 35cm

B) 30cm

C) 25cm

D) 20cm

The Answer is D

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio. We need to calculate it first.

Step 2: The initial voids ratio can be calculated based on,

$$e = \frac{V_v}{V_s} = \frac{V_v}{V - V_v} = \frac{45 \text{ cm}^3}{100 \text{ cm}^3 - 45 \text{ cm}^3} = 0.8$$

Therefore, we can know the settlement of this soil by,

$$\Delta H = \frac{5 \text{ m}}{1 + 0.8} \left[(0.1) \log \frac{400 \text{ kN/m}^2}{300 \text{ kN/m}^2} + (0.6) \log \frac{500 \text{ kN/m}^2}{400 \text{ kN/m}^2} \right] = 19.6 \text{ cm}$$

~20cm

103. A 3m soil is given. C_R is given as 0.1 and C_C is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 , Δp is equal to 200kN/m^2 , and p_c is 400kN/m^2 . In order to know the initial voids ratio, 100cm^3 initial soil is selected and we find that there is 75cm^3 soil solids. Determine the settlement of this soil.

A) 18cm

B) 16cm

C) 20cm

D) 14cm



The Answer is B

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio. We need to calculate it first.

Step 2: The initial voids ratio can be calculated based on,

$$e = \frac{V_v}{V_s} = \frac{V - V_s}{V_s} = \frac{100\text{cm}^3 - 75\text{cm}^3}{75\text{cm}^3} = 0.33$$

Therefore, we can know the settlement of this soil by,

$$\Delta H = \frac{3\text{m}}{1 + 0.33} \left[(0.1) \log \frac{400\text{kN/m}^2}{300\text{kN/m}^2} + (0.6) \log \frac{500\text{kN/m}^2}{400\text{kN/m}^2} \right] = 15.9\text{cm}$$

~16cm

104. A 3m soil is given. C_R is given as 0.1 and C_c is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 , Δp is equal to 200kN/m^2 , and p_c is 400kN/m^2 . It is known that it has 11cm settlement. What is the porosity of this soil for the initial state?

A) 0.25

B) 0.5

C) 0.75

D) 1

The Answer is B

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio. We can calculate it by,

$$\begin{aligned} \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right] \\ = \frac{3\text{m}}{1 + e} \left[(0.1) \log \frac{400\text{kN/m}^2}{300\text{kN/m}^2} + (0.6) \log \frac{500\text{kN/m}^2}{400\text{kN/m}^2} \right] = 11\text{cm} \end{aligned}$$

$$e = 1$$

Step 2: The initial porosity can be calculated based on,

$$n = \frac{e}{1 + e} = \frac{1}{1 + 1} = 0.5$$

105. A 3m soil is given. C_R is given as 0.1 and C_c is unknown. If the initial effective pressure p_0 is equal to 300kN/m^2 , Δp is equal to 200kN/m^2 , and p_c is 400kN/m^2 . It is known that it has 11cm settlement. If the initial porosity is 0.5, then what is the value of C_C ?

A) 0.5

B) 0.8

C) 0.6

D) 0.7

The Answer is C

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio and C_C . We can calculate voids ratio by,

$$e = \frac{n}{1 - n} = \frac{0.5}{1 - 0.5} = 1$$

Step 2: Based on the equation, we have the following as,

$$\begin{aligned} \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right] \\ = \frac{3\text{m}}{1 + 1} \left[(0.1) \log \frac{400\text{kN/m}^2}{300\text{kN/m}^2} + C_C \log \frac{500\text{kN/m}^2}{400\text{kN/m}^2} \right] = 11\text{cm} \end{aligned}$$

$$C_C = 0.6 \sim 0.6$$

106. A 3m soil is given. C_R is unknown and C_c is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 , Δp is equal to 200kN/m^2 , and p_c is 400kN/m^2 . It is known that it has 11cm settlement. If the initial porosity is 0.5, then what is the value of C_R ?

A) 0.2

B) 0.1

C) 0.3

D) 0.4



The Answer is B

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio and C_R . We can calculate voids ratio by,

$$e = \frac{n}{1 - n} = \frac{0.5}{1 - 0.5} = 1$$

Step 2: Based on the equation, we have the following as,

$$\begin{aligned} \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right] \\ = \frac{3\text{m}}{1 + 1} \left[C_R \log \frac{400\text{kN/m}^2}{300\text{kN/m}^2} + (0.6) \log \frac{500\text{kN/m}^2}{400\text{kN/m}^2} \right] = 11\text{cm} \end{aligned}$$

$$C_R = 0.1 \sim 0.1$$

107. A 3m soil is given. C_R is given as 0.1 and C_c is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 and p_c is 400kN/m^2 . It is known that it has 11cm settlement when adding pressure Δp . If the initial porosity is 0.5, then what is the value of Δp ?

- A) 120kN/m^2 B) 200kN/m^2 C) 160kN/m^2 D) 180kN/m^2

The Answer is B

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio and Δp . We can calculate voids ratio by,

$$e = \frac{n}{1 - n} = \frac{0.5}{1 - 0.5} = 1$$

Step 2: Based on the equation, we have the following as,

$$\begin{aligned} \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right] \\ = \frac{3\text{m}}{1 + 1} \left[(0.1) \log \frac{400\text{kN/m}^2}{300\text{kN/m}^2} + (0.6) \log \frac{300\text{kN/m}^2 + \Delta p}{400\text{kN/m}^2} \right] \\ = 11\text{cm} \end{aligned}$$

$$\Delta p = 200\text{kN/m}^2 \sim 200\text{kN/m}^2$$

108. A 3m soil is given. C_R is given as 0.1 and C_c is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 . It is known that it has 11cm settlement when adding pressure Δp . If the voids occupy 50% of total volume at initial state, then what is the value of Δp ?

- A) 200kN/m^2 B) 100kN/m^2 C) 150kN/m^2 D) 180kN/m^2

The Answer is A

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio and Δp . We can calculate voids ratio by,

$$e = \frac{V_v}{V_s} = \frac{50\%V}{50\%V} = 1$$

Step 2: Based on the equation, we have the following as,

$$\begin{aligned} \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right] \\ = \frac{3\text{m}}{1 + 1} \left[(0.1) \log \frac{400\text{kN/m}^2}{300\text{kN/m}^2} + (0.6) \log \frac{300\text{kN/m}^2 + \Delta p}{400\text{kN/m}^2} \right] \\ = 11\text{cm} \end{aligned}$$

$$\Delta p = 200\text{kN/m}^2 \sim 200\text{kN/m}^2$$

109. A 3m soil is given. It is known that the initial porosity is 0.5 and after ultimate consolidation settlement it become to 0.45. Determine the ultimate consolidation settlement in this soil layer.

- A) 0.56m B) 0.48m C) 0.33m D) 0.50m

The Answer is C

Step 1: Ultimate consolidation settlement in soil layer can be determined by the following equation,

$$S_{ULT} = \varepsilon_v H = \frac{e_0 - e_1}{1 + e_0} H$$

From this problem, we can know the thickness of this soil layer as 3m. We need to determine the voids ratio before and after ultimate consolidation settlement. They can be calculated based on,

$$e_0 = \frac{n_0}{1 - n_0} = \frac{0.5}{1 - 0.5} = 1$$

$$e_1 = \frac{n_1}{1 - n_1} = \frac{0.45}{1 - 0.45} = 0.8$$

Step 2: Based on the equation, we have the following as,

$$S_{ULT} = \frac{e_0 - e_1}{1 + e_0} H = \frac{1 - 0.8}{1 + 0.8} (3\text{m}) = 0.33\text{m} \sim 0.33\text{m}$$

110. A 3m soil is given. It is known that the initial voids ratio is 1 and after ultimate consolidation settlement it is unknown. Determine the final porosity of this soil if the ultimate consolidation settlement is 0.5m.

A) 0.5

B) 0.4

C) 0.3

D) 0.2

The Answer is B

Step 1: Ultimate consolidation settlement in soil layer can be determined by the following equation,

$$S_{ULT} = \epsilon_v H = \frac{e_0 - e_1}{1 + e_0} H$$

From this problem, we can know the thickness of this soil layer as 3m. We need to determine the voids ratio after ultimate consolidation settlement first. It can be obtained by,

$$\frac{e_0 - e_1}{1 + e_0} H = \frac{1 - e_1}{1 + 0.8} (3\text{m}) = 0.5\text{m}$$

$$e_1 = 0.7$$

Step 2: We can calculate the final porosity of this soil by,

$$n = \frac{e}{1+e} = \frac{0.7}{1+0.7} = 0.41 \sim 0.4$$

111. A 3m soil is given. It is known that the final voids ratio is 0.7 after ultimate consolidation settlement but the initial voids ratio is unknown. Determine the initial porosity of this soil if the ultimate consolidation settlement is 0.5m.

- A) 0.5 B) 0.4 C) 0.3 D) 0.2



The Answer is A

Step1:Ultimate consolidation settlement in soil layer can be determined by the following equation,

$$S_{ULT} = \varepsilon_v H = \frac{e_0 - e_1}{1 + e_0} H$$

From this problem, we can know the thickness of this soil layer as 3m. We need to determine the initial voids ratio first. It can be obtained by,

$$\frac{e_0 - e_1}{1 + e_0} H = \frac{e_0 - 0.7}{1 + e_0} (3m) = 0.5m$$

$$e_0 = 1$$

Step 2:We can calculate the initial porosity of this soil by,

$$n = \frac{e}{1+e} = \frac{1}{1+1} = 0.5 \sim 0.5$$

112. A soil layer is given. The thickness of this soil layer is unknown. It is known that the final voids ratio is 0.7 after ultimate consolidation settlement and the initial porosity is 0.5. Determine the thickness of this soil layer if the ultimate consolidation settlement is 0.5m.

- A) 2m B) 3m C) 4m D) 5m



The Answer is B

Step 1: Ultimate consolidation settlement in soil layer can be determined by the following equation,

$$S_{ULT} = \varepsilon_v H = \frac{e_0 - e_1}{1 + e_0} H$$

From this problem, we can know the initial porosity of this soil. We need to determine the initial voids ratio first. It can be obtained by,

$$e_0 = \frac{n_0}{1 - n_0} = \frac{0.5}{1 - 0.5} = 1$$

Step 2: We can calculate the thickness of this soil layer by,

$$S_{ULT} = \frac{e_0 - e_1}{1 + e_0} H = \frac{1 - 0.7}{1 + 0.7} H = 0.5m$$

$$H = 3m \sim 3m$$

113. A 3m soil is given. It is known that the initial porosity is 0.5 and after ultimate consolidation settlement it become to 0.45. Determine the approximate settlement in this soil layer if the average degree of consolidation is 70%.

- A) 0.34m B) 0.23m C) 0.18m D) 0.36m



The Answer is B

Step 1: To calculate the approximate settlement, we need to calculate the ultimate consolidation settlement in soil layer first. It can be determined by the following equation,

$$S_{ULT} = \varepsilon_v H = \frac{e_0 - e_1}{1 + e_0} H$$

From this problem, we can know the thickness of this soil layer as 3m. We need to determine the voids ratio before and after ultimate consolidation settlement. They can be calculated based on,

$$e_0 = \frac{n_0}{1 - n_0} = \frac{0.5}{1 - 0.5} = 1$$

$$e_1 = \frac{n_1}{1 - n_1} = \frac{0.45}{1 - 0.45} = 0.8$$

Step 2: The approximate settlement can be then calculated by,

$$S_T = U_{AV} S_{ULT} = U_{AV} \frac{e_0 - e_1}{1 + e_0} H = (70\%) \frac{1 - 0.8}{1 + 0.8} (3m) = 0.23m$$

~0.23m

114. A 3m soil is given. It is known that the initial voids ratio is 1 and after ultimate consolidation settlement it is unknown. Determine the final porosity of this soil if the approximate settlement is 0.2m and average degree of consolidation is 50%.

A) 0.34

B) 0.26

C) 0.38

D) 0.42

The Answer is D

Step 1: Approximate settlement can be calculated based on the equation as shown below,

$$S_T = U_{AV} S_{ULT} = U_{AV} \frac{e_0 - e_1}{1 + e_0} H$$

The final voids ratio based on this equation can be determined by,

$$U_{AV} \frac{e_0 - e_1}{1 + e_0} H = (50\%) \frac{1 - e_1}{1 + 1} (3m) = 0.2m$$

$$e_1 = 0.73$$

Step 2: The final porosity can be then calculated by,

$$n = \frac{e}{1+e} = \frac{0.73}{1+0.73} = 0.42 \sim 0.42$$

115. A 3m soil is given. It is known that the final voids ratio is 0.3 after ultimate consolidation settlement but the initial voids ratio is unknown. Determine the initial porosity of this soil if the approximate settlement is 0.2m and average degree of consolidation is 50%.

A) 0.5

B) 0.4

C) 0.3

D) 0.2

The Answer is A

Step 1: Approximate settlement can be calculated based on the equation as shown below,

$$S_T = U_{AV} S_{ULT} = U_{AV} \frac{e_0 - e_1}{1 + e_0} H$$

The final voids ratio based on this equation can be determined by,

$$U_{AV} \frac{e_0 - e_1}{1 + e_0} H = (50\%) \frac{e_0 - 0.73}{1 + e_0} (3m) = 0.2m$$

$$e_0 = 1$$

Step 2: The initial porosity can be then calculated by,

$$n = \frac{e}{1+e} = \frac{1}{1+1} = 0.5 \sim 0.5$$

116. It is known that initial effective consolidation stress, p_0 is equal to 200kN/m^2 and final effective consolidation stress, $p_0 + \Delta p$ is equal to 500kN/m^2 . The past maximum consolidation stress, p_c is 600kN/m^2 . For this soil in initial state, we know that it has 100pcf dry density, 40% water content and 80% degree of saturation. If the thickness of soil is 3m , and $C_R = 0.1$, $C_C = 0.6$, calculate the settlement of this soil.

- A) 44mm B) 55mm C) 66mm D) 33m



The Answer is C

Step 1: According to this problem, we know that,

$$p_0 = 200 \text{ kN/m}^2$$

$$p_0 + \Delta p = 500 \text{ kN/m}^2$$

$$p_c = 600 \text{ kN/m}^2 > p_0 + \Delta p \text{ and } p_0$$

Therefore, we can pick the equation to calculate the settlement as,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_0 + \Delta p}{p_0} \right]$$

From this problem, we can know everything except for the initial voids ratio. We need to calculate it first.

Step 2: Equation of void ratio of soil is,

$$e = \frac{V_v}{V_s}$$

To determine the water content from voids ratio, we need to determine the voids ratio equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{V_s}{S}} = \frac{\frac{wW}{\gamma_w S}}{\frac{V_s}{S}} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Step 3: To get the value of water content, we can substitute all known value into the equation above and we have,

$$e = \gamma_d \frac{w}{\gamma_w S} = (100 \text{ pcf}) \frac{40\%}{(62.5 \text{ ppcf})(80\%)} = 0.8$$

Step 4: Based on the equation above, we can calculate the settlement as,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_0 + \Delta p}{p_0} \right] = \frac{3 \text{ m}}{1 + 0.8} \left[(0.1) \log \frac{500 \text{ kN/m}^2}{200 \text{ kN/m}^2} \right] = 0.066 \text{ m} = 66 \text{ mm} \sim 66 \text{ mm}$$

117. It is known that initial effective consolidation stress, p_0 is equal to 200kN/m^2 and final effective consolidation stress, $p_0 + \Delta p$ is equal to 500kN/m^2 . The past maximum consolidation stress, p_c is 600kN/m^2 . For this soil in initial state, we know that it has 40% water content and 80% degree of saturation. If the thickness of soil is 3m, and $C_R = 0.1$, $C_C = 0.6$, calculate the dry density of this soil with 66mm settlement.

A) 100pcf

B) 110pcf

C) 120pcf

D) 130p

The Answer is A

Step 1: According to this problem, we know that,

$$p_0 = 200 \text{ kN/m}^2$$

$$p_0 + \Delta p = 500 \text{ kN/m}^2$$

$$p_c = 600 \text{ kN/m}^2 > p_0 + \Delta p \text{ and } p_0$$

Therefore, we can pick the equation to calculate the settlement as,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_0 + \Delta p}{p_0} \right]$$

From this problem, we can know everything except for the initial voids ratio. We need to calculate it first.

Step 2: Equation of void ratio of soil is,

$$e = \frac{V_v}{V_s}$$

To determine the water content from voids ratio, we need to determine the voids ratio equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{V_s}{S}} = \frac{\frac{wW}{\gamma_w S}}{\frac{V_s}{S}} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Step 3: To get the value of water content, we can substitute all known value into the equation above and we have,

$$e = \gamma_d \frac{w}{\gamma_w S} = \gamma_d \frac{40\%}{(62.5 \text{ pccf})(80\%)} = \frac{\gamma_d}{125}$$

Step 4: Based on the equation above, we can calculate the dry density as,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_0 + \Delta p}{p_0} \right] = \frac{3m}{1 + \frac{\gamma_d}{125}} \left[(0.1) \log \frac{500 \text{ kN/m}^2}{200 \text{ kN/m}^2} \right] = 0.066m$$

$$= 66 \text{ mm}$$

$$\gamma_d = 100 \text{ pcf} \sim 100 \text{ pcf}$$

118. A 5m soil is given. C_R is given as 0.1 and C_c is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 , Δp is equal to 200kN/m^2 , and p_c is 400kN/m^2 , determine the settlement of this soil when the initial dry density of soil is 100pcf , the water content is 36% and the degree of saturation is 54%.

- A) 17cm B) 20cm C) 23cm D) 15cm

The Answer is A

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio. We need to calculate it first.

Step 2: Equation of void ratio of soil is,

$$e = \frac{V_v}{V_s}$$

To determine the water content from voids ratio, we need to determine the voids ratio equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{V_s}{S}} = \frac{\frac{wW}{\gamma_w S}}{\frac{V_s}{S}} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Step 3: To get the value of water content, we can substitute all known value into the equation above and we have,

$$e = \gamma_d \frac{w}{\gamma_w S} = (100 pcf) \frac{36\%}{(62.5 ppcf)(54\%)} = 1.1.$$

Step 4: Based on the equation above, we can calculate the settlement as,

$$\Delta H = \frac{5m}{1 + 1.1} \left[(0.1) \log \frac{400 kN/m^2}{300 kN/m^2} + (0.6) \log \frac{500 kN/m^2}{400 kN/m^2} \right] = 16.8 cm$$

~17cm

119. A 3m soil is given. C_R is unknown and C_c is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 , Δp is equal to 200kN/m^2 , and p_c is 400kN/m^2 . It is known that it has 11cm settlement. If the initial dry density of soil is 110pcf , the water content is 38% and the degree of saturation is 66%, then what is the value of C_R ?

A) 0.1

B) 0.2

C) 0.3

D) 0.4

The Answer is A

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio and C_R . We need to calculate it first.

Step 2: Equation of void ratio of soil is,

$$e = \frac{V_v}{V_s}$$

To determine the water content from voids ratio, we need to determine the voids ratio equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{W}{\gamma_w S}} = \frac{\frac{wW}{\gamma_w S}}{\frac{W}{\gamma_w S}} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Step 3: To get the value of water content, we can substitute all known value into the equation above and we have,

$$e = \gamma_d \frac{w}{\gamma_w S} = (110 \text{ pcf}) \frac{38\%}{(62.5 \text{ ppcf})(66\%)} = 1$$

Step 4: Based on the equation, we have the following as,

$$\begin{aligned} \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_C \log \frac{p_0 + \Delta p}{p_c} \right] \\ = \frac{3 \text{ m}}{1 + 1} \left[C_R \log \frac{400 \text{ kN/m}^2}{300 \text{ kN/m}^2} + (0.6) \log \frac{500 \text{ kN/m}^2}{400 \text{ kN/m}^2} \right] = 11 \text{ cm} \end{aligned}$$

$$C_R = 0.1 \sim 0.1$$

120. A 3m soil is given. C_R is given as 0.1 and C_c is 0.6. If the initial effective pressure p_0 is equal to 300kN/m^2 and p_c is 400kN/m^2 . It is known that it has 11cm settlement when adding pressure Δp . If the initial dry density of soil is 110pcf , the water content is 38% and the degree of saturation is 66%, then what is the value of Δp ?

- A) 120kN/m^2 B) 200kN/m^2 C) 160kN/m^2 D) 180kN/m^2

The Answer is B

Step 1: Settlement for soil can be calculated based on the equation shown below,

$$\Delta H = \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_c \log \frac{p_0 + \Delta p}{p_c} \right]$$

From this problem, we can know everything except for the initial voids ratio and Δp . We need to calculate it first.

Step 2: Equation of void ratio of soil is,

$$e = \frac{V_v}{V_s}$$

To determine the water content from voids ratio, we need to determine the voids ratio equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{V_s}{S}} = \frac{\frac{wW}{\gamma_w S}}{\frac{V_s}{S}} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Step 3: To get the value of water content, we can substitute all known value into the equation above and we have,

$$e = \gamma_d \frac{w}{\gamma_w S} = (110 pcf) \frac{38\%}{(62.5 ppcf)(66\%)} = 1$$

Step 4: Based on the equation, we have the following as,

$$\begin{aligned} \frac{H_0}{1 + e_0} \left[C_R \log \frac{p_c}{p_0} + C_c \log \frac{p_0 + \Delta p}{p_c} \right] \\ = \frac{3m}{1 + 1} \left[(0.1) \log \frac{400 kN/m^2}{300 kN/m^2} + (0.6) \log \frac{300 kN/m^2 + \Delta p}{400 kN/m^2} \right] \\ = 11 cm \end{aligned}$$

$$\Delta p = 200 kN/m^2 \sim 200 kN/m^2$$

121. A highway embankment constructed with a sandy clay soil has a height $H = 20$ ft. The clay soil has a cohesion $c = 600$ lbf/ft², and friction angle $\phi = 20^\circ$. It is determined that the embankment slope will fail along a plane that has an angle $\alpha_s = 15^\circ$ from the horizontal surface, as shown in the figure below. The weight of the sliding soil wedge is 42,000 lbf per foot long of the embankment. Determine the available shearing resistance along the slip plane per per foot long of the embankment.

- A) $1,870 \frac{\text{lbf}}{\text{ft}}$ B) $10,870 \frac{\text{lbf}}{\text{ft}}$ C) $61,130 \frac{\text{lbf}}{\text{ft}}$ D) $20,078 \frac{\text{lbf}}{\text{ft}}$

The Answer is C**Step 1:**

The mobilized shear force along the slip plane is $T_{FF} = cL_s + W_M \cos(\alpha_s) \tan(\phi)$.

Step 2:

From the given information, it is known $W_M = 42,000 \text{ lbf/ft}$, $\alpha_s = 15^\circ$, $\phi = 20^\circ$, $c = 600 \text{ lbf/ft}^2$, and $H = 20 \text{ ft}$. Since

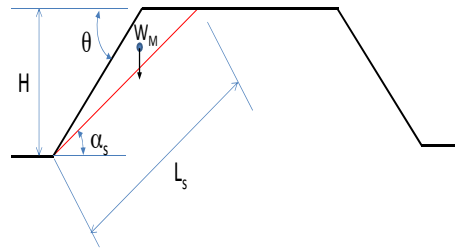
$$L_s = H/\sin(\alpha_s)$$

$$T_{FF} = cL_s + W_M \cos(\alpha_s) \tan(\phi) = cH/\sin(\alpha_s) + W_M \cos(\alpha_s) \tan(\phi)$$

$$= \frac{\left(600 \frac{\text{lbf}}{\text{ft}^2}\right) (20 \text{ ft})}{\sin(15^\circ)} + \left(42,000 \frac{\text{lbf}}{\text{ft}}\right) \cos(15^\circ) \tan(20^\circ)$$

$$\cong 61,130 \frac{\text{lbf}}{\text{ft}}$$

122. A highway embankment constructed with a clay soil has a height $H = 8$ m, and a slope angle $\theta = 25^\circ$. The clay soil has a cohesion $c = 60$ kPa, unit weight $\gamma = 19$ kN/m³, and friction angle $\phi = 30^\circ$. Determine the factor of safety against slope instability, FS, for a plane that has an angle $\alpha_s = 15^\circ$ from the horizontal surface.



- A) 1.4 B) 5.2 C) 6.9 D) 9.6

The Answer is D**Step 1:**

The top width of the sliding wedge is $w = \frac{H}{\tan(\alpha_s)} - \frac{H}{\tan(\theta)} = \frac{(8 \text{ m})}{\tan(15^\circ)} - \frac{(8 \text{ m})}{\tan(25^\circ)} = 12.70 \text{ m}$

Step 2:

The weight of the sliding soil wedge is

$$W_M = \gamma A = \gamma \frac{1}{2} w H = \left(19 \frac{\text{kN}}{\text{m}^3} \right) (0.5)(12.70 \text{ m})(8 \text{ m}) = 965.2 \text{ kN/m}$$

Step 3:

Since the length of the slip surface is $L_s = H/\sin(\alpha_s)$, the shear resistance along the slip plane is

$$\begin{aligned} T_{FF} &= cL_s + W_M \cos(\alpha_s) \tan(\phi) = cH/\sin(\alpha_s) + W_M \cos(\alpha_s) \tan(\phi) \\ &= \frac{(60 \text{ kPa})(8 \text{ m})}{\sin(15^\circ)} + (965.2 \text{ kN/m}) \cos(15^\circ) \tan(30^\circ) \\ &\cong 2393 \text{ kN/m} \end{aligned}$$

Step 4:

The mobilized shear force along the slip plane is

$$T_{MOB} = W_M \sin(\alpha_s) = (965.2 \text{ kN/m}) \sin(15^\circ) = 250 \text{ kN/m}$$

Step 5:

The factor of safety against slope instability, FS is

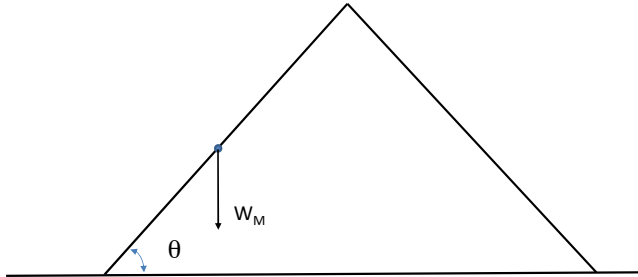
$$FS = \frac{T_{FF}}{T_{MOB}} = \frac{\left(\frac{2393 \text{ kN}}{\text{m}} \right)}{250 \frac{\text{kN}}{\text{m}}} \cong 9.6$$

123. A sandy soil is stockpiled on a ground. The soil is cohesionless with a unit weight $\gamma = 20 \text{ kN/m}^3$, and friction angle $\phi = 35^\circ$. What is the maximum slope angle θ the sand cone can reach?

- A) 10° B) 15° C) 20° D) 35°

The Answer is D

When the slope angle θ of the sand cone reaches its maximum, the sand on the slope surface is in equilibrium, $T_{MOB} = T_{FF}$.

**Step 1:**

The mobilized shear force along the slope surface is

$$T_{MOB} = W_M \sin(\theta)$$

The shearing resistance along the slope surface is

$$T_{FF} = cL_s + W_M \cos(\alpha_s) \tan(\phi) = 0 \cdot L_s + W_M \cos(\theta) \tan(\phi)$$

Step 2:

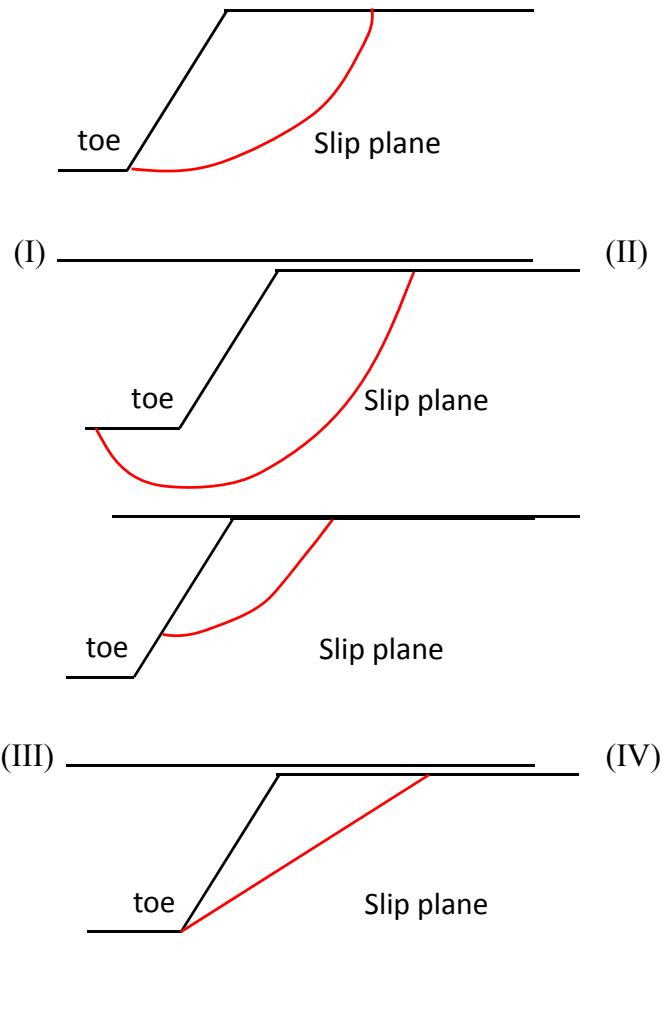
When $T_{MOB} = T_{FF}$, we have

$$W_M \sin(\theta) = W_M \cos(\theta) \tan(\phi)$$

Therefore, $\tan(\phi) = \frac{\sin(\theta)}{\cos(\theta)} = \tan(\theta)$. So

$$\theta = \phi = 35^\circ$$

124. Which of the following slope failure types are impossible?



A) I

B) II

C) III

D) None

The Answer is D

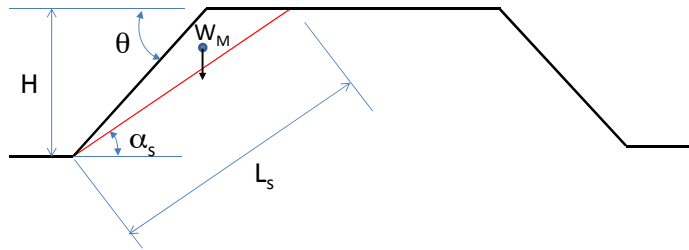
(A) is incorrect. Slope failure in (I) is a type of rotational failure (toe slide) that is common in homogeneous fine-grained soils.

(B) is incorrect. Slope failure in (II) is a type of rotational failure (base failure) that is common in homogeneous fine-grained soils. A soft soil layer resting on a stiff layer of soil is prone to base failure.

(C) is incorrect. Slope failure in (III) is a type of rotational failure (slope slide) that is common in homogeneous fine-grained soils.

(D) is correct. Slope failure in (IV) is a type of translational slide that is common in coarse-grained soils.

125. An embankment constructed with a sandy clay soil has a height $H = 10$ m, and a slope angle θ . The soil has a cohesion $c = 10$ kPa, unit weight $\gamma = 19$ kN/m³, and friction angle $\phi = 20^\circ$. The factor of safety against slope instability, FS, for a plane that has an angle $\alpha_s = 25^\circ$ from the horizontal surface, is 2. Determine the slope angle θ .



- A) 10° B) 15° C) 20° D) 31°

The Answer is D**Step 1:**

The top width of the sliding wedge is

$$w = \frac{H}{\tan(\alpha_s)} - \frac{H}{\tan(\theta)} = \frac{(10 \text{ m})}{\tan(25^\circ)} - \frac{(10 \text{ m})}{\tan(\theta)}$$

Step 2:

The weight of the sliding soil wedge is

$$\begin{aligned} W_M &= \gamma A = \gamma \frac{1}{2} wH = \left(19 \frac{\text{kN}}{\text{m}^3}\right) (0.5) \left(\frac{(10 \text{ m})}{\tan(25^\circ)} - \frac{(10 \text{ m})}{\tan(\theta)}\right) (10 \text{ m}) \\ &= \left(950 \frac{\text{kN}}{\text{m}}\right) \left(\frac{1}{\tan(25^\circ)} - \frac{1}{\tan(\theta)}\right) \end{aligned}$$

Step 3:

Since the length of the slip surface is $L_s = \frac{H}{\sin(\alpha_s)} = \frac{10 \text{ m}}{\sin(25^\circ)} = 23.66 \text{ m}$, the shear resistance along the slip plane is

$$\begin{aligned} T_{FF} &= cL_s + W_M \cos(\alpha_s) \tan(\phi) \\ &= (10 \text{ kPa})(23.66 \text{ m}) \\ &\quad + \left(\left(950 \frac{\text{kN}}{\text{m}}\right) \left(\frac{1}{\tan(25^\circ)} - \frac{1}{\tan(\theta)}\right)\right) \cos(25^\circ) \tan(20^\circ) \\ &= 236.6 \frac{\text{kN}}{\text{m}} + 313.38 \frac{\text{kN}}{\text{m}} \left(\frac{1}{\tan(25^\circ)} - \frac{1}{\tan(\theta)}\right) \end{aligned}$$

Step 4:

The mobilized shear force along the slip plane is

$$T_{MOB} = W_M \sin(\alpha_s) = \left(\left(950 \frac{\text{kN}}{\text{m}}\right) \left(\frac{1}{\tan(25^\circ)} - \frac{1}{\tan(\theta)}\right)\right) \sin(25^\circ)$$

Step 5:

The factor of safety against slope instability, FS is

$$FS = \frac{T_{FF}}{T_{MOB}} = \frac{236.6 \frac{\text{kN}}{\text{m}} + 313.38 \frac{\text{kN}}{\text{m}} \left(\frac{1}{\tan(25^\circ)} - \frac{1}{\tan(\theta)} \right)}{\left(\left(950 \frac{\text{kN}}{\text{m}} \right) \left(\frac{1}{\tan(25^\circ)} - \frac{1}{\tan(\theta)} \right) \right) \sin(25^\circ)} = 2$$

It is obtained that

$$\frac{1}{\tan(25^\circ)} - \frac{1}{\tan(\theta)} = 0.4833$$

$$\tan(\theta) = 0.6020 \cong 31^\circ$$

126. A highway is to be built in a region where soft soil exists in the ground, which will cause excessive settlement of the highway embankment. Which of the following methods may be used to reduce the settlement of the embankment during the service life of the highway?

- (I) Precompression of the soft soil
- (II) Installation of vertical drains in the soil under a surcharge load
- (III) Stabilization of the soil using admixtures
- (IV) Injection of grout into the ground

A) I and II B) III C) II and IV D) All of the above

The Answer is D

Soft soils are soils that have high porosity and water content, and under load, will produce large settlements over a very long time span. Soft soils can be improved by a variety of methods, such as precompression, vertical drains, in-situ densification, grouting, stabilization using admixtures, and reinforcement with geosynthetics.

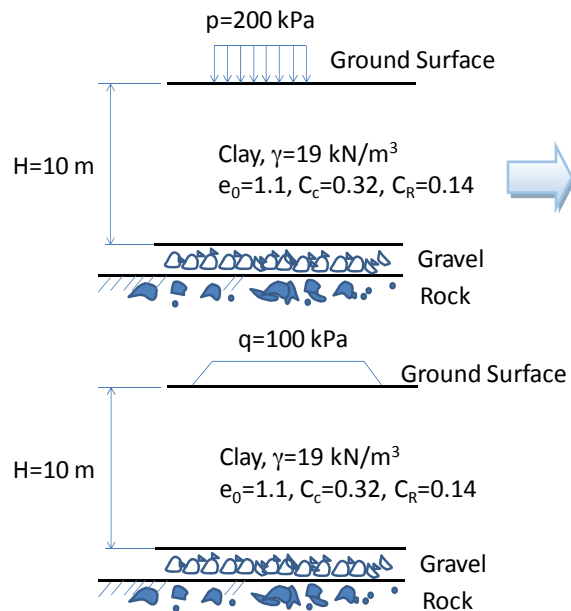
(A) is incorrect. (III) and (IV) are also correct. Stabilization of soil using admixtures can increase the strength and stiffness of the soil. Injection of cementitious or chemical grout into the ground may also increase the strength and stiffness of the soil.

(B) is incorrect. (I), (II) and (IV) are also correct. Precompression of the soft soil using a surcharge will consolidate the soil to a level that the future consolidation of the soil under highway loads is limited to an acceptable value.

(C) is incorrect. (I) and (II) are also correct.

(D) is correct.

127. A highway embankment is to be built on a deposit of clay with a depth of 10 m. The properties of the clay are shown in the figure below. Before the embankment is constructed, a uniformly distributed surcharge load $p = 200$ kPa is applied on the ground for an extended period of time. After the consolidation of clay under the surcharge is completed, the embankment is built. Assume the embankment pressure on the clay can be treated as a uniformly distributed $q = 100$ kPa. Determine the settlement of embankment due to the primary consolidation of the clay layer.



A) 0.120 m

B) 0.208 m

C) 0.342 m

D) 0.652 m

The Answer is B

The surcharge load pre-consolidates the clay soil. After the surcharge is removed, the settlement of the ground recovers to some extent. When the embankment pressure is applied, the clay is recompressed, following the reversed strain-pressure path of settlement recovery, to the previously reached maximum settlement, and then follow the normal consolidation curve to a final settlement.

Step 1:

In this problem, since the embankment pressure q is less than the preconsolidation pressure p , the compression of soil under embankment pressure is in the range of recompression. Therefore, the recompression index C_R should be used for calculation.

After the surcharge load is removed and before the embankment is constructed, the effective vertical stress at center of the clay layer is

$$\sigma_0 = \frac{1}{2} H \gamma = \frac{1}{2} (10 \text{ m}) \left(19 \frac{\text{kN}}{\text{m}^3} \right) = 95 \text{ kPa}$$

Step 3:

After the embankment is constructed, the effective vertical stress at center of the clay layer is

$$\sigma_1 = \sigma_0 + q = 95 \text{ kPa} + 100 \text{ kPa} = 195 \text{ kPa}$$

Step 4:

The settlement of the embankment is

$$\begin{aligned} S &= H \varepsilon_v = H \frac{\Delta e_1}{1 + e_0} = H \frac{C_R \log_{10} \left(\frac{\sigma_1}{\sigma_0} \right)}{1 + e_0} \\ &= (10 \text{ m}) \frac{(0.14) \log_{10} \left(\frac{195 \text{ kPa}}{95 \text{ kPa}} \right)}{1 + 1.1} \\ &= 0.208 \text{ m} \end{aligned}$$

128. A field density test was performed on compacted fill in accordance with ASTM D 1556 Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method. The following is a summary of the test data:

Weight of wet soil extracted from the sand cone hole	6.438 lb
Weight of sand needed to fill the hole (funnel correction already applied)	4.125 lb
The bulk density of the sand used in the sand cone apparatus	82.4 pcf

Water Content Test:

Empty cup	0.462lb
Cup plus wet soil	1.832lb
Cup plus dry soil	1.720lb

The laboratory maximum dry density performed on the same soil is equal to 130 pcf. Based on this data, the relative compaction of the fill is most nearly:

- A) 99% B) 95% C) 91% D) 63

The Answer is C

$$\text{Volume of hole} = 4.125 \text{ lb} / 82.4 \text{ pcf} = 0.0501 \text{ cf}$$

$$\gamma_t = 6.438 \text{ lb} / 0.0501 \text{ cf} = 128.6 \text{ pcf}$$

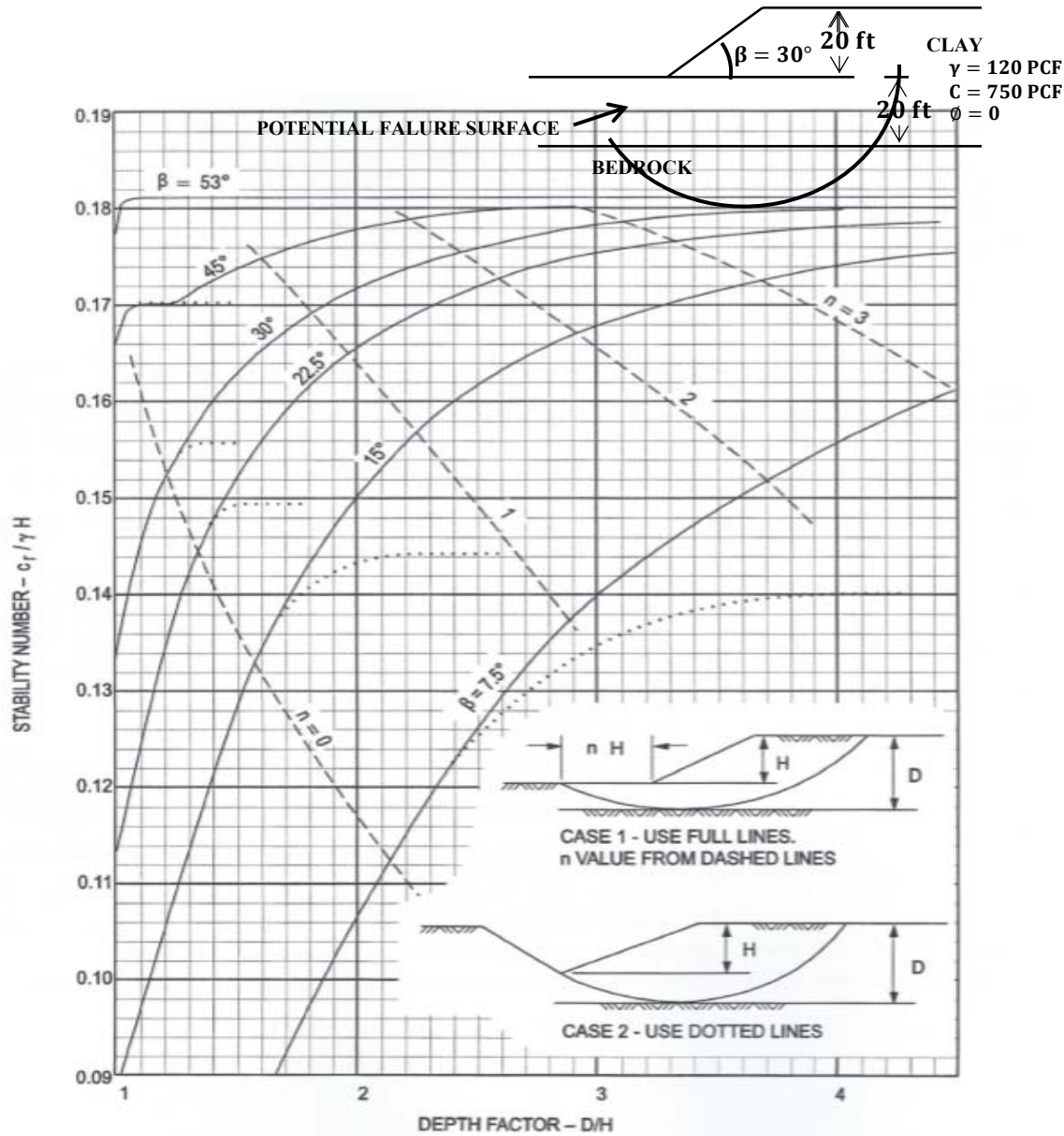
$$W = (1.832 - 1.720) / (1.720 - 0.462) = 0.112 / 1.258 = 0.0890$$

$$\gamma_d = 128.6 / (1 + 0.0890) = 118.0$$

$$\text{Relative compaction} = \gamma_d / \gamma_{d\max} = 118.0 / 130.0 = 0.908 \text{ or } 91\%$$

129. A temporary slope will be excavated to the dimensions shown in the figure. Laboratory testing has yielded the geotechnical parameters shown in the chart. The safety factor for the failure surface shown is most nearly:

- A) 1.4 B) 1.6 C) 1.8 D) 2.0



From *Fundamentals of Geotechnical Analysis*, Dunn, Anderson, Rieffer, John Wiley and Sons, New York, 1980. Used by permission.

The Answer is C

$$D = 20$$

$$D/H = 2$$

$$\beta = 30$$

$$FS = \frac{C}{C_r} = \frac{750}{413} = 1.82$$

$$\left. \begin{array}{l} D = 20 \\ D/H = 2 \\ \beta = 30 \end{array} \right\} \begin{array}{l} \text{Stability number} = 0.172 = \frac{C_r}{\gamma H} = \frac{C_r}{120 \times 20} \\ C_r \cong 413 \end{array}$$

130. During a standard penetration test (SPT), unusually low blow counts are encountered in a foundation expected to be medium-dense to dense sand. This is an indication that the following condition may be present:

- A) The sampler drive shoe is badly damaged or worn due to too many drivings to refusal.
- B) Cobbles are encountered.
- C) The sampler drive shoe is plugged.
- D) The groundwater in the borehole is much lower than in situ conditions immediately outside the bore hole.

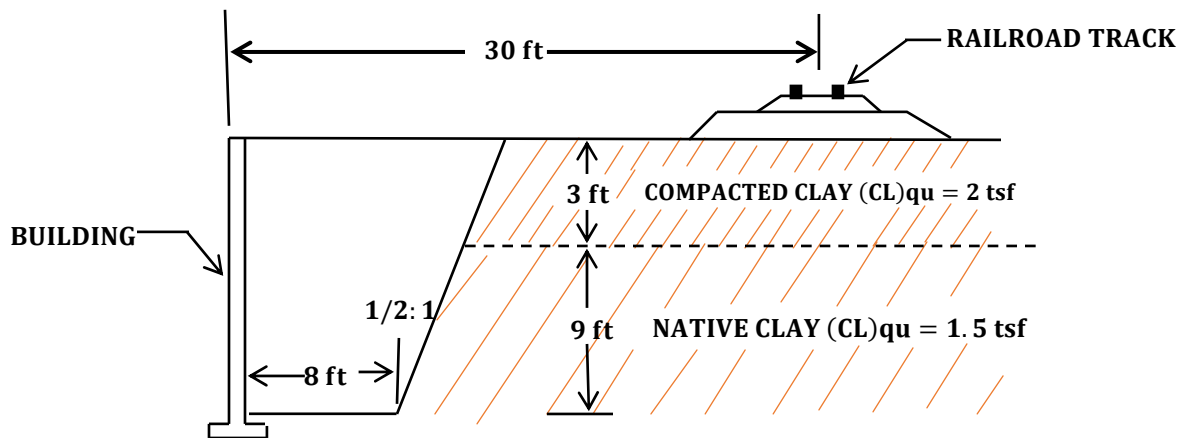
The Answer is D

Reference: Cheney and Chassie, Soils and Foundations Workshop Manual, 2nd ed., National Highway Institute, 1993, p. 23.

Groundwater coming up into the casing or hollow stem augers can cause the test zone to become "quick." This loosening will reduce the strength of the soil

131. The excavation shown in the figure is required to repair a broken fiber optic line. The excavation is to be open for 18 hours. A train will pass the excavation area each hour. As the competent person on-site, you must determine which of the following statements is most correct:

- A) OSHA Soil Type A--Excavation is safe for entry.
- B) OSHA Soil Type A--Excavation should be sloped at 3/4: 1 or flatter prior to entry.
- C) OSHA Soil Type B over A-- Excavation should be sloped at 3/4: 1 in upper 3 ft and 3/4: 1 in lower 9 ft prior to entry.
- D) OSHA Soil Type B-- Excavation should be sloped at 1:1 or flatter prior to entry.



The Answer is D

Reference: OSHA 1926, Subpart P, Appendix A and B, 2006, p. 334.

Per definition for Soil Type A Exception (ii), the soil cannot be classified as "A" as it will be subject to vibration from the railroad. Therefore, per definition of Soil Type B (iv), the soil should be classified as "B." Per Table B-1, Type B soils should be sloped at 1:1 or flatter. /

132. A soil layer is given. The thickness of this soil layer is unknown. It is known that the final voids ratio is 0.73 after ultimate consolidation settlement and the initial porosity is 0.5. Determine the thickness of this soil layer if the approximate settlement is 0.2m and average degree of consolidation is 50%.

- A) 2m B) 3m C) 4m D) 5m

The Answer is B

Step 1: First we need to calculate the initial voids ratio. It can be obtained by,

$$e_0 = \frac{n_0}{1 - n_0} = \frac{0.5}{1 - 0.5} = 1$$

Step 2: Approximate settlement can be calculated based on the equation as shown below,

$$S_T = U_{AV} S_{ULT} = U_{AV} \frac{e_0 - e_1}{1 + e_0} H$$

Based on this equation, only the thickness of soil layer is unknown. Then we can solve it by,

$$U_{AV} \frac{e_0 - e_1}{1 + e_0} H = (50\%) \frac{1 - 0.73}{1 + 1} H = 0.2m$$

$$H = 3m \sim 3m$$

133. A retaining wall is shown in the following figure. The soil profile is also shown in the figure. According to Rankine active state, calculate the moment the soil produces toward retaining wall.

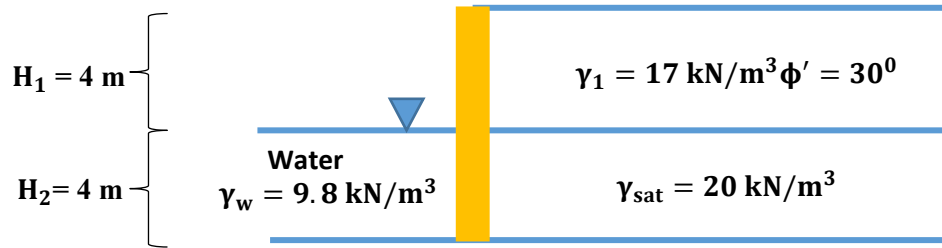


Fig. 19

- A) 372.4 kN-m
C) 542.5 kN-m

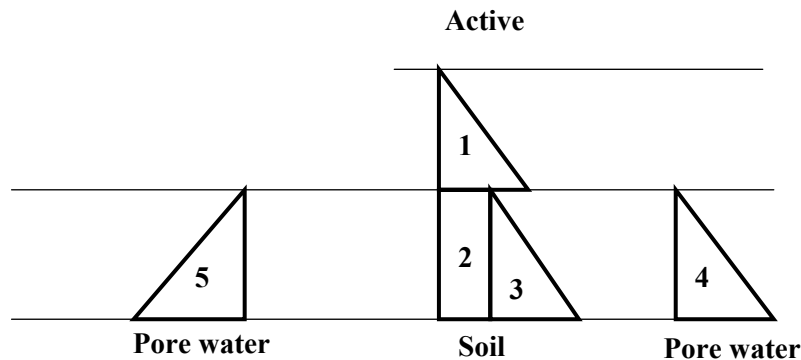
- B) 456.9 kN-m
D) 435.1 kN-m

The Answer is B

Step 1: Rankine active earth pressure coefficient,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: The lateral earth pressure distribution with depth is shown:



Step 3: Notice that area 4 and area 5 are the same with different direction forces toward retaining wall, there is no need to calculate them.

Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \left(\frac{1}{2} \right) \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (4 \text{ m})^2 = 45.3 \text{ kN}$$

$$\text{Location to base, } Z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (4 \text{ m}) + 4 \text{ m} = 5.3 \text{ m}$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (4 \text{ m}) (4 \text{ m}) = 90.7 \text{ kN}$$

$$\text{Location to base, } Z_2 = \frac{1}{2} H_2 = \frac{1}{2} (4 \text{ m}) = 2 \text{ m}$$

Lateral force for Area 3,

$$P_3 = \frac{1}{2} K_a (\gamma_{\text{sat}} - \gamma_w) H_2^2$$

$$= \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) (20 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (4 \text{ m})^2 = 27.2 \text{ kN}$$

$$\text{Location to base, } Z_3 = \frac{1}{3} H_2 = \frac{1}{3} (4 \text{ m}) = 1.3 \text{ m}$$

Step 4: Summing up moments at the base:

$$\begin{aligned} M &= P_1 Z_1 + P_2 Z_2 + P_3 Z_3 \\ &= (45.3 \text{ kN})(5.3 \text{ m}) + (90.7 \text{ kN})(2 \text{ m}) + (27.2 \text{ kN})(1.3 \text{ m}) \\ &= 456.9 \text{ kN} \end{aligned}$$

134. There is a frictionless wall shown in the following figure. Assume that the soil will slip along the dash line and the angle is 60° . Calculate the active lateral force per unit length.

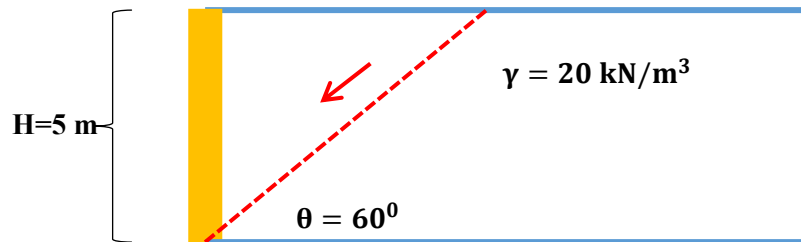


Fig. 20

A) 73.3 kN

B) 83.3 kN

C) 68.8 kN

D) 85.6 kN

The Answer is B

Step 1: Rankine active earth pressure coefficient can be given by,

$$K_a = \tan^2(90^\circ - \theta) = \tan^2(90^\circ - 60^\circ) = \frac{1}{3}$$

Step 2: The vertical stress at the surface for soil,

$$\sigma_v' = 0$$

The vertical stress at the base for soil,

$$\sigma_v' = \gamma' H = (20 \text{ kN/m}^3)(5 \text{ m}) = 100 \text{ kPa}$$

Step 3: The lateral stress at base for soil,

$$(\sigma_h')_a = K_a \gamma' z = K_a \sigma_v' = \frac{1}{3} (100 \text{ kPa}) = 33.3 \text{ kPa}$$

Step 4: The lateral force due to soil,

$$P_a = \frac{1}{2} (\sigma_h')_a H = \frac{1}{2} (33.3 \text{ kPa})(5 \text{ m}) = 83.3 \text{ kN}$$

135. There is a frictionless wall shown in the following figure. Assume that the retaining wall push the soil to slip along the dash line and the angle is 30° . Calculate the passive lateral force per unit length.

- A) 700 kN B) 820 kN C) 680 kN D) 750 kN

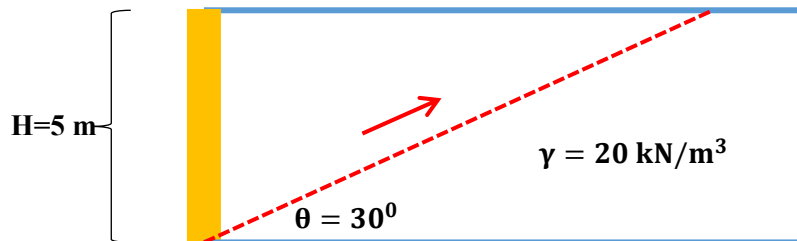


Fig. 21

The Answer is B

Step 1: Rankine passive earth pressure coefficient can be given by,

$$K_p = \tan^2(90^\circ - \theta) = \tan^2(90^\circ - 30^\circ) = 3$$

Step 2: The vertical stress at the surface for soil,

$$\sigma_v' = 0$$

The vertical stress at the base for soil,

$$\sigma_v' = \gamma' H = (20 \text{ kN/m}^3)(5 \text{ m}) = 100 \text{ kPa}$$

Step 3: The lateral stress at base for soil,

$$(\sigma_h')_p = K_p \gamma' z = K_p \sigma_v' = 3(100 \text{ kPa}) = 300 \text{ kPa}$$

Step 4: The lateral force due to soil,

$$P_p = \frac{1}{2} (\sigma_h')_p H = \frac{1}{2} (300 \text{ kPa})(5 \text{ m}) = 750 \text{ kN}$$

136. There is a frictionless wall shown in the following figure. Assume that the soil will slip along the dash line and the angle is 60° . Calculate moment soil produce to retaining wall.

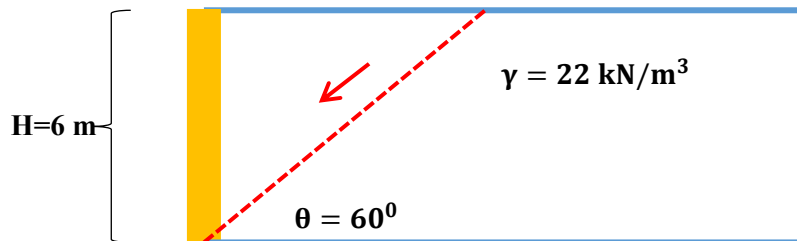


Fig. 22

- A) 328 kN-m
B) 264 kN-m
C) 196 kN-m
D) 236 kN-m

The Answer is B

Step 1: Rankine active earth pressure coefficient can be given by,

$$K_a = \tan^2(90^\circ - \theta) = \tan^2(90^\circ - 60^\circ) = \frac{1}{3}$$

Step 2: The vertical stress at the surface for soil,

$$\sigma_v' = 0$$

The vertical stress at the base for soil,

$$\sigma_v' = \gamma' H = (22 \text{ kN/m}^3)(6 \text{ m}) = 132 \text{ kPa}$$

Step 3: The lateral stress at base for soil,

$$(\sigma_h')_a = K_a \gamma' z = K_a \sigma_v' = \frac{1}{3}(132 \text{ kPa}) = 44 \text{ kPa}$$

Step 4: The lateral force due to soil,

$$P_a = \frac{1}{2}(\sigma_h')_a H = \frac{1}{2}(44 \text{ kPa})(6 \text{ m}) = 132 \text{ kN}$$

$$\text{Location to base, } Z = \frac{1}{3}H = \frac{1}{3}(6 \text{ m}) = 2 \text{ m}$$

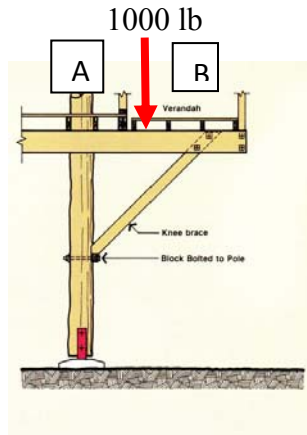
Step 5: Summing up moments at the base:

$$M = P_a Z = (132 \text{ kN})(2 \text{ m}) = 264 \text{ kN} - \text{m}$$

Part 4 : Structure

206 problems

- 1) Find the reaction in the knee bracing (diagonal member)? The load is in the middle of AB, where the length of AB is 10 ft., and the knee bracing has the 45 degree inclination.



- A) 500 lb
B) 707 lb
C) 1000 lb
D) 353 lb

Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(8-10)

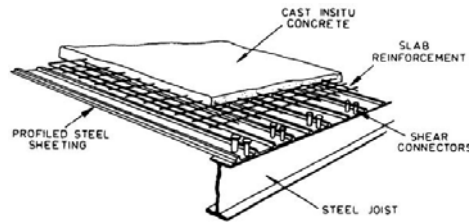
Step 1: AB is a simple beam, so determine the reaction at both end.

$$R_A = R_B = 1000/2 = 500 \text{ lb}$$

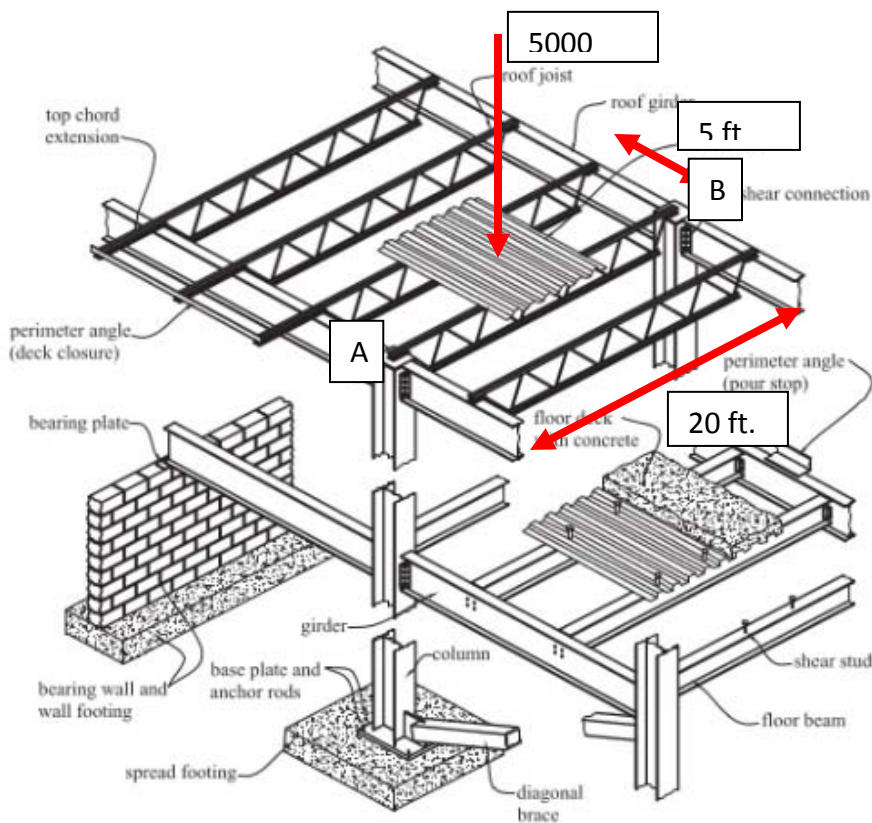
Step 2: The diagonal member has 45 degree, so the compression in the diagonal member is equal to:

$$F_{(\text{diagonal})} = R_B / \cos 45^\circ = 500 / \cos 45^\circ = 707.1 \text{ lb}$$

- 2) Find the total reaction forces of the Joist AB at point “A” if the amount of the concentrated live load is given equal to 5000 lbs. and the metal deck is used for the flooring system (i.e., dead load). Consider the amount of 4” as the average thickness for the concrete in the metal deck and 18 gage for the metal deck sheeting. Dead load is uniformly distributed on top of the joists. Neglect the dead weight of the joists and column. Use ASCE 7 code of practice to find the weight of metal deck and concrete.



Typical detail of metal deck, 4" concrete with steel sheeting (18 gage)



- A) 1250 lbs. B) 2650 lbs.
C) 3900 lbs. D) 4200 lbs.

Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.1, page 51-(1-6) and ASCE 7-10, Commentary C3, page 397.

Step 1: The concentrated live load will be divided by 2 and the reaction goes to the joist. Then half of that reaction will transfer to the column.

So:

$$P (\text{live}) = (5000/2)/2 = 1250 \text{ lbs.}$$

Step 2: Estimate the dead load of the metal deck in psf. According to the ASCE 7-10, table C3-1 page 399, the weight of 18 gage metal deck is equal to: 3 Psf

Table C3-1 Minimum Design Dead Loads^a

Component	Load (psf)
CEILINGS	
Acoustical fiber board	1
Gypsum board (per 1/8-in. thickness)	0.55
Mechanical duct allowance	4
Plaster on tile or concrete	5
Plaster on wood lath	8
Suspended steel channel system	2
Suspended metal lath and cement plaster	15
Suspended metal lath and gypsum plaster	10
Wood furring suspension system	2.5
COVERINGS, ROOF, AND WALL	
Asbestos-cement shingles	4
Asphalt shingles	2
Cement tile	16
Clay tile (for mortar add 10 psf)	
Book tile, 2-in.	12
Book tile, 3-in.	20
Ludowici	10
Roman	12
Spanish	19
Composition:	
Three-ply ready roofing	1
Four-ply felt and gravel	5.5
Five-ply felt and gravel	6
Copper or tin	1
Corrugated asbestos-cement roofing	4
Deck, metal, 20 gage	2.5
Deck, metal, 18 gage	3
Decking, 2-in. wood (Douglas fir)	5

The thickness of concrete is given equal to 4" so according to the ASCE 7-10, C3 the density of concrete is considered as: 150 pcf

Bronze	552	Masonry, ashlar stone	
Cast-stone masonry (cement, stone, sand)	144	Granite	165
Cement, portland, loose	90	Limestone, crystalline	165
Ceramic tile	150	Limestone, oolitic	135
Charcoal	12	Marble	173
Cinder fill	57	Sandstone	144
Cinders, dry, in bulk	45	Masonry, brick	
Coal		Hard (low absorption)	130
Anthracite, piled	52	Medium (medium absorption)	115
Bituminous, piled	47	Soft (high absorption)	100
Lignite, piled	47	Masonry, concrete ^a	
Peat, dry, piled	23	Lightweight units	105
Concrete, plain		Medium weight units	125
Cinder	108	Normal weight units	135
Expanded-slag aggregate	100	Masonry grout	140
Haydite (burned-clay aggregate)	90	Masonry, rubble stone	
Slag	132	Granite	153
Stone (including gravel)	144	Limestone, crystalline	147
Vermiculite and perlite aggregate, nonload-bearing	25-50	Limestone, oolitic	138
Other light aggregate, load-bearing	70-105	Marble	156
Concrete, reinforced		Sandstone	137
Cinder	111	Mortar, cement or lime	130
Slag	138	Particleboard	45
Stone (including gravel)	150	Plywood	36
Copper	556	Riprap (not submerged)	
Cork, compressed	14	Limestone	83
Earth (not submerged)		Sandstone	90
Clay, dry	63	Sand	
Clay, damp	110	Clean and dry	90
Clay and gravel, dry	100	River, dry	106
Silt, moist, loose	78	Slag	
		Bank	70

Dead weight of concrete = $4/12 * 150 = 50$ psf

Total dead weight = metal sheeting and concrete = $50+3 = 53$ psf

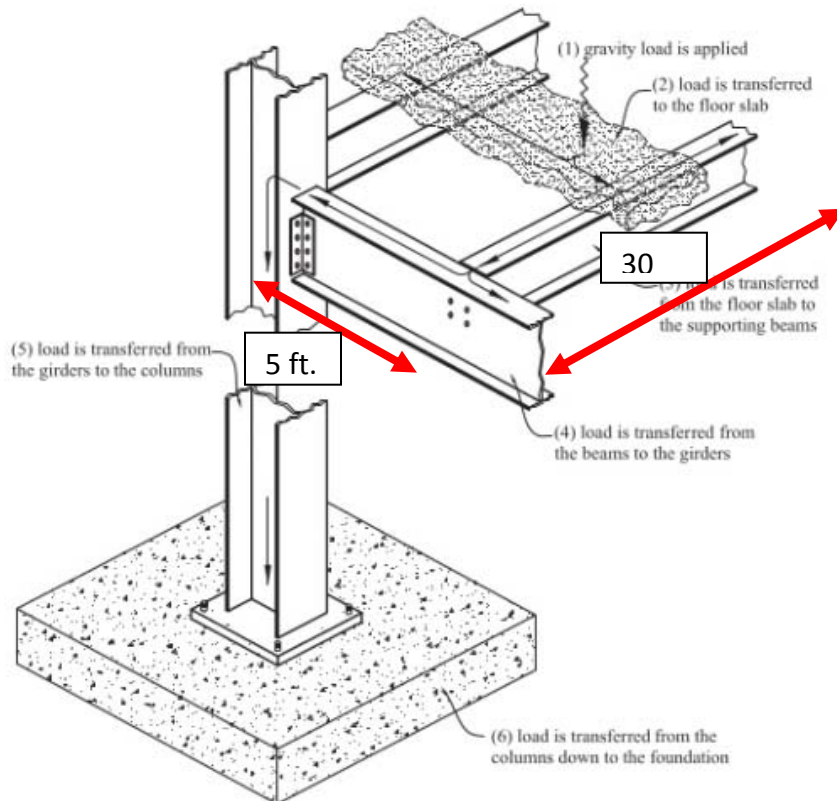
Step 3: find the dead load reaction, dead load act like a uniformly distributed load psf on whole structure, so the dead load on the joist = tributary width of the joist * dead load (psf) =

$5'(\text{distance between joists}) * 53 = 265$ lb/ft

Reaction of the joist at point "A" = $265 * 20 / 2 = 2650$ lbs.

Reaction forces at "A" = Dead + Live = $2650+1250 = 3900$ lbs.

- 3) Find the bending moment at the end of the cantilever beam if the composite flooring system is used with 4" concrete thickness and architectural finish floor is made of 1" granite stone over 1" of the cement mortar. The building will be used to carry the live load of the classroom at a university. Neglect the dead weight of the joists and column. Use ASCE 7 code of practice to find the weight of materials. (Consider that whole span covered by the composite slab.)



- A) 13.983.75 lb-ft B) 20358.75 lb-ft
C) 6375.53 lb-ft D) 28241.75 lb-ft

Answer is B

Photo captured from: Structural steel design, a practice oriented approach, by Abi Aghayer and Jason Vigil, published by Pearson educational Inc., 2009.

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.1, page 51-(1-6) and ASCE 7-10, Commentary C3, page 397.

Step 1: find the dead load on the flooring system

No.	Material	Thickness	Density	Weight
1	Concrete slab in Composite system	4" = 4/12 = 0.33ft	150 pcf	50 psf
2	Granite stone	1" = 1/12	165 pcf Masonry, ashlar stone → Granite 165 Limestone, crystalline 165 Limestone, oolitic 135 Marble 173	13.75 psf
3	Cement mortar	1" = 1/12	130 pcf Limestone, oolitic 138 Marble 156 Sandstone 137 → Mortar, cement or lime 130 Particleboard 45	10.83 psf
4	Total dead load			74.58 psf

Step 2: Live load shall be estimated from the following table in ASCE 7-10:

Table C4-2 Typical Live Load Statistics

Occupancy or Use	Survey Load		Transient Load		Temporal Constants			Mean Maximum Load ^a lb/ft ² (kN/m ²)
	m _s lb/ft ² (kN/m ²)	σ _s ^a lb/ft ² (kN/m ²)	m _t ^a lb/ft ² (kN/m ²)	σ _t ^a lb/ft ² (kN/m ²)	τ _s ^b (years)	v _e ^c (per year)	T ^d (years)	
Office buildings: offices	10.9 (0.52)	5.9 (0.28)	8.0 (0.38)	8.2 (0.39)	8	1	50	55 (2.63)
Residential								
renter occupied	6.0 (0.29)	2.6 (0.12)	6.0 (0.29)	6.6 (0.32)	2	1	50	36 (1.72)
owner occupied	6.0 (0.29)	2.6 (0.12)	6.0 (0.29)	6.6 (0.32)	10	1	50	38 (1.82)
Hotels: guest rooms	4.5 (0.22)	1.2 (0.06)	6.0 (0.29)	5.8 (0.28)	5	20	50	46 (2.2)
Schools: classrooms	12.0 (0.57)	2.7 (0.13)	6.9 (0.33)	3.4 (0.16)	1	1	100	34 (1.63)

^aFor 200 ft² (18.58 m²) area, except 1,000 ft² (92.9 m²) for schools.^bDuration of average sustained load occupancy.^cMean rate of occurrence of transient load.^dReference period.

Step 3: Total dead and live load = 34 + 74.58 = 108.58 psf

Step 4: Load on the joist = tributary width * load = 5/2 * 108.58 = 271.45 lb/ft

Reaction of the joist on the cantilever = 271.45*30/2 = 4071.75 lb.

M= reaction of joist * length of the cantilever = 4071.75 * 5 = 20358.75 lb-ft

- 4) Which statement shows a **wrong** meaning of the live load?
- A) Grandstands, stadiums, and similar assembly structures may be subjected to loads caused by crowds, jumping to their feet, or stomping. So, designers are cautioned that the possibility of such loads should be considered if they make a bigger reaction than the described live loads in the codes.
 - B) Elevator loads are changed over the time so a direct 100% impact factor shall be applied to this load to consider the dynamic effects of this load according to ASCE 7-10 and then it is classified as live load.
 - C) Live loads are non-permanent in which variations over time are not rare and classified as variable loads like light weight partitions in the office buildings.
 - D) Live loads are moving and dynamic in nature.

Answer is D

ASCE 7-10, chapter 4, page 13.

“A, B, and C” are correct. Live loads are variable over the time but this is not the meaning of dynamic in their nature. For example, partitions’ placements might be changed over the period of the design life but they are not dynamic or moving load in their nature. In addition, if the ratio of the dynamic load to the structure is low, in some several cases like the elevator, crane, or vehicle moving load the equivalent static load shall be considered as live load and just the impact factor (dynamic load factor) may be applied to that load like choice “B”. So this is not correct to consider live loads as dynamic loads since they are variable loads.

- 5) Which statement shows a **wrong** meaning of the moving load and influence line?
- A) Moving load is the load of vehicles or cranes and they are considered as live load.
 - B) Influence line is a method of dynamic analysis of beams which will show the internal forces coming from moving loads.
 - C) Influence line is a method of incremental static analysis of beams which will show the internal forces coming from moving loads.
 - D) Sometimes, the impact factor (dynamic load factor) applies to the influence line analysis results for the moving loads to show their dynamic effects.

Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.1, page 51-(1-6)

If a moving load happens on a building like the cranes' or vehicles' loads it is possible to use the influence line analysis techniques to estimate the internal forces' effects on structures. The influence line method is a static method of analysis which uses the flexural and shear force analysis procedures but for different points of the beams and for different moving steps.

On the other hand the influence line diagram for bending moment is the locus of maximum bending moment for a certain point in the beam when the moving load passes over the beam.

So, there is no dynamic analysis procedures to get the internal forces. In some cases the impact factor or dynamic load factor may be applied to the results of the analysis to demonstrate the dynamic effects without using a dynamic analysis. So, in fact an equivalent static analysis method is using in this case. So, choice "B" is not correct.

- 6) Which one **is not correct** about the variable loads and dynamic loads?
- A) Live loads are variable loads but they may be static or dynamic in their nature.
 - B) Since dynamic means changes over the time, so all variable loads are dynamic loads.
 - C) Dynamic loads are important when the frequency of changes over the time shows a meaningful ratio to the structure to create the resonance. So, all variable loads are not necessary classified as dynamic loads.
 - D) Dynamic analysis of loads requires a solution of loads, mass and structural stiffness together to find the maximum reactions, but variable loads are just need to be considered like the static loads.

Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.1, page 51-(1-6), ASCE 7-10, Commentary, Table C-4, page 407. And C7.4.7. Page 410.

According to the ASCE 7-10, chapter 2 live load classified as variable loads, but the meaning of variable does not necessary have the meaning of dynamics in nature. For example, changing the position of students in a classroom over the time, changing the position of partitions are some examples of the variation but the frequency of these changes are far from the structural system frequencies, so the dynamic response will not be a question in these cases. Dynamic loads are changing rapidly with a higher frequencies close to the structure. Then the mass, damping and stiffness of the structure will participate in the analysis to find the reactions of loads. That is why in the ASCE 7-10, all live loads are just classifies like the static loads with the certain amount of load in the table and the user should use those loads as the minimum required of the live load for the design purposes.

- 7) Which one is **not** a fact about the dead loads?
- A) Dead loads are not variable loads.
 - B) Walls and partitions are always classified as dead loads.
 - C) Snow loads are classified as live loads.
 - D) Although weight of equipment like HVAC systems are not variable but they are not considered as dead load.

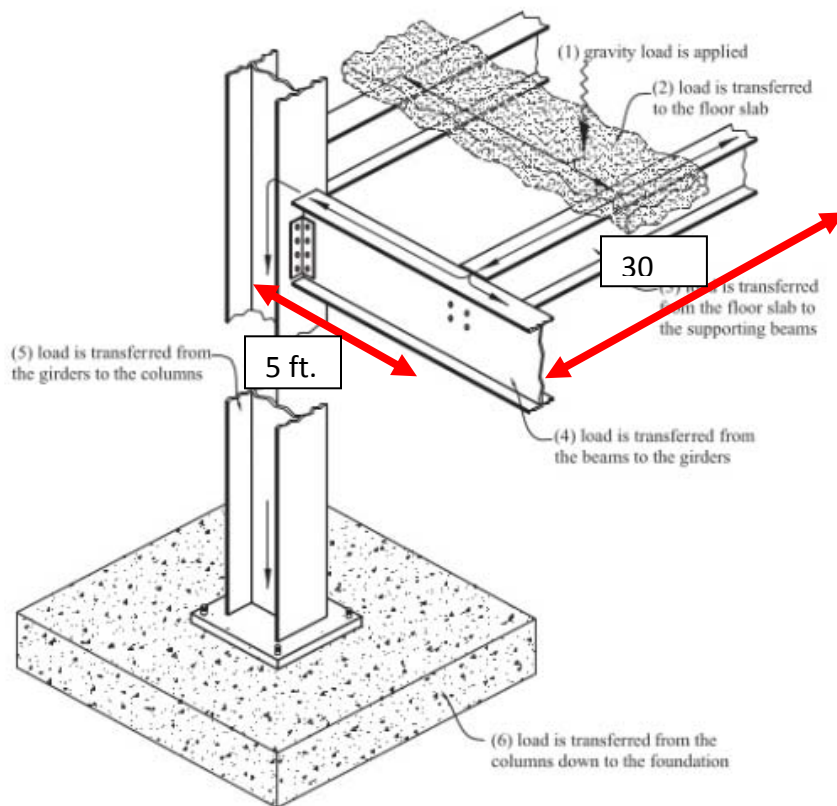
Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.1, page 51-(1-6), ASCE 7-10, Commentary, Table C-4, page 407. And C7.4.7. Page 410.

According to the ASCE 7-10, chapter 1 and 2 live load classified as variable loads, and dead load is considered as a permanent load. So “A” is a fact.

Snow load and equipment loads are also considered as live loads. Weight of walls are considered as dead load as well as the heavy partitions. However, the light weight partitions in the office buildings are considered as live loads. So this is not correct to consider partitions always as the dead load. So “B” is not a fact.

- 8) The dead load of the flooring system consists of joist slab with 3" thickness of concrete and the joists with 2.5 psf of weight. The live load is the truck wheel with 1000 and it can be considered everywhere. Find the reaction at the end of the cantilever beam using your guess for the worst case scenario of the moving load to find the maximum amount of the reaction.



- A) 1000 lbs. B) 1500 lbs.
C) 2500 lbs. D) 3500 lbs.

Answer is C

Photo captured from: Structural steel design, a practice oriented approach, by Abi Aghayer and Jason Vigil, published by Pearson educational Inc., 2009.

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.1, page 51-(1-6) and ASCE 7-10, C4.7.4, page 410.

According to the code, page 410 the garage loads would be considered as 50 psf as a live load and there is not a requirement to consider that load with the influence line. However, since the question tries to examine the understanding about the moving load the answer is estimated as follows:

Step 1: find the dead load on the flooring system

No.	Material	Thickness	Density	Weight
1	Concrete slab in Composite system	3" = 3/12 = 0.25ft	150 pcf	37.5 psf
2	Joists			2.5
4	Total dead load			40 psf

Step 2: Worst case scenario of the Live load shall be investigated with the geometry of the structure. If the live load is placed at any point of the cantilever beam, then the maximum reactions will occur and will be equal to the wheel load, all other positions will give less reaction since the force will be distributed to the other beams. So, $P_{live} = 1000$ lbs.

Step 3: Distribution of the dead load

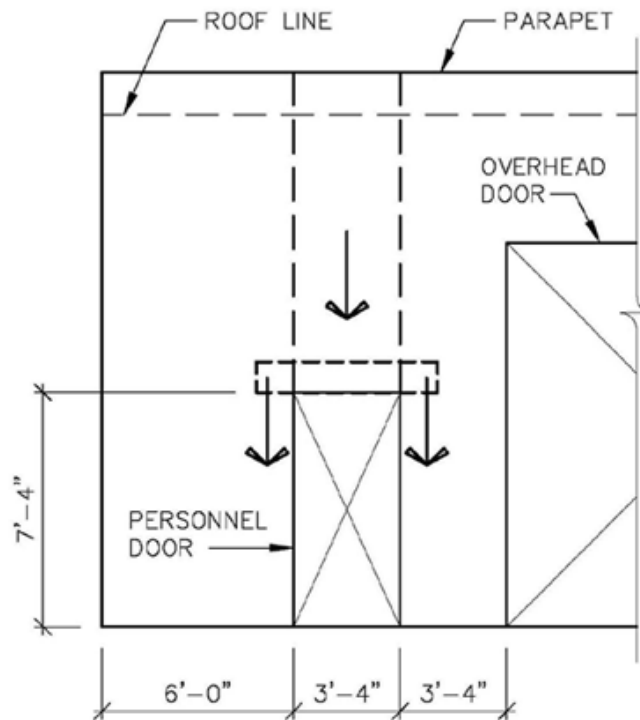
Load on the joist = tributary width * load = $5/2 * 40 = 100$ lb/ft

Reaction of the joist on the cantilever = $100 * 30/2 = 1500$ lbs.

Reaction at the end of cantilever = 1500 lbs.

Step 4: Total dead and live load reaction = $1500 + 1000 = 2500$ lbs.

- 9) Find the total amount of load on the lintel if the thickness of the masonry wall is given equal to 1' and the dead weight of the wall is equal to 30 psf.



- A) 52 lb
B) 325 lb
C) 202 lb
D) 101 lb

Answer is D

The amount of load can be estimated by the following rule if the height of the wall above lintel is bigger than the length of span. Then a triangle with the 45o will shows the weight of the wall on the lintel. (See the below picture.)

The length of span may be considered equal to $1.1 \times$ the clear length of span.

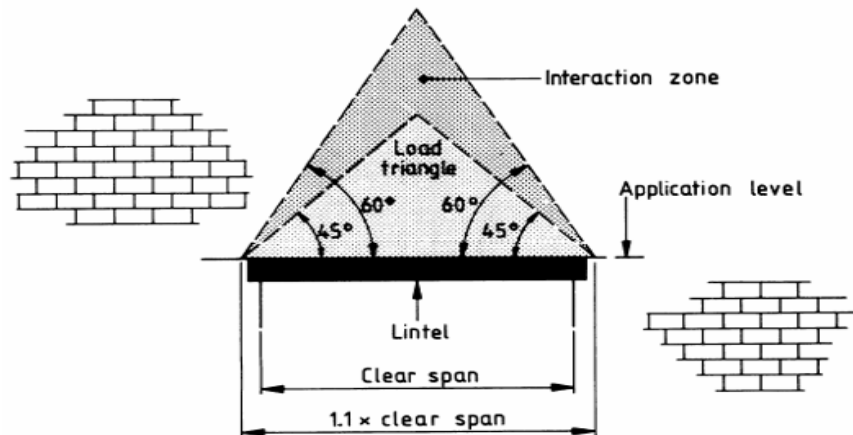
So, Length of lintel = $1.1 \times (3+4/12) = 3.67$ ft

Height of the triangle = $3.67/2$ (for the right angle) = 1.83 ft

Area of the triangle = $1.83 \times 3.67/2 = 3.36$ sf

Weight = $3.36 \times 30 = 100.92$ lbs

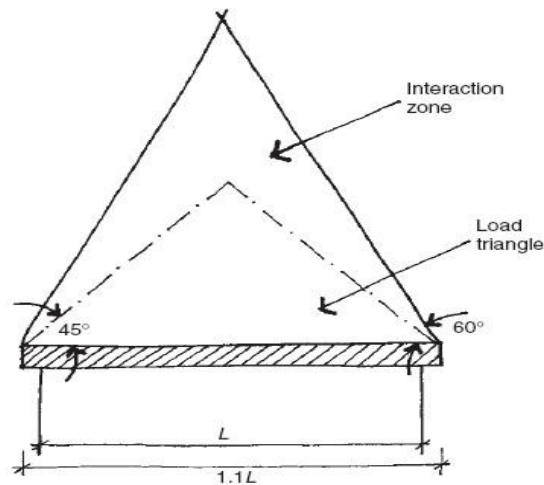
It is not required to consider the thickness because the dead weight has been given.



More reference for study:

Lintel design to BS 5977

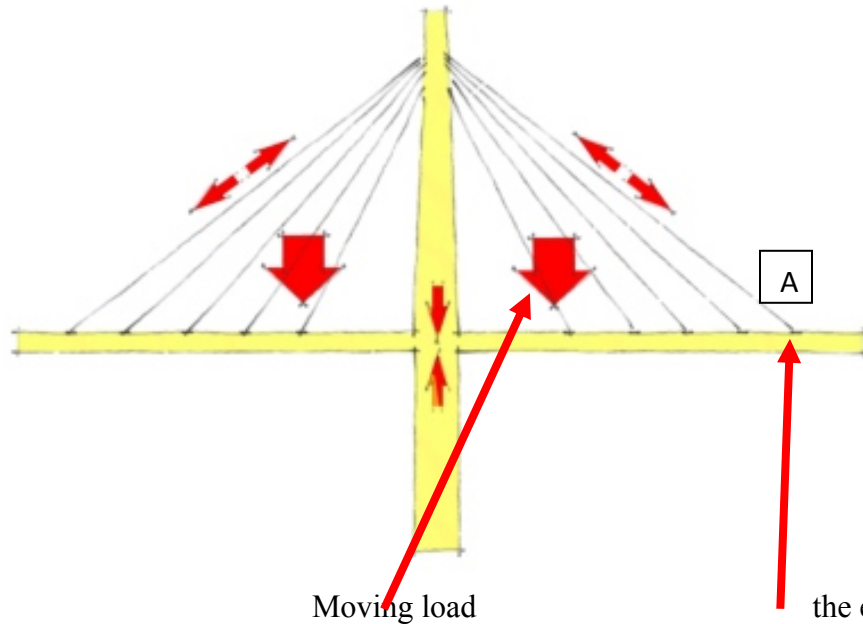
BS 5977 sets out the method for load assessment of lintels in masonry structures for openings up to 4.5m in single storey construction or up to 3.6m in normal domestic two to three storey buildings. The method assumes that the masonry over an opening in a simple wall will arch over the opening. The code guidance must be applied with common sense as building elevations are rarely simple and load will be channelled down piers between openings. Typically there should be not less than 0.6m or $0.2L$ of masonry to each side of the opening (where L is the clear span), not less than $0.6L$ of masonry above the lintel at midspan and not less than 0.6m of masonry over the lintel supports. When working on existing buildings, the effect of new openings on existing lintels should be considered.



Loading assumptions:

1. The weight of the masonry in the loaded triangle is carried on the lintel – not the masonry in the zone of interaction.
2. Any point load or distributed load applied within the load triangle is dispersed at 45° and carried by the lintel.
3. Half of any point, or distributed, load applied to the masonry within the zone of interaction is carried by the lintel.

- 10) Find the compressive force in the deck just for the expected strand when the moving load with 5000 lbs. stays at point A. (The strand stays with the angle of 45° .)



- A) 7071 lb
C) 14141 lb

- B) 10000 lb
D) 5000 lb

Answer is D

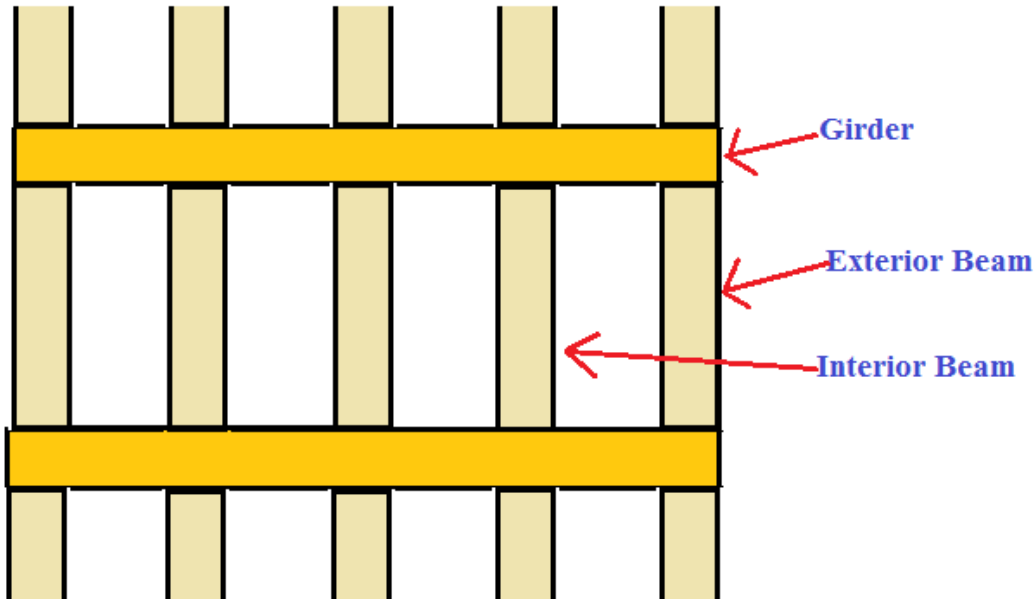
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-23

$$F_v = 5000 \text{ lbs}$$

$$T \text{ (in the strand)} = 5000 / \cos(45) = 9517.98 \text{ lbs.}$$

$$F_h = 9517.98 * \cos 45 = 5000 \text{ lbs.}$$

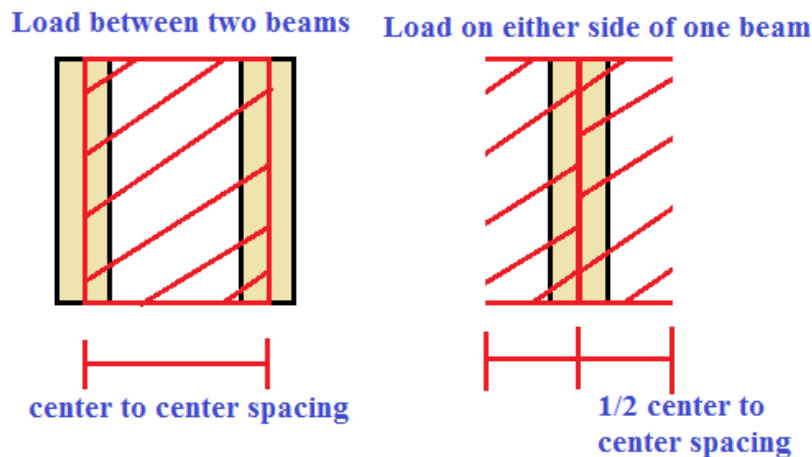
- 11) The floor system shown below has a system of beams supported by girders. Assume all connections are rigid. Each beam is 15 ft. in length and the center to center span between beams is 10 ft. There are only 5 beams supported on either side of each girder. An ultimate load of 60 psf is carried by the floor system. Determine the distributed load carried by the exterior and interior beams.



- A) interior: 0.5 kips/ft; exterior: 0.4 kips/ft
B) interior: 0.6 kips/ft; exterior: 0.3 kips/ft
C) interior: 0.6 kips/ft; exterior: 0.6 kips/ft
D) interior: 0.3 kips/ft; exterior: 0.3 kips/ft

Answer is B**IBC 2009, 1604.4, Page 306**

Step 1: Determine the tributary area for each beam. An exterior beam will carry a different load than an interior beam. To determine the tributary area consider that half of the load carried on either side of any given beam will be carried. The other half of this load will be carried by the adjacent beam. Thus an interior beam will carry half of a load on either side while an interior beam only carries half of a load on one side.



The load was given to be 60 psf on the floor system. Each Interior beam will carry this for half of the center to center spacing on either side. Each exterior beam will only carry this for half of the center to center spacing. Thus we can determine the tributary area.

$$\text{interior: } A = (2) \left(\frac{1}{2} \right) \text{center to center spacing}(L) = (10 \text{ ft})(15 \text{ ft})$$

$$\text{exterior: } A = \left(\frac{1}{2} \right) \text{center to center spacing}(L) = (5 \text{ ft})(15 \text{ ft})$$

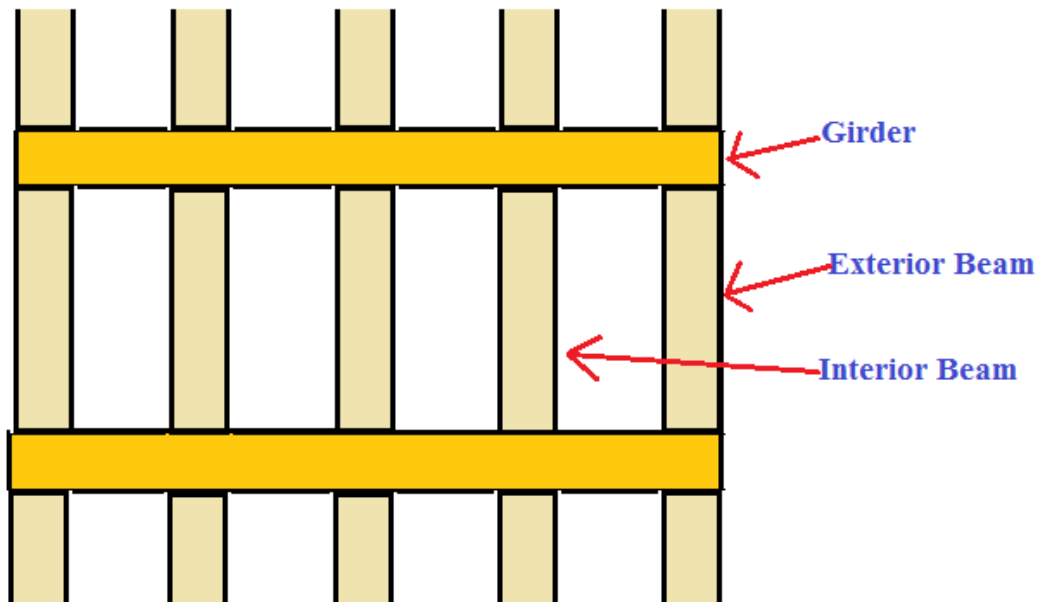
Step 2: Now determine the distributed load on the interior and exterior beams. Multiply by the distance on either side of the beam that the load acts to get the distributed load.

$$\text{Interior: } w = (60 \text{ psf})(10 \text{ ft}) \left(\frac{1 \text{ kip}}{1000 \text{ lb}} \right) = 0.6 \frac{\text{kips}}{\text{ft}}$$

$$\text{exterior: } w = (60 \text{ psf})(5 \text{ ft}) \left(\frac{1 \text{ kip}}{1000 \text{ lb}} \right) = 0.3 \frac{\text{kips}}{\text{ft}}$$

~interior: 0.6 kips/ft; exterior: 0.3 kips/ft

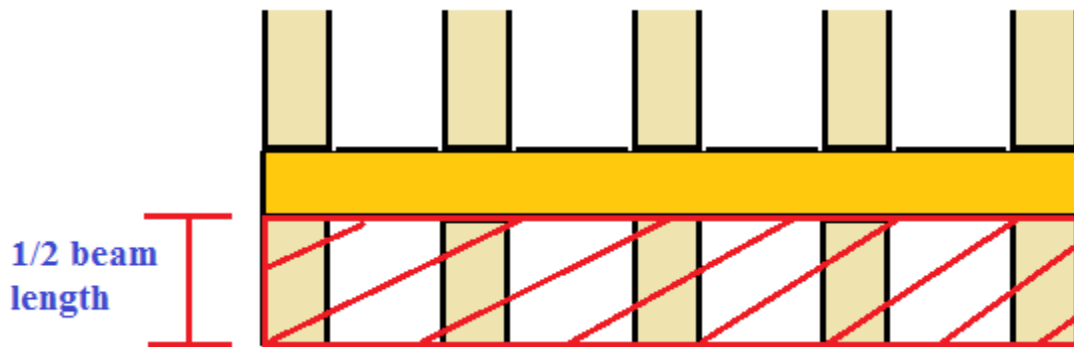
- 12) The floor system shown below has a system of beams supported by girders. Assume all connections are rigid. Each beam is 15 ft in length. There are only 5 beams supported on either side of each girder. Each interior beam and exterior beam carry a uniformly distributed load of 0.6 kips/ft and 0.3 kips/ft respectively. Determine the reaction force at either end of a girder.



- A) 14 kips
B) 22 kips
C) 16 kips
D) 18 kips

Answer is D**IBC 2009, 1604.4, Page 306**

Step 1: Determine the tributary area for the girder. An exterior beam carries a different load than an interior beam. To determine the tributary area consider that half of the load carried on either side of any given girder will be carried. The other half of this load will be carried by the adjacent girder. There are 3 interior beams on either side of the girder and 2 exterior beams.



The load was given to be 0.6 kips/ft for interior beams and 0.3 kips/ft on exterior beams. Each Interior girder will carry this for half of the beam length on either side. There are 5 beams total with 4 center to center spacing's between them. Thus we can determine the tributary area.

$$A = \left((2) \left(\frac{1}{2} \right) L \right) (4 \text{ center to center spacing}) = (15 \text{ ft})(40 \text{ ft})$$

Step 2: Now determine the load applied at each point along the girder by the beams. There are 3 interior beams and 2 exterior beams. Either support on the girder will support half the total load on the girder.

$$\text{Interior: } P = (15 \text{ ft}) \left(0.6 \frac{\text{kips}}{\text{ft}} \right) = 9 \text{ kips}$$

$$\text{exterior: } P = (15 \text{ ft}) \left(0.3 \frac{\text{kips}}{\text{ft}} \right) = 4.5 \text{ kips}$$

$$P_{\text{total}} = (3)(9 \text{ kips}) + (2)(4.5 \text{ kips}) = 36 \text{ kips}$$

$$\text{reaction} = \frac{P_{\text{total}}}{2} = 18 \text{ kips}$$

~18 kips

- 13) What factor may allow for an occupancy live load reduction on a structural member?
- (A) The frequency of foot traffic
 - (B) The Load size
 - (C) The Member capacity
 - (D) The loaded floor area

Answer is D**ASCE 7-10, 4.8.1, Page 10**

Answer "A" is not correct: The frequency of people walking over a floor area is already taken into account for particular structures. The table in the ASCE 7 for live floor load takes this factor into account.

Answer "B" is not correct: The load size does not dictate whether or not the live floor load may be reduced.

Answer "C" is not correct: The member capacity does not dictate whether or not the live floor load may be reduced.

Answer "D" is correct: The loaded floor area can determine how the load distributes through the floor system. Depending on the way in which the floor is supported the live load may be reduced

- 14) What is the factor that accounts for the design strength of members based on material properties and uncertainties?
- (A) Load factors
 - (B) Resistance factor
 - (C) Limit state
 - (D) Live load reduction factor

Answer is B**ACI 318-08, 9.3, Page 117**

Answer “A” is not correct: A load factor is applied to the loads that the structure is subject to.

These factors determine the design load that should be considered for the system.

Answer “B” is correct: A resistance factor is applied based on the material properties and uncertainties of a structural system. This defines the actual strength that will be used in design.

Answer “C” is not correct: Limit State is a condition in which a member becomes unsafe or is no longer useable.

Answer “D” is not correct: A live load reduction factor is a factor which allows you to reduce the occupancy load on a floor system.

15) What needs to be considered before determining the unit load at a point using an influence line?

- (A) The magnitude of the moving load
- (B) The distance the unit load will be traced across
- (C) Where the response is being considered
- (D) How the system is supported

Answer is C**IBC 2009, 1604.4, 306**

Answer "A" is not correct: The magnitude of the load is not important until the actual force at a point needs to be defined. The unit load at a point is defined with the application of a unit load traced across a structure.

Answer "B" is not correct: The distance between points on a structure will determine the value of the unit force at any given point. However this can change depending on where the response function is being considered.

Answer "C" is correct: The point at which a response is being considered will define the function for an influence line. If it is not known which point is being considered then there are several functions that could define the influence line and this would change the unit load at a point.

Answer "D" not is correct: How the system is supported will not determine the magnitude of an influence line at a point. If the system is stable then each support will have the same affect along the structure.

16) Find the volume of a concrete block (length $L_2 = 1$ m) to keep in balance a barrier of steel rod (Length $L_1 = 4$ m) with mass $M_1 = 5$ kg? The specific gravity of concrete is 2000 kg/m^3 .

- (A) 20 m^3 .
- (B) 1 m^3 .
- (C) 0.01 m^3 .
- (D) 0.4 m^3 .



Answer is C

ASCE 7-10, C3, Page 261

Step 1: From equilibrium, to keep in balance the rod, the total moment about the support shall be zero.

$$\frac{M_1 g L_1}{2} = \frac{M_2 g L_2}{2} \quad \Rightarrow \quad M_2 = 20 \text{ kg}$$

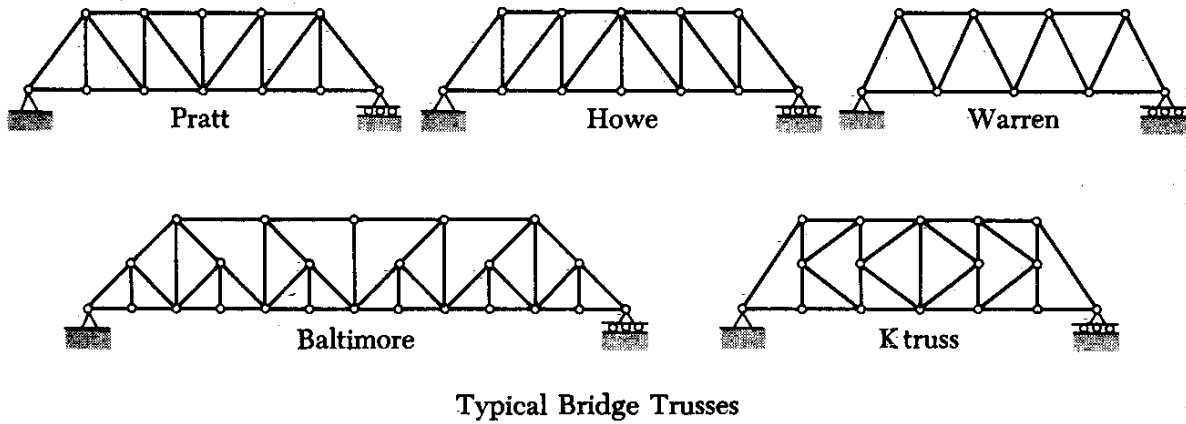
Step 2: Find the volume from definition of specific gravity.

$$\text{WC} = \text{volume} \times \text{specific gravity} \quad \rightarrow \quad \text{Volume} = \text{WC} / \text{specific gravity}$$

$$\text{Volume} = 20 \text{ Kg} / 2000 \text{ Kg/m}^3 = 0.01 \text{ m}^3$$

B. Trusses

- 17) Which truss has more tension members (and consequently more economic) for the effect of dead and live loads?



- (A) Pratt
- (B) Howe
- (C) Warren
- (D) K truss

The Answers is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-21

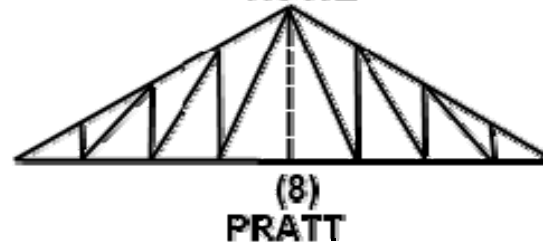
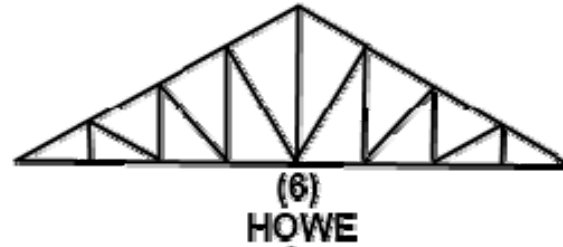
Choice B: is not correct because the transfer from the top chord effect on the diagonals in the form of compressive elements. So most of the elements including top chord and diagonals and vertical elements are in compression.

Choice C: The same as above but half of the diagonals are in tensions.

Choice D; the same as B.

Choice A: is the answer because the all diagonal members and bottom chord are in tension, so the number of tension members are typically at the most comparatively with the other type of trusses.

18) Which one is most likely has more compressive members if there are just dead and live loads?



- (A) Pratt
- (B) Howe
- (C) Cambered Fank
- (D) King-post truss

The Answers is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-21

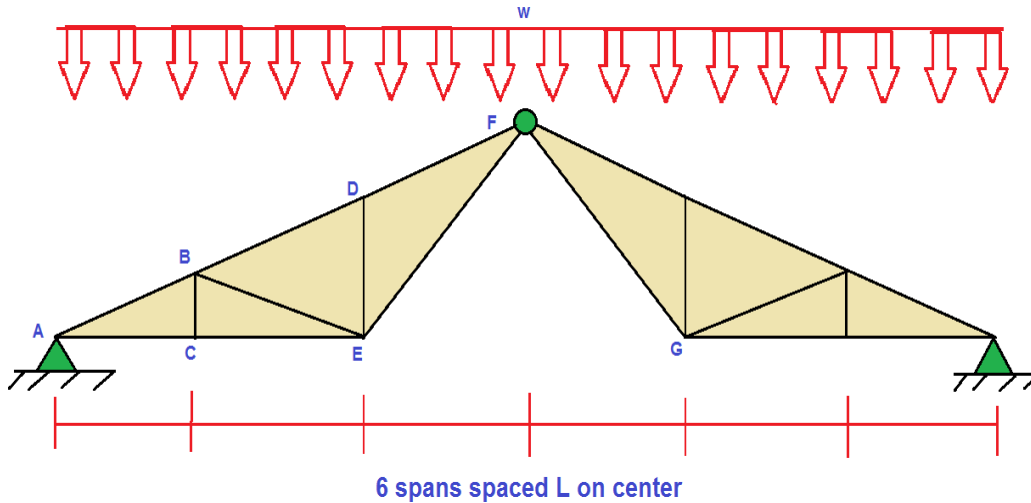
Choice B: is not correct because the transfer from the top chord effect on the diagonals in the form of tension elements. So most of the elements including top chord and diagonals and vertical elements are in compression.

Choice C: The same as above but half of the diagonals are in tensions.

Choice D; the same as B.

Choice A: is the answer because the all diagonal members and top chord are in compression, so the number of compression members are typically at the most comparatively with the other type of trusses.

- 19) A set of trusses with a central pin holds up a bay between them. The bay is 25 ft wide and spans across the length of the truss. The length L is 12 ft and the truss is 15 ft tall. The deck carries a uniform live load (w) of 25 psf. The dead weight of the deck and trusses can be neglected. No part of the deck load is transmitted to the central pin. All truss loads are axial. The vertical reaction at point A is 9000 lb. The force caused by the load at point A, C, and E are 1875, 3750, and 5625 lb respectively. Determine the horizontal reaction force at support A.



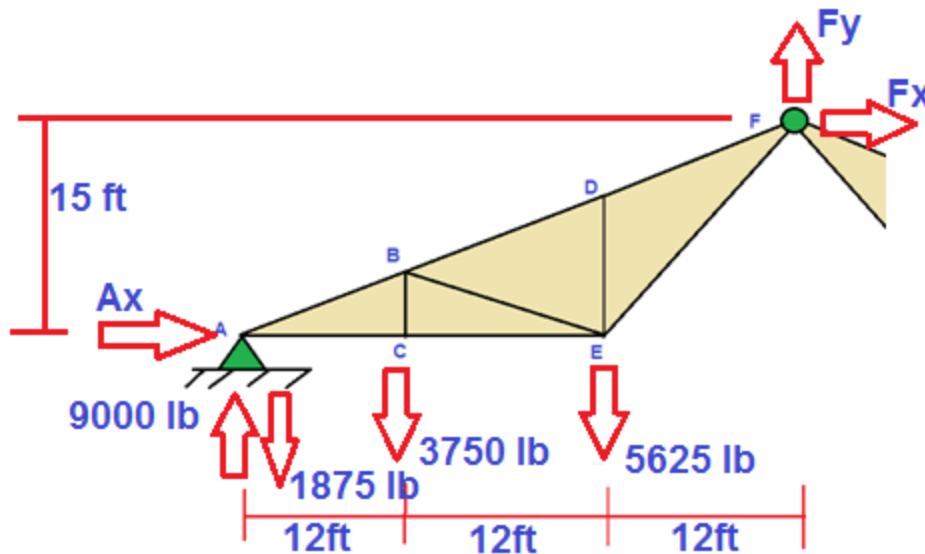
- A) 4.30 kips B) 9 kips
C) 6.60 kips D) 5.63 kips

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: In order to determine the horizontal reaction force for the pin support A we need to sum the moments about a different point. Given the forces at point A, C, and E we can sum a moment about point F. Determine the relative distance of each of these forces from point F. points A, C, and E are 3, 2, and 1 lengths of L away from point F. Thus points A, C and E are 36, 24, and 12 ft from point F respectively. The height of the truss was given to be 15 ft. Thus the horizontal reaction at support A will be 15 ft from point F.

Step 2: When summing moments keep in mind the direction of rotation of a moment. Specify a direction of rotation to be positive and a direction of rotation to be negative. Remain consistent when doing your calculations. I will be specifying counter-clockwise moments as positive and clockwise moments as negative.

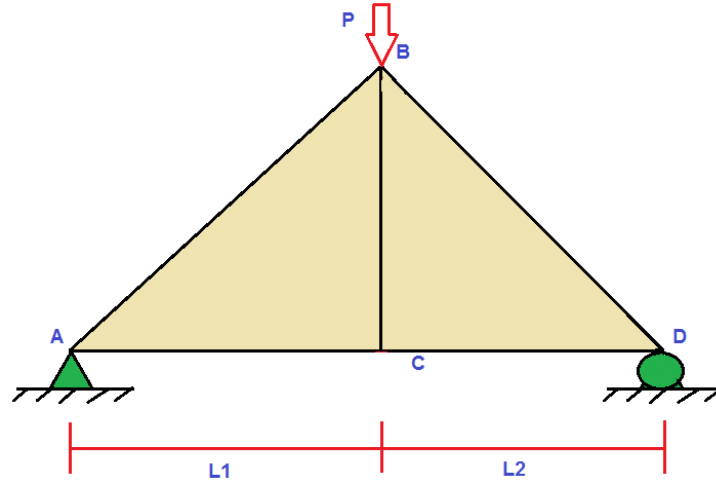


$$\sum M_F = 0 = A_x(15 \text{ ft}) + (1875 \text{ lb})(36 \text{ ft}) + (3750 \text{ lb})(24 \text{ ft}) + (5625 \text{ lb})(12 \text{ ft}) - (9000 \text{ lb})(36 \text{ ft})$$

$$A_x = 6600 \text{ lb}$$

6.6 kips

- 20) A statically determinate truss is subject to a point load of 150 k. L_1 and L_2 are 18 ft and 15 ft respectively and the truss is 10 ft tall. The reaction at supports A and D are 68 and 82 kips respectively. Determine the magnitude of the force in members AB and BD



- A) $F_{AB} = 150$ kips; $F_{BD} = 122$ kips
B) $F_{AB} = 68$ kips; $F_{BD} = 82$ kips
C) $F_{AB} = 140$ kips; $F_{BD} = 148$ kips
D) $F_{AB} = 125$ kips; $F_{BD} = 138$ kips

The Answers is C

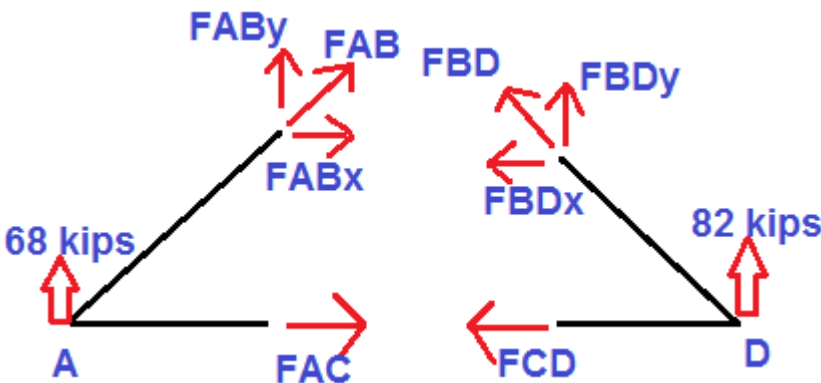
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: Given the reactions at support A and D we can determine the force in members AB and BD based on the lengths of the members. The reaction at either support acts vertically. It was given that the truss is 10 ft tall and we know the lengths to point C from either support. Using the Pythagorean Theorem we can determine the length of each member.

$$L_{AB} = \sqrt{(10\text{ft})^2 + (18\text{ft})^2} = 20.6 \text{ ft}$$

$$L_{BD} = \sqrt{(10\text{ft})^2 + (15\text{ft})^2} = 18 \text{ ft}$$

Step 2: Using the method of sections we can determine the forces in members AB and BD knowing the length of the members and the length the reaction forces at the supports act over.



The reaction forces act over the 10 ft height of the truss. Use the relationship between the length of the members and the height of the truss to find the force in member AB and BD.

$$F_{AB} = \frac{20.6 \text{ ft}}{10 \text{ ft}} (68 \text{ kips}) = 140 \text{ kips}$$

$$F_{BD} = \frac{18 \text{ ft}}{10 \text{ ft}} (82 \text{ kips}) = 148 \text{ kips}$$

$$\sim F_{AB} = 140 \text{ kips}; F_{BD} = 148 \text{ kips}$$

21) Which statement is correct concerning a plane truss?

- (A) A plane truss is a rigid framework.
- (B) The members of the truss lie in the same plane.
- (C) The members are connected at their ends by frictionless pins.
- (D) All of the above are correct.

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Answer "A" is correct: The members have no elastic deformation.

Answer "B" is correct: The members of the truss lie in the same plane.

Answer "C" is correct: The members are connected at the ends by pins which are frictionless.

Answer "D" is correct: Answers "A", "B" and "C" are correct.

22) Which statement is correct concerning external loads on a plane truss?

- (A) The external loads lie in the same plane as the truss.
- (B) The external loads are applied at the joints only.
- (C) None of the above.
- (D) Both (A) and (B) are correct.

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Answer "A" is correct: The external loads lie in the same plane as the plane truss. There are no external loads lying out of plane.

Answer "B" is correct: The external loads are applied at the joints only. There are no external loads applying on the body of members.

Answer "C" is not correct: Answers "A" and "B" are correct.

Answer "D" is correct: Answers "A" and "B" are correct.

23) A plane truss is statically determinate if

- (A) The truss reactions can be determined using the equations of equilibrium.
- (B) The member forces can be determined using the equations of equilibrium.
- (C) Neither (A) nor (B).
- (D) Both (A) and (B).

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Answer “A” is correct: A truss is statically determinate if the truss reactions and member forces can be determined using the equations of equilibrium.

Answer “B” is correct: A truss is statically determinate if the truss reactions and member forces can be determined using the equations of equilibrium.

Answer “C” is not correct: Answers “A” and “B” are correct.

Answer “D” is correct: Answers “A” and “B” are correct.

24) A plane truss is statically indeterminate if

- (A) The truss reactions can be determined using the equations of equilibrium.
- (B) The member forces can be determined using the equations of equilibrium.
- (C) Neither (A) nor (B).
- (D) Both (A) and (B).

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Answer “A” is not correct: A truss is statically indeterminate if the truss reactions and member forces cannot be determined using the equations of equilibrium.

Answer “B” is not correct: A truss is statically determinate if the truss reactions and member forces cannot be determined using the equations of equilibrium.

Answer “C” is correct: Answers “A” and “B” are not correct.

Answer “D” is not correct: Answers “A” and “B” are not correct.

25) Which statement is correct concerning the member force in a plane truss?

- (A) The member forces at the ends of the member must be directed along the axis of the member.
- (B) If the force tends to elongate the member, it is a tensile force.
- (C) If the force tends to shorten the member, it is a compressive force.
- (D) All of the above are correct.

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

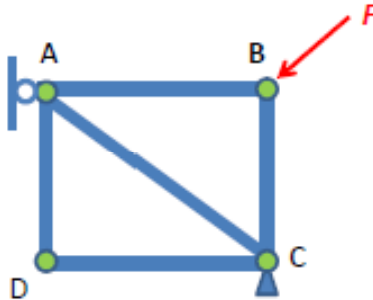
Answer "A" is correct: In a plane truss, each member acts as a two-force member, and therefore the forces at the ends of the members must be directed along the member's axis.

Answer "B" is correct: The member force is a tensile force if the force tends to elongate the member.

Answer "C" is correct: The member force is a compressive force if the force tends to shorten the member.

Answer "D" is correct: Answers "A", "B" and "C" are correct.

- 26) Find zero-force members in the following plane truss. Joint C is pinned, and joint A can slide vertically.



- (A) member AB.
- (B) member CD.
- (C) member AC.
- (D) None of the above.

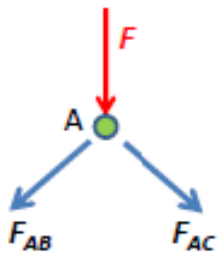
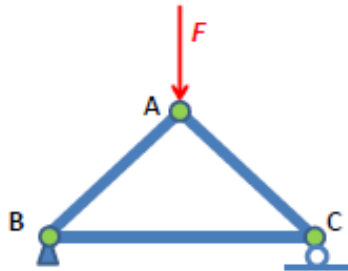
The Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

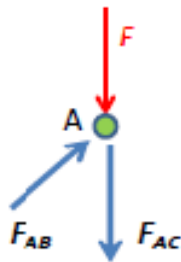
Step 1: In this problem, joint D is formed by member AD and member CD. There is no external force or support reaction applied to joint D. Therefore, members AD and CD are zero-force members.

Members AD and CD are zero-force members

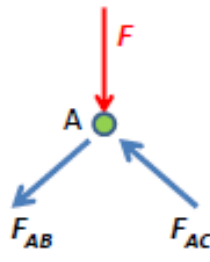
- 27) To analyze the following plane truss, the free-body diagram of joint A is needed to determine forces in members AB and AC when using the method of joints. Which one of the following free-body diagrams is correct? Assuming the unknown member forces acting on the joint's free-body diagram to be in tension. Joint B is pinned, and joint C can slide horizontally.



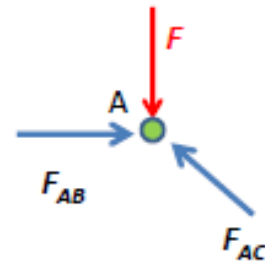
(A)



(B)



(C)



(D)

The Answer is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

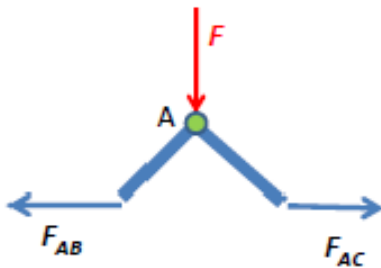
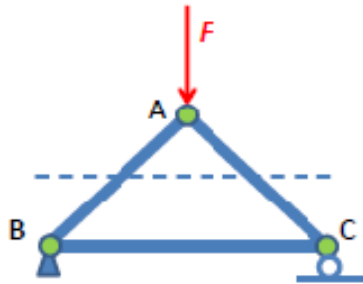
Step 1: The member force act on the joint must be directed along the axis of the member. In answer (B), member force FAC is not directed along the axis of member AC. In answer (D), member force FAB is not directed along the axis of member AB. Therefore, answers (B) and (D) are not correct free-body diagrams.

Step 2: In answer (C), member force FAC is "pushing" on the joint, and consequently FAC is a compressive force. Therefore, answer (C) is not a correct free-body diagram.

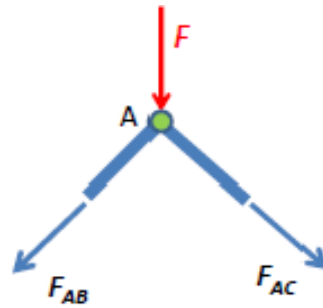
Step 3: In answer (A), member forces FAC and FAB are "pulling" on the joint, and consequently they are tensile forces.

(A) is the correct diagram

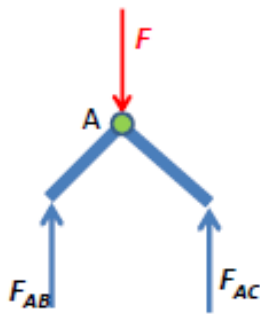
- 28) To analyze the following plane truss using the method of sections, the free-body diagram of upper section is needed to determine forces in members AB and AC. Which one of the following free-body diagrams is correct? Assuming the unknown member forces acting on the section's free-body diagram to be in tension. Joint B is pinned, and joint C can slide horizontally.



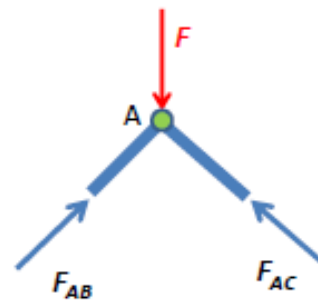
(A)



(B)



(C)



(D)

The Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

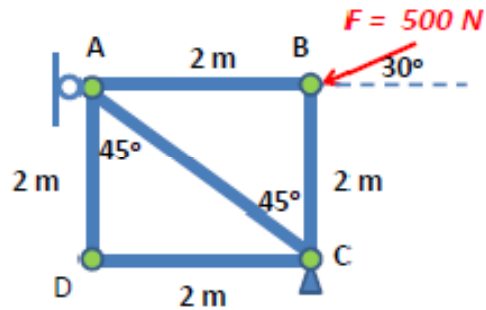
Step 1: The member force must be directed along the axis of the member. In answers (A) and (C), member force F_{AB} is not directed along the axis of member AB, and member force F_{AC} is not directed along the axis of member AC. Therefore, answers (A) and (C) are not correct free-body diagrams.

Step 2: In answer (D), member force F_{AB} is "pushing" member AB, and consequently F_{AB} is a compressive force. In addition, member force F_{AC} is "pushing" member AC, and consequently F_{AC} is a compressive force. Therefore, answer (D) is not a correct free-body diagram.

Step 3: In answer (B), member forces F_{AB} and F_{AC} are "pulling" members AB and AC respectively, and consequently they are tensile forces.

(B) is the correct diagram

- 29) Determine the force in member AD in the following plane truss. Joint C is pinned, and joint A can slide vertically.



- (A) 500 N (tension)
- (B) 250 N (tension)
- (C) 0
- (D) 250 N (compression)

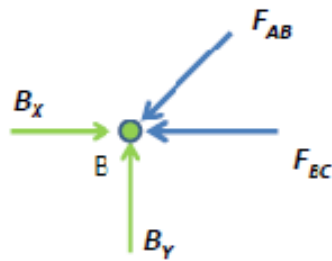
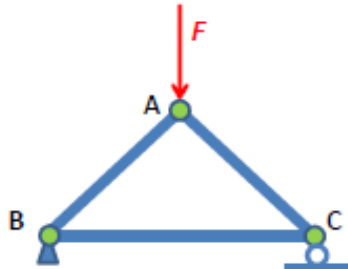
The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

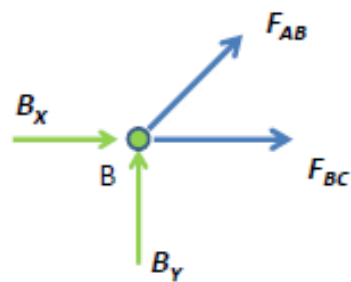
Step 1: In this problem, joint D is formed by member AD and member CD. There is no external force or support reaction applied to joint D. Therefore, members AD and CD are zero-force members.

$$F_{AD} = 0$$

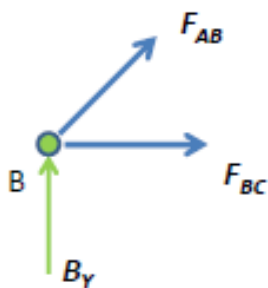
- 30) Find the correct free-body diagram of joint B. Assuming the unknown member forces acting on the joint's free-body diagram to be in tension. Joint B is pinned, and joint C can slide horizontally.



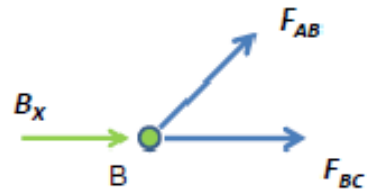
(A)



(B)



(C)



(D)

The Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

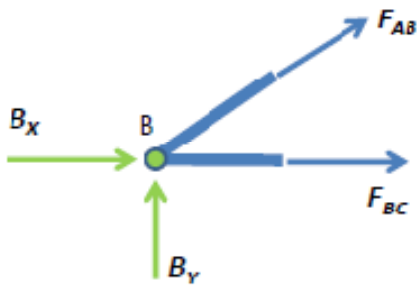
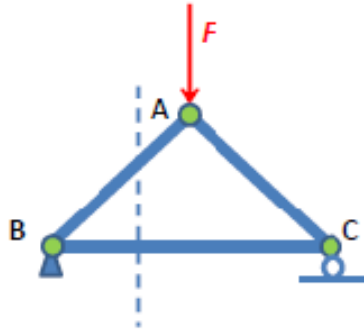
Step 1: Joint B is pinned. Therefore, there are horizontal and vertical support reactions on joint B. Answers (C) and (D) are not correct.

Step 2: In answer (A), member force FAB is "pushing" on the joint, and consequently FAB is a compressive force. In addition, member force FBC is "pushing" on the joint, and consequently FBC is a compressive force. Therefore, answer (A) is not a correct free-body diagram.

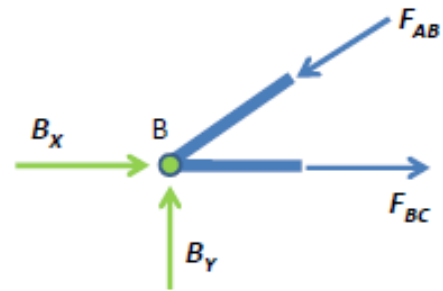
Step 3: In answer (B), member forces FAB and FAC are "pulling" on the joint respectively, and consequently they are tensile forces.

(B) is the correct diagram

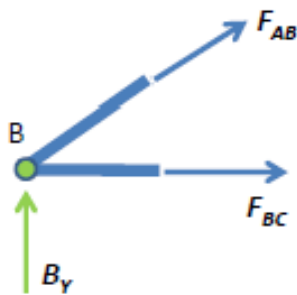
- 31) Determine the correct free-body diagram of left section. Assuming the unknown member forces acting on the section's free-body diagram to be in tension. Joint B is pinned, and joint C can slide horizontally.



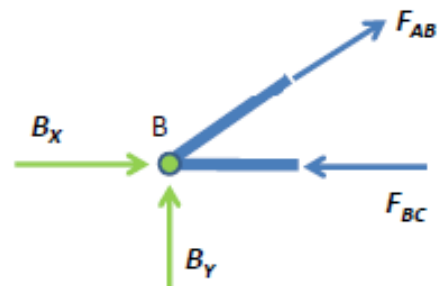
(A)



(B)



(C)



(D)

The Answer is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: Joint B is pinned. Therefore, there are horizontal and vertical support reactions on joint B. Answer (C) is not correct.

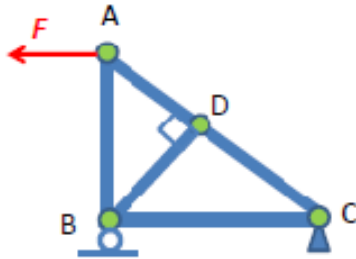
Step 2: In answer (B), member force F_{AB} is "pushing" member AB, and consequently F_{AB} is a compressive force. Answer (B) is not correct.

Step 3: In answer (D), member force F_{BC} is "pushing" member BC, and consequently F_{BC} is a compressive force. Answer (D) is not correct.

Step 4: In answer (A), member forces F_{AB} and F_{BC} are "pulling" members AB and BC respectively, and consequently they are tensile forces.

(A) is the correct free-body diagram

- 32) Determine the zero-force member in the plane truss shown below. Joint C is pinned and joint B can slide horizontally.



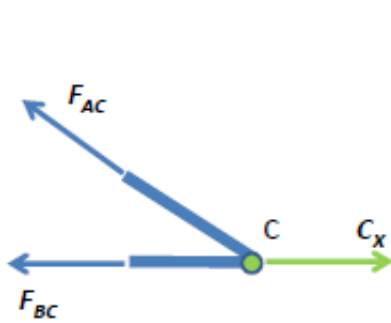
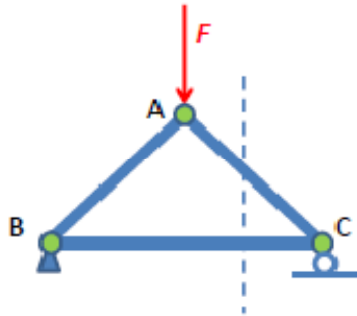
- (A) member AB
- (B) member BC
- (C) member BD
- (D) member CD

The Answer is C

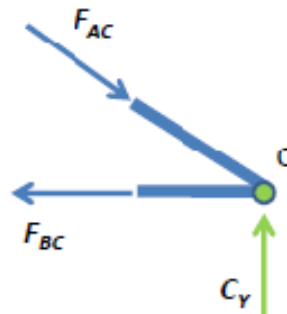
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: In this problem, joint D is formed by members AD, BD and CD. Two of the members, AD and CD are collinear. There is no external force and support reaction on joint D. Therefore, member BD is a zero-force member.

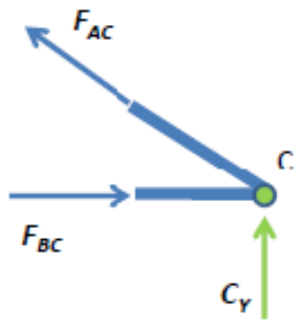
- 33) Determine the correct free-body diagram of right section. Assuming the unknown member forces acting on the section's free-body diagram to be in tension. Joint B is pinned, and joint C can slide horizontally.



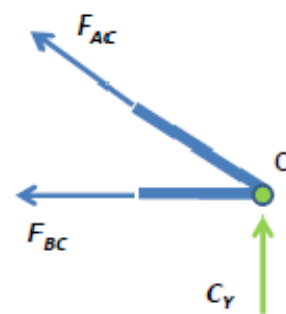
(A)



(B)



(C)



(D)

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19**

Step 1: Joint C can slide horizontally. Therefore, there is only a vertical support reaction on joint C. Answer (A) is not correct.

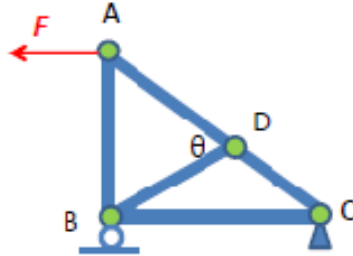
Step 2: In answer (B), member force F_{AC} is "pushing" member AC, and consequently F_{AC} is a compressive force. Answer (B) is not correct.

Step 3: In answer (C), member force F_{BC} is "pushing" member BC, and consequently F_{BC} is a compressive force. Answer (C) is not correct.

Step 4: In answer (D), member forces F_{AC} and F_{BC} are "pulling" members AC and BC respectively, and consequently they are tensile forces.

(D) is the correct free-body diagram

- 34) Determine the zero-force member in the plane truss shown below. Joint C is pinned and joint B can slide horizontally.



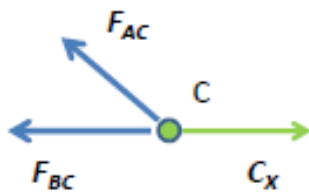
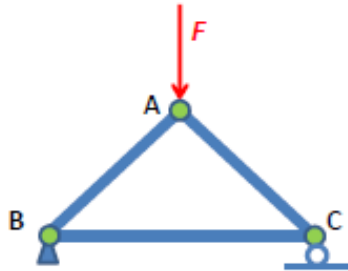
- (A) member AB
- (B) member BD
- (C) member AD
- (D) member BC

The Answer is B

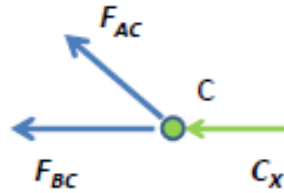
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: In this problem, joint D is formed by members AD, BD and CD. Two of the members, AD and CD are collinear. There is no external force and support reaction on joint D. Therefore, member BD is a zero-force member.

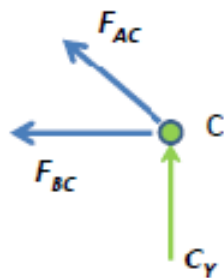
- 35) Find the correct free-body diagram of joint C. Assuming the unknown member forces acting on the joint's free-body diagram to be in tension. Joint B is pinned, and joint C can slide horizontally.



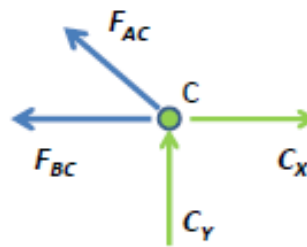
(A)



(B)



(C)



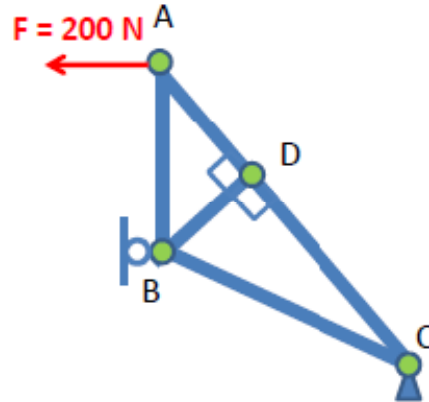
(D)

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19**

Step 1: Joint C can slide horizontally. Therefore, there is only vertical support reaction on joint C. Answers (A) (B) and (D) are not correct.

(C) is the correct diagram

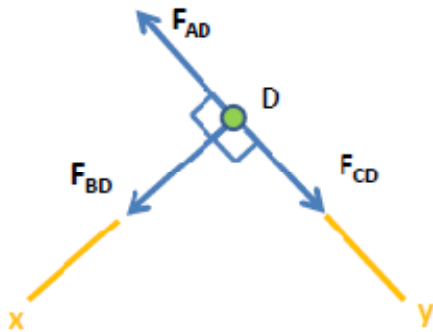
- 36) Determine the force in member BD in the plane truss shown below. Joint C is pinned and joint B can slide vertically.



- (A) 200 N
(B) 0
(C) 100 N
(D) 150 N

The Answer is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19**

Step 1: Plot the free-body diagram of joint D as below, and assume the unknown member forces are tensile forces. Set the Cartesian coordinate system with the original point at joint D.

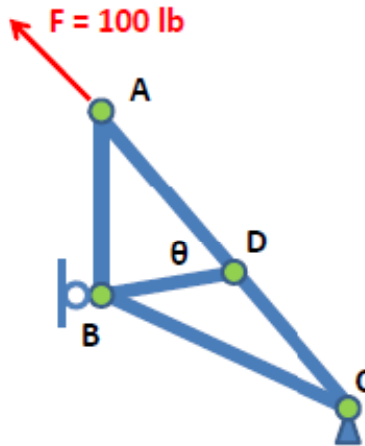


Step 2: Apply the force equilibrium equations

$$\sum F_X = 0 \Rightarrow F_{BD} = 0$$

$$F_{BD} = 0$$

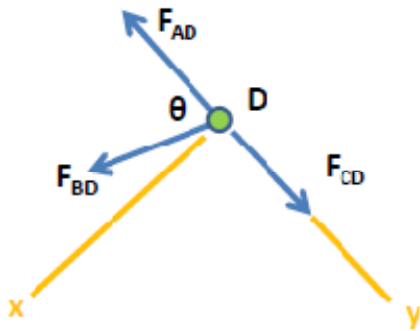
- 37) Determine the force in member BD in the plane truss shown below. Joint C is pinned and joint B can slide vertically.



- (A) 100 lb
- (B) $100 \sin \theta$ lb
- (C) 0
- (D) $100 \cos \theta$ lb

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19**

Step 1: Plot the free-body diagram of joint D as below, and assume the unknown member forces are tensile forces. Set the Cartesian coordinate system with the original point at joint D.

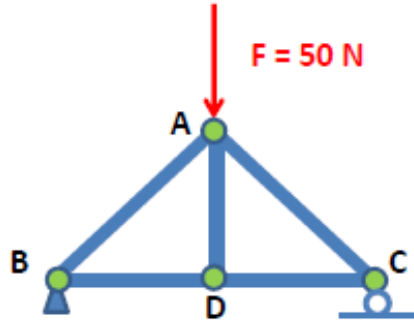


Step 2: Apply the force equilibrium equations

$$\sum F_X = 0 \Rightarrow F_{BD}\sin\theta = 0 \Rightarrow F_{BD} = 0$$

$$F_{BD} = 0$$

- 38) Determine the force in member AD in the plane truss shown below. Joint B is pinned and joint C can slide horizontally.

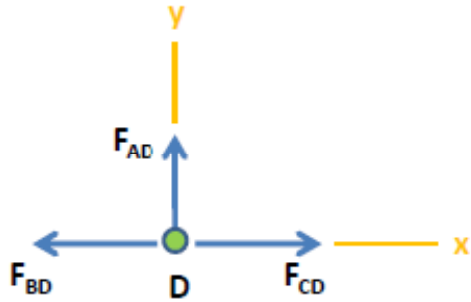


- (A) 0
- (B) 50 N (tension)
- (C) 25 N (compression)
- (D) 50 N (compression)

The Answer is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: Plot the free-body diagram of joint D as below, and assume the unknown member forces are tensile forces. Set the Cartesian coordinate system with the original point at joint D.

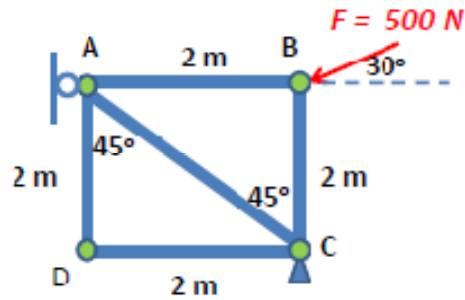


Step 2: Apply the force equilibrium equations

$$\sum F_Y = 0 \Rightarrow F_{AD} = 0$$

$$F_{AD} = 0$$

- 39) Determine the force in member CD in the plane truss shown below. Joint C is pinned and joint A can slide vertically.

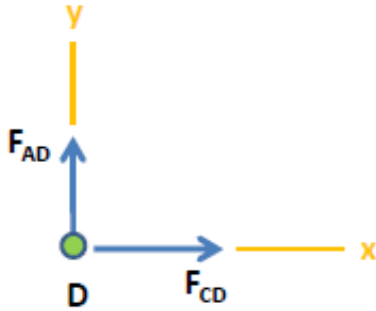


- (A) 500 N (tension)
- (B) 500 N (compression)
- (C) 250 N (compression)
- (D) 0

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: Plot the free-body diagram of joint D as below, and assume the unknown member forces are tensile forces. Set the Cartesian coordinate system with the original point at joint D.

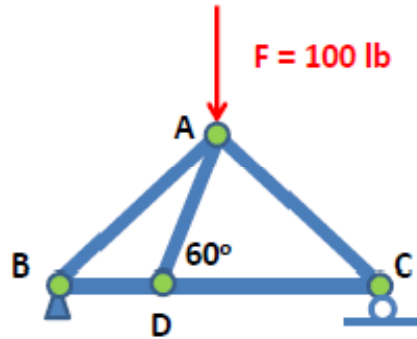


Step 2: Apply the force equilibrium equations

$$\sum F_x = 0 \Rightarrow F_{CD} = 0$$

$$F_{CD} = 0$$

- 40) Determine the force in member AD in the plane truss shown below. Joint B is pinned and joint C can slide horizontally.

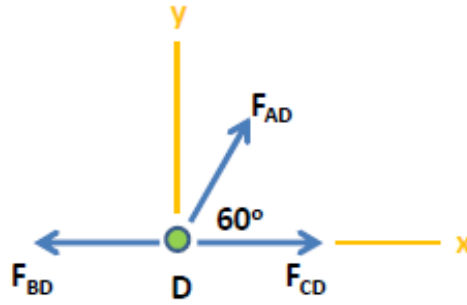


- (A) 100 lb (compression)
- (B) 50 lb (tension)
- (C) 0
- (D) 50 lb (compression)

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: Plot the free-body diagram of joint D as below, and assume the unknown member forces are tensile forces. Set the Cartesian coordinate system with the original point at joint D.

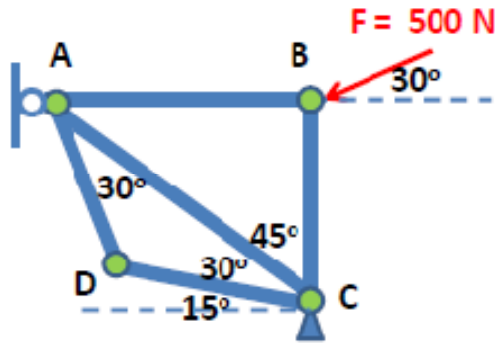


Step 2: Apply the force equilibrium equations

$$\sum F_Y = 0 \Rightarrow F_{AD} \sin 60^\circ = 0 \Rightarrow F_{AD} = 0$$

$$F_{AD} = 0$$

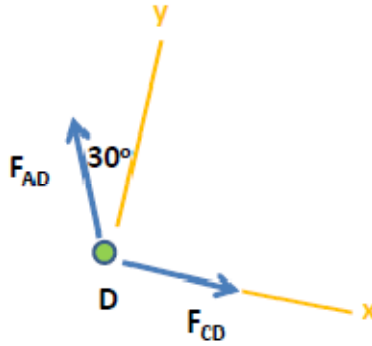
- 41) Determine the force in member CD in the plane truss shown below. Joint C is pinned and joint A can slide vertically.



- (A) 500 N (tension)
- (B) 0
- (C) 250 N (compression)
- (D) 500 N (compression)

The Answer is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19**

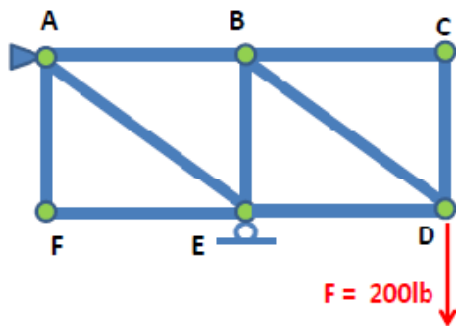
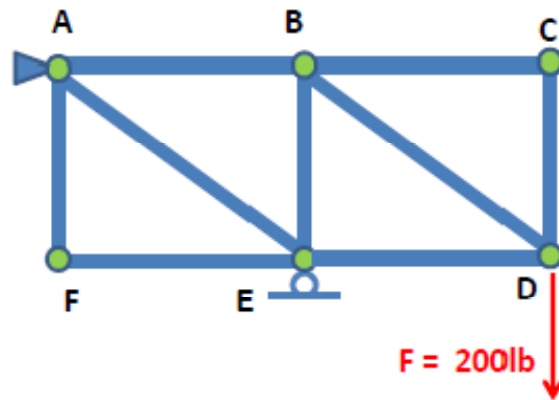
Step 1: Plot the free-body diagram of joint D as below, and assume the unknown member forces are tensile forces. Set the Cartesian coordinate system with the origin at joint D.



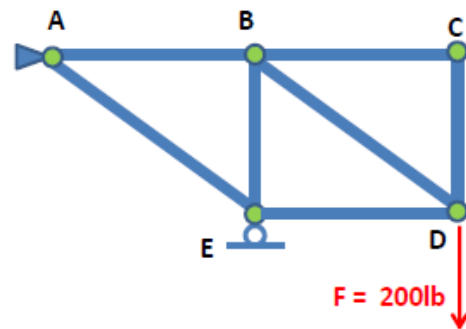
Step 2: Apply the force equilibrium equations

$$\begin{aligned}\sum F_Y = 0 &\Rightarrow F_{AD} \cos 30^\circ = 0 \Rightarrow F_{AD} = 0 \\ F_{CD} &= 0\end{aligned}$$

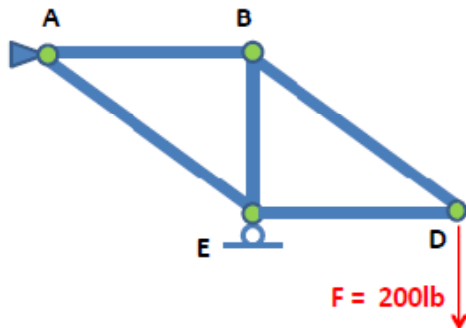
- 42) An engineer designed a plane truss subject to a downward external force at the right lower corner as shown below. Determine the equivalent plane truss without zero-force members.



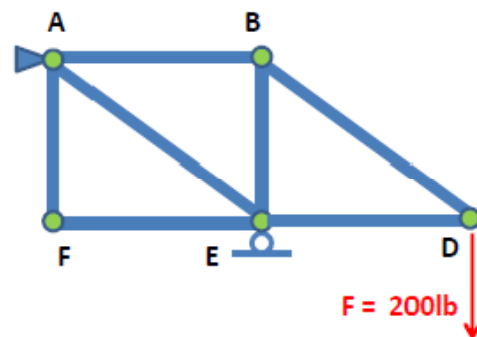
(A)



(B)



(C)



(D)

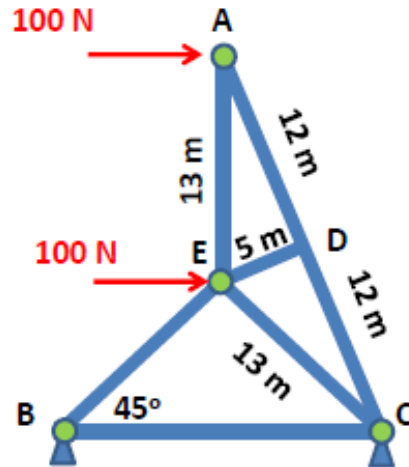
The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19**

Step 1: In this problem, joint C is formed by member BC and member CD. There is no external force or support reaction is applied to joint C. Therefore, members BC and CD are zero-force members.

Step 2: Joint F is formed by member AF and member EF. There is no external force or support reaction is applied to joint F. Therefore, members AF and EF are zero-force members.

Members BC, CD, AF and EF are zero-force members

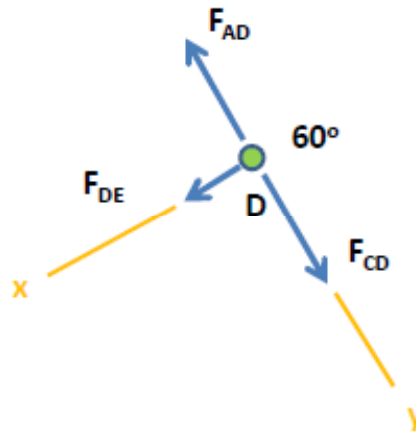
- 43) Determine the force in member DE in the plane truss shown below. Joints B and C are pinned.



- (A) 100 N (tension)
- (B) 100 N (compression)
- (C) 0
- (D) 200 N (compression)

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19**

Step 1: Plot the free-body diagram of joint D as below, and assume the unknown member forces are tensile forces. Set the Cartesian coordinate system with the original point at joint D.

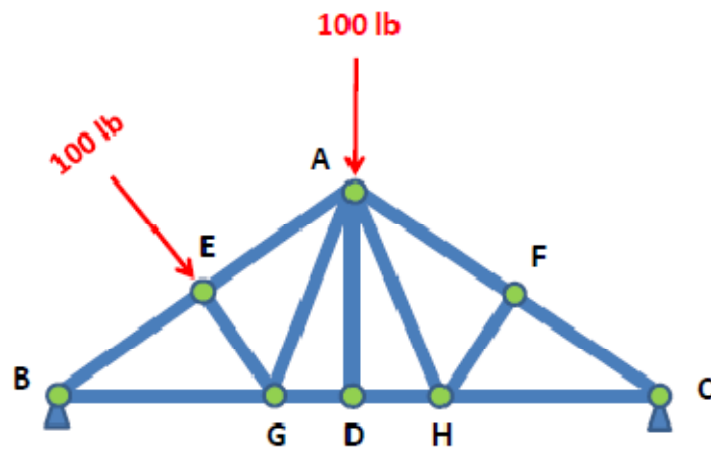


Step 2: Apply the force equilibrium equations

$$\sum F_x = 0 \Rightarrow F_{DE} = 0$$

$$F_{DE} = 0$$

44) Determine zero-force members in the plane truss shown below. Joints B and C are pinned.



- (A) FH
- (B) AH
- (C) AD
- (D) all of the above

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

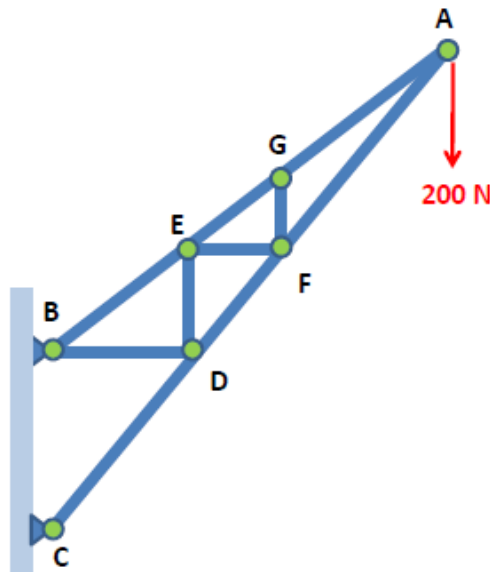
Step 1: In this problem, joint D is formed by members AD, DG and DH. Two of the members, DG and DH are collinear. There is no external force and support reaction on joint D. Therefore, member AD is a zero-force member.

Step 2: In this problem, joint F is formed by members AF, CF and FH. Two of the members, AF and CF are collinear. There is no external force and support reaction on joint F. Therefore, member FH is a zero-force member.

Step 3: Joint H is formed by members DH, AH, FH and CH. Member FH is a zero-force member. Two of the other members, DH and CH are collinear. There is no external force or support reaction is applied to joint H. Therefore, member AH is a zero-force member.

Members AD, FH and AH are zero-force members

45) Determine zero-force members in the plane truss shown below. Joints B and C are pinned.



- (A) FG
(B) EF
(C) DE
(D) BD
(E) All of the above

The Answer is E

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.3, page 47-19

Step 1: In this problem, joint G is formed by members AG, EG and FG. Two of the members, AG and EG are collinear. There is no external force and support reaction on joint G. Therefore, member FG is a zero-force member.

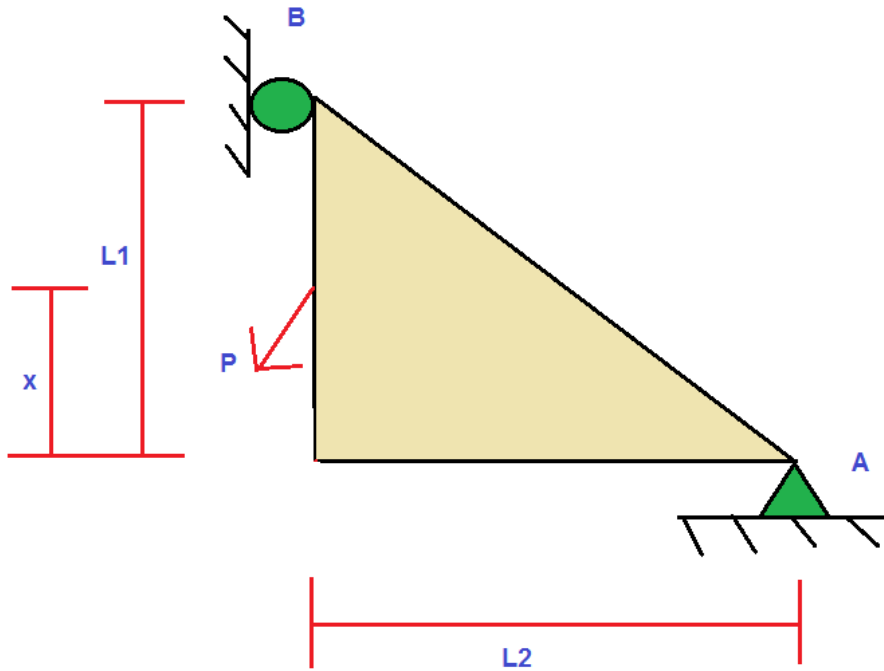
Step 2: In this problem, joint F is formed by members AF, DF, EF and FG. Member FG is a zero-force member. Two of the other members, AF and DF are collinear. There is no external force and support reaction on joint F. Therefore, member EF is a zero-force member.

Step 3: In this problem, joint E is formed by members EG, BE, DE and EF. Member EF is a zero-force member. Two of the other members, EG and BE are collinear. There is no external force and support reaction on joint E. Therefore, member DE is a zero-force member.

Step 4: In this problem, joint D is formed by members BD, CD, DF and DE. Member DE is a zero-force member. Two of the other members, CD and DF are collinear. There is no external force and support reaction on joint D. Therefore, member BD is a zero-force member.

Members FG, EF, DE and BD are zero-force members

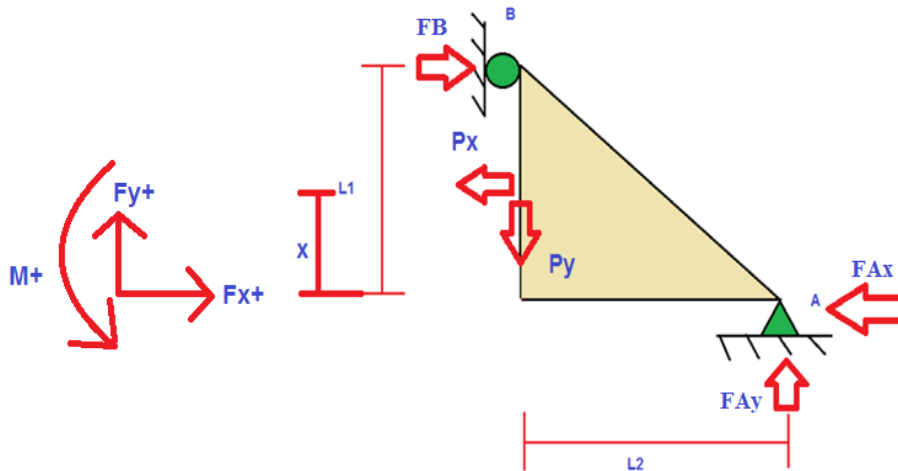
- 46) A statically determinate truss is subject to a point load of 550 kips at a 30 degree angle from vertical. The supports for the truss are on different axis. Reaction forces should be solved for based on the axis for each support. Perpendicular to plane will be considered the vertical direction and parallel to plane will be considered horizontal. $L1 = 10$ ft, $L2 = 15$ ft, $x = 5$ ft. Determine the reactions at the supports.



- A) $F_{Ax} = -575$ kips; $F_{Ay} = 475$ kips; $F_B = 850$ kips
B) $F_{Ax} = 0$ kips; $F_{Ay} = 550$ kips; $F_B = 0$ kips
C) $F_{Ax} = 75$ kips; $F_{Ay} = 475$ kips; $F_B = 200$ kips
D) $F_{Ax} = -350$ kips; $F_{Ay} = 475$ kips; $F_B = 625$ kips

The Answer is A

Step 1: Determine the components of the angled load and their distance from the pin support. To do this consider both the vertical and horizontal components of the load along with the direction. Since the angled load is at 30 degrees from vertical we need to consider both the sin and cosine of this angle to determine the components of the load. The cosine of this angle will determine the vertical component and the sin will determine the horizontal component.



$$P_x = \sin(30)(550 \text{ kips}) = 275 \text{ kips}$$

$$P_y = \cos(30)(550 \text{ kips}) = 476 \text{ kips}$$

The vertical component of this force will work downward and the horizontal component will work left.

Step 2: Use equilibrium equations to solve for the reactions at either support for the truss. There is force both vertically and horizontally so all equilibrium equations should be considered. Take the moment about the pin support to determine the reaction at the roller support. Assume a sign convention where a counterclockwise moment is positive and a clockwise moment is negative.

$$\sum F_x = 0 = F_{Ax} + F_B - P_x$$

$$\sum F_y = 0 = F_{Ay} - P_y$$

$$\sum M_A = 0 = P_y(15 \text{ ft}) + P_x(5 \text{ ft}) - F_B(10 \text{ ft})$$

$$F_B = \frac{P_y(15 \text{ ft}) + P_x(5 \text{ ft})}{10 \text{ ft}} = 851.5 \text{ kips}$$

$$F_{Ax} = P_x - F_B = -576.5 \text{ kips}$$

$$F_{Ay} = P_y = 476 \text{ kips}$$

$$\sim F_{Ax} = -575 \text{ kips}; F_{Ay} = 475 \text{ kips}; F_B = 850 \text{ kips}$$

C. Bending

- 47) Which of the following choice will not affect the value of design flexural strength ($\phi_b M_n$) of a steel beam?
- | | |
|--------------------------|---|
| A) Yield stress of steel | B) Plastic section modulus about the x-axis |
| C) Reduced factor | D) Elastic section modulus about the x-axis |

Answer is D**ACI 318-08, 9.3.1, page 117**

Step 1: $\phi_b M_n = \phi_b F_y Z_x$

From this equation, yield stress of steel (F_y), plastic section modulus about the x-axis (Z_x) and reduced factor (ϕ_b) are all shown except for elastic section modulus. Thus, elastic section modulus will not affect the value of design shear strength.

48) Find the thickness of a square beam so that under the moment of 400 KN-m the radius of curvature does not exceed 10 m. $E = 200 \text{ GPa}$.

- (A) 40 cm
- (B) 20cm
- (C) 5cm
- (D) 100 cm

Answer is A

ACI 318-08, 9.3.1, page 117

Step 1: Using moment-curvature relationship to find the moment of inertia, I:

$$\frac{1}{\rho} = \frac{M}{EI} \quad \Rightarrow \quad I = \frac{\rho M}{E} = \frac{(10m)(400 \text{ KN}\cdot\text{m})}{200 \times 10^6 \frac{\text{N}}{\text{m}^2}}$$

$$I = 0.002 \text{ m}^4$$

Step 2: Find thickness h:

$$I = \left(\frac{1}{12}\right) bh^3 = \left(\frac{1}{12}\right) (h)^4 \quad \rightarrow \quad h = \sqrt[4]{12 I}$$

$$h = \sqrt[4]{12 (0.002)}$$

$$h = 0.393 \text{ say } 0.4\text{m or } 40\text{cm}$$

- 49) In a design process we need to choose the smallest size for a **14 ft** long pole. This pole will resist a load that results moment of 34,300 lb·ft, and the maximum permissible bending stress is 1.6 kpsi, use the following table to select the proper size? Let $E = 1.5 \times 10^6$ psi.

Size (in × in)	Actual Size (in × in)	Area (in ²)	I (in ⁴)	Section Modulus (in ³)
6 × 10	5.5 × 9.5	52.25	393	82.7
6 × 12	5.5 × 11.5	63.25	697	121
6 × 14	5.5 × 13.5	74.25	1127	167
8 × 12	7.5 × 11.5	86.25	950	165
8 × 14	7.5 × 13.5	101.25	1537	228
8 × 16	7.5 × 15.5	116.25	2,327	300

- (A) **6 × 12 beam**
(B) **6 × 14 beam,**
(C) **8 × 16 beam,**
(D) **8 × 12 beam,**

Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-22

Step 1: Calculate the S, Section Modulus, from the maximum normal stress equation:

$$\sigma_{max} = \frac{M_{max}}{S} < 1600 \text{ psi}$$

$$S > \frac{M_{max}}{1600 \text{ psi}} = \frac{34300 \text{ ft}\cdot\text{lb} \left(12 \frac{\text{in}}{\text{ft}}\right)}{1600 \text{ psi}}$$
$$S > 257.25 \text{ in}^3$$

Step 2: Select the beam from the table:

(8 × 16) is the smallest cross section that matches the requirements.

Beam (8 × 16)

50) Calculate the Section modulus of a circular cross section with diameter of 20 cm.

- (A) 1000 cm³
- (B) 4000 cm³
- (C) 1570 cm³
- (D) 852.5 cm³

Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-22

Using the $S = \frac{I}{c}$: and

$$I = \pi r^4 / 2$$

$$c = r = 10 \text{ cm}$$

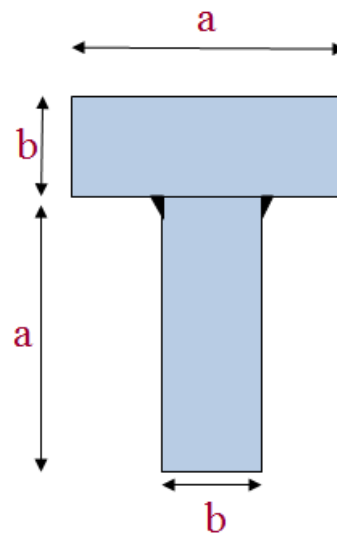
$$I = \pi 10^4 / 2$$

$$I = 15700 \text{ cm}^4$$

$$S = 15700 \text{ cm}^4 / 10 \text{ cm} = 1570 \text{ cm}^3$$

- 51) Two timbers attached to act as one T beam as shown below. Find the shear force on the joint. The timbers are $(10\text{ cm} \times 30\text{ cm})$ in cross section, the shear force acting on the cross section is 300 N.

(A) 36 kN/m (B) 65 kN/m (C) 3.6 kN/m (D) 220 kN-m



Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-22**

Step 1: Find moment of inertia, I:

$$I = \frac{bh^3}{12} = \left(\frac{1}{12}\right) (10 \text{ cm})(30 \text{ cm})^3 + \left(\frac{1}{12}\right) (30 \text{ cm})(10 \text{ cm})^3$$

$$I = 25 \times 10^3 \text{ cm}^4$$

Step 2: Find the shear force (i.e. transverse shear flow):

$$q = \frac{VQ}{I}$$

$$\bar{Y} = (A_1 \bar{y} + A_2 \bar{y}) / (A_1 + A_2) = (300 \times 15 + 300 \times 35) / 600$$

$$\bar{Y} = 25$$

$$\bar{y}' = (30 + 5 - 25 = 10 \text{ cm})$$

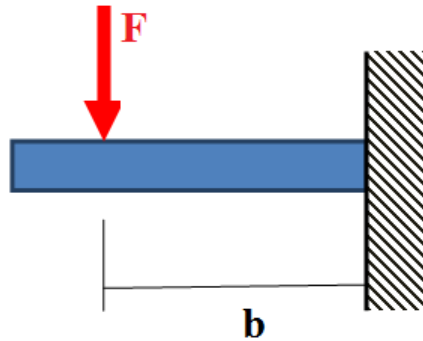
$$Q = A' \bar{y}' = (10 \text{ cm})(30 \text{ cm})(10 \text{ cm}) = 3\,000 \text{ cm}^3$$

$$q = \frac{(300 \text{ N})(3\,000 \text{ cm}^3)}{25 \times 10^3 \text{ cm}^4}$$

$$q = 36 \frac{\text{N}}{\text{cm}} \text{ or } 3.6 \text{ kN/m}$$

- 52) Determine the moment of inertia for the rectangular cross section of the cantilever shown below if $F = 6$ kN acting at 4 m from the support. The maximum compressive stress for the material is 8 MPa, and the maximum tensile stress is 6 MPa. The depth of cross section, $h = 10$ cm

- (A) $8 \times 10^{-5} \text{ m}^4$
(B) $12 \times 10^{-5} \text{ m}^4$
(C) $2 \times 10^{-4} \text{ m}^4$
(D) $24 \times 10^{-5} \text{ m}^4$



Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-22

Step 1: To have the safe cross-section, use the maximum bending moment that occurs at the fixed support:

$$M = F \times b = 6 \text{ KN} \times 4 \text{ m} = 24 \text{ KN}\cdot\text{m}$$

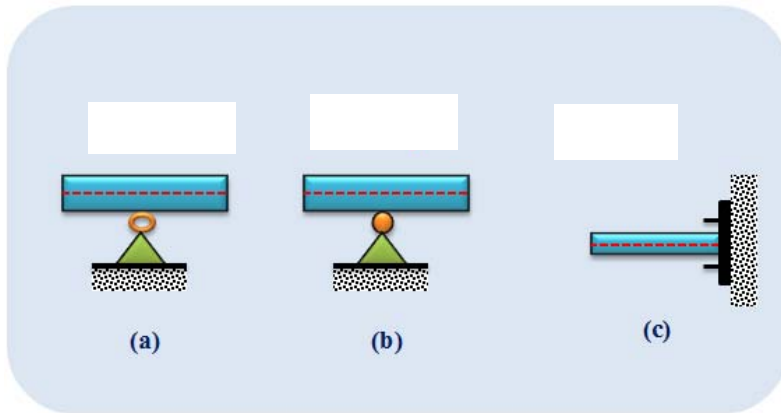
Step 2: select the safe stress: for the case of safety consider the lower stress among the compression and tension. Here the tensile stress is the lower amount so use (tensile stress): 6 MPa

Step 3: Now find moment of inertia, I:

$$\sigma_{max} = \frac{Mc}{I} \quad \rightarrow \quad I = \frac{Mc}{\sigma_{max}} = \frac{24 \text{ KN}\cdot\text{m} \times 0.05 \text{ m} \times 1000}{6 \times 10^6 \text{ N/m}^2} = 2 \times 10^{-4} \text{ m}^4$$

$$I = 2 \times 10^{-4} \text{ m}^4$$

53) A beam is going to carry moment at the ends, what supports do you suggest for the ends?



- (A) A Pin.
- (B) A sliding support.
- (C) Fixed for one end and the other sliding.
- (D) Fixed for both ends.

Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.1, page 47-2

Answer “A” is not correct: Moment is zero at the pinned support.

Answer “B” is not correct: Moment is zero at the sliding support.

Answer “C” is not correct: Moment is non-zero at the fixed support but zero at the sliding support

Answer “D” is correct: Moments are zero at the fixed supports.

54) If the strain in a point 3 cm apart from the neutral axis of a beam cross section is 0.003. Find the radius of curvature.

- (A) 20 cm
- (B) 12 m
- (C) 1 m
- (D) 10 m

Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.4, page 46-13

Step 1: Using the strain equation:

$$\varepsilon = \frac{\text{distance from neutral axis}}{\text{radius of curvature}}$$

$$\varepsilon = \frac{y}{\rho} \rightarrow \rho = \frac{y}{\varepsilon}$$

$$\rho = \frac{0.03 \text{ m}}{0.003}$$

$$\rho = 10 \text{ m}$$

55) If the stress at outermost fiber of a cross section of a beam is 200 KPa. Find the radius of curvature. The cross section of beam is 20cm width $\times$ 40cm depth, and $E=200$ GPa.

- (A) 200m (B) 100m (C) 25m (D) 15m

Answer is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-22

Step 1: Using the stress- curvature equation:

$$\sigma_x = -E y / \rho$$

$$\rho = -E \frac{y}{\sigma_x}$$

$$\rho = 200 \times (106) \times 0.2 / 200(103)$$

$$\rho = 200 \text{ m}$$

56) A beam with a rectangular cross section of width 0.1 m and height 0.3 m is subject to a bending moment of 100 kN·m. Determine the bending stress on the neutral axis.

- (A) 6.67 MPa
- (B) 3.33 MPa
- (C) 0
- (D) – 6.67 MPa

Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-22

The normal stress in a beam due to bending can be calculated as:

$$\sigma_x = -\frac{My}{I}$$

On the neutral axis, $y = 0$. Then,

$$\sigma_x = 0$$

57) A beam with a rectangular cross section of width 0.1 m and height 0.3 m is subject to a bending moment of 100 kN·m. Determine the normal stress on the top of the beam.

- (A) 66.67 MPa
- (B) 33.33 MPa
- (C) 0
- (D) – 66.67 MPa

Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-22**

The normal stress in a beam due to bending can be calculated as:

$$\sigma_x = -\frac{My}{I}$$

On the top of the beam, $y = 0.15 \text{ m}$. Then,

$$\sigma_x = -\frac{(100 \text{ KN}\cdot\text{m})(0.15 \text{ m})}{\frac{(0.1 \text{ m})(0.3 \text{ m})^3}{12}} = -66666.67 \frac{\text{KN}}{\text{m}^2} = -66.67 \text{ MPa}$$

PE Style

The normal stress in a beam due to bending moment is directly proportional to the distance y from the neutral axis and the bending moment M . When the beam is subject to a positive bending moment, compression occurs above the neutral axis while tension occurs below the neutral axis. The only answer with a negative normal stress (compression) on the top of the beam is D.

58) A beam with a rectangular cross section of width 0.2 m and height 0.4 m is subject to a shear force of 100 kN. Determine the transverse shear stress on the top of the beam.

- (A) 0
- (B) 1 MPa
- (C) 2 MPa
- (D) 4 MPa

Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-27**

The transverse shear stress in a beam due to shear force can be calculated as:

$$\tau_{xy} = \frac{VQ}{Ib} \quad \text{where} \quad Q = A'\bar{y}'$$

A' is the area above the layer (or plane) upon which the desired transverse shear stress acts.

On the top of the beam, $A' = 0$. Then,

$$\tau_{xy} = 0$$

PE- Style

The shear stress distribution in a beam depends on how Q/b varies. In standard-section beams, the maximum shear stress always occurs on the neutral axis while zero stress occurs on the top or bottom of the beam. The only answer with zero shear stress on the top of the beam is A.

59) A beam with a rectangular cross section is subject to a shear force. Determine the location where the maximum transverse shear stress occurs.

- (A) Top of the beam
- (B) Neutral axis
- (C) Bottom of the beam
- (D) None of the above

Answer is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-27**

The transverse shear stress in a beam due to shear force can be calculated as:

$$\tau_{xy} = \frac{vQ}{Ib} \quad \text{where} \quad Q = A'\bar{y}'$$

A' is the area above the layer (or plane) upon which the desired transverse shear stress acts.

$\bar{y}'$ is the distance from the neutral axis to the centroid of area A' .

PE- Style

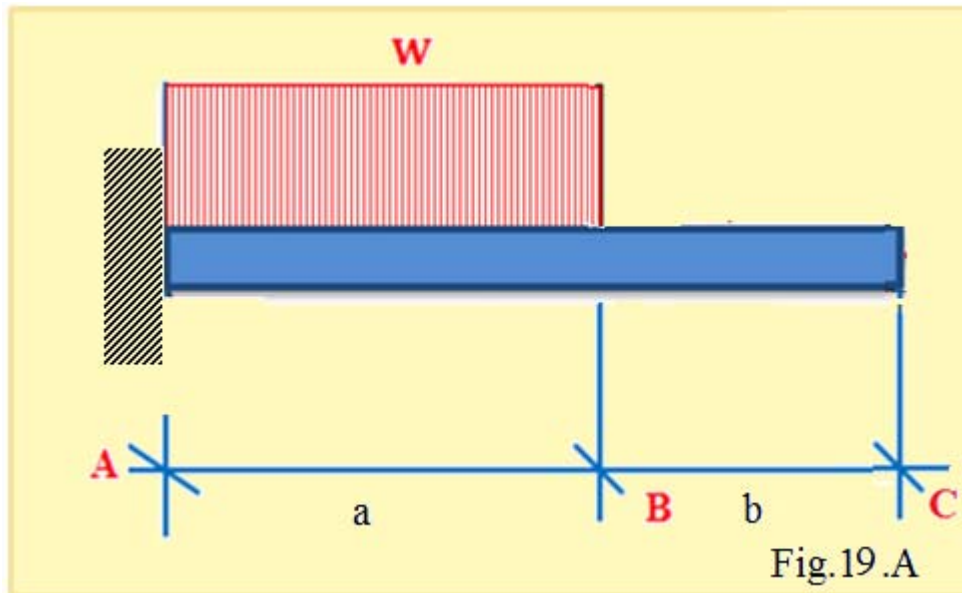
The shear stress distribution in a beam depends on how Q/b varies. In standard-section beams, the maximum shear stress always occurs on the neutral axis while zero stress occurs on the top or bottom of the beam. The only answer with the maximum transverse shear stress occurring on the neutral axis is B.

60) A cantilever concrete beam is loaded by $20 \frac{\text{kN}}{\text{m}}$, If the maximum axial stress at the beam is 160 MPa, find the proper cross section moment of inertia for the beam, assume $c = 10\text{cm}$.
 $a = 3\text{ m}$ $b = 1\text{ m}$

- (A) $0.16 \times 10^{-6} \text{ m}^4$
- (B) $60 \times 10^{-6} \text{ m}^4$
- (C) $5.625 \times 10^{-5} \text{ m}^4$
- (D) $180 \times 10^{-6} \text{ m}^4$

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23



Step 1: The maximum axial stress is resulted from the maximum moment that occurs at the support. Determine the moment at the support.

$$\sum M = 0$$

$$M = 20(3) (3/2)$$

$$\mathbf{M = 90 \text{ KN} \cdot \text{m}}$$

Step 2: Determine the moment of inertia of the section. Using equation:

$$\sigma_x = \frac{Mc}{I}$$

$$I = \frac{Mc}{\sigma_x} = \frac{(90 \text{ KN} \cdot \text{m})(0.1 \text{ m})}{160 \times 10^6 \text{ N/m}^2}$$

$$\mathbf{I = 5.625 \times 10^{-5} \text{ m}^4}$$

61) Find the thickness of a square beam so that under the moment of 400 KN-m the radius of curvature does not exceed 10 m. $E = 200 \text{ GPa}$.

- (A) 40 cm
- (B) 20cm
- (C) 5cm
- (D) 100 cm

The Answers is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23**Step 1: Using moment-curvature relationship to find the moment of inertia, I:

$$\frac{1}{\rho} = \frac{M}{EI} \quad \Rightarrow \quad I = \frac{\rho M}{E} = \frac{(10\text{m})(400 \text{ KN}\cdot\text{m})}{200 \times 10^6 \frac{\text{N}}{\text{m}^2}}$$

$$I = 0.002 \text{ m}^4$$

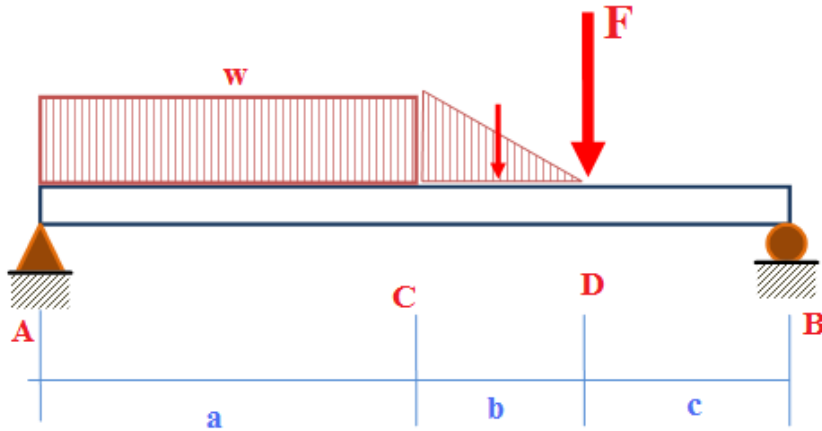
Step 2: Find thickness h:

$$I = \left(\frac{1}{12}\right) bh^3 = \left(\frac{1}{12}\right) (h)^4 \quad \rightarrow \quad h = \sqrt[4]{12 I}$$

$$h = \sqrt[4]{12 (0.002)}$$

$$h = 0.393 \text{ say } 0.4\text{m or } 40\text{cm}$$

- 62) For the following beam find the bending stress at the neutral axis at point C. $w = 20 \text{ kN/m}$
 $F = 200 \text{ kN}$



- (A) 200MPa.
- (B) 2GPa.
- (C) 200 GPa.
- (D) Zero

The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23

Step 1: Look up on the definition of bending stress in any mechanics of material book. it is found that the bending stress is zero at the neutral axis, therefore (D) is correct.

63) Determine the maximum moment for a 5 m long beam if the elastic section modulus is 600 cm³ and the maximum allowable bending stress (σ) in the beam is 200 MPa.

- (A) 1.0 kN·m
- (B) 1.5 kN·m
- (C) 200 kN·m
- (D) 120 kN·m

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23**

Step 1: the elastic section modulus is the ratio of the section area moment of inertia by the distance from the neutral axis to the outermost fibers in the beam.

$$S = I/c$$

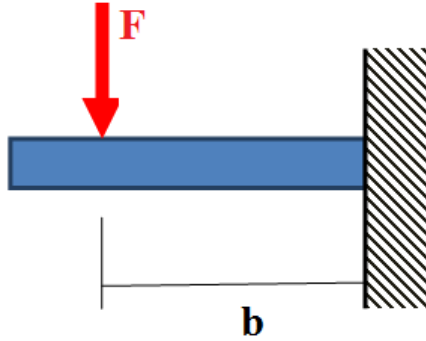
$$\sigma = Mc/I \rightarrow M = \sigma I/c \rightarrow M = (\sigma)(S)$$

$$M = 200 \times (10^6) \text{ N/m}^2 \times (0.0006) \text{ m}^3$$

$$M = 120000 \text{ N}\cdot\text{m} = 120 \text{ kN}\cdot\text{m}$$

- 64) Determine the moment of inertia for the rectangular cross section of the cantilever shown below if $F = 6$ kN acting at 4 m from the support. The maximum compressive stress for the material is 8 MPa, and the maximum tensile stress is 6 MPa. The depth of cross section, $h = 10$ cm

- (A) $8 \times 10^{-5} \text{ m}^4$
(B) $12 \times 10^{-5} \text{ m}^4$
(C) $2 \times 10^{-4} \text{ m}^4$
(D) $24 \times 10^{-5} \text{ m}^4$



The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23**

Step 1: To have the safe cross-section, use the maximum bending moment that occurs at the fixed support:

$$M = F \times b = 6 \text{ kN} \times 4 \text{ m} = 24 \text{ kN}\cdot\text{m}$$

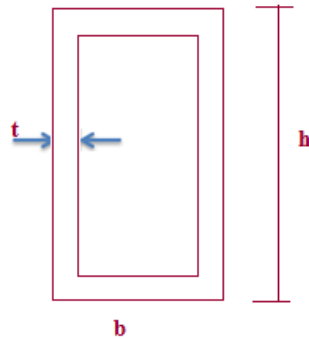
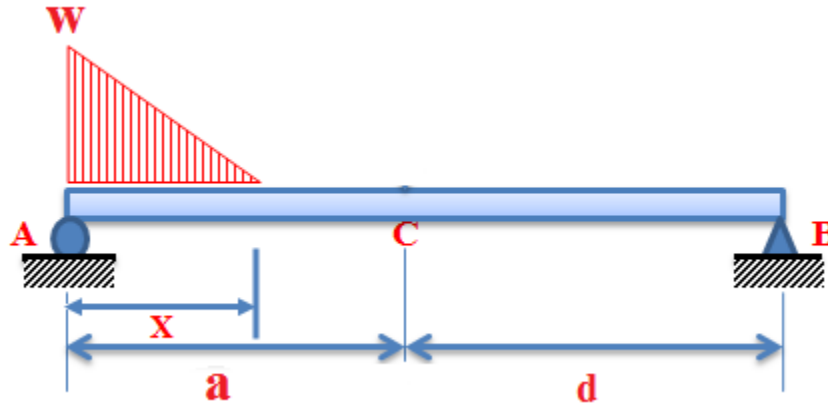
Step 2: select the safe stress: for the case of safety consider the lower stress among the compression and tension. Here the tensile stress is the lower amount so use (tensile stress): 6 MPa

Step 3: Now find moment of inertia, I:

$$\sigma_{\max} = \frac{Mc}{I} \rightarrow I = \frac{Mc}{\sigma_{\max}} = \frac{24 \text{ kN}\cdot\text{m} \times 0.05 \text{ m} \times 1000}{6 \times 10^6 \text{ N/m}^2} = 2 \times 10^{-4} \text{ m}^4$$

$$I = 2 \times 10^{-4} \text{ m}^4$$

- 65) Find the maximum tensile bending stress at point C on the beam shown below. $W = 10 \frac{\text{kN}}{\text{m}}$, $x = 3 \text{ m}$, $a = 5 \text{ m}$, and $d = 5 \text{ m}$. For the cross section: $b = 60 \text{ mm}$, $h = 120 \text{ mm}$ and $t = 10 \text{ mm}$.



(A) 169.6 MPa
(D) 660 MPa

(B)

100 MPa

(C) 120 MPa

The Answers is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23

Step 1: Find the reactions:

$$\Sigma M_{(A)} = 0$$

$$R_B (10) - w (x) (x/3) = 0$$

$$R_B = (10 (3^2/3)) / 10$$

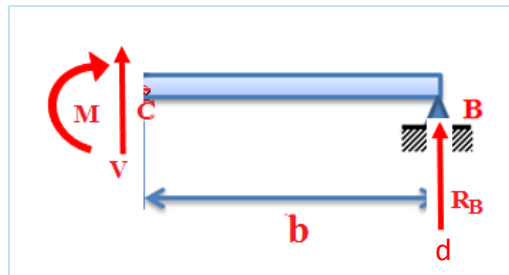
$$R_B = 3$$

Step 2: Find the moment at C:

$$\Sigma M_{(v)} = 0$$

$$M = R_B (d)$$

$$M = 3(5) = 15 \text{ KN-m}$$



Step 3: Calculate moment of inertia, I.

$$I = I (\text{outside rectangle}) - I (\text{inside rectangle})$$

$$I = \frac{bh^3}{12}$$

$$I = \left(\frac{1}{12}\right)(60)(120)^3 - \left(\frac{1}{12}\right)(40)(100)^3$$

$$I = 53.07 \times 10^5 \text{ mm}^4$$

Step 4: Calculate maximum stress:

$$\sigma = \frac{Mc}{I} = \frac{(15 \text{ kN} \cdot \text{m})(0.06 \text{ m})}{(53.07 \times 10^5 \text{ mm}^4) \left(\frac{1 \text{ m}}{1000 \text{ mm}}\right)^4}$$
$$= 16.96 \times 10^4 \text{ kPa} \quad (169.6 \text{ MPa})$$

- 66) Determine the maximum axial (normal) stress at the middle of the 8' long wood beam which is simply supported. The cross section of height 6 inches has moment of inertia $I = 36 \text{ in}^4$. The loading is as below.

$$C = 4 \text{ ft} \quad D = E = 2 \text{ ft} \quad w = 20 \frac{\text{lb}}{\text{ft}} \quad P = 200 \text{ lb}$$

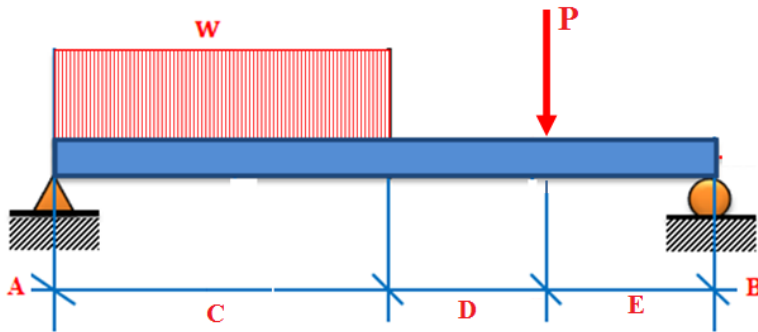
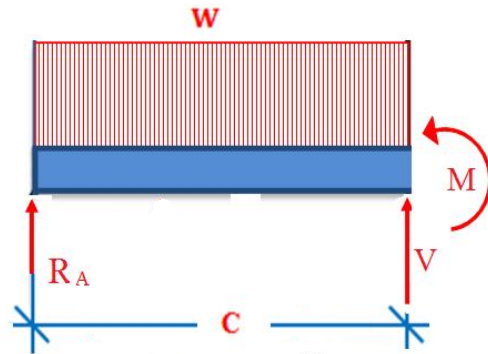


Fig.12.A

- (A) $1520 \frac{\text{lb}}{\text{in}^2}$.
(B) $1680 \frac{\text{lb}}{\text{in}^2}$
(C) $280 \frac{\text{lb}}{\text{in}^2}$
(D) $236 \frac{\text{lb}}{\text{in}^2}$

The Answers is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23****Step 1:** Determine the reactions:

$$\begin{aligned}\sum M_B &= 0 \\ 0 &= P(E) + w(C)(E+D+C/2) - R_A(8) \\ 200(2) + 20(4)(6) &= R_A(8) \\ R_A &= 110 \text{ lb}\end{aligned}$$

 $R_A = ($

the free body

Fig. 12.B

Step 2: Section from middle and draw diagram for the left section:**Step 3:** Find the Maximum moment:

$$\begin{aligned}\sum M_V &= 0 \\ 0 &= + w(C)(C/2) - R_A(C) + M \\ M &= -20(4)(2) + 110(4) \\ M &= 280 \text{ lb}\cdot\text{ft}\end{aligned}$$

Step 4: Determine the maximum axial (normal) stress that occurs at the outer most fibers.

$$\sigma_x = \frac{Mc}{I}$$

σ_x = axial (normal or bending) stress.
 c = distance from neutral axis to outer most

$$c = \frac{h}{2} = \frac{6'}{2} = 3 \text{ in}$$

$$\sigma_x = \frac{(280 \text{ lb}\cdot\text{ft})(3 \text{ in})(12 \frac{\text{in}}{\text{ft}})}{(36 \text{ in}^4)}$$

$$\sigma_x = 280 \text{ psi}$$

67) What amount of concentrated load can bear a simply supported wooden beam of (2" $\times$ 12") with 16 ft span at its center, If the maximum normal bending stress is 1000 psi?

- (A) 3000 lb (B) 2000 lb (C) 1000 lb (D) 500 lb

The Answers is C

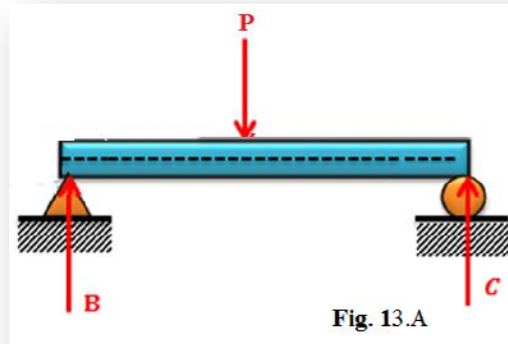
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23

Step 1: Determine the Moment for the section:

$$I = \left(\frac{1}{12}\right) bh^3$$

$$h = 12"$$

$$(2") (12")^3 = 288 \text{ in}^4$$



of inertia

$$b = 2"$$

$$I = \left(\frac{1}{12}\right)$$

Step 2: The maximum moment under the concentrated load. Therefore determine the moment

occurs

$$\sigma_x = \frac{Mc}{I} \Rightarrow c = \frac{h}{2} = \frac{12'}{2} = 6 \text{ in}$$

$$M = \frac{\sigma_x I}{c} = \frac{1000 \text{ psi}(288 \text{ in}^4)}{6 \text{ in}}$$

$$M = 48000 \text{ in-lb} = 4000 \text{ ft-lb}$$

Step 3: Find the load:

$$\sum M_V = 0 \Rightarrow -R_B \left(\frac{L}{2}\right) + M = 0$$

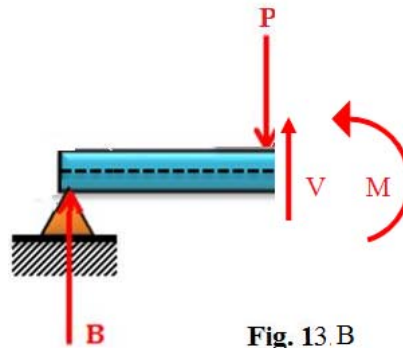
$$L = 16 \text{ ft}$$

$$R_B = P/2$$

$$M = P/2 \left(\frac{16}{2}\right)$$

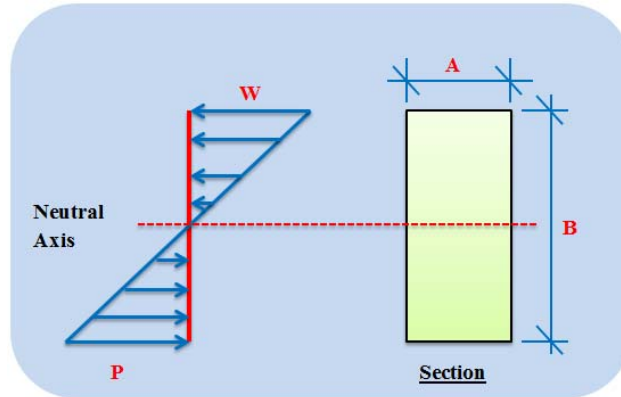
$$4000 = P\left(\frac{16}{4}\right)$$

$$P = 1000 \text{ lb}$$



- 68) The figure below shows the cross section of a beam and the maximum normal stress is $w = 350 \text{ KPa}$. What is the bending moment at this section? **$A = 10 \text{ cm}$ $B = 25 \text{ cm}$**

- (A) **$964 \text{ N} \cdot \text{m}$**
(B) **$152 \text{ N} \cdot \text{m}$**
(C) **$666 \text{ N} \cdot \text{m}$**
(D) **$364 \text{ N} \cdot \text{m}$**



The Answers is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23**

Step 1: Determine the Moment of inertia for the section:

$$I = \left(\frac{1}{12}\right) bh^3$$

$$b = 10 \text{ cm} \quad h = 25 \text{ cm}$$

$$I = \left(\frac{1}{12}\right) (10 \text{ cm}) (25 \text{ cm})^3 = 13020.8 \text{ cm}^4$$

Step 2: Determine the bending moment:

$$\sigma_{\max} = \frac{Mc}{I}$$

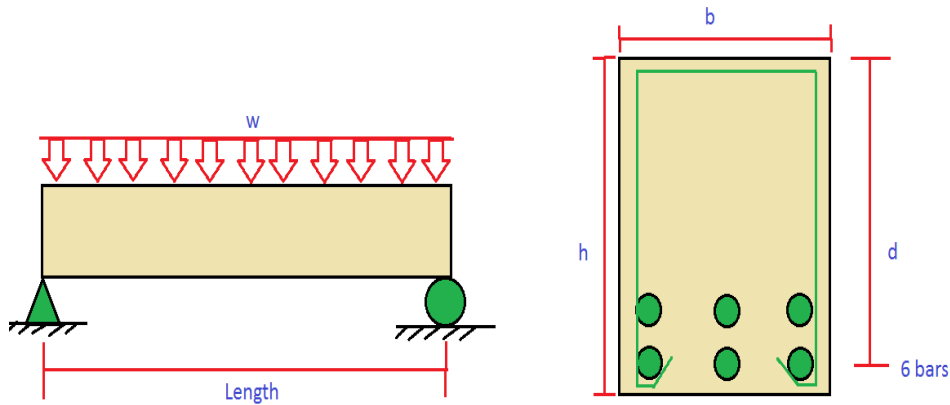
$\sigma_{\max}$ = maximum axial (normal or bending) stress.

c = distance from neutral axis to outer most

$$M = \frac{\sigma_{\max} I}{c} = \frac{(350 \text{ KPa})(13020.8 \text{ cm}^4)}{12.5 \text{ cm}}$$

$$M = 364 \text{ N} \cdot \text{m}$$

- 69) A 15 ft long reinforced concrete beam is subject to a dead and live load (w_D , w_L) of 1.2 kips/ft and 2.5 kips/ft respectively. The dead load does not include self-weight which is 150 lb/ft³. The beam cross section is 18 in by 28 in. The reinforcement is at a depth of 16 in. For a simple beam the max moment can be defined as $\frac{wL^2}{8}$. Determine the maximum moment on the system.



- A) 185 kip · ft B) 190 kip · ft
C) 160 kip · ft D) 170 kip · ft

The Answers is D

ACI 318-08, 10.2-10.4 & 10.5, page 129-135

Step 1: Determine the total distributed dead load on the system by summing the given dead load and the self-weight.

$$w_{\text{self}} = \left(150 \frac{\text{lb}}{\text{ft}^3}\right) \left(\frac{18}{12} \text{ft}\right) \left(\frac{28}{12} \text{ft}\right) \left(\frac{1 \text{kip}}{1000 \text{lb}}\right) = 0.525 \frac{\text{kip}}{\text{ft}}$$

$$w_D = 1.2 \frac{\text{kip}}{\text{ft}} + .525 \frac{\text{kip}}{\text{ft}} = 1.725 \frac{\text{kip}}{\text{ft}}$$

Now determine the total factored load on the beam.

Continuing:

To do this we need to consider a set of LRFD load factors. Find the LRFD load factors as designated by ASCE. For live and dead we can take the load combination of $1.2D + 1.6L + 0.5(L_r, S, R)$. The last factor represents live roof load snow or rain load and our system does not consider these so use $1.2D + 1.6L$. Substitute the distributed dead load (w_D) and distributed live load (w_L) in order to get the ultimate distributed load (w_u).

$$w_u = 1.2w_D + 1.6w_L = 1.2 \left(1.725 \frac{\text{kip}}{\text{ft}}\right) + 1.6 \left(2.5 \frac{\text{kip}}{\text{ft}}\right) = 6.07 \frac{\text{kip}}{\text{ft}}$$

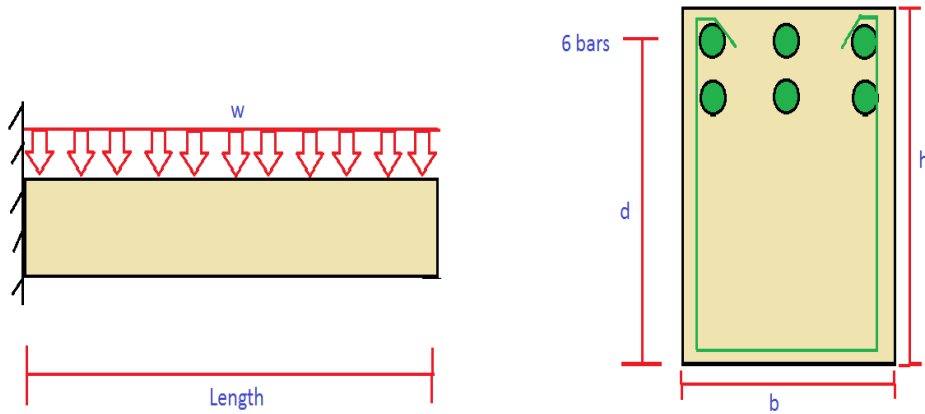
Step 2: Calculate the maximum moment using the given equation. Sub in the ultimate distributed load that you calculated.

$$M_u = \frac{wL^2}{8} = \frac{\left(6.07 \frac{\text{kip}}{\text{ft}}\right) (15 \text{ft})^2}{8} = 170.7 \text{kip} \cdot \text{ft}$$

Information on the depth of the reinforcement was not needed.

~170 kip · ft

- 70) A 15 ft long reinforced concrete beam is subject to a dead and live load (w_D , w_L) of 1.5 kips/ft and 3.5 kips/ft respectively. This dead load does not consider self-weight. The weight of concrete is 150 lb/ft³. The beam cross section is 18 in by 28 in. $f_y = 60000 \text{ psi}$; $f'_c = 4000 \text{ psi}$. The maximum moment for a cantilever beam is $\frac{wL^2}{2}$. Determine the maximum moment on the beam.



- A) 1000 kip · ft
B) 850 kip · ft
C) 900 kip · ft
D) 950 kip · ft

The Answer is C**ACI 318-08, 10.2-10.4 & 10.5, page 129-135**

Step 1: First determine the total dead load and then factor the loads to get the ultimate distributed load. The total dead load will be the load given plus the self weight of the beam. Multiply the concrete weight by the dimensions of the cross sectional area to get the distributed load over the beam.

$$w_{\text{self}} = \left(150 \frac{\text{lb}}{\text{ft}^3}\right) \left(\frac{18}{12} \text{ft}\right) \left(\frac{28}{12} \text{ft}\right) \left(\frac{1 \text{kip}}{1000 \text{lb}}\right) = .525 \frac{\text{kips}}{\text{ft}}$$

$$w_D = 1.5 \frac{\text{kips}}{\text{ft}} + .525 \frac{\text{kips}}{\text{ft}} = 2.025 \frac{\text{kips}}{\text{ft}}$$

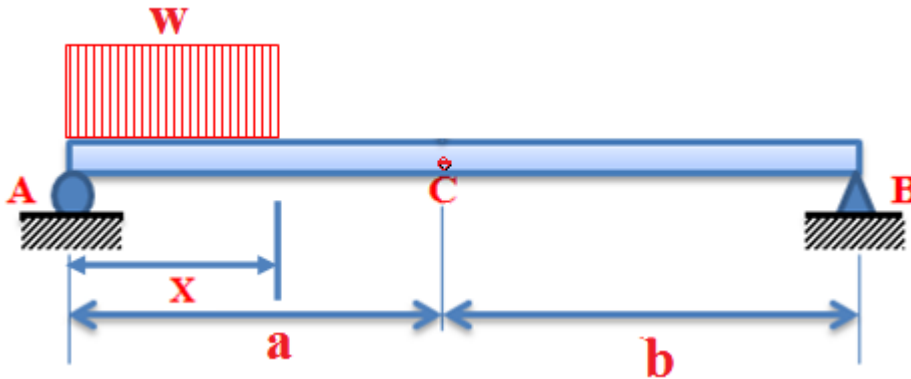
Now find the ultimate load. To do this we need to consider a set of LRFD load factors. Use the appropriate LRFD load factors as designated by ASCE. For live and dead we can take the load combination of $1.2D + 1.6L + 0.5(L_r, S, R)$. The last factor represents live roof load snow or rain load and our system does not consider these so use $1.2D + 1.6L$. Substitute the distributed dead load (w_D) and distributed live load (w_L) in order to get the ultimate distributed load (w_u).

$$w_u = 1.2w_D + 1.6w_L = 1.2 \left(2.025 \frac{\text{kip}}{\text{ft}}\right) + 1.6 \left(3.5 \frac{\text{kip}}{\text{ft}}\right) = 8 \frac{\text{kips}}{\text{ft}}$$

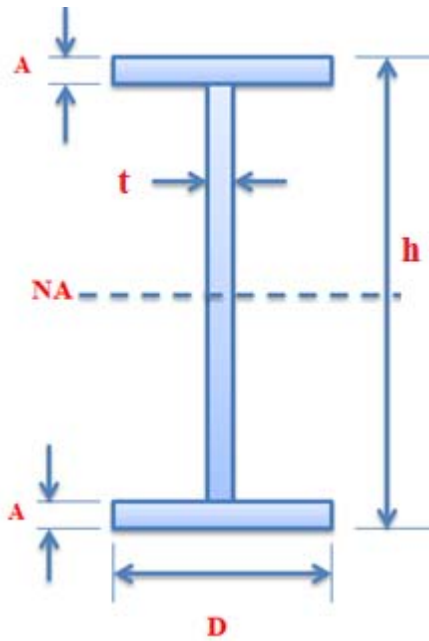
Step 2: Use the given equation to determine the max moment on the beam.

$$M_{\text{max}} = \frac{wL^2}{2} = \frac{\left(8 \frac{\text{kips}}{\text{ft}}\right)(15 \text{ft})^2}{2} = 900 \text{ kip} \cdot \text{ft} \quad 900 \text{ kip} \cdot \text{ft}$$

- 71) What is the maximum bending stress in the beam cross section at point C? $W = 10 \text{ kN/m}$, $X = 2 \text{ m}$, $a = 4 \text{ m}$ and $b = 6 \text{ m}$.



The beam cross section is shown as below, $A = 50 \text{ mm}$, $t = 60 \text{ mm}$, $D = 400 \text{ mm}$ and $h = 500 \text{ mm}$.



- (A) 25 MPa (B) 12 MPa (C) 3.9 KPa (D) 1.27 MPa

The Answers is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-23**Step 1: Find the reactions at A and B:

$$\Sigma M (B) = 0 \Rightarrow R_A (10) - 10 (2) (9) = 0$$

$$R_A = 180/10 = 18 \text{ KN}$$

$$\Sigma F_y = 0 \Rightarrow R_A - w(2) + R_B = 0 \Rightarrow 18 - 10(2) + R_B = 0$$

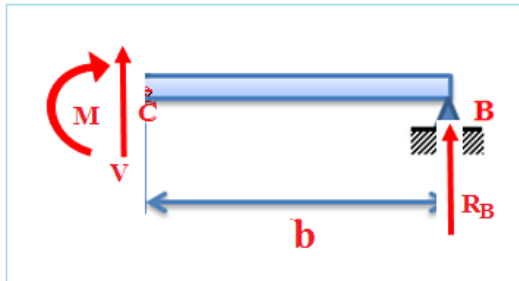
$$R_B = 2 \text{ KN}$$

Step 2: Find the moment at C:

$$\Sigma M (V) = 0$$

$$- M + R_B(b) = 0$$

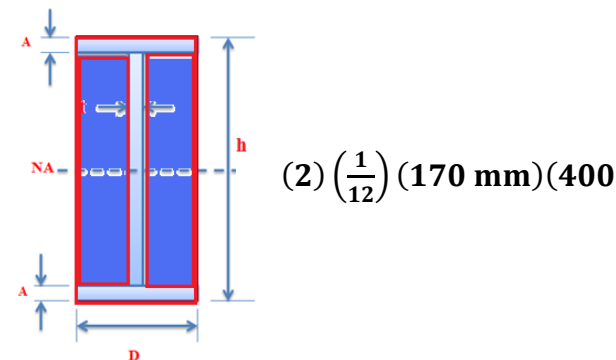
$$M = 2 (6) = 12 \text{ KN}\cdot\text{m}$$

Step 3: Find the moment of inertia about the neutral axis, NA:

$$I = I (\text{Red rectangle}) - 2 I (\text{Blue areas})$$

$$I = \left(\frac{1}{12} \right) (400 \text{ mm})(500 \text{ mm})^3 -$$

$$I = 23.54 \times 10^8 \text{ mm}^4$$

Step 4: Find the maximum Stress:

$$c = \left(\frac{1}{2} \right) h \quad c = 250 \text{ mm},$$

$$\sigma_{\max} = \frac{Mc}{I} = \frac{(12 \text{ kN} \cdot \text{m})(0.25 \text{ m})}{23.54 \times 10^{-4} \text{ m}^4}$$

$$\sigma_{\max} = 1274.4 \frac{\text{KN}}{\text{m}^2} \quad (1.27 \text{ MPa})$$

D. Shear

- 72) Which of the following choice will not affect the value of design shear strength ($\phi_v V_n$) of a steel beam?
- | | |
|--------------------------|---------------------|
| A) Yield stress of steel | B) Depth of section |
| C) Thickness of web | D) Length of span |

The Answer is D

AISC Manual, 13th edition, 2010, 16.1, page 65

Step 1: $\phi_v V_n = \phi_v 0.6 F_y (d t_w)$

From this equation, yield stress of steel (F_y), depth of section (d) and thickness of web (t_w) are all shown except for length of span. Thus, length of span will not affect the value of design shear strength.

73) A beam with a circular cross section of radius 0.1 m is subject to a shear force of 100 kN. Determine the transverse shear stress on the top of the beam.

- (A) 0
- (B) 2 MPa
- (C) 4 MPa
- (D) 8 MPa

The Answer is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-27

The transverse shear stress in a beam due to shear force can be calculated as:

$$\tau_{xy} = \frac{vQ}{Ib} \quad \text{where} \quad Q = A'\bar{y}'$$

A' is the area above the layer (or plane) upon which the desired transverse shear stress acts.

On the top of the beam, $A' = 0$. Then,

$$\tau_{xy} = 0$$

74) A beam with a circular cross section of radius 0.1 m is subject to a shear force of 100 kN. Determine the maximum transverse shear stress.

- (A) 8.5 MPa
- (B) 0
- (C) -4.25 MPa
- (D) 4.25 MPa

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-27

The transverse shear stress in a beam due to shear force can be calculated as:

$$\tau_{xy} = \frac{VQ}{Ib} \quad \text{where} \quad Q = A'\bar{y}'$$

A' is the area above the layer (or plane) upon which the desired transverse shear stress acts.

$\bar{y}'$ is the distance from the neutral axis to the centroid of area A' .

Here, the maximum shear stress occurs on the neutral axis of the circular cross section:

$$\tau_{max} = \frac{4V}{3A} = \frac{4(100 \text{ KN})}{3(\pi r^2)} = 4246.3 \text{ KN/m}^2$$

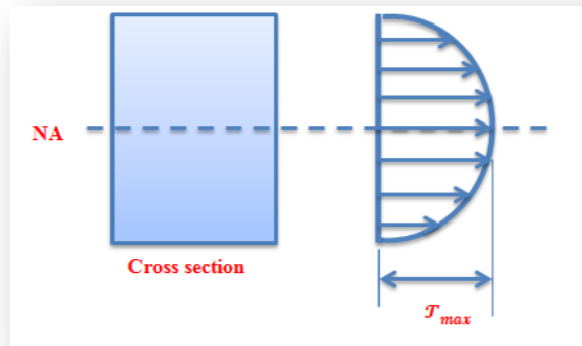
$$\tau_{max} = 4.25 \text{ MPa}$$

- 75) Which statement is true for a beam with a rectangular cross section under transverse (shear) loading?
- (A) The shear stress is uniform.
 - (B) The shear stress distribution changes as a straight line.
 - (C) The shear stress is maximum at the neutral axis and zero at edges.
 - (D) The shear stress is maximum at the edges and zero at neutral axis.

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-27

The shear distribution diagram in a cross section shows that the shear stress is maximum at the neutral axes and zero at edges as shown in figure below.



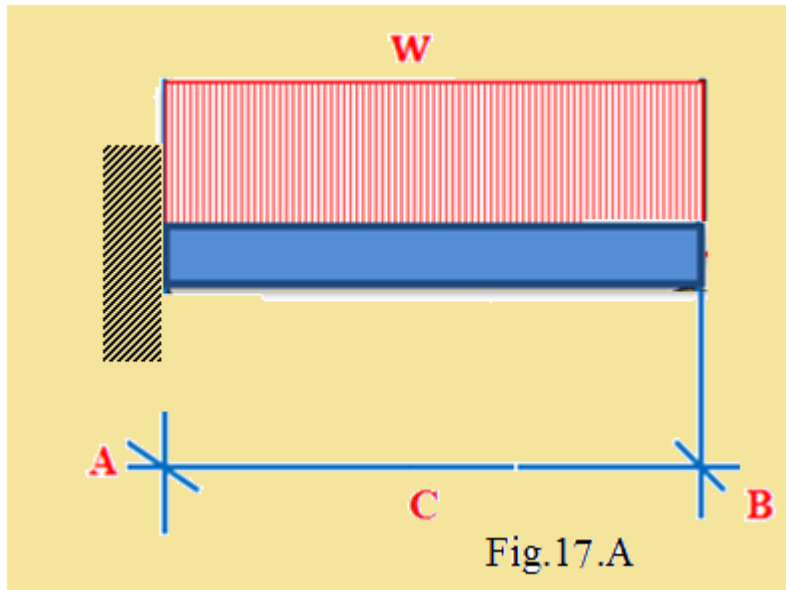
Answer “A” is not correct: The shear stress varies on the cross section.

Answer “B” is not correct: The shear stress distribution changes as a quadratic function.

Answer “C” is correct: The stress is maximum at the neutral axis and zero at edges.

Answer “D” is not correct: The stress is maximum at the neutral axis and zero at edges.

- 76) A 4 m cantilever beam is loaded with distributed load of 1 kN/m. If the beam had cross sectional dimensions (10 cm wide by 25 cm deep). Find the maximum shearing stress in at the support.



- (A) $\tau_{\max} = 260 \text{ kPa}$
(B) $\tau_{\max} = 270 \text{ kPa}$
(C) $\tau_{\max} = 240 \text{ kPa}$
(D) $\tau_{\max} = 380 \text{ kPa}$

The Answers is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-27****Step 1:** At a section right at the support find the shear force.

$$\begin{aligned}\sum F_y &= 0 \\ 0 &= -(1000 \text{ N})(4 \text{ m}) + V \\ V &= 4000 \text{ N}\end{aligned}$$

Step 2: Find Moment of inertia, I , A' = Area above the plane where the shear acts and $Q = A' y'_c$ (y'_c = Distance from neutral axis to centroid of A'):

$$I = \left(\frac{1}{12}\right) b(h^3)$$

$$= \left(\frac{1}{12}\right)(10 \text{ cm})(25 \text{ cm})^3$$

$$I = 13,020.8 \text{ cm}^4 = 0.0001302 \text{ m}^4$$

$$Q = A' y'_c$$

$$A' = \left(\frac{h}{2}\right)(b) = (0.125)(.1) = 0.0125 \text{ m}^2$$

$$y'_c = 0.0625 \text{ m}$$

$$Q = 0.0125 \text{ m}^2 (0.0625 \text{ m}) = 0.0007812 \text{ m}^3$$

Step 3: Find the shear stress:

$$\tau_{xy} = \frac{VQ}{(Ib)} = \frac{(4000 \text{ N})(0.0007812 \text{ m}^3)}{(0.0001302 \text{ m}^4)(0.1 \text{ cm})}$$

$$\tau_{xy} = 240,000 \frac{\text{N}}{\text{m}^2} = 240 \text{ KPa}$$

- 77) A load $P = 2 \text{ kN}$ placed at the middle of a 4 m long beam shown below. Calculate the value of the maximum shear stress and its location if the beam's cross section is 20 cm by 20 cm.

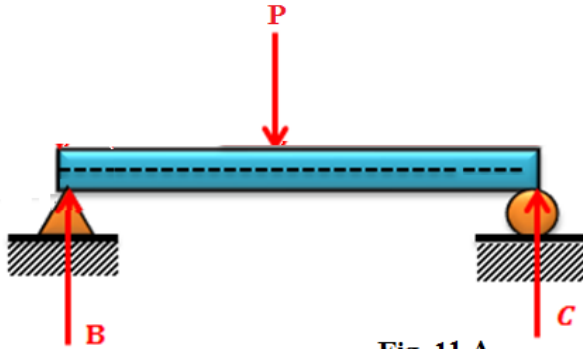


Fig. 11.A

- (A) 10 KPa at the ends. (B) 25 KPa at the center.
(C) 10 KPa constant in all length (D) 25 KPa constant in all length

The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-27

Step 1: From the shear diagram, for the concentrated loaded beam shear value is constant with variable sign of positive at right and negative at the left.

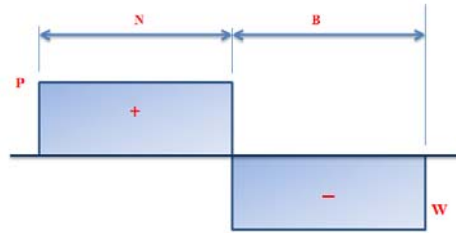
Step 2: Determine the Reactions at the supports and shear:
from symmetry

$$R = \frac{P}{2}$$

$$R = 1 \text{ KN}$$

$$\sum F_y = 0$$

$$V = 1 \text{ KN}$$

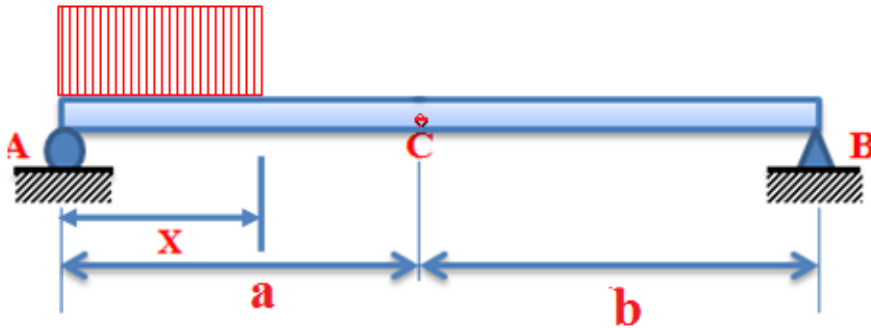


Step 3: Find shear stress, τ_{ave}

$$\tau_{ave} = \frac{V}{A} = \frac{1000 \text{ N}}{0.2 \text{ m} \times 0.2 \text{ m}}$$

$$\tau_{ave} = 25000 \text{ Pa or } 25 \text{ KPa}$$

- 78) An simply supported beam is loaded with load $w = 2 \text{ KN/m}$ as shown. Find the maximum shear stress at point C. the cross section is a rectangle of $(40 \times 60) \text{ cm}$, $x = 2 \text{ m}$, $a = 4 \text{ m}$, and $b = 4 \text{ m}$

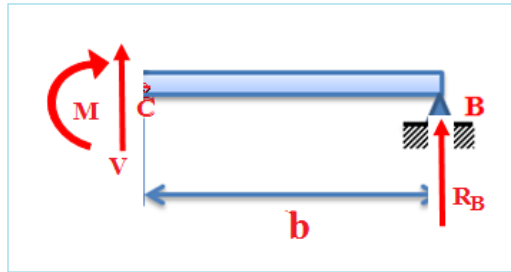


- (A) 3.125 KPa (B) 37 KPa (C) 400 KPa (D) 85 KPa

The Answers is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-27**Step 1: Find the reactions:

$$\sum M_A = -\frac{wx^2}{2} + R_B(a + b) = 0$$

$$R_B = 0.5 \text{ kN}$$

Step 2: Find the shear force at point C:

$$\sum F_y = 0$$

$$|V_C| = R_B = 0.5 \text{ kN}$$

Step 3: Now find the maximum shear stress

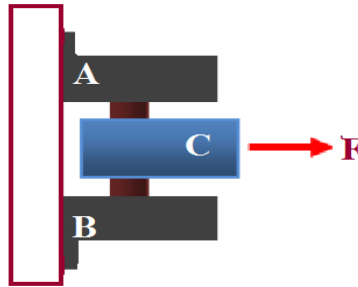
for solid rectangular section :

$$\tau = \frac{3V}{2A} = \frac{3 \times 0.5 \text{ kN}}{2(0.4 \text{ m} \times 0.6 \text{ m})}$$

$$\tau = 3.125 \text{ kN/m}^2$$

- 79) The rod C is attached to the fixed plate on points A and B by bolts. each nut had 12.5 mm diameter and a height of 11 mm. Find force F, if the ultimate strength of the nut material in shear is **103 MPa**, and safety factor =2.5?

- (A) 15.5 kN
(B) 17.8 kN
(C) 5.2 kN
(D) 45.2 kN



The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 48.11, page 48-58

Step 1: Determine the allowable shear stress from the safety factor equation:

$$\mathcal{T}_{\text{allowable}} = \frac{\mathcal{T}}{\text{SF}} \quad \mathcal{T}_{\text{allowable}} = \frac{103\,000 \frac{\text{kN}}{\text{m}^2}}{2.5} = 41\,200 \frac{\text{kN}}{\text{m}^2}$$

Step 2: Determine the allowable shear force on each nut using the shear stress equation in threads
(Note that the shear area in thread can be found by $\left(\frac{1}{2}\pi dh\right)$):

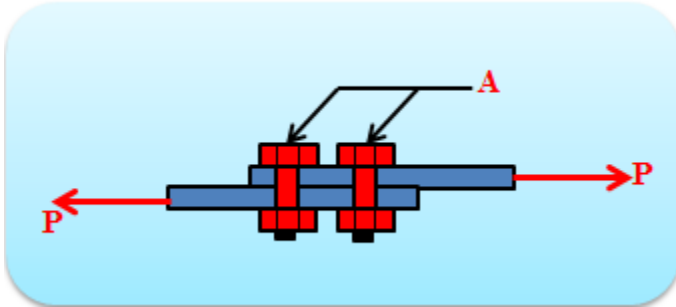
$$V = \mathcal{T}_{\text{allowable}} A = \mathcal{T}_{\text{allowable}} \left(\frac{1}{2}\pi dh\right)$$

$$V = \left(41\,200 \frac{\text{kN}}{\text{m}^2}\right) \left(\frac{1}{2}\pi\right) (0.0125 \text{ m})(0.011 \text{ m}) = 8.9 \text{ kN}$$

Step 3: Determine the Force F: the force F is supported by two nuts:

$$F = 2V = 2 \times 8.9 = 17.8 \text{ kN}$$

- 80) Find the cross-sectional area for the following bolts to resist the tensile force $P=30$ kN, if the bolt shear strength is 20 MPa. Use factor of safety of 1.4.



- (A) 1.5 cm^2 (B) 22.05 mm^2 (C) 8.05 mm^2 (D) 1.05 mm^2

The Answers is d**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 48.11, page 48-58**Step 1: Find the ultimate load using the safety factor equation:

$$P_{ult} = P \times SF$$

$$P_{ult} = 30\text{KN} \times 1.4 = 42 \text{ KN}$$

Step 2: Find the required area using the stress equation:

$$\sigma = \frac{P}{A} \quad \text{Solve for A} \rightarrow A = \frac{P}{\sigma}$$

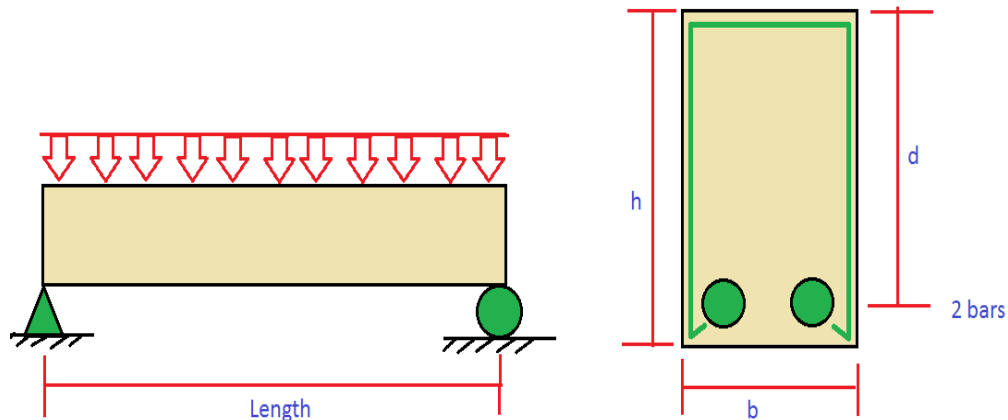
$$A = \frac{42 \text{ KN}}{20 \times 10^6 \frac{\text{KN}}{\text{m}^2}}$$

$$A = 2.1 \times 10^{-6} \text{m}^2 \quad \text{or} \quad A = 2.1 \times 10^{-6} \text{m}^2 \times \frac{10^6 \text{mm}^2}{1 \text{m}^2}$$

$$A = 2.1 \text{ mm}^2$$

$$\text{Area of each bolt} = A/2 = 2.1/2 = 1.05 \text{mm}^2$$

- 81) A 10 ft long simply supported concrete beam subject to a uniformly distributed load is reinforced with two #6 bars at a depth of 13 in. The cross section is 6 in wide and 16 in tall. the beam is subject to a 0.75 kip/ft live load and weighs 150 lb/ft³. $f_y = 60000$ psi, $f'_c = 4000$ psi. Determine the whether or not shear reinforcement is required and where it is required.



- A) Required at ends of this beam
- B) required at the mid-span
- C) Not required on this beam
- D) required on the entire beam but possibly more at both ends

The Answers is D

ACI 318, 11.2-11.4 & 11.4.6., Page 158-164

Short cut answer:

Technically according to 11.4.6. all reinforced concrete beam should have shear bars at the entire element, however shear force is maximized at both ends in the simple beams, so “D” is the answer.

Solution:

Step 1: First we need to determine the shear strength provided by the concrete. If the shear strength in the concrete is adequate then the beam will not require shear reinforcement. Use appropriate equation for defining the shear capacity of beams.

$$V_c = 2b_w d \sqrt{f'_c} = 2(6 \text{ in})(13 \text{ in}) \sqrt{4000 \text{ psi}} \left(\frac{1 \text{ kip}}{1000 \text{ lb}} \right) = 9.87 \text{ kips}$$

Step 2: Determine the maximum shear force carried by the beam and compare with the shear capacity of the concrete.

To do this we need to consider the self-weight as dead load and the given live load. Consider a set of LRFD load factors. For live and dead we can take the load combination of $1.2D + 1.6L + 0.5(L_r, S, R)$. The last factor represents live roof load snow or rain load and our system does not consider these so use $1.2D + 1.6L$. Substitute the distributed dead load (D) and distributed live load (L) in order to get the ultimate distributed load (U).

$$D = \left(150 \frac{\text{lb}}{\text{ft}^3} \right) \left(\frac{6}{12} \text{ ft} \right) \left(\frac{16}{12} \text{ ft} \right) \left(\frac{1 \text{ kip}}{1000 \text{ lb}} \right) = 0.1 \frac{\text{kip}}{\text{ft}}$$

$$U = 1.2 \left(0.1 \frac{\text{kip}}{\text{ft}} \right) + 1.6 \left(0.75 \frac{\text{kip}}{\text{ft}} \right) = 1.32 \frac{\text{kip}}{\text{ft}}$$

The max shear will be at either end of the beam. Either side will carry half of the total shear. The shear on the beam is simply the length times the distributed load we calculated.

$$V_u = \left(\frac{1}{2} \right) (10 \text{ ft}) \left(1.32 \frac{\text{kip}}{\text{ft}} \right) = 6.6 \text{ kips}$$

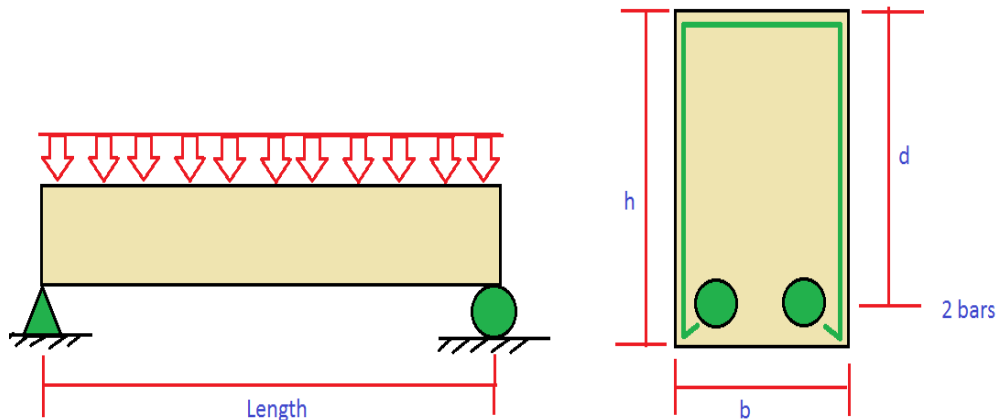
Use the appropriate equation to determine if shear reinforcement is required. according to the resistance factors table a factor of 0.75 is used for shear and torsion.

$$\frac{\phi V_c}{2} = \frac{(0.75)(9.87 \text{ kips})}{2} = 3.7 \text{ kips} < 6.6 \text{ kips}$$

Shear reinforcement is required at the supports. At the mid-span of the beam the shear will be essentially 0 and does not need reinforcement, so minimum reinforcement according to 11.4.6. shall be provided.

Required at the ends of the beam

- 82) A 12 ft long simply supported concrete beam subject to a uniformly distributed load is reinforced with two #6 bars at a depth of 18 in. The cross section is 12 in wide and 20 in tall. The beam is subject to a 0.7 kip/ft live load and weighs 150 lb/ft³. $f_y = 60000$ psi, $f'_c = 4000$ psi. Determine the whether or not shear reinforcement is required and where it is required.



- A) Required at ends of this beam
- B) Not required at all
- C) Not required on this beam so minimum shear reinforcement shall be provided
- D) Required on the entire beam

The Answers is C

ACI 318, 11.2-11.4 & 11.4.6., Page 158-164

Step 1: First we need to determine the shear strength provided by the concrete. If the shear strength in the concrete is adequate then the beam will not require shear reinforcement. Use the appropriate equations defining the shear capacity of beams.

$$V_c = 2b_w d \sqrt{f'_c} = 2(12 \text{ in})(18 \text{ in}) \sqrt{4000 \text{ psi}} \left(\frac{1 \text{ kip}}{1000 \text{ lb}} \right) = 27.3 \text{ kips}$$

Step 2: Determine the maximum shear force carried by the beam and compare with the shear capacity of the concrete.

To do this we need to consider the self-weight as dead load and the given live load. Consider a set of LRFD load factors. Check the LRFD load factors as designated by ASCE. For live and dead we can take the load combination of $1.2D + 1.6L + 0.5(L_r, S, R)$. The last factor represents live roof load snow or rain load and our system does not consider these so use $1.2D + 1.6L$. Substitute the distributed dead load (D) and distributed live load (L) in order to get the ultimate distributed load (U).

$$D = \left(150 \frac{\text{lb}}{\text{ft}^3} \right) \left(\frac{12}{12} \text{ ft} \right) \left(\frac{20}{12} \text{ ft} \right) \left(\frac{1 \text{ kip}}{1000 \text{ lb}} \right) = 0.25 \frac{\text{kip}}{\text{ft}}$$

$$U = 1.2 \left(0.25 \frac{\text{kip}}{\text{ft}} \right) + 1.6 \left(0.7 \frac{\text{kip}}{\text{ft}} \right) = 1.42 \frac{\text{kip}}{\text{ft}}$$

The max shear will be at either end of the beam. Either side will carry half of the total shear. The shear on the beam is simply the length times the distributed load we calculated.

$$V_u = \left(\frac{1}{2} \right) (12 \text{ ft}) \left(1.42 \frac{\text{kip}}{\text{ft}} \right) = 8.52 \text{ kips}$$

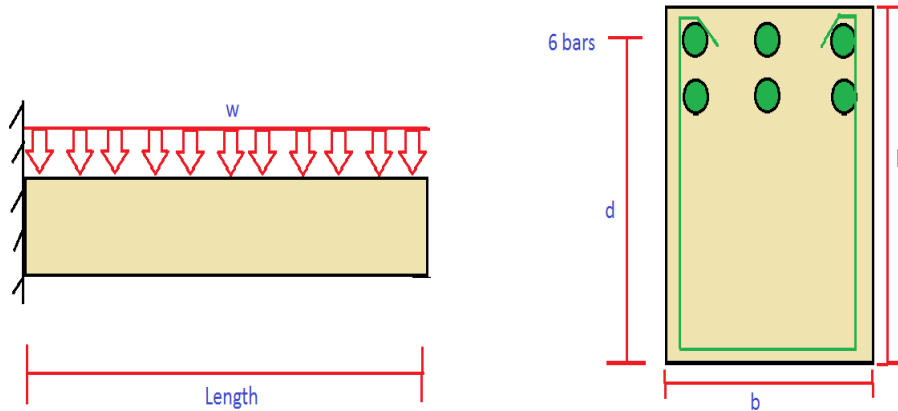
Use appropriate relation to determine if shear reinforcement is required. According to the resistance factors table a factor of 0.75 is used for shear and torsion.

$$\frac{\phi V_c}{2} = \frac{(0.75)(27.3 \text{ kips})}{2} = 10.24 \text{ kips} > 8.52 \text{ kips}$$

No shear reinforcement is required. So use minimum reinforcement according to 11.4.6.

Not required on this beam, So use minimum reinforcement according to 11.4.6.

- 83) A 15 ft long reinforced concrete beam is subject to a dead and live shear of 4 kips and 10 kips respectively. The beam is 18 in by 28 in with a reinforcement depth of 25 in. $f_y = 60000$ psi; $f'_c = 4000$ psi. Determine whether or not this beam requires shear reinforcement.



- A) Shear reinforcement is not needed so minimum reinforcement shall be used.
- B) Shear reinforcement required at the fixed end and mid-span
- C) Shear reinforcement required at the fixed end
- D) Shear reinforcement required at the free end and mid-span

The Answer is A

ACI 318, 11.2-11.4 & 11.4.6., Page 158-164

Short cut answer:

Technically according to 11.4.6. all reinforced concrete beam should have shear bars at the entire element, however shear force is maximized at both ends in the simple beams, so “D” is the answer.

Solution:

Step 1: First factor the given shear to determine ultimate shear. Consider a set of LRFD load factors. Use appropriate LRFD load factors as designated by ASCE. For live and dead we can take the load combination of $1.2D + 1.6L + 0.5(L_r, S, R)$. The last factor represents live roof load snow or rain load and our system does not consider these so use $1.2D + 1.6L$.

$$V_u = 1.2D + 1.6L = 1.2(4 \text{ kips}) + 1.6(10 \text{ kips}) = 20.8 \text{ kips}$$

Step 2: Determine the shear the concrete is able to carry and compare to the ultimate shear that we calculated. Use the appropriate equations for shear

$$V_c = 2b_w d \sqrt{f'_c} = 2(18 \text{ in})(25 \text{ in})\sqrt{(4000 \text{ psi})} \left(\frac{1 \text{ kip}}{1000 \text{ lb}} \right) = 56.9 \text{ kips}$$

If the ultimate shear is less than $\frac{\phi V_c}{2}$ then no shear reinforcement will be needed. a resistance factor needs to be applied and can be found in the resistance factors table. For shear use a factor of 0.75.

$$\frac{\phi V_c}{2} = \frac{(0.75)(56.9 \text{ kips})}{2} = 21.3 \text{ kips}$$

$$21.3 \text{ kips} > 20.8 \text{ kips}$$

Shear reinforcement is not needed, therefore minimum reinforcements according to 11.4.6. shall be provided.

E. Axial (e.g., forces and stresses)

84) There are two bars made with the same material, having the same cross sections and supporting the same tension forces. Bar A has a length of L and bar B has a length of $2L$. The elongation in bar A is equal to

- (A) the elongation in bar B.
- (B) 4 times of the elongation in bar B.
- (C) twice of the elongation in bar B.
- (D) half of the elongation in bar B.

The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.3, page 46-(5-10)

From the the equation of the Engineering Strain

$$\epsilon = \Delta L / L_0$$

Where ϵ = Engineering Strain

ΔL = Change in Length of the member

L_0 = Original Length of the member

For the two bars A and B have the same material (E is constant), same tension force and the same cross-sectional area so therefore the two bars have the same stress ($\sigma = P/A$) and strain ($\epsilon = \delta/L$).

$$\therefore \epsilon_A = \epsilon_B$$

$$\therefore (\delta/L)_A = (\delta/L)_B$$

$$L_A = L, L_B = 2L$$

$$\therefore \delta_A/L = \delta_B/2L$$

$$\therefore \delta_A = \delta_B/2$$

$\therefore$ Answer D is the correct statement

85) There are two bars made with the same material, having the same cross sections and supporting the same tension forces. The elongation in bar A is 3 times of the elongation in bar B. Which of the following statements is correct?

- (A) the strain in bar A is 3 times of the strain in bar B.
- (B) the strain in bar A is $1/3$ of the strain in bar B.
- (C) the length of bar A is 3 times of the length of bar B.
- (D) the length of bar A is $1/3$ of the length of bar B.

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.3, page 46-(5-10)**

From the equation of the Engineering Strain

$$\epsilon = \Delta L / L_0$$

Where ϵ = Engineering Strain

ΔL = Change in Length of the member

L_0 = Original Length of the member

For the two bars A and B have the same material (E is constant), same tension force and the same cross-sectional area so therefore the two bars have the same stress ($\sigma = P/A$) and strain ($\epsilon = \delta/L$).

$$\therefore \epsilon_A = \epsilon_B$$

$$\therefore (\delta/L)_A = (\delta/L)_B$$

$$\delta_A = 3 \times \delta_B$$

$$\therefore 3 \times \delta_B / L_A = \delta_B / L_B$$

$$\therefore 3 / L_A = 1 / L_B$$

$$\therefore L_A = 3 L_B$$

$\therefore$ Answer C is the correct statement

86) For a circular steel bar under an axial compression force, which of the following statements is false?

- (A) The normal strain is negative.
- (B) The percent elongation is negative.
- (C) The cross sectional area of the deformed bar is reduced.
- (D) The diameter of the deformed bar is increased.

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.3, page 46-(5-10)

From the equation of the Engineering Strain

$$\epsilon = \Delta L / L_o$$

Where ϵ = Engineering Strain

ΔL = Change in Length of the member

L_o = Original Length of the member

Also From the equation of Percent Elongation

$$\% \text{ Elongation} = \Delta L / L_o \times 100$$

Answer (A) is correct since as it is a compressive force the steel bar will be compressed causing a negative strain.

Answer (B) is correct since as it is a compressive force the steel bar will be shortened which cause negative elongation .

Answer (C) is incorrect since as a result of the compressive force the diameter of the bar will be increased causing increase in area.

Answer (D) is correct since as a result of the compressive force the diameter of the bar will be increased.

87) A pipe is subjected to a tension load; the inner and outer diameters of the unloaded pipe are $d_1 = 3.6$ in and $d_2 = 5.0$ in respectively. The inner and outer diameters of the loaded pipe are $d'_1 = 3.5998$ in and $d'_2 = 4.9996$ in, respectively. What is the most accurate percent reduction in area?

- (A) 0.01% (B) 0.02% (C) 0.001% (D) 0.002%

The Answers is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.3, page 46-(5-10)****Step (1):**

The Percent Reduction in Area from initial area A_i to final area A_f given by:

$$\% \text{ RA} = \left(\frac{A_i - A_f}{A_i} \right) \times 100$$

Step (2): Get the initial and final areas:

$$A_{\text{initial}} = \pi(D_o^2/4) - \pi(D_i^2/4)$$

$$A_{\text{initial}} = \pi(5^2/4) - \pi(3.6^2/4)$$

$$A_{\text{initial}} = 3.01\pi \text{ in}^2$$

$$A_{\text{Final}} = \pi(D_o^2/4) - \pi(D_i^2/4)$$

$$A_{\text{Final}} = \pi(4.9996^2/4) - \pi(3.5998^2/4)$$

$$A_{\text{Final}} = 3.0094\pi \text{ in}^2$$

Step (3): From the given equation:

$$\% \text{ RA} = \left(\frac{A_i - A_f}{A_i} \right) \times 100$$

$$\% \text{ RA} = \left(\frac{3.01\pi - 3.0094\pi}{3.01\pi} \right) \times 100$$

$$\% \text{ RA} = 0.02 \%$$

- 88) If an element is subject to uniaxial tension, the material failure will occur when the tensile stress exceeds the tensile strength according to the maximum-normal-stress theory. Consider a bar having the tensile strength of 200 MPa. If it is subject to uniaxial tension, which of the following tensile stresses will result in the material failure according to the maximum-normal-stress theory?
- (A) 70 MPa
 - (B) 90 MPa
 - (C) 110 MPa
 - (D) None of the above.

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.3, page 46-(10-12)

Step 1: According to the maximum-normal-stress theory, the material failure occurs whenever the greatest positive principal stress exceeds the tensile strength or whenever the greatest negative principal stress exceeds the compressive strength. If a member is subject to uniaxial tension, material failure occurs whenever

$$\sigma \geq S_{ut}$$

Step 2: In this problem, $S_{ut} = 200 \text{ MPa}$.

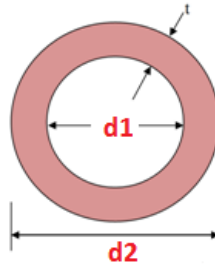
(A) $\sigma(70\text{MPa}) < S_{ut}(200 \text{ MPa})$

(B) $\sigma(90\text{MPa}) < S_{ut}(200 \text{ MPa})$

(C) $\sigma(110\text{MPa}) < S_{ut}(200 \text{ MPa})$

None of them will result in material failure

- 89) A metal pipe shaped column with outer diameter of 10 m, as shown, is subjected to an axial loading of 400 kN. If the allowable compressive stress is **14.4 kPa**. Find the most appropriate wall thickness, t .



- (A) **0.95 m** (B) **0.8 m** (C) **1 m** D) 2.2m

The Answers is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.3, page 46-(10-12)

Step 1: Calculate the cross sectional area according the load.

For axial stress,

$$\sigma = \frac{F}{A}$$

$$\rightarrow A = \frac{F}{\sigma} \rightarrow A = 400 / (14.4)$$

$$A = 27.77 \text{ m}^2$$

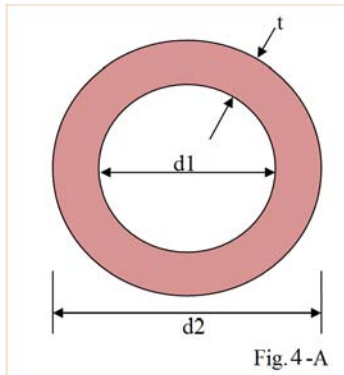
Step 2: Calculate the inner diameter and then thickness t:

$$A = \pi / 4 (d_2^2 - d_1^2) \rightarrow d_1 = 8.04$$

Therefore,

$$t = \frac{d_2 - d_1}{2} = \frac{10 \text{ m} - 8.04 \text{ m}}{2} = 0.98 \text{ m say (1 m)}$$

- 90) Find the wall thickness, t for the pipe with outer diameter of 10 m, as shown in Fig.4-A, that is subjected to an axial loading of 400 kN. The allowable compressive stress is **28 kPa**.



- (A) 97 cm (B) 48 cm (C) 24 cm (D) 34 cm

The Answers is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.3, page 46-(10-12)**

Step 1: Calculate the required cross sectional area according the load from the equation of stress.

$$\sigma = \frac{F}{A} \quad \text{Solve for A} \rightarrow A = \frac{F}{\sigma}$$

$$A = 400/(28)$$

$$A = 14.286 \text{ m}^2$$

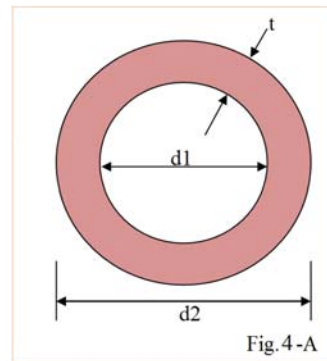
Step 2: Calculate the inner diameter and then thickness t , from the equation of area for hollow circle:

$$A = \pi/4 (d_2^2 - d_1^2) \rightarrow \text{solve for } d_1 \rightarrow d_1 = 9.04$$

Step 3: Calculate the thickness t :

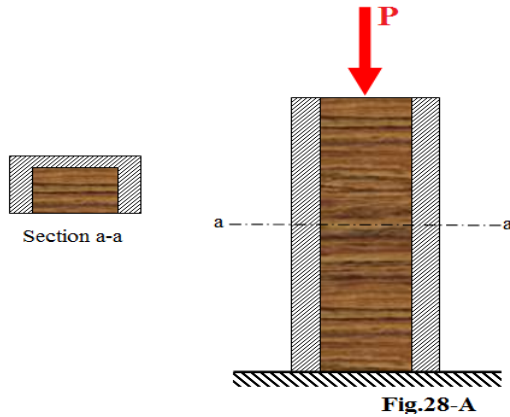
From the figure: $d_2 = d_1 + 2t$ solve for t :

$$t = \frac{d_2 - d_1}{2} = \frac{10 \text{ m} - 9.04 \text{ m}}{2} \\ = 0.48 \text{ m or } 48 \text{ cm}$$



91) On the 3 m composite column shown below load $P = 2600$ kN is placed, calculate the reduction in length of the column.

$$A_1 = 1500 \text{ mm}^2 \quad A_2 = 3000 \text{ mm}^2 \quad E_1 = 100 \text{ GPa} \quad E_2 = 20 \text{ GPa}$$



- (A) 5.2 mm
(D) 37 mm

- (B) 0.46 mm (C) 105 mm

The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.3, page 46-(5-12)

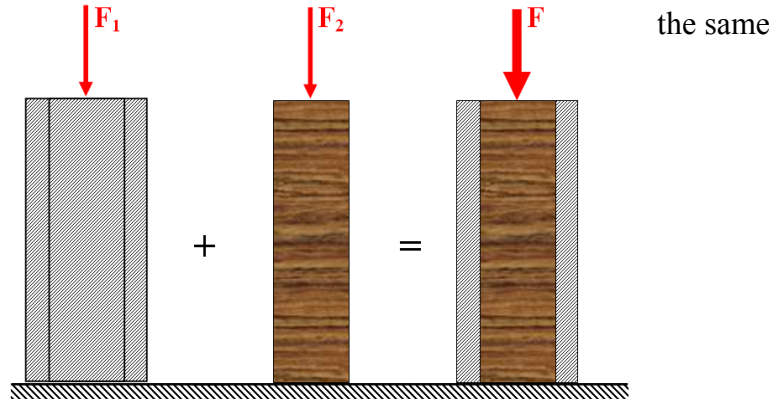
Step 1: Using the fact that both components of the column have elongation.

$$\epsilon_1 = \epsilon_2,$$

$$\epsilon_1 = \frac{\sigma_1}{E_1} \rightarrow \epsilon_1 = \frac{F_1}{A_1 E_1}$$

$$\epsilon_2 = \frac{\sigma_2}{E_2} \rightarrow \epsilon_2 = \frac{F_2}{A_2 E_2}$$

$$\frac{F_1}{A_1 E_1} = \frac{F_2}{A_2 E_2} \quad (I)$$



Step 2: From a force balance,

$$F_1 + F_2 = 2600 \text{ kN} \rightarrow F_1 = 2600 \text{ kN} - F_2 \quad (II)$$

$$I \text{ and } II \rightarrow 2600 \text{ kN} - F_2 = \left(\frac{A_1 E_1}{A_2 E_2} \right) F_2 \rightarrow 2600 \text{ kN} = \left(1 + \frac{A_1 E_1}{A_2 E_2} \right) F_2$$

$$F_2 = \frac{2600 \text{ kN}}{1 + \frac{A_1 E_1}{A_2 E_2}} \quad F_2 = \frac{2600 \text{ kN}}{1 + \frac{(1500 \text{ mm}^2)(100 \text{ GPa})}{(3000 \text{ mm}^2)(20 \text{ GPa})}}$$

$$F_2 = 742.86 \text{ kN}$$

$$\epsilon_2 = \frac{742.86 \text{ kN}}{(0.0030 \text{ m}^2)(20 \times 10^6 \frac{\text{kN}}{\text{m}^2})} \quad \epsilon_2 = 0.0124$$

Step 3: Find the reduction in length:

$$\delta = \epsilon L = 0.0124(3 \text{ m})$$

$$\delta = 0.037 \text{ m} \quad (37 \text{ mm})$$

F. Combined stresses

92) Which one shows the plastic moment capacity of the circular section if the yield stress is given as F_y and the elastic section modulus given as S_x ?

- (A) $M_p = 2/3 S_x \cdot F_y$ (B) $M_p = 3/2 S_x \cdot F_y$ (C) $M_p = 4/3 S_x \cdot F_y$ (D) $M_p = 3/4 S_x \cdot F_y$

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-24

Plastic bending Moment = $M_p = Z \cdot F_y$

Z = Plastic section modulus that for the rectangular sections is equal to $3/2 S_x$ and for circular sections = $4/3 S_x$.

S_x = Elastic section modulus = I/C

So, for the circular section: $Z = 4/3 S_x$.

Then, $M_p = 4/3 S_x \cdot F_y$

93) Which one shows the plastic moment capacity of the rectangular section if the yield stress is given as F_y and the elastic section modulus given as S_x ?

- (A) $M_p = 2/3 S_x \cdot F_y$ (B) $M_p = 3/2 S_x \cdot F_y$ (C) $M_p = 4/3 S_x \cdot F_y$ (D) $M_p = 3/4 S_x \cdot F_y$

The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-24

Plastic bending Moment = $M_p = Z \cdot F_y$

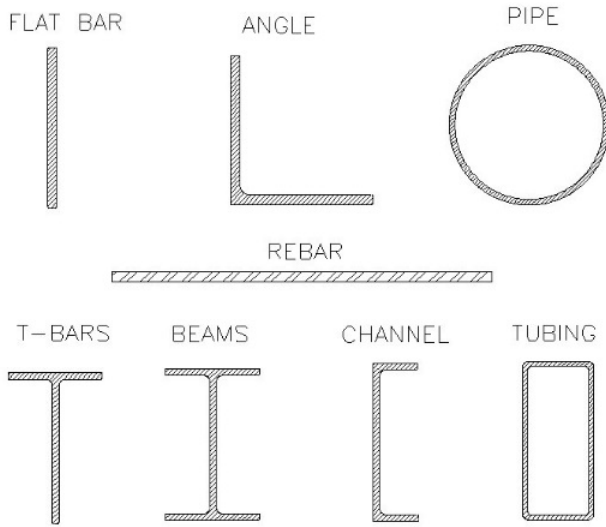
Z = Plastic section modulus that for the rectangular sections is equal to $3/2 S_x$ and for circular sections = $4/3 S_x$.

S_x = Elastic section modulus = I/C

So, for the rectangular section: $Z = 3/2 S_x$.

Then, $M_p = 3/2 S_x \cdot F_y$

94) Which one has shear center?



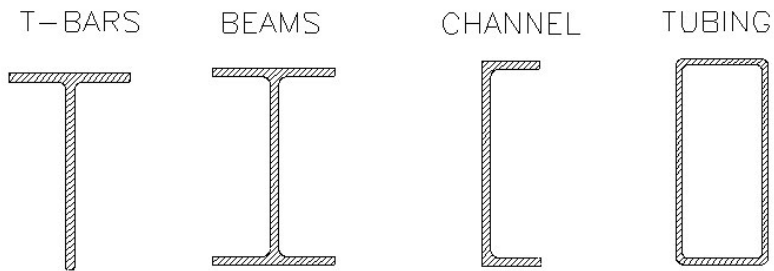
- (A) Pipe (B) Channel (C) I beams (D) Angle

The Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.7, page 46-33

According to the definition of the equation of equilibrium for stress distribution in the sections, just C type sections like channels have the center of shear.

95) Which one shows a better performance as a purlin on the sloped roofs at the same condition?

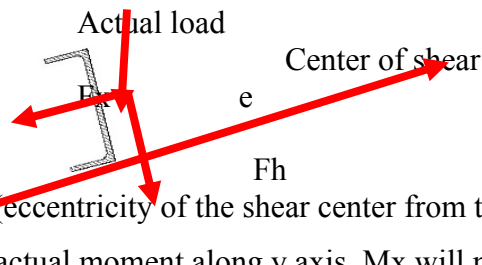


- (A) Pipe (B) Channel (C) I beams (D) Angle

The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-25

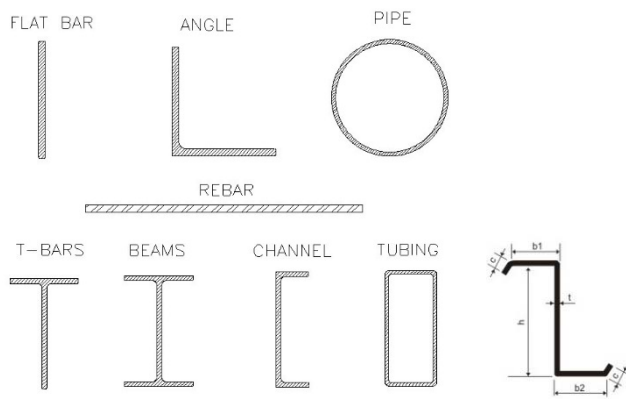
Sloped roof creates the biaxial bending moment in the purlins. This biaxial bending moment needs stronger sections. However, when a channel uses as purlin, the shear center will create a torsional bending moment in the opposite direction of the horizontal vector of the loads which necessarily reduce the amount of bending moment along the minor axis. Consequently, the results yields to smaller size of channels in compare with other sections. So, technically the channels show better behavior on the sloped roofs and that is why traditionally they are used more than I and box sections. So:



Torsional moment (upward) = $f_h * e$ (eccentricity of the shear center from the channel)

My due to Fx - Torsional moment = actual moment along y axis, Mx will not be changed.

96) Which two has more principle axis other than the orthogonal axis?



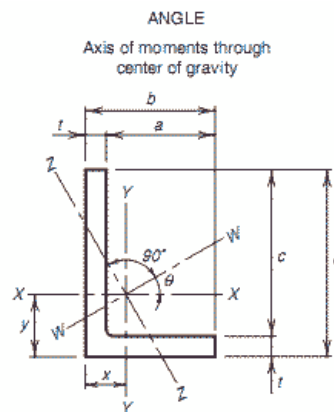
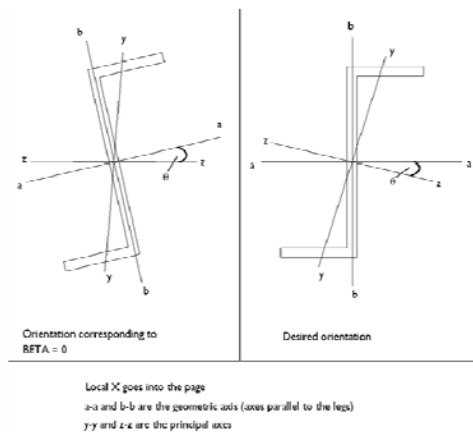
- (A) Pipe and tubing
- (B) Channel and T section
- (C) I beams and angle
- (D) Angle and Z section

The Answers is D

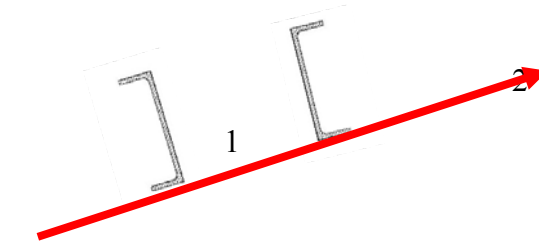
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-25

The asymmetric section have the principle axis other than the geometric orthogonal axis that creates the maximum and minimum stresses along those axis. In fact those principle axis represents their main axis. All calculation should be given along those principle axis.

In addition, these sections usually are defined by the angle between geometric axis and their principle axis. See below pictures:



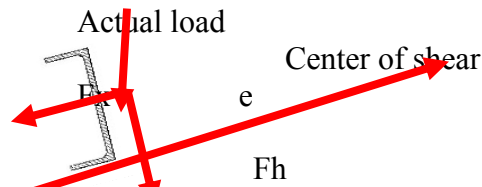
97) Which one shows the correct installation of the channel on the sloped roof?



- (A) #1
- (B) #2
- (C) Both positions may be used
- (D) None of them, Channels shall not be used on the sloped roofs

The Answer is A

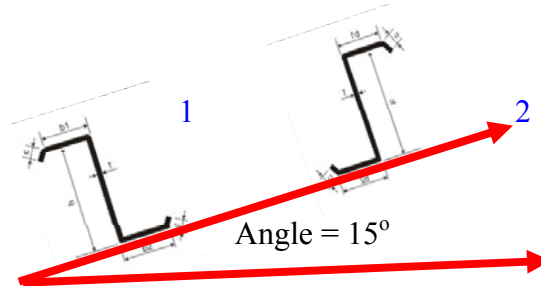
Sloped roof creates the biaxial bending moment in the purlins. This biaxial bending moment needs stronger sections. However, when a channel uses as purlin, the shear center will create a torsional bending moment in the opposite direction of the horizontal vector of the loads which necessary reduce the amount of bending moment along the minor axis. Consequently, the results yields to smaller size of channels in compare with other sections. So, technically the channels show better behavior on the sloped roofs and that is why traditionally they are used more than I and box sections. So:



Torsional moment (upward) = $f_h * e$ (eccentricity of the shear center from the channel)

My due to Fx - Torsional moment = actual moment along y axis, M_x will not be changed.

98) Which one shows the correct installation of the Z section on the sloped roof if the angle of the principle axis for the assumed Z section is given equal to 17° ?



- (A) #1
- (B) #2
- (D) None of them, Z sections shall not be used on the sloped roofs

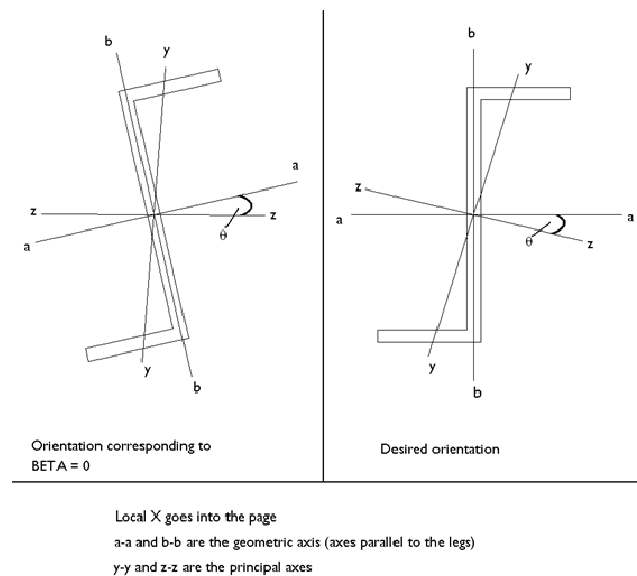
The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-25

The asymmetric section have the principle axis other than the geometric axis that creates the maximum and minimum stresses along those axis. In fact those principle axis represents their main axis. All calculation should be given along those axis.

These sections are usually defined by the angle between geometric axis and their principle axis.

See below pictures:



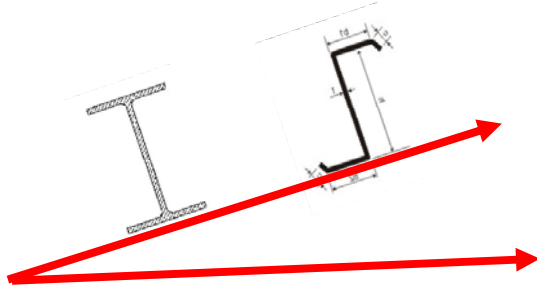
If the angle of the sloped roof is considered exactly equal to the principle axis angle, then the section will not experience the biaxial bending moment; because the major principle axis act parallel to the vertical loads and therefore, no horizontal force will act on the section. This is the most economic design of this section and will give the minimum size of the section in the design procedures. In this case both options of installation are possible.

If the principle axis is less than the roof angle, then case 1 is the answer because the principle axis will not help the section to decrease the biaxial bending moments, so option 1 will be a better choice for the easier installation.

If the principle axis is bigger than the roof slope, then case 2 will help to reduce the horizontal force and consequently the bending moment along the minor axis and will reduce the biaxial

effects of bending moments. So in this case: principle axis angle = $17^\circ >$ roof angle = 15° so choice B is correct.

- 99) What does make a difference for the amount of the combined bending moments when we use Z section with principle axis angle equal to the sloped roof and use the I beam sections?



Angle of sloped roof = Angle of the principle axis of Z

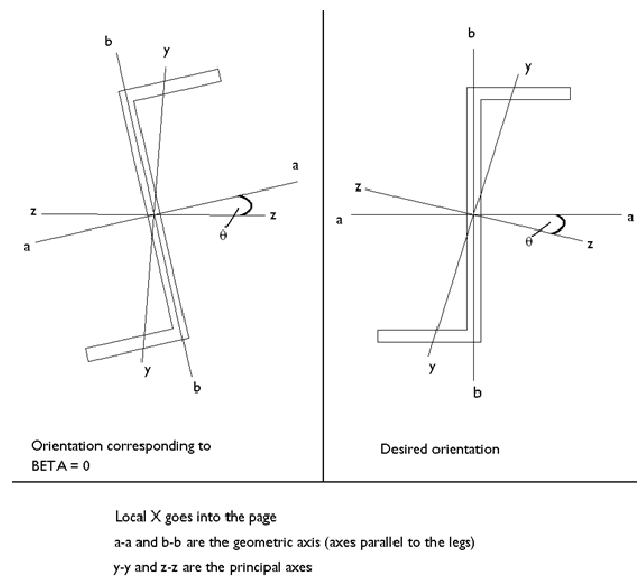
- (A) Both of them experience the biaxial bending moment.
- (B) I section experience biaxial bending moment but the Z section does not.
- (C) I section experience mono axial bending moment but the Z section experience biaxial one.
- (D) None of them experience biaxial bending moment.

The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-25

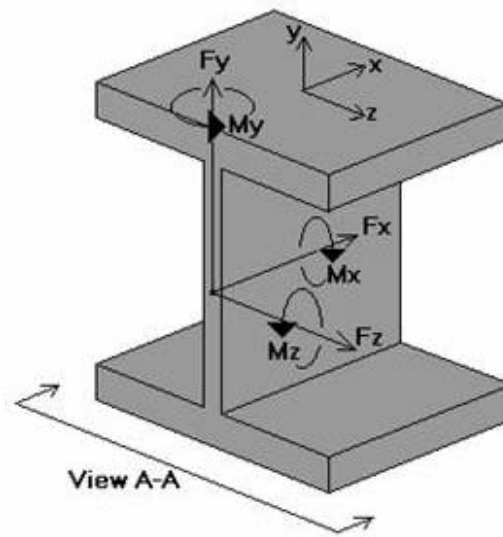
The asymmetric section have the principle axis other than the geometric axis that creates the maximum and minimum stresses along those axis. In fact those principle axis represents their main axis. All calculation should be given along those axis.

These sections usually represent by the angle between geometric axis and their principle axis. See below pictures:



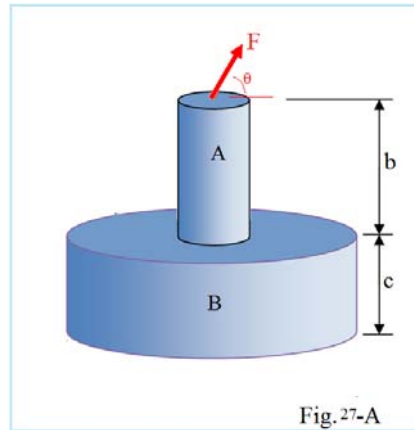
If the angle of the sloped roof is considered exactly equal to the principle axis angle, then the section will not experience the biaxial bending moment; because the major principle axis act parallel to the vertical loads and therefore, no horizontal force will act on the section. This is the most economic design of this section and will give the minimum size of the section in the design procedures. In this case both options are possible.

I sections are symmetric, so the geometric axis and principle axis are the same, consequently the section experience the biaxial bending moment.



So, choice B is the correct answer.

- 100) Find the elongation in system below. $F = 800 \text{ kN}$, $\theta = 30^\circ$. $b = 2c = 1 \text{ m}$, cross-sectional area $A = 0.02 \text{ m}^2$, and $B = 0.04 \text{ m}^2$? $E = 200 \text{ GPa}$.



(A) 0.001mm
mm

(B)

1mm

(C) 10 mm

(D) 100

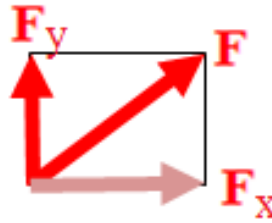
The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-24****Step 1:** Draw a free-body diagram:The component that causes elongation is F_y :

$$\sum F_y = 0$$

$$F_y = F \sin \theta$$

$$F_y = 800 \text{ kN} (\sin 30^\circ)$$

$$F_y = 400 \text{ kN}$$

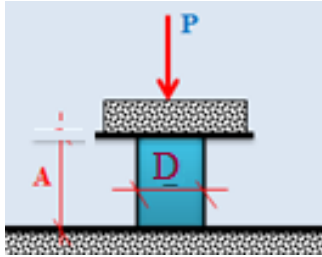
**Step 2:** Determine the elongation

$$\delta_{\text{total}} = \delta_A + \delta_B = \frac{F_y L_1}{EA_A} + \frac{F_y L_2}{EA_B}$$

$$= \frac{(400 \text{ kN})(2 \text{ m})}{(0.02 \text{ m}^2) \left(200 \times 10^9 \frac{\text{kN}}{\text{m}^2} \right)} + \frac{(400 \text{ kN})(1 \text{ m})}{(0.4 \text{ m}^2) \left(200 \times 10^9 \frac{\text{kN}}{\text{m}^2} \right)}$$

$$= 1 \times 10^{-5} \text{ m} \quad \text{or } 0.001 \text{ mm}$$

- 101) The cylindrical member shown below is loaded by $P=120\text{KN}$. Find the diameter after compression, if the Poisson's ratio, $\nu = 0.42$, $E = 8\text{ GPa}$ and $D = 8\text{ cm}$.



- (A) 7.9 cm (B) 14 cm (C) 8.01 cm (D) 7.99 cm

The Answers is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 46.6, page 46-24**

Step 1: Find the longitudinal strain as function of force and area by combination of Hook's law, and stress equations:

$$\text{Combine } \sigma = E \epsilon_{\text{long}} \text{ and } \sigma = \frac{P}{A} \rightarrow \frac{P}{A} = E \epsilon_{\text{long}} \text{ Solve for } \epsilon_{\text{long}} \rightarrow \epsilon_{\text{long}} = \frac{P}{AE} \quad (\text{I})$$

Step 2: Find the longitudinal strain as function of lateral strain:

$$\text{Poisson's ratio} = \nu = -\frac{\epsilon_{\text{lateral}}}{\epsilon_{\text{long}}} \text{ solve for } \rightarrow \epsilon_{\text{long}} = -\frac{\epsilon_{\text{lateral}}}{\nu} \quad (\text{II})$$

Step 3: Compare the two equations (I) and (II) to obtain lateral strain as function of force and area:

$$\text{Compare (I) and (II)} \rightarrow -\frac{\epsilon_{\text{lateral}}}{\nu} = \frac{P}{AE} \text{ Solve for } \epsilon_{\text{lateral}} \rightarrow$$

$$\epsilon_{\text{lateral}} = -\nu \frac{P}{AE}$$

Step 4: Find the lateral strain:

$$\epsilon_{\text{lateral}} = -0.42 \frac{-120 \text{ kN}}{\frac{\pi}{4} 0.08^2 \times 8 \times 10^6 \frac{\text{KN}}{\text{m}^2}} = 1.253 \times 10^{-3}$$

Step 5: By having the lateral strain find the change in diameter:

$$\epsilon_{\text{lateral}} = \frac{\delta}{D} \text{ solve for } \delta \rightarrow \epsilon_{\text{lateral}} D = \delta$$

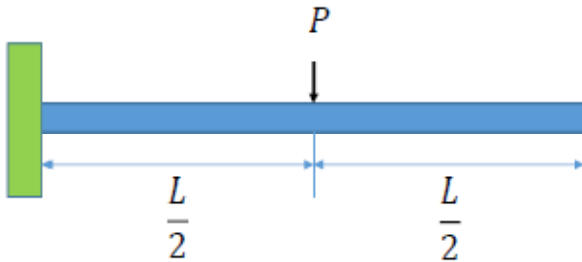
$$\delta = 1.253 \times 10^{-3} (8 \text{ cm}) = 0.01 \text{ cm}$$

Step 6: Now find the diameter after compression:

$$\text{Diameter(after compression)} = D + \delta = 8.01 \text{ cm}$$

G. Deflection

102) If the beam shown below has the deflection not exceeding 0.005 m, which material shall be chosen?



$$I = 0.0002 \text{ m}^4 \quad P = 100 \text{ kN} \quad L = 2 \text{ m}$$

- (A) 20 GPa
- (B) 50 GPa
- (C) 100 GPa
- (D) Both (B) and (C)

The Answer is C**AISC 13th ed., 2010, 3-211-226**

Step 1: The deflection at the end of cantilever beam is the maximum, it can be calculated as

$$y_{max} = \frac{-5PL^3}{48EI} > -0.005 \text{ m}$$

Step 2: Determine the Young's modulus

$$E > \frac{-5PL^3}{(-0.005\text{m})48I} = \frac{-5(100\text{kN})(2\text{ m})^3}{(0.005\text{m})(48)(0.0002\text{ m}^4)}$$

$$E > 83333333.3 \frac{\text{KN}}{\text{m}^2} = 83.33 \text{ GPa}$$

103) Find the maximum slope for the beam shown below.



$$I = 5 \text{ in}^4 \quad E = 200 \text{ kpsi} \quad M_0 = 10000 \text{ lb} \cdot \text{in} \quad L = 2 \text{ ft}$$

- (A) -0.08 rad
- (B) 0.008 rad
- (C) 4.59 rad
- (D) -4.59 rad

The Answer is D**AISC 13th ed., 2010, 3-211-226**

Step 1: The maximum slope occurs at the left end where the bending moment is applied. The angle is clockwise so that the slope is negative. It can be calculated as

$$\theta_{max} = \frac{-M_0 L}{3EI} = \frac{-(10000 \text{ lb} \cdot \text{in})(24 \text{ in})}{(3) \left(200 \times 10^3 \frac{\text{lb}}{\text{in}^2} \right) (5 \text{ in}^4)}$$

$$\theta_{max} = -0.08 \text{ rad} \quad \text{or} \quad -4.59 \text{ degree}$$

104) A cantilever beam is subject to a uniform load. Which statement of the following is correct?

- (A) The slope at the fixed end is maximum.
- (B) The deflection at the free end is zero.
- (C) The slope at the free end is maximum.
- (D) None of the above.

The Answers is C**AISC 13th ed., 2010, 3-211-226**

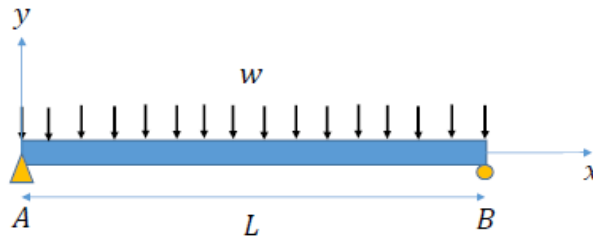
Answer "A" is not correct: The slope at the fixed end is zero.

Answer "B" is not correct: The deflection at the free end is the maximum.

Answer "C" is correct: The slope at the free end is the maximum.

Answer "D" is not correct: The slope at the free end is the maximum.

- 105) A simply supported beam is subject to a uniform load as shown below. Which statement of the following is correct?



- (A) The slope at the left end is the maximum, and it is negative.
- (B) The slope at the right end is the maximum, and it is negative.
- (C) The slope at the midpoint is the maximum.
- (D) None of the above.

The Answer is A**AISC 13th ed., 2010, 3-211-226**

Answer "A" is correct: The slope at the left end is the maximum, and it is negative because the angle is clockwise.

Answer "B" is not correct: The slope at the right end is the maximum, and it is positive because the angle is counterclockwise.

Answer "C" is not correct: The slope at the midpoint is zero.

Answer "D" is not correct: Answer "A" is correct.

- 106) A cantilever beam is subject to a bending moment at the free end as shown below. Which statement of the following is correct?



- (A) The slope at the fixed end is zero.
- (B) The slope at the free end is the maximum, and it is positive.
- (C) The slope at the free end is the maximum, and it is negative.
- (D) Both (A) and (B).

The Answer is D**AISC 13th ed., 2010, 3-211-226**

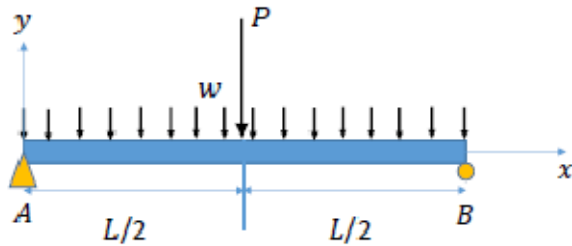
Answer "A" is correct: The slope at the fixed end is zero.

Answer "B" is correct: The slope at the free end is the maximum, and it is positive because the angle is counterclockwise.

Answer "C" is not correct: The slope at the free end is the maximum, and it is positive.

Answer "D" is correct: Both answers "A" and "B" are correct.

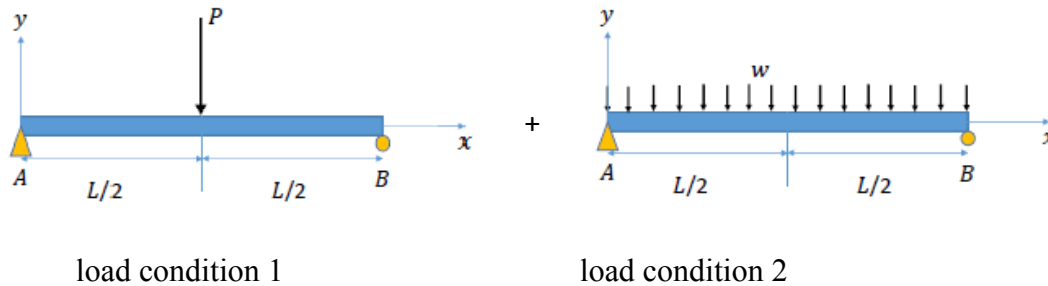
- 107) Find the maximum deflection of the beam shown below. Given: $I = 50 \text{ cm}^4$, $L = 4 \text{ m}$, $P = 2 \text{ kN}$, $w = 1 \text{ kN/m}$ and $E = 100 \text{ GPa}$.



- (A) -12 cm (B) 12 cm (C) -5.33 cm (D) -6.67 cm

The Answer is A**AISC 13th ed., 2010, 3-211-226**

Step 1: Using superposition method, the original load condition can be viewed as the sum of the following two conditions:



In both cases, the maximum deflections occur at the midpoint of the cantilever beam. Therefore, the total maximum deflection is

$$y_{\max} = y_{\max(P)} + y_{\max(w)}$$

Step 2: Using any mechanics of material book, calculate the maximum deflection

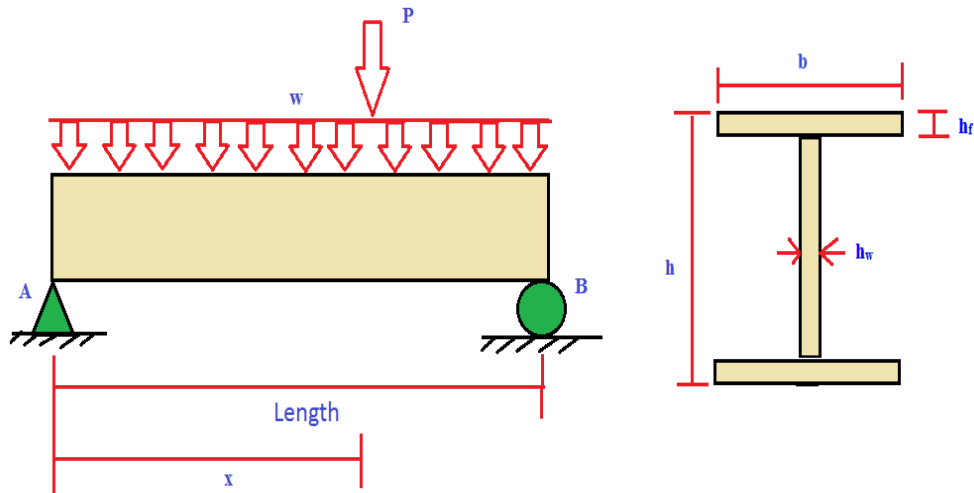
$$y_{\max(P)} = \frac{-PL^3}{48EI} = \frac{-(2000 \text{ N})(4 \text{ m})^3}{48 \left(100 \times 10^9 \frac{\text{N}}{\text{m}^2}\right) (50 \times 10^{-8} \text{ m}^4)} = -0.0533 \text{ m}$$

$$y_{\max(w)} = \frac{-5wL^4}{384EI} = \frac{-5 \left(1000 \frac{\text{N}}{\text{m}}\right) (4 \text{ m})^4}{384 \left(100 \times 10^9 \frac{\text{N}}{\text{m}^2}\right) (50 \times 10^{-8} \text{ m}^4)} = -0.0667 \text{ m}$$

$$y_{\max} = y_{\max(P)} + y_{\max(w)} = -0.12 \text{ m}$$

$$y_{\max} = -0.12 \text{ m} \quad \text{or} \quad -12 \text{ cm}$$

- 108) A 30 ft long simply supported I-beam is subject to a uniformly distributed load and a point load of 3.5 kips/ft and 85 kips respectively. Height of the beam(h) is 18 in, Flange width(b_f) is 1.5 in, section width(b) is 14 in, and The web width(h_w) is 0.5 in. The distance x is 22 ft. $E = 29000$ ksi. Determine the max deflection on the beam.



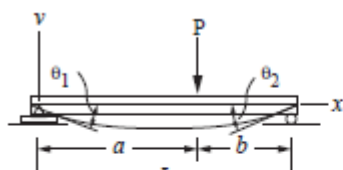
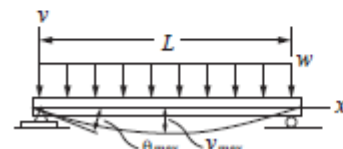
- A) 1.31 in B) 1.13 in
C) 0.77 in D) 1.65 in

The Answers is A**AISC 13th ed., 2010, 3-211-226**

Step 1: Determine the moment of inertia for the cross section. To do this we need to consider the dimensions for each section of the I-beam. The area of the top and bottom flange will be the same. Use the relationship for the moment of inertia of a rectangle. Only strong axis bending needs to be considered. We can take the moment of inertia for the section as a rectangle and then subtract out the areas for the empty space on either side of the I-beam.

$$I = \frac{bh^3 - (h - 2h_f)^3(b - h_t)}{12} = \frac{(14 \text{ in})(18 \text{ in})^3 - (18 \text{ in} - 2(1.5 \text{ in}))^3(14 \text{ in} - 0.5 \text{ in})}{12} = 3007 \text{ in}^4$$

Step 2: Use the relationships in any mechanics of materials book to calculate max deflection. Our beam is simply supported with a distributed load and a point load. The deflections from each load will sum to the total deflection. This is called the principle of superposition.

	$\theta_1 = \frac{-Pab(L+b)}{6EI}$ $\theta_2 = \frac{Pab(L+a)}{6EI}$	$v _{x=a} = \frac{-Pba}{6EI}(L^2 - b^2 - a^2)$
	$\theta_{\max} = \frac{-wL^3}{24EI}$	$v_{\max} = \frac{-5wL^4}{384EI}$

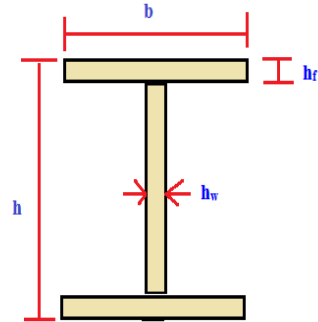
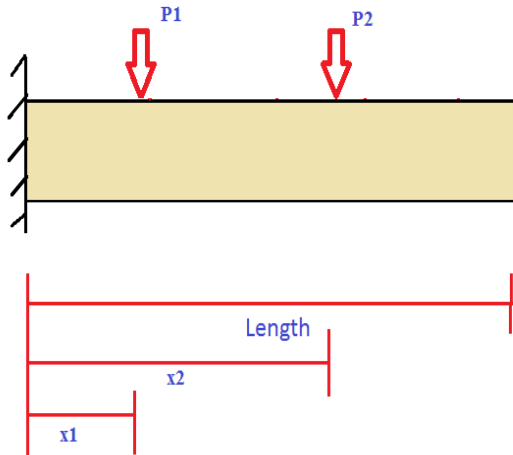
$$b = L - x = 30 \text{ ft} - 22 \text{ ft} = 8 \text{ ft}$$

$$v_{\max} = \frac{-5wL^4}{384EI} + \frac{-Pba}{6EI}(L^2 - b^2 - a^2) =$$

$$\frac{-5\left(3.5 \frac{\text{kips}}{\text{ft}}\right)(30 \text{ ft})^4\left(\frac{12 \text{ in}}{1 \text{ ft}}\right)^3}{384(29000 \text{ ksi})(3007 \text{ in}^4)} + \frac{-(85 \text{ kips})(8 \text{ ft})(22 \text{ ft})}{6(29000 \text{ ksi})(3007 \text{ in}^4)(30 \text{ ft})}((30 \text{ ft})^2 - (8 \text{ ft})^2 - (22 \text{ ft})^2)\left(\frac{12 \text{ in}}{\text{ft}}\right)^3 =$$

$$-1.31 \text{ in} \quad .31 \text{ in}$$

- 109) A 20 ft long cantilever I-beam is subject to two point loads P_1 and P_2 of 80 kips and 95 kips respectively. Height of the beam (h) is 30 in, Flange width (b_f) is 2.5 in, section width (b) is 22 in, and The web width (h_w) is 0.75 in. The distances x_1 and x_2 are 8 ft and 16 ft respectively. $E = 29000$ ksi. Determine the max deflection on the beam.



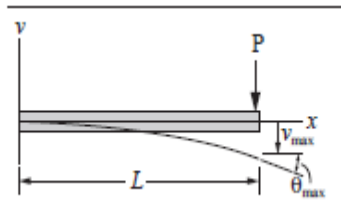
- A) 1.12 in B) 0.38 in
C) 0.61 in D) 1.34 in

The Answers is C**AISC 13th ed., 2010, 3-211-226**

Step 1: Determine the moment of inertia for the cross section. To do this we need to consider the dimensions for each section of the I-beam. The area of the top and bottom flange will be the same. Use the relationship for the moment of inertia of a rectangle. Only strong axis bending needs to be considered. We can take the moment of inertia for the section as a rectangle and then subtract out the areas for the empty space on either side of the I-beam.

$$I = \frac{bh^3 - (h - 2h_f)^3(b - h_t)}{12} = \frac{(22 \text{ in})(30 \text{ in})^3 - (30 \text{ in} - 2(2.5 \text{ in}))^3(22 \text{ in} - 0.75 \text{ in})}{12} = 21831 \text{ in}^4$$

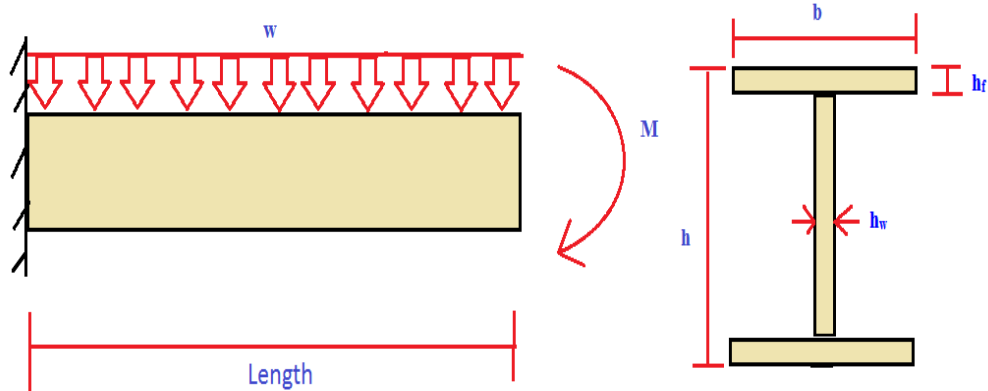
Step 2: Use the relationships listed on any mechanics of material book to calculate max deflection. Our beam is a cantilever system with two point loads. We can sum the deflection caused by each point load to get the total deflection. This is the principle of superposition.

	$\theta_{\max} = \frac{-PL^2}{2EI}$	$v_{\max} = \frac{-PL^3}{3EI}$	$v = \frac{-Px^2}{6EI}(3L - x)$
--	-------------------------------------	--------------------------------	---------------------------------

$$x_1 = 8 \text{ ft}; x_2 = 16 \text{ ft}$$

$$v_{\max} = \frac{-P_1 x_1^2}{6EI}(3L - x_1) + \frac{-P_2 x_2^2}{6EI}(3L - x_2) = \left[\frac{-(80 \text{ kips})(8 \text{ ft})^2}{6(29000 \text{ ksi})(21831 \text{ in}^4)}(3(20 \text{ ft}) - 8 \text{ ft}) + \frac{-(95 \text{ kips})(16 \text{ ft})^2}{6(29000 \text{ ksi})(21831 \text{ in}^4)}(3(20 \text{ ft}) - 16 \text{ ft}) \right] \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^3 = -0.61 \text{ in } 0.61 \text{ in}$$

- 110) A 20 ft long cantilever I-beam is subject to a distributed load and end moment of 12 kips/ft and 600 **kip · ft** respectively. Height of the beam(h) is 30 in, Flange width(h_f) is 2.5 in, section width(b) is 22 in, and The web width(h_w) is 0.75 in. $E = 29000$ ksi. Determine the max deflection on the beam.



- A) 0.65 in B) 0.38 in
C) 0.98 in D) 1.34 in

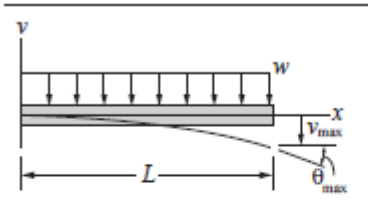
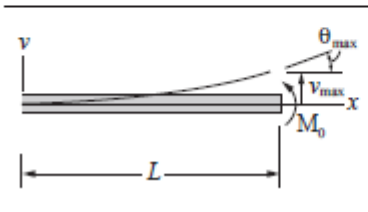
The Answers is C**AISC 13th ed., 2010, 3-211-226**

Step 1: Determine the moment of inertia for the cross section. To do this we need to consider the dimensions for each section of the I-beam. The area of the top and bottom flange will be the same. Use the relationship for the moment of inertia of a rectangle. Only strong axis bending needs to be considered. We can take the moment of inertia for the section as a rectangle and then subtract out the areas for the empty space on either side of the I-beam.

$$I = \frac{bh^3 - (h - 2h_f)^3(b - h_t)}{12} = \frac{(22 \text{ in})(30 \text{ in})^3 - (30 \text{ in} - 2(2.5 \text{ in}))^3(22 \text{ in} - 0.75 \text{ in})}{12}$$

$$= 21831 \text{ in}^4$$

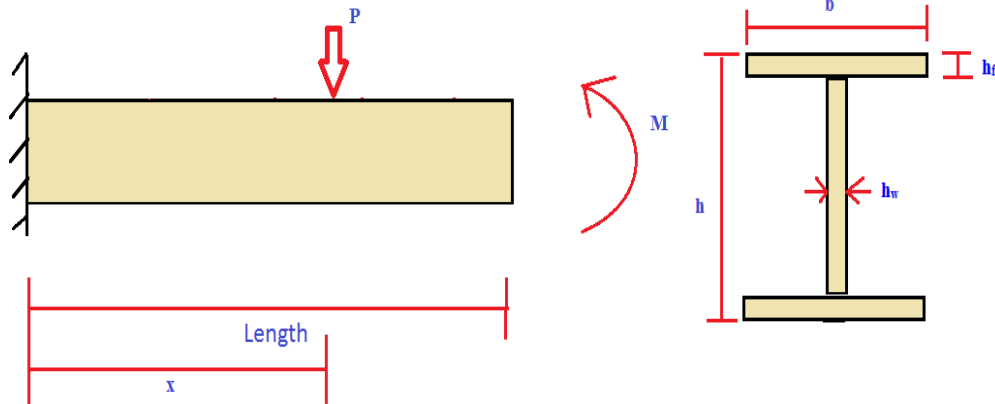
Step 2: Use the relationships listed on any mechanics of materials book to calculate max deflection. Our beam is a cantilever system with a distributed load and an end moment. The end moment and the distributed load will both cause downward deflection thus we can sum the deflection caused by each. This is the principle of superposition.

	$\theta_{\max} = \frac{-wL^3}{6EI}$	$v_{\max} = \frac{-wL^4}{8EI}$
	$\theta_{\max} = \frac{M_0L}{EI}$	$v_{\max} = \frac{M_0L^2}{2EI}$

$$v_{\max} = \frac{-wL^4}{8EI} - \frac{ML^2}{2EI} = \left[\frac{\left(12 \frac{\text{kips}}{\text{ft}}\right)(20 \text{ ft})^4}{8(29000 \text{ ksi})(21831 \text{ in}^4)} + \frac{(600 \text{ kip}\cdot\text{ft})(20 \text{ ft})^2}{2(29000 \text{ ksi})(21831 \text{ in}^4)} \right] \left(\frac{12 \text{ in}}{1 \text{ ft}}\right)^3 = -0.98 \text{ in}$$

0.98 in

- 111) A 20 ft long cantilever I-beam is subject to a point load and end moment of 250 kips and 300 **kip · ft** respectively. Height of the beam(h) is 30 in, Flange width(h_f) is 2.5 in, section width(b) is 22 in, and The web width(h_w) is 0.75 in. The distance x is 16 ft. $E = 29000$ ksi. Determine the max deflection on the beam.



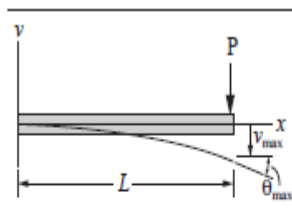
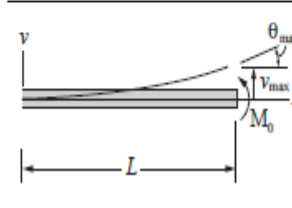
- A) 1.44 in B) 1.12 in
C) 0.98 in D) 0.73 in

The Answers is B**AISC 13th ed., 2010, 3-211-226**

Step 1: Determine the moment of inertia for the cross section. To do this we need to consider the dimensions for each section of the I-beam. The area of the top and bottom flange will be the same. Use the relationship for the moment of inertia of a rectangle. Only strong axis bending needs to be considered. We can take the moment of inertia for the section as a rectangle and then subtract out the areas for the empty space on either side of the I-beam.

$$I = \frac{bh^3 - (h - 2h_f)^3(b - h_t)}{12} = \frac{(22 \text{ in})(30 \text{ in})^3 - (30 \text{ in} - 2(2.5 \text{ in}))^3(22 \text{ in} - 0.75 \text{ in})}{12} = 21831 \text{ in}^4$$

Step 2: Use the relationships listed below to calculate max deflection. Our beam is a cantilever system with a point load and end moment. The end moment creates deflection in the opposite direction of the point load. Take the point load deflection as positive and the end moment deflection as negative and sum the deflections to get the total. This is the principle of superposition.

	$\theta_{\max} = -\frac{PL^2}{2EI}$	$v_{\max} = -\frac{PL^3}{3EI}$	$v = -\frac{Px^2}{6EI}(3L - x)$
	$\theta_{\max} = \frac{M_0L}{EI}$	$v_{\max} = \frac{M_0L^2}{2EI}$	$v = \frac{M_0x^2}{2EI}$

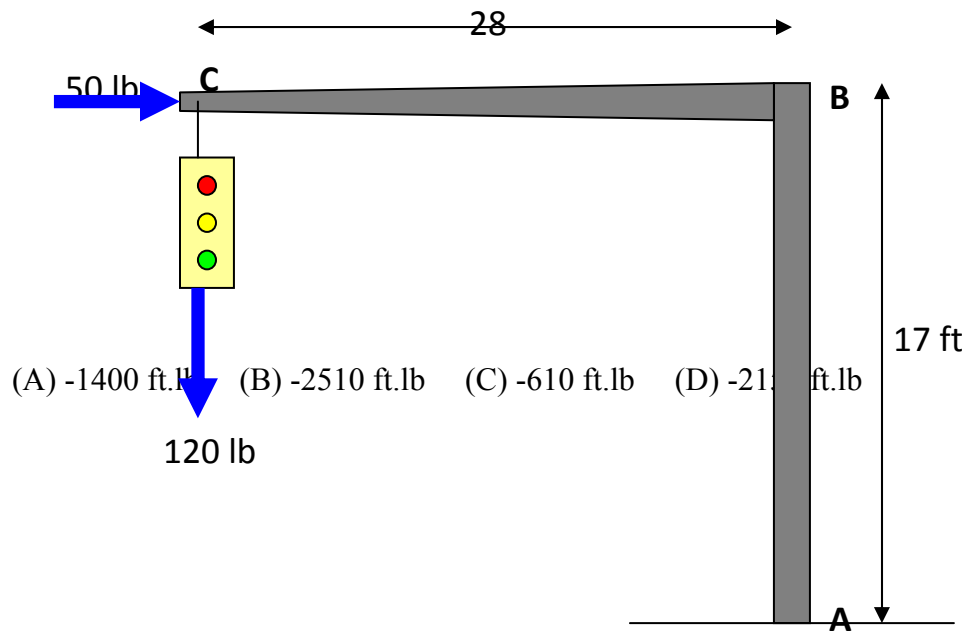
$$x = 16 \text{ ft}$$

$$v_{\max} = \frac{-Px^2}{6EI}(3L - x) + \frac{ML^2}{2EI} =$$

$$\left[\frac{(250 \text{ kips})(16 \text{ ft})^2}{6(29000 \text{ ksi})(21831 \text{ in}^4)}(3(20 \text{ ft}) - 16 \text{ ft}) - \frac{(300 \text{ kip}\cdot\text{ft})(20 \text{ ft})^2}{2(29000 \text{ ksi})(21831 \text{ in}^4)} \right] \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^3 = 1.12 \text{ in} \quad 1.12 \text{ in}$$

H. Beams

- 112) A traffic signal experiences its self-weight and a force due to wind. Determine the resulting moment about the base of the structure.



The Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

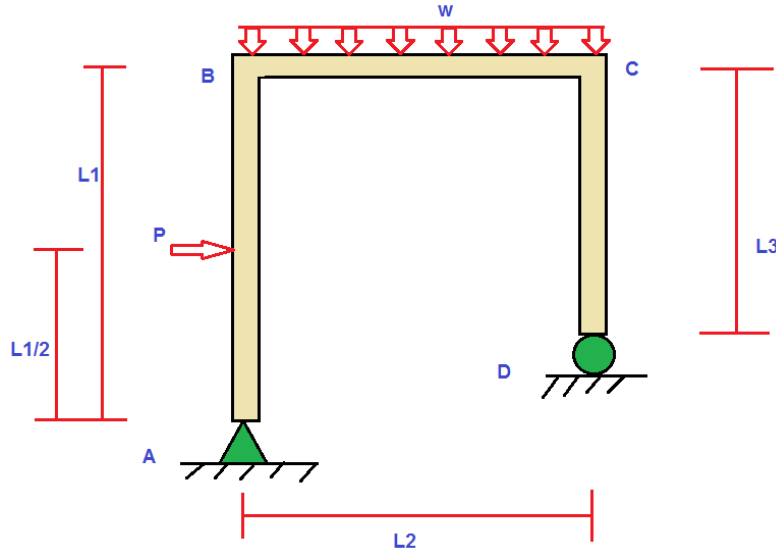
Sum the moments about point A to determine the reaction at support A. Notice that the two given forces in the figure rotate in opposite directions from one another. The 50 lb force rotates clockwise and the 120 lb force rotates counter-clockwise. Designate one as positive and the other as negative. The reaction at support A must be equal to the sum of the moments caused by the applied forces to remain in equilibrium.

$$M_{R_A} = \sum M_A$$

$$M_{R_A} = 50 \text{ lb}(17 \text{ ft}) - 120 \text{ lb}(28 \text{ ft})$$

$$M_{R_A} = -2510 \text{ ft} \cdot \text{lb}$$

- 113) A plane frame consisting of 3 rigid members is simply supported with a pin and roller support as shown. The frame is subject to a dead uniformly distributed load and a live point load of 2.5 kips/ft and 250 kips respectively. $L_1 = 18$ ft, $L_2 = 18$ ft, and $L_3 = 14$ ft. Determine the reactions at the supports.



- A) $F_{Ax} = -250$ kips; $F_{Ay} = 102.5$ kips $F_D = 147.5$ kip
B) $F_{Ax} = 250$ kips; $F_{Ay} = -102.5$ kips $F_D = 147.5$ kip
C) $F_{Ax} = -400$ kips; $F_{Ay} = 173$ kips $F_D = 227$ kips
D) $F_{Ax} = -400$ kips; $F_{Ay} = -173$ kips $F_D = 227$ kips

The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15) and ASCE 7-10, chapter 2, page 5-6

Step 1: Determine the factored loads on the system. To do this we need to consider a set of LRFD load factors. For live and dead we can take the load combination of $1.2D + 1.6L + 0.5(L_r, S, R)$. The last factor represents live roof load snow or rain load and our system does not consider these so use $1.2D + 1.6L$.

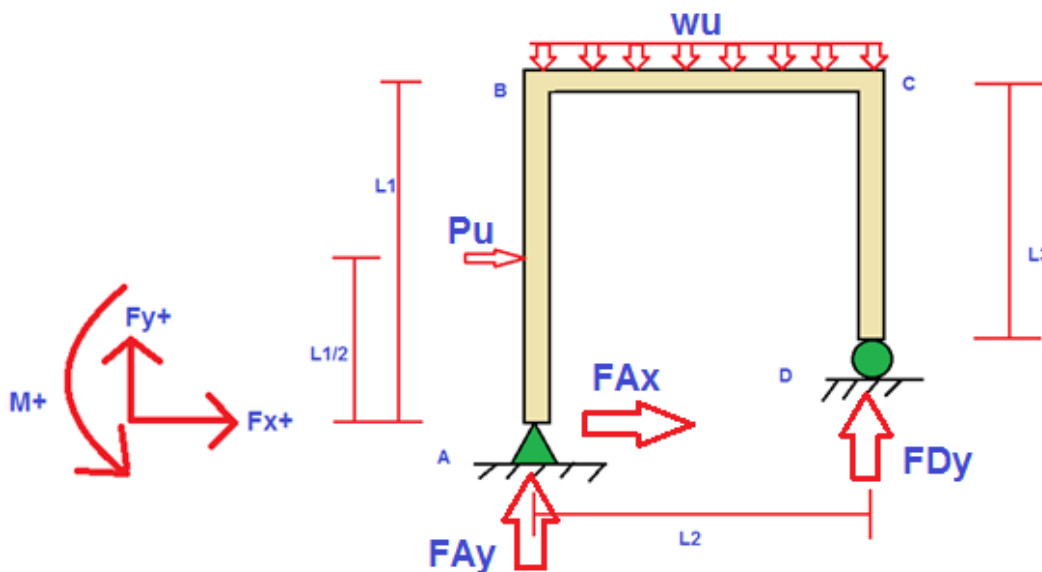
Continuing:

Substitute the distributed dead load (w_D) in order to get the ultimate distributed dead load (w_u). Also the live load will need to be considered separately with its appropriate factor.

$$w_u = 1.2 \left(2.5 \frac{\text{kips}}{\text{ft}} \right) = 3 \frac{\text{kips}}{\text{ft}}$$

$$P_u = 1.6(250 \text{ kips}) = 400 \text{ kips}$$

Step 2: Determine the reactions at supports A and D. To do this take the equilibrium equations for the entire system and solve for the reactions at the supports. The point load acts at half the length of member AB and the distributed load will act at half the length of member BC.



$$\sum M_A = 0 = F_{Dy}(L2) - (w_u L2) \left(\frac{L2}{2} \right) - P_u \left(\frac{L1}{2} \right)$$

$$F_{Dy} = \frac{(w_u L2) \left(\frac{L2}{2} \right) + P_u \left(\frac{L1}{2} \right)}{L2} = 227 \text{ kips}$$

$$\sum F_x = 0 = F_{Ax} + P_u$$

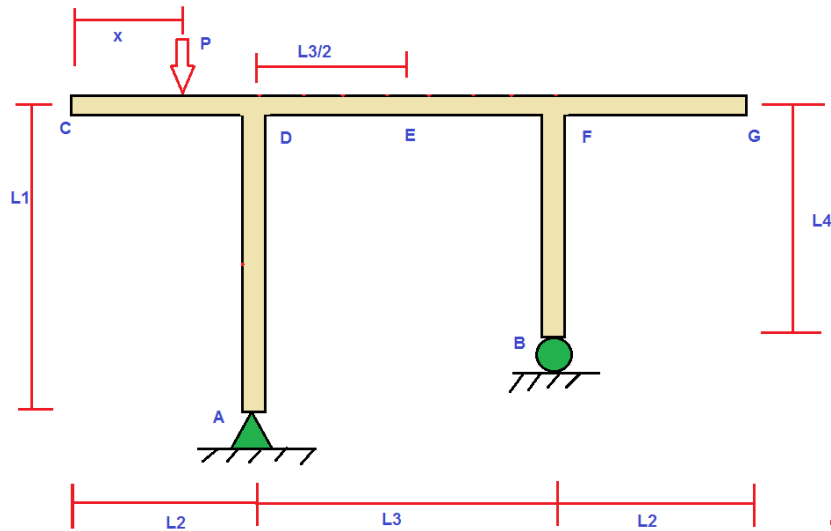
$$\sum F_y = 0 = F_{Dy} + F_{Ay} - (w_u L2)$$

$$F_{Ax} = -400 \text{ kips}; F_{Ay} = -173 \text{ kips}$$

$$\sim F_{Ax} = -400 \text{ kips}; F_{Ay} = -173 \text{ kips}$$

$$F_D = 227 \text{ kips}$$

- 114) A plane frame consisting of 5 rigid members and is simply supported with a pin and roller support as shown. The frame is subject to a moving point load of 250 kips. $L_1 = 18$ ft, $L_2 = 10$ ft, $L_3 = 18$ ft, and $L_4 = 14$ ft. Determine the force carried by point E considering the influence line for the vertical reaction at support A.



- A) $F_E = 125$ kips
B) $F_E = 250$ kips
C) $F_E = 390$ kips
D) $F_E = 140$ kips

The Answers is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

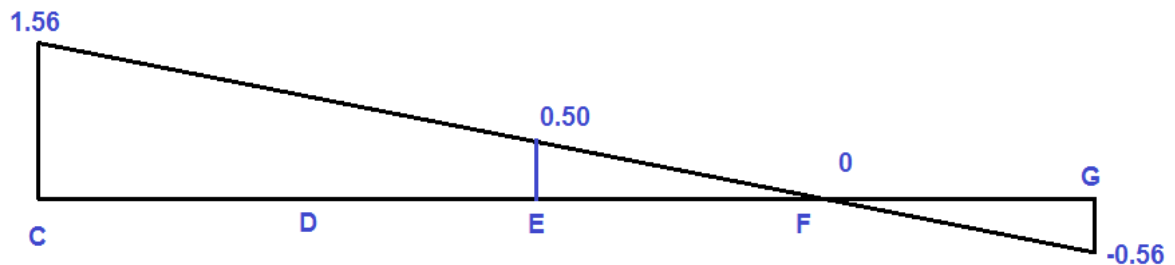
Step 1: First apply a unit load to the frame and trace the reaction at A as the point load moves across the frame. When the point load is directly over member AD the influence lines value will be 1. This will decrease linearly to 0 at support B. When the point load is on segment CD support A will take more of the load. Similarly on segment FG support B will take more of the load.

The value of the influence line will decrease from 1.0 to 0 from support A to support B which is a distance of 18 ft. So the load will decrease or increase linearly by a value of $x/18$ on segments CD and FG. If we sum the moments about support B we can determine a value for the reaction at support A in terms of x .

$$\sum M_B = 0 = (1)((L2 + L3) - x) - A_y L3$$

$$A_y = \frac{(1)((L2 + L3) - x)}{L3} = 1.56 - \frac{x}{18 \text{ ft}}$$

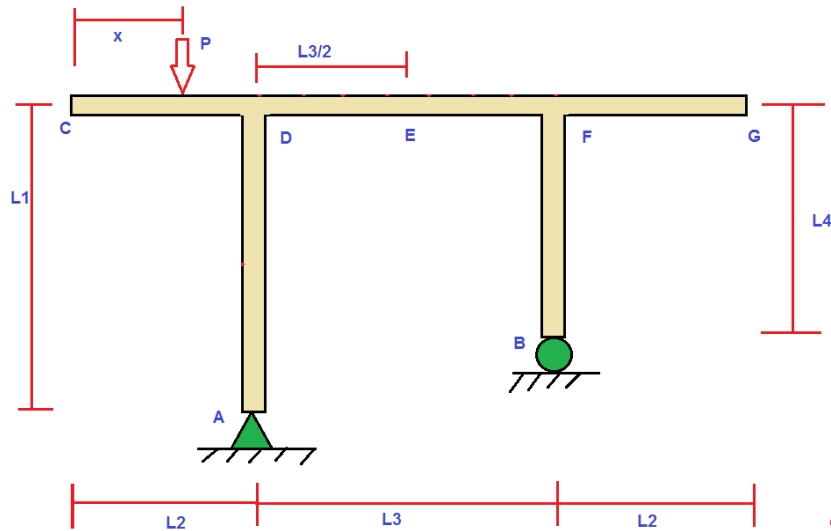
Step 2: Draw the influence line for the system and determine the magnitude of the load at point E. To do this simply multiply the unit force on the influence line by the magnitude of the force given in the problem description.



$$F_E = 0.50(250 \text{ kips}) = 125 \text{ kips}$$

$$\sim F_E = 125 \text{ kips}$$

- 115) A plane frame consisting of 5 rigid members and is simply supported with a pin and roller support as shown. The frame is subject to a moving point load of 250 kips. $L_1 = 18$ ft, $L_2 = 10$ ft, $L_3 = 18$ ft, and $L_f = 14$ ft. Determine the force carried by point C considering the influence line for the vertical reaction at support B.



- A) $F_E = -125$ kips
 B) $F_E = -250$ kips
 C) $F_E = 390$ kips
 D) $F_E = -140$ kips

The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: First apply a unit load to the frame and trace the reaction at B as the point load moves across the frame. When the point load is directly over member FB the influence lines value will be 1. This will decrease linearly to 0 at support A. When the point load is on segment CD support A will take more of the load. Similarly on segment FG support B will take more of the load. The value of the influence line will decrease from 1.0 to 0 from support B to support A which is a distance of 18 ft. So the load will decrease or increase linearly by a value of $x/18$ on segments CD and FG. If we sum the moments about support B we can determine a value for the reaction at support A in terms of x . Then sum the vertical forces to determine the reaction at support B in terms of x

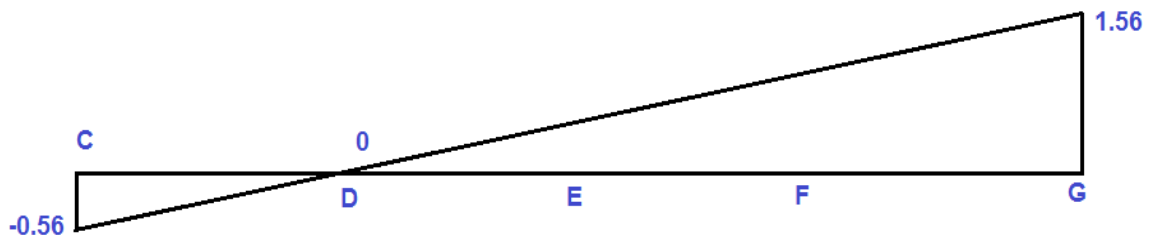
$$\sum M_B = 0 = (1)((L2 + L3) - x) - A_y L3$$

$$A_y = \frac{(1)((L2 + L3) - x)}{L3} = 1.56 - \frac{x}{18 \text{ ft}}$$

$$\sum F_y = 0 = A_y + B_y - 1$$

$$B_y = 1 - A_y = \frac{x}{18 \text{ ft}} - 0.56$$

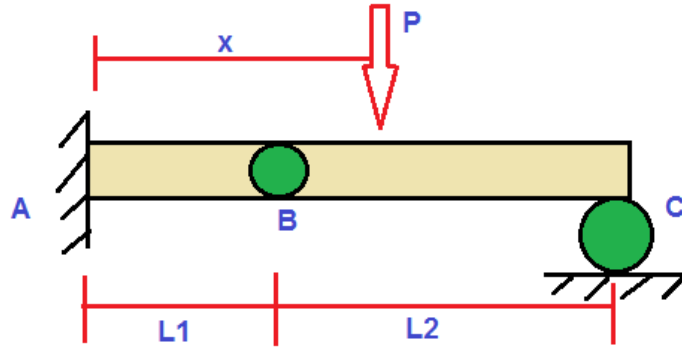
Step 2: Draw the influence line for the system and determine the magnitude of the load at point E. To do this simply multiply the unit force on the influence line by the magnitude of the force given in the problem description.



$$F_C = -0.56(250 \text{ kips}) = -140 \text{ kips}$$

$$\sim F_E = -140 \text{ kips}$$

- 116) A cantilever beam has a roller support at one end to prevent rotation from an internal pin support. The beam is subject to a point load of 150 kips which moves along the beam. L_1 and L_2 are 5 ft and 10 ft respectively. Determine the ordinate of the influence line for the reaction at support C when the distance x is 3 ft.

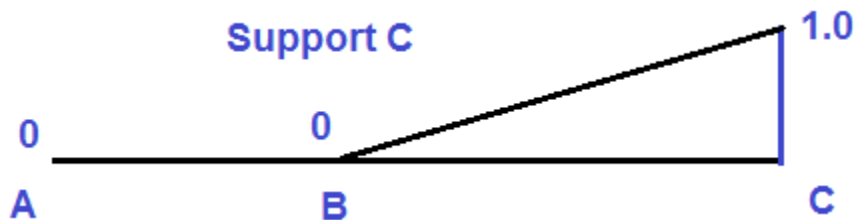


- A) 0.33 B) 0.75
C) 0 D) 1.0

The Answer is C

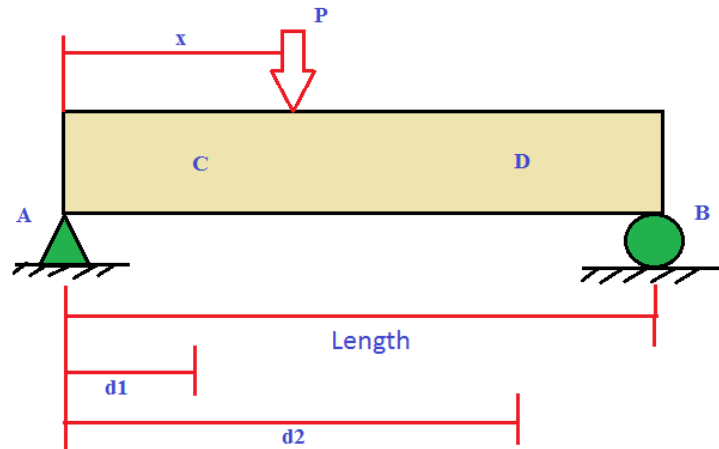
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: First determine the influence line for the system relative to the reaction at point C. If the force is located between the fixed support A and the pinned support B then these supports will carry all the load and the reaction at C will be 0. As the force moves beyond support B the force carried at support C will increase from 0 to the total load on the system. We can draw an influence line that is representative of this. Use a unit load of 1.0 to represent this system.



Step 2: Now determine the ordinate for the influence line for the reaction at point C. It was given that the load is located 3 ft from support A. The distance from support A to B is 5 ft. Thus the load is located between points A and B. This means that these supports will carry the entire load and support C will not receive any. Thus the reaction at support C will be 0.

- 117) A 30 ft long simply supported beam is subject to a point load that is not fixed. The load can be located at any point on the beam and has a magnitude of 150 kips. The distances d_1 and d_2 are 8 ft and 24 ft respectively. Determine the shear at point C considering the influence line for point D.



- A) $V_C = 40$ kips
B) $V_C = 60$ kips
C) $V_C = -40$ kips
D) $V_C = -60$ kips

The Answers is C

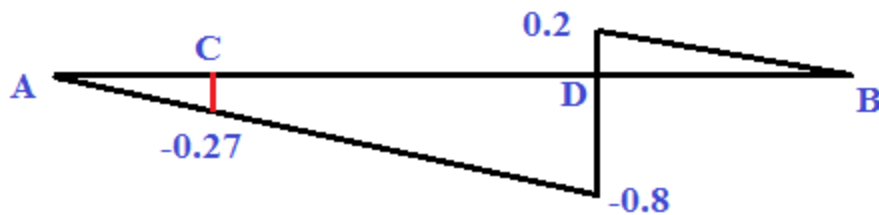
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: First draw the influence line for the shear point D. To do this we will consider a unit load of 1 and trace the shear to either support. The total shear at point D will be 1 and it will decrease to 0 at either support. There will be a positive shear and negative shear at B which will sum to a total of 1. Start by determining the unit shear at point D by tracing the load from support A to support B as shown in the figure. The farther the point is from point A the more it will decrease. Point D will decrease by the change in distance to B divided by the length of the beam.

$$D = \frac{(30 \text{ ft} - 24 \text{ ft})}{30 \text{ ft}} (1.0) = 0.2$$

The positive shear will be 0.2 and this will decrease to 0 as it moves towards support B. The negative shear must be 0.8 since the total shear at B will be 1.0. This will increase to 0 at support A. The unit shear at point C will be the distance to point C divided by the distance to point D times the unit shear at point D.

$$C = \frac{8 \text{ ft}}{24 \text{ ft}} (-0.8) = -0.27$$

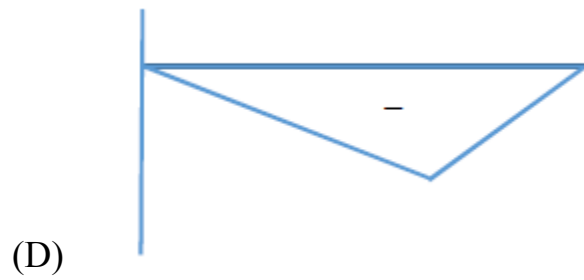
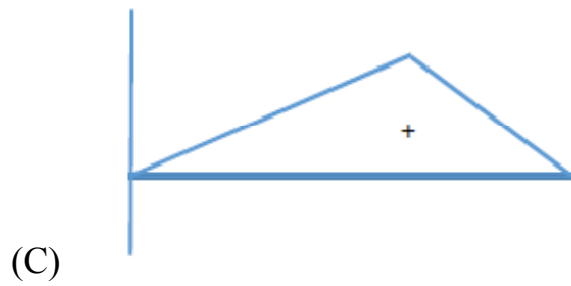
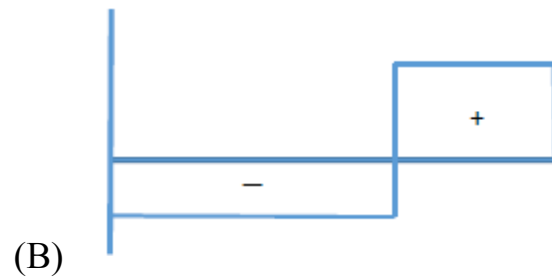
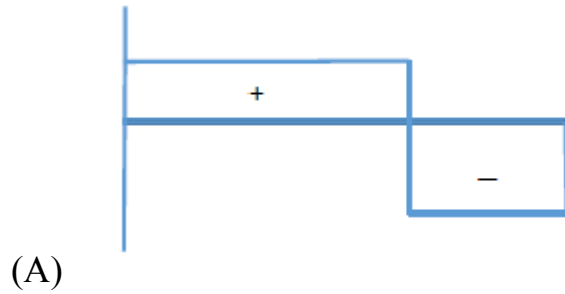
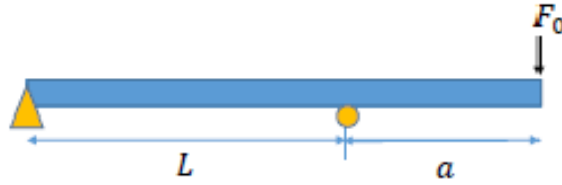


Step 2: To get the shear at points C simply multiply the unit value on the influence line by the magnitude of the load given in the problem description.

$$V_C = -0.27(150 \text{ kips}) = -40.5 \text{ kips}$$

$$\sim V_C = -40 \text{ kips}$$

- 118) Determine the moment diagram which can represent the beam with F_0 acting at the right end of the beam:



The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Draw a free-body diagram, determine the support reactions R_A and R_B

$$R_B = \frac{F_0(L+a)}{L} \quad \text{and} \quad R_A = -\frac{F_0a}{L}$$

Step 2: Choose the beam's left end as the origin, draw a free-body diagram at position x ($x < L$)

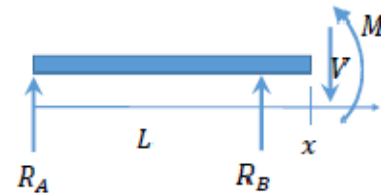
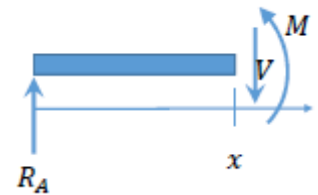
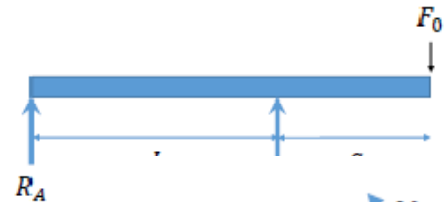
$$\sum M(V) = 0 \Rightarrow -R_A x + M = 0 \Rightarrow M = -\frac{F_0 a x}{L}$$

Step 3: Choose the beam's left end as the origin, draw a free-body diagram at position x ($x > L$)

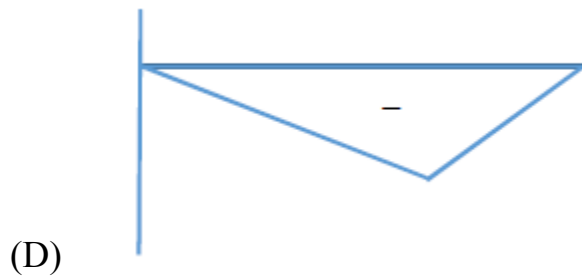
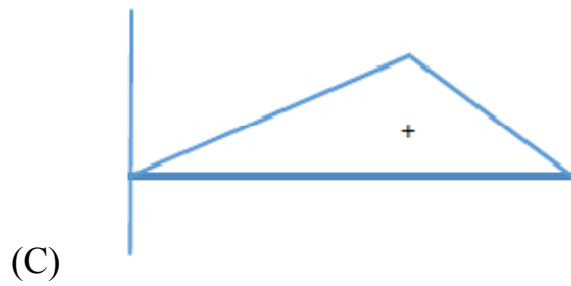
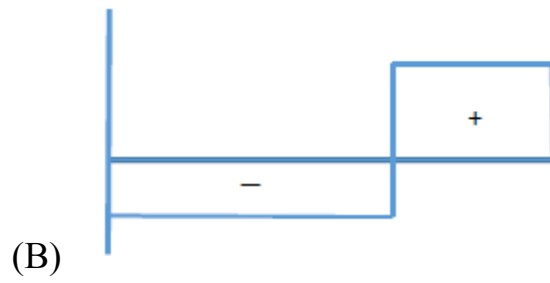
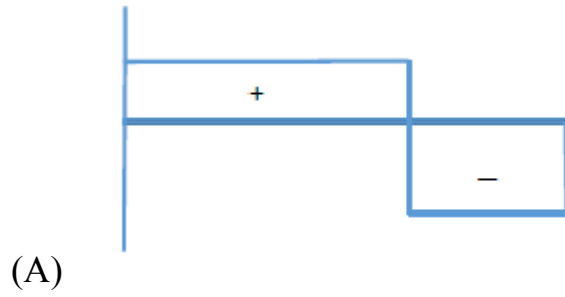
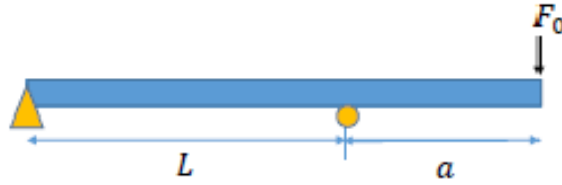
$$\sum M(V) = 0 \Rightarrow -R_A x - R_B(x - L) + M = 0$$

$$\Rightarrow M = F_0(x - L - a)$$

$$M = \begin{cases} -\frac{F_0 a x}{L} & x < L \\ F_0(x - L - a) & x \geq L \end{cases}$$



- 119) Determine the shear diagram which can represent the beam with F_0 acting at the right end of the beam:



The Answers is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)**

Step 1: Draw a free-body diagram, determine the support reactions R_A and R_B

$$R_B = \frac{F_0(L+a)}{L} \quad \text{and} \quad R_A = -\frac{F_0a}{L}$$

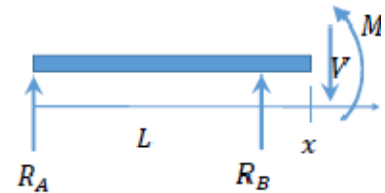
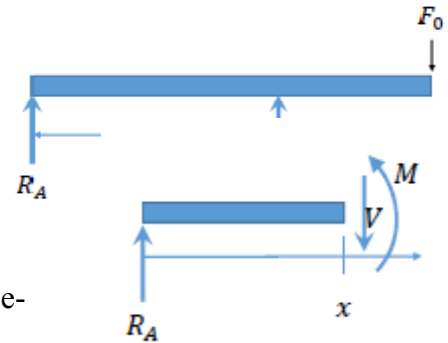
Step 2: Choose the beam's left end as the origin, draw a free-body diagram at position x ($x < L$)

$$\sum F_y = 0 \Rightarrow R_A - V = 0 \Rightarrow V = -\frac{F_0a}{L}$$

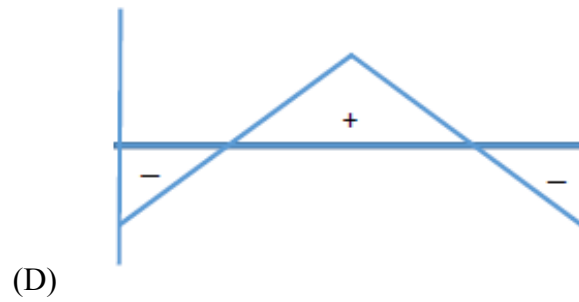
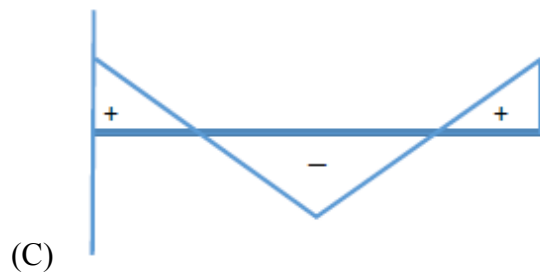
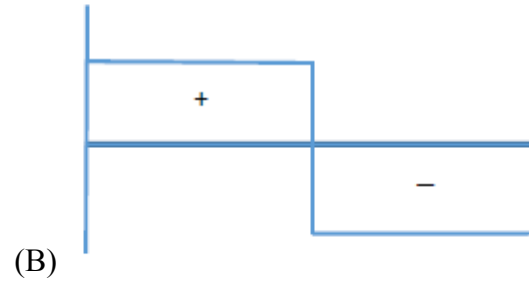
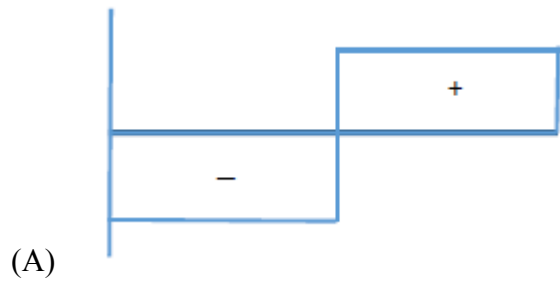
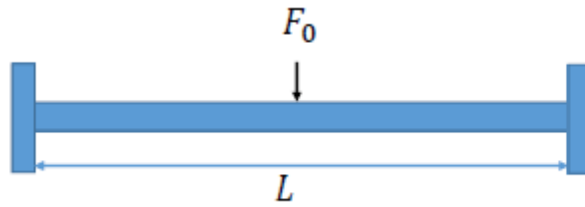
Step 3: Choose the beam's left end as the origin, draw a free-body diagram at position x ($x > L$)

$$\sum F_y = 0 \Rightarrow R_A + R_B - V = 0 \Rightarrow V = F_0$$

$$V = \begin{cases} -\frac{F_0a}{L} & x < L \\ F_0 & x \geq L \end{cases}$$



- 120) Determine the shear diagram which can represent the beam with F_0 acting at the middle of the beam:



The Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Draw a free-body diagram, determine the support reactions R_A and R_B

$$R_A = R_B = \frac{F_0}{2}$$

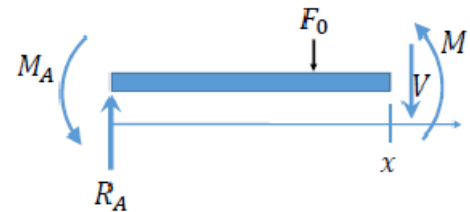
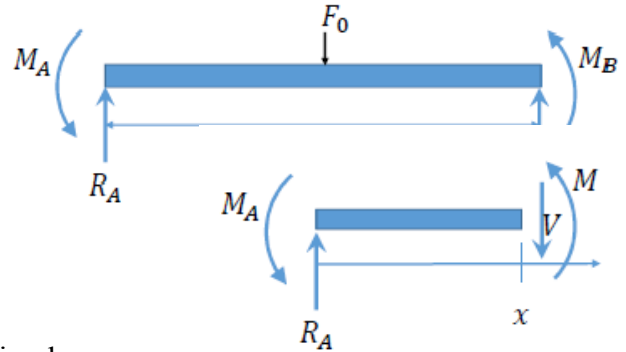
Step 2: Choose the beam's left end as the origin, draw a free-body diagram at position x ($x < 0.5 L$)

$$\sum F_y = 0 \Rightarrow R_A - V = 0 \Rightarrow V = \frac{F_0}{2}$$

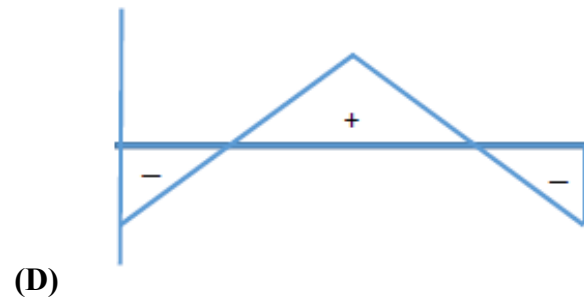
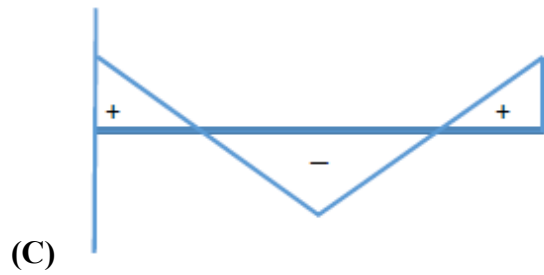
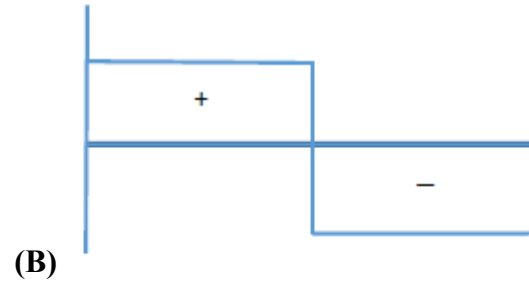
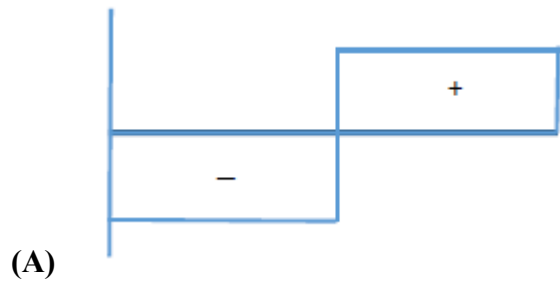
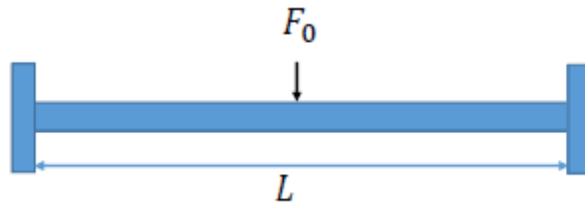
Step 3: Choose the beam's left end as the origin, draw a free-body diagram at position x ($x > 0.5 L$)

$$\sum F_y = 0 \Rightarrow R_A - F_0 - V = 0 \Rightarrow V = -\frac{F_0}{2}$$

$$V = \begin{cases} \frac{F_0}{2} & x < 0.5 L \\ -\frac{F_0}{2} & x \geq 0.5 L \end{cases}$$



- 121) Determine the moment diagram which can represent the beam with F_0 acting at the middle of the beam:



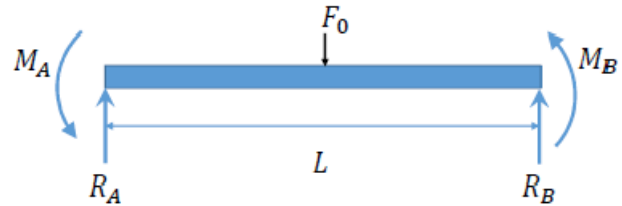
The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Draw a free-body diagram, determine the support reactions

$$R_A = R_B = \frac{F_0}{2}$$

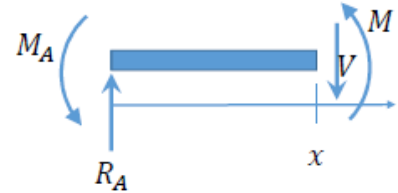
Due to symmetry, $M_A = -M_B = M_0$



Step 2: Choose the beam's left end as the origin, draw a free-body diagram at position x ($x < 0.5 L$)

$$\sum M(V) = 0 \Rightarrow M_A - R_A x + M = 0$$

$$\Rightarrow M = -M_0 + \frac{F_0 x}{2}$$

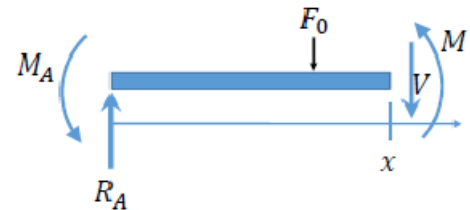


Step 3: Choose the beam's left end as the origin, draw a free-body diagram at position x ($x > 0.5 L$)

$$\sum M(V) = 0$$

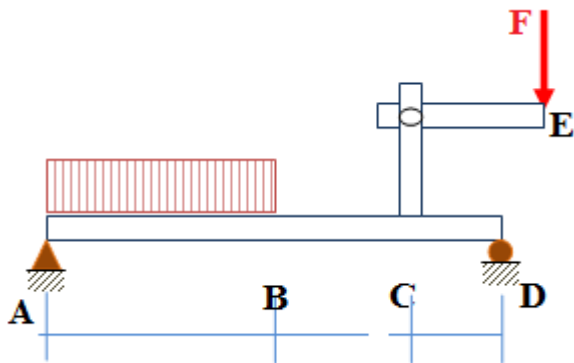
$$\Rightarrow M_A - R_A x + F_0 \left(x - \frac{L}{2} \right) + M = 0$$

$$\Rightarrow M = -M_0 - \frac{F_0(x-L)}{2}$$



$$M = \begin{cases} -M_0 + \frac{F_0 x}{2} & x < 0.5 L \\ -M_0 - \frac{F_0(x-L)}{2} & x \geq 0.5 L \end{cases}$$

122) The shear diagram for the following beam is:



- (A)
- (B)
- (C)
- (D)

The Answer is A

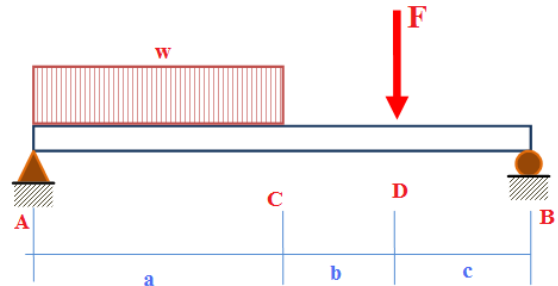
Step1: The force F acting on point E will transfer on point C as a concentrated load, therefore the shear diagram has a jump at point C .

Step2: Using the fact that the shear diagram is integration of load diagram, we can find that shear diagram on part AB is an inclined line. And constant between B and C .

Step3: Therefore the shear diagram is inclined at AB and decreases at point C , so (A) is correct.

123) For the following beam if load $w=3\text{KN/m}$ and reaction at $A=6\text{ KN}$, where the maximum moment occurs? $a=2b$ $b=c=1\text{m}$

- (A) 1 m from A.
- (B) 1.5 m from A.
- (C) Between D and B.
- (D) Between C and D.



The Answers is D

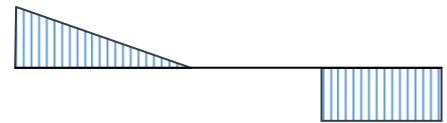
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step1: Using the fact that the moment diagram is the integration of shear diagram, we can find that the maximum moment occurs where the shear is Zero.

Step2: By drawing a sketch for the shear diagram it is found that shear is minimum around center, so let's find shear between C and D.

Step3: Make a section between C and D.

Step4: Find shear:



$$\sum F_y = 0 \quad R_A - wa + V = 0 \rightarrow V = -R_A + wa \rightarrow V = (2)(3) - 6 \rightarrow V = 0$$

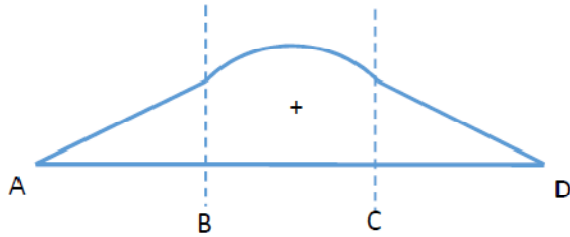
Therefore the maximum moment occurs between C and D.

124) From the following statements which one belongs the moment diagram for the beam shown?

- (A) The moment diagram is increasing linearly to maximum and then decreases.
- (B) The moment between A and B is zero
- (C) The moment between A and B is constant.
- (D) The moment diagram is parabolic between B and C.

The Answers is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)**

Answer "A" is not correct: The bending moment, shown as below, increases linearly from A to B and decreases linearly from B to C. The diagram is parabolic between B and C. The maximum moment occurs at the midpoint of BC.



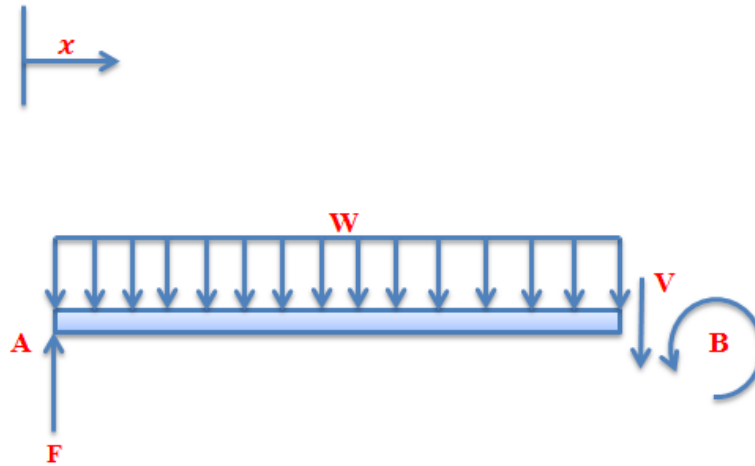
Answer "B" is not correct: The bending moment increases linearly from A to B.

Answer "C" is not correct: The bending moment is not constant between A and B.

Answer "D" is correct: The bending moment diagram is parabolic between B and C.

- 125) From the following expressions which one gives moment for the following simply supported beam of length L ? If x is the distance from the left pinned support. And $W = w \frac{kN}{m}$

- (A) $M = \frac{wL}{2}(L - x)$
(B) $M = \frac{wx^2}{2}(L - x)$
(C) $M = \frac{w}{2}(L - x)$
(D) $M = \frac{wx}{2}(L - x)$



The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Find the reactions:

$$F = \frac{wL}{2}, \quad M_A = 0$$

Step 2: Find the shear force at point B:

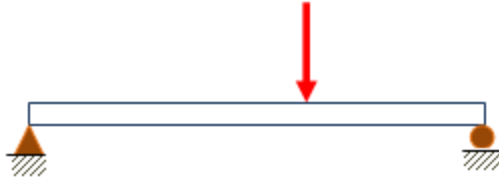
$$V = F - \int_0^x w dx = \frac{wL}{2} - wx$$

Step 3: Find the moment at point B:

$$M = M_A - \int_0^x \left(\frac{wL}{2} - wx \right) dx = 0 + \frac{wLx}{2} - \frac{wx^2}{2} = \frac{wx}{2}(L - x)$$

$$M = \frac{wx}{2}(L - x)$$

126) The shear diagram for the following beam is:



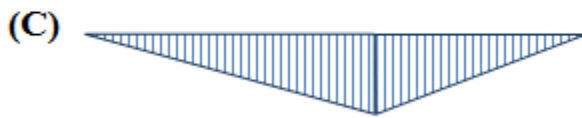
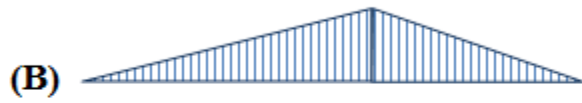
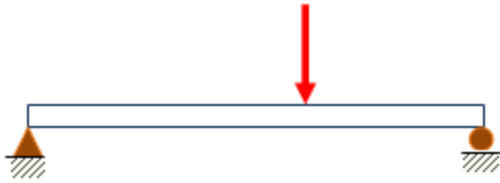
- (A) A shear diagram consisting of two rectangular blocks. The first block, on the left, is filled with vertical lines and is above the horizontal axis. The second block, on the right, is also filled with vertical lines and is below the horizontal axis.
- (B) A shear diagram consisting of two triangular blocks. The first block, on the left, is filled with vertical lines and has a linear variation from a maximum value at the left end to zero at the center. The second block, on the right, is also filled with vertical lines and has a linear variation from zero at the center to a maximum value at the right end.
- (C) A shear diagram consisting of two rectangular blocks. The first block, on the left, is filled with vertical lines and is above the horizontal axis. The second block, on the right, is also filled with vertical lines and is below the horizontal axis.
- (D) Non of the above.

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step1: By a glance at the reactions we know that at both supports there are reactions and they are upward (positive). Therefore the shear diagram should increase at both supports. Therefore the diagram (C) is the shear diagram of the beam.

127) The moment diagram for the following beam is:

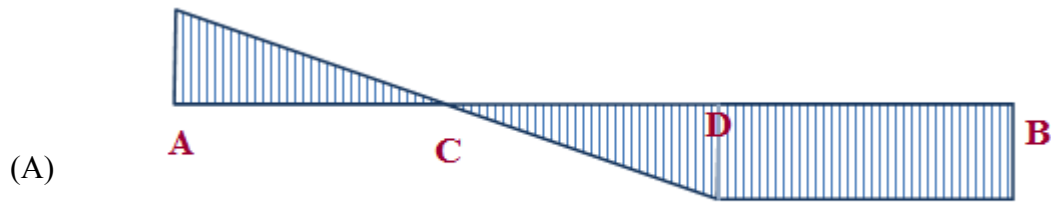


The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step1: By a glance we can find that the moment at reactions are zero because the supports cannot support moment, and shear diagram is a straight line so the moment diagram should have slope and is maximum under the load and is positive, (B) is the correct moment diagram for the beam.

128) The following shear diagram represents the beam with loading:



Concentrated load at D.

(B) Distributed load from C to B.

(C) Uniform distributed load from A to D.

(D) (A) and (C).

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Look at the shape of shear diagram,

Step 2: There is not jump on the diagram, which means there is not a concentrated load.

Step 3: The inclined diagram between A and D shows the uniform distributed load on A to D, because the shear diagram for uniform distributed load is an inclined line.

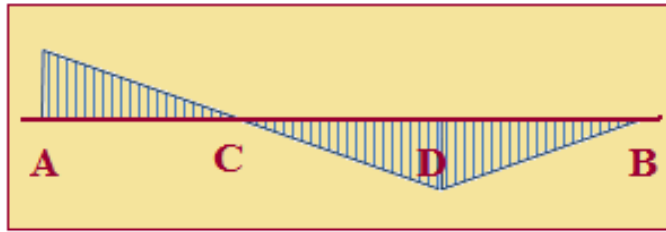
Answer "A" is not correct: There is no concentrated load.

Answer "B" is not correct: There is uniformly distributed load between A and D.

Answer "C" is correct: There is uniformly distributed load between A and D.

Answer "D" is not correct: There is no concentrated load.

129) The following moment diagram represents the member with loading:



- (A) One
- (B) from C
- (C) One moment at A.
- (D) (A) and (C)

concentrated load at D.
Uniformly Distributed load
to B.

The Answers is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Look up on the shape of moment diagram,

Step 2: The inclined line of moment diagram shows the load is concentrated and it is applied at D because the maximum moment is at that point.

Step 3: The jump at A shows a that a moment applied at A.

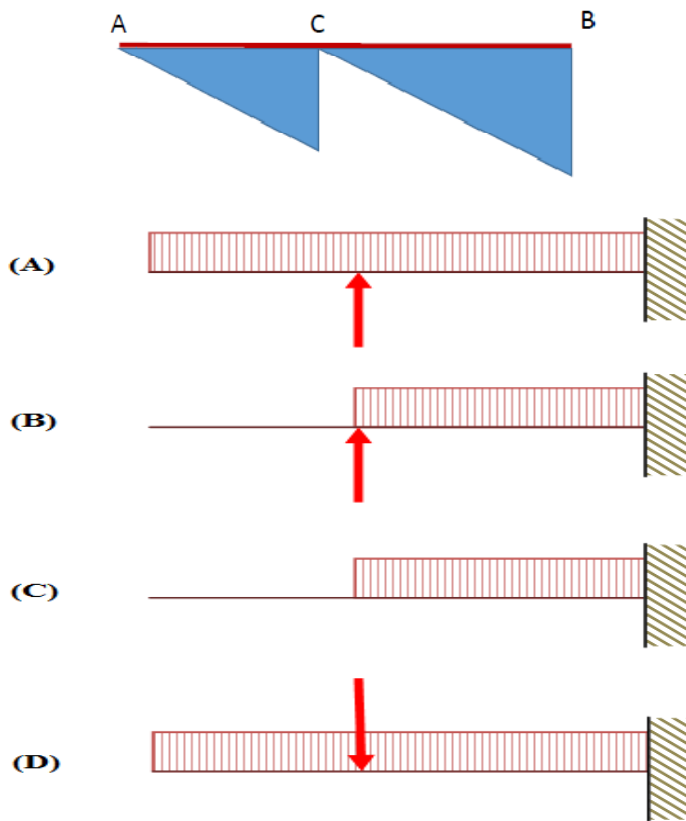
Answer "A" is not correct: There are one concentrated load at D and one moment at A.

Answer "B" is not correct: There is no uniformly distributed load between C and B.

Answer "C" is not correct: There are one concentrated load at D and one moment at A.

Answer "D" is correct: There are one concentrated load at D and one moment at A.

130) The following shear diagram represents the beam with loading:



The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)**

Step 1: From the fact that the shear diagram is the integration of load, the shear diagram is an inclined line, so the loading should be uniformly distributed load. The slope of shear diagram shows that shear zero at point A if there was not a jump at C; therefore, uniform load is distributed on full length.

Step 2: Upward jump at C shows that an upward concentrated load is applied at C.

Step 3: Therefore the shear diagram represents beam (A).

131) Find the distance (a) from the right support at which the shear force is zero. The beam is 20 m long as shown. $W = 10 \frac{\text{kN}}{\text{m}}$ and $b = 8 \text{ m}$

(A) 9 m

(B) 6.4 m

(C) 18 m

(D) 8 m

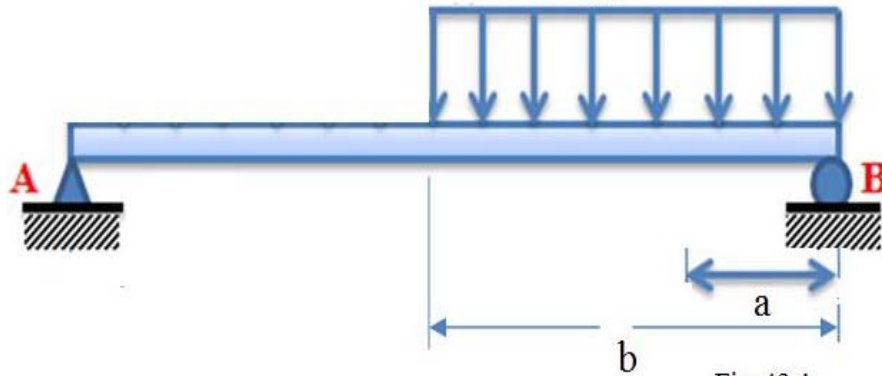


Fig. 43.A

The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Find the reactions at A by taking moment at B and then Reaction at B:

$$\sum M_B = 0$$

$$0 = -R_A(L) + W(b) \left(\frac{b}{2}\right)$$

$$R_A = \frac{10(8) \left(\frac{8}{2}\right)}{20} = 16 \text{ KN}$$

$$\sum F_y = 0$$

$$R_B = W(b) - R_A = 10(8) - 16 = 64 \text{ KN}$$

Step 2: Section the beam at **a**, from free-body

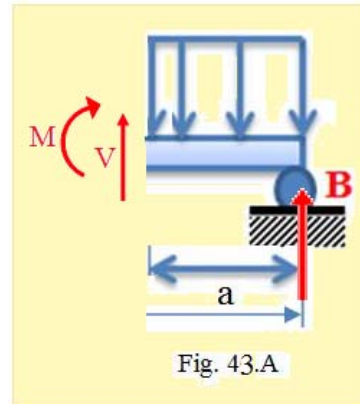
$$\sum F_v = 0$$

$$R_B - w(a) + V = 0$$

$$R_B = 10(a) - 0$$

$$a = 64/10$$

$$a = 6.4 \text{ m}$$



diagram

- 132) Find the shear force at distance $a = 6$ from the left support in the 20 m long beam shown, $W = 10 \frac{\text{kN}}{\text{m}}$ and $b = 8$ m.

(A) -90 kN (B) 80 kN (C) +16 kN (D) -16 kN

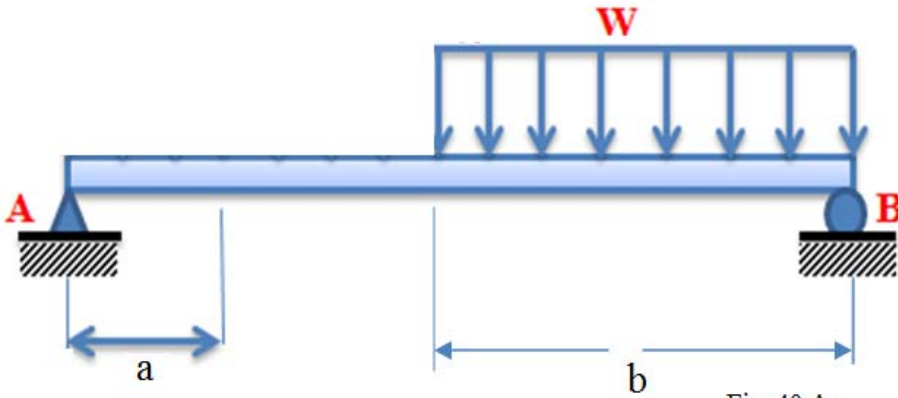


Fig. 40.A

The Answers is D

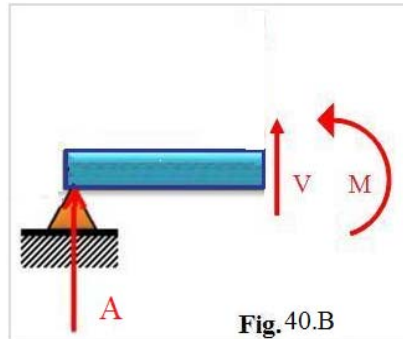
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Find the reaction force at A by taking moment at B:

$$\sum M_B = 0$$

$$0 = -R_A(L) + w(b) \left(\frac{b}{2} \right)$$

$$R_A = \frac{10(8) \left(\frac{8}{2} \right)}{20} = 16 \text{ KN}$$



Step 2: Section the beam at **a**, from free-body diagram

$$\sum F_y = 0$$

$$R_A + V = 0$$

$$V = -16 \text{ KN}$$

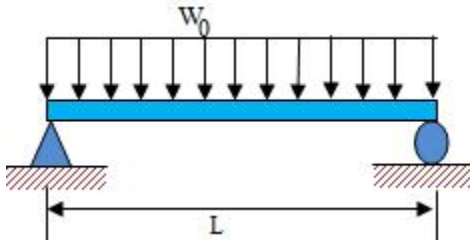
133) In Fig. below, find the maximum shear and moment in a simply supported beam that is loaded with uniform load of w_0 .

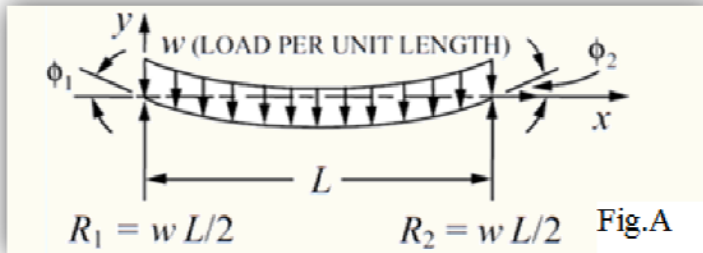
(A) $V = w_0L$, $M = w_0L^2/4$

(B) $V = w_0L/2$, $M = w_0L/8$

(C) $V = w_0L/2$, $M = w_0L^2/8$

(D) $V = 2 w_0L$, $M = w_0L^2/12$



The Answers is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)****Step 1:** Free body diagram:**Step 2:** Find the reactions:from symmetry, $R_1 = R_2$. And summing vertical forces:

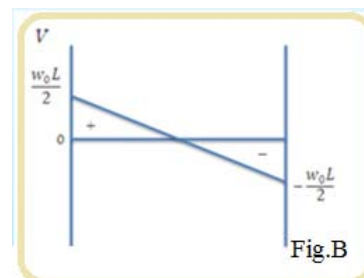
$$\sum F_y = 0, \quad R_1 = R_2 = w_0 L / 2$$

Step 3: Draw the Shear diagramThe load $q = -w_0$ and Shear diagram is result of load diagram (Fig. B):

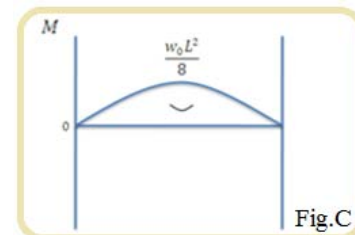
$$V = V_0 - \int_0^x w_0 dx = \frac{w_0 L}{2} - w_0 x$$

So the maximum shear occurs when $x=0$

$$V = w_0 L / 2$$



integration of



Step 4: Draw the Moment diagram; Moment diagram is result of integration of shear diagram. Nothing that the moment at $x = 0$ is zero.

$$M = M_0 - \int_0^x \left(\frac{w_0 L}{2} - w_0 x \right) dx = 0 + \frac{w_0 Lx}{2} - \frac{w_0 x^2}{2} = \frac{w_0 x}{2} (L - x)$$

It can be seen that the maximum bending moment occurs at the center of the beam where the shear stress is zero (at the middle of the beam).

$$\text{When } x = \frac{L}{2} \quad M = \frac{w_0 L^2}{8}$$

- 134) Determine the shear and moment at the middle of the beam shown below. $L = 8\text{ m}$, $w = 20 \frac{\text{KN}}{\text{m}}$.

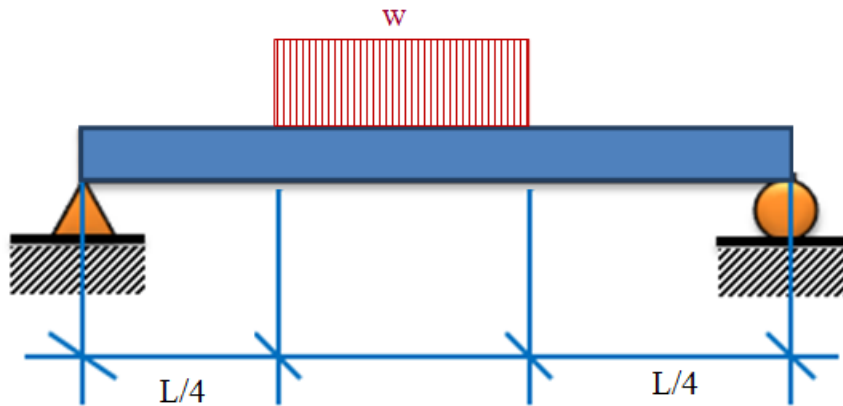


Fig. 1.A

- (A) $M = 40$ $V = 0$,
(B) $M = 40$ $V = 40$
(C) $M = 220$ $V = 0$
(D) $M = 0$ $V = 40$

The Answers is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: Find the Reactions:

$$\sum M_B = 0$$

$$R_A (8) - w(L/2)(L/2) = 0$$

$$R_A = 40 \text{ KN}$$

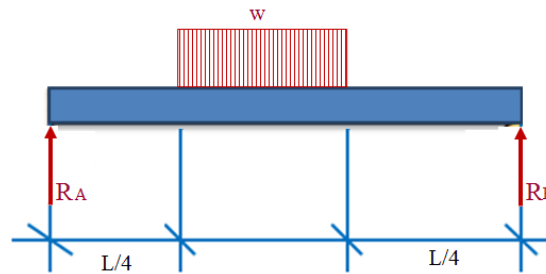


Fig. 1. B

Step 2: Draw Free-Body-Diagram. At a section of beam at the middle, apply the equilibrium equations

$$\sum M_V = 0$$

$$M = R_A (L/2) - w(L/4)(L/4 + L/8)$$

$$M = 40(4) - 20(2)(2+1)$$

$$M = 40 \text{ KN-m}$$

$$\sum F_y = 0$$

$$R_A - w(L/4) + V = 0$$

$$V = 40 - 40 = 0$$

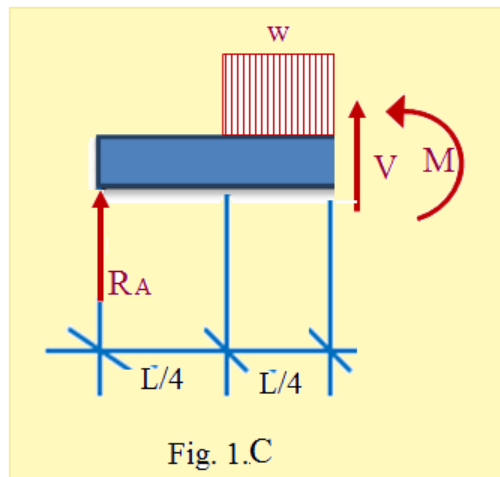


Fig. 1.C

135) Determine the moment at the middle of the beam shown. If $w = 1 \frac{\text{KN}}{\text{m}}$ and $C = D = 2\text{m}$

- (A) $-2 \text{ KN} \cdot \text{m}$
- (B) $1 \text{ KN} \cdot \text{m}$
- (C) $-1 \text{ KN} \cdot \text{m}$
- (D) $4 \text{ KN} \cdot \text{m}$

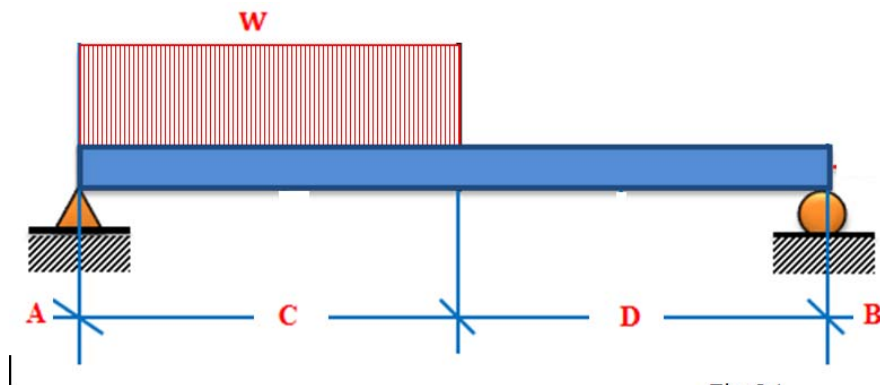


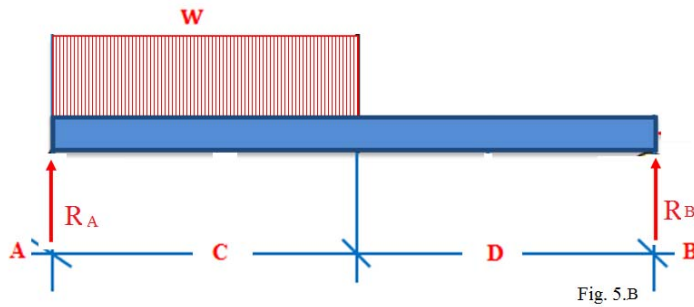
Fig. 5.A

The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1:

Draw free-body diagrams for beam to find reaction forces.



Step 2:

Calculate the Reaction forces:

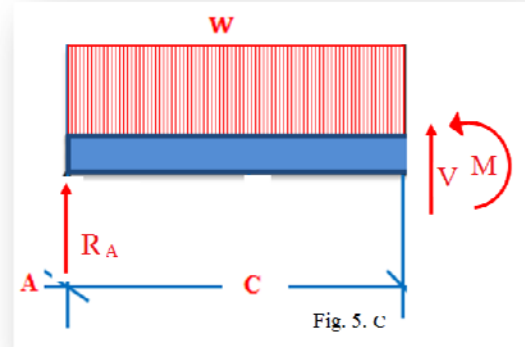
R_A :

$$\sum M(B) = 0 \rightarrow W(C)(D+C/2) - R_A(C+D) = 0$$

$$1(2)(3) - R_A(4) = 0$$

$$\rightarrow R_A = 1.5 \text{ KN}$$

Step 3: Draw free-body diagram at a section at the middle, as shown below.

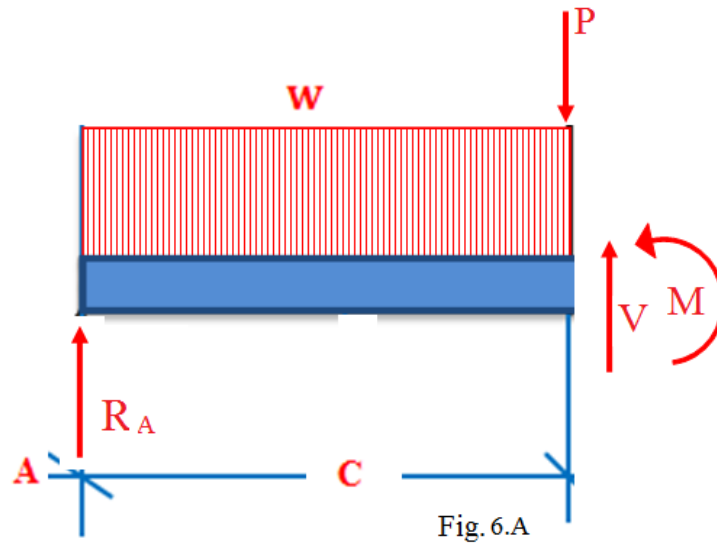


Step 4: Find M :

$$\sum M = 0 \quad \rightarrow \quad M + W(C)(C/2) - R_A(C) = 0$$
$$M + 1(2)(2/2) - 1.5(2) = 0$$

$$M = 1 \text{ KN}\cdot\text{m}$$

- 136) A beam at the section shown can carry moment $M = 12 \text{ KN}\cdot\text{m}$ and negative shear force $V = 5 \text{ KN}$ find the uniform load w . If $P = 2 \text{ KN}$ and $C = 2 \text{ m}$



- (A) 5 KN
- (B) 15KN
- (C) -5 KN
- (D) 9 KN

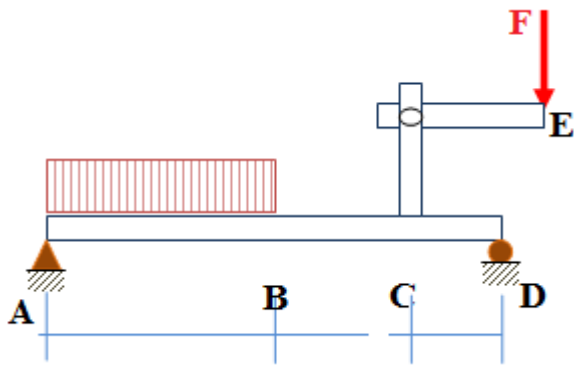
The Answers is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)**Step 1: Using the Free-Body-Diagram:

$$\sum M_{(A)} = 0 \quad 0 = M - w (C)(C/2) - P (C) + V (C)$$

$$0 = 12 - w(2)(1) - 2(2) + (5)(2)$$

$$w = (12+6)/2 = 9 \text{ KN}$$

137) In the following beam the moment diagram:



- (A) is parabolic through beam AD.
- (B) is parabolic through beam AC.
- (C) is parabolic through beam AB and straight line to D.
- (D) is parabolic through beam AB and had jump on C.

The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step1: The Force F acting on point E creates moment at point C , therefore the moment diagram has a jump at point C .

Step2: Using the fact that the moment diagram is the integration of shear diagram, and the shear diagram is integration of load diagram, we can find that shear diagram on part AB is an inclined line and the moment diagram should be parabolic between A and B .

Step3: There is no external force acting between B and C . The shear force is constant in beam BC , and the moment diagram is a straight line.

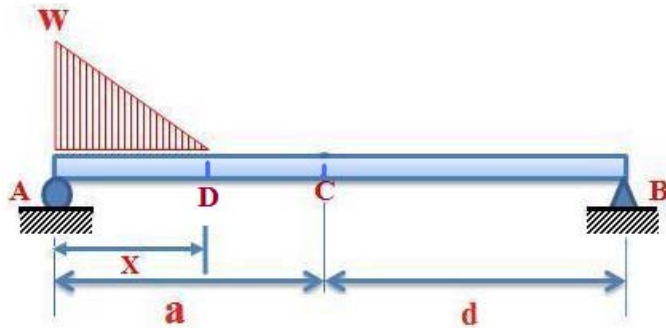
Answer "A" is not correct: The moment diagram is parabolic only through beam AB .

Answer "B" is not correct: The moment diagram is parabolic only through beam AB , and then straight line through beam BC .

Answer "C" is not correct: The moment diagram has a jump on C .

Answer "D" is correct: The moment diagram is parabolic only through beam AB , and then has a jump on C .

138) For the following beam the maximum moment is: $a=d$



- (A) At
- (B) At point D
- (C) Between D and B
- (D) Between A and D.

point C.

The Answer is D

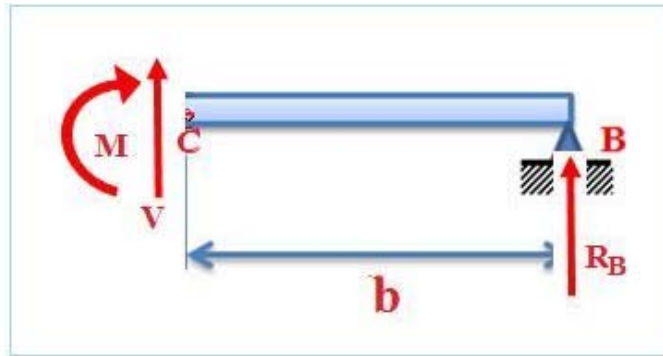
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step1: Look up on the shape of shear diagram, and find the location that shear can be zero. That will be the maximum point for the moment diagram.

Step2: Make a section at a point D and B (take the side that does have load)

between
not

Step3: Draw a free-body as shown:



diagram

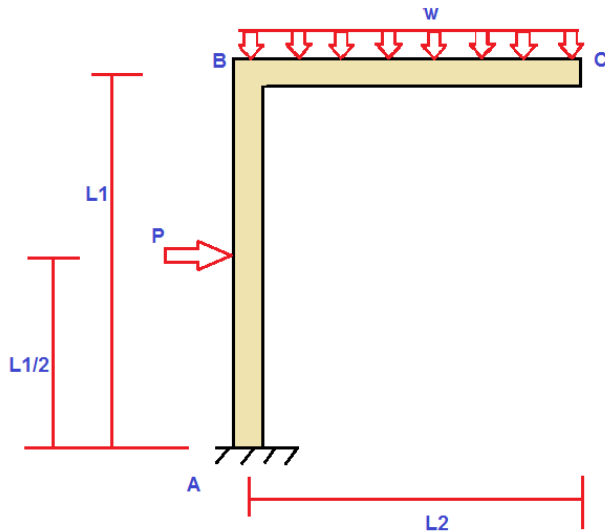
Step4: Find the shear by applying the equilibrium equations:

$$\sum F_y = 0 \rightarrow V = R_B$$

Which is constant, and cannot be zero, therefore maximum moment cannot occur between D and B, so (A), (B) and (C) are not correct.

Therefore (D) is correct.

- 139) A plane frame consisting of 2 rigid members is supported by a fixed connection at the base. The frame is subject to a uniformly distributed load and point load of 2 kips/ft and 25 kips respectively. $L1 = 20$ ft, and $L2 = 18$ ft. Determine the forces in the rigid connection B.



- A) $F_{By} = 36$ kips; $F_{Bx} = 25$ kips $M_B = 324$ kip · ft
B) $F_{By} = 36$ kips; $F_{Bx} = 25$ kips $M_B = 574$ kip · ft
C) $F_{By} = 36$ kips; $F_{Bx} = 0$ kips $M_B = 574$ kip · ft
D) $F_{By} = 36$ kips; $F_{Bx} = 0$ kips $M_B = 324$ kip · ft

The Answers is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)

Step 1: First determine the reactions at support A then we can trace the load to connection B. To do this take the equilibrium equations for the entire system and solve for the reactions at the supports. The point load acts at half the length of member AB and the distributed load will act at half the length of member BC. A fixed connection resists moment.

$$\sum M_A = 0 = M_A - (wL2) \left(\frac{L2}{2} \right) - P \left(\frac{L1}{2} \right)$$

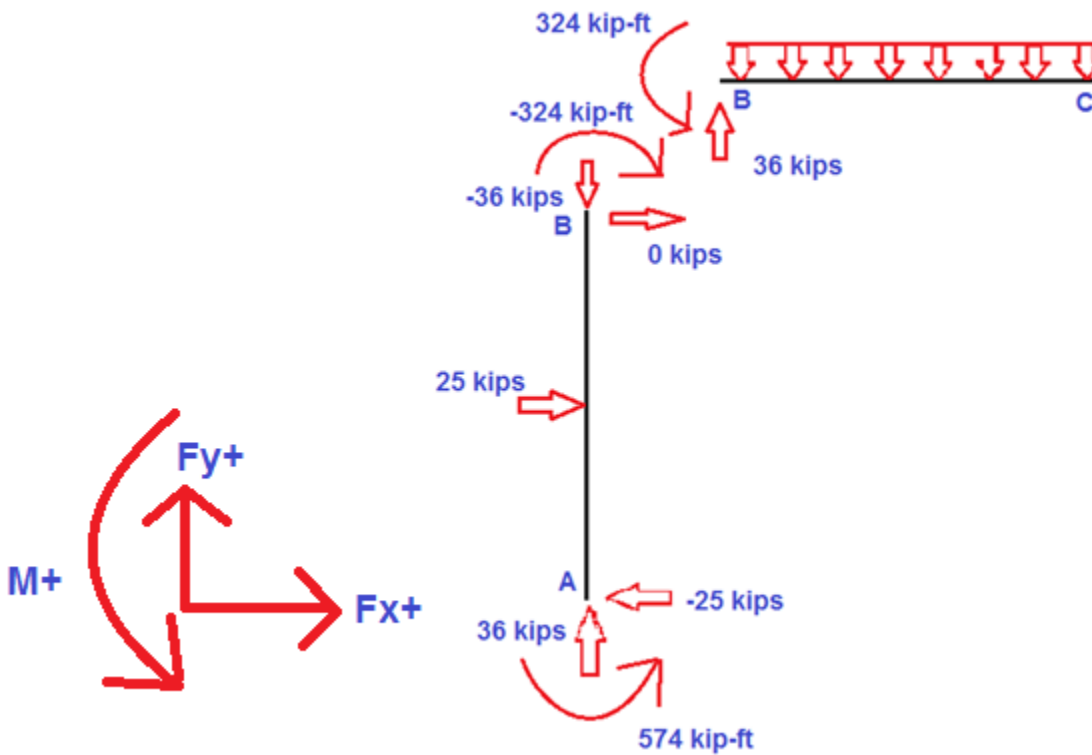
$$M_A = (wL2) \left(\frac{L2}{2} \right) + P \left(\frac{L1}{2} \right) = 574 \text{ kip} \cdot \text{ft}$$

$$\sum F_x = 0 = F_{Ax} + P$$

$$\sum F_y = 0 = F_{Ay} - (wL2)$$

$$F_{Ax} = -25 \text{ kips}; F_{Ay} = 36 \text{ kips}$$

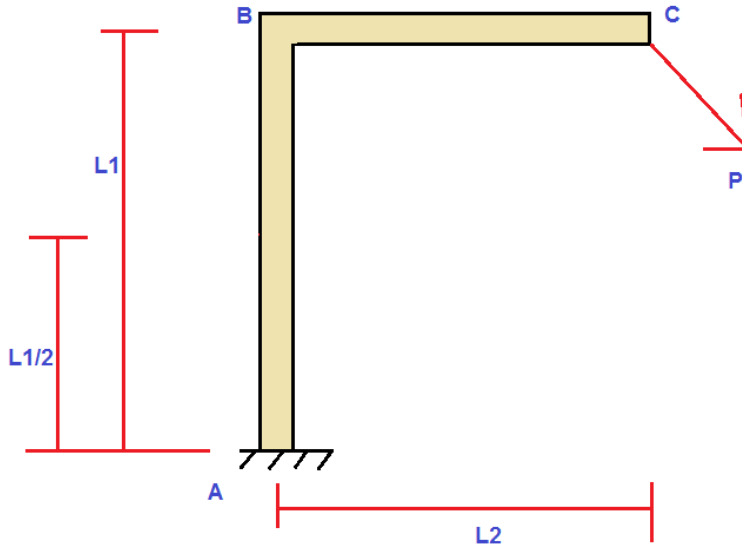
Step 2: Since this frame is rigidly connected it can support moments at its connections. Because of this the moment caused by any loads not in line with the member need to be considered. For each member we can determine the forces at each location on the structure. Each member must be in equilibrium individually. Thus if there is a force at one end of the member than an identical force considering other loads across the member will be at the other end. See the diagram below.



The moment at connection A will be transferred to connection B. This moment is reduced by the 25 kip force at mid-span of member AB. The moment at A rotates counterclockwise and the moment caused by the 25 kip load rotates clockwise thus the moment caused by the load subtracts from the moment at the support. The final magnitude of the moment at connection B is **324kip · ft**.

$$\sim F_{By} = 60.3 \text{ kips}; M_B = 324 \text{ kip} \cdot \text{ft}$$

- 140) A plane frame consisting of 2 rigid members is supported by a fixed connection at the base. The frame is subject to an angled point load of 130 kips at 45 degrees from vertical. $L_1 = 20$ ft, and $L_2 = 18$ ft. Determine the forces in the rigid connection B.



- A) $F_{By} = 91.9$ kips; $F_{Bx} = 91.9$ kips $M_B = 3492$ kip · ft
B) $F_{By} = 91.9$ kips; $F_{Bx} = 0$ kips $M_B = 3492$ kip · ft
C) $F_{By} = 91.9$ kips; $F_{Bx} = 91.9$ kips $M_B = 2854$ kip · ft
D) $F_{By} = 91.9$ kips; $F_{Bx} = 0$ kips $M_B = 2854$ kip · ft

The Answers is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 47.2, page 47-(5-15)**

Step 1: First determine the reactions at support A then we can trace the load to connection B. To do this take the equilibrium equations for the entire system and solve for the reactions at the supports. In order to take equilibrium equations we need to determine the horizontal and vertical components of our point load. At 45 degrees these components are equal.

$$P_x = P_y = (130 \text{ kips})\sin(45) = (130 \text{ kips})\cos(45) = 91.9 \text{ kips}$$

$$\sum M_A = 0 = M_A - P_x(L2) - P_y(L1)$$

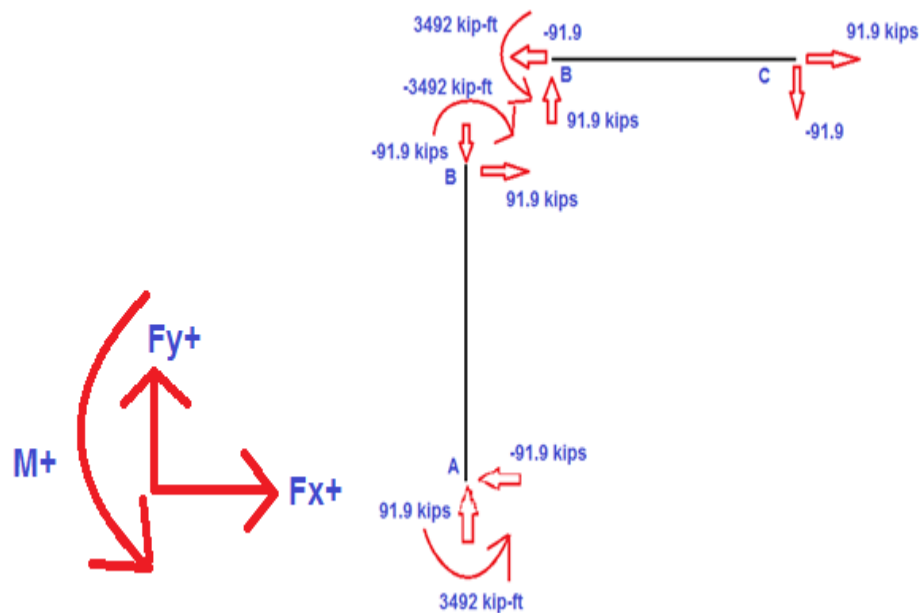
$$M_A = P_x(L2) + P_y(L1) = 3492 \text{ kip} \cdot \text{ft}$$

$$\sum F_x = 0 = F_{Ax} + P_x$$

$$\sum F_y = 0 = F_{Ay} - P_y$$

$$F_{Ax} = -91.9 \text{ kips}; F_{Ay} = 91.9 \text{ kips}$$

Step 2: Since this frame is rigidly connected it can support moments at its connections. Because of this the moment caused by any loads not in line with the member need to be considered. For each member we can determine the forces at each location on the structure. Each member must be in equilibrium individually. Thus if there is a force at one end of the member than an identical force considering other loads across the member will be at the other end. See the diagram below.

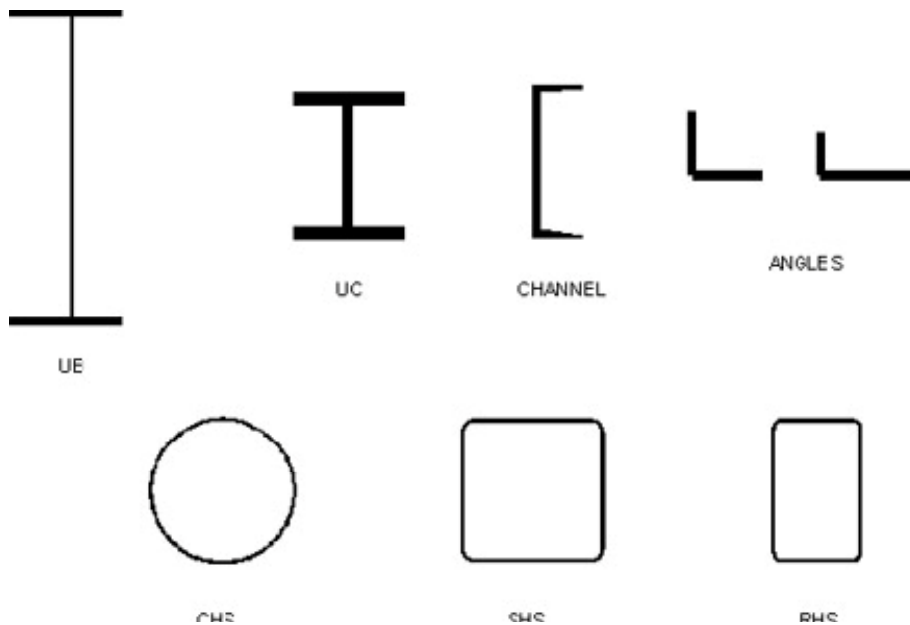


The moment at connection A will be transferred to connection B. There are no forces specifically applied to this member so the magnitude of the moment at connection B will be equal to the moment at support A. The final magnitude of the moment at connection B is **324 $\text{kip} \cdot \text{ft}$** . All forces in the horizontal and vertical direction are conserved throughout the system.

$$\sim F_{By} = 91.9 \text{ kips kips}; M_B = 3492 \text{ kip} \cdot \text{ft} \quad F_{Bx} = 91.9 \text{ kips}$$

I. Columns (Column and slenderness)

- 141) For the same length and support conditions, which section shows less stability against buckling? (Assume the same weight or area for the sections).

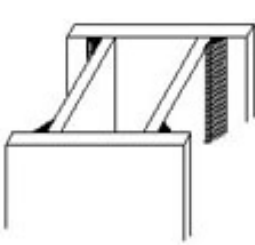


- (A) UC
- (B) SHS
- (C) RHS
- (D) CHS

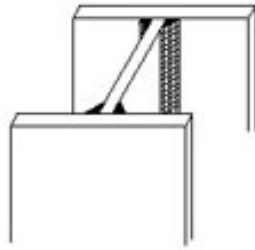
The Answer is A**AISC 13th ed, 2010, 16.1-227-243 & 16.1-32**

To minimize the buckling effects, it is required to maximize the radius of gyration along both axis. All RHS, SHS, and CHS shows almost the same radius of gyration and I sections do not have an equal radius of gyration, so for the effect of vertical force choice "A" will give the less strength for the buckling when the area of all sections are given the same.

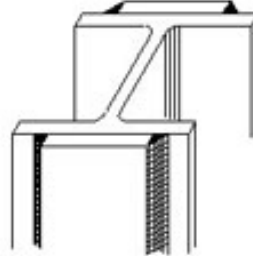
- 142) For the shown columns, which section most likely shows minimum sensitivity to the buckling at both direction when the minimum weight of column is the target?



Welded box
section



Welded
H-section



Welded cover plate
on hot-rolled
H-section

- (A) Welded cover plate
- (B) Welded H section
- (C) Welded box section
- (D) They have the same behavior

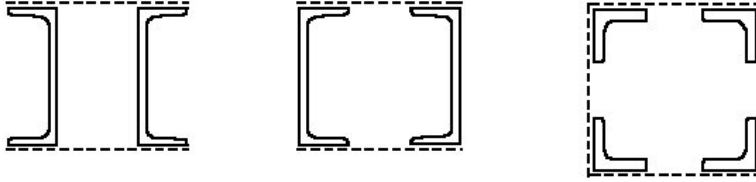
The Answer is C**AISC 13th ed, 2010, 16.1-227-243 & 16.1-32**

To minimize the buckling effects, it is required to maximize the radius of gyration along both axis. The best section is considered with equal radius of gyration along both axis. So, technically, welded box sections can maintain this fact with the minimum materials.

Two other sections also can maintain the requirement but by using more material. Because the moment of inertia of the box section is estimated with this expression: $I = bh^3/12 + Ad^2$ but the other two sections can maintain the moment of inertia by, $I = bh^3/12$. Missing the term of Ad^2 will consequently create lesser moment of inertia and thus the radius of gyration.

So, choice C is correct.

- 143) Which one of the following build-up sections in the below figure can create the same strength and stability as a column along both axis?



U or angle sections used as main components



- (A) None of them, just use the I section for columns
- (B) Columns with Channels and angle sections
- (C) Columns with I sections
- (D) All of them

The Answer is D**AISC 13th ed, 2010, 16.1-227-243 & 16.1-32**

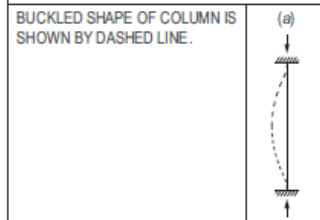
To minimize the buckling effects, it is required to maximize the radius of gyration along both axis. The best section is considered with equal radius of gyration along both axis. So, technically, all these sections can maintain equal radius of gyration by adjusting the distance between sections.

So, choice D is correct.

- 144) In the below figure which shows a real buckled column after an earthquake which statement is likely correct?



- (A) The column can be considered with both end pinned connections.
- (B) The column can be considered with both end almost fixed connections.
- (C) The column can be considered with Fixed at base and free on top.
- (D) The column can be considered with pinned at base and almost fixed on top.

The Answer is B**AISC ASD-LRFD, 13th edition, 16.1.240**

Comparing the buckling behavior in the above figure and the real column shows a both end fixed column. So, choice B is correct.

- 145) In the below figure which shows a real buckled bracing with the angle section after an earthquake; which statement is likely correct?



- (A) The bracing was not designed for the effect of buckling, because the tension member shows enough strength and it is almost in the straight line.
- (B) The angle buckled along its principle minor axis and not along the geometrical axis.
- (C) The tension elements created a flexible support at the connection of both sections, so the buckling length was not the whole length of the bracing.
- (D) All of them are correct.

The Answers is D

AISC 13th ed, 2010, 16.1-227-243 & 16.1-32

All of them are applied.

- 146) In the below figures which were taken after an earthquake, the tie plate on the build-up column was broken (right figure)? The columns did not experience the bending moment but this failure was happened. What would likely make this failure in that build-up section according to the codes?



- (A) The number of ties were not sufficient to resist against longitudinal shear force.
- (B) The welding of the tie beams neither have sufficient length nor size.
- (C) The shear force in the column was influenced form the 0.02 axial force and the overturning of building was increased this force, so the shear force increased and consequently the ties were failed.
- (D) All answers are possible

The Answers is D

AISC 13th ed, 2010, 16.1-227-243 & 16.1-32

All of them are applied.

- 147) In the below figure the stepped columns are shown. If we assume no bending moment distribution from the roof truss why the columns have extremely bigger size than the other side?



- (A) They are the seat for the crane girders
- (B) Columns have shorter free length in the longitudinal directions, so they need less radius of gyration along the minor axis.
- (C) Columns have the fixed base along the major axis and have more length than the longitudinal direction.
- (D) All of them are the reasons for this design.

The Answer is D

AISC 13th ed, 2010, 16.1-227-243 & 16.1-32

All of them are definitions.

148) Which end condition results in the strongest column. Assume that the columns are made of the same material and have the same dimension.

(E) Pinned-pinned

(F) Fixed-fixed

(G) Fixed-pinned

(H) Fixed-free

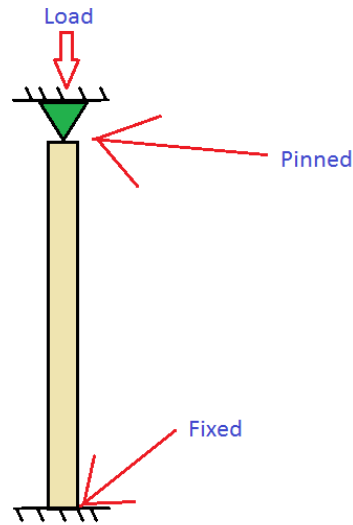
The Answer is B**AISC 13th ed, 2010, 16.1-227-243 & 16.1-32**

Step 1: The critical axial load for a long column subject to buckling can be calculated as:

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} \text{ where } K \text{ is the effective-length factor.}$$

Step 1: The fixed-fixed end condition gives the smallest effective-length factor $K = 0.5$. It results in the largest critical load if the columns have the same dimension and material.

- 149) The column shown is a W 21 X 50 steel member. The applied dead load is 7 kips. Neglect self-weight. The column length is 12 ft tall. The column is not braced along the length. Assume $F_y = 50$ ksi. Determine the effective length of the column.



- A) 12 ft B) 14.4 ft
C) 24.2 ft D) 9.5 ft

The Answers is D

AISC ASD-LRFD, 13th edition, 16.1.240

TABLE C-C2.2 APPROXIMATE VALUES OF EFFECTIVE LENGTH FACTOR, K						
BUCKLED SHAPE OF COLUMN IS SHOWN BY DASHED LINE.	(a)	(b)	(c)	(d)	(e)	(f)
						
THEORETICAL K VALUE	0.5	0.7	1.0	1.0	2.0	2.0
RECOMMENDED DESIGN VALUE WHEN IDEAL CONDITIONS ARE APPROXIMATED	0.65	0.80	1.2	1.0	2.10	2.0
END CONDITION CODE	 ROTATION FIXED AND TRANSLATION FIXED  ROTATION FREE AND TRANSLATION FIXED  ROTATION FIXED AND TRANSLATION FREE  ROTATION FREE AND TRANSLATION FREE					

Step 1: We first need to determine an effective length factor for the column. The top end of our system is fixed in translation but free in rotation, and the bottom end is fixed in both translation and rotation. See the appropriate table which is shown above. For our system select an effective length factor of 0.8 for design.

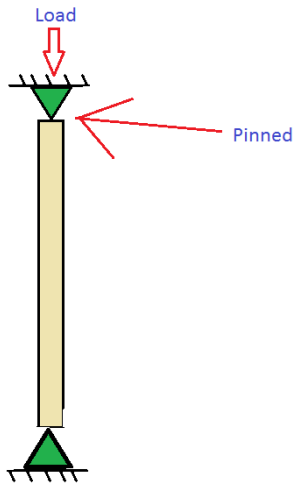
Step 2: Calculate the effective length for our column.

$$KL = (0.8)(12 \text{ ft}) = 9.6 \text{ ft}$$

The column size and applied load are extra information.

~9.5 ft

- 150) A column pinned at both ends is subject to an axial dead load of 60 kips and an unspecified live load. The column is composed of three plates forming a W-shape. Each flange is a 1in X 9in plate and the web is a 1/4in X 15in plate. The column is 15 ft in length and un-braced. Determine the slenderness ratio for this column.



- A) 80 B) 71
C) 65 D) 76

The Answers is D

AISC ASD-LRFD, 13th edition, 16.1.240

TABLE C-C2.2 APPROXIMATE VALUES OF EFFECTIVE LENGTH FACTOR, K						
BUCKLED SHAPE OF COLUMN IS SHOWN BY DASHED LINE.	(a)	(b)	(c)	(d)	(e)	(f)
						
THEORETICAL K VALUE	0.5	0.7	1.0	1.0	2.0	2.0
RECOMMENDED DESIGN VALUE WHEN IDEAL CONDITIONS ARE APPROXIMATED	0.65	0.80	1.2	1.0	2.10	2.0
END CONDITION CODE	 ROTATION FIXED AND TRANSLATION FIXED  ROTATION FREE AND TRANSLATION FIXED  ROTATION FIXED AND TRANSLATION FREE  ROTATION FREE AND TRANSLATION FREE					

Step 1: First determine some section properties for the section. Since this is a custom section they need to be calculated. Use appropriate equation to calculate the moment of inertia and area. Since the unbraced length is the same in both direction weak axis buckling will control for design.

$$I_y = \frac{2(1.00 \text{ in})(9.00 \text{ in})^3}{12} + \frac{(15 \text{ in})(0.25 \text{ in})^3}{12} = 121.5 \text{ in}^4$$

$$A = 2(9.00 \text{ in})(1.00 \text{ in}) + (15 \text{ in})(0.25 \text{ in}) = 21.75 \text{ in}^2$$

Now we can calculate the radius of gyration for the weak axis. Use the following equation.

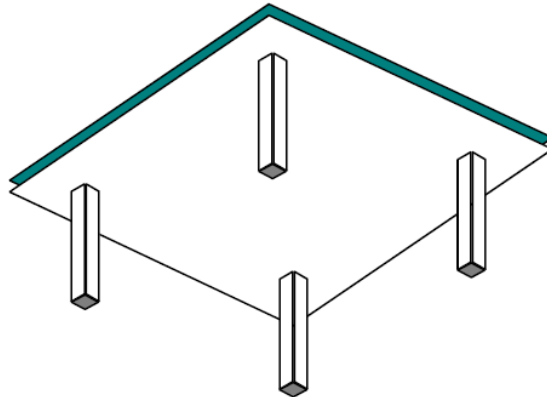
$$r_y = \sqrt{\frac{I_y}{A}} = \sqrt{\frac{121.5 \text{ in}^4}{21.75 \text{ in}^2}} = 2.36 \text{ in}$$

Step 2: Calculate the slenderness ratio. To do this we only need the effective length for the column. See table shown above. For pinned-pinned end conditions use $k = 1.0$. Slenderness equations can be seen below.

$$\frac{kL}{r} = \frac{(1.0)(15 \text{ ft})(12 \frac{\text{in}}{\text{ft}})}{2.36 \text{ in}} = 76.27 \quad \sim 76$$

J. Slabs

151) In the flat slabs which statement is correct?



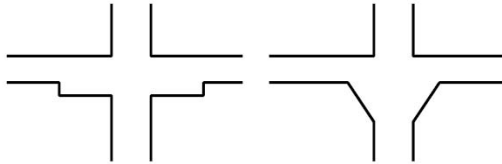
- A) There is no beams in this system
- B) The beam strip in the slab resists against lateral loads
- C) The thickness of slab is a parameter that can save the slab against punching shear stress.
- D) All of them are correct

The Answer is D

ACI 318-08, 13.1-13.7, page 233-250

All of them are definitions.

152) In the flat slabs when should a drop panel be used?



Slab with drops

Flared column head

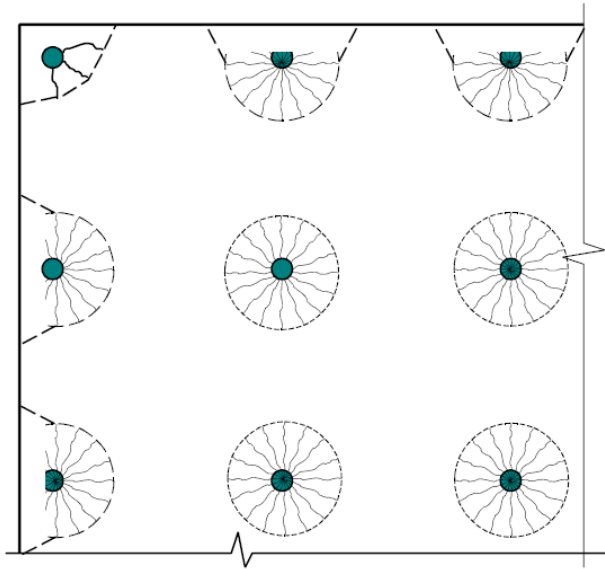
- A) When the architect ask for aesthetics.
- B) It will help for more flexural strength at the column locations.
- C) It is just help to increase the punching shear strength.
- D) It helps to increase the shear capacity of slab.

The Answer is C

ACI 318-08, 21.3.6., page 331

This is the definition.

- 153) The following figure shows the plan view of the flat slab, if the axial load in columns are given equal (i.e., equally transferred shear force from the slab) and the slab has the uniform thickness, then which columns may need drop panel due to the amount of the transferred shear force?



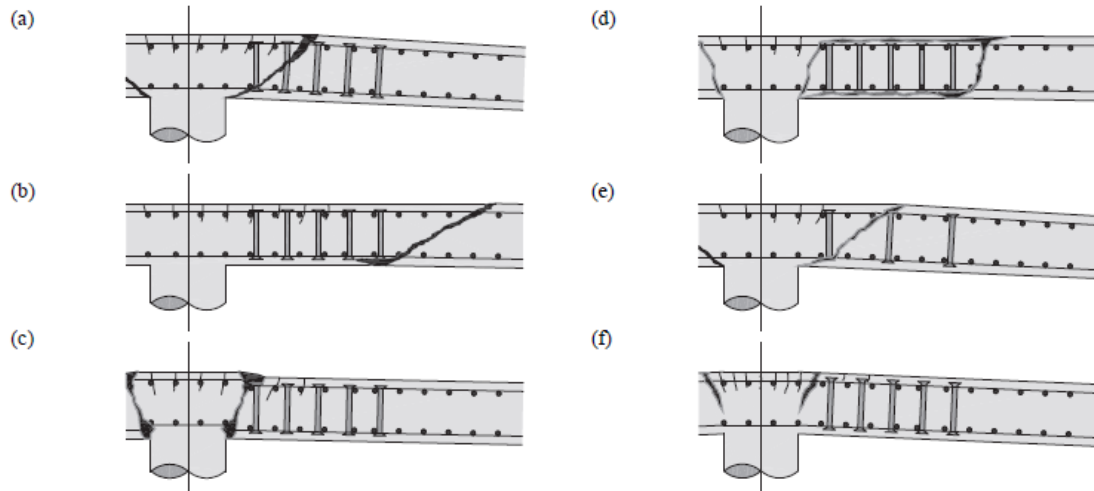
- A) Corners columns
B) side columns
C) middle columns
D) drop panel is not required

The Answer is A

ACI 318-08, 21.3.6., page 331

If the amount of shear force is equal for all columns then, the critical column has the minimum punching shear area. In this case the corner column has the minimum punching shear area, so "A" is correct.

154) Which one is not showing the shear failure in the flat slabs?



A) e

B) b

C) a

D) c & f

The Answer is A

ACI 318-08, 21.3.6., page 331

The picture and explanation captured from: Punching of flat slabs with large amount of shear reinforcements, by "Ecole Polytechnique Federale De Lausanne", Thesis No. 5409, 2012

Possible failure modes are: a, b, & e are different modes of shear failure. "c" shows the crushing of concrete due to high pressure, d is the delamination of concrete core, and f is the flexural failure.

155) For the shown concrete slab with surrounding beams punching shear should be:



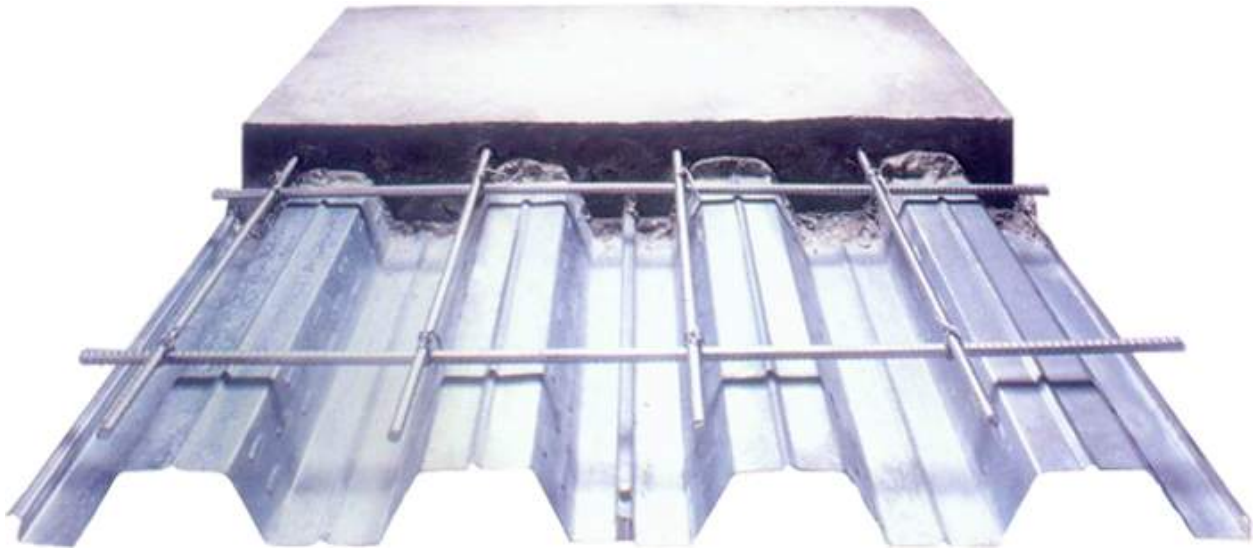
- A) Neglected
- B) calculated
- C) no punch shear will occur
- D) both A and C are correct

The Answer is D

ACI 318-08, 21.3.6., page 331

The slabs with the surrounding beams do not have punching shear and the beams will carry the shear force. So, D is correct.

156) What is the role of steel corrugated sheets in the metal deck system?



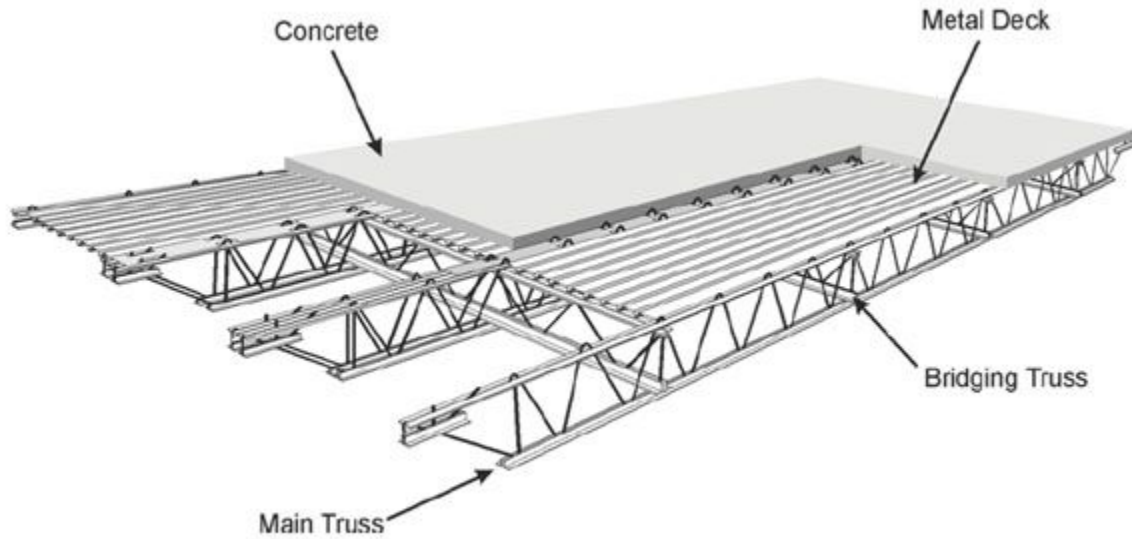
- A) Formwork
- B) creates more tensile stress
- C) Act like the reinforcements and a composite section
- D) All of them

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.2, page 51-(5-15)

All of them are definitions

- 157) In the composite metal deck supporting on the truss type joists, according to the following figure, the load of deck directly goes to:



- A) Bridging truss B) Main truss C) Concrete D) Columns

The Answer is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.2, page 51-(5-15)

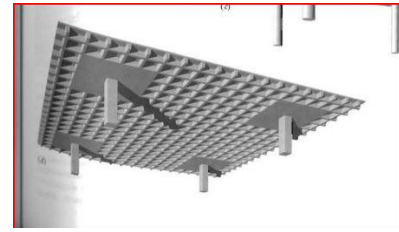
Metal decks carry the load parallel to the trapezoids and they are one way slab. So, in this figure, the direction of metal deck shows that the load transfers to the bridging truss and the bridging truss transfer it to the main truss.

So, “A” is correct.

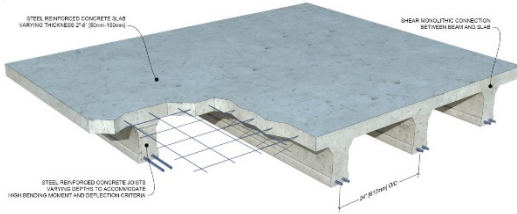
158) Which one can be classified with the two ways load distribution?



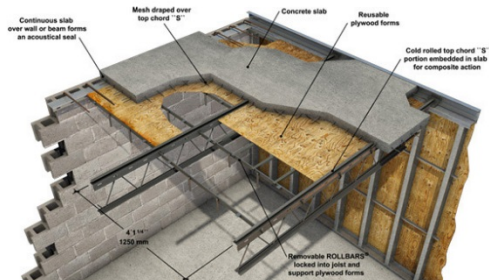
A



B



C



D

A) A

B) B

C) C

D) D

The Answers is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 51.2, page 51-(5-15),

ACI 318-08, 13.6.1.2.

“A” shows a joist floor with pre stressed or reinforced concrete joists. This is a clear example of the one way load distribution.

C and D are classified with the same concept as “A” but with reinforced concrete beam and steel truss joists respectively.

“B” shows waffle slab which can represent the two way load distribution system.

159) For the following waffle slab in the Louvre museum (Paris) which statement is correct?

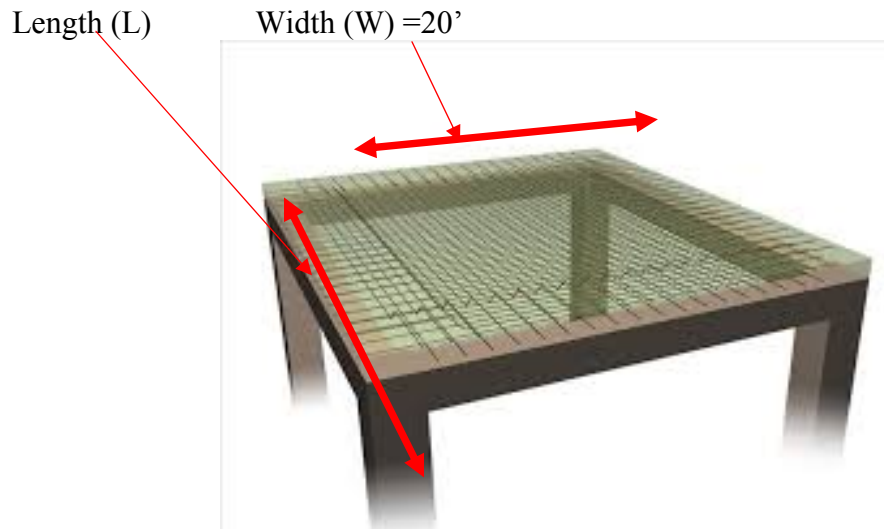


- A) Waffle slabs are made of the grid of concrete joists
- B) Depends on their size and length of span it can distribute load in two ways or in one way.
- C) Shear force can be transferred directly to the columns, or by drop panel or surrounding beam.
- D) All of them.

The Answer is D

This is the definition.

- 160) For the reinforced concrete slabs like the following picture, if the width is given equal to 20', then what would be the maximum length of the slab if the designer wants to have two way slab and have the following reinforcement arrangement?

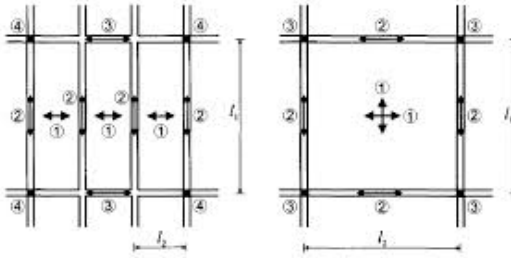


- A) 40' B) 10' C) 20' D) 50'

The Answer is A

ACI 318-08, 13.6.1.2., page 242 : Panels shall be rectangular with a ratio of longer to shorter span less than or equal to 2 for the 2 way slabs. (Also see below picture)

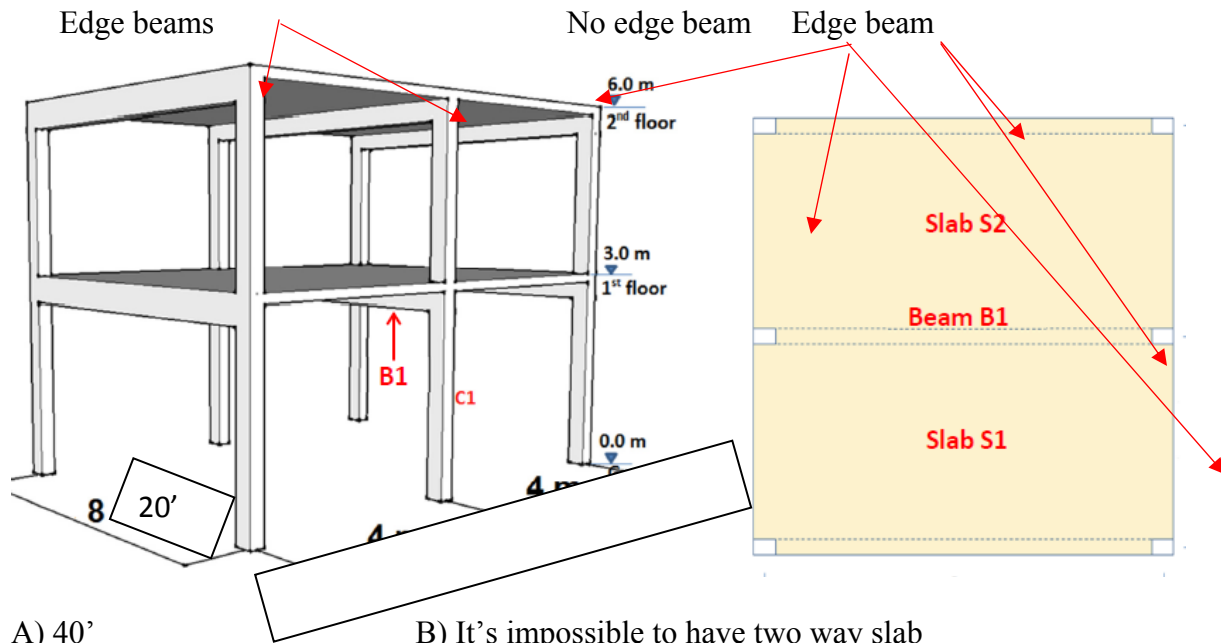
According to the code, $\frac{L}{W} \leq 2$



So, $L = 2 * 20' = 40'$

The slab can have 40' to be considered as a two way slab. Thus the distribution of bending moment in both direction needs reinforcement for negative and positive bending moment in both directions.

- 161) For the reinforced concrete slabs like the following picture, if the width is given equal to 20', then what would be the maximum length of the slab if the designer wants to have two way slab and have the following reinforcement arrangement? (The slab supported by edge beam in two sides and there is no edge beam at the other sides.)



- A) 40'
B) It's impossible to have two way slab
C) 20'
D) 55

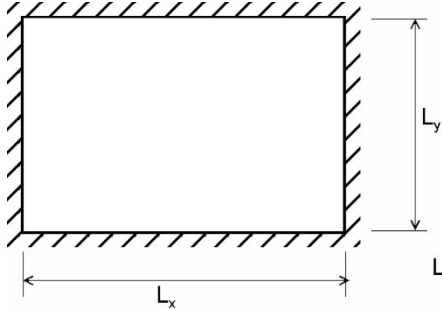
The Answers is B

ACI 318-08, 13.6.1.6. & 7, page 246: For panel with beams between supports on all sides moment distribution is permitted.

According to the code the stiffness of beams will create the support on all sides, therefore the two way slab can be created. Panels with beams on two parallel edges is the one way slab so at no circumstances the designer cannot create the two way slab unless the designer add two beams on other edges or design the slab without beams on all edges. (i.e., according to the code 13.2.4., beams should be monolithic (or fully composite with slab) to make the two way slab, or according to 13.2.1., the slabs without beams on all sides the slab can have the two way slab behavior.

So, in this question the slab is the one way slab at all conditions.

- 162) The Figure below shows a plan of a rectangular reinforced concrete slab simply supported on four edges. The slab carries uniform load of 100 lb/sf . Find the load distribution on the beam at L_x side if $L_y=6.0'$ and $L_x=8'$.

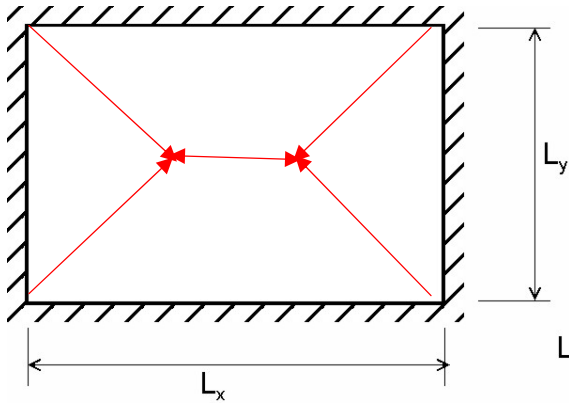


- A) Trapezoidal, $q_{\max}=300$ lb/ft
- B) Uniform, $q= 300$ lb/ft
- C) Triangular, $q_{\max}= 300$ lb/ft
- D) Uniform, $q= 600$ lb/ft

The Answer is A

ACI 318-08, Figure R13.6.8. page 247

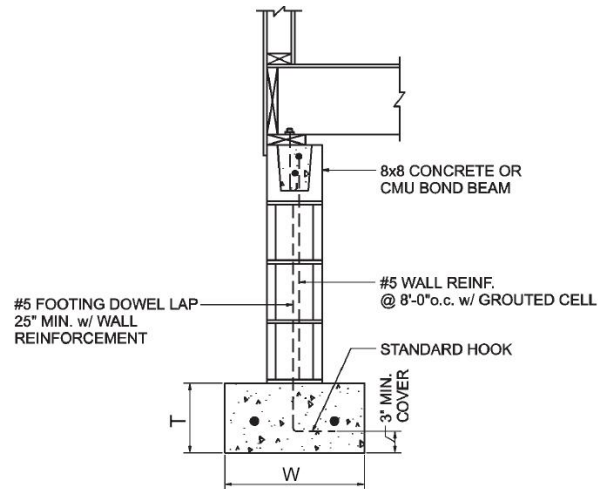
According to the code the tributary area shall be considered with 45° .



Therefore, the beam on L_x edge will carry the trapezoidal load and the beam on the L_y edge will carry the triangular load. The maximum load on triangular and trapezoidal load are equal and can be calculated as: $L_y/2 * \text{Load} = 6/2 * 100 = 300 \text{ lb/ft}$

K. Footings

- 163) Find the actual stress under the footing for the given wood structure if the transferred load is given equal to 50000 lbs. and $L=W=3$ ft.:



- A) 39 PSI B) 57 PSI
C) 14 PSI D) 28 PSI

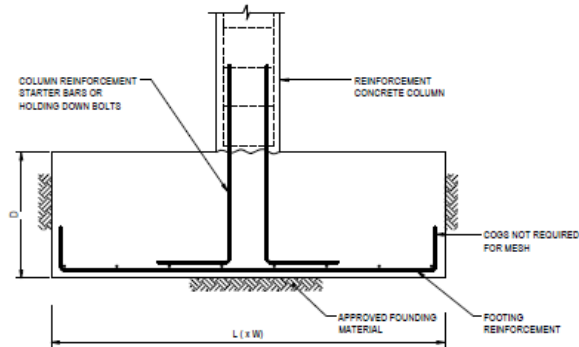
The Answer is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)

Since there is no bending moment: $\text{Stress} = P / A$

$\text{Stress} = 50000 \text{ lb} / 3 \times 3 = 5555.55 \text{ psf} = 5555.55 / 12 / 12 = 38.5 \text{ PSI}$

- 164) Find the actual stress under the footing if the actual bending moment is equal to 3000 lb-ft and the axial load is given equal to 5000 lbs. ($B=D=3$ ft.).



PAD FOOTING DETAIL
SCALE 1/10 REFER TO PLAN FOR SIZE

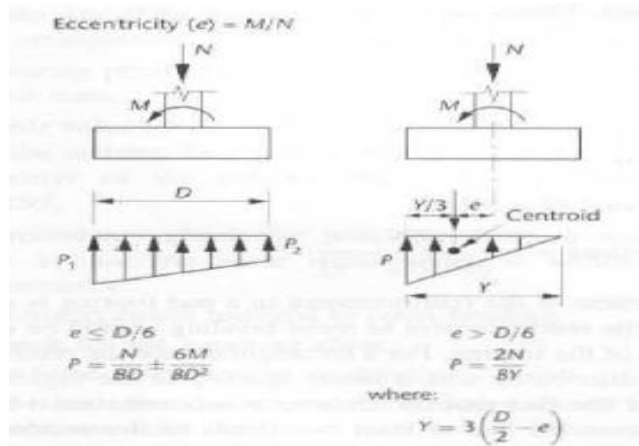
- A) 3.85 PSI
C) 8.6 PSI

- B) 8.4 PSI
D) 5.3 PSI

The Answers is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)

The first step is to estimate the eccentricity of the load, then one of the two conditions shall be applied:

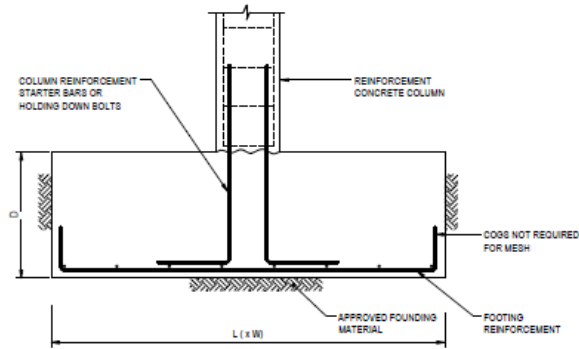


$$N=5000 \text{ lbs, } M=120 \text{ lb-ft, } e = M/N = 3000/5000 = 0.6 \text{ ft} > 1/6W = 3/6 = 0.5 \text{ ft}$$

$$Y = 3(D/2 - e) = 3(3/2 - 0.6) = 2.7$$

$$P = 2N/BY = 2*5000/(3*2.7) = 1234.5 \text{ psf} = 8.57 \text{ PSI}$$

- 165) Find the actual stress under the footing for if the actual bending moment is equal to 1500 lb-ft and the axial load is given equal to 5000 lbs. ($B=D=3$ ft.).



PAD FOOTING DETAIL
SCALE 1/10 REFER TO PLAN FOR SIZE

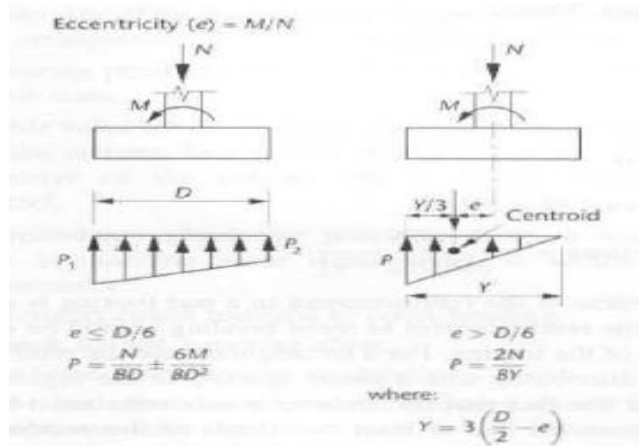
- A) 3.85 PSI
C) 6.2 PSI

- B) 8.4 PSI
D) 5.3 PSI

The Answers is C

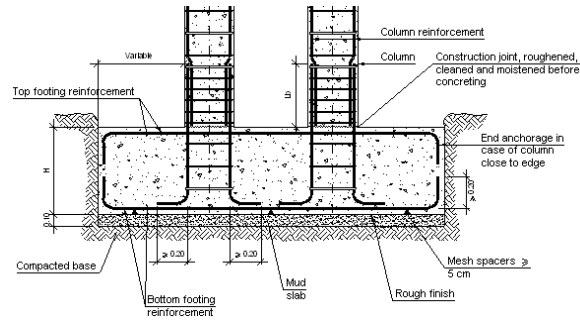
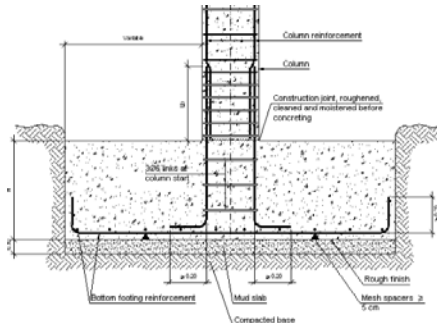
The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)

The first step is to estimate the eccentricity of the load, then one of the two conditions shall be applied:



$N=5000$ lbs, $M= 120$ lb-ft, $e = M/N = 1500/5000 = 0.3$ ft $< 1/6W = 3/6 = 0.5$ ft
 $P= N/(BD) \cdot (1+6e/D) = 888.88$ psf $= 6.17$ PSI

166) For the shown foundations, which statement is likely correct?



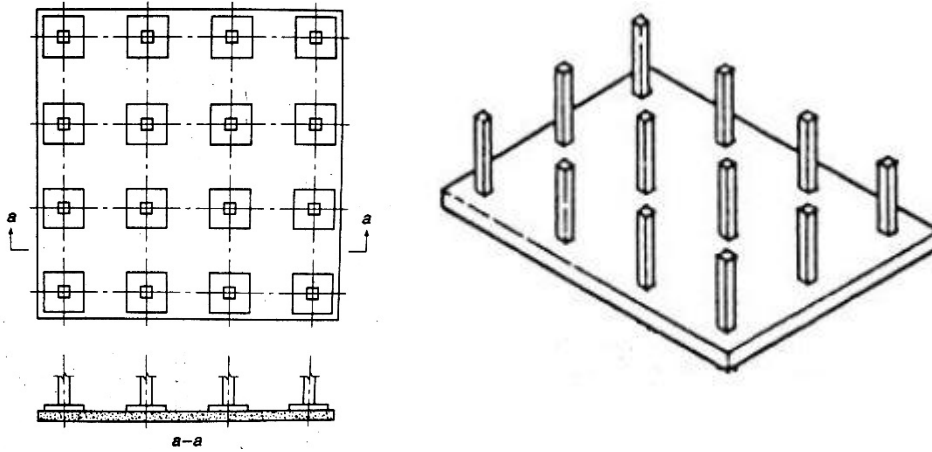
- A) The left figure shows a pad footing
- B) The right figure needs top reinforcements because this is a strip footing and there is a negative bending moment.
- C) Pad foundation does not require top reinforcements unless for the shrinkage and thermal effects in the huge size of pad footings.
- D) All choices

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)

This is the definition.

167) For the shown foundations, which statement is likely correct?



- A) Left figure shows a raft footing with the drop panel to increase the punching shear capacity
- B) The right figure shows a mat foundation
- C) Both of them may be classified as raft or mat foundations.
- D) All of the above

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)**

This is the definition. The raft and mat foundations are very similar to each other and it is hard to distinguish by observation. Raft foundations have more flexibility, so the thickness is usually less than the mat foundations and on the stiff bed they need less reinforcements. Mat foundations are designed as the semi-rigid foundation, so they have more thickness.

168) Which type of deep foundation transfers the load to the soil just through the surface of the pile?

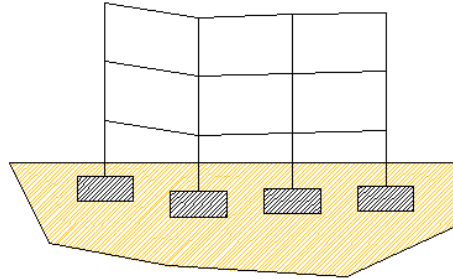
- A) Bearing Pile
- B) Friction cum End Bearing Pile
- C) Friction Pile
- D) None of the above

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)

The load-carrying capacity is dependent on both end-bearing and shaft friction. So, the pile resistance is the sum of both choice A and B.

169) What type of settlement has occurred in the building below?



A) Total Uniform Settlement
C) Tilting Settlement

B) Differential Settlement
D) None of the above

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(5-11)

Different portions of the building have gone through different settlement rates. Therefore, differential settlement has occurred.

170) You are asked to design a foundation system for a building. After testing the soil, you find out there is a shallow layer of very loose fill and there is a stiff layer of sand beneath that layer. What type of foundation would you suggest as the best option if the building does not have large loads and you do not have the option to remove the fill layer?

A) Friction Piles

B) Mat Footing

C) Friction Cum End Bearing Piles

D) Bearing Piles

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)**

Since there is a shallow layer of very loose fill, we can easily reach to the stiff layer, so we do not need to use friction piles.

If you had the option to remove the very loose layer, you could have removed this layer and used the mat footing on the stiff layer of sand. However, this is not an option as mentioned in the question.

Since the building loads are not very large, we do not need to use Friction Cum End Bearing piles to support the loads partly by friction and partly by bearing.

Therefore, Bearing piles are the best suggestion.

171) Assume you have to design a deep foundation for a building. This building is located on a layer of loose to medium sand and there is no stiff layer available at an accessible depth. What type of foundation would you use?

A) Friction Piles

B) Mat Footing

C) Friction Cum End Bearing Piles

D) Bearing Piles

The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)**

Since we cannot easily reach a stiff layer, we cannot use the bearing capacity and transfer the load through the end of the pile. So we need to use friction piles and transfer the load through the friction between the pile surface and the soil.

172) If all factors are the same except for the type of foundation and for the equal length of pile, which one of the following foundations can support larger loads compared to other ones?

A) Bearing Piles

B) Friction Cum End Bearing Piles

C) Friction Piles

D) All of the above can support equal loads

The Answer is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)**

Friction Cum End Bearing Piles use the friction capacity and the bearing capacity of the soil at the same time. Therefore, if all the factors are the same, this type of foundation is capable of supporting larger loads.

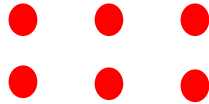
173) For design of a footing with the trapezoidal footing concept, which one is likely correct?

- A) A wall
- B) A combination of walls and columns
- C) Two columns which are close together with two different amount of loads
- D) Four columns in a row

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)**

Trapezoidal footing is generally a type of compound footing. Compound footings are designed for supporting two columns which are close together and have two different loads. The bigger side shall be under the bigger load.

174) Assume we have 6 columns with the following plan? What type of foundation system can be used to support them?



A) Three Compound Footings

B) One mat footing

C) Two Strip Footings

D) All of the above

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)

All of the above-mentioned options are possible, as shown below:

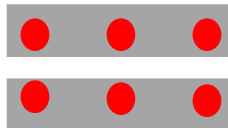
Three compound footings



One mat footing



Two strip footings



175) Which Type of settlement is more likely to cause structural damage in the building?

- A) Uniform Total Settlement
- B) Differential Settlement
- C) Tilting Settlement
- D) All of the above

The Answer is B

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(5-11)

Differential Settlement causes cracks and structural problems.

176) Which one causes vertical movement in the foundation?

- | | |
|-----------------------------|----------------------------|
| A) Uniform Total Settlement | B) Differential Settlement |
| C) Tilting Settlement | D) All of the above |

The Answer is D

In all types of settlement, a vertical movement occurs in the foundation.

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(5-11)

177) To support the 5 columns in a row, what type of foundation would you propose to support these columns?

A) Mat Footing

B) Batter Piles

C) Strip Footing

D) Compound Footing

The Answer is C

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)

Strip footing is usually used to support a row of columns or a wall.

- 178) Which of the following foundation types are deep foundation? I) Spread footings
II) Piles III) Wall footings IV) Mats V) Raft foundation

A) II B) II and IV C) III and V D) I, II, and V

The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)**

When the term deep foundations is used, it invariably means pile foundations. A pile is a long structural member installed in the ground to transfer loads to soils at some significant depths.

(A) is correct.

(B) is incorrect. Mats are shallow foundations.

(C) is incorrect. Both wall footings and mats are shallow foundations.

(D) is incorrect. Spread footings and raft foundations are both shallow

Raft foundations are the same as mat foundations however, rafts are designed with more flexibility than mat foundations.

179) Through which is almost all the structural load on a friction pile transferred to the soil?

- A) The bottom end of the pile
- B) Skin friction along the length of the pile
- C) Both A and B
- D) None of the above

The Answer is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)**

Friction pile is one that transfers almost all the structural load to the soil by skin friction along a substantial length of the pile.

(A) is incorrect. A pile that transfers almost all the structural load to the soil at the bottom end of the pile is named an end bearing or point bearing pile.

(B) is correct.

(C) is incorrect, since A is incorrect.

(D) is incorrect, since B is correct.

180) What type of shallow foundation should be used when the allowable soil pressure is low or where an array of columns and/or walls are so close that individual footings would overlap or nearly touch each other?

- | | |
|------------------|-------------------|
| A) Strip footing | B) Spread footing |
| C) Pile | D) Mat foundation |

The Answer is D

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 23, page 23-(1-33)

when the allowable soil pressure is low or where an array of columns and/or walls are so close that individual footings would overlap or nearly touch each other, a mat or raft foundation is required.

(A) is incorrect. A strip footing, also known as continuous footing, is used for a load-bearing wall, or for a row of columns which are closely spaced. It cannot be used for an array of columns.

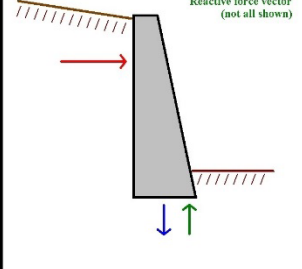
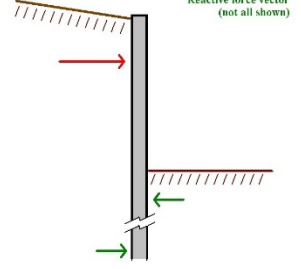
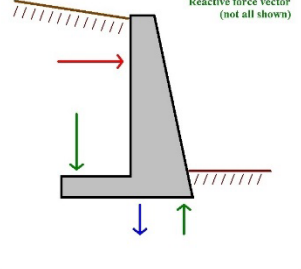
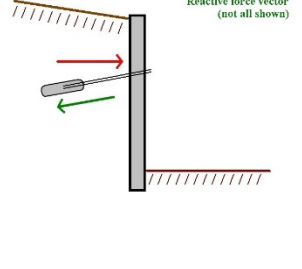
(B) is incorrect. A spread footing (or isolated or pad) footing is provided to support an individual column. A spread footing is typically circular, square or rectangular slab of uniform thickness.

(C) is incorrect. Piles are deep foundations, not shallow foundations.

(D) is correct. A mat or raft foundation is a large slab supporting a number of columns and walls under the entire structure or a large part of the structure.

L. Retaining walls

181) Which concept may be used when there is a limitation in the depth of excavation and the less pressure on the soil is expected?

Gravity wall  Earth pressure vector Gravity vector (of wall) Reactive force vector (not all shown)	Piling wall  Earth pressure vector Gravity vector (of wall) Reactive force vector (not all shown)	Cantilever wall  Earth pressure vector Gravity vector (of wall) Reactive force vector (not all shown)	Anchored wall  Earth pressure vector Gravity vector (of wall) Reactive force vector (not all shown)
Standard wall type that holds the earth mainly through its own weight. Can pivot and topple relatively easily, as the internal leverage of the earth pressure is very high.	Using long piles, this wall is fixed by soil on both sides of its lower length. If the piles themselves can resist the bending forces, this wall can take high loads.	The cantilever wall (which may also extend in the other direction) uses the same earth pressure trying to topple it to stabilize itself with a second lever arm.	This wall keeps itself from toppling by having cables driven into the soil or rock, fixed by expanding anchors (can be combined with other types of walls).

A) Gravity wall

B) Piling wall

C) Cantilever wall

D) Anchored wall

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Gravity and cantilever walls create pressure on soil under their foundations and need excavation for their foundation and piling wall needs a deep root for the wall.

Anchored wall is the only concept that does not have pressure on soil or needs more excavation.

182) Which concept gives a better performance to the retaining walls?

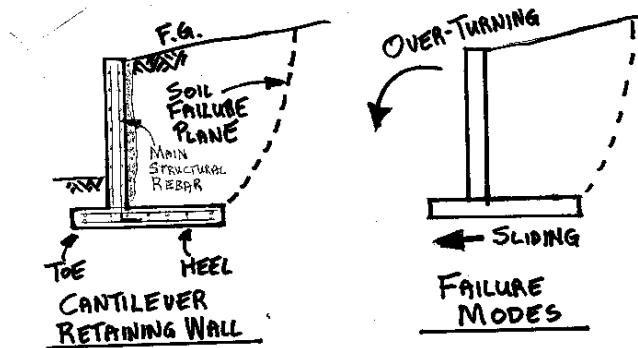
- A) Cantilever wall with foundation spread in the side of soil (heel).
- B) Cantilever wall with foundation in the toe side.
- C) There is no difference between "A" and "B".
- D) Cantilever walls are always classified as the best concept

The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

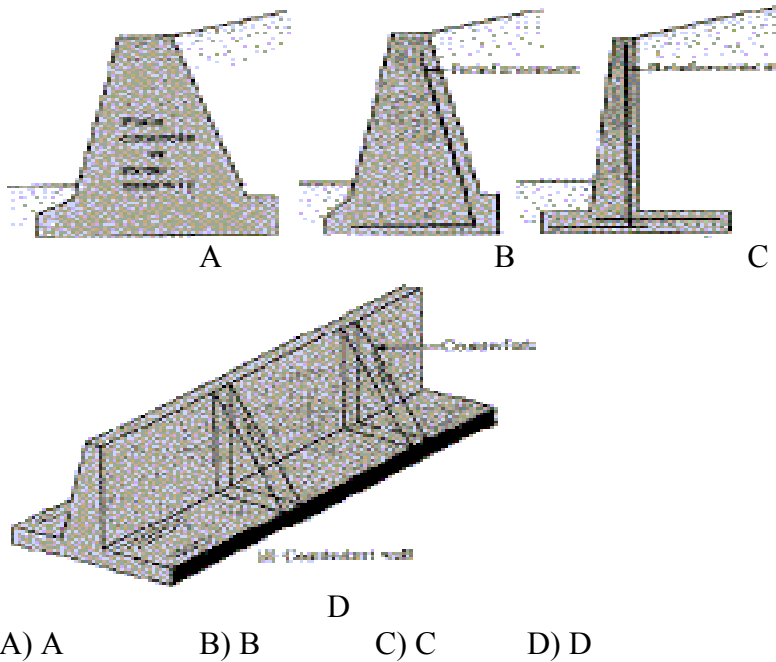
The most part of the foundation should be under the soil (i.e., in the heel side). This will help to increase the overturning moment and friction force for the effect of sliding.

So choice “A” is correct.

See the following figures.



183) For an extremely tall retaining wall which concept shall be used

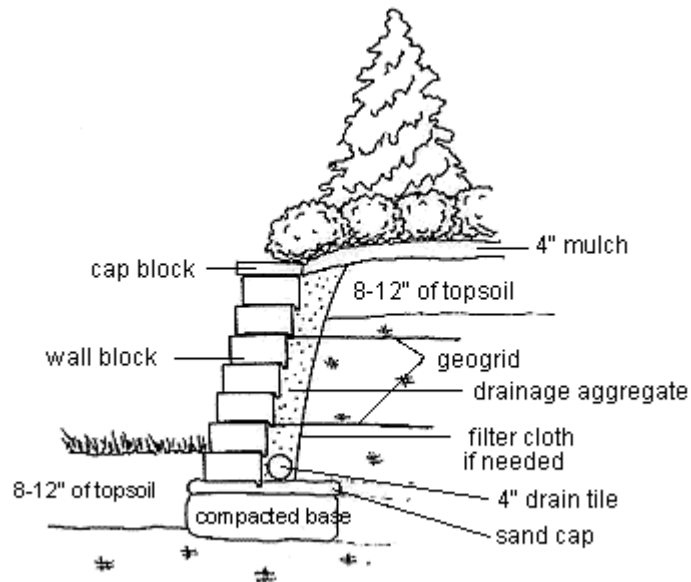


The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Both “A” and “B” are classified as gravity walls which are useful for the short walls. “C” is a cantilever wall which is useful for short to medium size walls.

For the tall retaining walls, the counterfort walls shall be used. So choice “D” is correct.

184) Which one is a better explanation for the following wall system?



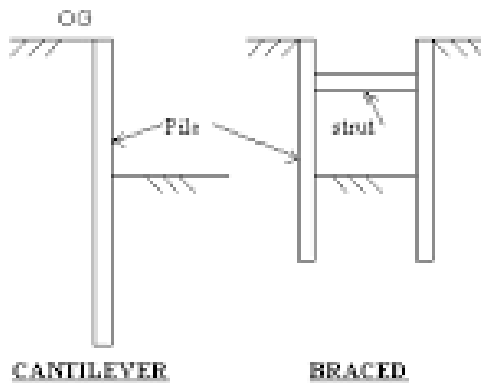
- A) This is a Cantilever wall
- B) This is an anchored wall with drilled nails
- C) This is the reinforced soil wall with geo-grid layers act like the anchors
- D) This is the reinforced soil wall with geo-grid layers but the anchor strips keep the wall

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

The picture does not show any anchorages either nail or strips, so choices C and B are not correct. Clearly this is not a cantilever wall.

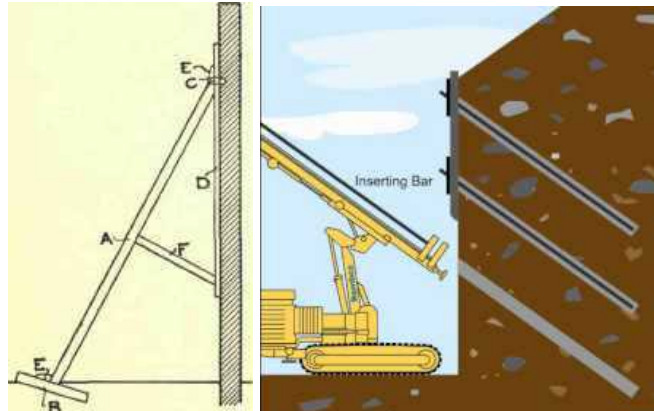
In this system the geo-grid layer keep the wall like the anchorages.

- 185) For the purpose of excavation a temporary retaining wall is required. Which concept may provide more safety? If the soil is cohesion-less and the risk of flood during the excavation is likely.



A

B



C

D

- A) A
- B) B
- C) C
- D) D

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

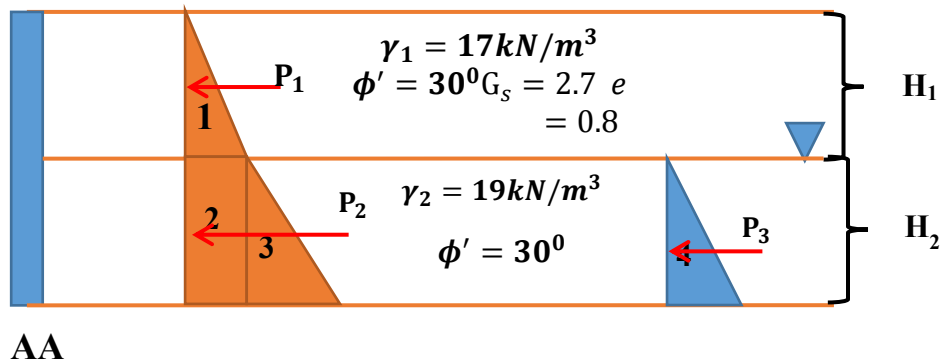
For the choices “B” and “C” the excavation should be finished and then the temporary structure shall be installed, which is not possible for the loose soil and the risk of flood during excavation.

Choice “A” may be the answer, but in the loose sand the use of piles is very expensive and scouring under the pile during the flood may occur.

The best answer is “D” because the required operations for the excavation and nailing the temporary structure are almost happen at the same time and there is not the risk of scouring, or soil failure during the excavation.

So “D” is the best answer.

- 186) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 3m and H_2 is 3m. Calculate the lateral force P_1 shown in the following diagram.



A) 35kN

B) 30kN

C) 25kN

D) 40kN

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

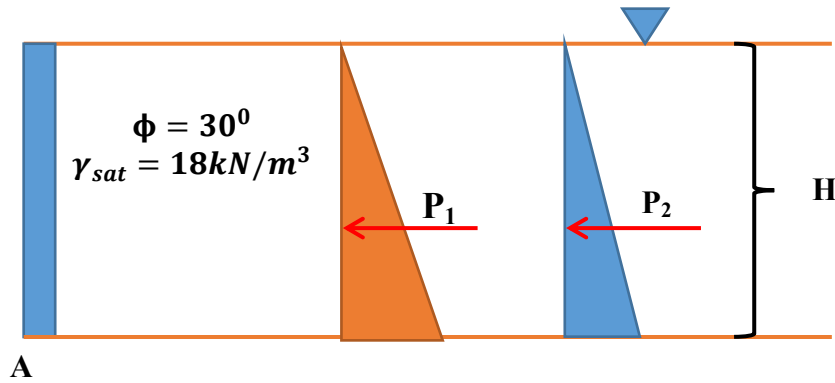
Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: The lateral force P_1 is equal to,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m})^2 = 25.5 \text{ kN}$$

- 187) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H is 5m. Calculate the value of horizontal force from soil only, P_1 .



- A) 30kN/m B) 35kN/m C) 40kN/m D) 45kN/m

The Answer is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Step 1: From the diagram, we can see that horizontal force P_1 acting towards the wall.

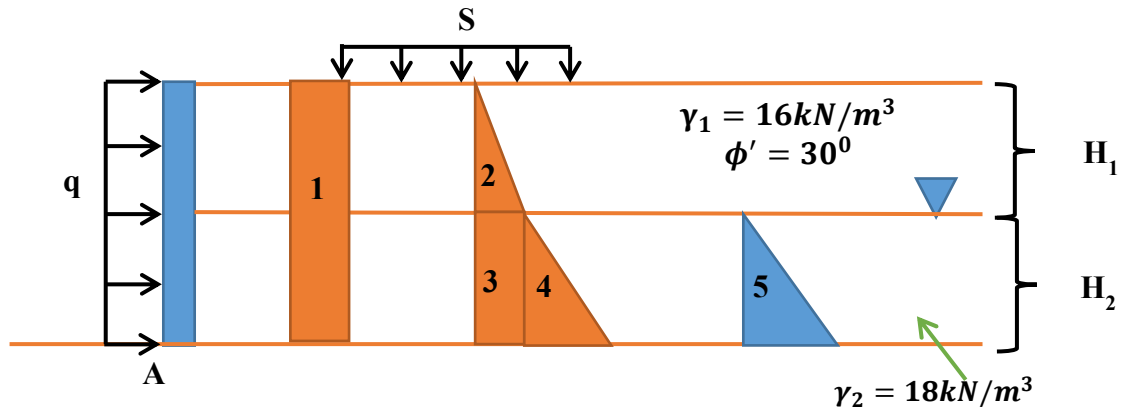
Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

According to the equation, we can get that the horizontal force P_1 is equal to,

$$P_1 = \frac{1}{2} K_a (\gamma_{sat} - \gamma_w) H^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) (5 \text{ m})^2 = 34 \text{ kN}$$

- 188) A retaining wall is shown in the following diagram. Assume that $S=10\text{kN/m}$. $H_1=2\text{m}$. $H_2=2\text{m}$. Determine the value of q if lateral resultant force is equal to zero.



- A) 16.3kN/m
 B) 17.6kN/m
 C) 18.2kN/m
 D) 15.4kN/m

The Answer is B**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = K_a S(H_1 + H_2) = \left(\frac{1}{3} \right) (10kN/m)(2m + 2m) = 13.3kN$$

Lateral force for Area 2,

$$P_2 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (16kN/m^3)(2m)^2 = 10.7kN$$

Lateral force for Area 3,

$$P_3 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) (16kN/m)(2m)(2m) = 21.3kN$$

Lateral force for Area 4,

$$P_4 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (18kN/m^3 - 9.8kN/m^3)(2m)^2 = 5.5kN$$

Lateral force for Area 5,

$$P_5 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8kN/m^3)(2m)^2 = 19.6kN$$

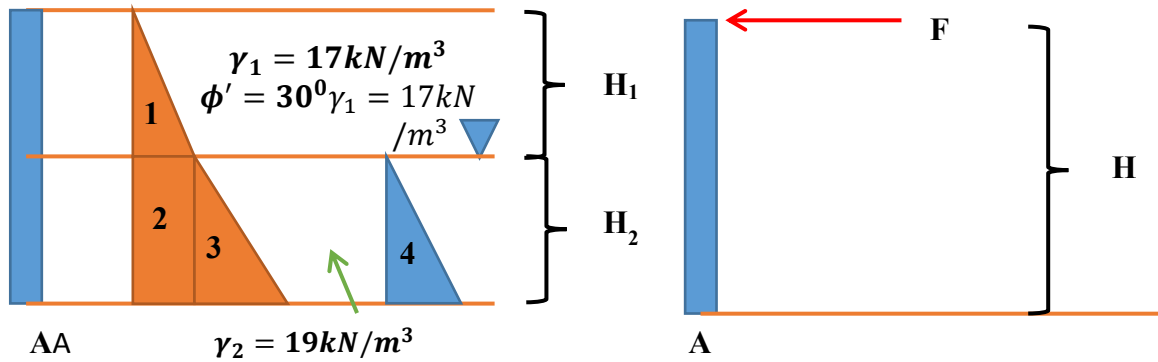
Step 3: The resultant force is equal to zero,

$$\begin{aligned} F &= P_1 + P_2 + P_3 + P_4 + P_5 - q(H_1 + H_2) \\ &= 13.3kN + 10.7kN + 21.3kN + 5.5kN + 19.6kN - q(2m + 2m) \\ &= 0 \end{aligned}$$

Solve this equation,

$$q = 17.6kN/m$$

- 189) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H_1 is 3m. H is 6m. $F = 40\text{kN}$. If moment at point A in the left figure is the same as that in the right figure, calculate the value of H_2 .



- A) 3m B) 4m C) 5m D) 6m

The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**Step 1: Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

Step 2: Lateral force for Area 1,

$$P_1 = \frac{1}{2} K_a \gamma_1 H_1^2 = \frac{1}{2} \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m})^2 = 25.5 \text{ kN}$$

$$z_1 = \frac{1}{3} H_1 + H_2 = \frac{1}{3} (3 \text{ m}) + H_2 = 1 + H_2$$

Lateral force for Area 2,

$$P_2 = K_a \gamma_1 H_1 H_2 = \left(\frac{1}{3} \right) (17 \text{ kN/m}^3) (3 \text{ m}) H_2 = 17 H_2$$

$$z_2 = \frac{1}{2} H_2 = 0.5 H_2$$

$$P_3 = \frac{1}{2} K_a (\gamma_2 - \gamma_w) H_2^2 = \frac{1}{2} \left(\frac{1}{3} \right) (19 \text{ kN/m}^3 - 9.8 \text{ kN/m}^3) H_2^2$$

$$z_3 = \frac{1}{3} H_2$$

$$P_4 = \frac{1}{2} \gamma_w H_2^2 = \frac{1}{2} (9.8 \text{ kN/m}^3) (H_2)^2$$

$$z_4 = \frac{1}{3} H_2$$

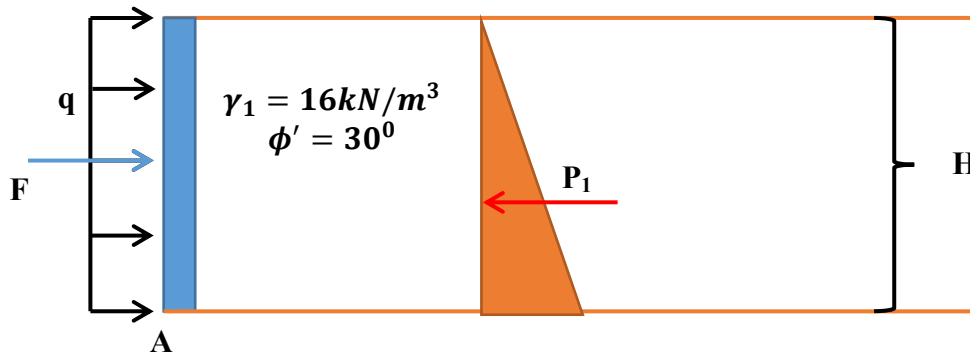
Step 2: The moment at point A can be calculated by,

$$\begin{aligned} M &= P_1 z_1 + P_2 z_2 + P_3 z_3 + P_4 z_4 \\ &= (25.5 \text{ kN})(1 + H_2) + (17 H_2)(0.5 H_2) + (1.53 H_2^2) \left(\frac{1}{3} H_2 \right) \\ &\quad + (4.9 H_2^2) \left(\frac{1}{3} H_2 \right) = (40 \text{ kN})(6 \text{ m}) \end{aligned}$$

Solve this equation,

$$H_2 = 3 \text{ m}$$

- 190) There is a frictionless wall shown in the following figure. The information is provided in this figure. Assume that H is 5m and $q = 5\text{kN/m}$. Also, there is a concentration force acting at the middle of the wall. Calculate the value of this force so that the horizontal resultant force of this wall is equal to zero.



- A) 36kN B) 48kN C) 42kN D) 54kN

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Step 1: From the diagram, we can see that there is only one horizontal force acting towards the wall. We need to calculate this force first. From the problem, Rankine active earth pressure coefficient can be calculated by,

$$K_a = \tan^2 \left(45^\circ - \frac{\phi'}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = \frac{1}{3}$$

The lateral force can be calculated by,

$$P_1 = \frac{1}{2} K_a \gamma_1 H^2 = \frac{1}{2} \left(\frac{1}{3} \right) (16 \text{ kN/m}^3) (5 \text{ m})^2 = 67 \text{ kN}$$

Step 2: In order to make the resultant force zero we got,

$$qH + F = P_1 = 67 \text{ kN} = (5 \text{ kN/m})(5 \text{ m}) + F$$

Solve this equation,

$$F = 42 \text{ kN}$$

191) Pore-water pressure can affect the value of effective horizontal stress. Which of the following parameter is necessary to calculate pore water pressure directly?

- A) Distribution of effective vertical stress
- B) Rankine active/passive earth pressure coefficient
- C) Location of depth of soil
- D) Location of water level

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Answer "A" is not correct: Distribution of effective vertical stress can be calculated from total vertical stress and pore water pressure. However, it cannot help to calculate pore water pressure directly.

Answer "B" is not correct: Rankine active or passive earth pressure coefficient is needed to calculate the value of horizontal stress from vertical stress. However, pore water pressure is the same for both horizontal and vertical direction. There is no need to consider it.

Answer "C" is not correct: Location of depth of soil is important to calculate soil stress. However, it is not necessary to calculate pore water pressure.

Answer "D" is correct: To calculate pore water pressure, the only thing we need to know is the water level.

192) Surcharge sometimes can be found on the top of soil with constant value. It will change the distribution of effective vertical and horizontal stress. Which of the following parts is the main difference comparing to the same case without surcharge?

- A) Value of vertical and horizontal stress from surcharge will decrease with the depth and distance.
- B) Value of effective vertical and horizontal stress will increase more under deeper location
- C) Value of effective vertical and horizontal stress will increase less under deeper location
- D) There is no difference between the one with constant surcharge value and the one without surcharge

The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Answer "A" is correct: From the Boussinesq diagram we can see that distribution of effective vertical and horizontal stress decrease with the depth and distance.

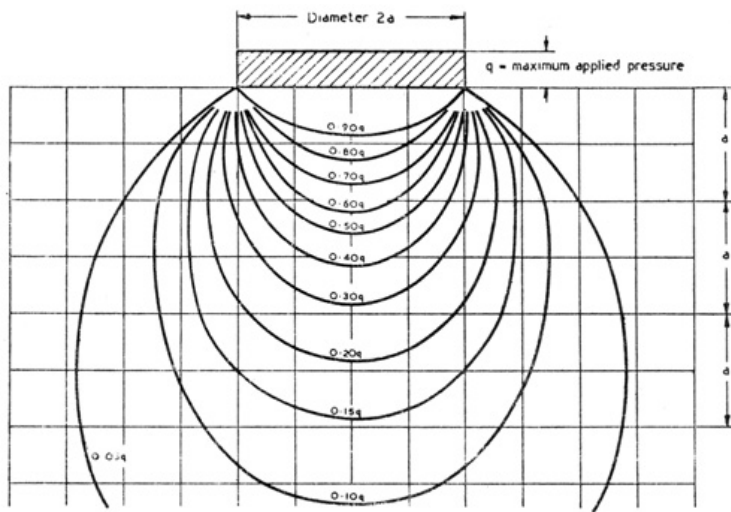


FIG. 22-10 BULB OF PRESSURE UNDER UNIFORM CIRCULAR LOAD

Answer "B" is not correct: The increasing value should be same at any location.

Answer "C" is not correct: The increasing value should be same at any location.

Answer "D" is not correct: They have difference for sure. The value of this difference should be equal to the value of surcharge.

193) Comparing the value of horizontal force caused by Rankine active state, F_1 , horizontal force caused by Rankine passive state, F_2 and vertical force, F_3 located at the same point, which of the following statement is correct?

- A) $F_1 > F_2 > F_3$
- B) $F_2 > F_1 > F_3$
- C) $F_2 > F_3 > F_1$
- D) $F_1 = F_2 > F_3$

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Answer "A" is not correct: Rankine active earth pressure coefficient should be smaller than Rankine passive earth pressure. Therefore, F_1 should be smaller than F_2

Answer "B" is not correct: Due to the fact that Rankine active earth pressure coefficient should be less than 1, it cannot be higher than vertical force, F_3 .

Answer "C" is correct: Horizontal force caused by Rankine passive state should be larger than the vertical force. The vertical force should be larger than horizontal force caused by Rankine active state.

Answer "D" is not correct: Rankine active earth pressure coefficient should be less than 1. Rankine passive earth pressure coefficient should be larger than 1. Therefore, F_1 cannot be equal to F_2 .

194) Stability of retaining walls is very important for retaining walls. The stability is determined by horizontal pressure. Which of the following statement is correct about horizontal pressure at a certain point?

- A) Soil with larger internal friction angle will have larger horizontal pressure
- B) Soil with smaller internal friction angle will have larger horizontal pressure
- C) Internal friction angle of soil will not determine the value of horizontal pressure
- D) Soil with larger internal friction angle will have larger horizontal pressure at active state and smaller horizontal pressure at passive state

The Answer is A

The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)

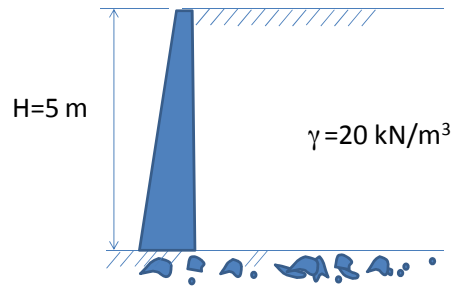
Answer "A" is correct: Horizontal pressure is determined by Rankine earth pressure coefficient. Soil with larger internal friction angle will have larger horizontal pressure for both active and passive state.

Answer "B" is not correct: Soil with larger internal friction angle should have larger horizontal pressure

Answer "C" is not correct: There are relations between horizontal pressure and internal friction angle. Internal friction angle will determine the value of Rankine earth pressure coefficient and thus determine the value of horizontal force.

Answer "D" is not correct: For both cases, Rankine earth pressure coefficient will increase with larger internal friction angle and thus increase the value of horizontal pressure.

- 195) For the frictionless retaining wall shown in the figure below, the active lateral earth pressure on the wall at the base of the wall is 32 kPa. The soil behind of the wall is clean sand with an in-situ total unit weight of 20 kN/m³. Determine the passive lateral earth pressure on the wall at the base of the wall.



- a) 32 kPa b) 21 kPa c) 26 kPa d) 312 kPa

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**Step 1:

Since there is no underground water in the soil behind the wall, the effective vertical stress at the base of the wall is $\sigma'_v = \gamma H = \left(20 \frac{\text{kN}}{\text{m}^3}\right) (5 \text{ m}) = 100 \text{ kPa}$.

Step 2:

The active lateral earth pressure at the base of the wall, σ_a , is 32 kPa. Since $\sigma_a = K_A \sigma'_v$,

$$K_A = \frac{\sigma_a}{\sigma'_v} = \frac{32 \text{ kPa}}{100 \text{ kPa}} = 0.32$$

Therefore, $K_A = \tan^2 \left(45^\circ - \frac{\phi}{2}\right) = 0.32$, from which it can be calculated that $\phi = 2 \times [45 - \arctan(\sqrt{0.32})] = 31^\circ$

Step 3:

The Rankine passive earth pressure coefficient, K_P , is

$$K_P = \tan^2 \left(45^\circ + \frac{\phi}{2}\right) = \tan^2 \left(45^\circ + \frac{31^\circ}{2}\right) = 3.12$$

The active lateral earth pressure at the base of the wall, σ_p , is

$$\sigma_p = K_P \sigma'_v = 3.12 \times (100 \text{ kPa}) = 312 \text{ kPa}$$

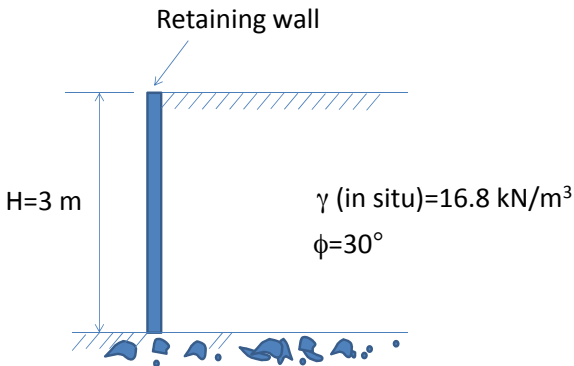
Short Cut

Note $K_A = \tan^2 \left(45^\circ - \frac{\phi}{2}\right) = \frac{1 - \sin(\phi)}{1 + \sin(\phi)}$ and $K_P = \tan^2 \left(45^\circ + \frac{\phi}{2}\right) = \frac{1 + \sin(\phi)}{1 - \sin(\phi)}$, $K_P = 1/K_A$. From

Step 2, it is easy to get $K_P = \frac{1}{K_A} = \frac{1}{0.32} = 3.12$. Therefore, $\sigma_p = K_P \sigma'_v = 3.12 \times (100 \text{ kPa}) = 312 \text{ kPa}$.

Or since generally the passive earth pressure is significantly larger than the active earth pressure, only option (D) satisfies this condition.

- 196) A 3 m high smooth retaining wall extends from the top of bedrock to the ground surface. The soil behind of the retaining wall is homogeneous and cohesionless, has an in-situ total unit weight of 16.8 kN/m^3 , and angle of internal friction of 30° . Based on the Rankine theory, what is the total active resultant lateral earth force per unit length of retaining wall?



- A) 25.2 kN/m B) 35.2 kN/m C) 50.4 kN/m D) 75.6 kN/m

The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Step 1: Based on the Rankine theory, the Rankine active earth pressure coefficient $K_A =$

$$\tan^2 \left(45^\circ - \frac{\phi}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = 0.333$$

Step 2:

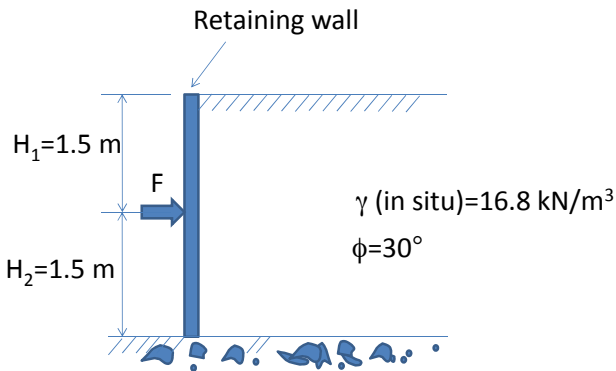
The active lateral earth pressure distribution is linear. The active lateral earth pressure at any depth, h , below the ground surface is calculated by $\sigma_a = K_A \sigma_v = K_A \gamma h$.

Step 3:

The total active lateral earth force per unit length of retaining wall is

$$P_a = \frac{1}{2} K_A \gamma H^2 = \frac{1}{2} (0.333) (16.8 \text{ kN/m}^3) (3 \text{ m})^2 = 25.2 \text{ kN/m}$$

- 197) A 3 m high smooth retaining wall extends from the top of bedrock to the ground surface. The soil behind the retaining wall is homogeneous and cohesionless, with an in-situ total unit weight of 16.8 kN/m^3 , and angle of internal friction of 30° . A lateral force F is applied on the opposite side of the wall. Based on the Rankine theory, what is the total passive resultant lateral earth force per unit length of retaining wall?



- A) 25.2 kN/m
- B) 35.2 kN/m
- C) 50.4 kN/m
- D) 226.8 kN/m

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**

Step 1: Based on the Rankine theory, the Rankine passive earth pressure coefficient $K_p =$

$$\tan^2 \left(45^\circ + \frac{\phi}{2} \right) = \tan^2 \left(45^\circ + \frac{30^\circ}{2} \right) = 3$$

Step 2:

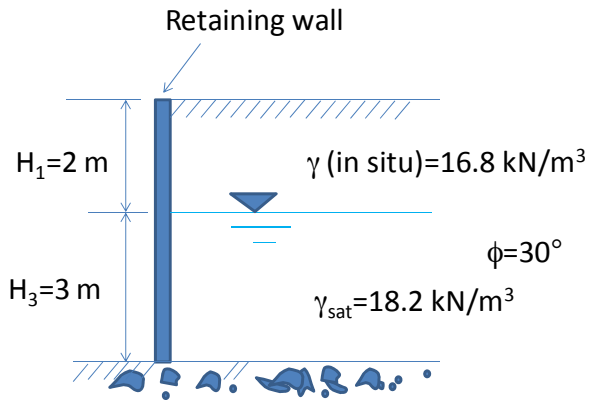
The passive lateral earth pressure distribution is linear. The passive lateral earth pressure at any depth, h , below the ground surface is calculated by $\sigma_p = K_p \sigma_v = K_p \gamma h$.

Step 3:

The total active lateral earth force per unit length of retaining wall is

$$P_p = \frac{1}{2} K_p \gamma H^2 = \frac{1}{2} (3) (16.8 \text{ kN/m}^3) (3 \text{ m})^2 = 226.8 \text{ kN/m}$$

- 198) A 5 m high smooth retaining wall extends from the top of bedrock to the ground surface. The soil behind of the retaining wall is homogeneous and cohesionless, has an in-situ total unit weight of 16.8 kN/m^3 , and angle of internal friction of 30° . Groundwater level is 2 m below the surface of the soil, as shown in the figure below. The saturated unit weight of the soil is 18.2 kN/m^3 . Based on the Rankine theory, what is the total active resultant lateral earth force per unit length of retaining wall?



- A) 25.2 kN/m
- B) 35.2 kN/m
- C) 57.4 kN/m
- D) 75.6 kN/m

The Answer is C**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**Step 1: Based on the Rankine theory, the Rankine active earth pressure coefficient $K_A =$

$$\tan^2 \left(45^\circ - \frac{\phi}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = 0.333$$

Step 2:

The total vertical stress σ_{vH1} at the groundwater level is calculated by $\sigma_{vH1} = \gamma H_1 = (16.8 \text{ kN/m}^3)(2 \text{ m}) = 33.6 \text{ kPa}$. Since the pore water pressure at the groundwater level is 0, the effective vertical stress $\sigma'_{vH1} = \sigma_{vH1} - u_{H1} = 33.6 \text{ kPa} - 0 = 33.6 \text{ kPa}$.

Step 3:

At the bottom level of the retaining wall, the total vertical stress σ_{vH2} is calculated by

Continuing:

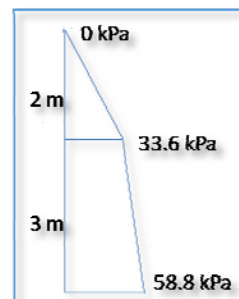
$$\sigma_{vH2} = \gamma H_1 + \gamma_{\text{sat}} H_2 = (16.8 \text{ kN/m}^3)(2 \text{ m}) + (18.2 \text{ kN/m}^3)(3 \text{ m}) = 88.2 \text{ kPa}.$$

The pore water pressure u is calculated by $u_{H2} = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(3 \text{ m}) = 29.4 \text{ kPa}$.

The effective vertical stress $\sigma'_{vH2} = \sigma_{vH2} - u_{H2} =$

$$88.2 \text{ kPa} - 29.4 \text{ kPa} = 58.8 \text{ kPa}.$$

The distribution of effective vertical stress along shown in the plot below:



depth, σ'_v , is then

Step 4:

The active lateral earth pressure at any depth, h , below the ground surface is calculated by

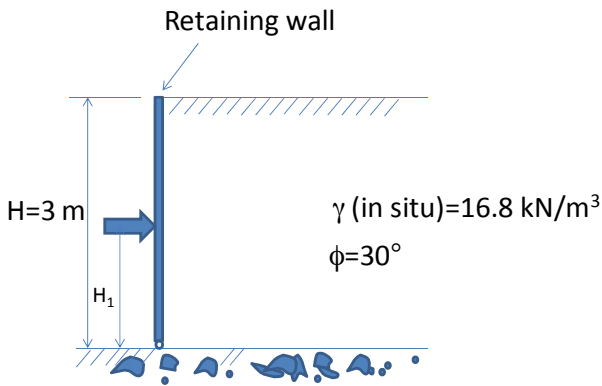
$$\sigma_a = K_A \sigma'_v.$$

Step 5:

The total active lateral earth force per unit length of retaining wall is

$$\begin{aligned} P_a &= K_A \left[\frac{1}{2} (0 + \sigma'_{vH1}) H_1 + \frac{1}{2} (\sigma'_{vH1} + \sigma'_{vH2}) H_2 \right] \\ &= (0.333) \left[\frac{1}{2} (0 + 33.6 \text{ kPa})(2 \text{ m}) + \frac{1}{2} (33.6 \text{ kPa} + 58.8 \text{ kPa})(3 \text{ m}) \right] \\ &= 57.4 \text{ kN/m} \end{aligned}$$

- 199) A 3 m high smooth retaining wall extends from the top of bedrock to the ground surface. The wall is connected to the bedrock by a frictionless hinge. The soil behind of the retaining wall is homogeneous and cohesion less, has an in-situ total unit weight of 16.8 kN/m^3 , and angle of internal friction of 30° . A lateral force is applied at the opposite side of the wall, at a height of $H_1=1.5 \text{ m}$ from the top of bedrock. Based on Rankine theory, what is the minimum force per unit length of retaining wall required to resist the overturning moment?



- A) 16.8 kN/m
- B) 35.2 kN/m
- C) 50.4 kN/m
- D) 75.6 kN/m

The Answer is A**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**Step 1: Based on the Rankine theory, the Rankine active earth pressure coefficient $K_A =$

$$\tan^2 \left(45^\circ - \frac{\phi}{2} \right) = \tan^2 \left(45^\circ - \frac{30^\circ}{2} \right) = 0.333$$

Step 2:

The active lateral earth pressure distribution is linear. The active lateral earth pressure at any depth, h , below the ground surface is calculated by $\sigma_a = K_A \sigma_v = K_A \gamma h$.

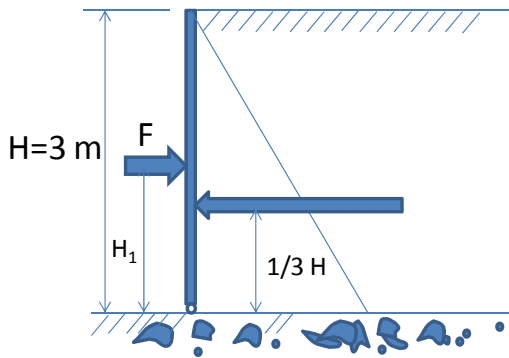
Step 3:

The total active lateral earth force per unit length of retaining wall is

$$P_a = \frac{1}{2} K_A \gamma H^2 = \frac{1}{2} (0.333) (16.8 \text{ kN/m}^3) (3 \text{ m})^2 = 25.2 \text{ kN/m}$$

P_a , acts at a height above the bedrock of

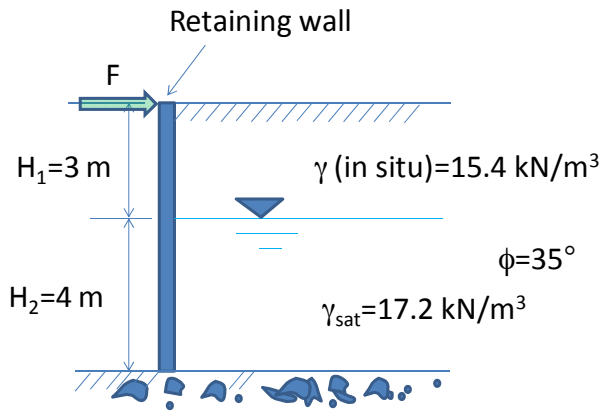
$$\frac{H}{3} = \frac{3 \text{ m}}{3} = 1 \text{ m}.$$

Step 4:

Summing moments on the retaining wall about the hinge gives

$$F = \frac{P_a \frac{H}{3}}{H_1} = \frac{\left(25.2 \frac{\text{kN}}{\text{m}} \right) (1 \text{ m})}{1.5 \text{ m}} = 16.8 \text{ kN/m}$$

- 200) A 7 m high smooth retaining wall extends from the top of bedrock to the ground surface. The soil behind of the retaining wall is homogeneous and cohesion less, has an in-situ total unit weight of 15.4 kN/m^3 and angle of internal friction of 35° . Groundwater level is 3 m below the surface of the soil, as shown in the figure below. The saturated unit weight of the soil is 17.2 kN/m^3 . A resisting force F is applied on the opposite side of the wall at the top of the wall so that the soil behind the wall reaches the passive earth pressure condition. Based on the Rankine theory, what is the total passive resultant lateral earth force per unit length of retaining wall?



- A) 252 kN/m
- B) 352 kN/m
- C) 574 kN/m
- D) 1156 kN/m

The Answer is D**The Civil Engineering Handbook, 2nd ed. Chen and Liew, 22, page 22-(1-12)**Step 1: Based on the Rankine theory, the Rankine passive earth pressure coefficient $K_p =$

$$\tan^2 \left(45^\circ + \frac{\phi}{2} \right) = \tan^2 \left(45^\circ + \frac{35^\circ}{2} \right) = 3.69.$$

Step 2:

The total vertical stress σ_{vH1} at the groundwater level is calculated by $\sigma_{vH1} = \gamma H_1 = (15.4 \text{ kN/m}^3)(3 \text{ m}) = 46.2 \text{ kPa}$. Since the pore water pressure at the groundwater level is 0, the effective vertical stress $\sigma'_{vH1} = \sigma_{vH1} - u_{H1} = 46.2 \text{ kPa} - 0 = 46.2 \text{ kPa}$.

Step 3:

At the bottom level of the retaining wall, the total vertical stress σ_{vH2} is

$$\sigma_{vH2} = \gamma H_1 + \gamma_{\text{sat}} H_2 = (15.4 \text{ kN/m}^3)(3 \text{ m}) + (17.2 \text{ kN/m}^3)(4 \text{ m}) = 115 \text{ kPa}.$$

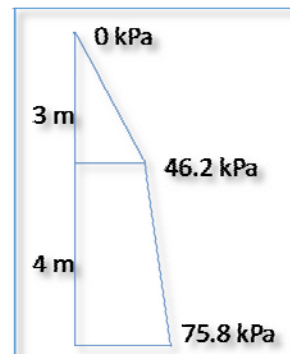
The pore water pressure u is calculated by

$$u_{H2} = \gamma_w H_2 = (9.8 \text{ kN/m}^3)(4 \text{ m}) = 39.2 \text{ kPa}.$$

The effective vertical stress

$$\sigma'_{vH2} = \sigma_{vH2} - u_{H2} = 115 \text{ kPa} - 39.2 \text{ kPa} = 75.8 \text{ kPa}.$$

The distribution of effective vertical stress along depth, σ'_v , shown in the plot:



is then

Step 4:

The passive lateral earth pressure at any depth, h , below the ground surface is calculated by

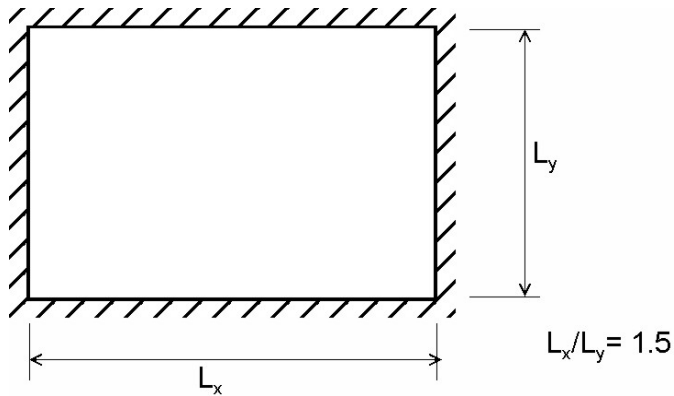
$$\sigma_p = K_p \sigma'_v.$$

Step 5:

The total passive lateral earth force per unit length of retaining wall is

$$\begin{aligned} P_p &= K_p \left[\frac{1}{2} (0 + \sigma'_{vH1}) H_1 + \frac{1}{2} (\sigma'_{vH1} + \sigma'_{vH2}) H_2 \right] \\ &= (3.69) \left[\frac{1}{2} (0 + 46.2 \text{ kPa})(3 \text{ m}) + \frac{1}{2} (46.2 \text{ kPa} + 75.8 \text{ kPa})(4 \text{ m}) \right] \\ &= 1156 \text{ kN/m} \end{aligned}$$

- 201) The Figure below shows a plan of a rectangular reinforced concrete slab simply supported on four edges. The slab carries uniform dead load of 100 lb/sf and uniform live load of 30 lb/sf. If the simple analysis method gives 0.2 as the bending moment coefficient along longitudinal direction (L_x) and 0.08 along the L_y which is equal to 10', then find the required reinforcements for this slab. (Thickness of slab is 8" and Assume cover for reinforcements for 1", $F_y=60$ KSI, and $F'_c= 3$ KSI.)



- A) $A_{sy}=0.042$, $A_{sx}=0.24$ Reinforcements are on top layer
- B) $A_{sy}=0.075$, $A_{sx}=0.24$ Reinforcements are on Bottom layer
- C) $A_{sy}=0.042$, $A_{sx}=0.24$ Reinforcements are on Bottom layer
- D) $A_{sy}=0.075$, $A_{sx}=0.24$ Reinforcements are on top layer

This slab is simply supported slab which means the negative bending moment is zero. For the positive bending moment the reinforcement should be installed at the bottom layer of the slab.

The Answers is B

$L_y = 10'$, then $L_x/L_y = 1.5$ $L_x = 1.5 * 10 = 15'$

Also, $L_x/L_y = 1.5 < 2$, then It is 2 way slab.

$M_{uy} = 0.2qlx^2 = 0.2 * 168 * 15^2 = 7560 \text{ lb-ft}$

$M_{ux} = 0.08qly^2 = 0.08 * 168 * 10^2 = 1344 \text{ lb-ft}$

$M_n = M_u/\Phi$; $M_{nx} = 1344/0.9 = 1493.3 \text{ lb-ft}$, $b = 1' = 12''$ $d = 8'' - 1''(\text{cover}) = 7''$

$$A_{sy} = \frac{M_n}{f_y (d - \lambda)} = \frac{1493.3 * 12(\text{inch})}{60000 * 7} = 0.042 \text{ in}^2$$

$M_n = M_u/\Phi$; $M_{ny} = 7560/0.9 = 8400 \text{ lb-ft}$, $b = 1' = 12''$ $d = 8'' - 1''(\text{cover}) = 7''$

$$A_{sx} = \frac{M_n}{f_y (d - \lambda)} = \frac{8400 * 12(\text{inch})}{60000 * 7} = 0.24 \text{ in}^2$$

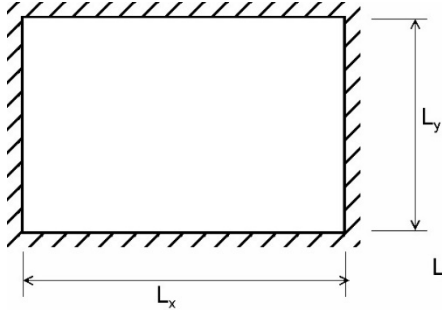
ACI 318-08, 10.5.4., required for the minimum reinforcement in the slabs:

$A_{smin} \text{ for slabs} = 50\% * 0.0018 bd = .5 * 0.0018 * 7 * 12 = 0.0756$

So, for A_{sy} we have to use the minimum reinforcement given equal to $0.0756 > 0.042$ and $A_{sx} = 0.24 \text{ in}^2$

A & D are not correct because for the simply supported slab, the reinforcement should be in bottom and C is not correct because the minimum reinforcement should not be less than the minimum allowed bars according to the code.

202) The Figure below shows a plan of a rectangular reinforced concrete slab simply supported on four edges. The slab carries uniform load of 100 lb/sf . Find the load distribution on the beam at L_x side if $L_y=6.0'$ and $L_x=8'$.

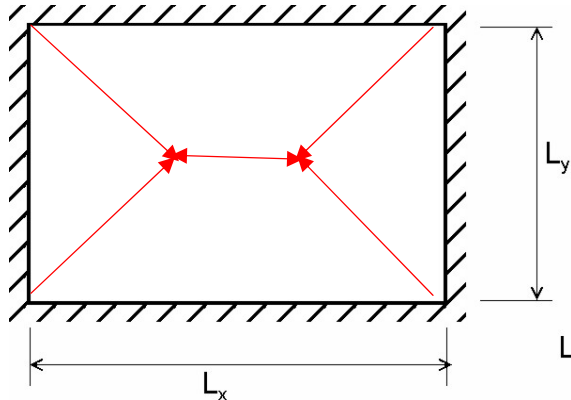


- A) Trapezoidal, $q_{\max}=300$ lb/ft
- B) Uniform, $q=300$ lb/ft
- C) Triangular, $q_{\max}=300$ lb/ft
- D) Uniform, $q=600$ lb/ft

The Answer is A

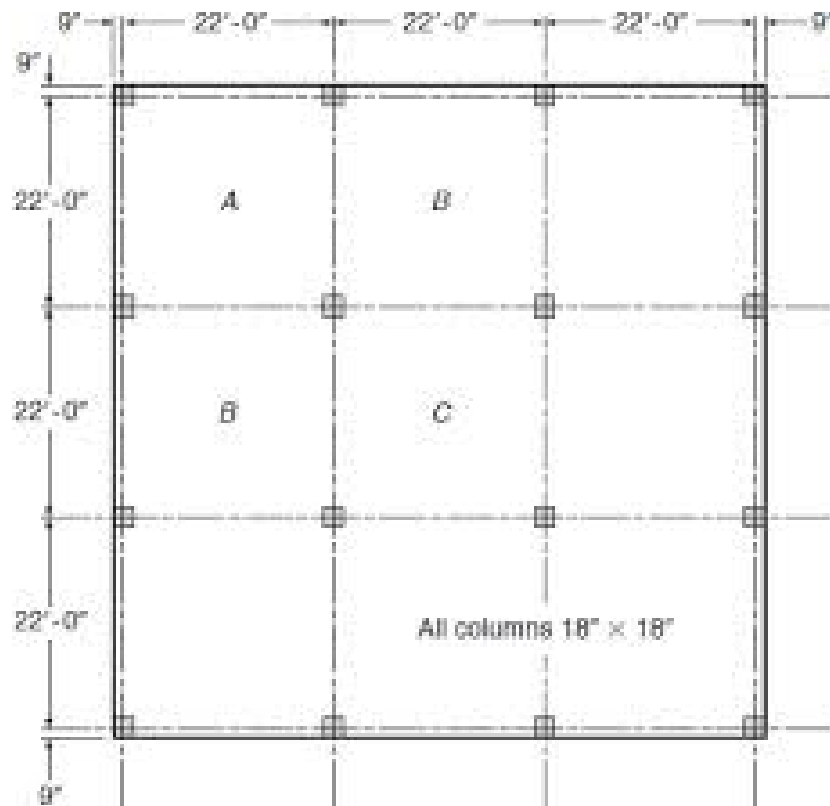
ACI 318-08, Figure R13.6.8.

According to the code the tributary area shall be considered with 45°.



Therefore, the beam on L_x edge will carry the trapezoidal load and the beam on the L_y edge will carry the triangular load. The maximum load on triangular and trapezoidal load are equal and can be calculated as: $L_y/2 * \text{Load} = 6/2 * 100 = 300$ lb/ft

203) A flat plate is supported on 18 in x 18in columns that are spaced 22' in both sides. Assume cover for reinforcements for 1", $F_y=60$ KSI, and $F'_c= 3$ KSI. The minimum thickness if the edge beam is eliminated is mostly nearly:



- A) $d_{min} = 9''$
C) $d_{min} = 7''$

- B) $d_{min} = 12''$
D) $d_{min} = 8.2''$

The Answers is D

ACI 318-08, 9.5.3.3 & table 9.5.(C)

According to the table for the flat slab without interior beams, edge beam, and drop panel the minimum thickness is given equal to:

$$F_y = 60 \text{ KSI}, d_{\min} = \frac{L_n}{30}$$

Where L_n is the clear span length, so:

$$L_n = \text{Column face to the column face} = 22' - (18''/12) = 20.5'$$

$$\text{So, } d_{\min} = \frac{20.5}{30} = 0.68' = 8.2 \text{ in}$$

204) Assume the question number 7 with edge beams with $b_w = 30''$ and total height including slab thickness is given equal to $36''$. Thickness of slab is given equal to 8 inches. Find the relative torsional stiffness β_t for the edge beams if :



- A) $\beta_t = 5$ B) $\beta_t = 0.7$ C) $\beta_t = 0.5$ D) $\beta_t = 7$

The Answers is D**ACI 318-08, 13.2.6.**

This clause gives the required dimensions for the beam. So the actual size of the section shall be considered as:

**ACI 318-08, 13.2.6. – Equation 13-6**

The torsional constant “C” is given equal to:

$$C = \sum \left(1 - 0.63 \frac{x}{y} \right) * \left(\frac{x^3 y}{3} \right)$$

Where, x and y are the dimension of each rectangular part of the section and the $x \leq y$.

$$C = \sum \left(1 - 0.63 * \frac{8}{28} \right) * \left(\frac{8^3 * 28}{3} \right) + \left(1 - 0.63 * \frac{30}{36} \right) * \left(\frac{30^3 * 36}{3} \right)$$

$$= 157818.507 \text{ in}^4$$

Equation 13-5 will give the relative torsional constant

$$\beta_t = \frac{E_{cb} C}{2 E_{cs} I_s}$$

. b= width of slab and h is the thickness of slab for the edge beam which is equal to 22’ for each span.

$$I_s = \text{moment of inertia} = \frac{bh^3}{12} = 22.12 * \frac{8^3}{12} = 11264 \text{ in}^4$$

ACI 318-08, 8.5.1.

$$E_{cb} = 57000 \sqrt{f'_c} = 57000 * \sqrt{3} = 98726 \text{ psi} = E_{cs}$$

$$\beta_t = \frac{157818.507}{2 * 11264} = 7.005$$

205) For the question number 7, considering an edge beam; the minimum thickness for deflection control is most nearly:

- A) $d_{min} = 9''$
C) $d_{min} = 8''$

- B) $d_{min} = 12''$
D) $d_{min} = 7.5''$

The Answer is D

ACI 318-08, 9.5.3.3 & table 9.5.(C)

According to the table for the flat slab without interior beams, and drop panel but with the edge beam, the minimum thickness is given equal to:

$$F_y = 60 \text{ KSI}, d_{\min} = \frac{L_n}{33}$$

Where L_n is the clear span length, so:

$$L_n = \text{Column face to the column face} = 22' - (18''/12) = 20.5'$$

$$\text{So, } d_{\min} = \frac{20.5}{33} = 0.62' = 7.5 \text{ in}$$

206) For the question number 7, if drop panel that meet ACI 318 requirements are provided, considering an edge beam; the minimum slab thickness for deflection control outside of the drop panel is most nearly:

- A) $d_{min} = 9''$
- C) $d_{min} = 8''$

- B) $d_{min} = 12''$
- D) $d_{min} = 7.0''$

The Answer is D

ACI 318-08, 9.5.3.3 & table 9.5.(C)

According to the table for the flat slab without interior beams, but with the edge beam, and drop panel the minimum thickness is given equal to:

$$F_y = 60 \text{ KSI}, d_{\min} = \frac{L_n}{36}$$

Where L_n is the clear span length, so:

$$L_n = \text{Column face to the column face} = 22' - (18''/12) = 20.5'$$

$$\text{So, } d_{\min} = \frac{20.5}{36} = 0.57' = 6.84 \text{ in}$$

Part 5 : Hydraulics and Hydrology

106 Problems

1) An open channel has a rectangular cross section that is 10 ft wide, and depth is 2.5 ft. The channel slope is 1.5% and Manning $n=0.019$. What is the average velocity in this channel?

- (A) 4.47 ft/sec
- (B) 6.43 ft/sec
- (C) 9.81 ft/sec
- (D) 13.5 ft/sec

The Answer is D

Step 1: The Manning equation is $V = \frac{K}{n} R_h^{2/3} S^{1/2}$, where V= average velocity, K=1 for metric units, K=1.49 for USCS units, n = Manning roughness coefficient, $R_h = A/P$ = hydraulic radius, A = cross sectional area, P = wetted perimeter, and S = channel slope.

Step 2: Here $A = (2.5)(10) = 25 \text{ ft}^2$, $P = 10 + 2.5 + 2.5 = 15 \text{ ft}$, $R_h = 25/15 = 1.67 \text{ ft}$, $S = 1.5/100 = 0.015$, so $V = (1.49/0.015) (1.67)^{2/3} (0.015)^{1/2} = 13.5 \text{ ft/sec}$.

2) A gutter at the side of a street has a curb height of 8 inches, slope along the length of the street of 0.7%, and Manning $n=0.032$. The pavement slopes from the center of the street to the gutter at 5%. What is the flow capacity of this gutter?

- (A) 4.9 cfs
- (B) 8.0 cfs
- (C) 11.1 cfs
- (D) 14.7 cfs

The Answer is B

Step 1: When the gutter is full, the water extends a distance $= 8/0.05 = 160$ inches $= 13.3$ feet from the curb. The cross sectional area $A = 0.5(13.3)(8/12) = 4.44 \text{ ft}^2$, and wetted perimeter $P = 8/12 + 13.3 = 14 \text{ ft}$, hydraulic radius $R_h = (4.44)/(14) = 0.317 \text{ ft}$.

Step 2: Manning's equation for the discharge $Q = \frac{K}{n} A R_h^{2/3} S^{1/2}$, where $K=1$ for metric units, $K=1.49$ for USCS units, n = Manning roughness coefficient
 $R_h = A/P$ = hydraulic radius, A = cross sectional area, P = wetted perimeter, S = channel slope, so $Q = (1.49/0.032)(4.44)(0.317)^{2/3} (.7/100)^{1/2} = 8.0 \text{ cfs}$.

3) Three rain gages are located within or nearby a watershed, whose total area is 1100 acres. Using the Thiessen polygon method, the portions of the watershed associated with gages A, B, and C are 520, 310, and 270 acres, respectively. For a particular storm, it is determined that the excess rainfall for these 3 portions of the watershed is 0.95, 1.23, and 1.44 inches, respectively. What is the volume of runoff from this watershed for this storm?

- (A) 19.2 MG
- (B) 28.3 MG
- (C) 34.3 MG
- (D) 44.7 MG

The Answer is C

Step 1: The volume of runoff is equal to the product of the watershed area and the excess rainfall depth.

Step 2: Here the volume = $(0.95 \text{ in})(1/12 \text{ ft/in})(520 \text{ acres})(43560 \text{ ft}^2/\text{acre}) + (1.23 \text{ in})(1/12 \text{ ft/in})(310 \text{ acres})(43560 \text{ ft}^2/\text{acre}) + (1.44 \text{ in})(1/12 \text{ ft/in})(270 \text{ acres})(43560 \text{ ft}^2/\text{acre}) = 4.59 \times 10^6 \text{ ft}^3 = 34.3 \text{ MG (million gallons)}$

4) A watershed of interest is characterized by area of 2200 acres, runoff coefficient $C=0.65$ and time of concentration $T_C=3$ hours. The following are 10-year return period rainfall intensities for various durations: 2.35 inches/hour for 1 hour, 2.15 inches/hour for 2 hours, 1.95 inches/hour for 3 hours, and 1.70 inches/hour for 4 hours. What is the 10-year return period peak runoff from this watershed?

- (A) 2,430 cfs
- (B) 2,790 cfs
- (C) 3,070 cfs
- (D) 3,360 cfs

The Answer is B

Step 1: The rational formula is $Q = C I A$, where Q = peak discharge in cfs, C = runoff coefficient, I = average rainfall intensity in inches/hour, and A = watershed area in acres. The rainfall intensity I is the intensity averaged over the time of concentration of the watershed.

Step 2: The 10-year return period rainfall intensity for this watershed is $I = 1.95$ in/hr, so $Q = C I A = (0.65)(1.95 \text{ in/hr})(2200 \text{ acres}) = 2,790 \text{ cfs}$.

5) Which of the following is not an advantage of providing dead storage in a storm water detention pond?

- (A) Encourages settling of particles
- (B) Reduction of scour from the basin bottom
- (C) Encourage flushing of pollutants from the pond
- (D) Reduction of peak discharge

The Answer is C

Step 1: All detention ponds provide some amount of dead storage, which is storage below the level of the outlet structure.

Step 2: All of the above are advantages of dead storage except C. The dead storage should not be so large or deep so that pollutants from the bottom material of the pond are released back into the overlying water.

6) A horizontal pipeline carries water at a discharge of 13.5 cfs. Upstream of a contraction the pipe diameter is 24 inches and pressure is 14 psi, while downstream of the contraction the diameter is 18 inches. Neglecting head loss, what is the pressure downstream?

- (A) 11.2 psi
- (B) 13.7 psi
- (C) 14.2 psi
- (D) 15.9 psi

The Answer is B

Step 1: The Bernoulli equation states that $z_1 + \frac{p_1}{\gamma} + \frac{V_1^2}{2g} = z_2 + \frac{p_2}{\gamma} + \frac{V_2^2}{2g}$, where z = elevation, p = pressure, V = average velocity, g = acceleration of gravity, and γ = specific weight of water. Let point 1 be upstream of the contraction, and point 2 downstream.

Step 2: Since pipe is horizontal, $z_1 = z_2$. Area $A_1 = \pi D^2/4 = 3.14 \text{ ft}^2$, $A_2 = 3.14(18/12)^2/4 = 1.77 \text{ ft}^2$. $V_1 = Q_1/A_1 = (13.5)/(3.14) = 4.3 \text{ ft/sec}$, $V_2 = Q_2/A_2 = (13.5)/(1.77) = 7.63 \text{ ft/sec}$. Also $p_1 = 14 \text{ psi} = 14(144) = 2020 \text{ lbs/ft}^2$, $p_1/\gamma = 4900/62.4 = 32.3 \text{ ft}$.

Step 3: Rearranging the Bernoulli Eq gives $\frac{p_2}{\gamma} = \frac{p_1}{\gamma} + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} = 32.3 + (4.3)^2/2/32.2 - (7.63)^2/2/32.2 = 31.7 \text{ ft}$; $p_2 = 31.7(62.4 \text{ lbs/ft}^3)/144 = 13.7 \text{ psi}$.

7) Two 18-inch diameter pipes carry water flow into a pipe junction, and a single 30-inch diameter pipe carries flow out of the junction. The velocity in the two 18-inch pipes is 2.5 and 3.8 ft/sec. What is the velocity in the 30 inch pipe?

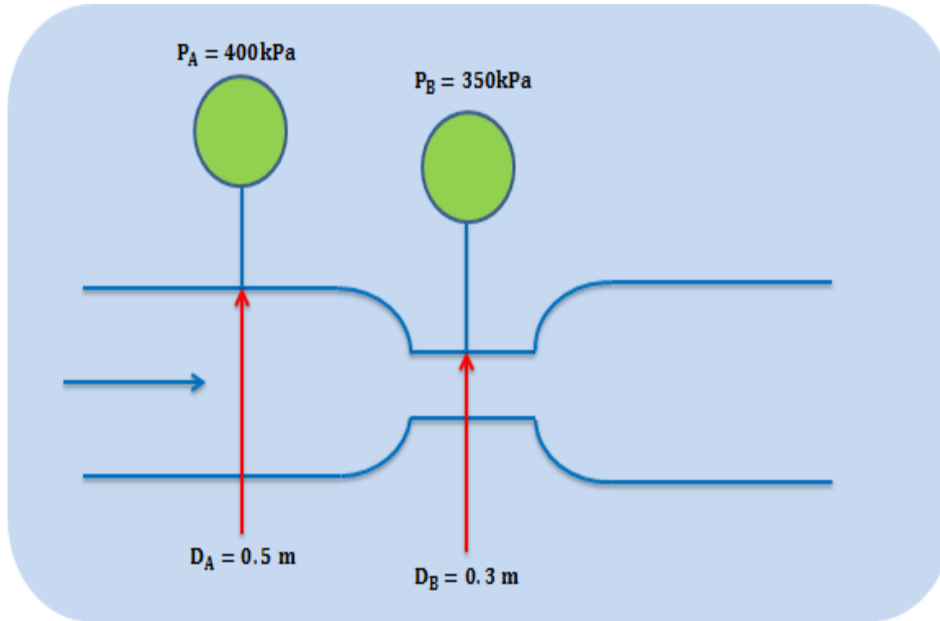
- (A) 1.98 ft/sec
- (B) 2.27 ft/sec
- (C) 3.15 ft/sec
- (D) 4.27 ft/sec

The Answers is B

Step 1: Pipes 1 and 2 are the 18-inch pipes, and pipe 3 is the 30-inch pipe. The continuity equation states that $V_1A_1 + V_2A_2 = V_3A_3$.

Step 2: Here $A_1 = A_2 = \pi D^2/4 = 3.14(18/12)^2/4 = 1.77 \text{ ft}^2$, and $A_3 = 3.14(30/12)^2/4 = 4.91 \text{ ft}^2$. Also $V_1 = 2.5 \text{ ft/sec}$, $V_2 = 3.8 \text{ ft/sec}$. Solving the continuity equation for $V_3 = (V_1A_1 + V_2A_2)/A_3 = 2.27 \text{ ft/sec}$.

8) Using the venture meter below, determine the mass flow rate of the water through the pipe (@ B).



A) 758 kg/s
C) 993 kg/s

B) 507 kg/s
D) 842 kg/s

The Answers is A

$$\begin{cases} Q_{in} = Q_{out} \\ Q = Av \end{cases} \quad A_A v_A = A_B v_B \quad (\pi/4)(\frac{d_a^2}{4})v_A = (\pi/4)(\frac{d_b^2}{4})v_B$$

$$v_A = 0.36v_B$$

$$\frac{P_2}{\gamma} + \frac{v_2^2}{2g} + z_2 = \frac{P_1}{\gamma} + \frac{v_1^2}{2g} + z_1 \quad \text{or}$$

$$\frac{P_2}{\rho} + \frac{v_2^2}{2g} + z_2 g = \frac{P_1}{\rho} + \frac{v_1^2}{2g} + z_1 g$$

$$\frac{P_A}{\rho} + \frac{v_A^2}{2g} + z_A g = \frac{P_B}{\rho} + \frac{v_B^2}{2g} + z_B g$$

$$\Delta z = 0$$

$$P_A - P_B = 400\text{kPa} - 350\text{kPa} = 50\text{kPa}$$

$$\frac{v_B^2 - v_A^2}{2} = \frac{50000\text{Pa}}{1000 \frac{\text{kg}}{\text{m}^3}}$$

$$v_B^2 - v_A^2 = 100 \frac{\text{m}^2}{\text{s}^2}$$

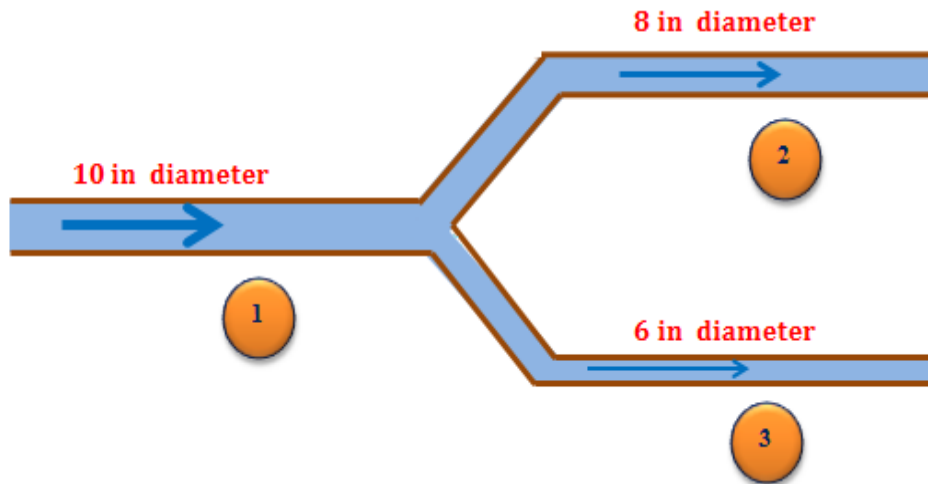
$$v_B^2 - (0.36v_B)^2 = 100 \frac{\text{m}^2}{\text{s}^2}$$

$$v_B^2(1 - (0.36)^2) = 100 \frac{\text{m}^2}{\text{s}^2}$$

$$v_B = 10.72 \frac{\text{m}}{\text{s}}$$

$$\dot{m} = \rho V_B A_B = \left(1000 \frac{\text{kg}}{\text{m}^3}\right) \left(10.72 \frac{\text{m}}{\text{s}}\right) \pi (0.15\text{m})^2 = 758 \frac{\text{kg}}{\text{s}}$$

9) In the figure below, the velocity in pipe 1 is 2 m/s and the velocity in pipe 2 is 1.5 m/s is. What is the flow rate in pipe 3?

A) 2.38 m³/sB) .053 m³/sC) 3.01 m³/sD) 2.87 m³/s

The Answers is B

$$Q1 = Q2 + Q3$$

$$Q1 = v_1 A_1$$

$$Q2 = v_2 A_2$$

$$v_1 A_1 = v_2 A_2 + Q3 \qquad Q3 = v_1 A_1 - v_2 A_2$$

$$A1 = \pi/4(d1)^2$$

$$A_2 = \pi/4(d_2)^2$$

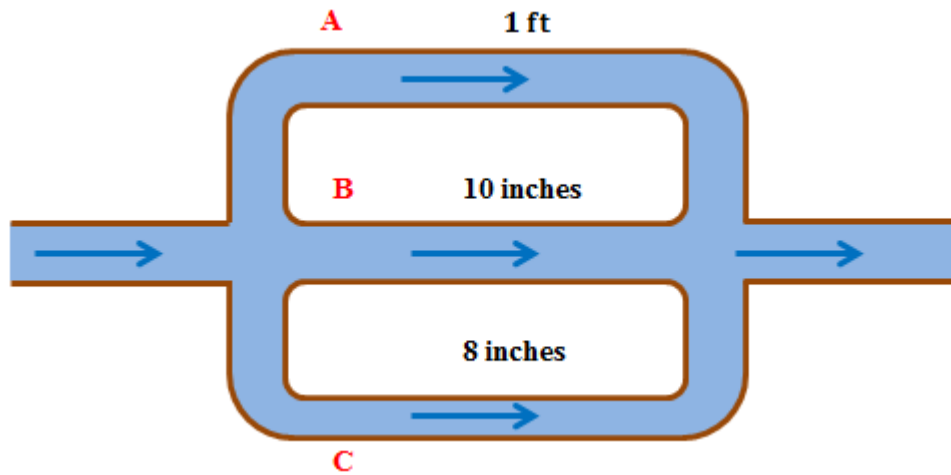
$$Q3 = v_1 (\pi/4)(d_1)^2 - v_2 (\frac{\pi}{4})(d_2)^2$$

$$d1 = 10 \text{in} \left(\frac{1 \text{ft}}{12 \text{in}} \right) \left(\frac{1 \text{m}}{3.281 \text{ft}} \right) = 0.254 \text{m}$$

$$d2 = 8 \text{in} \left(\frac{1 \text{ft}}{12 \text{in}} \right) \left(\frac{1 \text{m}}{3.281 \text{ft}} \right) = 0.203 \text{m}$$

$$Q3 = \left(\frac{2 \text{m}}{\text{sec}} \right) [\pi/4(0.254 \text{m})^2] - \left(\frac{1.5 \text{m}}{\text{sec}} \right) [\pi/4(0.203 \text{m})^2] = .053 \text{ m}^3/\text{s}$$

10) All three pipes in the figure below have the same length and are connected in parallel. The diameters are given. Determine the ratio of velocities in the branches. Assume the friction factor for all the lines to be the same.



A) V_C to V_B to $V_A = 1$ to 1.223 to 1.684

B) V_C to V_B to $V_A = 1$ to 2.3 to 3.1

C) V_C to V_B to $V_A = 1$ to 1.118 to 1.224

D) V_A to V_B to $V_C = 1$ to 1.118 to 1.224

The Answers is C

$$h_{L,A} = h_{L,B} = h_{L,C}$$

$$h_L = f \left(\frac{L}{D} \right) \left(\frac{V^2}{2g} \right)$$

$$f_A \left(\frac{L_A}{D_A} \right) \left(\frac{V_A^2}{2g} \right) = f_B \left(\frac{L_B}{D_B} \right) \left(\frac{V_B^2}{2g} \right) = f_C \left(\frac{L_C}{D_C} \right) \left(\frac{V_C^2}{2g} \right)$$

$$\left(\frac{V_A^2}{D_A} \right) = \left(\frac{V_B^2}{D_B} \right) = \left(\frac{V_C^2}{D_C} \right)$$

$$\left(\frac{V_A^2}{1} \right) = \left(\frac{V_B^2}{0.8333} \right) = \left(\frac{V_C^2}{0.667} \right)$$

$$1 V_A^2 = 1.2 V_B^2 = 1.5 V_C^2$$

$$\text{Let } V_C = 1$$

Then

$$1.5 (1)^2 = 1.2 V_B^2$$

$$V_B = \left[\frac{1.5 (1)^2}{1.2} \right]^{\frac{1}{2}}$$

$$V_B = 1.118$$

Also

$$1.5 (1)^2 = V_A^2$$

$$V_A = \left[\frac{1.5 (1)^2}{1.2} \right]^{\frac{1}{2}}$$

$$V_A = 1.224$$

$$V_C \text{ to } V_B \text{ to } V_A = 1 \text{ to } 1.118 \text{ to } 1.224$$

11) Determine the friction factor for flow in a circular pipe with a Reynold's number of 984.

A) 0.028

C) 0.041

B) 0.084

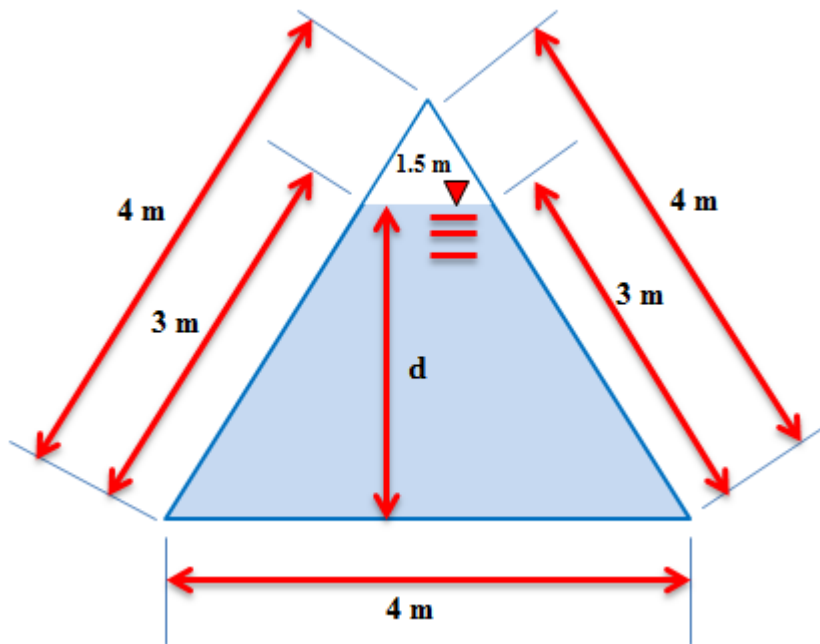
D) 0.065

The Answer is D

A careful scrutinizing of the moody diagram indicates when $Re < 4 \times 10^3$ we may use the following equation;

$$f = \frac{64}{Re} = \frac{64}{984} = 0.065$$

12) Determine the hydraulic radius for the figure below.



A) 0.53m

B) 0.71m

C) 0.96m

D) 0.22m

The Answers is B

$$R_H = \frac{\text{cross - sectional area}}{\text{wetted perimeter}}$$

$$\text{Cross-sectional area} = \left(\frac{1.25\text{m} + 4\text{m}}{2} \right) d$$

$$d = \sqrt{(3\text{m})^2 - (1.25\text{m})^2} = 2.73\text{m}$$

$$\text{Cross-sectional area} = \left(\frac{5.25}{2} \text{m} \right) (2.73\text{m}) = 7.16\text{m}^2$$

$$R_H = \frac{\text{cross-sectional area}}{\text{wetted perimeter}} = \frac{7.16\text{m}^2}{3\text{m} + 3\text{m} + 4\text{m}} = 0.71\text{m}$$

13) A concrete pipe has water flowing through it with a flow rate of $12 \text{ m}^3/\text{min}$. The pipe has a diameter of 0.2 m and a coefficient of friction of 0.021 . Determine the pressure drop over a length of 22 m .

- | | |
|-----------------------|-----------------------|
| A) 28.4 kPa | C) 51.7 kPa |
| B) 33.5 kPa | D) 46.9 kPa |

The Answers is D

$$\dot{V} = \frac{12 \frac{\text{m}^3}{\text{min}}}{60 \frac{\text{s}}{\text{min}}} = 0.2 \frac{\text{m}^3}{\text{s}}$$

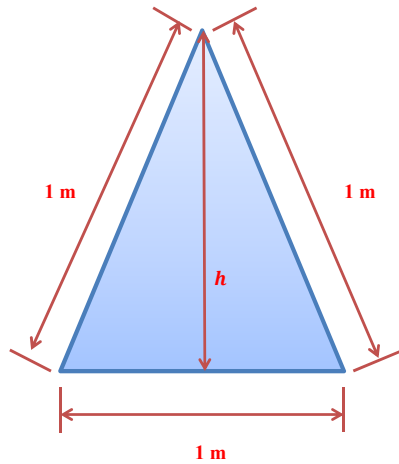
$$v = \frac{\dot{V}}{A} = \frac{\dot{V}}{\pi r^2} = \frac{0.2 \frac{\text{m}^3}{\text{s}}}{\pi \left(\frac{0.2 \text{m}}{2} \right)^2} = 6.37 \text{ m/s}$$

$$h_f = f \frac{L}{D} \frac{v^2}{2g} = (0.021) \left(\frac{22 \text{m}}{0.2 \text{m}} \right) \left(\frac{\left(6.37 \frac{\text{m}}{\text{s}} \right)^2}{(2) \left(9.81 \frac{\text{m}}{\text{s}^2} \right)} \right) = 4.78 \text{m}$$

$$\Delta h = \frac{\Delta P}{\rho g}$$

$$\Delta p = \rho g \Delta h = \left(1000 \frac{\text{kg}}{\text{m}^3} \right) \left(9.81 \frac{\text{m}}{\text{s}^2} \right) (4.78 \text{m}) = 46,892 \text{ Pa} = 46.9 \text{ kPa}$$

14) Shown in the figure below is a concrete equilateral triangular pipe. If the volumetric flow rate is $25 \text{ m}^3/\text{min}$ and the friction factor is 0.02, what is the pressure drop in the 34m length of pipe?



A) 2.2 kPa

B) 4.7 kPa

C) 6.8 kPa

D) 1.3 kPa

The Answers is A

$$h = \sqrt{(1 \text{ m})^2 - \left(\frac{1 \text{ m}}{2}\right)^2} = 0.866 \text{ m}$$

$$\text{cross-sectional area} = \frac{1}{2}bh = \left(\frac{1}{2}\right)(1 \text{ m})(0.866 \text{ m}) \\ = 0.433 \text{ m}^2$$

$$D_H = 4 \left(\frac{\text{cross-sectional area}}{\text{wetted perimeter}} \right) = (4) \left(\frac{0.433 \text{ m}^2}{1 \text{ m} + 1 \text{ m} + 1 \text{ m}} \right) = 0.144 \text{ m}$$

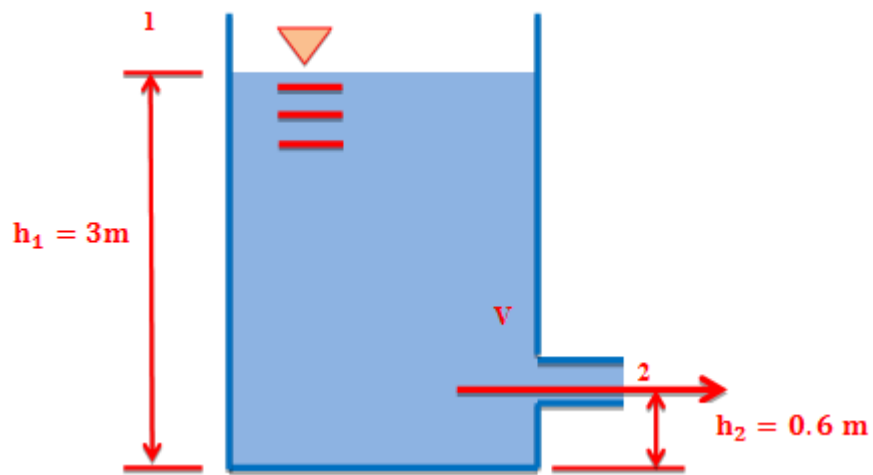
$$\dot{V} = \frac{25 \frac{\text{m}^3}{\text{min}}}{60 \frac{\text{sec}}{\text{min}}} = 0.417 \frac{\text{m}^3}{\text{s}}$$

$$V = \frac{\dot{V}}{A} = \frac{0.417 \frac{\text{m}^3}{\text{s}}}{0.433 \text{ m}^2} = 0.96 \frac{\text{m}}{\text{s}}$$

$$h_f = f \frac{L}{D} \frac{V^2}{2g} = (0.02) \left(\frac{34 \text{ m}}{0.144 \text{ m}} \right) \left(\frac{\left(0.96 \frac{\text{m}}{\text{s}}\right)^2}{(2) \left(9.81 \frac{\text{m}}{\text{s}^2}\right)} \right) = 0.222 \text{ m}$$

$$\Delta p = \rho g \Delta h = \gamma \Delta h = \left(1000 \frac{\text{kg}}{\text{m}^3}\right) \left(9.81 \frac{\text{m}}{\text{s}^2}\right) (0.222 \text{ m}) = 2.2 \text{ kPa}$$

15) Using the figure below, determine the velocity of the water at the exit pipe.



- A) 6.9 m/s B) 8.4 m/s C) 3.7 m/s D) 7.2 m/s

The Answer is A

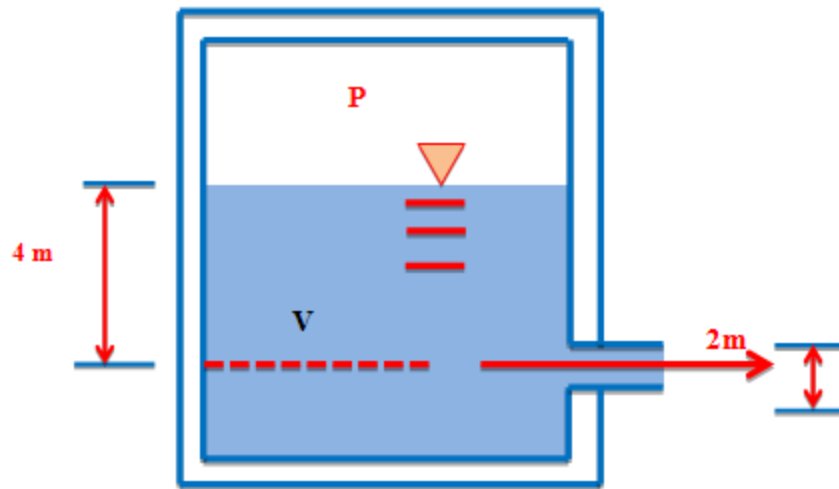
$$\frac{P_2}{\gamma} + \frac{v_2^2}{2g} + z_2 = \frac{P_1}{\gamma} + \frac{v_1^2}{2g} + z_1$$

**Note: @ point 1, $v = 0$ & $P = 0$. @point 2: $P = 0$

$$\frac{P_2}{\gamma} + \frac{v_2^2}{2g} + z_2 = \frac{P_1}{\gamma} + \frac{v_1^2}{2g} + z_1 \quad 0 + \frac{v_2^2}{2g} + z_2 = 0 + 0 + z_1$$

$$v_2 = \sqrt{2g(z_1 - z_2)} = \sqrt{(2)(9.81 \frac{\text{m}}{\text{s}^2})(3\text{m} - 0.6\text{m})} = 6.9 \text{ m/s}$$

- 16) The tank below is pressurized and filled with a fluid that has a density of 965 kg/m^3 . If the pressure of the air above the fluid is 500 kPa , determine the exit velocity.



- A) 27 m/s B) 30 m/s C) 15 m/s D) 21 m/s

The Answers is B

$$\frac{P_2}{\gamma} + \frac{v_2^2}{2g} + z_2 = \frac{P_1}{\gamma} + \frac{v_1^2}{2g} + z_1$$

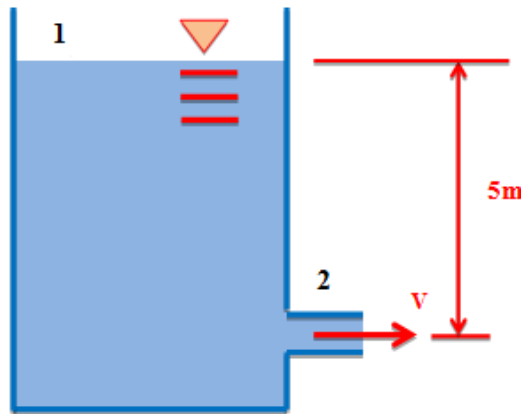
$$\frac{P_{\text{tank}}}{\rho} + \frac{v_1^2}{2} + gz_1 = \frac{P_{\text{atm}}}{\rho} + \frac{v_2^2}{2} + gz_2$$

**the velocity, $v_1 = 0$ & $z_2 = 0$

$$\frac{P_{\text{tank}}}{\rho} + 0 + gz_1 = \frac{P_{\text{atm}}}{\rho} + \frac{v_2^2}{2} + 0$$

$$v_2 = \sqrt{2g \left(\frac{P_{\text{tank}} - P_{\text{atm}}}{\rho g} + z_1 \right)} = \sqrt{(2)(9.81 \frac{\text{m}}{\text{s}^2}) \left[\frac{500 \frac{\text{kN}}{\text{m}^2} - 101 \frac{\text{kN}}{\text{m}^2}}{\left(\frac{965 \text{kg}}{\text{m}^3} \right) (9.81 \frac{\text{m}}{\text{s}^2}) \left(\frac{1}{1000 \text{N}} \right)} + 4 \text{m} \right]} = 30 \text{ m/s}$$

17) In the figure shown below, the volume remains constant. Determine the velocity of the water exiting into the atmosphere.



A) 7.1 m/s

B) 9.9 m/s

C) 5.6 m/s

D) 10.4 m/s

The Answer is C

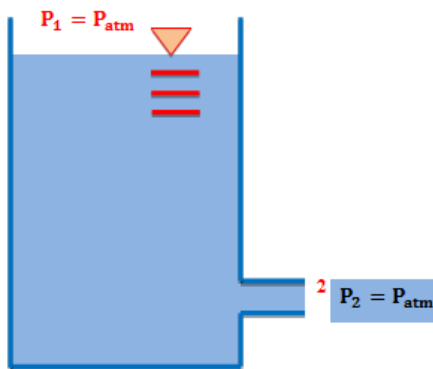
$$\frac{P_2}{\gamma} + \frac{v_2^2}{2g} + z_2 = \frac{P_1}{\gamma} + \frac{v_1^2}{2g} + z_1 \text{ or}$$

$$\frac{P_2}{\rho} + \frac{v_2^2}{2g} + z_2 g = \frac{P_1}{\rho} + \frac{v_1^2}{2g} + z_1 g$$

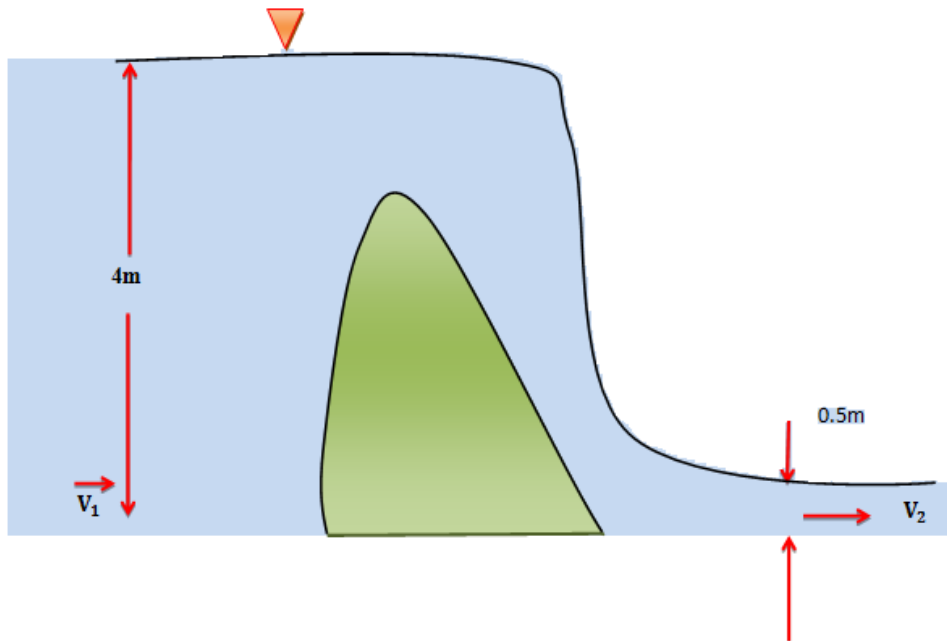
**Note: The pressure at point 1 & 2 are equal, the velocity at point 1 is zero

$$\frac{v_2^2}{2g} + z_2 g = 0 + z_1 g$$

$$v_2 = \sqrt{2g(z_1 - z_2)} = \sqrt{(2)(9.81 \frac{\text{m}}{\text{s}^2})(5\text{m})} = 9.9 \text{ m/s}$$



18) In the figure below, water flows over a spillway. Determine the velocity, v_2 . Assume a unit width.



A) 6.31 m/s

B) 5.09 m/s

C) 2.76 m/s

D) 8.32 m/s

The Answers is D

$$\frac{P_2}{\gamma} + \frac{v_2^2}{2g} + z_2 = \frac{P_1}{\gamma} + \frac{v_1^2}{2g} + z_1 \quad 0 + \frac{v_1^2}{2g} + 4\text{m} = 0 + \frac{v_2^2}{2g} + 0.5\text{ m}$$

$$A_1 v_1 = A_2 v_2 \quad (4\text{m})(1\text{m})v_1 = (0.5\text{m})(1\text{m})v_2 \quad v_2 = 8v_1$$

$$\frac{v_1^2}{2(9.81 \frac{\text{m}}{\text{s}^2})} + 4\text{m} = \frac{(8v_1)^2}{2(9.81 \frac{\text{m}}{\text{s}^2})} + 0.5 \quad mv_1 = 1.04 \text{ m/s}$$

$$v_2 = 8v_1 = 8(1.04 \text{ m/s}) = 8.32 \text{ m/s}$$

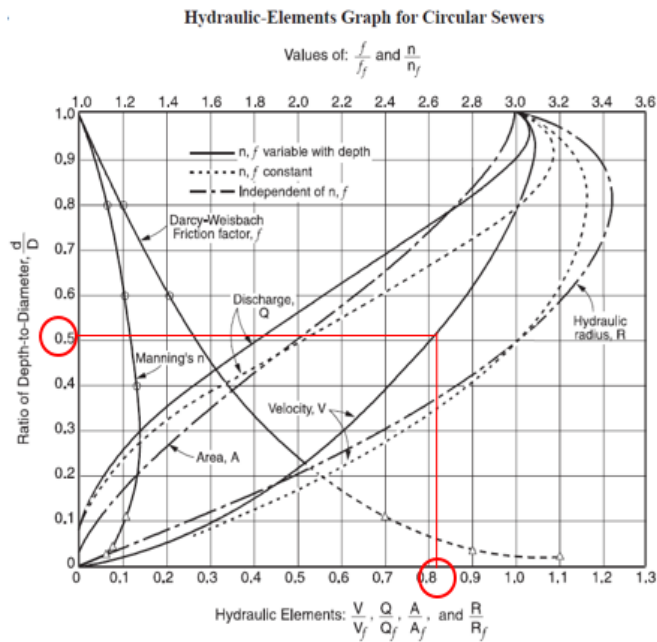
19) An 18-inch storm sewer has a full flow rate of 8.15 cfs and an average velocity of 4.61 ft/sec. What is the average velocity of the storm sewer when it is flowing at a depth of 9.4 inches? Assume f is variable with depth.

- A) 0.5 ft/s
- B) 2.4 ft/s
- C) 3.8 ft/s
- D) 4.1 ft/s

The Answer is C

: From the “Hydraulic-Elements Graph for Circular Sewers” on any hydraulic book. .
 Depth/diameter ratio = $d/D = 9.4''/18'' = 0.52$. Finding the d/D of 0.52 on the y-axis and where this intersects the velocity line (f varying with depth). Bringing this point down to the x-axis we can determine that $V/V_{full} = 0.82$.

$$V = (V/V_{full})(V_{full}) = 0.82 \cdot 4.61 \text{ ft/s} = 3.78 \text{ ft/s}$$



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20) Determine the velocity of stormwater flowing through a pipe with a slope of 0.006. Using Manning's equation and a roughness coefficient of $n = 0.04$, $R_h = 0.5$ for the flow.

- A) 1.2 ft/s
- B) 2.3 ft/s
- C) 3.1 ft/s
- D) 1.6 ft/s

The Answers is A

1. Manning's Equation $Q = (K * A * R_H^{2/3} S^{1/2}) / n$
2. Divide by A to solve for V, $V = (K * R_H^{2/3} S^{1/2}) / n$
- a. $K = 1$ SI units
- b. $S = 0.006$
- c. $n = 0.04$
- d. $R_h = 0.5$
3. $V = (1 * 0.5^{2/3} 0.006^{1/2}) / 0.04 = 1.2 \text{ ft/s}$

21) A full pipe has a velocity of 3 ft/s and is 36" in diameter. The concrete pipe has an $n = 0.013$. What is the slope of the pipe necessary to meet these conditions?

- A) 0.001
- B) 0.002
- C) 0.0015
- D) 0.0025

The Answers is A

1. Manning's Equation $Q = (K * A * R_H^{2/3} S^{1/2}) / n$
2. Divide by A to solve for V, $V = (K * R_H^{2/3} S^{1/2}) / n$
- a. $K_1 = 1.486$ for US units
- b. Hydraulic Radius for a full pipe $R = A/P = \pi(D/2)^2 / (\pi D) = D/4$
3. $V = 3 \text{ ft/s} = (1.486 * (3 \text{ ft}/4)^{2/3} S^{1/2}) / 0.013$
4. $S = 0.001$

22) Irrigation water flows at a depth of 4 m in a 15 m wide, concrete, rectangular open channel with a 0.5% slope. The Manning roughness coefficient for this channel is 0.015. The water velocity is most nearly

- (A) $7.5 \frac{\text{m}}{\text{s}}$
- (B) $8.9 \frac{\text{m}}{\text{s}}$
- (A) 9.8 m/s
- (C) 10 m/s

The Answers is B

The Manning equation for open channel flow is

$$V = \frac{1}{n} R^{\frac{2}{3}} \sqrt{S}$$

Determine the hydraulic radius.

$$R = \frac{A}{P} = \frac{(15 \text{ m})(4 \text{ m})}{4 \text{ m} + 15 \text{ m} + 4 \text{ m}} \\ = 2.61 \text{ m}$$

Solve for the velocity.

$$V = \frac{1}{n} R^{\frac{2}{3}} \sqrt{S} \\ = \left(\frac{1}{0.015} \right) (2.61 \text{ m})^{\frac{2}{3}} \sqrt{0.005} \\ = 8.94 \frac{\text{m}}{\text{s}} \quad \left(8.9 \frac{\text{m}}{\text{s}} \right)$$

23) A property consists of 7500 m² of lawn area with a runoff coefficient of 0.20, 2000 m² of gravel roadway with a runoff coefficient of 0.80.

The overall runoff coefficient for the entire property is most nearly

- A) 0.15
- B) 0.22
- C) 0.38
- D) 0.80

The Answer is B

The overall runoff coefficient is determined by weighting each runoff coefficient by the area it covers.

$$\begin{aligned} A_{\text{total}} &= A_{\text{lawn}} + A_{\text{roadway}} + A_{\text{roof}} \\ &= 7,500 \text{ m}^2 + 2,000 \text{ m}^2 + 500 \text{ m}^2 \\ &= 10,000 \text{ m}^2 \end{aligned}$$

The overall runoff coefficient is

$$\begin{aligned} C_{\text{overall}} &= \frac{(A_{\text{lawn}}C_{\text{lawn}} + A_{\text{roadway}}C_{\text{roadway}} + A_{\text{roof}}C_{\text{roof}})}{A_{\text{total}}} \\ &= \frac{(7,500 \text{ m}^2)(0.20) + (2,000 \text{ m}^2)(0.15) + (500 \text{ m}^2)(0.80)}{10,000 \text{ m}^2} \\ &= 0.22 \end{aligned}$$

24) A property consists of 7500 m² of lawn area with a runoff coefficient of 0.20, 2000 m² of gravel roadway with a runoff coefficient of 0.80.

The overall runoff coefficient for the entire property is most nearly

- (A) 0.15
- (B) 0.22
- (C) 0.38
- (D) 0.80

The Answers is B

The overall runoff coefficient is determined by weighting each runoff coefficient by the area it covers.

$$\begin{aligned}A_{\text{total}} &= A_{\text{lawn}} + A_{\text{roadway}} + A_{\text{roof}} \\&= 7,500 \text{ m}^2 + 2,000 \text{ m}^2 + 500 \text{ m}^2 \\&= 10,000 \text{ m}^2\end{aligned}$$

The overall runoff coefficient is

$$\begin{aligned}C_{\text{overall}} &= \frac{(A_{\text{lawn}}C_{\text{lawn}} + A_{\text{roadway}}C_{\text{roadway}} + A_{\text{roof}}C_{\text{roof}})}{A_{\text{total}}} \\&= \frac{(7,500 \text{ m}^2)(0.20) + (2,000 \text{ m}^2)(0.15) + (500 \text{ m}^2)(0.80)}{10,000 \text{ m}^2} \\&= 0.22\end{aligned}$$

25) Water from a 175-ac light industrial watershed is collected and drained by a trapezoidal open channel. The channel (Manning' roughness coefficient, $n = 0.02$) has a 4.5-ft-wide bottom and 1:1 sides. The channel direction is perpendicular to a road where twin, side-by-side 54-in-diameter corrugated metal pipe (CMP) culverts take the water under the roadway. The average slope of the channel and culverts is 0.75% (i.e., 0.0075 ft/ft).

The time for runoff from the farthest part of the watershed to begin contributing to the flow is 35 min.

- What rational-method runoff coefficient would you recommend?

- | | |
|--------|--------|
| A) .65 | B) .5 |
| C) 0.8 | D) 0.3 |

The Answer is A

A runoff coefficient for light industrial areas is between 0.5 and 0.8. for this problem an average value of 0.65 will be used.

26) Water from a 175-ac light industrial watershed is collected and drained by a trapezoidal open channel. The channel (Manning' roughness coefficient, $n = 0.02$) has a 4.5-ft-wide bottom and 1:1 sides. The channel direction is perpendicular to a road where twin, side-by-side 54-in-diameter corrugated metal pipe (CMP) culverts take the water under the roadway. The average slope of the channel and culverts is 0.75% (i.e., 0.0075 ft/ft).

The time for runoff from the farthest part of the watershed to begin contributing to the flow is 35 min.

- Using the rational method and assuming the intensity after 35 min is 2 in/hr, what is the runoff?

A) $308 \frac{\text{ft}^3}{\text{sec}}$

B) $180 \frac{\text{ft}^3}{\text{sec}}$

C) $228 \frac{\text{ft}^3}{\text{sec}}$

D) $340 \frac{\text{ft}^3}{\text{sec}}$

The Answer is C

The runoff is given by the rational formula.

$$Q_p = CIA_d = (0.65) \left(2 \frac{\text{in}}{\text{hr}} \right) (175 \text{ ac}) = 228 \frac{\text{ft}^3}{\text{sec}}$$

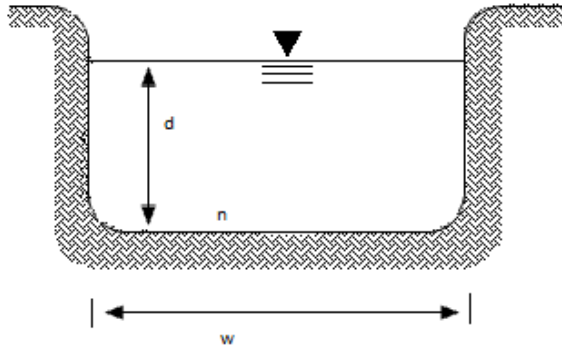
27) A section of an existing natural channel (Manning's $n = 0.030$) drops 470 ft as it meanders 8.4 mi through an area scheduled to be developed commercially. The existing channel is basically rectangular in shape, is 25 ft wide, and flows (at maximum) with a depth of 12 ft. The developers want to install an artificial channel 3.7 mi shorter over a more direct route, starting and ending at the same points as the natural channel. The artificial channel will be trapezoidal in shape with a base of 20 ft and sides inclined at 1:1, and it will be earth-line.

- Will erosion be more or less of a problem with the new channel than with the old channel? Why?

- A) Is More
- B) is less
- C) they are equal
- D) We cannot answer with given info.

The Answer is A

The capacity of the section can be calculated assuming uniform flow.



$$d = 12 \text{ ft} \quad n = 0.030 \quad w = 25 \text{ ft}$$

The erosion potential of the flow can be assessed from the mean velocity.

For the natural channel,

$$v = \frac{Q}{A} = \frac{5134 \frac{\text{ft}^3}{\text{sec}}}{300 \text{ ft}^2} = 17.1 \text{ ft/sec}$$

In the artificial channel, the maximum velocity occurs when the flow becomes uniform. At this point,

$$A = d_o(w + md_o) = (7.3 \text{ ft})[20 \text{ ft} + (1)(7.3 \text{ ft})] = 199.3 \text{ ft}^2$$

$$v = \frac{Q}{A} = \frac{5134 \frac{\text{ft}^3}{\text{sec}}}{199.3 \text{ ft}^2} = 25.8 \text{ ft/sec}$$

Erosion problems are probably more severe in the artificial channel. However, both channels are well above maximum permissible velocities for the bed material.

28) The SCS Curve Number for a small watershed is $CN=80$. The watershed area is 110 acres. If this watershed receives 0.8 inches of rainfall, what will be the resulting runoff volume from the watershed?

- (A) 1000 feet³ (B) 11000 feet³ (C) 12,800 feet³ (D) 15,000 feet³

The Answer is C

Step 1: The maximum basin retention is $S = 1000/CN - 10 = 1000/80 - 10 = 2.5$ inches

Step 2: The runoff volume Q is given by

$$Q = (P - 0.2 S)^2 / (P + 0.8 S) = (0.8 - 0.2 * 2.5)^2 / (0.8 + 0.8 * 2.5) = 0.032 \text{ inches}$$

In this equation, Q is the runoff volume per unit watershed area.

Step 3: To get the actual runoff volume, multiply by the watershed area

$$\begin{aligned} Q &= 0.032 \text{ inches} / (12 \text{ inch/foot}) (110 \text{ acres}) (43560 \text{ sq foot/acre}) \\ &= 12,800 \text{ cubic feet} \end{aligned}$$

29) Runoff from a 3-acre site is to be drained by a channel. The time of concentration for this site is 40 minutes. The site has a runoff coefficient $C=0.2$. Rainfall quantities to be used for design are 0.5 inches for a storm of duration 20 minutes, 0.7 inches for duration 40 minutes, and 0.9 inches for duration 60 minutes. For what discharge should this channel be designed?

- (A) 0.7 cfs (B) 0.63 cfs (C) 1.04 cfs (D) 40 cfs

The Answers is B

Step 1: Apply the rational formula:

$$Q = C i A$$

where Q is the discharge in cfs, C is the runoff coefficient for the watershed, i is the rainfall intensity in inches/hour, and A is the watershed area in acres.

Step 2: In applying the rational method, rain falling over a time period equal to the time of concentration of the watershed should be used. In this case, the time of concentration is given as 40 minutes, or 0.67 hours, and the corresponding rainfall amount is 0.7 inches.

So the rainfall intensity is

$$i = (0.7 \text{ inches}) / (0.67 \text{ hour}) = 1.04 \text{ inches/hour}$$

Step 3: Solver for the discharge $Q = C i A = (0.2) (1.04) (3) = 0.63 \text{ cfs}$

30) A parking lot adjoining a shopping center has a surface area of 3.5 acres. A rainstorm that delivers rainfall at a rate of 2.5 inches/hour occurs. The parking area has a runoff coefficient $C=0.7$. What is the peak runoff from the parking lot during this rainstorm?

- (A) 0.62 cfs (B) 2.5 cfs (C) 6.13 cfs (D) 25 cfs

The Answer is C

Step 1: The rational formula is

$$Q = C i A$$

where Q is the discharge, C is the runoff coefficient, i is the rainfall intensity in inches/hour, and A is the drainage area in acres.

Step 2: Solve for $Q = (0.7) (2.5) (3.5) = 6.13$ cfs

31) A horizontal pipeline of uniform diameter has a pressure decrease of 3 psi over a length of 250 feet. What is the head loss in the pipe over this same 250 foot length?

- (A) 3 feet (B) 3.5 feet (C) 0.0026 feet (D) 6.9 feet

The Answer is D

Step 1: The Bernoulli equation states that for a horizontal pipe of constant diameter, the head loss is equal to the decrease in pressure head.

Step 2: Convert the pressure decrease to head:

$$\Delta p = (3 \text{ pounds/inch}^2) (144 \text{ inch}^2/\text{foot}^2) = 432 \text{ pounds/foot}^2$$

Specific weight of water $\gamma = 62.4 \text{ lbs/ft}^3$

$$\text{Pressure head } \Delta p/\gamma = (432 \text{ pounds/foot}^2)/(62.4 \text{ pounds/foot}^3) = 6.9 \text{ feet}$$

Step 3: Head loss $= \Delta p/\gamma = 6.9 \text{ feet}$

32) A horizontal pipeline of uniform diameter has a pressure decrease of 3 psi over a length of 250 feet. What is the head loss in the pipe over this same 250 foot length?

- (A) 3 feet (B) 3.5 feet (C) 0.0026 feet (D) 6.9 feet

The Answer is D

Step 1: The Bernoulli equation states that for a horizontal pipe of constant diameter, the head loss is equal to the decrease in pressure head.

Step 2: Convert the pressure decrease to head:

$$\Delta p = (3 \text{ pounds/inch}^2) (144 \text{ inch}^2/\text{foot}^2) = 432 \text{ pounds/foot}^2$$

Specific weight of water $\gamma = 62.4 \text{ lbs/ft}^3$ (pg. 62)

$$\text{Pressure head } \Delta p/\gamma = (432 \text{ pounds/foot}^2)/(62.4 \text{ pounds/foot}^3) = 6.9 \text{ feet}$$

Step 3: Head loss $= \Delta p/\gamma = 6.9 \text{ feet}$

33) An open channel has a rectangular cross section, width = 60 cm. The channel carries a discharge $Q=0.1 \text{ meter}^3/\text{sec}$. At the downstream end of the channel, there is a large vertical drop. What is the depth of flow at the downstream end of the channel?

- (A) 0 meters (B) 14 centimeters (C) 1.4 meters (D) 24 centimeters

The Answers is B

Step 1: If there is a large vertical or abrupt drop in the bottom elevation of an open channel flow, the depth of flow at the drop will be equal to the critical depth y_c .

Step 2: The channel width $B = 60 \text{ cm} = 0.6 \text{ meter}$. The unit discharge, or discharge per unit width $q = Q / B = 0.1 / 0.6 = 0.17 \text{ meter}^2/\text{sec}$.

Step 3: The critical depth for a rectangular channel is given by
 $y_c = (q^2/g)^{1/3} = (0.17^2/9.81)^{1/3} = 0.14 \text{ meters} = 14 \text{ cm}$

34) In a rectangular open channel of width $b=5$ feet, the depth of flow is 1.5 feet and the discharge $Q=1.5$ cfs. Compute the alternate depth at which the specific energy is the same as that for these given conditions.

- (A) 0.031 feet (B) 0.51 feet (C) 1.5 feet (D) 1.8 feet

The Answers is A

Step 1: The specific energy $E = V^2/2g + y$.

For the given conditions, $A = y b = 5 \times 1.5 = 7.5 \text{ feet}^2$

$V = Q/A = 1.5/7.5 = 0.2 \text{ feet/sec}$

$E = V^2/2g + y = (0.2)(0.2)/2/32.2 + 1.5 = 1.5006 \text{ feet}$

Step 2: We seek another value of y that yields the same value of $E = 1.5006$, this is the alternate depth. We seek another value of y such that:

$E = V^2/2g + y = Q^2/2gA^2 + y = 1.5^2/2/32.2/(5y)^2 + y = 1.5006$

or $(1.40 \times 10^{-3})/y^2 + y = 1.5006$. Solving for y by trials, $y = 0.031 \text{ feet}$

35) A sanitary sewer serves a population of 4000 people. The sewer should be designed for an average wastewater flow of 100 gallons per capita per day. What is the peak flow rate for which the sewer should be designed?

- (A) 0.26 mgd (B) 0.4 mgd (C) 1.6 mgd (D) 1.9 mgd

The Answers is C

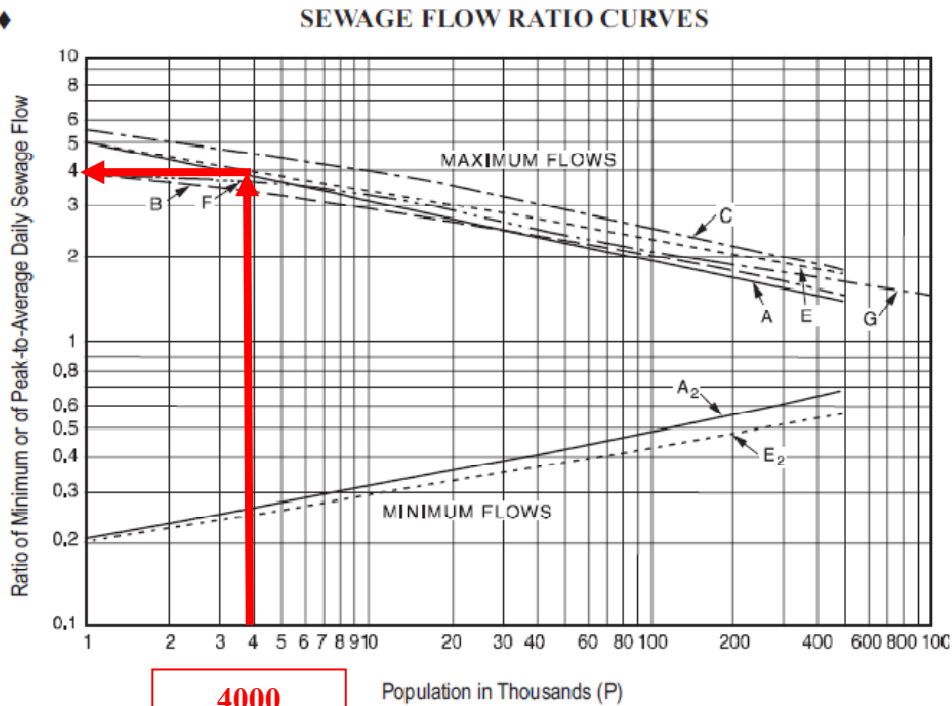
Step 1: The average wastewater flow rate

= 100 gallons per capita (per person) per day * 4000 people

= 400,000 gallons/day = 0.4 mgd

Step 2: From Figure at top of page 160 (see below), for population of 4000, ratio of peak to average flow is equal to 3.5 to 4.5 for the various curves. Choose average value of 4.

Peak wastewater flow = (0.4 mgd)(4) = 1.6 mgd (million gallons/day)



36) A sanitary sewer 15 inches in diameter, carries an open channel flow. The downstream end of the pipe is open, and the water flow from the pipe forms a waterfall. At the end of the pipe, the depth is equal to half of the pipe diameter. Compute the discharge Q in the pipe.

- (A) 0.62 cfs (B) 1.23 cfs (C) 2.41 cfs (D) 25 cfs

The Answer is C

Step 1: At the downstream end of a channel where the water descends in a waterfall, the depth of flow at the downstream end is the critical depth. Under these conditions, the following condition applies:

$$Q^2/g = A^3/T$$

where A is the cross sectional area and T is the top width at the downstream end of the channel (or pipe in this case).

Step 2: For a full pipe,

$A_f = \pi (15/12/2)^2 = 1.23 \text{ feet}^2$. If the pipe is half full, then the area is half the full value, or $A = 0.61 \text{ feet}^2$. For a half full circular pipe, the top width T is equal to the pipe diameter $D = 1.25 \text{ feet}$.

Step 3: Solving the above equation for Q gives

$$Q^2 = gA^3/T = 32.2 (0.61)^3/1.25 = 6.0$$

$$Q = 2.41 \text{ cfs}$$

37) A pump is located 5 feet above a pond. The outlet of the pump discharges directly to the atmosphere through an 8 inch diameter outlet. The velocity of water at the pump outlet is 15 feet/second. Neglect the head loss in the short pipe from the pond to the pump. At what rate is energy being added to the water flow by the pump?

- | | |
|---------------------|---------------------|
| (A) 0.62 horsepower | (B) 5.05 horsepower |
| (C) 62.4 horsepower | (D) 2780 horsepower |

The Answers is B

Step 1: Bernoulli equation between point 1 (reservoir) and point 2 (discharge from pump):

$$z_1 + p_1/\gamma + V_1^2/2g + H = z_2 + p_2/\gamma + V_2^2/2g + h_f$$

In the reservoir, $p_1=0$ and $V_1=0$. Neglect head loss, so $h_f=0$. Since the pump discharges into the atmosphere, $p_2=0$.

Solving for pump head $H = (z_2 - z_1) + V_2^2/2g = 5 + 15^2/32.2 = 8.5$ feet

Step 2: At pump outlet, $A=0.349$ feet²; discharge $Q=V A = (15)(0.349) = 5.2$ cfs

Step 3: Specific weight of water $\gamma= 62.4$ lbs/ft³ (pg. 62)

The rate of energy addition to the flow $= \gamma Q H$

$$=(62.4 \text{ pounds/ft}^3)(5.2 \text{ cfs})(8.5 \text{ feet}) = 2780 \text{ ft-pounds/sec}$$

$$=(2780 \text{ ft-pounds/sec}) / (550 \text{ ft-pounds/sec per horsepower}) = 5.05 \text{ horsepower}$$

38) A given watershed has 3 inches of precipitation during a 2-hr storm. The change in watershed storage is approximately 1.2 inches. The average infiltration rate is 0.2 in/hr. What is the approximate runoff of the watershed in inches?

- A) 0.1 inches B) 0.7 inches C) 1.4 inches D) 3 inches

The Answer is C

Solution: Use the water balance equation to solve.

Step 1: Define all variables

Assume evaporation to be zero.

$$\begin{array}{lll} P = 3 \text{ in} & E = 0 \text{ in} & I = 0.2 \frac{\text{in}}{\text{hr}} \times 2 \text{ hr} = 0.4 \text{ in} \\ R = ? & \Delta S = 1.2 \text{ in} & \end{array}$$

Step 2: Solve

$$P - E - I - R - \Delta S = 0 \rightarrow 3 - 0 - 0.4 - R - 1.2 = 0$$

$$R = 1.4 \text{ inches}$$

39) A given watershed is currently storing approximately 3 inches of water when a 3-hr storm with the following precipitation occurs. The average infiltration rate is 0.2 in/hr. Runoff is 0.9 in. What is the approximate total storage of the watershed at the end of the storm?

Time	8:00	9:00	10:00
Rainfall (in)	0.5	0.5	1.0

- A) 0.5 inches B) 3.0 inches C) 3.5 inches D) 4.0 inches

The Answer is C

Solution: Use the water balance equation to solve.

Step 1: Define all variables

Assume evaporation to be zero.

$$P = 0.5 + 0.5 + 1.0 = 2.0 \text{ in}$$

$$E = 0 \text{ in}$$

$$I = 0.2 \frac{\text{in}}{\text{hr}} \times 3 \text{ hr} = 0.6 \text{ in}$$

$$R = 0.9 \text{ in} \quad \Delta S = ?$$

Step 2: Solve water balance equation

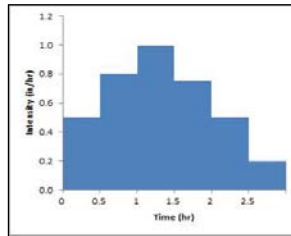
$$P - E - I - R - \Delta S = 0$$

$$\Delta S = 2 - 0 - 0.6 - 0.9 = 0.5 \text{ in}$$

Step 3: Solve for total storage after the storm.

$$S_{\text{total}} = 3 + 0.5 = 3.5 \text{ inches}$$

40) Given the following hyetograph what is the total precipitation?



A) 1.4 inches

B) 1.9 inches

C) 3.8 inches

D) 5.0 inches

The Answers is B

Solution: The total precipitation is the sum of all the rainfall over a given period of time for a watershed.

Step 1: Determine the time interval
The time frame given is in half hour increments.

Step 2: Determine the rainfall for each interval

Time (hr)	0-0.5	0.5-1.0	1.0-1.5	1.5-2.0	2.0-2.5	2.5-3.0
Intensity (in/hr)	0.5	0.8	1.0	0.8	0.5	0.2
Rainfall (in)	0.25	0.40	0.50	0.38	0.25	0.10
Total Rainfall (in)	0.25	0.65	1.15	1.53	1.78	1.88

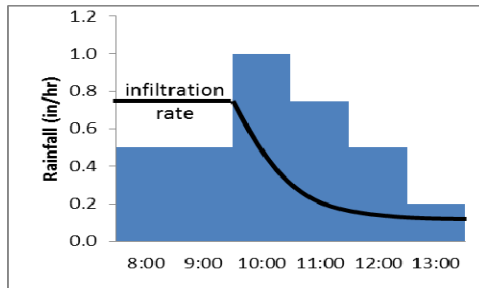
$$\text{Rainfall (hr 0-0.5)} = 0.5 \frac{\text{in}}{\text{hr}} * 0.5 \text{ hr} = 0.25 \text{ in}$$

$$\text{Rainfall (hr 0.5-1.0)} = 0.8 \frac{\text{in}}{\text{hr}} * 0.5 \text{ hr} = 0.40 \text{ in etc.}$$

Step 3: Determine total rainfall

$$\text{Total rainfall} = 0.25 + 0.40 + 0.50 + 0.38 + 0.25 + 0.10 = 1.88 \text{ inches}$$

41) What is the approximate runoff for a watershed with the following rainfall and infiltration?

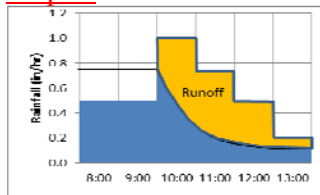


- A) 0 inches B) 0.4 inches C) 1.5 inches D) 4.4 inches

The Answer is C

Solution: Use the water balance equation to solve. The runoff is the rainfall minus infiltration.

Step 1: Determine the area above the infiltration rate.



42) A 7 acre basin has the following curve numbers for the associated areas. What is the curve number for the basin?

Area (acres)	Curve Number
2	60
3	80
2	50

- A) 47.1 B) 58.2 C) 60.5 D) 65.7

The Answer is D

Solution: The curve number must be weighted with its associated area

$$(CN)_{\text{weighted}} = \frac{\sum (CN)_i A_i}{A_{\text{total}}} = \frac{2*60 + 3*80 + 2*50}{2+3+2} = 65.7$$

43) A 10 acre basin stores approximately 7.0 inches of water. What is the runoff for the given basin for a 2-hr storm with an average of 0.5 in/hr of rainfall?

- A) 0 inches B) 0.02 inches C) 0.05 inches D) 0.10 inches

The Answer is B

Solution: Use the NRCS Rainfall-Runoff method to determine the amount of runoff.

Step 1: Define all variables

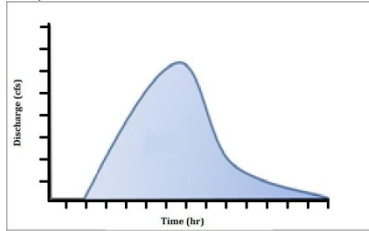
$$S = 7.0 \text{ inches}$$

$$P = 2 \text{ hr} * 0.5 \frac{\text{in}}{\text{hr}} = 1 \text{ in}$$

Step 2: Solve for runoff (Q)

$$Q = \frac{(P-0.2S)^2}{P+0.8S} = \frac{(1-0.2*7)^2}{1+0.8*7} = \frac{0.16}{7.6} = 0.02 \text{ inches}$$

44) What is the runoff volume for the given unit hydrograph?



A) 0 ft^3

B) 1 ft^3

C) 4.7 ft^3

D) $23,500 \text{ ft}^3$

The Answer is B

Solution: Runoff volume is the area under a hydrograph. The area under a unit hydrograph is one unit.

45) A 4 acre basin has the following runoff coefficients for the associated areas. What is the runoff coefficient for the basin?

Area (Acres)	Runoff Coefficient
0.5	0.26
1.5	0.28
2.0	0.89

- A) 0.26 B) 0.28 C) 0.58 D) 0.89

The Answer is C

Solution: The runoff coefficient must be weighted with its associated area

$$(C)_{\text{weighted}} = \frac{\sum(C)_i A_i}{A_{\text{total}}} = \frac{0.5 \cdot 0.26 + 1.5 \cdot 0.28 + 2.0 \cdot 0.89}{0.5 + 1.5 + 2.0} = 0.58$$

46) A 5 acre sub-basin with the following runoff coefficients and an average rainfall intensity of 0.2 in/hr. What is the peak discharge of the sub-basin?

Area (Acres)	Runoff Coefficient
1	0.24
2.5	0.32
1.5	0.96

A) 0.28 cfs

B) 0.50 cfs

C) 0.56 cfs

D) 0.82 cfs

The Answers is B

Solution: Use the rational method to determine the peak discharge.

Step 1: Define all variables

$$(C)_{\text{weighted}} = \frac{\sum(C)_i A_i}{A_{\text{total}}} = \frac{1 \cdot 0.24 + 2.5 \cdot 0.32 + 1.5 \cdot 0.96}{1 + 2.5 + 1.5} = 0.50$$

$$I = 0.2 \frac{\text{in}}{\text{hr}} \quad A = 5 \text{ acre}$$

Step 2: Solve

Tip: Remember the runoff coefficient contains the unit conversion.

$$Q = CIA = 0.50 * 0.2 \frac{\text{in}}{\text{hr}} * 5 \text{ acre} = 0.50 \text{ cfs}$$

47) The discharge from a weir is $5.4 \text{ m}^3/\text{s}$ with a maximum allowable water depth of 1 m. What would be the weir length for a rectangular contracted weir?

- A) 1.8 m B) 2.4 m C) 2.8 m D) 3.1 m

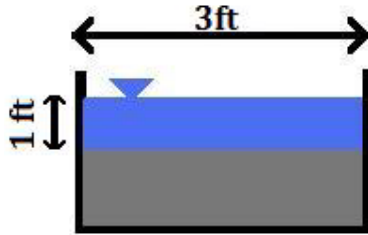
The Answer is D

Solution: This is a rectangular contracted weir with metric (SI) units.

$$C = 1.84$$

$$Q = C(L - 0.2H)H^{3/2} \rightarrow L = \frac{Q}{H^{3/2}C} + 0.2H = \frac{5.4}{1^{3/2} * 1.84} + 0.2 * 1 = 3.1 \text{ m}$$

48) What is the discharge in ft^3/s for the following rectangular weir?



- A) $1.0 \frac{\text{ft}^3}{\text{s}}$ B) $4.2 \frac{\text{ft}^3}{\text{s}}$ C) $5.5 \frac{\text{ft}^3}{\text{s}}$ D) $9.9 \frac{\text{ft}^3}{\text{s}}$

The Answer is D

Solution: Use weir formulas to solve

Step 1: Identify the type of weir and units used

This is a rectangular suppressed weir with USCS units.

Step 2: Use proper to determine equation and coefficient to be used

$$C = 3.33 \quad \text{and} \quad Q = CLH^{3/2}$$

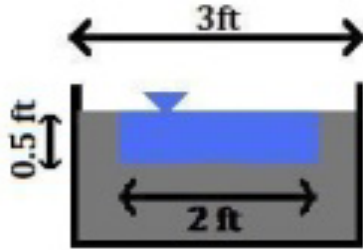
Step 3: Define all variables

$$H = 1 \text{ ft} \quad L = 3 \text{ ft}$$

Step 4: Solve

$$Q = CLH^{3/2} = 3.33 * 3 * 1^{3/2} = 9.9 \text{ ft}^3/\text{s}$$

49) What is the discharge in ft^3/s for the following rectangular weir?



- A) $1.0 \frac{\text{ft}^3}{\text{s}}$ B) $1.3 \frac{\text{ft}^3}{\text{s}}$ C) $2.2 \frac{\text{ft}^3}{\text{s}}$ D) $5.5 \frac{\text{ft}^3}{\text{s}}$

The Answer is C

Solution: Use weir formulas to solve

Step 1: Identify the type of weir and units used

This is a rectangular contracted weir with USCS units.

Step 2: Use Appropriate to determine equation and coefficient to be used

$$C = 3.33 \quad \text{and} \quad Q = C(L - 0.2H)H^{3/2}$$

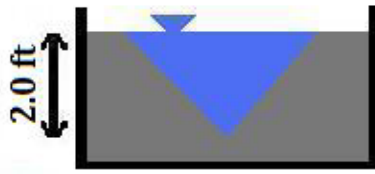
Step 3: Define all variables

$$H = 0.5 \text{ ft} \quad L = 2 \text{ ft}$$

Step 4: Solve

$$Q = C(L - 0.2H)H^{3/2} = 3.33 * (2 - 0.2 * 0.5) * 0.5^{3/2} = 2.2 \text{ ft}^3/\text{s}$$

50) What is the discharge in ft^3/s for the following v-notch weir?



- A) $2.2 \frac{\text{ft}^3}{\text{s}}$ B) $2.5 \frac{\text{ft}^3}{\text{s}}$ C) $7.2 \frac{\text{ft}^3}{\text{s}}$ D) $14.4 \frac{\text{ft}^3}{\text{s}}$

The Answer is D

Solution: Use weir formulas to solve

Step 1: Identify the type of weir and units used

This is a v-notch weir with USCS units.

Step 2: Use Appropriate to determine equation and coefficient to be used

$$C = 2.54 \quad \text{and} \quad Q = CH^{\frac{5}{2}}$$

Step 3: Define all variables

$$H = 2.0 \text{ ft}$$

Step 4: Solve

$$Q = CH^{5/2} = 2.54 * 2.0^{5/2} = 14.4 \text{ ft}^3/\text{s}$$

51) The discharge from a 5 foot rectangular suppressed weir is found to be 510 gal/min. What would be the height of water running over the weir?

- A) 0.5 in B) 1.7 in C) 2.0 in D) 3.2 in

The Answer is C

Solution: Use weir formulas to solve

Step 1: Identify the type of weir and units used

This is a rectangular suppressed weir with USCS units

Step 2: Use Appropriate to determine equation and coefficient to be used

$$C = 3.33 \quad \text{and} \quad Q = CLH^{3/2}$$

Step 3: Define all variables

$$L = 5.0 \text{ ft}$$

$$Q = 510 \frac{\text{gal}}{\text{min}} * 0.134 \frac{\text{ft}^3}{\text{gal}} * \frac{1 \text{ min}}{60 \text{ s}} = 1.14 \frac{\text{ft}^3}{\text{s}}$$

Step 4: Solve for height

$$Q = CLH^{3/2} \rightarrow H = \left(\frac{Q}{CL} \right)^{2/3} = \left(\frac{1.17}{3.33 * 5} \right)^{2/3} = 0.17 \text{ ft} * 12 \frac{\text{in}}{\text{ft}} = 2 \text{ in}$$

52) The discharge from a weir is $5.4 \text{ m}^3/\text{s}$ with a maximum allowable water depth of 1 m. What would be the weir length for a rectangular contracted weir?

- A) 1.8 m B) 2.4 m C) 2.8 m D) 3.1 m

The Answer is D

Solution: Use weir formulas to solve

Step 1: Identify the type of weir and units used

This is a rectangular contracted weir with metric (SI) units.

Step 2: Use Appropriate to determine equation and coefficient to be used

$$C = 1.84 \quad \text{and} \quad Q = C(L - 0.2H)H^{3/2}$$

Step 3: Define all variables

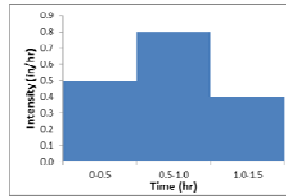
$$H = 1.0 \text{ ft} \quad Q = 5.4 \text{ m}^3/\text{s}$$

Step 4: Solve

$$Q = C(L - 0.2H)H^{3/2}$$
$$\rightarrow L = \frac{Q}{H^{3/2}C} + 0.2H = \frac{5.4}{1^{3/2} * 1.84} + 0.2 * 1 = 3.1 \text{ m}$$

53) A 6.6 acre basin has the following curve numbers for the associated areas and rainfall. What is the runoff for the basin?

Area (acres)	Curve Number
1.2	65
3.2	80
2.2	50



- A) 0.05 inches B) 0.5 inches C) 1.5 inches D) 4 inches

The Answer is A

Solution: Use the NRCS method to determine runoff (Q).

Step 1: Determine the weighted curve number

$$(CN)_{\text{weighted}} = \frac{\sum (CN)_i A_i}{A_{\text{total}}} = \frac{1.2*65 + 3.2*80 + 2.2*50}{1.2 + 3.2 + 2.2} = 67.3$$

Step 2: Determine the watershed storage

$$S = \frac{1000}{CN} - 10 = \frac{1000}{67.3} - 10 = 4.9 \text{ inches}$$

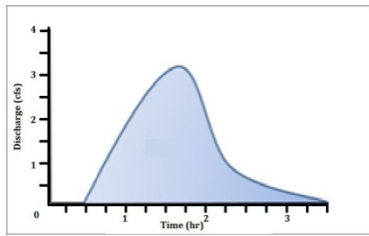
Step 3: Determine the amount of precipitation

Using the chart $P = 0.5 + 0.7 + 0.3 = 1.5$ inches

Step 4: Determine the runoff for the watershed

$$Q = \frac{(P - 0.2S)^2}{P + 0.8S} = \frac{(1.5 - 0.2*4.9)^2}{1.5 + 0.8*4.9} = \frac{0.27}{5.42} = 0.05 \text{ inches}$$

54) What is the approximate runoff for a watershed with the given hydrograph?

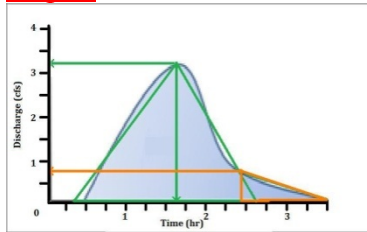


- A) 4.7 ft^3 B) $14,500 \text{ ft}^3$ C) $17,000 \text{ ft}^3$ D) $23,500 \text{ ft}^3$

The Answer is C

Solution: Runoff volume is the area under a hydrograph

Step 1: Divide the area under the hydrograph into easily calculable areas



$$\text{Triangle 1} = \frac{3.5 \text{ cfs} * (1.6 - 0.3) \text{ hr}}{2} * 3600 \frac{\text{s}}{\text{hr}} = 8190 \text{ ft}^3$$

$$\text{Triangle 2} = \frac{3.5 \text{ cfs} * (2.6 - 1.6) \text{ hr}}{2} * 3600 \frac{\text{s}}{\text{hr}} = 6300 \text{ ft}^3$$

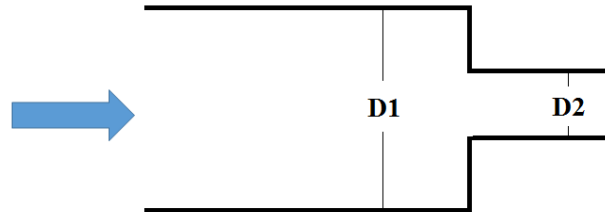
$$\text{Triangle 3} = \frac{1.5 \text{ cfs} * (3.5 - 2.5) \text{ hr}}{2} * 3600 \frac{\text{s}}{\text{hr}} = 2700 \text{ ft}^3$$

Step 2: Determine the total area

$$\text{Total} = 8190 + 6300 + 2700 = 17190 \text{ ft}^3$$

This is an approximate due to triangle approximation.

55) Air flows in a circular pipe with diameter $D_1=1\text{m}$, then suddenly, the pipe experience a contraction to $D_2=0.5\text{m}$, see the figure below. What is the head loss if air velocity is 4m/s ?



A) 3.88m

B) 2.62m

C) 0.86m

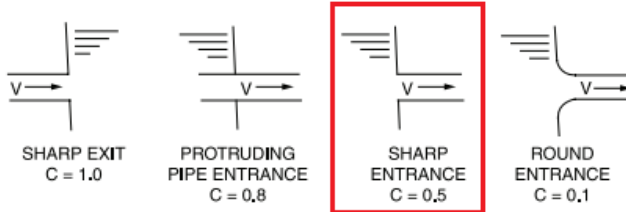
D) 0.41m

The Answers is D

Head losses occur when pipe experience sudden contractions.

$$h_f = \frac{Cv^2}{2g}$$

Coefficient C can be found in any standard Fluid mechanics book.



Substitute values, got

$$h_f = \frac{Cv^2}{2g} = \frac{0.5 \times 4^2}{2 \times 9.8} = \frac{8}{19.6} = 0.41\text{m}$$

So the head loss is 0.41m

55) Air flows in an unsmooth circular cast iron pipe with diameter 0.2m. Air flow velocity 1.4m/s, density is 1.2kg/m^3 , kinematic viscosity $15 \times 10^{-6}\text{m}^2/\text{s}$, what is the head loss Stanton friction factor?

- A) 0.028
C) 0.046

- B) 0.015
D) 0.07

The Answers is C

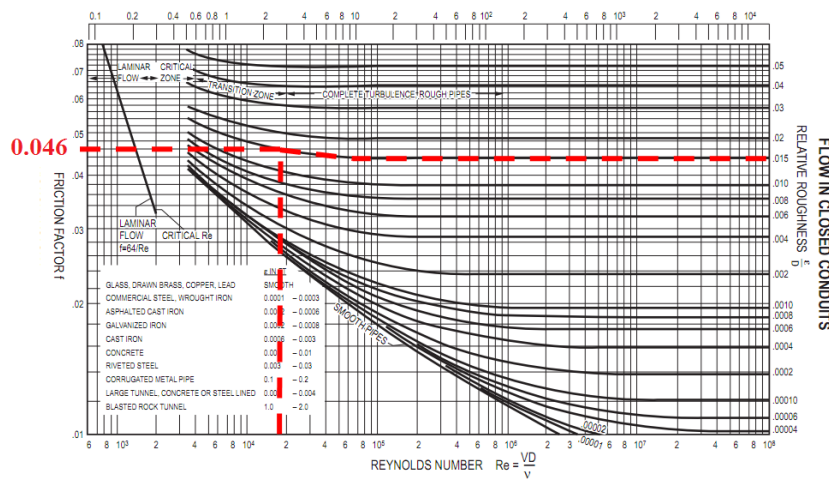
Air flows in a circular pipe, the Reynolds number

$$Re = \frac{vd}{\nu} = \frac{1.4 \times 0.2}{15 \times 10^{-6}} = 18667$$

$Re > 10000$, flow is turbulent. Cast iron, surface roughness 0.003, so the relative roughness is

$$\frac{\varepsilon}{d} = \frac{0.003}{0.2} = 0.015$$

The head loss Stanton friction factor can be determined from Moody chart in p111, $f = 0.046$



56) Air flows through a 300m horizontal pipe, the flow velocity is 0.6m/s, head loss is 0.4m. It is known that, air is in laminar flow with Re 1800. What is the diameter of the pipe?



- A) 0.82m B) 1.13m
C) 0.51m D) 0.24m

The Answers is C

Air flows in horizontal pipe, the head loss due to flow

$$h_f = \frac{fL}{d} \frac{v^2}{2g} = f \times \frac{300}{d} \times \frac{0.6^2}{2 \times 9.8} = 0.4\text{m}$$

Got Eq(1)

$$\frac{f}{d} = 0.07$$

From Moody chart for laminar flow

$$f = \frac{64}{\text{Re}} = \frac{64}{1800} = 0.036$$

Combine Eq(1) and (2), got

$$d = 0.51\text{m}$$

57) Air flows in an unsmooth cast iron circular pipe with diameter 0.2m, length 800m. Air flow velocity 1.4m/s, density is 1.22kg/m^3 , kinematic viscosity $15 \times 10^{-6}\text{m}^2/\text{s}$, what is the head loss due to flow?

- A) 21.24m
- C) 18.4m

- B) 11.22m
- D) 4.82m

The Answer is C

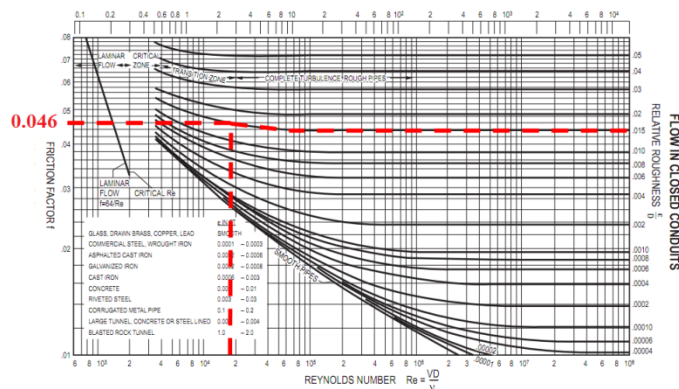
Air flows in a circular pipe, the Reynolds number

$$Re = \frac{vd}{\nu} = \frac{1.4 \times 0.2}{15 \times 10^{-6}} = 18667$$

$Re > 10000$, flow is turbulent. Surface roughness of cast iron is 0.003, so relative roughness

$$\frac{\epsilon}{d} = \frac{0.003}{0.2} = 0.015$$

The head loss Stanton friction factor can be determined from Moody chart in p111, $f = 0.046$



Head loss due to flow

$$h_f = \frac{fL}{d} \frac{v^2}{2g} = 0.046 \times \frac{800}{0.2} \times \frac{1.4^2}{2 \times 9.8} = 18.4\text{m}$$

58) Air flows in a circular pipe with diameter 0.2m. Air flow velocity 14m/s, density is 1.2kg/m^3 , kinematic viscosity $15 \times 10^{-6}\text{m}^2/\text{s}$, the head loss Stanton friction factor $f=0.042$, what is the material of the pipe?

A) Concrete

B) cast iron

C) Glass

D) Copper

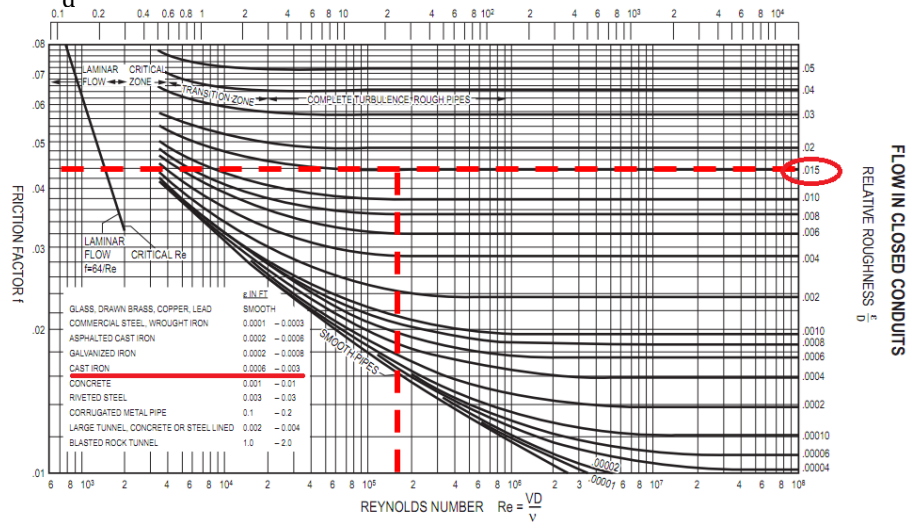
The Answers is B

Air flows in a circular pipe, the Reynolds number

$$Re = \frac{vd}{\nu} = \frac{14 \times 0.2}{15 \times 10^{-6}} = 186667$$

$Re > 10000$, flow is turbulent. From Moody chart in p111, $f=0.042$, $Re=186667$,

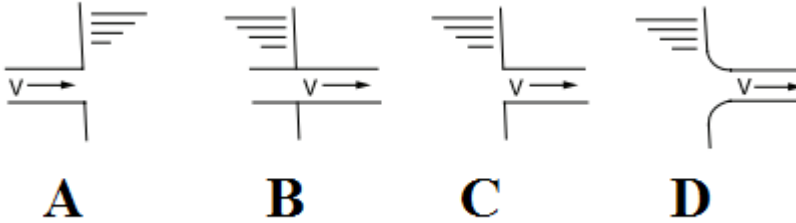
Got $\frac{\varepsilon}{d} = 0.015$



So, $\varepsilon = 0.015d = 0.015 \times 0.2 = 0.003$

From Moody chart, we know that the pipe is made of cast iron

59) Air flows in a duct, the sudden change of duct dimension would lead to a head loss. Which one of the following contractions and expansions would result in the biggest head loss with the same air velocity?



A) Layout D
C) Layout A

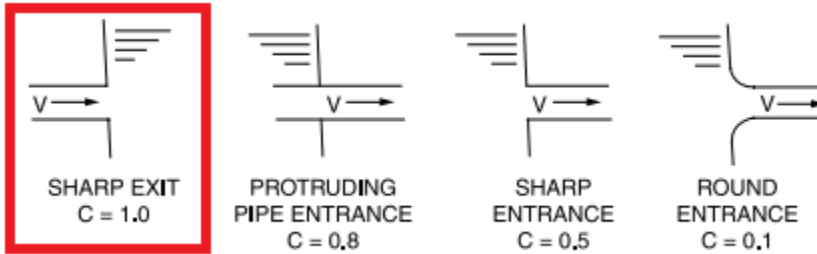
B) Layout B
D) Layout C

The Answers is C

Head losses occur when pipe experience sudden contractions and expansions.

$$h_f = \frac{Cv^2}{2g}$$

Coefficient C can be found in p102, column 1.



Layout A has the biggest head loss with the same velocity.

60) A 0.305m diameter pipe flows full and has an allowable head loss of 4.5 m for a length of 250 m. Assuming the $C = 130$, what is the velocity using the Hazen-Williams equation.

- A. 2.8m/s
- B. 4.4m/s
- C. 2.5m/s
- D. 1.4m/s

The Answers is A

1. $V = k_1 C R_H^{0.63} S^{0.54}$
2. $S = h_f / L = 4.5\text{m} / 200\text{m} = 0.0225$
3. $K = 0.849$ for SI units
4. Hydraulic Radius for a full pipe $R = A/P = \pi(D/2)^2 / (\pi D) = D/4 = 0.305\text{m} / 4 = 0.076$
5. $V = 0.849 * 130 * (0.076\text{m})^{0.63} 0.0225^{0.54}$
6. $V = 2.8\text{m/s}$

61) Determine the discharge in a full-flowing pipe with a diameter of 18" and a slope of 0.003. Use the Hazen Williams equation assuming the $C=120$ for the pipe.

- A) 6.5 cfs
- B) 14.0 cfs
- C) 4.2 cfs
- D) 9.0 cfs

The Answers is A

1. $Q = V \cdot A$
2. $Q = (k_1 C R_H^{0.63} S^{0.54}) \cdot (\pi(D/2)^2)$
3. $K = 1.318$ for USCS units
4. Convert diameter into feet: $18\text{in} \cdot 1\text{ft}/12\text{in} = 1.5\text{ft}$
5. Hydraulic Radius for a full pipe $R = A/P = \pi(D/2)^2 / (\pi D) = D/4$
6. Substituting $Q = 1.318 \cdot C \cdot (D/4)^{0.63} S^{0.54} \cdot (\pi \cdot D^2/4)$ simplifies the equation to $Q = 0.432 C D^{2.63} S^{0.54}$ for USCS units
7. $Q = 0.432 \cdot 120 \cdot (1.5\text{ft})^{2.63} (0.003)^{0.54} = 6.5 \text{ cfs}$

62) Two pipelines carry water from a common starting point to a common end point. The two pipes have the same friction factor f and diameter, but pipe 1 is twice as long as pipe 2. What fraction of the discharge between the start and end points flows through pipe 1?

(A) 34%

(B) 41%

(C) 50%

(D) 67%

The Answers is B

Step 1: For 2 pipes in parallel, the head loss in each pipe is equal.

Step 2: Using the Darcy Weisbach equation, equating the head losses in pipes 1 and 2 gives $f_1 (L_1/D_1) V_1^2/2g = f_2 (L_2/D_2) V_2^2/2g$

Step 3: Given that $f_1 = f_2$, $D_1 = D_2$, we get that $V_1^2/V_2^2 = (L_2/L_1)=0.5$, or $V_1/V_2 = (0.5)^{1/2} = 0.71$. Since the diameters and pipe areas are the same $Q_1/Q_2 = 0.71$. Then $Q_1/(Q_1+Q_2) = 1/(1+1/0.71)=0.41 = 41\%$

63) Two pipelines carry water from a common starting point to a common end point. The two pipes have the same friction factor f and length, but pipe 1 has a diameter that is twice as that of pipe 2. What fraction of the discharge between the start and end points flows through pipe 1?

- (A) 17% (B) 50% (C) 67% (D) 85%

The Answers is D

Step 1: For 2 pipes in parallel, the head loss in each pipe is equal.

Step 2: Using the Darcy Weisbach equation, equating the head losses in pipes 1 and 2 gives $f_1 (L_1/D_1) V_1^2/2g = f_2 (L_2/D_2) V_2^2/2g$

Step 3: Given that $f_1 = f_2$, $L_1 = L_2$, we get that $V_1^2/V_2^2 = (D_1/D_2)=2$, or $V_1/V_2 = (2)^{1/2} = 1.41$. The ratio of the cross sectional areas $A_1/A_2 = (D_1/D_2)^2 = 4$, and $Q_1/Q_2 = (V_1/V_2)(A_1/A_2) = 1.41 * 4 = 5.65$ since the diameters and areas are the same $Q_1/Q_2 = 0.71$. Then $Q_1/(Q_1+Q_2) = 1/(1+1/5.65) = 0.85 = 85\%$

64) Two pipelines carry water from a common starting point to a common end point. The two pipes have the same length and diameter, but pipe 1 is concrete with friction factor $f=0.027$, and pipe 2 is steel with $f=0.018$. What fraction of the discharge between the start and end points flows through pipe 1?

- (A) 33% (B) 45% (C) 50% (D) 78%

The Answers is B

Step 1: For 2 pipes in parallel, the head loss in each pipe is equal.

Step 2: Using the Darcy Weisbach equation, equating the head losses in pipes 1 and 2 gives $f_1 (L_1/D_1) V_1^2/2g = f_2 (L_2/D_2) V_2^2/2g$

Step 3: Given that $f_1 = f_2$, $L_1 = L_2$, we get that $V_1^2/V_2^2 = (f_2/f_1)=0.018/0.027=0.67$, or $V_1/V_2=(0.67)^{1/2} = 0.82$. The ratio of the cross sectional areas $A_1/A_2=1$, and $Q_1/Q_2=(V_1/V_2) = 0.82$. Then $Q_1/(Q_1+Q_2)= 1/(1+1/0.82)=0.45 = 45\%$

65) Two pipelines are to carry water from a common starting point to a common end point. The two pipes have the friction factor f , but pipe 1 is 30% longer than pipe 2. If it is desired to have each pipe carry the same discharge Q , what is the required ratio of diameters of pipe 1 to pipe 2?

- (A) 1.068 (B) 1.121 (C) 1.232 (D) 1.303

The Answer is A

Step 1: For 2 pipes in parallel, the head loss in each pipe is equal.

Step 2: Using the Darcy Weisbach equation, equating the head losses in pipes 1 and 2 gives $f_1 (L_1/D_1) V_1^2/2g = f_2 (L_2/D_2) V_2^2/2g$

Step 3: Given that $f_1 = f_2$, $L_1 = 1.3L_2$, we get that $V_1^2/V_2^2 = (L_2/L_1) = 1/1.3 = 0.77$. $V_1/V_2 = (0.77)^{1/2} = 0.877$. $Q_1/Q_2 = 1 = (V_1/V_2)(A_1/A_2) = 0.877(D_1^2/D_2^2)$. So $(D_1/D_2) = (1/0.877)^{1/2} = 1.068$

66) A pipeline is 300 meters long, diameter 25 m, and friction factor $f=0.022$. If the head loss h_f in this pipeline is expressed as $h_f=KQ^2$, what is the value of the head loss coefficient K ?

- (A) $43.1 \text{ s}^2/\text{m}^5$ (B) $127 \text{ s}^2/\text{m}^5$ (C) $338 \text{ s}^2/\text{m}^5$ (D) $558 \text{ s}^2/\text{m}^5$

The Answer is D

Step 1: The Darcy Weisbach equation is $h_f = f(L/D) V^2/2g$

Step 2: Substituting $V=Q/A = Q/(\pi D^2/4)$ gives $h_f = [f(L/D^5) 1/2g/(\pi/4)^2] Q^2$. So $K = [f(L/D^5) 1/2g/(\pi/4)^2] = 0.022(300/0.25^5) 1/(2)(9.81)/(\pi/4)^2 = 558 \text{ s}^2/\text{m}^5$

67) A pipeline connecting two reservoirs consists of a pipe entrance (minor loss coefficient $C=0.5$), 65 feet of 12-inch diameter pipe (friction factor $f=0.020$), and exit to the downstream reservoir (minor loss coefficient $C=1.0$). If the total head loss for this pipe is to be expressed as KQ^2 , what is the value of the head loss coefficient K ?

- (A) $0.043 \text{ s}^2/\text{ft}^5$ (B) $0.071 \text{ s}^2/\text{ft}^5$ (C) $0.129 \text{ s}^2/\text{ft}^5$ (D) $32.2 \text{ s}^2/\text{ft}^5$

The Answer is B

Step 1: The pipe area $A = (\pi D^2/4) = (\pi (1.0)^2/4) = 0.785 \text{ ft}^2$

Step 2: The total head loss for this pipeline is the sum of the head loss of the 3 components, or $h_f = C_{\text{entrance}} V^2/2g + f(L/D) V^2/2g + C_{\text{exit}}$

$$= (C_{\text{entrance}} + f(L/D) + C_{\text{exit}}) V^2/2g = [(C_{\text{entrance}} + f(L/D) + C_{\text{exit}})/2g/A^2] Q^2$$

$$K = (C_{\text{entrance}} + f(L/D) + C_{\text{exit}})/2g/A^2 = (0.5 + 0.02(65/1) + 1)/2(32.2)/(0.785)^2 = 0.071 \text{ s}^2/\text{ft}^5$$

68) Using the Hazen Williams equation, determine the head loss in a full flowing 20 cm concrete pipe with a C of 130 that has a velocity of 0.75m/s and is 45 m long.

- A)0.24
- B)0.05
- C)0.29
- D)0.144

The Answers is D

1. $V = k_1 C R_H^{0.63} S^{0.54}$
2. $K = 0.849$ for SI units
3. Hydraulic Radius for a full pipe $R = A/P = \pi(D/2)^2 / (\pi D) = D/4 = 0.2\text{m}/4 = 0.05\text{m}$
4. Solve the Hazen-Williams for slope
5. $S = (V / (k C R_H^{0.63}))^{1/0.54} = ((0.75\text{m/s}) / (0.849 * 130 * 0.05^{0.63}))^{1.85} = 0.0032$
6. $S = \text{head loss/pipe length}$
7. $H_L = 0.0032(45\text{m}) = 0.144$

69) Determine the head loss from a full flowing pipe that is 1000 m long and diameter of 20". The flow in the pipe is $0.2 \text{ m}^3/\text{s}$. Use the Hazen Williams equation, assuming $C = 130$.

- A) 1.3m
- B) 0.8m
- C) 1.8m
- D) 3.0m

The Answers is A

1. $Q = V \cdot A$
2. $Q = (k_1 C R_H^{0.63} S^{0.54}) \cdot (\pi(D/2)^2)$
3. $K = 0.849$ for SI units
4. Convert diameter to SI units: $20 \text{ in} \cdot 2.54 \text{ cm/in} \cdot 0.01 \text{ m/cm} = .508 \text{ m}$
5. Hydraulic Radius for a full pipe $R = A/P = \pi(D/2)^2 / (\pi D) = D/4$
6. Substituting $Q = 0.849 \cdot C \cdot (D/4)^{0.63} S^{0.54} \cdot (\pi \cdot D^2/4)$ simplifies the equation to $Q = 0.278 C D^{2.63} S^{0.54}$ for SI units
7. $0.2 \text{ m}^3/\text{s} = 0.278 \cdot 130 \cdot (0.508 \text{ m})^{2.63} (h_f/1000 \text{ m})^{0.54}$
8. $(h_f/1000 \text{ m})^{0.54} = 0.0329$
9. $h_f = 1.8 \text{ m}$

70) Water is flowing through 35 feet of 6" pipe at a rate of 300 gpm. The water also travels through 2 gate valves ($K = 0.11$) and 2 elbows ($K = 0.29$). The losses through the pipe are known to be 0.7 ft per 100 feet of pipe. What are the total friction losses in the system?

- A) 0.18 ft
- B) 1.1 ft
- C) 0.32 ft
- D) 0.25 ft

The Answers is C

Step 1: Determine friction losses in pipe

$$\text{Friction losses in pipe} = 0.7\text{ft}/100\text{ft pipe} * 35\text{ft} = 0.245\text{ft}$$

Step 2: Determine velocity

$$Q = 300 \text{ gpm} * 0.134\text{ft}^3/\text{gallon} * 1\text{min}/60\text{s} = 0.67\text{cfs}$$

$$\text{Area pipe} = \pi * (3/12')^2 = 0.196\text{ft}^2$$

$$= V * A, V = 0.67\text{cfs}/0.196\text{ft}^2 = 3.42\text{ft/s}$$

Step 3: Determine velocity head

$$\text{Velocity head} = V^2/(2g) = (3.42\text{ft/s})^2/(2 * 32.2\text{ft/s}^2) = 0.18\text{ft}$$

Step 4: Determine friction loss from velocity head

$$\text{Friction loss in 2 valves} = 0.18 * 0.11 * 2 = 0.04$$

$$\text{Friction loss in 2 elbows} = 0.18 * 0.29 * 2 = 0.1$$

Step 5: Sum friction loss

$$\text{Total friction losses} = 0.18\text{ft} + 0.04\text{ft} + 0.1\text{ft} = 0.32\text{ft}$$

71) Water is flowing through 55 feet of 8" pipe at a rate of 250gpm. The water also travels through 3 check valves ($K = 0.3$) and 2 gate valves ($K=0.15$). The losses through the pipe are known to be 0.65ft per 100 feet of pipe. What are the total friction losses in the system?

- A) 0.41ft
- B) 0.8ft
- C) 0.32ft
- D) 0.21ft

The Answers is A

Step 1: Determine friction losses in pipe

$$\text{Friction losses in pipe} = 0.65\text{ft}/100\text{ft pipe} * 55\text{ft} = 0.358\text{ft}$$

Step 2: Determine velocity

$$Q = 250 \text{ gpm} * 0.134\text{ft}^3/\text{gallon} * 1\text{min}/60\text{s} = 0.56\text{cfs}$$

$$\text{Area pipe} = \pi * (4/12')^2 = 0.35\text{ft}^2$$

$$Q = V * A, V = 0.56\text{cfs}/0.35\text{ft}^2 = 1.6\text{ft/s}$$

Step 3: Determine velocity head

$$\text{Velocity head} = V^2/(2g) = (1.6\text{ft/s})^2/(2*32.2\text{ft/s}^2) = 0.04\text{ft}$$

Step 4: Determine friction loss from velocity head

$$\text{Friction loss in 3 check valves} = 0.04 * 0.3 * 3 = 0.036$$

$$\text{Friction loss in 2 gate valves} = 0.04 * 0.15 * 2 = 0.012$$

Step 5: Sum friction loss

$$\text{Total friction losses} = 0.358\text{ft} + 0.036\text{ft} + 0.012\text{ft} = 0.41\text{ft}$$

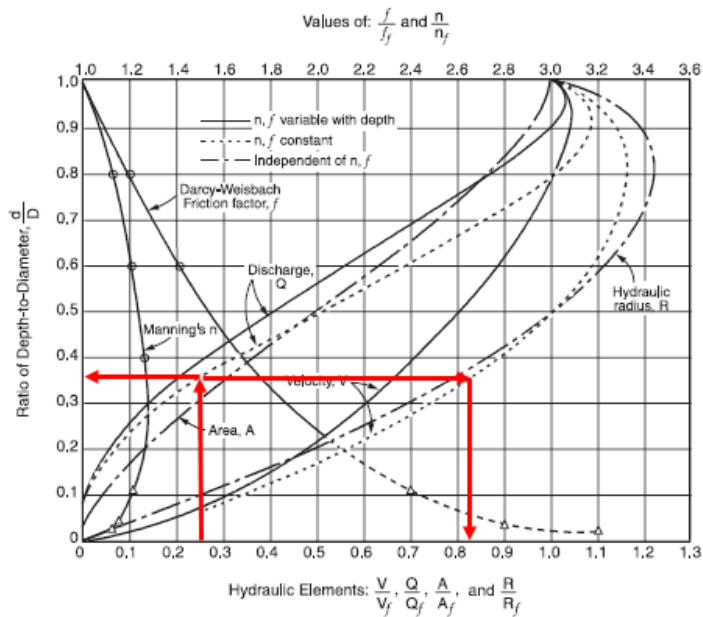
72) A sanitary sewer pipe is 24 inch diameter, Manning $n=0.026$ and slope = 0.004. The minimum discharge in this sewer is 1.8 cfs. The minimum velocity to prevent deposition of solids in a sanitary sewer pipe is 2.0 ft/sec. Is the minimum velocity criteria met in this pipe at the minimum discharge?

- (A) Yes (B) No (C) Cannot tell from given information

The Answers is B

Step 1: When the pipe is full: $A_f = \pi D^2/4 = 3.14(24/12)^2/4 = 3.14 \text{ ft}^2$; $R_f = D/4 = (24/12)/4 = 0.5 \text{ ft}$; $Q_f = (1.49/n) A_f R_f^{2/3} S^{1/2} = (1.49/0.026) (3.14) (0.5)^{2/3} (0.004)^{1/2} = 7.17 \text{ cfs}$, and $V_f = Q_f/A_f = (7.17)/(3.14) = 2.28 \text{ ft/sec}$

Step 2: From diagram (see below), for $Q/Q_f = (1.8)/(4.81) = 0.25$, $d/D = 0.36$. Then for $d/D = 0.36$, $V/V_f = 0.82$, so $V = 0.82(2.28) = 1.87 \text{ ft/sec}$. Does not meet criteria.



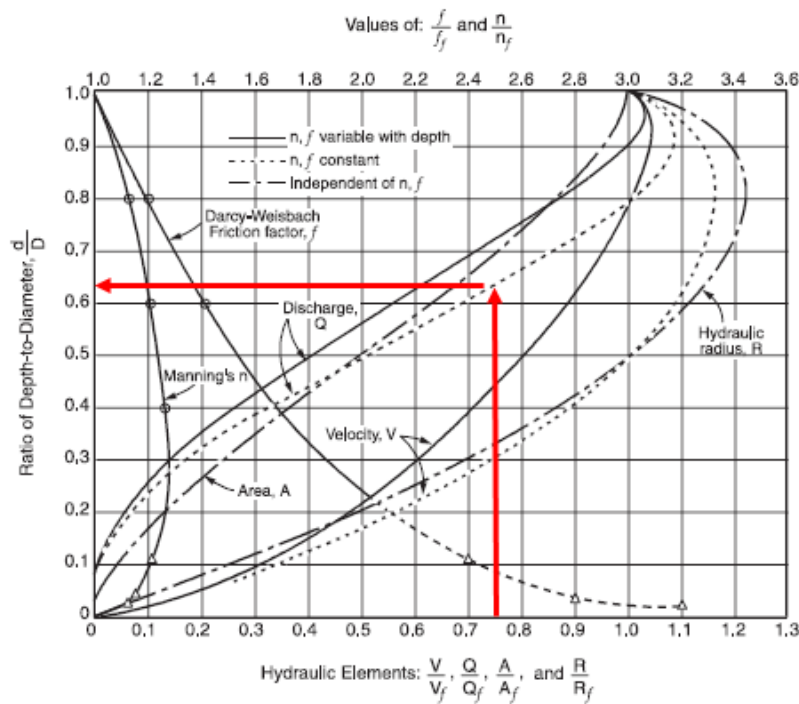
73) A sanitary sewer pipe is 40 cm diameter, Manning $n=0.028$ and slope = 0.003. The maximum discharge in this sewer is $0.04 \text{ m}^3/\text{sec}$. To prevent surcharge, the depth at the maximum discharge should be less than 70% of the diameter. Is this criteria met in this pipe at the maximum discharge?

- (A) Yes (B) No (C) Cannot tell from given information

The Answer is A

Step 1: When the pipe is full: $A_f = \pi D^2/4 = 3.14(40/100)^2/4 = 0.126 \text{ m}^2$; $R_f = D/4 = (40/100)/4 = 0.1 \text{ m}$; $Q_f = (1/n) A_f R_f^{2/3} S^{1/2} = (1/0.028) (0.126) (0.1)^{2/3} (0.003)^{1/2} = 0.053 \text{ m}^3/\text{sec}$

Step 2: From diagram (see below), for $Q/Q_f = (0.04)/(0.053) = 0.75$, $d/D = 0.63$, which is less than 0.7, so the criteria is met.



74) The outlet of a culvert is considered a free outlet under which of the following conditions?

- (A) tailwater elevation is 1 pipe diameter below the pipe invert
- (B) tailwater elevation is below the pipe invert
- (C) tailwater elevation above the invert is less than the critical depth
- (D) tailwater elevation above the invert is less than the uniform depth

The Answer is C

Step 1: If the tailwater elevation above the pipe invert is less than the critical depth, the outlet is considered free. If the tailwater elevation is lowered below this level, it will have no effect on the flow in the culvert.

75) If a culvert is described as having a steep slope, which of the following conditions is true?

- (A) critical depth is greater than the uniform depth
- (B) critical depth is less than the uniform depth
- (C) critical depth is less than the pipe diameter
- (D) uniform depth is less than the pipe diameter

The Answers is A

Step 1: A steep slope is where the critical depth is greater than the uniform flow depth; if the opposite is the case, the slope is mild.

76) At the design discharge, an existing single circular culvert has the following characteristics: headwater elevation = 5.5 ft above invert, culvert diameter = 2.5 ft, critical depth = 1.9 ft, uniform depth = 1.7 ft, and the outlet is free. Which of the following describes the culvert flow under these conditions?

- (A) inlet control, inlet is not submerged, critical flow at the inlet
- (B) inlet control, inlet is submerged with orifice flow control
- (C) outlet control, inlet is submerged
- (D) outlet control, inlet is not submerged

The Answer is B

Step 1: Since the critical depth is greater than the uniform depth, the culvert slope is steep. Therefore the culvert is under inlet control.

Step 2: Since the headwater elevation above the invert is more than 1.2 times the pipe diameter, the inlet will be submerged, and an orifice flow condition exists at the inlet.

77) A rectangular culvert cross section is 2.5 ft high and 5.5 ft wide. At the design discharge of 135 cfs, the culvert is subject to inlet control with a submerged inlet. What is the headwater elevation above the culvert invert?

- (A) 1.6 ft
- (B) 2.8 ft
- (C) 3.9 ft
- (D) 5.4 ft

The Answer is D

Step 1: For a culvert with inlet control and a submerged inlet, the following orifice flow equation applies: $Q = C_D A \sqrt{2g(HW - b/2)}$, where Q =discharge, C_D =orifice discharge coefficient=0.6, A = culvert cross sectional area, g =acceleration of gravity, HW =headwater elevation above invert, and b =culvert height or diameter.

Step 2: Here $Q=135$ cfs, $A=(2.5)(5.5)=13.75$ ft², $g=32.2$ ft/sec², and $b=2.5$ ft. Solve for $HW=5.4$ ft.

78) A rectangular culvert is 1.5 meters high and 2.5 meters wide with a design discharge of $17.5 \text{ m}^3/\text{sec}$. The culvert has slope of 1.5% and Manning $n=0.024$. What is the critical depth under these conditions?

- (A) 1.13 m
- (B) 1.29 m
- (C) 1.71 m
- (D) not defined for these conditions

The Answer is D

Step 1: For a rectangular channel, the critical depth $y_c = \left(\frac{Q^2}{gb^2} \right)^{1/3}$, where Q is the culvert discharge, g= acceleration of gravity=9.81 m/sec², and b=culvert width.

Step 2: Here Q=17.5 m³/sec and b=2.5 meters, yielding $y_c=1.71$ meters. However, this depth exceeds the height of the culvert (1.5 m), so the critical depth does not exist for these conditions.

79) At a free outlet from a 40-cm diameter circular culvert, the depth of flow is 32 cm. The culvert slope is 0.9% and Manning $n=0.028$. What is the discharge in the pipe?

- (A) $0.20 \text{ m}^3/\text{sec}$
- (B) $0.47 \text{ m}^3/\text{sec}$
- (C) $0.98 \text{ m}^3/\text{sec}$
- (D) $1.18 \text{ m}^3/\text{sec}$

The Answers is A

Step 1: At a free outlet, the flow is critical, and the discharge $Q = \sqrt{gA^3 / T}$, where g =acceleration of gravity, A = cross sectional area, and T =top width.

Step 2: For a circular pipe, the following equations can be used to compute the top width T and cross sectional area A , given the pipe diameter D and depth y : Angle

$\theta = \cos^{-1}\left(1 - 2\frac{y}{D}\right)$, must be in radians ($0 \leq \theta \leq \pi$); cross sectional area

$A = \frac{D^2}{4}(\theta - \sin\theta\cos\theta)$; and top width $T = D\sin\theta$.

Step 3: Here $y=32$ cm, $D=40$ cm, so $\theta=2.21$ radians, $A=0.108$ m², $T=0.32$ m, and discharge $Q=0.20$ m³/sec.

80) A 42-inch diameter culvert with slope 1.5% and Manning $n=0.025$ is subject to outlet control, and has a free outlet. What is the maximum discharge that this culvert can convey without surcharging?

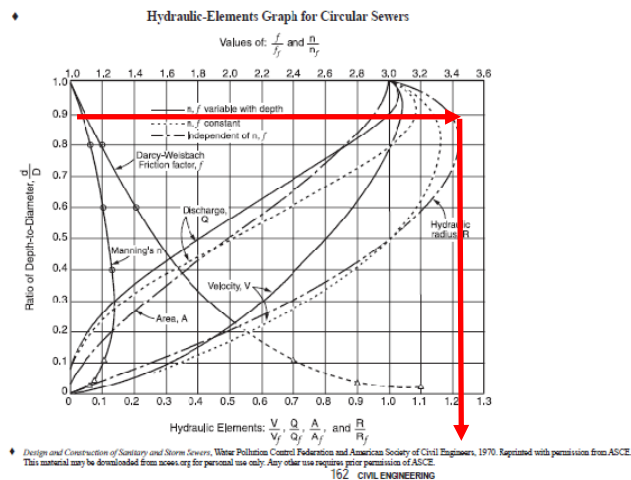
- (A) 33.9 cfs
- (B) 47.2 cfs
- (C) 64.2 cfs
- (D) 69.4 cfs

The Answers is D

Step 1: If the culvert is subject to outlet control and has a free outlet, then critical depth will occur at the outlet, and uniform flow will occur within the culvert itself.

Step 2: For uniform flow, the maximum discharge occurs at a depth of 92% of the diameter; at this point $Q=1.08Q_f$, where Q_f is the discharge when flowing full (see figure below)

Step 3: Full pipe calculation: $A = \frac{\pi D^2}{4} = 3.24(42/12)^2/4 = 9.62 \text{ ft}^2$. $R_h = D/4 = 0.875 \text{ ft}$.
 $Q_f = (1.49/n) A R_h^{2/3} S^{1/2} = 64.2 \text{ cfs}$. $Q = 1.08 Q_f = 69.4 \text{ cfs}$



81) A double box culvert (rectangular cross section) is submerged at both ends at a discharge of 325 cfs. Each box is 5.5 ft wide and 2.0 ft high, 75 ft long, with slope of 1.3% and Manning $n=0.029$, entrance and exit loss coefficients of 0.3 and 1.0. If the upstream water surface elevation is 234.5 ft above datum, what is the downstream elevation?

- (A) 209.9 ft
- (B) 214.1 ft
- (C) 217.3 ft
- (D) 224.6 ft

The Answers is C

Step 1: The discharge in each culvert = $325/2 = 162.5$ cfs. Area $A = (2)5.5 = 11$ ft², wetted perimeter $P = 2*2 + 2*7.5 = 19$ ft, hydraulic radius $R_h = A/P = 0.58$ ft.

Step 2: Rearranging the Manning equation, $S^{1/2} = Q n / (1.49 A R_h^{2/3}) = 0.41$, $S = 0.171$, so culvert head loss = $S L = (0.171)(75 \text{ ft}) = 12.8$ ft.

Step 3: Velocity $V = Q/A = (162.5)/(11) = 14.8$ ft/sec, $V^2/2g = (14.8)^2/2/32.2 = 3.39$ ft. Entrance and exit head loss = $(0.3 + 1.0) V^2/2g = 4.4$ ft. Total head loss = $12.8 + 4.4 = 17.2$ ft. Downstream elevation = upstream elevation – head loss = $234.5 - 17.2 = 217.3$ ft

82) A culvert consists of a single 36-inch diameter circular pipe has an inlet that is not submerged, mild slope, and a free (unsubmerged) outlet. The pipe slope is 0.85% and Manning $n=0.031$. If the depth at the outlet is 17 inches, what is the culvert discharge?

- (A) 13.1 cfs
- (B) 19.5 cfs
- (C) 22.6 cfs
- (D) 27.8 cfs

The Answers is B

Step 1: Under the conditions of a unsubmerged inlet, mild slope, and free outlet, the flow is under outlet control, and critical depth will occur at the outlet, where the depth is 17 inches. At the outlet, critical flow means that the discharge $Q = \sqrt{gA^3 / T}$, where g =acceleration of gravity, A = cross sectional area, and T =top width.

Step 2: For a circular pipe, the following equations can be used to compute the top width T and cross sectional area A , given the pipe diameter D and depth y : Angle

$\theta = \cos^{-1}\left(1 - 2\frac{y}{D}\right)$, must be in radians ($0 \leq \theta \leq \pi$); cross sectional area

$A = \frac{D^2}{4}(\theta - \sin\theta\cos\theta)$; and top width $T = D\sin\theta$.

Step 3: Here $y=17/12=1.42$ ft, $D=36/12= 3.0$ ft, so $\theta=1.515$ radians, $A=3.28$ ft², $T=3.0$ ft, and discharge $Q=19.5$ ft³/sec.

83) A twin-box culvert (each box is 90 cm high and 300 cm wide) is subject to inlet control. Both the upstream and downstream ends are unsubmerged, and the culvert slope is steep. The depths at the upstream and downstream ends are 65 and 52 cm, respectively. The slope and Manning n of each box is $S=1.1\%$ and $n=0.027$. What is the discharge under these conditions?

- (A) $1.7 \text{ m}^3/\text{sec}$
- (B) $2.7 \text{ m}^3/\text{sec}$
- (C) $3.8 \text{ m}^3/\text{sec}$
- (D) $4.9 \text{ m}^3/\text{sec}$

The Answer is D

Step 1: For the conditions of inlet control, unsubmerged at both ends, with steep slope, the control condition at the upstream end is critical depth. For each box the discharge $Q = \sqrt{gb^2y^3}$, where g = acceleration of gravity, b = box width, and y = critical depth.

Step 2: Here $Q = [9.81(300/100)^2(65/100)^3]^{1/2} = 4.9 \text{ m}^3/\text{sec}$

84) A rectangular channel has a depth of 3.5 ft and width of 15 ft. If the discharge is 34 cfs, what is the flow condition?

- (A) subcritical
- (B) critical
- (C) supercritical

The Answer is A

Step 1: Critical flow is evaluated by computing the Froude number. For a rectangular channel, the Froude No. $Fr = V / \sqrt{gy}$, where V is the average velocity, g = acceleration of gravity, and y = channel depth.

Step 2: Here cross sectional area $A = yB = (3.5)(15) = 52.5 \text{ ft}^2$. Velocity $V = Q/A = (34)/(52.5) = 0.65 \text{ ft/sec}$. Then $Fr = (0.65)/[(32.2)(3.5)]^{1/2} = 0.061$.

Step 3: Since $Fr < 1$, flow is subcritical.

85) A trapezoidal channel has a bottom width of 3 meters and 45-degree side slopes. If the depth of flow is 1.5 meters and the discharge = $4.5 \text{ m}^3/\text{sec}$, what is the state of flow?

- (A) subcritical
- (B) critical
- (C) supercritical

The Answer is A

Step 1: The state of flow is determined by the Froude number; for a non-rectangular channel the Froude Number $Fr = V / \sqrt{g y_h}$ where V = average velocity, g =acceleration of gravity, $y_h=A/T$, A =cross sectional area, and T =top width.

Step 2: Here $A=(3)(1.5)+2*0.5(1.5)(1.5)=6.75 \text{ m}^2$, $T=3+2(1.5)=6 \text{ m}$, $y_h=(6.75)/(6)=1.13 \text{ m}$. $V=Q/A=(4.5)/(6.75)=0.67 \text{ m/sec}$. $Fr=(0.67)/[(9.81)(1.13)]^{1/2}=0.20$. Since $Fr<1$, flow is subcritical.

86) Water flows at a discharge of 12 cfs in a rectangular channel that is 5 feet wide. What is the critical depth for these conditions?

- (A) 0.34 ft
- (B) 0.56 ft
- (C) 0.73 ft
- (D) 0.98 ft

The Answer is B

Step 1: In a rectangular channel, the critical depth $y_c = \left(\frac{Q^2}{gB^2} \right)^{1/3}$, where Q=discharge, g=acceleration of gravity, and B=channel width.

Step 2: Here Q=12 cfs, g=32.2 ft/sec², and B=5 feet, so $y_c=0.56$ ft

87) Water flows in a rectangular open channel of width 2.5 meters at a depth of 0.45 m and discharge of $2.3 \text{ m}^3/\text{sec}$. Under these conditions, which of the following statements is true regarding the alternate depth?

- (A) less than 0.45 m
- (B) equal to 0.45 m
- (C) greater than 0.45 m
- (D) cannot say from given information

The Answer is A

Step 1: To answer this, evaluate the Froude Number. For a rectangular channel, the Froude No. $Fr = V / \sqrt{gy}$, where V is the average velocity, g = acceleration of gravity, and y = channel depth.

Step 2: Here cross sectional area $A = yB = (0.45)(2.5) = 1.13 \text{ m}^2$. Velocity $V = Q/A = (2.3)/(1.13) = 2.04 \text{ m/sec}$. Then $Fr = (2.04)/[(9.81)(0.45)]^{1/2} = 0.97$.

Step 3: The Froude number Fr is less than 1. SO at the alternate depth, Fr is greater than 1 and the depth will be less than the given subcritical depth of 0.45 m

88) A circular sewer pipe is 24 inches in diameter and contains a discharge of 2.5 cfs. It has been suggested that the critical depth for these conditions is 12 inches (half full). Which of the following statements is true?

- (A) critical depth < 12 inches
- (B) critical depth $= 12$ inches
- (C) critical depth > 12 inches
- (D) cannot say from given information

The Answer is A

Step 1: This question can be answered by evaluating the Froude number; for a non-rectangular channel the Froude Number $Fr = V / \sqrt{g y_h}$ where V = average velocity, g =acceleration of gravity, $y_h=A/T$, A =cross sectional area, and T =top width.

Step 2: Since the circular pipe is half full, $A=0.5 \pi D^2/4 = 0.5(3.14)(24/12)^2/4 = 1.57 \text{ ft}^2$, $T=D=2 \text{ ft}$, $y_h=(1.57 \text{ ft}^2)/(2 \text{ ft})=0.785 \text{ ft}$. Average velocity $V=Q/A=(2.5 \text{ cfs})/(1.57 \text{ ft}^2)=1.59 \text{ ft/sec}$. $Fr=(1.59)/[(32.2)(0.785)]^{1/2}=0.317$. At a depth of 12 inches, the flow is subcritical ($Fr<1$), so at critical flow the depth must be less than 12 inches.

89) Water flows out of a reservoir and enters a short channel of rectangular cross section that is 2.2 meters wide, has zero slope, and discharges $2.4 \text{ m}^3/\text{sec}$. At the downstream end of this channel, there is a vertical drop and water spills out of the channel. What is the depth of flow at the downstream end of this channel?

- (A) 0.28 m
- (B) 0.39 m
- (C) 0.50 m
- (D) 0.67 m

The Answer is C

Step 1: Since the channel is horizontal, there is no uniform flow depth; at the downstream end of the channel, where water spills out, critical flow will occur.

Step 2: For a rectangular channel, the critical depth $y_c = \left(\frac{Q^2}{gB^2} \right)^{1/3}$, where Q=discharge, g=acceleration of gravity, and B=channel width.

Step 3: Here $Q=2.4 \text{ m}^3/\text{sec}$, $g=9.81 \text{ m/sec}^2$, and $B=2.2 \text{ m}$, so $y_c=0.50 \text{ m}$

90) Water flows in a long, uniform rectangular channel of width 2.5 feet, which ends in a free overfall. In the channel upstream of the free overfall, uniform flow occurs at a depth of 0.75 feet. The channel discharge is 13.5 cfs. What is the depth of flow at the free overfall?

- (A) 0.75 ft
- (B) 0.85 ft
- (C) 0.96 ft
- (D) 1.26 ft

The Answer is A

Step 1: The answer depends, in the region upstream where the flow is uniform, is the flow subcritical or supercritical. If the upstream flow is supercritical, then the depth at the overfall will not change, and will be equal to 1.75 ft. If the upstream flow is subcritical, the depth at the overfall will decrease to the critical depth.

Step 2: Channel area $A=yB=(0.75)(2.5)=1.88 \text{ ft}^2$. Average velocity $V=Q/A=(13.5 \text{ cfs})/(1.88)=7.18 \text{ ft/sec}$. Froude number $Fr = V / \sqrt{gy} = (7.18)/[(32.2)(0.75)]^{1/2}=1.46$. Since the flow is supercritical ($Fr>1$), the depth will remain at 0.75 as the flow approaches and spills over the free overfall.

91) At a point in an open channel flow, the depth y decreases with distance downstream x , so that $dy/dx < 0$. If it is known that the local depth y is less than the uniform flow depth, which of the following described the state of flow?

- (A) subcritical
- (B) critical
- (C) supercritical
- (D) cannot tell from given information

The Answer is A

Step 1: The differential equation describing nonuniform flow in an open channel is

$\frac{dy}{dx} = \frac{S - S_f}{1 - Fr^2}$, where y = depth, x = position along length of channel, S = channel slope, S_f = friction slope, and Fr = Froude number.

Step 2: If the local depth is less than the uniform flow depth, then the numerator ($S - S_f$) < 0 . So if $dy/dx < 0$ as given, then the denominator $(1 - Fr^2) > 0$, or $Fr^2 < 1$, or the flow is subcritical.

92) A hydraulic jump is a standing wave in an open channel, where the depth downstream of the jump (or wave) is always greater than the depth upstream. Given this, which of the following statements is true?

- (A) the flow downstream is always supercritical
- (B) the flow downstream is always uniform
- (C) the flow upstream is always supercritical
- (D) the flow upstream is always subcritical

The Answer is C

Step 1: A hydraulic jump is governed by $\frac{y_2}{y_1} = \frac{1}{2} \left(\sqrt{1 + 8Fr_1^2} - 1 \right)$, where y_1 = depth upstream of the jump, y_2 = depth downstream of the jump, and Fr_1 = is the Froude number upstream of the jump.

Step 2: If this equation is rearranged to solve for $Fr_1^2 = \frac{1}{8} \left[\left(2 \frac{y_2}{y_1} + 1 \right)^2 - 1 \right]$, so if $y_2/y_1 > 1$ then $Fr_1^2 > 1$ and the flow upstream is supercritical.

93) An open channel has a triangular shape with a 60-degree angle at the channel base. If the critical depth in this channel is 1.2 ft, what is the discharge associated with this depth?

- (A) 1.12 cfs
- (B) 1.87 cfs
- (C) 2.56 cfs
- (D) 3.65 cfs

The Answer is D

Step 1: For a triangular shape with a 60 degree angle at the base and a depth of 1.2 ft, the top width $T=2(1.2)\tan(30)=1.39$ ft. The cross sectional area then is $A=0.5(1.39)(1.2)=0.831$ ft², and the hydraulic depth $y_h=A/T=(0.831)/(1.39)=0.60$ ft

Step 2: If the flow is critical, then the Froude number $Fr=V/(g y_h)^{1/2}=1$, or $V = (g y_h)^{1/2}=(32.2*0.60)^{1/2}=4.39$ ft/sec, discharge $Q=VA=(4.39$ ft/sec)(0.831 ft²)=3.65 cfs

94) Which of the following is true regarding a unit hydrograph?

- (A) it is the hydrograph from a watershed of 1 percent slope
- (B) it is the hydrograph from a watershed of 1 unit area (e.g. 1 acre)
- (C) it is the hydrograph from 1 unit (e.g. 1 inch) of excess rainfall
- (D) it is the hydrograph from 1 unit (e.g. 1 inch) of rainfall

The Answers is C

Step 1: The unit hydrograph is a characteristic of a particular watershed. It is the the hydrograph resulting from 1 unit (typically 1 inch) of excess rainfall. Excess rainfall is the rainfall remaining after abstractions such as infiltration, detention storage, evaporation.

95) If rainfall were to fall uniformly over a watershed at a constant rate for a duration equal to 1.5 times the time of concentration of the watershed, which of the following is true regarding the resulting runoff hydrograph?

- (A) the runoff hydrograph would rise to sharp peak and then decrease
- (B) the maximum runoff would be sustained for a period of time
- (C) after a period equal to 2 times of concentration, direct runoff would end
- (D) the time interval from the start of the storm to peak runoff would be 60% of the time of concentration

The Answer is B

Step 1: If rainfall is sustained for a period equal to the time of concentration, then all parts of the watershed are contributing runoff to the outlet. If the rainfall is sustained for an additional time, the runoff would not increase further and would remain constant until the rainfall decreases.

96) Which of the following is not a basic assumption of the unit hydrograph approach?

- (A) for a storm of a given duration, the duration of runoff does not depend on rainfall depth
- (B) rainfall is spatially uniform over the entire watershed area
- (C) runoff is linearly related to rainfall
- (D) the unit hydrograph includes both direct runoff and baseflow

The Answer is D

Step 1: The unit hydrograph does not consider or include baseflow, which is the streamflow occurring after direct runoff has ceased.

97) A unit hydrograph has a peak discharge of 45 cfs, and a discharge duration of 2.5 hours. This unit hydrograph is associated with 1 inch of excess rainfall occurring over 20 minutes. Based on this unit hydrograph, if 1 inch of excess rainfall occurs over a 40 minute period, which of the following describes the resulting peak discharge?

- (A) less than 45 cfs
- (B) 45 cfs
- (C) greater than 45 cfs
- (D) cannot say from given information

The Answer is A

Step 1: The peak runoff resulting from the just the first 20 minutes of the 40-minute storm will be half of 45, or 22.5 cfs. The peak runoff from the second 20 minutes will also be 22.5 cfs, but will occur at a later time. Superimposing these two responses, the peak at any time will be less than 45 cfs.

98) During and shortly after a rainfall event, streamflow is assumed to be made up of direct runoff and base flow. At some later time, the direct runoff ends and the stream flow is entirely base flow. In hydrograph analysis, how is the time at which direct runoff ends commonly defined?

- (A) when the total stream flow returns to its pre-storm value
- (B) at the inflection point of the recession limb of the hydrograph
- (C) a time period equal to the time of concentration after the start of the storm
- (D) a time period equal to the time of concentration after the hydrograph peak

The Answer is B

Step 1: The inflection point of the falling limb, or recession limb of the hydrograph is commonly assumed to be the point or time at which direct runoff ends.

99) What is the relationship between the time of concentration of a watershed and duration of runoff for a unit hydrograph for the same watershed?

- (A) (time of concentration) < (unit hydrograph duration)
- (B) (time of concentration) = (unit hydrograph duration)
- (C) (time of concentration) > (unit hydrograph duration)

The Answer is A

Step 1: The time of concentration is the time it takes for all parts of a watershed to contribute runoff to a downstream point. The unit hydrograph contains both a rising and falling limb, and thus considers the time for runoff to increase in response to rainfall, and then decrease to zero after rainfall has ceased. So, the duration of runoff in a unit hydrographs is longer.

100) Using the Soil Conservation Service (SCS) triangular unit hydrograph, if the time of concentration of a watershed is 3 hours, what should be the rainfall duration which generates the unit hydrograph?

- (A) 10 minutes
- (B) 24 minutes
- (C) 45 minutes
- (D) 90 minutes

The Answer is B

Step 1: For the SCS triangular unit hydrograph, T_C is the time of concentration and ΔD is the duration of the unit rainfall that generates the unit hydrograph. These two quantities are related by $\Delta D = 0.133 T_C$.

Step 2: Here $T_C = 3$ hours, so $\Delta D = 0.133 T_C = 0.133(3) = 0.40$ hours = 24 minutes

101) A reservoir is formed by a dam, and has a fixed spillway that controls the outflow from the dam. Over the course of a rainfall event, the hydrographs of the inflow to and spill from the dam are measured. Which of the following statements is true when comparing the spill hydrograph to the inflow hydrograph?

- (A) the spill peak is smaller and occurs earlier in time
- (B) the spill peak is larger and occurs earlier in time
- (C) the spill peak is smaller and occurs later in time
- (D) the spill peak is larger and occurs later in time

The Answer is C

Step 1: A reservoir with a fixed spillway will attenuate the inflow hydrograph, meaning that the peak spillway outflow will be lower than the peak of the inflow hydrograph, and will be delayed in time

102) A hydrograph has a triangular shape, with a peak flow of 125 cfs. The time to peak is 1.5 hours, and the recession time is 3.5 hours. What is the volume of runoff for this hydrograph in million gallons (MG)?

- (A) 4.2 MG
- (B) 8.4 MG
- (C) 16.8 MG
- (D) 24.1 MG

The Answer is B

Step 1: Since the hydrograph has a triangular shape, the runoff volume $V = 0.5 Q_p T$, where Q_p = the peak runoff, and T = duration of the hydrograph.

Step 2: Here $Q_p = 125$ cfs, and $T = 1.5 + 3.5 = 5.0$ hours = 18,000 sec, so $V = 0.5(125 \text{ cfs})(18000 \text{ sec}) = 1.125 \times 10^6 \text{ ft}^3 = 8.4 \text{ MG}$ (million gallons)

103) A unit hydrograph for a watershed is associated with 1 inch of excess rainfall occurring over a 20-minute period. The unit hydrograph has a triangular shape, with peak flow of 72 cfs, time to peak of 50 minutes, and recession time of 190 minutes. Based on this unit hydrograph, if 2.5 inches of excess rain falls over a 60-minute, what will be the duration of direct runoff?

- (A) 240 min
- (B) 280 min
- (C) 500 min
- (D) 720 min

The Answer is B

Step 1: The duration of runoff for 1 inch of excess rain over 20 minutes = (time to peak)+(recession time)=50+190=240 minutes.

Step 2: For a 60 minute duration, the duration will be 240 plus an additional 20 minutes for each additional 20 minutes of rainfall = $24+20+20=280$ min

104) A detention pond is to be used to reduce peak runoff from a 50 acre watershed. The detention pond is to be designed to store the first $\frac{1}{2}$ inch of runoff from the watershed. What is the required volume of the pond?

- | | |
|-----------------------------|-------------------------------|
| (A) 91,000 ft ³ | (B) 109,000 ft ³ |
| (C) 345,000 ft ³ | (D) 1,090,000 ft ³ |

The Answer is A

Step 1: Water volume can be determined from runoff depth by multiplying by the drainage area, with correction for units.

Step 2: The required volume = $(0.5 \text{ inch})(1/12 \text{ ft/inch})(50 \text{ acres})(43560 \text{ ft}^2/\text{acre}) = 91,000 \text{ ft}^3$.

105) A sharp crested weir is to be used as the outflow control from a detention pond. The pond serves a drainage area of 80 acres. If the maximum water surface elevation above the weir crest is to be 1 foot, what is an estimate of the required length of the weir for these conditions?

- | | |
|-----------|-----------|
| (A) 4 ft | (B) 8 ft |
| (C) 12 ft | (D) 16 ft |

The Answers is C

Step 1: A preliminary design estimate for detention ponds is the maximum pond outflow (cfs) = 0.2 (watershed area in acres).

Step 2: For an 80-acre watershed, the estimated maximum outflow is 40 cfs. Using the weir equation $Q = 3.33 L H^{3/2}$ where Q is discharge in cfs, L is the weir crest length (ft), and H = weir head (ft). Solving for $L = Q / (3.33 H^{3/2}) = 40 / (3.33 * 1^{3/2}) = 12$ ft.

106) For the design storm, the runoff volume from a 120-acre watershed is currently 2.3 inches. Due to housing construction, the volume of runoff from a design storm is expected to increase by 15%. If a detention pond is to be designed to store the additional runoff generated by development, what is the required pond volume?

- (A) 45,000 ft³ (B) 150,000 ft³
(C) 350,000 ft³ (D) 1,000,000 ft³

The Answer is B

Step 1: The current runoff from the design storm = $(2.3 \text{ inch})(1/12 \text{ ft/inch})(120 \text{ acres})(43560 \text{ ft}^2/\text{acre}) = 1.0 \times 10^6 \text{ ft}^3$

Step 2: The increase in runoff volume is $0.15 (1.0 \times 10^6 \text{ ft}^3) = 150,000 \text{ ft}^3$

Part 6 : Geometrics

52 Problems

1. An equal-tangent crest vertical curve is to connect grades of +2.0% and -3%. The design sight distance on the curve is 645 ft. Determine the minimum length of the curve to meet the sight distance requirement. (Assume the height of driver's eyes above the roadway surface is 3.5 ft, and the height of object above the roadway surface is 2.0 ft.)

- A) 689.50 ft B) 963.91 ft C) 734.30 ft D) 878.40 ft

The Answer is B

P 3-152, AASHTO Geometric Design-Green Book 2011, 2011, 6th Ed.

Step 1:

If we assume that the curve length (L) is greater than the sight distance (S), under the standard criteria of the heights of driver's eyes and objects above the roadway surface, the minimum curve length is

$$L = \frac{AS^2}{2,158}$$

Step 2:

The grades of the two tangents are $g_1 = +2.0\%$ and $g_2 = -3.0\%$, respectively. Therefore, the absolute value of algebraic difference in grades (%) is

$$A = |g_1 - g_2| = |2 - (-3)| = 5$$

Step 3:

The minimum curve length is

$$L = \frac{AS^2}{2,158} = \frac{(5)(645)^2}{2,158} \cong 963.91 \text{ ft}$$

which is greater than the sight distance, so the assumption that $L > S$ is correct.

2. A 600 ft long equal-tangent crest vertical curve connects grades of +4.0% and -2.5%. The point of vertical intersection (PVI) is located at station 123+00 with an elevation of 62.80 ft. What is the elevation of the PVT?

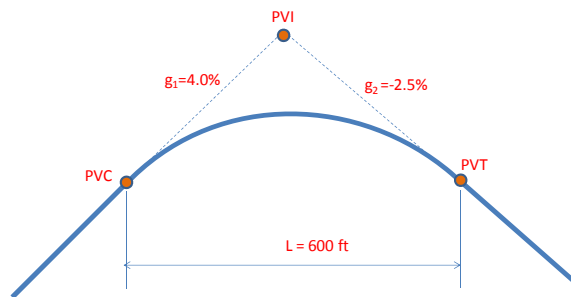
- A) 55.30 ft B) 70.30 ft C) 73.50 ft D) 65.30 ft

The Answer is A

[P3-151, AASHTO Geometric Design-Green Book 2011, 2011, 6th ed](#)

Step 1:

The horizontal distance between the vertex (PVI) and the Point of Vertical Tangency (PVT) is half of the curve length.



$$\frac{L}{2} = \frac{600 \text{ ft}}{2} = 300 \text{ ft}$$

Step 2:

The grade between PVI and PVT is $g_2 = -2.5\%$. Therefore, the vertical distance between PVI and PVT is

$$\Delta h = g_2 \left(\frac{L}{2} \right) = (-2.5\%) \times 300 \text{ ft} = -7.50 \text{ ft}$$

Step 3:

The elevation of PVT is

$$\text{PVI elevation} + \Delta h = 62.80 \text{ ft} - 7.50 \text{ ft} = 55.30 \text{ ft}$$

3. An equal-tangent crest vertical curve is to connect grades of +1.0% and -1%. The design sight distance on the curve is 910 ft. Determine the minimum length of the curve to meet the sight distance requirement. (Assume the height of driver's eyes above the roadway surface is 3.5 ft, and the height of object above the roadway surface is 2.0 ft.)

- A) 767.47 ft B) 963.91 ft C) 741.00 ft D) 878.40 ft

The Answer is C

[P3-152, AASHTO Geometric Design-Green Book 2011, 2011, 6th ed](#)

Step 1:

The absolute value of algebraic difference in grades (%) is

$$A = |g_1 - g_2| = |1 - (-1)| = 2$$

Step 2:

If we assume that the curve length (L) is greater than the sight distance (S), under the standard criteria of the heights of driver's eyes and objects above the roadway surface, the minimum curve length is

$$L = \frac{AS^2}{2,158} = \frac{(2)(910)^2}{2,158} \cong 767.47 \text{ ft}$$

which is smaller than S , so the assumption that $L > S$ is incorrect.

Step 3:

Using the formula for $L < S$, the minimum curve length is

$$L = 2S - \frac{2158}{A} = 2 \times 910 - \frac{2158}{2} = 741 \text{ ft}$$

which is smaller than the sight distance.

4. An equal-tangent sag vertical curve is to connect grades of -2.0% and +3.0%. The design sight distance on the curve is 645 ft. Determine the minimum curve length based on the stopping sign distance requirements following the standard headlight criteria.

- A) 889.50 ft B) 782.74 ft C) 654.50 ft D) 934.52 ft

The Answers is B

[P3-158, AASHTO Geometric Design-Green Book 2011, 2011, 6th ed](#)

Step 1:

The absolute value of algebraic difference in grades (%) is

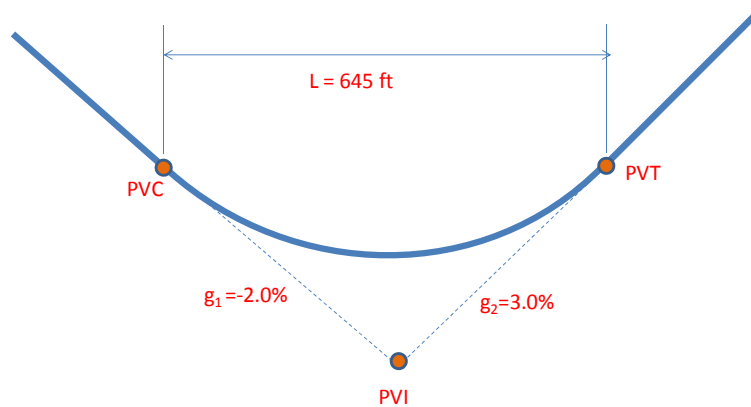
$$A = |g_1 - g_2| = |-2 - (3)| = 5$$

Step 2:

If we assume that the curve length (L) is greater than the sight distance (S), under the standard headlight criteria, the minimum curve length is

$$L = \frac{AS^2}{400 + 3.5S} = \frac{(5)(645)^2}{400 + 3.5 \times 645} \cong 782.74 \text{ ft}$$

which is greater than S, so the assumption that $L > S$ is correct.



5. An equal-tangent sag vertical curve is to connect grades of -1.0% and $+1.0\%$. The design sight distance on the curve is 910 ft. Determine the minimum curve length based on the stopping sight distance requirements following the standard headlight criteria.

- A) 27.50 ft B) 64.74 ft C) 434.50 ft D) 161.62 ft

The Answers is A

[P3-158, AASHTO Geometric Design-Green Book 2011, 2011, 6th ed](#)

Step 1:

The absolute value of algebraic difference in grades (%) is

$$A = |g_1 - g_2| = |-1 - (1)| = 2$$

Step 2:

If we assume that the curve length (L) is greater than the sight distance (S), under standard headlight criteria, the minimum curve length is

$$L = \frac{AS^2}{400 + 3.5S} = \frac{(2)(910)^2}{400 + 3.5 \times 910} \cong 461.98 \text{ ft}$$

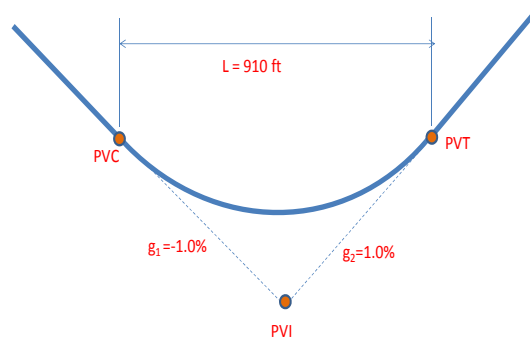
which is smaller than S, so the assumption that $L > S$ is incorrect.

Step 3:

Using the formula for $L < S$, the minimum curve length is

$$L = 2S - \left(\frac{400 + 3.5S}{A} \right) = 2 \times 910 - \left(\frac{400 + 3.5 \times 910}{2} \right) = 27.5 \text{ ft}$$

which is smaller than the sight distance.



the

6. An equal-tangent sag vertical curve is to connect grades of -1.0% and $+1.0\%$. The design speed on the curve is 80 mph. Determine the minimum curve length based on the riding comfort criteria.

- A) 27.50 ft B) 64.74 ft C) 275.27 ft D) 161.62 ft

The Answer is C

[P3-160, AASHTO Geometric Design-Green Book 2011, 2011, 6th ed](#)

Step 1:

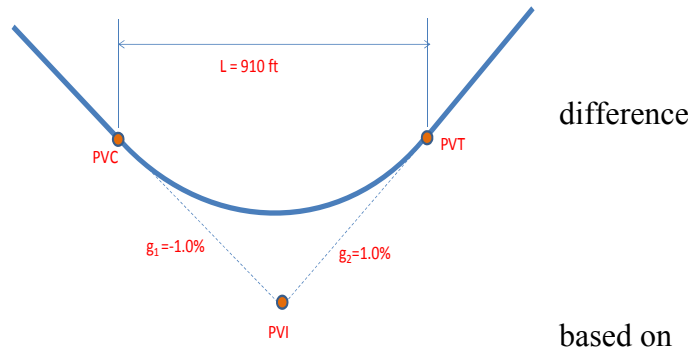
The absolute value of algebraic
in grades (%) is

$$A = |g_1 - g_2| = |-1 - (1)| = 2$$

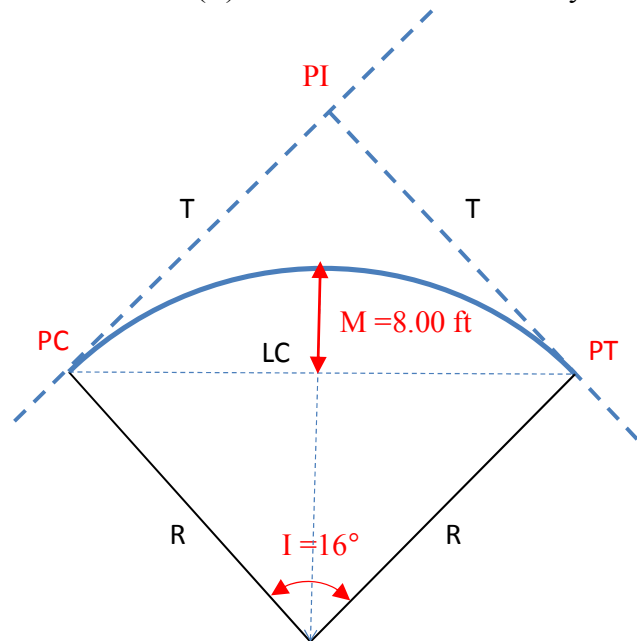
Step 2:

The minimum curve length (L)
the riding comfort criteria is

$$L = \frac{AV^2}{46.5} = \frac{(2)(80)^2}{46.5} \cong 275.27 \text{ ft}$$



7. A horizontal circular curve has an intersection angle of 16° . The length of middle ordinate (M) is 8.00 ft. The radius (ft) of the curve is most nearly



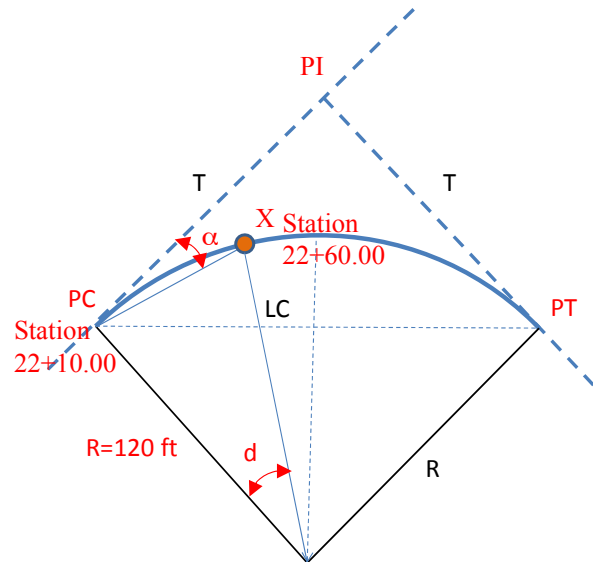
- A) 401.9 B) 786.4 C) 4.0 D) 822.0

The Answers is D

The relationship between the length of middle ordinate (M), radius of the curve (R), and intersection angle (I) is $M = R \left[1 - \cos \left(\frac{I}{2} \right) \right]$, which can be arranged into

$$R = \frac{M}{1 - \cos \left(\frac{I}{2} \right)} = \frac{8 \text{ ft}}{1 - \cos \left(\frac{16^\circ}{2} \right)} \cong 822.04 \text{ ft}$$

8. A horizontal circular curve has a radius of 120 ft. The station of PC is sta 22+10.00. At point X (station 22+60.00) on the curve, a stake is to be placed. The deflection angle between tangent PI-PC and chord X-PC is most nearly



- A) $14^{\circ}56'12''$
B) $13^{\circ}06'12''$
C) $11^{\circ}56'12''$
D) $11^{\circ}44'43''$

The Answer is C**Step 1:**

The curve length between X and PC is

$$l = (\text{sta } 22 + 60.00) - (\text{sta } 22 + 10.00) = 50 \text{ ft}$$

Step 2:

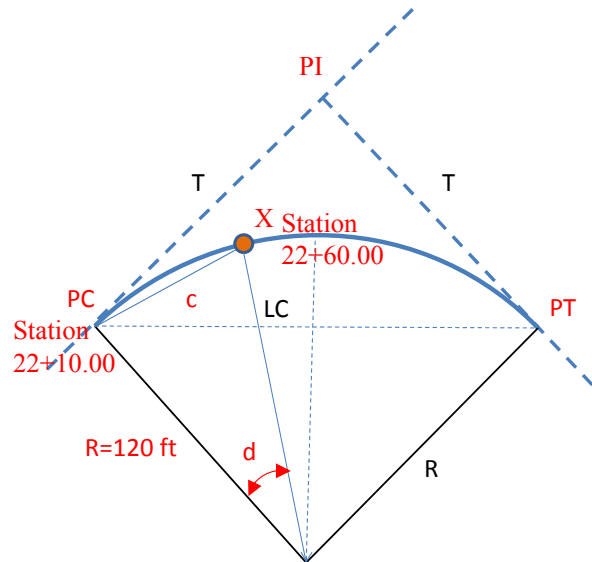
The angle for curve length l is $d = \frac{180^\circ l}{R\pi} = \frac{180^\circ(50 \text{ ft})}{(120 \text{ ft})\pi} = 23.873^\circ$.

Step 3:

The deflection angle between tangent PI-PC and chord X-PC is half of the angle for curve length l

$$\alpha = \frac{1}{2}d = \frac{23.873^\circ}{2} = 11.9366^\circ \cong 11^\circ 56' 12''$$

9. A horizontal circular curve has a radius of 120 ft. The station of PC is sta 22+10.00. At point X (station 22+60.00) on the curve, a stake is to be placed. The length of chord X-PC is most nearly (ft)



- A) 43.65 B) 49.64 C) 65.32 D) 44.69

The Answers is B**Step 1:**

The curve length between X and PC is

$$l = (\text{sta } 22 + 60.00) - (\text{sta } 22 + 10.00) = 50 \text{ ft}$$

Step 2:

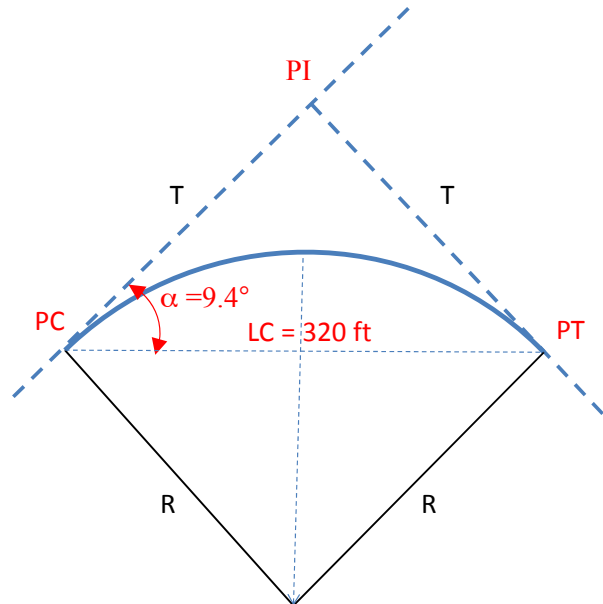
The angle for curve length l is $d = \frac{180^\circ l}{R\pi} = \frac{180^\circ(50 \text{ ft})}{(120 \text{ ft})\pi} = 23.873^\circ$.

Step 3:

The length of chord X-PC is

$$c = 2R\sin\left(\frac{d}{2}\right) = (2)(120 \text{ ft})\sin\left(\frac{23.873^\circ}{2}\right) \cong 49.64 \text{ ft}$$

10. A horizontal circular curve has a length of long chord (LC) of 320 ft. The deflection angle between tangent PI-PC and chord PT-PC is 9.4° . The radius of the curve is most nearly (ft)



- A) 890 B) 980 C) 672 D) 760

The Answers is B**Step 1:**

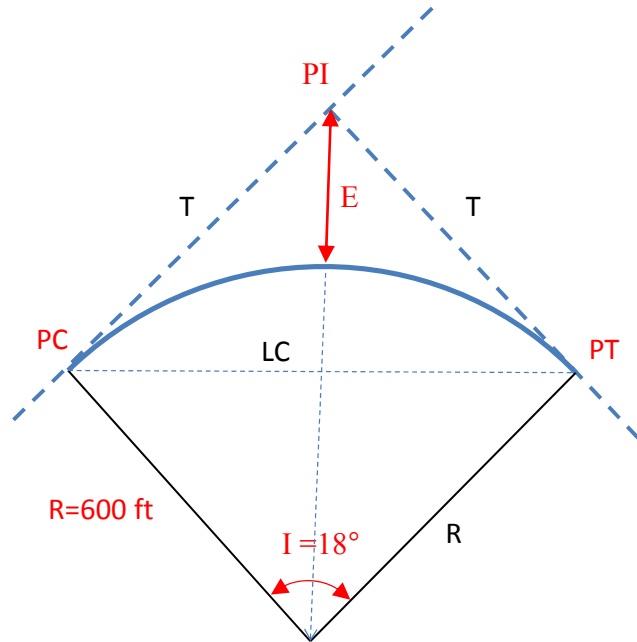
The deflection angle between tangent PI-PC and chord PT-PC is half of the intersection angle (I). Therefore, $\frac{I}{2} = 9.4^\circ$.

Step 2:

The relationship between the length of long chord (LC), radius of the curve (R), and intersection angle (I) is

$$R = \frac{LC}{2\sin\left(\frac{I}{2}\right)} = \frac{320 \text{ ft}}{2\sin(9.4^\circ)} \cong 979.64 \text{ ft}$$

11. A horizontal circular curve has a radius of 600 ft and an intersection angle of 18° . The external distance (E) is most nearly (ft)



- A) 7.48 B) 8.47 C) 4.78 D) 7.84

The Answers is A

The external distance (E) is

$$E = R \left[\frac{1}{\cos\left(\frac{I}{2}\right)} - 1 \right] = (600 \text{ ft}) \left[\frac{1}{\cos\left(\frac{18^\circ}{2}\right)} - 1 \right] \cong 7.48 \text{ ft}$$

12. In highway geometric design, which of the following are usually used for vertical curves?

- A) circular curves
- B) spiral curves
- C) equal tangent parabolic curves
- D) parabolic curves with unequal tangents

The Answer is C

Equal tangent parabolic curves are usually used for the vertical curves in highway geometric design.

(A) is incorrect. Circular curves are typically used for horizontal curves.

(B) is incorrect. Spiral curves are used to produce a gradual transition from horizontal tangents to horizontal circular curves in roadways.

(C) is correct.

(D) is incorrect. A curve with unequal tangents is only used occasionally for vertical curves.

13. A 420 ft long equal-tangent crest vertical curve connects grades of +1.0% and -2.0%. The point of vertical intersection (PVI) is located at station 103+20 with an elevation of 87.40 ft. What is the horizontal distance from PVC to maximum elevation on curve?

- A) 140 ft B) 210 ft C) 79 ft D) 120 ft

The Answer is A

The horizontal distance from PVC to maximum elevation on curve is

$$x_m = \frac{g_1 L}{g_1 - g_2} = \frac{(1.0\%)(420 \text{ ft})}{1.0\% - (-2.0\%)} = 140 \text{ ft}$$

14. A 500 ft long equal-tangent sag vertical curve connects grades of -1.0% and +2.5%. The point of vertical curvature (PVC) is located at station 82+40 with an elevation of 43.50 ft. A stake is to be placed at station 83+10. What is the curve elevation at that place?

- A) 43.75 ft B) 43.55 ft C) 42.97 ft D) 42.80 ft

The Answers is C**Step 1:**

The horizontal distance of the point of interest from PVC is

$$x = (\text{sta } 83 + 10) - (\text{sta } 82 + 40) = 8310 \text{ ft} - 8240 \text{ ft} = 70 \text{ ft}$$

Step 2:

The curve elevation at a horizontal distance x from PVC is

$$\begin{aligned} Y_{\text{PVC}} + g_1x + \frac{g_2 - g_1}{2L}x^2 &= (43.50 \text{ ft}) + (-1.0\%)(70 \text{ ft}) + \frac{+2.5\% - (-1.0\%)}{2(500 \text{ ft})}(70 \text{ ft})^2 \\ &\cong 42.97 \text{ ft} \end{aligned}$$

15. A 700 ft long equal-tangent sag vertical curve connects grades of -2.0% and +1.0%. The elevation of vertical curvature (PVC) is 233.80 ft. What is the lowest curve elevation?

- A) 204.43 ft B) 234.33 ft C) 219.13 ft D) 229.13 ft

The Answer is D**Step 1:**

The horizontal distance from PVC to minimum elevation on curve is

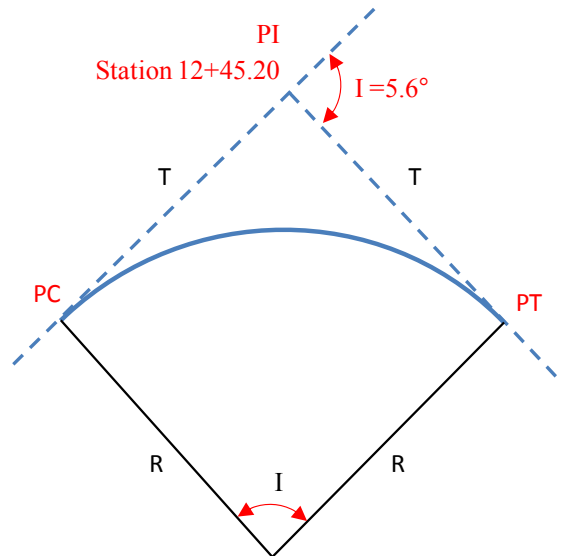
$$x_m = \frac{g_1 L}{g_1 - g_2} = \frac{(-2.0\%)(700 \text{ ft})}{-2.0\% - (1.0\%)} \cong 466.67 \text{ ft}$$

Step 2:

The minimum curve elevation is

$$\begin{aligned} Y_{\text{PVC}} + g_1 x_m + \frac{g_2 - g_1}{2L} x_m^2 \\ = (233.80 \text{ ft}) + (-2.0\%)(466.67 \text{ ft}) + \frac{+1.0\% - (-2.0\%)}{2(700 \text{ ft})} (466.67 \text{ ft})^2 \\ \cong 229.13 \text{ ft} \end{aligned}$$

16. A horizontal circular curve has an interior angle of 5.6° . The station of Point of Intersection (PI) is 12+45.20. The degree of curve (D) is 4° . Determine the station of Point of Tangent (PT).



- A) sta 14 + 03.88 B) sta 13 + 15.14
C) sta 12 + 03.88 D) sta 13 + 64.28

The Answers is B**Step 1:**

The radius of the curve, R, is

$$R = \frac{5729.58}{D} = \frac{5729.58}{4^\circ} = 1432.395 \text{ ft}$$

Step 2:

The Tangent Distance (T) from PI to PC is

$$T = R \tan\left(\frac{I}{2}\right) = (1432.395 \text{ ft}) \times \tan\left(\frac{5.6^\circ}{2}\right) \cong 70.06 \text{ ft}$$

The length of the arc (L) from PC to PT is

$$L = R I \frac{\pi}{180^\circ} = (1432.395 \text{ ft})(5.6^\circ) \frac{\pi}{180^\circ} = 140.00 \text{ ft}$$

Step 3:

The PC station is $PC = PI \text{ station} - T = (\text{sta } 12 + 45.20) - (\text{sta } 0 + 70.06) = \text{sta } 11 + 75.14$.

Step 4:

The PT station is

$$PT = PC \text{ station} + L = (\text{sta } 11 + 75.14) + (\text{sta } 1 + 40.00) = \text{sta } 13 + 15.14.$$

17. A sag vertical curve with a length of 800 ft is to connect grades of -4.0% and +5.0%. The vertex is located at station 11+30, and it has an elevation of 124 ft. What is the vertical curve elevation at station 12+50?

- A) 110.05 ft B) 112.10 ft C) 120.01 ft D) 125.05 ft

The Answer is C

For a sag curve, the elevation of the curve at any distance x from the PVC is equal to the tangent elevation plus the tangent offset.

Step 1:

The horizontal distance from PVC (sta 11 + 30) to the location of interest (sta 12 + 50) is $x = 1250 \text{ ft} - 1130 \text{ ft} = 120 \text{ ft}$.

The parabola constant a is

$$a = \frac{g_2 - g_1}{2L} = \frac{5\% - (-4\%)}{2 \times 800 \text{ ft}} = 5.625 \times 10^{-5} \text{ ft}^{-1}$$

Step 2:

The tangent offset at $x = 120 \text{ ft}$ from the PVC is

$$y = ax^2 = 5.625 \times 10^{-5} \text{ ft}^{-1} \times (120 \text{ ft})^2 = 0.81 \text{ ft}$$

Step 3:

The tangent elevation 120 ft from the PVC is

$$\text{PVC elevation} + g_1 x = 124 \text{ ft} + (-4.0\%) \times (120 \text{ ft}) = 119.2 \text{ ft}$$

Step 4:

The elevation of the curve at station 12+50 is

$$\text{Curve elevation} = y + \text{tangent elevation} = 0.81 \text{ ft} + 119.2 \text{ ft} = 120.01 \text{ ft}$$

18. An equal-tangent crest vertical curve is to connect grades of +2.5% and -2%. The design sight distance on the curve is 730 ft. The grades intersect at station 23+20. Determine the stationing of the PVC for the minimum curve length based on stopping sight distance requirements. (Assume the height of driver's eyes above the roadway surface is 3.5 ft, and the height of object above the roadway surface is 2.0 ft.)
- A) sta 12 + 68.76 B) sta 21 + 08.76
C) sta 12 + 08.76 D) sta 36 + 12.64

The Answers is C

[P3-152, AASHTO Geometric Design- 2011, 2011, 6th ed](#)

Step 1:

If we assume that the curve length (L) is greater than the sight distance (S), under the standard criteria of the heights of driver's eyes and objects above the roadway surface, the minimum curve length is $L = \frac{AS^2}{2,158}$. The grades of the two tangents are $g_1 = +2.5\%$ and $g_2 = -2.0\%$, respectively. Therefore, the absolute value of algebraic difference in grades (%) is $A = |g_1 - g_2| = |2.5 - (-2)| = 4.5$.

Step 2:

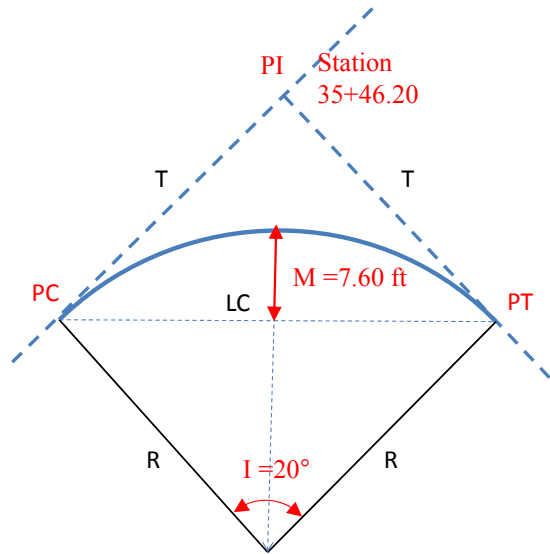
The minimum curve length is $L = \frac{AS^2}{2,158} = \frac{(4.5)(730)^2}{2,158} \cong 1111.24$ ft,

which is greater than the sight distance, so the assumption that $L > S$ is correct.

Step 3:

The station of the PVC is the station at PVI minus half of the curve length, which is $2320 - 1111.24 = 1208.76$ ft (station 12 + 08.76)

19. A horizontal circular curve has an intersection angle of 20° . The length of middle ordinate (M) is 7.60 ft. The station of Point of Intersection (PI) is 35+46.20. The station of Point of Tangent (PT) is most nearly



- A) sta 35 + 32.41 B) sta 36 + 34.41
C) sta 37 + 20.82 D) sta 36 + 32.41

The Answers is D**Step 1:**

The relationship between the length of middle ordinate (M), radius of the curve (R), and intersection angle (I) is $M = R \left[1 - \cos \left(\frac{I}{2} \right) \right]$, which can be arranged into $R = \frac{M}{1 - \cos \left(\frac{I}{2} \right)} =$

$$\frac{7.6 \text{ ft}}{1 - \cos \left(\frac{20^\circ}{2} \right)} \cong 500.26 \text{ ft.}$$

Step 2:

The tangent distance T is

$$T = R \tan \left(\frac{I}{2} \right) = (500.26 \text{ ft}) \tan \left(\frac{20^\circ}{2} \right) \cong 88.21 \text{ ft}$$

The curve length L is

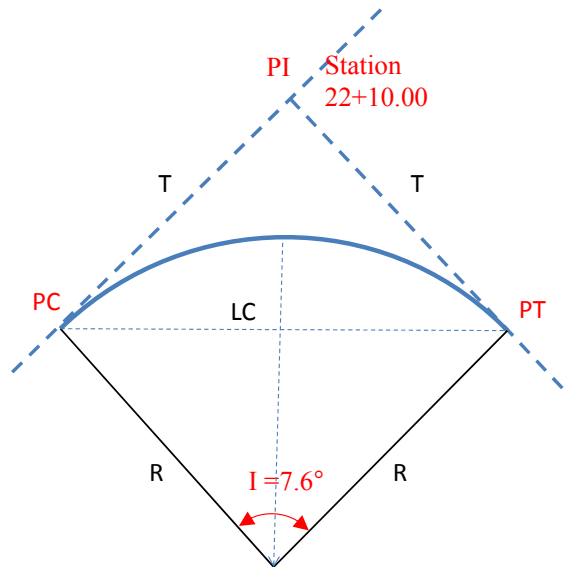
$$L = R I \frac{\pi}{180^\circ} = (500.26 \text{ ft})(20^\circ) \frac{\pi}{180^\circ} \cong 174.62 \text{ ft}$$

Step 3:

The station of PT is

$$\begin{aligned} PT = PI \text{ station} - T + L &= 3546.20 \text{ ft} - 88.21 \text{ ft} + 174.62 \text{ ft} = 3632.41 \text{ ft} \\ &= (\text{sta } 36 + 32.41) \end{aligned}$$

20. An interior angle of 7.6° is specified for a horizontal circular curve. The degree of curve (D) is 3. The station of Point of Intersection (PI) is 22+10.00. The station of Point of Tangent (PT) is most nearly



- A) sta 23 + 36.48 B) sta 23 + 34.42
C) sta 32 + 20.82 D) sta 23 + 63.48

The Answers is A**Step 1:**

The radius of the curve (R) is

$$R = \frac{5729.58}{D} = \frac{5729.58}{3} \cong 1909.86 \text{ ft.}$$

Step 2:

The tangent distance T is

$$T = R \tan\left(\frac{I}{2}\right) = (1909.86 \text{ ft}) \tan\left(\frac{7.6^\circ}{2}\right) \cong 126.85 \text{ ft}$$

The curve length L is

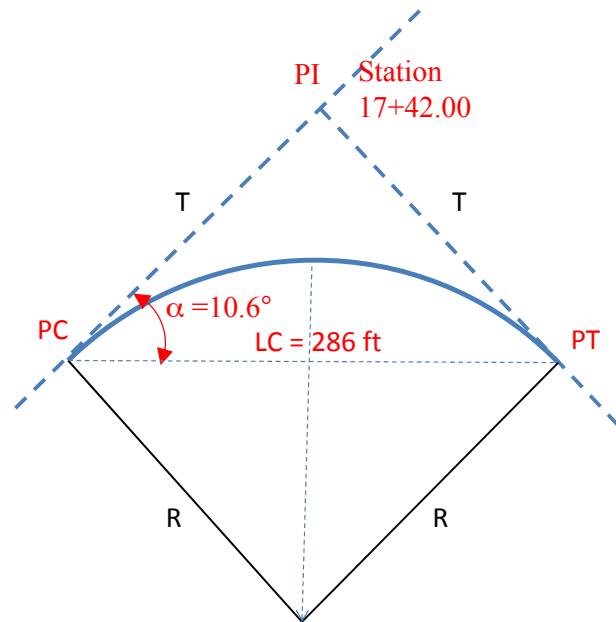
$$L = R I \frac{\pi}{180^\circ} = (1909.86 \text{ ft})(7.6^\circ) \frac{\pi}{180^\circ} \cong 253.33 \text{ ft}$$

Step 3:

The station of PT is

$$\begin{aligned} PT &= PI \text{ station} - T + L = 2210.00 \text{ ft} - 126.85 \text{ ft} + 253.33 \text{ ft} = 2336.48 \text{ ft} \\ &= (\text{sta } 23 + 36.48) \end{aligned}$$

21. A horizontal circular curve has a length of long chord (LC) of 286 ft. The deflection angle between tangent PI-PC and chord PT-PC is 10.6° . The station of Point of Intersection (PI) is sta 17 + 42.00. The station of Point of Tangent (PT) is most nearly



- A) sta 17 + 36.48 B) sta 18 + 48.61
C) sta 18 + 84.16 D) sta 18 + 54.48

The Answers is C**Step 1:**

The deflection angle between tangent PI-PC and chord PT-PC is half of the intersection angle (I). Therefore, $\frac{I}{2} = 10.6^\circ$.

Step 2:

The relationship between the length of long chord (LC), radius of the curve (R), and intersection angle (I) is

$$R = \frac{LC}{2\sin\left(\frac{I}{2}\right)} = \frac{286 \text{ ft}}{2\sin(10.6^\circ)} \cong 777.38 \text{ ft}$$

Step 3:

The tangent distance T is

$$T = R \tan\left(\frac{I}{2}\right) = (777.38 \text{ ft}) \tan(10.6^\circ) \cong 145.48 \text{ ft}$$

The curve length L is

$$L = R I \frac{\pi}{180^\circ} = (777.38 \text{ ft})(2 \times 10.6^\circ) \frac{\pi}{180^\circ} \cong 287.64 \text{ ft}$$

Step 4:

The station of PT is

$$\begin{aligned} PT &= \text{PI station} - T + L = 1742.00 \text{ ft} - 145.48 \text{ ft} + 287.64 \text{ ft} = 1884.16 \text{ ft} \\ &= (\text{sta } 18 + 84.16) \end{aligned}$$

22. A 660 ft long equal-tangent crest vertical curve connects grades of +1.0% and g_2 . The absolute value of elevation difference between point of vertical curvature (PVC) and maximum elevation on curve is 0.90 ft. What is the grade g_2 ?

A) -2.0%

B) 2.7%

C) -2.7%

D) -1.5%

The Answers is C**Step 1:**

The horizontal distance from PVC to maximum elevation on curve is

$$x_m = \frac{g_1 L}{g_1 - g_2}$$

Step 2:

The absolute value of elevation difference between PVC and maximum elevation on curve is

$$g_1 x_m + \frac{g_2 - g_1}{2L} x_m^2 = g_1 \frac{g_1 L}{g_1 - g_2} + \frac{g_2 - g_1}{2L} \left(\frac{g_1 L}{g_1 - g_2} \right)^2 = 0.90 \text{ ft}$$

We have

$$(1\%) \frac{(1\%)(660 \text{ ft})}{(1\%) - g_2} + \frac{g_2 - (1\%)}{2(660 \text{ ft})} \left(\frac{1\% \times 660 \text{ ft}}{1\% - g_2} \right)^2 = 0.90 \text{ ft}$$

$$\frac{0.066}{(1\%) - g_2} - \frac{0.0001 \times 330}{1\% - g_2} = 0.90$$

$$(1\%) - g_2 = \frac{0.033}{0.90} \cong 0.036667$$

$$g_2 = 1\% - 0.036667 = -0.026667 \cong -2.7\%$$

23. A 420 ft long equal-tangent crest vertical curve connects grades of +1.0% and -2.0%. The point of vertical intersection (PVI) is located at station 103+20 with an elevation of 87.40 ft. What is the maximum elevation on curve?

A) 86 ft

B) 88 ft

C) 84 ft

D) 85 ft

The Answers is A**Step 1:**

The horizontal distance from PVC to maximum elevation on curve is

$$x_m = \frac{g_1 L}{g_1 - g_2} = \frac{(1.0\%)(420 \text{ ft})}{1.0\% - (-2.0\%)} = 140 \text{ ft}$$

Step 2:

The curve elevation at a horizontal distance x from PVC is

$$Y_{PVC} + g_1 x + \frac{g_2 - g_1}{2L} x^2$$

When $x = x_m$, the maximum curve elevation is

$$\begin{aligned} Y_{PVC} + g_1 x_m + \frac{g_2 - g_1}{2L} x_m^2 &= Y_{PVI} - \frac{g_1 L}{2} + g_1 x_m + \frac{g_2 - g_1}{2L} x_m^2 \\ &= (87.40 \text{ ft}) - \frac{(1.0\%)(420 \text{ ft})}{2} + (1.0\%)(140 \text{ ft}) \\ &\quad + \frac{-2.0\% - 1.0\%}{2(420 \text{ ft})} (140 \text{ ft})^2 = 86 \text{ ft} \end{aligned}$$

24) The following information is known about a vertical curve underneath a bridge:

Length of vertical curve = 1500 ft

Vertical Clearance between curve and bridge = 30 ft

Beginning grade = -2.7%

End grade = +1.3%

Assuming an eye height of 8.0 ft for a truck driver and an object height of 2.0 ft for taillights of a vehicle, determine the sight distance.

- A) 3050 ft B) 3150 ft C) 3250 ft D) 3350 ft

The Answer is C

P3-163, AASHTO Geometric Design-Green Book 2011, 2011, 6th ed.

Step 1:

If we assume sight distance greater than length of vertical curve, under standard eye height and object height, the equation is

$$L = 2S - \frac{800(C - 5)}{A}$$

Step 2:

The grades of the two tangents are $g_1 = -2.7\%$ and $g_2 = +1.3\%$, respectively. Therefore, the absolute value of algebraic difference in grades (%) is

$$A = |g_1 - g_2| = |(-2.7) - (+1.3)| = 4.0$$

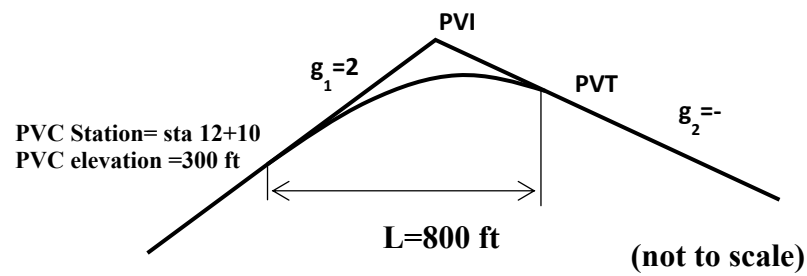
Step 3:

Put the numbers into the equation in step 1 to get

$$1500 = 2S - \frac{800(30 - 5)}{4}$$

Therefore $S = 3250$ ft, so the assumption $S > L$ is correct

25) In the vertical curve shown below, what is the elevation of point of vertical tangency (PVT)?



- A) 300 ft B) 320 ft C) 340 ft D) 295 ft

The Answers is A

The elevation of the point at the same horizontal distance from PVC as PVT but on the back tangent line is

$$\text{PVC elevation} + g_1 L = 300 \text{ ft} + 2\% \times (800 \text{ ft}) = 316 \text{ ft}$$

The vertical distance from the previous point to PVT is tangent offset y , which can be calculated by

$$y = \frac{g_2 - g_1}{2L} L^2 = \frac{-2\% - 2\%}{2} \times 800 \text{ ft} = -16 \text{ ft}$$

The elevation of PVT is

$$316 \text{ ft} + y = 316 \text{ ft} - 16 \text{ ft} = 300 \text{ ft}$$

Short cut

Since the slope of the back tangent is the same as the slope of the forward tangent, the vertical curve is symmetric about the vertical line through PVI. Therefore, the elevation of PVT is the same as the elevation of PVC, which is 300 ft.

26) Given the following traffic count data:

Time Interval	No. of Vehicles
8:00-8:15	1400
8:15-8:30	1600
8:30-8:45	2200
8:45-9:00	1800

The peak hour factor is closest to:

A) 0.795

B) 0.880

C) 0.650

D) 0.945

The Answers is A

Page 4-4, Equations 4-2 and 4-3, HCM 2010

Equation 4-2

$$PHF = \frac{V}{4 \times V_{15}}$$

where

 PHF = peak hour factor, V = hourly volume (veh/h), and V_{15} = volume during the peak 15 min of the analysis hour (veh/15 min).

Solution for Question 1:

$$PHF = \frac{V}{4 \times V_{15}} = \frac{1400 + 1600 + 2200 + 1800}{4 \times 2200} = 0.795,$$

27) Given the following traffic count data:

Time Interval	No. of Vehicles
8:00-8:15	1400
8:15-8:30	1600
8:30-8:45	2200
8:45-9:00	1800

The peak rate of flow (veh/hr) is closest to:

- A) 6800 B) 7000
C) 8800 D) 8000

The Answer is C

The peak rate of flow (veh/hr)= $4 \times 2200 = 8800$ veh/hr.

28) Given the following traffic information:

AADT= 3000 veh/day

K-factor=0.135

D-factor=0.70,

The directional design-hour volume (DDHV) is closest to:

A) 284

B) 386

C) 185

D) 234

The Answer is A

Page 3-11, Equation 3-1, HCM 2010

$$DDHV = AADT \times K \times D$$

where

 $DDHV$ = directional design-hour volume (veh/h), $AADT$ = annual average daily traffic (veh/day), K = proportion of $AADT$ occurring in the peak hour (decimal), and D = proportion of peak-hour traffic in the peak direction (decimal).

Equation 3-1

$$DDHV = 3000 \times 0.135 \times 0.70 = 284 \text{ (veh/h)}$$

29) A four-lane freeway (two lanes in each direction) has an average lane width of 11 feet and right-side lateral clearance of 2 feet. Find the adjustments to FFS for average lane width and right-side lateral clearance, respectively.

A) 1.9, 2.4

B) 6.6, 2.4

C) 1.9, 1.8

D) 6.6, 0.8

The Answers is A

Exhibit 11-8, Adjustment to FFS for Average Lane Width; P11-11, HCM 2010

Average Lane Width (ft)	Reduction in FFS, f_{LW} (mi/h)
≥ 12	0.0
$\geq 11-12$	1.9
$\geq 10-11$	6.6

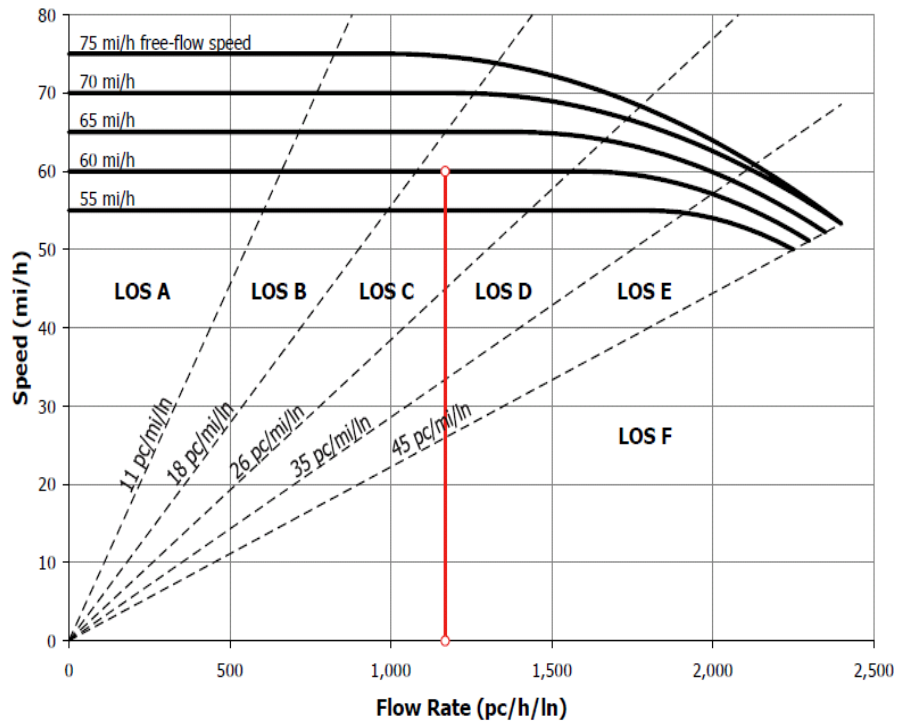
Exhibit 11-9, Adjustment to FFS for Right-Side Lateral Clearance; P11-12, HCM 2010

Right-Side Lateral Clearance (ft)	Lanes in One Direction			
	2	3	4	≥ 5
≥ 6	0.0	0.0	0.0	0.0
5	0.6	0.4	0.2	0.1
4	1.2	0.8	0.4	0.2
3	1.8	1.2	0.6	0.3
2	2.4	1.6	0.8	0.4
1	3.0	2.0	1.0	0.5
0	3.6	2.4	1.2	0.6

30) The free-flow speed (FFS) of a four-lane freeway (two lanes in each direction) is 60.8 mi/h, and the demand flow rate under equivalent base conditions is 1169 pc/h/ln. What is the expected level of service (LOS)?

- A) LOS A
- C) LOS C

- B) LOS B
- D) LOS D

The Answer is C

31) In last 20 years the traffic volume in a 4-lane divided highway has been increased by 50%. Find the annual growth rate.

- A) 2%
- C) 4%

- B) 50%
- D) 6%

The Answers is A

$$F = p(1+i)^n$$

$$F = 1.5P$$

$$1.5P = p(1+i)^n \quad \text{So } 1.5 = (1+i)^{20} \quad \text{or } i = .02$$

$$I = 2\%$$

33) The traffic counts between 8am-9am at a busy area in San Jose CA is reported as follows:

Time Interval	Left Turn	Right turn	ST Trucks	ST cars
8:00-8:15am	120	90	80	400
8:15-8:30 am	70	100	90	450
8:30-8:45 am	60	80	110	400
8:45-9:00 am	50	70	60	400

If a truck is equal to 2.5 passenger cars and a right-turn is 1.5 passenger car, and if a left-turn is equal to 2.0 passenger cars, then calculate the peak hour volume, peak hour factor (PHF)

- A) 0.88
C) 0.93

- B) 0.79
D) 0.83

The Answers is C

The first step is to find the total traffic volume for each 15 minute period

Time Interval	Left Turn	Right turn	ST Trucks	ST cars	Total
8:00-8:15am	120X2=240	90X1.5=135	80X2.5=200	400	975
8:15-8:30 am	70X2=140	100X1.5=150	90x2.5=225	450	965
8:30-8:45 am	60X2=120	80x1.5=120	110x2.5=275	400	915
8:45-9:00 am	50X2=100	70x1.5=105	60x2.5=150	400	775

HCM 2010, Eq4-1. Page 4-3

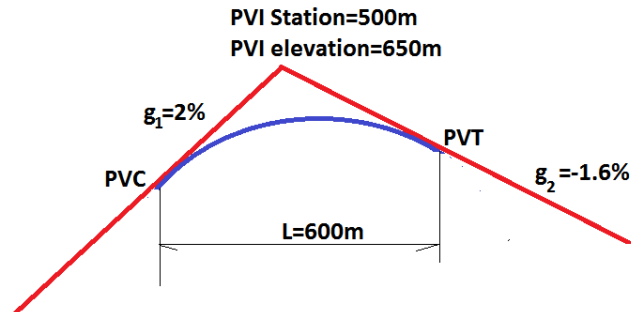
$$V_{15} = 975$$

$$V=975+965+915+775=3630$$

$$PHF = \frac{V}{4V_{15}}$$

$$PHF = \frac{3630}{4 \times 975} = 0.93$$

34. In the vertical curve shown below, what is the station of the highest point on the vertical curve?



- A) sta 5+45.33 B) sta 5+23.00 C) sta 5+23.33 D) sta 5+33.33

The Answers is D**Step 1:**

The station of PVC is

$$\text{PVC station} = \text{PVI station} - \frac{L}{2} = 500 \text{ m} - \frac{600 \text{ m}}{2} = 200 \text{ m (or 2 + 00)}.$$

Step 2:

The distance (x_m) between the maximum elevation point and PVC is

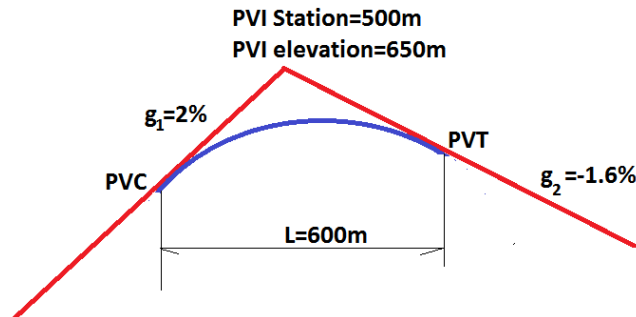
$$x_m = \frac{g_1 L}{g_1 - g_2} = \frac{(2\%) \times (600 \text{ m})}{(2\% - (-1.6\%))} = 333.33 \text{ m}$$

Step 3:

The station of the maximum elevation point on the vertical curve is

$$\text{PVC station} + x_m = 200 \text{ m} + 333.33 \text{ m} = 533.33 \text{ m (or 5 + 33.33)}.$$

35. In the vertical curve shown below, what is the elevation of the highest point on the vertical curve?



- A) 635.35 m B) 648.36 m C) 645.34 m D) 647.34 m

The Answers is D**Step 1:**

The distance (x_m) between the maximum elevation point and PVC is

$$x_m = \frac{g_1 L}{g_1 - g_2} = \frac{(2\%) \times (600 \text{ m})}{(2\% - (-1.6\%))} = 333.33 \text{ m}$$

Step 2:

The distance (x_{ml}) between the maximum elevation point and PVI is

$$x_{ml} = x_m - \frac{L}{2} = 333.33 \text{ m} - \frac{600 \text{ m}}{2} = 33.33 \text{ m}$$

Step 3:

Working along the tangent with a grade of $g_1=2\%$, the elevation (ele_{gra}) corresponding to the maximum elevation point is

$$ele_{gra} = \text{PVI elevation} + g_1 x_{ml} = 650 \text{ m} + 2\% \times 33.33 \text{ m} = 650.67 \text{ m}$$

Step 4:

The tangent offset (y) at the maximum elevation point is

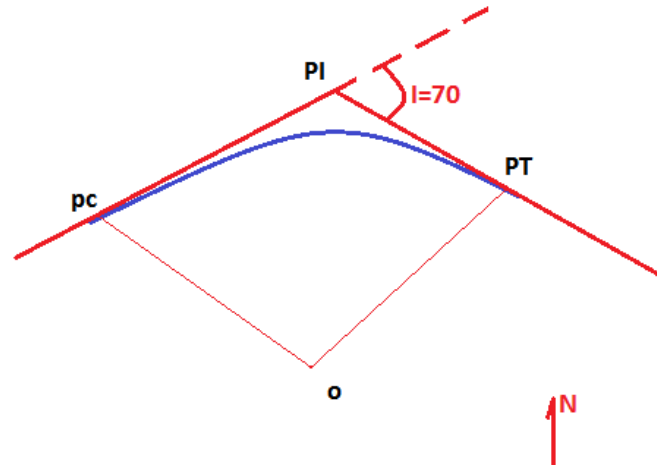
$$y = \frac{g_2 - g_1}{2L} x_m^2 = \frac{-1.6\% - 2\%}{2 \times 600 \text{ m}} (333.33 \text{ m})^2 = -3.33 \text{ m}$$

Step 5:

The elevation of the highest point on the vertical curve is

$$ele_{gra} + y = 650.67 \text{ m} - 3.33 \text{ m} = 647.34 \text{ m}$$

36. The centerline of a four-lane roadway is a horizontal circular curve, as shown below. It is known that the PC station is sta 8+60. The curve radius is 2480 ft. The intersection angle I is 70° . What is most nearly the PT station?



- A) sta 38+15 B) sta 38+60 C) sta 39+89 D) sta 38+90

The Answers is D**Step 1:**

The length of curve from PC to PT is

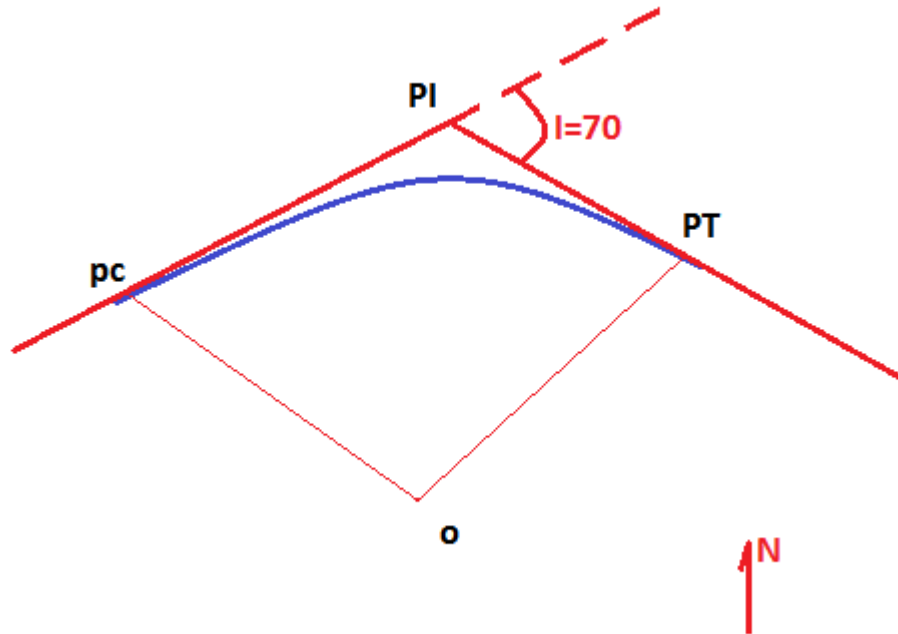
$$L = \frac{R\Delta\pi}{180^\circ} = \frac{(2480 \text{ ft}) \times (70^\circ) \times \pi}{180^\circ} \cong 3029.89 \text{ ft.}$$

Step 2:

The station of PT is

$$\text{Sta PT} = \text{sta PC} + L = (\text{sta 8} + 60) + 3029.89 \text{ ft} = \text{sta 38} + 89.89$$

37. The centerline of a two-lane roadway is a horizontal circular curve, as shown below. It is known that the PC station is sta 12+30. The curve radius is 2080 ft. The intersection angle I is 70° . What is most nearly the PI station?



- A) sta 20+89 B) sta 21+29 C) sta 20+82 D) sta 20+28

The Answers is C**Step 1:**

The tangent distance (T) from PC to PI is

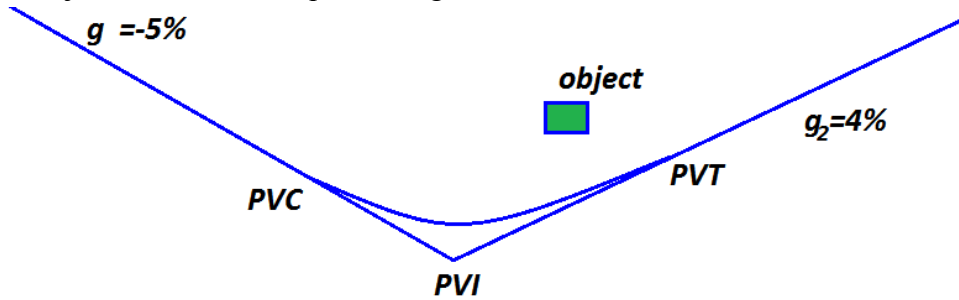
$$T = R \tan \left(\frac{I}{2} \right) = 2080 \text{ ft} \times \tan \left(\frac{70^\circ}{2} \right) \cong 852.29 \text{ ft.}$$

Step 2:

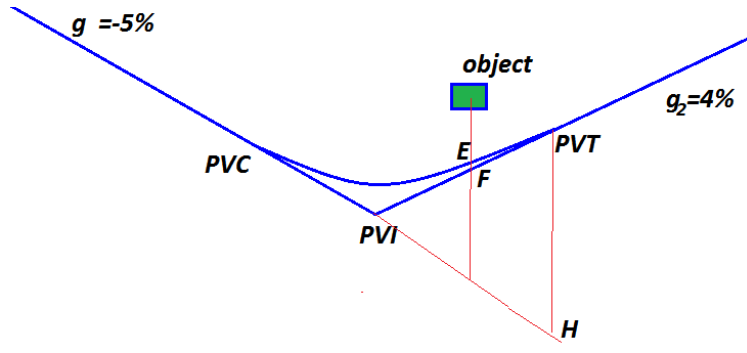
The station of PI is

$$\text{Sta PI} = \text{sta PC} + T = (\text{sta 12} + 30) + 852.29 \text{ ft} = \text{sta 20} + 82.29$$

38. In the sag vertical curve shown below, it is known that the PVC station is sta 12+30, and the PVC elevation is 234.4 ft. At sta 18+30, there is an object (overpass) with an underside elevation of 240.50 ft. If the curve is designed to have a clearance of 15 ft under the object, what is the required length of the curve?



- A) 758.75 ft B) 767.77 ft C) 765.77 ft D) 777.77 ft

The Answers is B**Step 1:**

The elevation at point E is

$$240.50 \text{ ft} - 15 \text{ ft} = 225.50 \text{ ft}.$$

Step 2:

The horizontal distance between PVC and point G is $x_1 = (\text{sta } 18 + 30) - (\text{sta } 12 + 30) = \text{sta } 6 = 600 \text{ ft}.$

The elevation at point G is

$$\text{PVC elevation} + g_1 x_1 = 234.4 \text{ ft} - 5\% \times 600 \text{ ft} = 204.4 \text{ ft}$$

The vertical distance between point E and point G is $y_1 = \text{Point E elevation} -$

$$\text{Point G elevation} = 225.50 \text{ ft} - 204.4 = 21.1 \text{ ft}$$

Step 3:

Assuming the curve length is L, the distance between point H and PVT is $y_2 = \frac{L}{2} \times$

$$(4\% - (-5\%)) = 4.5\%L$$

Step 4:

Since tangent offset (y) is proportional to the square of horizontal distance from PVC

(x^2), that is, for point E, we have $y_1 = ax_1^2$, and for point PVT, we have $y_2 =$

aL^2 . Therefore,

$$\frac{y_1}{y_2} = \frac{ax_1^2}{aL^2}, \text{ or } \frac{21.1 \text{ ft}}{4.5\%L} = \frac{(600 \text{ ft})^2}{L^2},$$

$$\text{So, } L = \frac{(600 \text{ ft})^2}{\frac{21.1 \text{ ft}}{4.5\%}} \cong 767.77 \text{ ft}$$

39. The following travel times were measured for vehicles as they traversed a 1-mile segment of freeway at constant speeds. What is the space mean speed for this data?

Vehicle ID	1	2	3	4
Tavel Time (s)	50	55	54	48

A) 69.78 mph

B) 70.57 mph

C) 69.57 mph

D) 68.78 mph

The Answers is C

Space mean speed is the average speed of all vehicles occupying a given section of highway or lane over some specified time period.

Step 1:

The speeds of the four vehicles in the 1-mile freeway segment are

$$\text{Vehicle 1: } v_1 = \frac{1 \text{ mile}}{50 \text{ s}} = \frac{1 \text{ mile}}{50 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} = 72 \text{ mph}$$

$$\text{Vehicle 2: } v_2 = \frac{1 \text{ mile}}{55 \text{ s}} = \frac{1 \text{ mile}}{55 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} \cong 65.45 \text{ mph}$$

$$\text{Vehicle 3: } v_3 = \frac{1 \text{ mile}}{54 \text{ s}} = \frac{1 \text{ mile}}{54 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} \cong 66.67 \text{ mph}$$

$$\text{Vehicle 4: } v_4 = \frac{1 \text{ mile}}{48 \text{ s}} = \frac{1 \text{ mile}}{48 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} = 75 \text{ mph}$$

Step 2:

The space mean speed is

$$v_s = \left(\frac{\sum_{i=1}^4 \frac{1}{v_i}}{4} \right)^{-1} = \left(\frac{\frac{1}{72 \text{ mph}} + \frac{1}{65.45 \text{ mph}} + \frac{1}{66.67 \text{ mph}} + \frac{1}{75 \text{ mph}}}{4} \right)^{-1} \\ \cong 69.57 \text{ mph}$$

40. The following travel times were measured for vehicles as they traversed a 1-mile segment of freeway at constant speeds. What is the time mean speed for this data?

Vehicle ID	1	2	3	4
Tavel Time (s)	50	55	54	48

A) 69.78 mph

B) 70.57 mph

C) 69.57 mph

D) 68.78 mph

The Answers is A

Time mean speed is the average speed of all vehicles passing a point on a highway or lane over some specified time period.

Step 1:

The speeds of the four vehicles in the 1-mile freeway segment are

$$\text{Vehicle 1: } v_1 = \frac{1 \text{ mile}}{50 \text{ s}} = \frac{1 \text{ mile}}{50 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} = 72 \text{ mph}$$

$$\text{Vehicle 2: } v_2 = \frac{1 \text{ mile}}{55 \text{ s}} = \frac{1 \text{ mile}}{55 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} \cong 65.45 \text{ mph}$$

$$\text{Vehicle 3: } v_3 = \frac{1 \text{ mile}}{54 \text{ s}} = \frac{1 \text{ mile}}{54 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} \cong 66.67 \text{ mph}$$

$$\text{Vehicle 4: } v_4 = \frac{1 \text{ mile}}{48 \text{ s}} = \frac{1 \text{ mile}}{48 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} = 75 \text{ mph}$$

Step 2:

The time mean speed is

$$v_t = \frac{\sum_{i=1}^4 v_i}{4} = \frac{72 \text{ mph} + 65.45 \text{ mph} + 66.67 \text{ mph} + 75 \text{ mph}}{4} \cong 69.78 \text{ mph}$$

41. Assume that a speed-density study has resulted in the following calibrated relationship between space mean speed (v , mph) and density (k , veh/mi/ln): $v = 55 - 0.45k$. what are the jam density and free-flow speed?

A) 0 veh/mi/ln, 55 mph

B) 122.2 veh/mi/ln, 0 mph

C) 122.2 veh/mi/ln, 55 mph

D) 120 veh/mi/ln, 50 mph

The Answer is C**Step 1:**

The jam density occurs when the speed is 0 mph, or:

$$v = 55 - 0.45k = 0$$

$$k = \frac{55}{0.45} = 122.2 \text{ veh/mi/ln}$$

Step 2:

The free-flow speed occurs when the density is 0 veh/mi/ln, or:

$$v = 55 - 0.45 \times 0 = 55 \text{ mph}$$

42. Assume that for a highway section, a speed-density study has resulted in the following calibrated relationship between space mean speed (v , mph) and density (k , veh/mi/ln): $v = 55 - 0.45k$. what is the capacity of the highway section?

- A) 1860 veh/hr/ln
- C) 1568 veh/hr/ln

- B) 5400 veh/hr/ln
- D) 1680 veh/hr/ln

The Answers is D

Capacity is found by determining the peak of the speed-flow or flow-density curves.

Step 1:

The three macroscopic measures of the state of a given traffic stream are related by

$$q = kv \quad (1)$$

in which q is rate of flow (veh/hr/ln), v is speed (mph), and k is density (veh/mi/ln).

Since it is given that

$$v = 55 - 0.45k \quad (2)$$

combining equations (1) and (2) gives

$$q = 55.0k - 0.45k^2 \quad (3)$$

Step 2:

From equation (3), the peak q occurs when

$$\frac{dq}{dk} = 0 = 55.0 - 0.90k, \text{ or } k = \frac{55.0}{0.90} = 61.1 \text{ veh/mi/ln}$$

Step 3:

From equation (3), when $k = 61.1$ veh/mi/ln, q is

$$q = 55.0 \times 61.1 - 0.45(61.1)^2 \cong 1680 \text{ veh/hr/ln}$$

43. Assume that for a highway section, a speed-density study has resulted in the following calibrated relationship between space mean speed (v , mph) and density (k , veh/mi/ln): $v = 60.5e^{-0.015k}$. what is the capacity of the highway section?

- A) 1460 veh/hr/ln
- C) 1468 veh/hr/ln

- B) 1400 veh/hr/ln
- D) 1484 veh/hr/ln

The Answers is D

Capacity is found by determining the peak of the speed-flow or flow-density curves.

Step 1:

The three macroscopic measures of the state of a given traffic stream are related by

$$q = kv \quad (1)$$

in which q is rate of flow (veh/hr/ln), v is speed (mph), and k is density (veh/mi/ln).

Since it is given that

$$v = 60.5e^{-0.015k} \quad (2)$$

combining equations (1) and (2) gives

$$q = k60.5e^{-0.015k} \quad (3)$$

Step 2:

From equation (3), the peak q occurs when

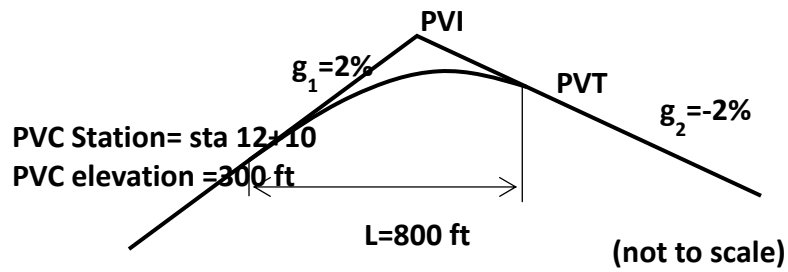
$$\frac{dq}{dk} = 0 = 60.5e^{-0.015k} + (-0.015)60.5ke^{-0.015k}, \text{ or } k = \frac{1}{0.015} \cong 66.67 \text{ veh/mi/ln}$$

Step 3:

From equation (3), when $k = 66.67$ veh/mi/ln, q is

$$q = 66.67 \times 60.5e^{-0.015 \times 66.67} \cong 1484 \text{ veh/hr/ln}$$

44. In the vertical curve shown below, what is the elevation of point of vertical tangency (PVT)?



- A) 300 ft B) 320 ft C) 340 ft D) 295 ft

The Answers is A

The elevation of the point at the same horizontal distance from PVC as PVT but on the back tangent line is

$$\text{PVC elevation} + g_1 L = 300 \text{ ft} + 2\% \times (800 \text{ ft}) = 316 \text{ ft}$$

The vertical distance from the previous point to PVT is tangent offset y , which can be calculated by

$$y = \frac{g_2 - g_1}{2L} L^2 = \frac{-2\% - 2\%}{2} \times 800 \text{ ft} = -16 \text{ ft}$$

The elevation of PVT is

$$316 \text{ ft} + y = 316 \text{ ft} - 16 \text{ ft} = 300 \text{ ft}$$

Short Cut

Since the slope of the back tangent is the same as the slope of the forward tangent, the vertical curve is symmetric about the vertical line through PVI. Therefore, the elevation of PVT is the same as the elevation of PVC, which is 300 ft.

45. The following travel times were measured for vehicles as they traversed a 3-mile segment of freeway at constant speeds. What is the difference between the time mean speed and space mean speed for this data?

Vehicle ID	1	2	3	4
Tavel Time (s)	210	218	226	202

A) 0.09 mph

B) -0.02 mph

C) -1.3 mph

D) -0.23 mph

The Answers is A**Step 1:**

The speeds of the four vehicles in the 3-mile freeway segment are

$$\text{Vehicle 1: } v_1 = \frac{3 \text{ miles}}{210 \text{ s}} = \frac{3 \text{ miles}}{210 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} \cong 51.43 \text{ mph}$$

$$\text{Vehicle 2: } v_2 = \frac{3 \text{ miles}}{218 \text{ s}} = \frac{3 \text{ miles}}{218 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} \cong 49.54 \text{ mph}$$

$$\text{Vehicle 3: } v_3 = \frac{3 \text{ miles}}{226 \text{ s}} = \frac{3 \text{ miles}}{226 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} \cong 47.79 \text{ mph}$$

$$\text{Vehicle 4: } v_4 = \frac{3 \text{ miles}}{202 \text{ s}} = \frac{3 \text{ miles}}{202 \text{ s}} \times 3600 \frac{\text{s}}{\text{hr}} \cong 53.47 \text{ mph}$$

Step 2:

The time mean speed is

$$v_t = \frac{51.43 + 49.54 + 47.79 + 53.47}{4} \text{ mph} \cong 50.56 \text{ mph}$$

The space mean speed is

$$v_s = \left(\frac{\sum_{i=1}^4 \frac{1}{v_i}}{4} \right)^{-1} = \left(\frac{\frac{1}{51.43 \text{ mph}} + \frac{1}{49.54 \text{ mph}} + \frac{1}{47.79 \text{ mph}} + \frac{1}{53.47 \text{ mph}}}{4} \right)^{-1} \cong 50.47 \text{ mph}$$

Step 3:

The difference between time mean speed and space mean speed is $50.56 - 50.47 = 0.09 \text{ mph}$

Short Cut

Since the space mean speed weights slower vehicles more heavily, based on the amount of time they occupy a highway section. Therefore, space mean speed is always lower than or equal to time mean speed. So the difference between time mean speed and space mean speed should be nonnegative. Only option (a) satisfies this condition.

46. A sag vertical curve has an elevation of 534.43 ft at station 12+15 and an elevation of 345.23 ft at the point of vertical curvature (PVC). For an overpass crossing the vertical curve's roadway at station 12+15, the required vertical clearance is 14.8 ft. What is the minimum elevation of the overpass?

- A) 360.03 ft B) 542.23 ft C) 549.23 ft D) 558.23 ft

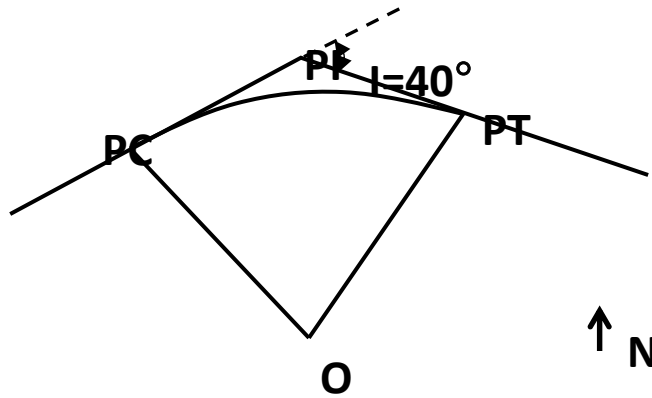
The Answer is C

The elevation of the overpass should be no less than the elevation of the vertical curve plus the required vertical clearance. Since at Station 12+15, the elevation is 534.43 ft and the required vertical clearance is 14.8 ft, the minimum elevation of the overpass is

$$534.43 \text{ ft} + 14.8 \text{ ft} = 549.23 \text{ ft}$$

Note: the information of the vertical curve shape and the elevation at the PVC does not affect the solution procedure.

47. The centerline of a two-lane roadway is a horizontal circular curve, as shown below. Each lane is 12 ft wide and there is a shoulder of 3 ft wide on both directions. It is known that the PC station is sta 7+50. The curve radius is 1840 ft. The intersection angle I is 40° . What is the station of Point of Tangent (PT)?



- A) sta 20+34.56 B) sta 20+60.56 C) sta 20+56.34 D) sta 20+35.46

The Answers is A**Step 1:**

The length of curve from PC to PT is

$$L = \frac{R\theta\pi}{180^\circ} = \frac{(1840 \text{ ft}) \times (40^\circ) \times \pi}{180^\circ} \cong 1284.56 \text{ ft.}$$

Step 2:

The station of PT is

$$\text{Sta PT} = \text{sta PC} + L = (\text{sta 7} + 50) + 1284.56 \text{ ft} = \text{sta 20} + 34.56$$

Note: the information of the wide of the lanes and the existence of shoulder and shoulder width does not affect the station of PT.

48. The intersection of Elk Street and Deer Street is signalized with a cycle length of 60 s, and effective green time of 32 s for the one-lane westbound approach of Elk Street. If at a specific time, it was observed that the westbound approach of the Elk Street had 12 vehicles over a length of 1 mile, among which 6 vehicles were at a speed of 30 mph and the other 6 vehicles at a speed of 28 mph, what is the flow rate at that time on the westbound approach of the Elk Street?

A) 350.6 veh/hr/ln

B) 374.6 veh/hr/ln

C) 347.6 veh/hr/ln

D) 367.4 veh/hr/ln

The Answer is C

Step 1:

The three macroscopic measures of the state of a given traffic stream are related by $q = kv$, in which q is rate of flow (veh/hr/ln), v is speed (mph), and k is density (veh/mi/ln).

Step 2:

From the given information, it is known that the density (k) is 12 veh/mi/ln. The space mean speed (v) is

$$v = \left(\frac{\sum_{i=1}^N \frac{1}{v_i}}{12} \right)^{-1} = \left(\frac{\frac{6}{30 \text{ mph}} + \frac{6}{28 \text{ mph}}}{12} \right)^{-1} \cong 28.97 \text{ mph}$$

Step 3:

The flow rate (q) on the westbound approach of the Elk Street is

$$q = kv = 12 \frac{\text{veh}}{\text{mi/ln}} \times 28.97 \text{ mph} \cong 347.6 \text{ veh/hr/ln}$$

Note: the signal cycle length and effective green time affect capacity of the Elk Street approach but do not affect the flow rate.

49. If it was observed that the saturation headway is 2.4 s/vehicle/lane on a freeway segment, and at a specific time the density is 20 veh/mi/ln and the space mean speed is 65 mph on that segment. What is the capacity per lane of that freeway segment?

- | | |
|----------------------|----------------------|
| A) 1,800 veh/hr/lane | B) 1,500 veh/hr/lane |
| C) 1300 veh/hr/lane | D) 1,600 veh/hr/lane |

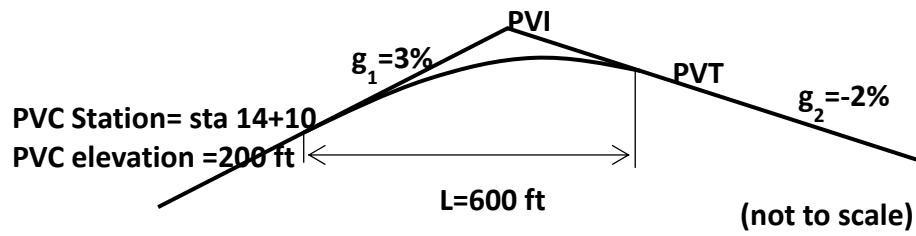
The Answers is B

Saturation headway is the headway when the traffic volume reaches the freeway capacity. Since the saturation headway (h) is 2.4 s/veh/lane, the flow rate (s) (maximum number of vehicles that can traverse the freeway segment per hour) is $s = \frac{3600 \text{ s/hr}}{h} = \frac{3600 \text{ s/hr}}{2.4 \text{ s/veh/lane}} \cong$

1500 veh/hr/lane.

Note: the highway capacity is maximum flow rate. It does not change with the density and space mean speed measured at any time. The information of density and space mean speed observed at any time cannot be used to calculate capacity.

50. In the vertical curve shown below, what is the elevation of point of vertical tangency (PVT)?



- A) 203 ft B) 210 ft C) 197 ft D) 205 ft

The Answer is A**Step 1:**

The elevation of the point at the same horizontal distance from PVC as PVT but on the back tangent line is

$$\text{PVI elevation} + g_1 L = 200 \text{ ft} + 3\% \times (600 \text{ ft}) = 218 \text{ ft}$$

Step 2:

The vertical distance from the previous point to PVT is tangent offset y , which can be calculated by

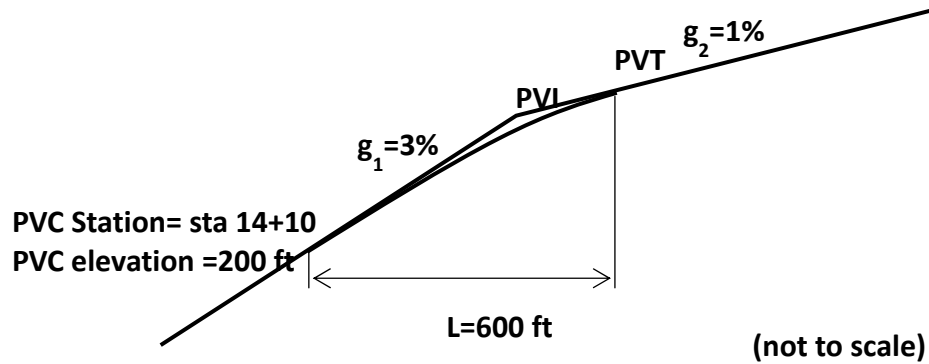
$$y = \frac{g_2 - g_1}{2L} L^2 = \frac{-2\% - 3\%}{2} \times 600 \text{ ft} = -15 \text{ ft}$$

Step 3:

The elevation of PVT is

$$218 \text{ ft} + y = 218 \text{ ft} - 15 \text{ ft} = 203 \text{ ft}$$

51. In the vertical curve shown below, what is the elevation of the highest point on the vertical curve?



- A) 285 ft B) 245 ft C) 218 ft D) 212 ft

The Answers is D**Step 1:**

Note in this problem both slopes of the back tangent and forward tangent are positive.

The distance (x_m) between the maximum elevation point and PVC is

$$x_m = \frac{g_1 L}{g_1 - g_2} = \frac{(3\%) \times (600 \text{ ft})}{(3\% - 1\%)} = 900 \text{ ft}$$

Since x_m is larger than the length of the vertical curve (L), this highest point has exceeded the range of vertical curve. Therefore, the highest point on the vertical curve is PVT.

Step 2:

The elevation of the point at the same horizontal distance from PVC as PVT but on the back tangent line is

$$\text{PVI elevation} + g_1 L = 200 \text{ ft} + 3\% \times (600 \text{ ft}) = 218 \text{ ft}$$

Step 3:

The vertical distance from the previous point to PVT is tangent offset y , which can be calculated by

$$y = \frac{g_2 - g_1}{2L} L^2 = \frac{1\% - 3\%}{2} \times 600 \text{ ft} = -6 \text{ ft}$$

Step 4:

The elevation of PVT is

$$218 \text{ ft} + y = 218 \text{ ft} - 6 \text{ ft} = 212 \text{ ft}$$

Therefore, the elevation of the highest point on the vertical curve is 212 ft.

52. If at a specific time, a freeway lane were observed to have 25 vehicles over a length of 1 mile, and 12 vehicles were at a speed of 60 mph, and the other 13 vehicles were at a speed of 50 mph, what is the flow rate at that time in the lane?

- A) 1,500 veh/hr/ln
C) 1,358.7 veh/hr/ln

- B) 1,370 veh/hr/ln
D) 1,538.7 veh/hr/ln

The Answers is C**Step 1:**

The three macroscopic measures of the state of a given traffic stream are related by $q = kv$, in which q is rate of flow (veh/hr/ln), v is speed (mph), and k is density (veh/mi/ln).

Step 2:

From the given information, it is known that the density (k) is 25 veh/mi/ln. The space mean speed (v) is

$$v = \left(\frac{\sum_{i=1}^N \frac{1}{v_i}}{N} \right)^{-1} = \left(\frac{\frac{12}{60 \text{ mph}} + \frac{13}{50 \text{ mph}}}{25} \right)^{-1} \cong 54.35 \text{ mph}$$

Step 3:

The flow rate (q) in the lane is

$$q = kv = 25 \frac{\text{veh}}{\text{mi}} \times 54.35 \frac{\text{mi}}{\text{hr}} \cong 1358.7 \text{ veh/hr/ln}$$

Part 7 : Materials

97 Problems

1) A sieve analysis on a non-organic soil reveals that 82% of the soil passes No. 200 sieve (0.075 mm). The liquid limit of the soil is 24%, and the soil has a medium toughness and a very slow dilatancy. Classify the soil according to the Unified Soil Classification system (USCS).

- A) GC B) ML C) CL D) CH

The Answer is C**Step 1:**

Because 82% of the soil is smaller than No. 200 sieve size, which is larger than 50%, the soil is fine-grained according to the Unified Soil Classification table

Step 2:

Because the liquid limit (24%) is less than 50%, the soil is low plastic. Therefore, the second letter of the USCS group symbol is L.

Step 3:

Because the soil is non-organic, the first letter of the USCS group symbol is not O. Because the soil has a medium toughness and a very slow dilatancy, it contains more clay than silt. The first letter of the USCS group symbol is C according to the Unified Soil Classification table.

Therefore, the soil is classified as CL according to USCS.

- 2) Particle size analysis was carried out on a soil with the results shown in the following table. Classify the soil according to the Unified Soil Classification system (USCS).

Size	Percent Passing
12.5 mm	100
9.5 mm	60
No. 4 (4.75 mm)	40
No. 20 (0.85 mm)	30
No. 60 (0.25 mm)	10
No. 200 (0.075 mm)	4

- A) GC B) GP C) SP D) SM

The Answers is B**Step 1:**

Because only 5% of the soil is smaller than No. 200 sieve size, the soil is coarse-grained according to the Unified Soil Classification table

Step 2:

The gravel fraction of the soil is those larger than No. 4 sieve size, which is $(100-40)=60$. The sand fraction of the soil is those larger than No. 200 sieve size but less than No. 4 sieve size, which is $(40-5)=35$. Since $60>35$, there are more gravels than sands in the coarse fraction of the soil. The first letter of the USCS group symbol is G according to the Unified Soil Classification table

Step 3:

Because the clay+silt fraction of the soil (fraction smaller than No. 200 sieve size) is very small (4%), the gravel contains little fines (clean). Therefore, the second letter of the USCS group symbol is either W or P.

Step 4:

From the gradation curve table, it is known that the grain size corresponding to 10% finer on grain size curve (D_{10}) is 0.25 mm, the grain size corresponding to 30% finer on grain size curve (D_{30}) is 0.85 mm, the grain size corresponding to 60% finer on grain size curve (D_{60}) is 9.5 mm. Therefore, the coefficient of uniformity (C_U) is

$$C_U = \frac{D_{60}}{D_{10}} = \frac{9.5 \text{ mm}}{0.25 \text{ mm}} = 38$$

The coefficient of curvature (C_C) is

$$C_C = \frac{(D_{30})^2}{D_{10}D_{60}} = \frac{(0.85 \text{ mm})^2}{(0.25 \text{ mm})(9.5 \text{ mm})} = 0.30$$

Step 5:

Because C_c is outside of the range of 1 to 3, the soil is poorly graded. Therefore, the second letter of the USCS group symbol is P.

Therefore, the soil is classified as GP according to USCS.

- 3) The results of a particle size analysis of a soil are given in the following table. No Atterberg limit tests were conducted. Classify the soil according to the AASHTO system.

Size	Percent Passing
9.5 mm	100
No. 4 (4.75 mm)	88
No. 10 (2.00 mm)	70
No. 20 (0.850mm)	60
No. 40 (0.425 mm)	52
No. 100 (0.150 mm)	29
No. 200 (0.075 mm)	4

- A) A-3 B) A-2-5 C) A-2-7 D) A-4

The Answer is A

Step 1:

Since the soil fraction passing No. 200 sieve size is very small (4%), the soil is coarse-grained. Atterberg limit tests are not needed for this soil.

Step 2:

Since the fraction passing No. 40 sieve (52%) is greater than 51%, and the fraction passing No. 200 sieve (4%) is less than 10%, the soil is A-3.

GENERAL CLASSIFICATION	GRANULAR MATERIALS (35% OR LESS PASSING 0.075 SIEVE)							SILT-CLAY MATERIALS (LESS THAN 35% PASSING 0.075 SIEVE)			
GROUP CLASSIFICATION	A-1		A-3	A-2				A-4	A-5	A-6	A-7-5 A-7-6
	A-1-a	A-1-b		A-2-4	A-2-5	A-2-6	A-2-7				
SIEVE ANALYSIS, PERCENT PASSING: 2.00 mm (No. 10) 0.425 mm (No. 40) 0.075 mm (No. 200)	≤ 50 ≤ 30 ≤ 15	— ≤ 50 ≤ 25	— ≥ 51 ≤ 10	— — ≤ 35	— — ≤ 35	— — ≤ 35	— — ≤ 35	— — ≥ 36	— — ≥ 36	— — ≥ 36	— — ≥ 36
CHARACTERISTICS OF FRACTION PASSING 0.425 SIEVE (No. 40): LIQUID LIMIT PLASTICITY INDEX *	— 6 max		— NP	≤ 40 ≤ 10	≥ 41 ≤ 10	≤ 40 ≤ 11	≥ 41 ≥ 11	≤ 40 ≤ 10	≥ 41 ≤ 10	≤ 40 ≥ 11	≥ 41 ≥ 11
USUAL TYPES OF CONSTITUENT MATERIALS	STONE FRAGMNTS, GRAVEL, SAND		FINE SAND	SILTY OR CLAYEY GRAVEL AND SAND				SILTY SOILS		CLAYEY SOILS	
GENERAL RATING AS A SUBGRADE	EXCELLENT TO GOOD							FAIR TO POOR			

4) The results of a particle size analysis of a soil are given in the following table. No Atterberg limit tests were conducted. Classify the soil according to the USCS system.

Size	Percent Passing
9.5 mm	100
No. 4 (4.75 mm)	88
No. 10 (2.00 mm)	70
No. 20 (0.850mm)	60
No. 40 (0.425 mm)	52
No. 100 (0.150 mm)	29
No. 200 (0.075 mm)	4

- A) GC B) GP C) SP D) SM

The Answer is C**Step 1:**

Because only 4% of the soil is smaller than No. 200 sieve size, the soil is coarse-grained according to the Unified Soil Classification table.

Step 2:

The gravel fraction of the soil (fraction larger than No. 4 sieve size) is $(100-88)=12$. The sand fraction of the soil (fraction larger than No. 200 sieve size but smaller than No. 4 sieve size) is $(88-4)=84$. Since $12 < 84$, there are more sands than gravels in the coarse fraction of the soil. The first letter of the USCS group symbol is S according to the Unified Soil Classification table.

Step 3:

Because the clay+silt fraction of the soil (fraction smaller than No. 200 sieve size) is very small (4%), the gravel contains little fines (clean). Therefore, the second letter of the USCS group symbol is either W or P.

Step 4:

From the gradation table, it is known that the grain size corresponding to 10% finer on grain size curve (D_{10}) is between 0.15 mm and 0.075 mm, the grain size corresponding to 30% finer on grain size curve (D_{30}) is slightly larger than 0.15 mm, the grain size corresponding to 60% finer on grain size curve (D_{60}) is 0.85 mm. Therefore, the coefficient of uniformity (C_U) is

$$C_U = \frac{D_{60}}{D_{10}} = \left\{ \frac{0.85 \text{ mm}}{0.15 \text{ mm}}, \frac{0.85 \text{ mm}}{0.075 \text{ mm}} \right\} = \{5.7, 11.3\}$$

That is, C_U is between 5.7 and 11.3, which is larger than 4.

The coefficient of curvature (C_C) is

$$C_C = \frac{(D_{30})^2}{D_{10}D_{60}} \approx \left\{ \frac{(0.15 \text{ mm})^2}{(0.15 \text{ mm})(0.85 \text{ mm})}, \frac{(0.15 \text{ mm})^2}{(0.075 \text{ mm})(0.85 \text{ mm})} \right\} = \{0.176, 0.353\}$$

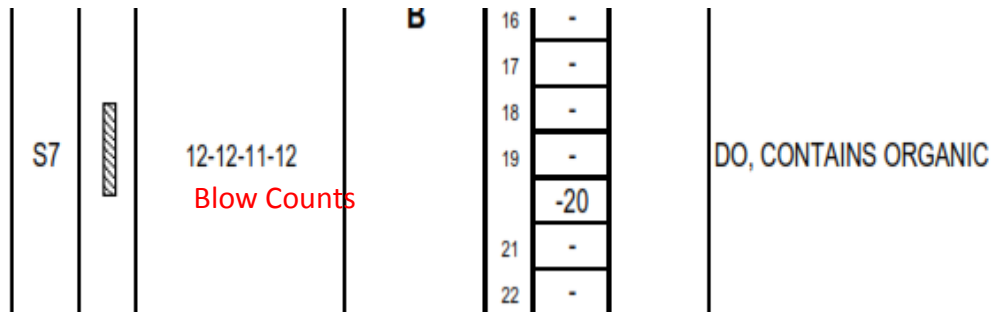
That is, C_C is approximately in the range of 0.176 to 0.353, which is smaller than 1.

Step 5:

Because C_C is outside of the range of 1 to 3, the soil is poorly graded. Therefore, the second letter of the USCS group symbol is P.

Therefore, the soil is classified as SP according to USCS.

5) The following picture shows a part of a boring log. Considering the number of blow counts, what is the N-value for this particular layer?



A) 24

B) 23

C) 12

D) 11

The Answers is B

The blow counts are:

12-12-11-12

Before, N-Value is calculated by summing number of blows required for the second and third 6 inches of penetration.

Therefore:

$$\text{N-Value} = 12 + 11 = 23$$

6) A pressure of 6.9 MPa is required to penetrate the CBR piston into a standard well-graded soil for 0.1 inch. We have taken a sample from the field and have run the CBR test on it. If we need to apply a 4.3 MPa pressure to penetrate 0.1 inch into this soil, how much is the California Bearing Ratio?

A) 62%

B) 38%

C) 160%

D) 59%

The Answer is A

The following formula is used to calculate the CBR value:

$$\text{CBR}(\%) = 100 \times \frac{P}{P_s} = 100 \times \frac{4.3}{6.9} = 62\%$$

- 7) The subgrade layer of a pavement section is to be constructed from a soil compacted to a dry unit weight of 19 kN/m^3 . The soil is transported from a borrow pit where the soil has a bulk unit weight of 19 kN/m^3 and a natural water content of 3%. Determine the volume of soil from the borrow pit required for 1 cubic meter of subgrade layer.
- A) 0.97 m^3 B) 0.99 m^3 C) 1.00 m^3 D) 1.03 m^3

The Answer is D**Step 1:**

Determine the dry unit weight of the soil in the borrow pit.

$$(\gamma_D)_{\text{borrow pit}} = \frac{\gamma}{(1 + \omega)} = \frac{19 \text{ kN/m}^3}{(1 + 3\%)} = 18.4466 \text{ kN/m}^3$$

Step 2:

Determine the volume of soil from the borrow pit required for 1 cubic meter of subgrade layer with a dry unit weight of 19 kN/m^3 .

$$V_{\text{borrow pit}} = \frac{(\gamma_D)_{\text{subgrade}} V_{\text{subgrade}}}{(\gamma_D)_{\text{borrow pit}}} = \frac{19 \text{ kN/m}^3 \times 1 \text{ m}^3}{18.4466 \text{ kN/m}^3} = 1.03 \text{ m}^3$$

Short cut

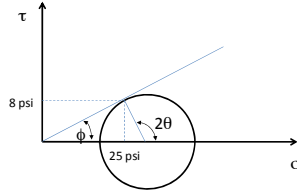
Since the dry unit weight of the subgrade soil is greater than the dry unit weight of the soil in the borrow pit, more than 1 m^3 borrow pit soil will be needed for 1 m^3 subgrade soil. Among the four options, only (d) satisfies this condition.

8) A triaxial shear test is performed on a well-drained sand soil sample. At failure, the normal stress on the failure plane was 25 psi and the shear stress on the failure plane was 8 psi. What is the angle of the failure plane with respect to the plane of major principal stress?

- A) 30° B) 40° C) 45° D) 54°

The Answer is D**Step 1:**

For a well-drained sand sample, cohesion is 0. Use Mohr's circle,



The angle of internal friction is

$$\phi = \arctan\left(\frac{8 \text{ psi}}{25 \text{ psi}}\right) = \arctan(0.32) = 17.7^\circ$$

Step 2:

Let θ be the angle of the failure plane with respect to the plane of major principal stress, then

$$\theta = \frac{1}{2}(90^\circ + \phi) = \frac{1}{2}(90^\circ + 17.7^\circ) = 53.9^\circ$$

Short cut

Since $\theta = 45^\circ + 0.5\phi$ and ϕ (angle of internal friction) is non-negative, θ should be no less than 45° . Among the four options, only (d) satisfies this condition.

- 9) Particle size analysis on a soil reveals the gradation as shown in the table below. The liquid limit of the soil fraction passing No. 40 sieve is 46% and the plastic limit is 28%. Classify the soil according to the AASHTO system.

Size	Percent Passing
12.5 mm	100
9.5 mm	80
No. 4 (4.75 mm)	40
No. 30 (0.60 mm)	35
No. 100 (0.15 mm)	30
No. 200 (0.075 mm)	26

- A) A-3 B) A-2-5 C) A-2-7 D) A-4

The Answer is C**Step 1:**

Given the liquid limit (LL) and the plastic limit (PL), the plasticity index (PI) can be calculated by $\text{PI} = \text{LL} - \text{PL} = 46\% - 28\% = 18\%$.

Step 2:

Since the LL and PI information is available from the soil, the only other information that is needed to classify the soil according to AASHTO is the percent passing of No. 200 sieve size, which is 26% as given in the table.

Step 3:

Since $26\% < 35\%$, $46\% (\text{LL}) > 41\%$, and $18\% (\text{PI}) > 11\%$, the soil is A-2-7.

0

10) A liquid limit test was conducted on a soil sample in the Casagrande's cup device. A best-fit straight line was drawn from the test data on a semilogarithmic plot of water content versus number of blows. On this best-fit line, water contents corresponding to different number of blows are given in the table below. Two determinations for the plastic limit gave water contents of 20.5% and 20.7%. Determine the liquid limit of the soil.

Number of blows	Water content (%)
40	25
30	36
25	38
20	45

- A) 45 B) 38 C) 25 D) 17

The Answer is B

In the Casagrande cup method, the liquid limit is defined as the water content at which the groove cut into the soil will close over a distance of 12.5 mm following 25 blows. From the given table, it can be seen that the water content corresponding to 25 blows is 38%. Therefore, the liquid limit is 38%.

Note: liquid limit is independent of plastic limit. Therefore, the plastic limit test data is not needed to determine the liquid limit.

11) A soil has a total unit weight of 120 lbf/ft^3 and a dry unit weight of 105 lbf/ft^3 . It is known that the soil solids have a specific gravity of 2.65. Determine the water content of the soil.

- A) 10.4% B) 14.3% C) 25.2% D) 30.5%

The Answer is B**Step 1:**

Water content (ω) is defined as the ratio of water mass (W_W) and soil solids mass (W_S):

$$\omega = \frac{W_W}{W_S} \times 100\%$$

Step 2:

Assume the soil has a total volume of V , then $W_S = \gamma_D V$, and $W_S = \gamma V - \gamma_D V$.

Therefore,

$$\omega = \frac{W_W}{W_S} \times 100\% = \frac{\gamma V - \gamma_D V}{\gamma_D V} \times 100\% = \frac{\gamma - \gamma_D}{\gamma_D} \times 100\% = \frac{120 \frac{\text{lb}_f}{\text{ft}^3} - 105 \frac{\text{lb}_f}{\text{ft}^3}}{105 \frac{\text{lb}_f}{\text{ft}^3}} \times 100\% \cong$$

14.3%.

Note: the specific gravity of soil solids is not needed since no volume calculation is involved in determining water content.

12) The subgrade of a highway is compacted from a natural soil layer. It is known that the natural soil has a total unit weight of 20 kN/m^3 , and a dry unit weight of 18 kN/m^3 . It needs to be compacted to attain a dry unit weight of 19 kN/m^3 and a water content of 15%. What is the total unit weight of the compacted soil?

A) $21.85 \frac{\text{kN}}{\text{m}^3}$

B) $22.85 \frac{\text{kN}}{\text{m}^3}$

C) $23.85 \frac{\text{kN}}{\text{m}^3}$

D) $24.85 \frac{\text{kN}}{\text{m}^3}$

The Answer is A**Step 1:**

From the given information, it is known that after compaction the subgrade soil has a dry unit weight (γ_D) of 19 kN/m^3 and a water content (ω) of 15%.

Step 2:

The total unit weight of the compacted soil is

$$\gamma = \gamma_D(1 + \omega) = 19 \frac{\text{kN}}{\text{m}^3} \times (1 + 15\%) = 21.85 \frac{\text{kN}}{\text{m}^3}$$

Note: It can be calculated to see that the water content of the compacted soil is higher than that of the uncompacted soil, therefore water must be added during compaction. However, this is irrelevant to the question in this problem.

13) In a permeability test, head of water, sample height, and area of specimen are 100 cm, 35 cm, and 25 cm², respectively. 200 cm³ water is discharged in 1.5 minutes. Calculate the coefficient of permeability:

A) 0.052

B) 0.038

C) 0.048

D) 0.031

The Answer is D

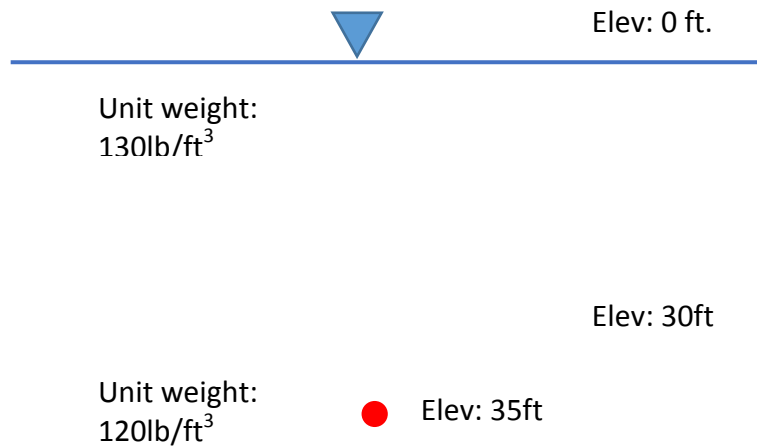
First, we have to calculate the hydraulic gradient:

$$i = \frac{H}{L} = \frac{100}{35} = 2.86$$

Then, we can calculate the coefficient of permeability using the following equation:

$$K = \frac{Q}{i.A.t} = \frac{200}{2.86 \times 25 \times 90} = 0.031 \text{ cm/s}$$

14) The clay stratum is shown in the profile below. It is known that the voids ratio of the red point is 0.9. When subjected to 4000lb/ft^2 , the voids ratio is 0.8. Determine the compression index.



A) 0.48

B) 0.38

C) 0.42

D) 0.45

The Answer is C

Step1: Initial effective consolidation stress can be calculated based on the information mentioned above,

$$p_0 = \gamma_1' H_1 + \gamma_2' H_2$$

$$p_0 = (130\text{lb}/\text{ft}^3 - 62.5\text{lb}/\text{ft}^3)(30\text{ft}) + (120\text{lb}/\text{ft}^3 - 62.5\text{lb}/\text{ft}^3)(35\text{ft} - 30\text{ft})$$

$$= 2313\text{lb}/\text{ft}^2$$

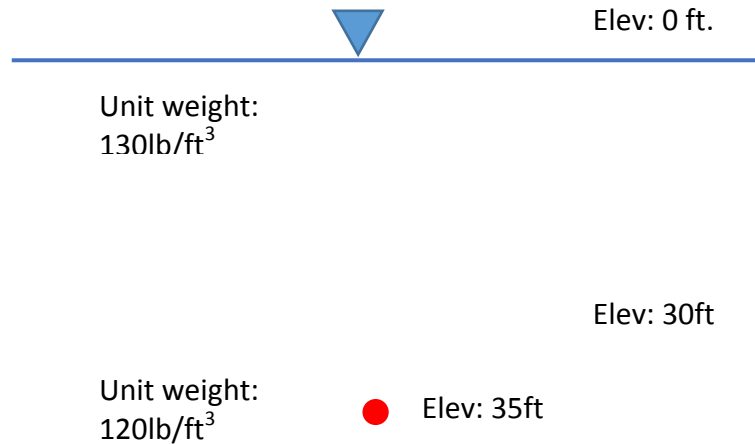
Step2: Based on the equation shown below, the compression index can be solved by,

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)} = \frac{0.9 - 0.8}{\log \left(\frac{4000\text{lb}/\text{ft}^2}{2313\text{lb}/\text{ft}^2} \right)} = 0.42$$

$$\sim 0.42$$

$$\sim 2600\text{lb}/\text{ft}^2$$

15) The clay stratum is shown in the profile below. It is known that the voids ratio of the red point is 0.9. Assume the compression index is known to be 0.42. Determine the voids ratio when subjecting to 4000 lb/ft^2 .



A) 0.5

B) 0.6

C) 0.7

D) 0.8

The Answer is D

Step 1: Initial effective consolidation stress can be calculated based on the information mentioned above,

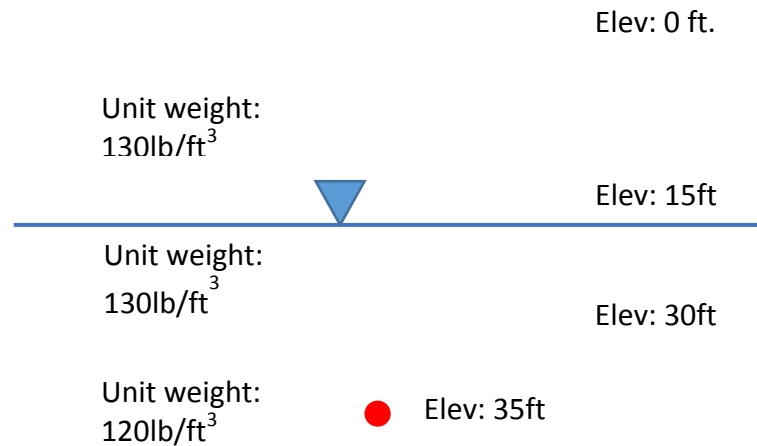
$$p_0 = \gamma_1' H_1 + \gamma_2' H_2$$

$$p_0 = (130\text{lb}/\text{ft}^3 - 62.5\text{lb}/\text{ft}^3)(30\text{ft}) + (120\text{lb}/\text{ft}^3 - 62.5\text{lb}/\text{ft}^3)(35\text{ft} - 30\text{ft}) = 2313\text{lb}/\text{ft}^2$$

Step 2: Based on the equation shown below, the voids ratio can be solved by,

$$e_2 = e_1 - C_c \log \left(\frac{p_2}{p_1} \right) = 0.9 - (0.42) \left(\log \left(\frac{4000\text{lb}/\text{ft}^2}{2313\text{lb}/\text{ft}^2} \right) \right) = 0.8 \sim 0.8$$

16) The clay stratum is shown in the profile below. It is known that the voids ratio of the red point is 0.9. When subjected to 5000lb/ft^2 , the voids ratio is 0.8. Determine the compression index.



A) 0.48

B) 0.38

C) 0.42

D) 0.53

The Answer is D

Step1: Initial effective consolidation stress can be calculated based on the information mentioned above,

$$p_0 = \gamma_1' H_1 + \gamma_2' H_2 + \gamma_3' H_3$$

$$p_0 = (130\text{lb}/\text{ft}^3)(15\text{ft}) + (130\text{lb}/\text{ft}^3 - 62.5\text{lb}/\text{ft}^3)(15\text{ft}) + (120\text{lb}/\text{ft}^3 - 62.5\text{lb}/\text{ft}^3)(5\text{ft}) = 3250\text{lb}/\text{ft}^2$$

Step2: Based on the equation shown below, the compression index can be solved by,

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_2}{\log\left(\frac{p_2}{p_1}\right)} = \frac{0.9 - 0.8}{\log\left(\frac{5000\text{lb}/\text{ft}^2}{3250\text{lb}/\text{ft}^2}\right)} = 0.53 \sim 0.53 \sim 2600\text{lb}/\text{ft}^2$$

17) Assume that the soil is in the range of virgin compression. Under 3000 lb/ft^2 , the porosity of soil is 0.5. Under 6000 lb/ft^2 , the porosity of soil is 0.45. Calculate the compression index.

A) 0.33

B) 0.45

C) 0.65

D) 0.28

The Answer is C

Step 1: Based on the equation shown below,

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)}$$

From this problem, information about two points with pressure is directly given. The voids ratios need to be calculated.

Step 2: Under 3000lb/ft², porosity is given as 0.5. Therefore, voids ratio can be calculated by,

$$e_1 = \frac{n_1}{1 - n_1} = \frac{0.5}{1 - 0.5} = 1$$

Voids ratio under 6000lb/ft² can be calculated by,

$$e_2 = \frac{n_2}{1 - n_2} = \frac{0.45}{1 - 0.45} = 0.8$$

Step 3: The compression index can be then calculated by,

$$C_c = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)} = \frac{1.0 - 0.8}{\log \left(\frac{6000 \text{ lb/ft}^2}{3000 \text{ lb/ft}^2} \right)} = 0.66 \sim 0.65$$

18) Assume that the soil is in the range of virgin compression. The total volume of this soil is equal to 100cm^3 . Under 3000lb/ft^2 , the volume of voids is 47cm^3 . Under 6000lb/ft^2 , the volume of voids is 44cm^3 . Calculate the compression index.

A) 0.33

B) 0.45

C) 0.65

D) 0.28

The Answer is A

Step 1: Based on the equation shown below,

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_2}{\log\left(\frac{p_2}{p_1}\right)}$$

From this problem, information about two points with pressure is directly given. The voids ratios need to be calculated.

Step 2: Under 3000lb/ft², the volume of voids is 47cm³. Therefore, voids ratio can be calculated by,

$$e_1 = \frac{V_v}{V_s} = \frac{V_v}{V - V_v} = \frac{47\text{cm}^3}{100\text{cm}^3 - 47\text{cm}^3} = 0.9$$

Voids ratio under 6000lb/ft² can be calculated by,

$$e_2 = \frac{V_v}{V_s} = \frac{V_v}{V - V_v} = \frac{44\text{cm}^3}{100\text{cm}^3 - 44\text{cm}^3} = 0.8$$

Step 3: The compression index can be then calculated by,

$$C_c = \frac{e_1 - e_2}{\log\left(\frac{p_2}{p_1}\right)} = \frac{0.9 - 0.8}{\log\left(\frac{6000\text{lb/ft}^2}{3000\text{lb/ft}^2}\right)} = 0.33 \sim 0.33$$

19) It is known that the past maximum consolidation stress is 400 kN/m^2 . Under 300 kN/m^2 , the porosity of soil is 0.44. Under 500 kN/m^2 , the porosity of soil is 0.43. Also we know that $C_R = 1/6 C_c$. Calculate the compression index, C_c .

- A) 0.6 B) 0.4 C) 0.8 D) 0.2



The Answer is A

The Answer is B

Step 1: During consolidation, the soil is actually in both range of recompression and virgin compression. For recompression range, the index can be found by,

$$C_R = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_c}{\log \left(\frac{p_c}{p_1} \right)}$$

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_c - e_2}{\log \left(\frac{p_2}{p_c} \right)}$$

Step 2: It is known that $C_R = 1/6 C_c$, therefore,

$$6 \frac{e_1 - e_c}{\log \left(\frac{p_c}{p_1} \right)} = \frac{e_c - e_2}{\log \left(\frac{p_2}{p_c} \right)}$$

$$6 \frac{e_1 - e_c}{\log \left(\frac{p_c}{p_1} \right)} = \frac{e_c - e_2}{\log \left(\frac{p_2}{p_c} \right)}$$

From the problem, we know the information of n_1 and n_2 . Therefore,

$$e_1 = \frac{n_1}{1 - n_1} = \frac{0.44}{1 - 0.44} = 0.8$$

$$e_2 = \frac{n_2}{1 - n_2} = \frac{0.43}{1 - 0.43} = 0.75$$

The equation should be equal to,

$$6 \frac{0.8 - e_c}{\log \left(\frac{400 \text{ kN/m}^2}{300 \text{ kN/m}^2} \right)} = \frac{e_c - 0.75}{\log \left(\frac{500 \text{ kN/m}^2}{400 \text{ kN/m}^2} \right)}$$

$$e_c = 0.79$$

Step 3: To calculate the compression index, we need to use equation as shown above to answer this question as,

$$C_c = \frac{e_c - e_2}{\log\left(\frac{p_2}{p_c}\right)} = \frac{0.79 - 0.75}{\log\left(\frac{500\text{kN/m}^2}{400\text{kN/m}^2}\right)} = 0.4$$

20) Assume that the soil is in the range of virgin compression. Under 300kN/m^2 , the water content of soil is 20%, total volume is 40cm^3 and total mass is 68g. Under 500kN/m^2 , the water content of soil is 16%, total volume is 38cm^3 and total mass is 66g. Also we know that $G_s = 2.7$. Calculate the compression index, C_c .

A) 0.28

B) 0.32

C) 0.45

D) 0.

The Answer is C

Step 1: Based on the equation shown below,

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)}$$

From the problem, we know the value of p_1 and p_2 . To solve this problem, we still need to solve two voids ratio

Step 2: Equation of void ratio of soil is shown as,

$$e = \frac{V_v}{V_s}$$

To determine the value of void ratio, we need to find volume of voids, V_v , and volume of solids, V_s . To determine V_s and V_v , we can deduce equations as shown below,

$$V_s = \frac{M_s}{\rho_s} = \frac{M}{\left(1 + \frac{M_w}{M_s}\right)\rho_s} = \frac{M}{(1 + w)\rho_s} = \frac{M}{(1 + w)G_s\rho_w}$$

$$V_v = V - V_s$$

Substitute all known value into these equations shown above, we can get the value of them as shown below,

$$V_{s,1} = \frac{M_1}{(1 + w_1)G_s\rho_w} = \frac{68g}{(1 + 20\%)(2.7)(1g/cm^3)} = 21m^3$$

$$V_{v,1} = V_1 - V_{s,1} = 40cm^3 - 21cm^3 = 19cm^3$$

$$e_1 = \frac{V_{v,1}}{V_{s,1}} = \frac{19cm^3}{21cm^3} = 0.9$$

The same process, we can get,

$$V_{s,2} = \frac{M_2}{(1 + w_2)G_s\rho_w} = \frac{66g}{(1 + 16\%)(2.7)(1g/cm^3)} = 21m^3$$

$$V_{v,2} = V_2 - V_{s,2} = 38cm^3 - 21cm^3 = 17cm^3$$

$$e_1 = \frac{V_{v,1}}{V_{s,1}} = \frac{17\text{cm}^3}{21\text{cm}^3} = 0.8$$

Step 3: The compression index can be then calculated by,

$$C_c = \frac{e_1 - e_2}{\log\left(\frac{p_2}{p_1}\right)} = \frac{0.9 - 0.8}{\log\left(\frac{500\text{kN/m}^2}{300\text{kN/m}^2}\right)} = 0.45$$

22) Assume that the soil is in the range of virgin compression. Under 125kN/m^2 , the water content of soil is 20%, dry density is 100pcf and degree of saturation is 40%. Under 500kN/m^2 , the water content of soil is 15%, dry density is 100pcf and degree of saturation is 30%. Calculate the compression index, C_c .

A) 0.28

B) 0.32

C) 0.50

D) 0.54

The Answer is C

Step 1: We can easily find the equation shown below,

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)}$$

From the problem, we know the value of p_1 and p_2 . To solve this problem, we still need to solve two voids ratio

Step 2: Equation of void ratio of soil is shown as,

$$e = \frac{V_v}{V_s}$$

To determine the value of void ratio, we need to find volume of voids, V_v , and volume of solids, V_s . To determine the voids ratio, we can deduce its equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{V_s} = \frac{\frac{W_w}{\gamma_w S}}{V_s} = \frac{\frac{wW}{\gamma_w S}}{V_s} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Substitute all known value into these equations shown above, we can get the value of them as shown below,

$$e_1 = \gamma_d \frac{w_1}{\gamma_w S_1} = (100 \text{pcf}) \frac{40\%}{(62.5 \text{pcf})(60\%)} = 1.1$$

The same process, we can get,

$$e_2 = \gamma_d \frac{w_2}{\gamma_w S_2} = (100 \text{pcf}) \frac{15\%}{(62.5 \text{pcf})(30\%)} = 0.8$$

Step 3: The compression index can be then calculated by,

$$C_c = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)} = \frac{1.1 - 0.8}{\log \left(\frac{500 \text{kN/m}^2}{125 \text{kN/m}^2} \right)} = 0.5 \sim 0.5$$

23) Assume that the soil is in the range of virgin compression. Under 125kN/m^2 , the water content of soil is 20%, and degree of saturation is 40%. Under 500kN/m^2 , the water content of soil is 15%, and degree of saturation is 30%. If the compression index, C_c is known to be 0.5, what is the dry density of this soil?

A) 100pcf

B) 120pcf

C) 110pcf

D) 90pcf

The Answer is A

Step 1: We can easily find the equation shown below,

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)}$$

From the problem, we know the value of p_1 and p_2 . However, we still need to know the value of voids ratio, e_1 and e_2 .

Step 2: Equation of void ratio of soil can be shown as,

$$e = \frac{V_v}{V_s}$$

To determine the value of void ratio, we need to find volume of voids, V_v , and volume of solids, V_s . To determine the voids ratio, we can deduce its equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{W}{\gamma_w S}} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Substitute all known value into these equations shown above, we can get the value of them as shown below,

$$e_1 = \gamma_d \frac{w_1}{\gamma_w S_1} = \gamma_d \frac{40\%}{(62.5 \text{ pcf})(60\%)} = \frac{4\gamma_d}{375}$$

The same process, we can get,

$$e_2 = \gamma_d \frac{w_2}{\gamma_w S_2} = \gamma_d \frac{15\%}{(62.5 \text{ pcf})(30\%)} = \frac{\gamma_d}{125}$$

Step 3: Based on the compression equation, we have,

$$C_c = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)} = \frac{\frac{4\gamma_d}{375} - \frac{\gamma_d}{125}}{\log \left(\frac{500 \text{ kN/m}^2}{125 \text{ kN/m}^2} \right)} = 0.5$$

$$\gamma_d = 100 \text{ pcf} \sim 100 \text{ pcf}$$

24) In the range of virgin compression, a soil with 50cm^3 total volume is subjected to two different stress. Under 200kN/m^2 , the voids ratio is unknown. Under 400kN/m^2 , the voids ratio is 0.8. Determine the value of water content under 200kN/m^2 if $C_c = 0.6$, degree of saturation is 30%, and dry density is 110pcf .

- A) 17% B) 19% C) 15% D) 21%

The Answer is B

Step 1: We can easily find the equation shown below,

$$C_c = \frac{\Delta e}{\Delta \log p} = \frac{e_1 - e_2}{\log \left(\frac{p_2}{p_1} \right)}$$

From this problem, information about two points with pressure is directly

given. However, we only know the first void ratios. To determine the voids ratio for the second time, we can calculate it by,

$$e_1 = e_2 + C_c \log \left(\frac{p_2}{p_1} \right) = 0.8 + (0.6) \left(\log \left(\frac{400 \text{ kN/m}^2}{200 \text{ kN/m}^2} \right) \right) = 1.0$$

Step 2: Equation of void ratio of soil can be shown as,

$$e = \frac{V_v}{V_s}$$

To determine the water content from voids ratio, we need to determine the voids ratio equation by,

$$e = \frac{V_v}{V_s} = \frac{\frac{V_w}{S}}{\frac{V_s}{S}} = \frac{\frac{W_w}{\gamma_w S}}{\frac{V_s}{S}} = \frac{\frac{wW}{\gamma_w S}}{\frac{V_s}{S}} = \frac{W}{V_s} \frac{w}{\gamma_w S} = \gamma_d \frac{w}{\gamma_w S}$$

Step 3: To get the value of water content, we can substitute all known value into the equation above and we have,

$$e = \gamma_d \frac{w}{\gamma_w S} = (100 \text{ pcf}) \frac{w}{(62.5 \text{ pcf})(30\%)} = 1$$

$$w = 18.75\% \sim 19\%$$

25) In a falling head permeability test on a soil sample, the following data are available:

Cross-sectional area of soil = 60 cm^2

Length of soil = 10 cm

Initial head = 120 cm

Final head = 108 cm

Duration of test = 20 minutes

Diameter of tube = 8 mm

Determine the coefficient of permeability of the soil, k .

A) 7.4 cm/sec

B) $7.36 \times 10^{-6} \text{ cm/sec}$

C) $5.8 \times 10^{-6} \text{ cm/sec}$

D) $4.60 \times 10^{-6} \text{ mm/sec}$

The Answer is B**Step 1:**

From a falling head test, coefficient of permeability, k , is calculated by

$$k = \frac{2.303aL}{At_E} \log_{10} \left(\frac{h_1}{h_2} \right)$$

where a = area of reservoir tube; L = length of flow; A = cross-sectional area of soil;
 t_E = elapsed time during falling head test; h_1 = initial head; h_2 = final head.

Step 2:

From the given information, it is known that

$$a = \frac{\pi D^2}{4} = \frac{\pi (8 \text{ mm})^2}{4} = 0.503 \text{ cm}^2, L = 10 \text{ cm}, A = 60 \text{ cm}^2, t_E = 20 \text{ minutes} =$$

$$20 \text{ minutes} \times 60 \frac{\text{seconds}}{\text{minute}} = 1200 \text{ seconds}, h_1 = 120 \text{ cm}, h_2 = 108 \text{ cm}.$$

Therefore

$$k = \frac{2.303aL}{At_E} \log_{10} \left(\frac{h_1}{h_2} \right) = \frac{2.303(0.503 \text{ cm}^2)(10 \text{ cm})}{(60 \text{ cm}^2)(1200 \text{ seconds})} \log_{10} \left(\frac{120 \text{ cm}}{108 \text{ cm}} \right)$$

$$= 7.36 \times 10^{-6} \text{ cm/sec}$$

26) A permeability test is conducted with a sample of soil that is 80 mm in diameter and 140 mm long. The head is kept constant at 240 mm. The flow is 1.8 mL in 10 minutes. What is the coefficient of permeability?

- A) 4.38×10^{-6} cm/sec B) 3.48×10^{-5} cm/sec
C) 5.8×10^{-6} cm/sec D) 3.48×10^{-6} cm/sec

The Answer is B**Step 1:**

From a constant head test, coefficient of permeability, k , is calculated by

$$k = \frac{Q}{iAt_E}$$

where Q = total quantity of water collected over time t_E ; i = hydraulic gradient = h/L ;

A = cross-sectional area of soil; t_E = elapsed time during falling head test; h = head; L = length of soil.

Step 2:

From the given information, it is known that

$$Q = 1.8 \text{ mL}, h = 240 \text{ mm}, L = 140 \text{ mm}, A = \frac{\pi D^2}{4} = \frac{\pi (80 \text{ mm})^2}{4} = 50.3 \text{ cm}^2, t_E =$$

$$10 \text{ minutes} = 10 \text{ minutes} \times 60 \frac{\text{seconds}}{\text{minute}} = 600 \text{ seconds},$$

Therefore,

$$k = \frac{Q}{iAt_E} = \frac{(1.8 \text{ mL})}{\left(\frac{240 \text{ mm}}{140 \text{ mm}}\right) (50.3 \text{ cm}^2) (600 \text{ seconds})} = 3.48 \times 10^{-5} \text{ cm/sec}$$

27) Deformations of soils are a function of

- A) Effective stresses
- B) Total stresses
- C) Pore water pressure
- D) All of the above

The Answer is A

Based on the principal of effective stress, effective stress is equal to total stress minus pore water pressure. It is the most important principle in soil mechanics. Deformations of soils are a function of effective stresses not total stresses.

(A) is correct.

(B) is incorrect. Deformations of soils are not a function of total stresses.

(C) is incorrect. Pore water pressure is isotropic and only can cause volumetric changes of soil solids, which is nearly incompressible. Pore water pressure does not cause displacement of soil solids, or deformation of soils.

(D) is incorrect, since options (B) and (C) are incorrect.

28) An oedometer test in a normally consolidated clay gave the following results:

Average effective pressure (psi)	Void ratio
20	0.80
40	0.68

Determine the compression index C_c .

A) 0.20

B) 0.40

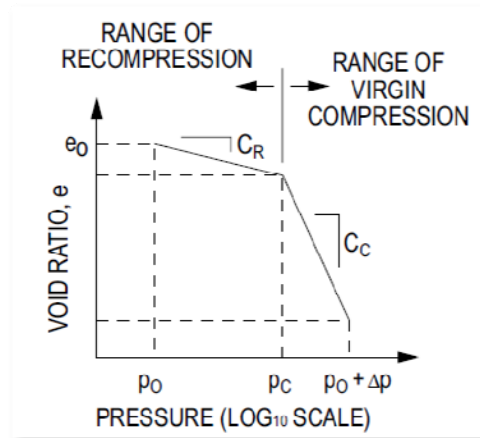
C) 0.60

D) 0.80

The Answer is B

The compression index can be determined from the soil consolidation curve obtained from laboratory tests, as shown in the figure below.

$$\begin{aligned}
 C_c &= \frac{\Delta e}{\Delta \log_{10} p} \\
 &= \frac{e_0 - e_1}{\log_{10} \left(\frac{p_1}{p_0} \right)} \\
 &= \frac{0.80 - 0.68}{\log_{10} \left(\frac{40 \text{ psi}}{20 \text{ psi}} \right)} \\
 &\cong 0.40
 \end{aligned}$$



29) As an engineer you are given a field log for a boring done with an SPT sampler with the following information:

Depth (ft)	1st Blows (6")	2nd Blows (6")	3rd Blows (6")	Description
0	5	8	10	Fill - Gravel (dry)
5	4	7	8	Fill - Sand & Gravel (moist)
10	5	9	9	Silt (wet)
15	8	8	9	Clayey Silt (moist)
20	7	12	15	Sandy Silt (wet) - (3" orange layer)
25	12	15	17	Sandy Silt (wet)
30	18	20	22	Sand
35	15	19	17	Sand
40	20	22	20	Sand
45	18	20	19	Sand
50	15	29	Refusal 100/6"	No Recovery

30) What range of relative density might you expect for the sand at a depth of 40 feet?

(A) 64% - 69%

(B) 75% - 80%

(C) 87% - 92%

(D) 94% - 99%

The Answer is C**Step 1**

It is first necessary to determine an appropriate range of effective stresses for this depth. Information about the location of the water table should be interpreted from the descriptions. That the soil is wet is not enough of an indicator of where the water table occurs. Water can be held over clay or layers of low permeability. The water could also be due to drilling methods used. The orange layer is a usual indicator of oxidation which is a result of fluctuation of the water table. It is most likely that the water table occurs at a depth of 20 feet.

Step 2

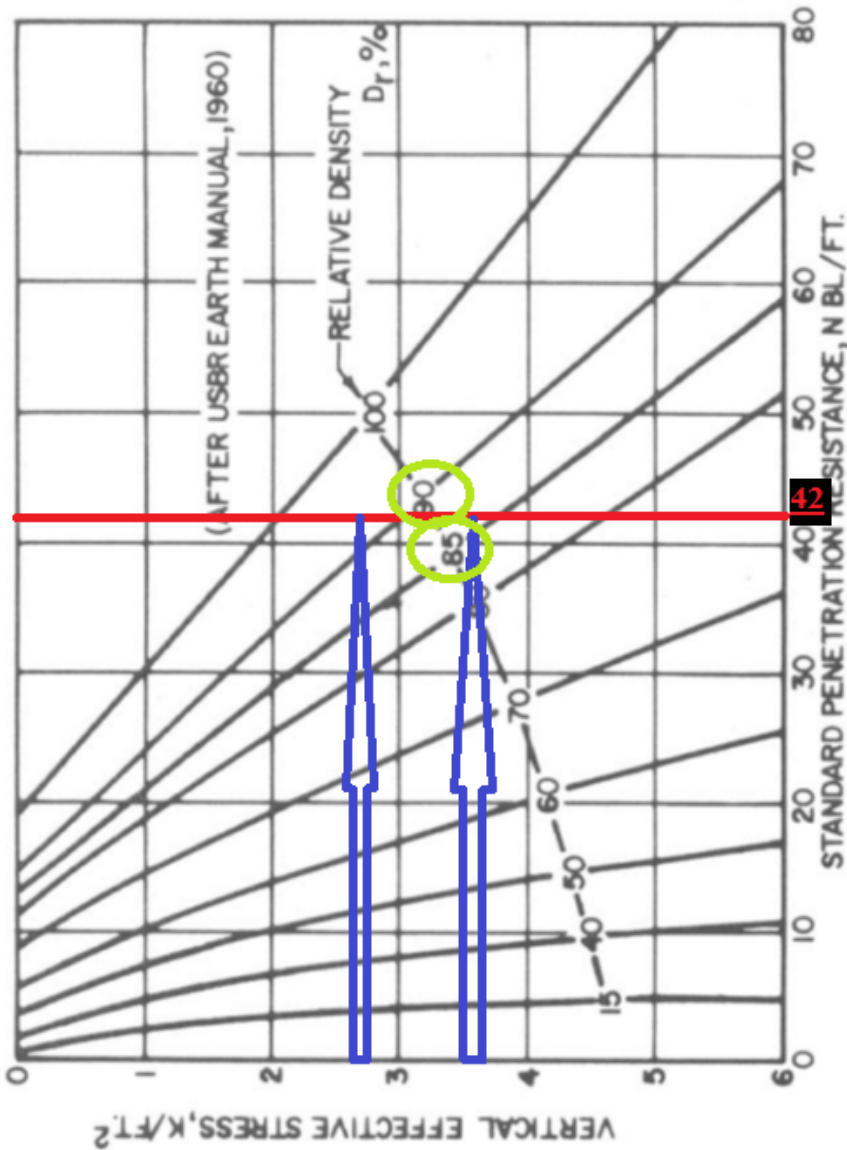
Calculating a likely range using relatively high and low values for unit weight and the ground water table at a depth of 20 feet to calculate a range of effective stresses.

$$100\text{pcf} \times 20 \text{ ft} + (100\text{pcf} - 62.4\text{pcf}) \times 20 \text{ ft} = 2752 \text{ psf} = 2.75 \text{ ksf}$$

$$120\text{pcf} \times 20 \text{ ft} + (120\text{pcf} - 62.4 \text{ pcf}) \times 20 \text{ ft} = 3552 \text{ psf} = 3.55 \text{ ksf}$$

Step 3

With these values of effective overburden stress and the SPT of 42 blows per foot (sum the 2nd and 3rd blow counts of 20 and 22) we can now use the chart developed by Gibbs & Holtz.



We arrive at a range of values for relative density greater than 85% but less than 95%.

Common Mistakes

Ignoring the ground water table will give a value of effective stress that is too high (likely between 4ksf and 5 ksf) which will result in a range between 75% and 80% leading to Choice B.

Interpreting the log description to say the ground water table is at a depth of 10 feet will result in calculating an effective stress that is too low (likely between 2ksf and 3ksf) which will result in a range of relative densities well greater than 90% leading to Choice D.

31) Prior to making a boring location plan your boss asks you to look through some very old boring logs that were given to you by the client. The borings were done with different equipment than we use now and the strength of the sample that was taken at a depth of 20 feet is of importance. Convert the blow count to $N_{1,60}$ assuming the soil above has a unit weight of 110 pcf.

Hammer: R-P Donut

Rod Length: 22 feet

Hole Diameter: 2-1/2 inches

Depth (ft)	1st Blows (6")	2nd Blows (6")	3rd Blows (6")	Description
0	5	8	10	Fill - Gravel
5	3	9	12	Fill - Sand & Gravel
10	5	4	9	Silt
15	4	5	4	Clayey Silt
20	13	14	17	Sand
25	14	13	19	Sand
30	18	20	22	Sand (groundwater table at 31 ft)
35	15	19	17	Sand
40	20	22	20	Sand
45	18	20	19	Sand
50	15	29	Refusal 100/6"	No Recovery

(A) 21

(B) 24

(C) 31

(D) 33

The Answer is C

Step 1:

To adjust blow counts the following equation should be used:

$$N_{1,60} = C_N \times N_{\text{FIELD}} \times \eta_1 \times \eta_2 \times \eta_3 \times \eta_4$$

To determine the correction factors use the table from Bowles, 1996.

Factors η for Blow Count Correction*

Hammer for η_1					Remarks
Average energy ratio, E_r					
Country	Donut		Safety		R-P = Rope-pulley or cathead $\eta_1 = E_r/E_{rb} = E_r/70$ For U.S. trip/auto w/ $E_r = 80$ $\eta_1 = 80/70 = 1.14$
	R-P	Trip	R-P	Trip/Auto	
United States/ North America	45	--	70-80	80-100	
Japan	67	78	--	--	
United Kingdom	--	--	50	60	
China	50	60	--	--	
Rod length correction η_2					
	Length	> 33 ft	η_2	= 1.00	N is too high for $L < 33$ ft
		20 - 33		= 0.95	
		13 - 20		= 0.85	
		0 - 13		= 0.75	
Sampler correction η_3					
	Without Liner	η_3	= 1.00		Base value
	With liner: Dense sand, clay		= 0.80		N is too high with liner
	Loose sand		= 0.90		
Sampler correction η_4					
	Hole diameter: [†]	2.4 - 4.8 in	η_4	= 1.00	Base value; N is too small when there is an oversize hole
		6 in		= 1.05	
		8 in		= 1.15	

* Data synthesized from Riggs (1986), Skempton (1986), Schmertmann (1978) and Seed et al. (1985).

[†] $\eta_4 = 1.00$ for all diameter hollow-stem augers where SPT is taken through the stem.

As no liner is mentioned we should assume no liner was used.

$$\eta_1 = 45/60 = 0.75$$

$$\eta_2 = 0.95$$

$$\eta_3 = 1.00$$

$$\eta_4 = 1.00$$

Step 2

Determine the effective overburden to use in calculating C_N . The ground water table is at a depth of 31 feet, therefore the effective stress is the total stress at 20 feet. Using the unit weight of 110 pcf:

$$p'_o = 20 \text{ ft} \times 110 \text{ pcf} = 2200 \text{ psf} = 1.1 \text{ tsf}$$

Calculating C_N :

$$C_N = 0.77 \times \log_{10} \left(\frac{20}{p'_o} \right)$$

(Ref Page 1-3)

$$C_N = 0.77 \times \log_{10} \left(\frac{20}{1.1} \right) = 0.97$$

Step 3

The field blow count is the sum of the 2nd and 3rd counts for 6" increments.

$$N_{\text{field}} = 14 + 17 = 31 \text{ blows/ft}$$

Calculating $N_{1,60}$:

$$\begin{aligned} N_{1,60} &= 0.97 \times 31 \text{ blows/ft} \times 0.75 \times 0.95 \times 1.00 \times 1.00 \\ &= 21.4 \text{ blows/ft} \rightarrow 21 \text{ blows/ft} \end{aligned}$$

32) Your company is doing consulting work on a job in Japan and you have just received boring logs from the client. The borings were done with different equipment than used by the standards your company designs with. The strength of the soil that was taken at a depth of 35 feet is of importance. Convert the blow count to $N_{1,60}$ assuming the soil above has a total unit weight of 105 pcf.

Hammer: Trip Donut Rod Length: 30 feet Hole Diameter: 2-1/2 inches Liner: none

Depth (ft)	1st Blows (6")	2nd Blows (6")	3rd Blows (6")	Description
2	5	15	19	Gravel
5	3	14	12	Gravel & Clay
10	8	12	12	Gravel & Clay
15	15	14	16	Sandy Gravel
20	10	11	12	Sand (groundwater table at 20 ft)
25	8	12	13	Sand
30	13	12	17	Sand
35	12	15	15	Sand
40	24	25	30	Decomposed Rock

- (A) 15 (B) 28 (C) 33 (D) 36

The Answer is C

Step 1

To adjust blow counts the following equation should be used:

$$N_{1,60} = C_N \times N_{\text{FIELD}} \times \eta_1 \times \eta_2 \times \eta_3 \times \eta_4$$

To determine the correction factors use the table from Bowles, 1996.

Factors η for Blow Count Correction*

Hammer for η_1					Remarks
Average energy ratio, Er					
Country	Donut		Safety		
	R-P	Trip	R-P	Trip/Auto	
United States/ North America	45	--	70-80	80-100	R-P = Rope-pulley or cathead
Japan	67	78	--	--	$\eta_1 = Er/Er_b = Er/70$
United Kingdom	--	--	50	60	For U.S. trip/auto w/Er = 80
China	50	60	--	--	$\eta_1 = 80/70 = 1.14$
Rod length correction η_2					
Length		> 33 ft	η_2	= 1.00	N is too high for L < 33 ft
		20 - 33		= 0.95	
		13 - 20		= 0.85	
		0 - 13		= 0.75	
Sampler correction η_3					
Without Liner		η_3	= 1.00		Base value
With liner: Dense sand, clay			= 0.80		N is too high with liner
Loose sand			= 0.90		
Sampler correction η_4					
Hole diameter:†	2.4 - 4.8 in	η_4	= 1.00		Base value; N is too small when there is an oversize hole
	6 in		= 1.05		
	8 in		= 1.15		

* Data synthesized from Riggs (1986), Skempton (1986), Schmertmann (1978) and Seed et al. (1985).

† $\eta_4 = 1.00$ for all diameter hollow-stem augers where SPT is taken through the stem.

$$\eta_1 = 78/60 = 1.30$$

$$\eta_2 = 0.95$$

$$\eta_3 = 1.00$$

$$\eta_4 = 1.00$$

Step 2

Determine the effective overburden to use in calculating C_N . The ground water table is at a depth of 20 feet. Calculating C_N :

$$C_N = 0.77 \times \log_{10} \left(\frac{20}{p'_o} \right)$$
$$C_N = 0.77 \times \log_{10} \left(\frac{20}{1.37} \right) = 0.90$$

Step 3

The field blow count is the sum of the 2nd and 3rd counts for 6" increments.

$$N_{\text{field}} = 15 + 15 = 30 \text{ blows/ft}$$

Calculating $N_{1,60}$:

$$N_{1,60} = 0.90 \times 30 \text{ blows/ft} \times 1.30 \times 0.95 \times 1.00 \times 1.00$$
$$= 33.3 \text{ blows/ft} \rightarrow 33 \text{ blows/ft}$$



33) Your company is doing consulting work on a job in the U.S. and you have just received boring logs from the field inspector. The borings were done with the standard equipment. The strength of the soil that was taken at a depth of 60 feet is of importance. Convert the blow count to $N_{1,60}$ assuming the soil above has a total unit weight of 110 pcf. You have been instructed to make conservative assumptions if assumptions are necessary.

Hammer: Auto/Safety Rod Length: 35 feet Hole Diameter: 2-1/2 inches Liner: none

Depth (ft)	1st Blows (6")	2nd Blows (6")	3rd Blows (6")	Description
5	5	10	10	Gravel
10	7	12	14	Gravel & Sand
15	8	12	12	Gravel & Sand
20	6	8	8	Fat Clay
25	10	11	12	Fat Clay (groundwater table at 25 ft)
30	8	12	13	Fat Clay
40	13	12	17	Fat Clay
50	12	15	15	Clay & Sand
60	24	25	30	Silty Sand and Gravel

- (A) 43 (B) 54 (C) 60 (D) 67

The Answer is B

Step 1

To adjust blow counts the following equation should be used:

$$N_{1,60} = C_N \times N_{\text{FIELD}} \times \eta_1 \times \eta_2 \times \eta_3 \times \eta_4$$

To determine the correction factors use the table from Bowles, 1996.

Factors η for Blow Count Correction*

Hammer for η_1					Remarks
Average energy ratio, E_r					
Country	Donut		Safety		
	R-P	Trip	R-P	Trip/Auto	
United States/ North America	45	--	70-80	80-100	R-P = Rope-pulley or cathead $\eta_1 = E_r/E_{rb} = E_r/70$ For U.S. trip/auto w/ $E_r = 80$ $\eta_1 = 80/70 = 1.14$
Japan	67	78	--	--	
United Kingdom	--	--	50	60	
China	50	60	--	--	
Rod length correction η_2					
Length	> 33 ft	η_2	= 1.00		N is too high for $L < 33$ ft
	20 - 33		= 0.95		
	13 - 20		= 0.85		
	0 - 13		= 0.75		
Sampler correction η_3					
	Without Liner	η_3	= 1.00		Base value
	With liner: Dense sand, clay		= 0.80		N is too high with liner
	Loose sand		= 0.90		
Sampler correction η_4					
Hole diameter:†	2.4 - 4.8 in	η_4	= 1.00		Base value; N is too small
	6 in		= 1.05		when there is an oversize hole
	8 in		= 1.15		

* Data synthesized from Riggs (1986), Skempton (1986), Schmertmann (1978) and Seed et al. (1985).

† $\eta_4 = 1.00$ for all diameter hollow-stem augers where SPT is taken through the stem.

The conservative assumption for the energy ratio E_r is 80.

$$\eta_1 = 80/60 = 1.33$$

$$\eta_2 = 1.00$$

$$\eta_3 = 1.00$$

$$\eta_4 = 1.00$$

Step 2

Determine the effective overburden to use in calculating C_N . The ground water table is at a depth of 20 feet. Using the unit weight of 105 pcf:

Calculating C_N :

$$C_N = 0.77 \times \log_{10} \left(\frac{20}{p'_o} \right)$$
$$C_N = 0.77 \times \log_{10} \left(\frac{20}{2.21} \right) = 0.74$$

Step 3

The field blow count is the sum of the 2nd and 3rd counts for 6" increments.

$$N_{\text{field}} = 25 + 30 = 55 \text{ blows/ft}$$

Calculating $N_{1,60}$:

$$N_{1,60} = 0.74 \times 55 \text{ blows/ft} \times 1.33 \times 1.00 \times 1.00 \times 1.00$$
$$= 54.1 \text{ blows/ft} \rightarrow 54 \text{ blows/ft}$$

34) A boring log has been given to you by a new field inspector. After looking at the cores in the core box and reading the logs you realize something is wrong. The recovery is 100%, and there are no breaks other than 2 mechanical breaks which the field inspector marked near the end (you don't have time to go measure the lengths). The core barrel itself is 60 inches, but the inspector noted the core barrel was stuck at 55 inches. The RQD is written as 92%. What might the field inspector's error be?

- (A) He/she incorrectly measured the length of the core sample.
- (B) He/she incorrectly assumed the RQD is a fraction of the core barrel length.
- (C) He/she incorrectly assumed that mechanical breaks are to be considered in measuring for RQD.
- (D) B or C

The Answer is D

RQD is defined as the percentage of total core run length consisting of pieces greater than 4 inches.

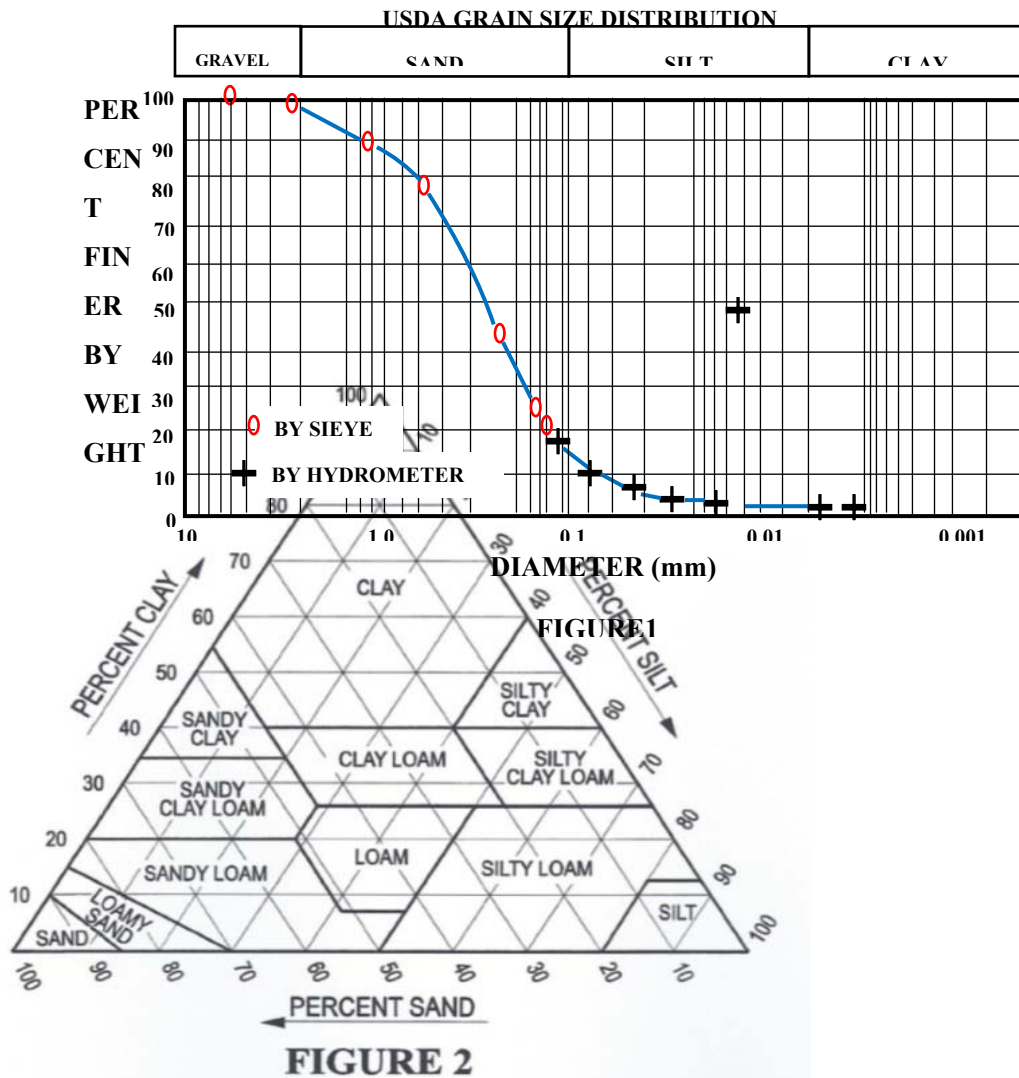
$$\text{RQD} = \frac{\sum \text{length of pieces longer than 4"}}{\text{total core run length}} \times 100\%$$

Therefore it could be B.

Given there are 2 mechanical breaks, this could account for up to (but less than) 8 inches taken off of the RQD by the field inspector. Therefore it could be C.

35) Figure 1 shows a gradation curve for a soil. Using Figure 2, you determine the best USDA textural classification of this soil is:

- A) loam B) loamy sand C) silt D) silty loam`



The Answer is B

The material percentages are:

Gravel 2%

Sand 80%

Silt 17%

Clay 1%

Using the USDA chart in Figure 2, loamy sand is the correct answer.

36) Based on the boring log shown below, the buoyant (submerged) unit weight (pcf) of soil at a depth of 12 ft is most nearly:

A) 33

B) 56

C) 95

D) 119

The Answer is C

Assume soil below water table is saturated and $\gamma = \gamma_{\text{Sat}} = \gamma_{\text{dry}}(1 + w)$.

$$\gamma_{\text{Sat}} = 95 (1 + 0.25) = 118.8 \text{ pcf}$$

$$\gamma_b = \gamma_{\text{Sat}} - \gamma_w$$

$$= 118 - 62.4 = 56.4 \text{ pcf}$$

DEPTH (ft)	N VALUE	UNIFID SOIL CLASSIFICATION	DEPTH TO WATER(FT) 5.0	
			MOISTURE CONTENT (%)	DRY DENSITY (PCF)

0 5 6	10	SM	10%	105
10 12	7	CL	27%	90
15 19 20	8	CL	25%	95
	9	CL	21%	100

37) A soil specimen is obtained from below the groundwater table. The soil has a void ratio of 0.72 and a specific gravity of solids of 2.65. The buoyant (submerged) unit weight (pcf) is most nearly:

- A) 60 B) 76 C) 96 D) 122

The Answer is A

$$\begin{aligned}\gamma_b &= \frac{(\text{sp gr} - 1)}{1 + e} \gamma_w \\ &= \frac{(2.65 - 1)}{1.72} (62.4 \text{ lb/ft}^3) \\ &= 60 \text{ lb/ft}^3\end{aligned}$$

Alternate solution:

$$e = \frac{V_w}{V_s} = 0.72 \Rightarrow V_w = 0.72 V_s$$

$$\text{Say } V_w + V_s = 1.0 \text{ ft}^3$$

$$0.72 V_s + V_s = 1.0 \text{ ft}^3$$

$$1.72 V_s = 1.0 \text{ ft}^3$$

$$V_s = 1/1.72 = 0.58 \text{ ft}^3$$

$$V_w = 1.0 \text{ ft}^3 - 0.58 \text{ ft}^3 = 0.42 \text{ ft}^3$$

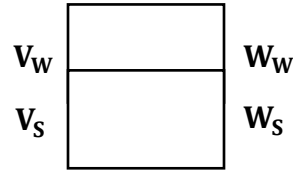
$$W_s = 2.65 (62.4 \text{ lb/ft}^3) V_s = 95.9 \text{ lb}$$

$$V_w = (62.4 \text{ lb/ft}^3) V_w = 26.2 \text{ lb}$$

$$W_t = 122.1 \text{ lb}$$

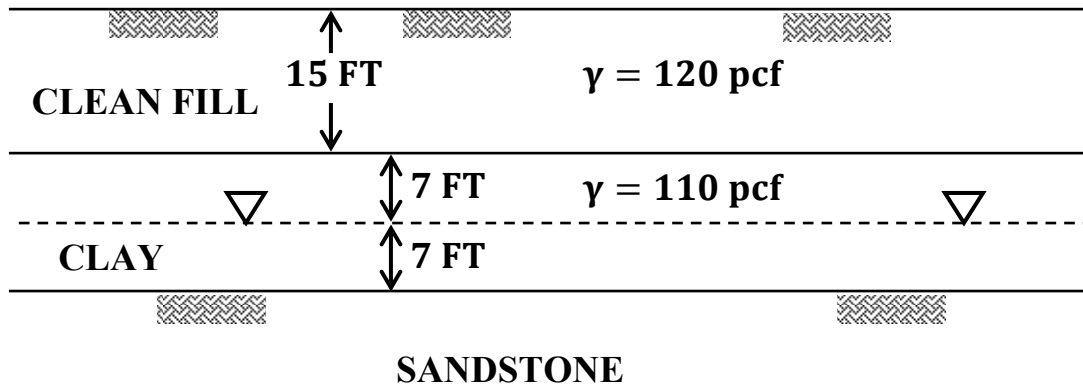
$$\gamma_t = \frac{122.1 \text{ lb}}{1.0 \text{ ft}^3} = 122.1 \text{ lb/ft}^3$$

$$\gamma_b = 122.1 \text{ lb/ft}^3 - 62.4 \text{ lb/ft}^3 = 59.7 \text{ lb/ft}^3$$



38) A soil profile is shown in the figure. The effective vertical stress (pst) at mid-height of the clay layer is most nearly:

- A) 770 B) 2,130 C) 2,570 D) 2,900



The Answer is C

$$\begin{aligned} &120 \text{ pcf}(15 \text{ ft}) + 110 \text{ pcf}(7 \text{ ft}) \\ &1,800 \text{ psf} + 770 \text{ psf} = 2,570 \text{ psf} \end{aligned}$$

39) A researcher will use 275 lbs of aggregate in a 3ft^3 concrete batch to occupy 52% of the mixture. The density of water is 62.4 lbs/ft^3 . What is the bulk specific gravity of the aggregate?

- (A) 2.826
- (B) 2.842
- (C) 2.856
- (D) 2.933

The Answer is A

Step1. Determine the bulk unit weight of the aggregate.

$$\gamma_{\text{aggregate}} = \frac{\text{Mass}_{\text{aggregate}}}{\text{Volume}_{\text{aggregate}}}$$
$$\gamma_{\text{aggregate}} = \frac{275}{3} = 91.7 \frac{\text{lbs}}{\text{ft}^3}$$

Step2. Determine the bulk specific gravity of the aggregate by rearranging the following equation:

$$\% \text{Mix} = \frac{\gamma_{\text{aggregate}}}{G_{\text{aggregate}} * \gamma_{\text{water}}} * 100$$
$$G_{\text{aggregate}} = \frac{\gamma_{\text{aggregate}}}{\% \text{Mix} * \gamma_{\text{water}}}$$
$$G_{\text{aggregate}} = \frac{91.7}{52 * 62.4} * 100 = 2.826$$

40) A concrete mixture has a 60:40 ratio of coarse aggregates to fine aggregates. When mixed separately, 250lbs of coarse aggregates are capable of fitting in a 3ft^3 container and 200lbs of fine aggregates are capable of fitting in a 2ft^3 container. Determine the bulk density of the concrete mixture with a 50:50 ratio.

- (A) $80\text{lb}/\text{ft}^3$
- (B) $87\text{lb}/\text{ft}^3$
- (C) $90\text{lb}/\text{ft}^3$
- (D) $97\text{lb}/\text{ft}^3$

The Answer is C

Step1. Determine the bulk unit weights of each type of aggregate with the following formula:

$$\gamma_{\text{aggregate}} = \frac{\text{Mass}}{\text{Volume}}$$
$$\gamma_{\text{coarse}} = \frac{250\text{lb}}{3\text{ft}^3} = \frac{83\text{lb}}{\text{ft}^3}$$
$$\gamma_{\text{fine}} = \frac{200\text{lb}}{2\text{ft}^3} = \frac{100\text{lb}}{\text{ft}^3}$$

Step2. Since it's a 60:40 ratio, simply take the percentage of each density and divided by 2.

$$\gamma_{\text{mixture}} = 0.6(83) + 0.4(100) = 90 \frac{\text{lbs}}{\text{ft}^3}$$

41) A researcher is trying to determine how much coarse aggregate to place in a 2ft^3 batch of concrete if he wants it to occupy 70% of the mixture. The bulk specific gravity of the aggregate is 2.800. The density of water is 62.4 lbs/ft^3 . How much coarse aggregate should be added?

- (A) 105 lbs
- (B) 128 lbs
- (C) 163 lbs
- (D) 210 lbs

The Answer is C

Step1. Calculate the bulk unit weight of the coarse aggregate needed by rearranging the following equation:

$$\% \text{Mix} = \frac{\gamma_{\text{coarse}}}{G_{\text{coarse}} * \gamma_{\text{water}}} * 100$$

$$\gamma_{\text{coarse}} = \frac{\% \text{Mix} * G_{\text{coarse}} * \gamma_{\text{water}}}{100}$$

$$\gamma_{\text{coarse}} = \frac{60 * 2.800 * 62.4}{100} = 105 \frac{\text{lbs}}{\text{ft}^3}$$

Step2. Determine how much weight is needed for a 2ft³ batch.

$$\text{Weight}_{\text{needed}} = \gamma_{\text{coarse}} * \text{Volume}_{\text{coarse}}$$

$$\text{Weight}_{\text{needed}} = 105 * 2 = 210 \text{lbs}$$

42) An engineer specifies that the concrete structure must have a strength to withstand an object with a diameter of 0.75 inches and a force of 2500 lbs. Determine the required average compressive strength for a plant where the standard deviation is unknown given the equations. The following table may be consulted:

$$f'_{cr} = f'_c + 1.34s \quad f'_{cr} = f'_c + 2.33s - 500$$

Specified Compressive Strength, f'_c (psi)	Required Average Compressive Strength, f'_{cr} (psi)
< 3000	$f'_c + 1000$
3000 to 5000	$f'_c + 1200$
> 5000	$f'_c + 1400$

- (A) 5659 psi
- (B) 6259 psi
- (C) 6659 psi

(D) 7059 psi

The Answer is D

Step1. Determine the pressure exerted on the structure with the following equation:

$$f'_c = \frac{\text{Force}_{\text{object}}}{\text{Area}_{\text{object}}}$$

$$f'_c = \frac{2500\text{lbs}}{\frac{\pi * 0.75^2}{4}} = 5659 \text{ psi}$$

Step2. Use the chart provided to calculate the required average compressive strength.
NOTE: $f'_c > 5000\text{psi}$.

$$f'_{cr} = f'_c + 1400$$

$$f'_{cr} = 5659 + 1400 = 7059\text{psi}$$

43) The required average compressive strength for a concrete structure was 2900psi for a plant where the standard deviation is unknown. Determine the maximum load that the structure can withstand for a 2in^2 area. The following table may be consulted:

Specified Compressive Strength, f'_c (psi)	Required Average Compressive Strength, f'_{cr} (psi)
< 3000	$f'_c + 1000$
3000 to 5000	$f'_c + 1200$
> 5000	$f'_c + 1400$

- (A) 3500 lbs
- (B) 3800 lbs
- (C) 4100 lbs
- (D) 4300 lbs

The Answer is B

Step1. Determine the specified compressive strength of the structure. NOTE: The required average compressive strength is less than 3000 psi, so the specified compressive strength must be less than 3000 psi.

$$f'_{cr} = f'_c + 1000$$

$$f'_c = f'_{cr} - 1000$$

$$f'_c = 2900 - 1000 = 1900\text{psi}$$

Step2. Determine the load by rearranging the following equation:

$$f'_c = \frac{\text{Force}_{\text{object}}}{\text{Area}_{\text{object}}}$$

$$\text{Force}_{\text{object}} = \text{Area}_{\text{object}} * f'_c$$

$$\text{Force}_{\text{object}} = 2 * 1900 = 3800\text{lbs}$$

- 44) The required average compressive strength for a concrete structure is 4500psi for a plant where the standard deviation is unknown. Determine the maximum area that a load of 2000 lbs can be subjected to. The following table may be consulted:

Specified Compressive Strength, f'_c (psi)	Required Average Compressive Strength, f'_{cr} (psi)
< 3000	$f'_c + 1000$
3000 to 5000	$f'_c + 1200$
> 5000	$f'_c + 1400$

- (A) 0.2 in^2
- (B) 0.4 in^2
- (C) 0.6 in^2
- (D) 0.8 in^2

The Answer is C

Step1. Determine the specified compressive strength of the structure. NOTE: The required average compressive strength is 4500 psi.

$$f'_{cr} = f'_c + 1200$$

$$f'_c = f'_{cr} - 1000$$

$$f'_c = 4500 - 1200 = 3300\text{psi}$$

Step2. Determine the load by rearranging the following equation:

$$f'_c = \frac{\text{Force}_{\text{object}}}{\text{Area}_{\text{object}}}$$

$$\text{Area}_{\text{object}} = \frac{\text{Force}_{\text{object}}}{f'_c}$$

$$\text{Area}_{\text{object}} = \frac{2000}{3300} = 0.6\text{in}^2$$

45) An engineer specified a concrete strength of 5500 psi for a plant where the standard deviation of 21 tests results was 600 psi. Determine the modified standard deviation value. The following table may be consulted.

Number of tests	Modification Factor, F
15	1.16
20	1.08
25	1.03
30 or more	1.00

- (A) 607 psi
- (B) 602 psi
- (C) 600 psi
- (D) 594 psi

The Answer is A

Step1. Determine the modification factor by interpolating the values from the provided table.

$$F = 1.08 - \left(\frac{1.08 - 1.03}{25 - 20} \right) (21 - 20) = 1.07$$

Step2. Multiply standard deviation by modification factor:

$$s' = sF$$

$$s' = 600 * 1.07 = 642\text{psi}$$

- 46) Determine the minimum number of tests needed to have a modification factor less than 1.055 for a concrete structure. The following table may be consulted.

Number of tests	Modification Factor, F
15	1.16
20	1.08
25	1.03
30 or more	1.00

- (A) 20
- (B) 21
- (C) 22
- (D) 23

The Answer is D

Step 1. Iterate the modification factors to determine the number of tests. Note: since we are looking for a modification factor less than 1.06 we know that the number of tests falls between 20 and 25.

$$F = 1.08 - \left(\frac{1.08 - 1.03}{25 - 20} \right) (21 - 20) = 1.07$$

$$F = 1.08 - \left(\frac{1.08 - 1.03}{25 - 20} \right) (22 - 20) = 1.06$$

$$F = 1.08 - \left(\frac{1.08 - 1.03}{25 - 20} \right) (23 - 20) = 1.05$$

47) Determine the standard deviation for a concrete plant if you know that they ran 27 tests and have a modified standard deviation value of 650. The following table may be consulted.

Number of tests	Modification Factor, F
15	1.16
20	1.08
25	1.03
30 or more	1.00

- (A) 625 psi
- (B) 637 psi
- (C) 645 psi
- (D) 650 psi

The Answer is B

Step1. Determine the modification factor by interpolating the values from the provided table.

$$F = 1.03 - \left(\frac{1.03 - 1.00}{30 - 25} \right) (27 - 25) = 1.02$$

Step2. Rearrange the following equation to determine the standard deviation of the concrete plant.

$$s' = sF$$

$$s = \frac{s'}{F}$$

$$s = \frac{650}{1.02} = 637\text{psi}$$

48) The specified compressive strength of a concrete structure can be estimated from the rebound hammer test with the following equation, where RHT is the number obtained from the test:

$$f'_c = 200 * RHT - 500$$

Determine the average compressive strength needed for the concrete mix if the plant producing the mixture historically produces concrete with a standard deviation of 500 psi and the value obtained from the rebound hammer test is 30.3 given the following equations.

$$f'_{cr} = f'_c + 1.34s \quad f'_{cr} = f'_c + 2.33s - 500$$

- (A) 6115 psi
- (B) 6180 psi
- (C) 6230 psi
- (D) 6420 psi

The Answer is C

Step1. Determine the specified compressive strength with the given equation:

$$f'_c = 200 * RHT - 500$$

$$f'_c = 200 * 30.3 - 500 = 5560\text{psi}$$

Step2. Determine the average compressive strength by choosing the higher of the two equations:

$$f'_{cr} = f'_c + 1.34s$$

$$f'_{cr} = f'_c + 2.33s - 500$$

$$f'_{cr} = 5560 + 1.34 * 500 = 6230\text{psi}$$

$$f'_{cr} = 5560 + 2.33 * 500 - 500 = 6225\text{psi}$$

49) The specified compressive strength of a concrete structure can be estimated from the rebound hammer test with the following equation, where RHT is the number obtained from the test:

$$f'_c = 200 * RHT - 500$$

Determine the number obtained on the rebound hammer if you know the average compressive strength required is 2800psi. The following table may be consulted:

Specified Compressive Strength, f'_c (psi)	Required Average Compressive Strength, f'_{cr} (psi)
< 3000	$f'_c + 1000$
3000 to 5000	$f'_c + 1200$
> 5000	$f'_c + 1400$

- (A) 4.1
- (B) 6.5
- (C) 8.2
- (D) 10.7

The Answer is B

Step1. Determine the specified compressive strength by rearranging the equation provided in the table. Note: Since the required average compressive strength is less than 3000psi, the specified compressive strength will also be less than 3000psi.

$$f'_{cr} = f'_c + 1000$$

$$f'_c = f'_{cr} - 1000$$

$$f'_c = 2800 - 1000 = 1800\text{psi}$$

Step2. Determine the rebound hammer test number by rearranging the formula provided in the problem:

$$f'_c = 200 * \text{RHT} - 500$$

$$\text{RHT} = \frac{f'_c - 500}{200}$$

$$\text{RHT} = \frac{1800 - 500}{200} = 6.5$$

50) The specified compressive strength of a concrete structure can be estimated from the rebound hammer test with the following equation, where RHT is the number obtained from the test:

$$f'_c = 200 * RHT - 500$$

If the rebound hammer test yielded a number of 21.2, determine the required average compressive strength. The standard deviation is unknown, so the following table may be consulted:

Specified Compressive Strength, f'_c (psi)	Required Average Compressive Strength, f'_{cr} (psi)
< 3000	$f'_c + 1000$
3000 to 5000	$f'_c + 1200$
> 5000	$f'_c + 1400$

- (A) 4940 psi
- (B) 5160 psi
- (C) 5570 psi
- (D) 5820 psi

The Answer is A

Step1. Determine the specified compressive strength by utilizing the equation given:

$$f'_c = 200 * RHT - 500$$

$$f'_c = 200 * 21.2 - 500 = 3740\text{psi}$$

Step2. Determine the required average compressive strength from the equation given in the table:

$$f'_{cr} = f'_c + 1200$$

$$f'_{cr} = 3740 + 1200 = 4940\text{psi}$$

51) The modulus of elasticity of concrete is given by the following equation:

$$E = 39700\sqrt{f'_{cr}}$$

The specified compressive strength for a concrete structure is 4500psi for a plant where the standard deviation is unknown. Determine the modulus of elasticity for the concrete. The following table may be consulted:

Specified Compressive Strength, f'_c (psi)	Required Average Compressive Strength, f'_{cr} (psi)
< 3000	$f'_c + 1000$
3000 to 5000	$f'_c + 1200$
> 5000	$f'_c + 1400$

- (A) 2.0×10^6 psi
- (B) 2.5×10^6 psi
- (C) 3.0×10^6 psi
- (D) 3.5×10^6 psi

The Answer is C

Step1. Determine the required average compressive strength with the equation given in the table.

$$f'_{cr} = f'_c + 1200$$

$$f'_{cr} = 4500 + 1200 = 5700\text{psi}$$

Step2. Determine the modulus of elasticity given the equation in problem:

$$E = 39700\sqrt{f'_{cr}}$$

$$E = 39700\sqrt{5700} = 3.0 * 10^6\text{psi}$$

52) The modulus of elasticity of concrete is given by the following equation:

$$E = 39700\sqrt{f'_{cr}}$$

The specified compressive strength for a concrete structure is 5500psi for a plant where the standard deviation is 750psi for 100 tests. Determine the modulus of elasticity for the concrete given the following equations.

$$f'_{cr} = f'_c + 1.34s \quad f'_{cr} = f'_c + 2.33s - 500$$

- (A) 3.3×10^6 psi
- (B) 3.7×10^6 psi
- (C) 4.0×10^6 psi
- (D) 4.5×10^6 psi

The Answer is A

Step1. Determine the required average compressive strength by choosing the higher of the two equations:

$$f'_{cr} = f'_c + 1.34s$$

$$f'_{cr} = f'_c + 2.33s - 500$$

$$f'_{cr} = 5500 + 1.34 * 750 = 6505\text{psi}$$

$$f'_{cr} = 5500 + 2.33(750) - 500 = 6748\text{psi}$$

Step2. Determine the modulus of elasticity given the equation in problem:

$$E = 39700\sqrt{f'_{cr}}$$

$$E = 39700\sqrt{6748} = 3.3 * 10^6\text{psi}$$

- 53) The modulus of elasticity of concrete is given by the following equation:

$$E = 39700\sqrt{f'_{cr}}$$

The specified compressive strength for a concrete structure is 3500psi for a plant where the standard deviation is 400 psi. Due to the lack of tests, the modification factor was calculated as 1.07. Determine the modulus of elasticity for the concrete given the equations.

$$f'_{cr} = f'_c + 1.34s \quad f'_{cr} = f'_c + 2.33s - 500$$

- (A) 2.0×10^6 psi
- (B) 2.5×10^6 psi
- (C) 3.0×10^6 psi
- (D) 3.5×10^6 psi

The Answer is B

Step1. Determine the modified standard deviation.

$$s' = sF$$

$$s' = 400 * 1.07 = 428\text{psi}$$

Step2. Determine the required average compressive strength by choosing the higher of the two equations.

$$f'_{cr} = f'_c + 1.34s$$

$$f'_{cr} = f'_c + 2.33s - 500$$

$$f'_{cr} = 3500 + 1.34 * 428 = 4074\text{psi}$$

$$f'_{cr} = 3500 + 2.33(428) - 500 = 3997\text{psi}$$

Step3. Determine the modulus of elasticity given the equation in problem:

$$E = 39700\sqrt{f'_{cr}}$$

$$E = 39700\sqrt{3997} = 2.5 * 10^6\text{psi}$$

- 54) The modulus of elasticity of concrete is given by the following equation:

$$E = 39700\sqrt{f'_{cr}}$$

It was determined that the modulus of elasticity for the concrete was 3.0×10^6 psi. Determine the specified strength for the concrete if the standard deviation is 400psi over 50 tests. Consider the following equation:

$$f'_{cr} = f'_c + 1.34s$$

- (A) 4777 psi
- (B) 4980 psi
- (C) 5035 psi
- (D) 5174 psi

The Answer is B

Step1. Determine the required average compressive strength by rearranging the following equation:

$$E = 39700\sqrt{f'_{cr}}$$

$$f'_{cr} = \left(\frac{E}{39700}\right)^2$$

$$f'_{cr} = \left(\frac{3.0 * 10^6}{39700}\right)^2 = 5710\text{psi}$$

Step2. Determine the specified compressive strength by rearranging the following equation:

$$f'_{cr} = f'_c + 1.34s$$

$$f'_c = f'_{cr} - 1.34s$$

$$f'_c = 5710 - 1.34 * 400 = 5174\text{psi}$$

55) The modulus of elasticity of concrete is given by the following equation:

$$E = 39700\sqrt{f'_{cr}}$$

It was determined that the modulus of elasticity for the concrete was 2.7×10^6 psi.
Determine the standard deviation for the concrete if specified strength of concrete is 3000psi.

$$f'_{cr} = f'_c + 2.33s - 500$$

- (A) 417 psi
- (B) 681 psi
- (C) 926 psi
- (D) 1341 psi

The Answer is D

Step1. Determine the required average compressive strength by rearranging the following equation:

$$E = 39700\sqrt{f'_{cr}}$$

$$f'_{cr} = \left(\frac{E}{39700}\right)^2$$

$$f'_{cr} = \left(\frac{2.7 * 10^6}{39700}\right)^2 = 5625\text{psi}$$

Step2. Determine the standard deviation by rearranging the following equation:

$$f'_{cr} = f'_c + 2.33s - 500$$

$$s = \frac{f'_{cr} - f'_c + 500}{2.33}$$

$$s = \frac{5625 - 3000 + 500}{2.33} = 1341\text{psi}$$

- 56) An engineer wants to incorporate air entrainers into the concrete mix using the manufacturer's specification of 0.1 fluid ounces per 1% air content per 100 lbs cement. The concrete mixture you are currently working with has a 0.4 water-cementitious materials ratio, 244 lb/yd³ of water, and 3% air content. Determine the amount of air entrainer needed.
- (A) 1.2 fluid ounces
 - (B) 1.5 fluid ounces
 - (C) 1.8 fluid ounces
 - (D) 2.1 fluid ounces

The Answer is C

Step1. Determine the weight of cement per unit volume in the mixture by rearranging the following equation:

$$\begin{aligned} \text{W:C Ratio} &= \frac{W_{\text{water}}}{W_{\text{cement}}} \\ W_{\text{cement}} &= \frac{W_{\text{water}}}{\text{W:C Ratio}} \\ W_{\text{cement}} &= \frac{244}{0.4} = 610 \frac{\text{lbs}}{\text{yd}^3} \end{aligned}$$

Step2. Determine the amount of air entrainer needed by using the manufacturer's specifications:

$$\begin{aligned} \text{Air Entrainment} &= 0.1 * \text{Air Content} * \left(\frac{W_{\text{cement}}}{100} \right) \\ \text{Air Entrainment} &= 0.1 * 3 * \frac{610}{100} = 1.8 \text{ fluid ounces} \end{aligned}$$

57) The amount of air entrainer specified for a mix is 0.2 fluid ounces per 1% air content per 100 lbs cement. For this given mix, 3.7 fluid ounces of air entrainer was added. Consider 250 lbs/yd^3 of water added to the mixture and 5% air content, determine the water to cement ratio of the mix.

- (A) 0.54
- (B) 0.68
- (C) 0.72
- (D) 0.76

The Answer is B

Step1. Determine the weight of cement for the mixture by rearranging the manufacturer's specifications:

$$\text{Air Entrainment} = 0.2 * \text{Air Content} * \left(\frac{W_{\text{cement}}}{100} \right)$$

$$W_{\text{cement}} = 100 * \frac{\text{Air Entrainment}}{0.2 * \text{Air Content}}$$

$$W_{\text{cement}} = 100 * \left(\frac{3.7}{0.2 * 5} \right) = 370 \frac{\text{lbs}}{\text{yd}^3}$$

Step2. Determine the water to cement ratio with the following equation.

$$\text{W:C Ratio} = \frac{W_{\text{water}}}{W_{\text{cement}}}$$

$$\text{W:C Ratio} = \frac{250}{370} = 0.68$$

58) A concrete mixture contains the following materials: cement, water, gravel, and sand. The weights for the cement, water, and gravel are 550 lbs/yd³, 250 lbs/yd³, and 2100 lbs/yd³ respectively. The specific gravity of the cement and gravel are 3.1 and 2.6 respectively. Considering the mixture has 4% air content, determine the volume of sand in the concrete mixture.

- (A) 0.158 yd³
- (B) 0.201 yd³
- (C) 0.228 yd³
- (D) 0.249 yd³

The Answer is B

Step1. Determine the volume of cement in the mixture with the following equation:

$$V_{\text{cement}} = \frac{W_{\text{cement}}}{G_{\text{cement}} * \gamma_{\text{water}}}$$
$$V_{\text{cement}} = \frac{550}{3.1 * 1684.8} = 0.105 \text{yd}^3$$

Step2. Determine the volume of water in the mixture with the following equation.

$$V_{\text{water}} = \frac{W_{\text{water}}}{\gamma_{\text{water}}}$$
$$V_{\text{water}} = \frac{250}{1684.8} = 0.148 \text{yd}^3$$

Step3. Determine the volume of gravel in the mixture with the following equation:

$$V_{\text{gravel}} = \frac{W_{\text{gravel}}}{G_{\text{gravel}} * \gamma_{\text{water}}}$$
$$V_{\text{gravel}} = \frac{2100}{2.6 * 1684.8} = 0.479 \text{yd}^3$$

Step4. Determine the volume of sand in the mixture with the following formula:

$$V_{\text{sand}} = 1 - V_{\text{cement}} - V_{\text{water}} - V_{\text{gravel}} - V_{\text{air}}$$
$$V_{\text{sand}} = 1 - 0.105 - 0.148 - 0.479 - 0.040 = 0.228 \text{yd}^3$$

59) A concrete mixture contains the following materials: cement, water, gravel, and sand. The following volumes are already known: volume of cement = 0.10 yd^3 , volume of water = $.18 \text{ yd}^3$, volume of sand = 0.25 yd^3 , and volume of air = 0.04 yd^3 . Given the specific gravity of gravel as 2.8, determine the weight of gravel in the mixture.

- (A) 1801 lbs/yd^3
- (B) 1846 lbs/yd^3
- (C) 1988 lbs/yd^3
- (D) 2028 lbs/yd^3

The Answer is D

Step1. Determine the volume of gravel with the following equation:

$$V_{\text{gravel}} = 1 - V_{\text{cement}} - V_{\text{water}} - V_{\text{sand}} - V_{\text{air}}$$

$$V_{\text{gravel}} = 1 - 0.10 - 0.18 - 0.25 - 0.04 = 0.43 \text{ yd}^3$$

Step2. Determine the weight of gravel by rearranging the following equation:

$$V_{\text{gravel}} = \frac{W_{\text{gravel}}}{G_{\text{gravel}} * \gamma_{\text{water}}}$$

$$W_{\text{gravel}} = V_{\text{gravel}} * G_{\text{gravel}} * \gamma_{\text{water}}$$

$$W_{\text{gravel}} = 0.43 * 2.8 * 1684.8 = 2028 \frac{\text{lbs}}{\text{yd}^3}$$

60) A designer intends to use 70 pounds of coarse aggregate in one cubic foot batch of concrete. The bulk specific gravity of the aggregate is 2.800. Determine the air content of the mixture if the volume of sand, cement, and water are 25%, 19% and 10% respectively.

- (A) 4%
- (B) 5%
- (C) 6%
- (D) 7%

The Answer is C

Step1. Determine the percent of aggregate by mixture volume with the following equation:

$$\% \text{Mix} = \frac{W_{\text{agg}}}{G_{\text{agg}} * \gamma_{\text{water}}} * 100$$

$$\% \text{Mix} = \frac{70}{2.800 * 62.4} * 100 = 40\%$$

Step2. Determine the air content with the following equation:

$$\text{Air Content} = 100 - V_{\text{agg}} - V_{\text{cement}} - V_{\text{sand}} - \text{Vol}_{\text{water}}$$

$$\text{Air Content} = 100 - 40 - 19 - 25 - 10 = 6\%$$

61) A designer is trying to calculate how much aggregate to place in one cubic foot batch of concrete. The bulk specific gravity of the aggregate is 2.700. Determine how much aggregate should be added to the mixture if the volume of sand, cement, water, and air are 0.27 ft^3 , 0.18 ft^3 , 0.11 ft^3 , 0.04 ft^3 respectively.

- (A) 61 lbs/ft^3
- (B) 67 lbs/ft^3
- (C) 72 lbs/ft^3
- (D) 77 lbs/ft^3

The Answer is B

Step1. Determine the volume of aggregate in the mixture with the following equation:

$$V_{\text{agg}} = 1 - V_{\text{sand}} - V_{\text{cement}} - V_{\text{water}} - V_{\text{air}}$$

$$V_{\text{agg}} = 1 - 0.27 - 0.18 - 0.11 - 0.04 = 0.40 \text{ ft}^3$$

Step2. Determine the weight of aggregate that should be added to the mixture by rearranging the following equation:

$$V_{\text{agg}} = \frac{W_{\text{agg}}}{G_{\text{agg}} * \gamma_{\text{water}}}$$

$$W_{\text{agg}} = V_{\text{agg}} * G_{\text{agg}} * \gamma_{\text{water}}$$

$$W_{\text{agg}} = 0.40 * 2.7 * 62.4 = 67 \frac{\text{lbs}}{\text{ft}^3}$$

62) The design of a concrete mix requires 1200 kg/m^3 of aggregate in dry condition and 600 kg/m^3 of sand in dry condition. The aggregate has a moisture content of 1.0% and absorption of 1.5%. The sand has a moisture content of 0.8% and absorption of 1.5%. Determine the cumulative amount of aggregate and sand needed in the mixture.

- (A) 1800 kg/m^3
- (B) 1807 kg/m^3
- (C) 1817 kg/m^3
- (D) 1825 kg/m^3

The Answer is C

Step1. Determine the amount of aggregate needed in the mixture with the following equation:

$$W_{\text{agg}} = W_{\text{dry}} * (1 + \text{Moisture Content})$$

$$W_{\text{agg}} = 1200 * (1 + 0.01) = 1212 \frac{\text{kg}}{\text{m}^3}$$

Step2. Determine the amount of sand needed in the mixture with the following equation:

$$W_{\text{sand}} = W_{\text{dry}} * (1 + \text{Moisture Content})$$

$$W_{\text{agg}} = 600 * (1 + 0.008) = 605 \frac{\text{kg}}{\text{m}^3}$$

Step3. Take the sum of the two weights needed:

$$W_{\text{total}} = 1212 + 605 = 1817 \frac{\text{kg}}{\text{m}^3}$$

63). Structural fill is required to support a slab-on-grade in an area with a deep frost line. To avoid potential problems with frost heave, the best material for structural fill would be:

- A) low-plasticity cohesive soil compacted dry of optimum (CL)
- B) Inelastic silt (ML)
- C) Silty sand (SM)
- D) Well-graded sand (SW)

The Answer is C

Well-graded sand is the least susceptible to frost heave.

64) The design of a concrete mix requires 1200 kg/m^3 of aggregate in dry condition and 160 kg/m^3 of free water. The aggregate has a moisture content of 1.0% and absorption of 1.5%. Determine the mass of water that should be used for the concrete mix.

- (A) 166 kg/m^3
- (B) 160 kg/m^3
- (C) 154 kg/m^3
- (D) 150 kg/m^3

The Answer is A

Step1. Determine the amount of water that the aggregate will absorb or release with the following equation:

$$W_{\text{aggwater}} = W_{\text{dry}} * (\text{Moisture Content} - \text{Absorption})$$

$$W_{\text{aggwater}} = 1200 * (0.010 - 0.015) = -6 \frac{\text{kg}}{\text{m}^3}$$

Step2. Determine the amount of water needed for the concrete mixture with the following equation:

$$W_{\text{water}} = W_{\text{waterneeded}} - W_{\text{aggwater}}$$

$$W_{\text{water}} = 160 - (-6) = 166 \frac{\text{kg}}{\text{m}^3}$$

65) While one person and a wheel of powered equipment is working on the 8in/1ft sloped roof of a house. What can you reduce the Personnel and Equipment Construction load to?

- A. 2500lbs B. 250 lbs C. 2000 lbs D. 1000 lbs

The Answer is C

Step 1: Figure out what is the construction load.

Using Table 1 in ASCE 37-02

- each person = 250lb
 - wheel of powered equipment = 2000lbs
- construction load = 2500lbs

Step 2: Figure out the reduction factor as per ASCE 37-02.

Using paragraph 4.8.3.3 Personnel and equipment loads on sloping roofs. Use

$$R = 1.2 - .05F = 1.2 - .05(8\text{in}/1\text{ft}) = 1.2 - .4 = .8$$

Step 3: Calculate the allowable design construction load.

- Allowable construction load = 2500lbs * .8 = 2000 lbs

66) Given a soil fill sample with a weight of 62 lbs and a total volume of 864 in^3 and a water content of 15%, determine the percent relative compaction of the sample if the maximum dry unit weight is 115pcf.

- (A) 93.8% (B) 95.2%
- (C) 102% (D) 127%

The Answer is A**Step 1**

The relative compaction cannot exceed 100%. That being known answers C and D are eliminated. To determine the relative compaction we must determine the dry unit weight of the fill sample. Then we can compare it to the maximum dry unit weight.

$$\gamma_d = \frac{\frac{W_T}{V_T}}{1 + \frac{\omega, \%}{100\%}} = \frac{\frac{62 \text{ lb}}{864 \text{ in}^3 \left(\frac{1 \text{ ft}}{12 \text{ in}}\right)^3}}{1 + \frac{15\%}{100\%}} = \frac{\frac{62 \text{ lb}}{0.5 \text{ ft}^3}}{1.15} = 107.8 \text{ pcf}$$

Step 2

The relative compaction is the dry unit weight of the sample divided by the maximum unit weight:

$$\frac{\gamma_d}{\gamma_{d, \max}} \times 100\% = \frac{107.8 \text{ pcf}}{115 \text{ pcf}} \times 100\% = 93.8\%$$

67) The in situ moisture content of a soil is 23% and the moist unit weight is 110 lb/ft^3 . The Specific Gravity of the soil solids is 2.73. If the specifications call for the soil to be compacted to a minimum dry unit weight of 104 lb/ft^3 at the same moisture content of 23%. Approximately how many cubic yards of soil from the excavation site are needed to produce 12,000 cubic yards of compacted fill?

- (A) $10,300 \text{ yd}^3$ (B) $12,400 \text{ yd}^3$
(C) $14,000 \text{ yd}^3$ (D) $17,000 \text{ yd}^3$

The Answer is C**Step 1**

The total mass must remain the same. Therefore the product of volume and unit weight must remain the same:

$$V_T \gamma_d = V_{T,comp} \gamma_{d,min}$$

Step 2

We know $V_{T,comp} = 12,000 \text{ yd}^3$ and $\gamma_{d,min} = 104 \text{ pcf}$.

We must calculate γ_d .

$$\gamma_d = \frac{\gamma_T}{1 + \frac{\omega, \%}{100\%}} = \frac{110 \text{ pcf}}{1 + \frac{23\%}{100\%}} = 89.4 \text{ pcf}$$

Now solving for the volume of soil from the excavation.

$$V_T = V_{T,comp} \frac{\gamma_{d,min}}{\gamma_d} = 12000 \text{ yd}^3 \times \frac{104 \text{ pcf}}{89.4 \text{ pcf}} = 13960 \text{ yd}^3 \approx 14000 \text{ yd}^3$$

68) The in situ moisture content of a soil is 15% and a void ratio of 0.5. The Specific Gravity of the soil solids is 2.65. If the specifications call for the soil to be compacted to a minimum dry unit weight of 115 lb/ft^3 at the same moisture content of 15%.

Approximately how many cubic yards of soil from the excavation site are needed to produce 20,000 cubic yards of compacted fill?

- (A) 19,200 yd^3 (B) 20,900 yd^3
(C) 22,600 yd^3 (D) 25,000 yd^3

The Answer is B**Step 1**

The total mass must remain the same. Therefore the product of volume and unit weight must remain the same:

$$V_T \gamma_d = V_{T,comp} \gamma_{d,min}$$

Step 2

We know $V_{T,comp} = 20,000 \text{ yd}^3$ and $\gamma_{d,min} = 115 \text{ pcf}$.

We must calculate γ_d .

$$\gamma_d = \frac{G_s \gamma_w}{1 + e} = \frac{2.65 \times 62.4 \text{ pcf}}{1 + 0.5} = 110.24 \text{ pcf}$$

Now solving for the volume of soil from the excavation.

$$V_T = V_{T,comp} \frac{\gamma_{d,min}}{\gamma_d} = 20000 \text{ yd}^3 \times \frac{115 \text{ pcf}}{110.24 \text{ pcf}} = 20,863 \text{ yd}^3 \approx 20,900 \text{ yd}^3$$

Common Mistakes

If $V_T = V_{T,comp} \frac{\gamma_d}{\gamma_{d,min}}$ is used instead of the appropriate $V_T = V_{T,comp} \frac{\gamma_{d,min}}{\gamma_d}$ the result of 19,172 cubic yards will lead to selection of Answer A.

69) A proposed embankment fill requires 7,000 yd³ of compacted soil. The void ratio of the compacted fill is specified as 0.65. Four borrow pits are available as described in the following table, which lists the respective void ratios of the soil and the cost per cubic yard for moving the soil to the proposed construction site. Select the pit from which the soil should be bought to minimize the cost.

Borrow Pit	Void Ratio	Cost (\$/yd ³)
A	0.85	9.00
B	0.75	10.00
C	0.95	7.00
D	1.20	6.00

(A) A

(B) B

(C) C

(D) D

The Answer is D

Step 1

First we must determine how to get the costs using the void ratios.

$$e \equiv \frac{V_v}{V_s}$$

$$e = \frac{V_v}{V_s} = \frac{V_T - V_s}{V_s} = \frac{V_T}{V_s} - 1 \rightarrow \frac{V_T}{V_s} = 1 + e \rightarrow V_s = \frac{V_T}{1 + e}$$

Step 2

From this we obtain two relations:

$$V_s = \frac{V_{\text{fill}}}{1 + e_{\text{fill}}} \text{ and } V_s = \frac{V_{T,\text{pit}}}{1 + e_{\text{pit}}} \text{ where,}$$

$V_{T,\text{pit}} \equiv$ total volume taken from given pit

$e_{\text{pit}} \equiv$ void ratio of given pit

$V_{\text{fill}} \equiv$ required volume of the embankment

$e_{\text{pit}} \equiv$ void ratio of the embankment soil

Step 3

We can determine the required volume from each pit from the second relation as:

$$(1 + e_{\text{pit}}) V_s = V_{T,\text{pit}}$$

From the first relation we can determine V_s as: $V_s = \frac{7000\text{yd}^3}{1 + 0.65}$

Step 4

Now we can determine the total volume and total cost for each pit knowing: Total Cost = Unit Cost $\times V_{T,\text{pit}}$.

Now solving for the volume of soil required from each borrow pit:

$$\text{Borrow Pit A: } V_{T,\text{pit}} = (1 + 0.85) \left(\frac{7000\text{yd}^3}{1.65} \right) = 7848.5\text{yd}^3$$

$$\text{Borrow Pit B: } V_{T,\text{pit}} = (1 + 0.75) \left(\frac{7000\text{yd}^3}{1.65} \right) = 7424.2\text{yd}^3$$

$$\text{Borrow Pit C: } V_{T,\text{pit}} = (1 + 0.95) \left(\frac{7000\text{yd}^3}{1.65} \right) = 8272.7\text{yd}^3$$

$$\text{Borrow Pit D: } V_{T,\text{pit}} = (1 + 1.20) \left(\frac{7000\text{yd}^3}{1.65} \right) = 9333.3\text{yd}^3$$

Now solving for the cost of soil given the volume from each borrow pit:

$$\text{Borrow Pit A: Total Cost} = 7848.5\text{yd}^3 \times \$9.00/\text{yd}^3 = \$70,637$$

Borrow Pit B: Total Cost = $7424.2\text{yd}^3 \times \$10.00/\text{yd}^3 = \$74,242$

Borrow Pit C: Total Cost = $8272.7\text{yd}^3 \times \$7.00/\text{yd}^3 = \$57,909$

Borrow Pit D: Total Cost = $9333.3\text{yd}^3 \times \$6.00/\text{yd}^3 = \$56,000$

The total cost is the lowest for Borrow Pit D.

70) A new roadway construction requires the sub base soil to have a dry density of 125 pcf and optimum moisture content (OMC) of 12.5%. A smooth drum roller will be used to compact the soil in 4-inch-thick lifts while the width is 32 ft. The soil has been tested in place and the results show moisture content of 6%. The water must be added to the stationing length of 100ft to obtain the required moisture content for compaction. How many gallons per yard must be added to meet the requirements?

- (A) 3.42 gal/yd²
- (B) 5.63 gal/yd²
- (C) 2.34 gal/yd²
- (D) 2.93 gal/yd²

The Answer is D

Gallons of Water =

$$\begin{aligned} &= \text{Desired dry density (pcf)} \times \frac{\text{Compacted cubic feet of soil}}{8.33 \text{ lb/gal}} \\ &\quad \times \frac{\text{Goal water content\%} - \text{Existing water content \%}}{100} \\ &= 125 \text{ pcf} \times \frac{32 \text{ ft} \times 100 \text{ ft (sta.)} \times (4 \text{ in} / 12 \text{ in/ft})}{8.33 \text{ lb/gal}} \times \frac{12.5\% - 6\%}{100} = 1040.42 \text{ gal /sta.} \\ \text{Gallonsyd}^2 &= \frac{1040.42 \text{ gal/sta.}}{(32 \text{ ft} \times 100 \text{ ft/sta}) / 9 \text{ ft}^2 / \text{yd}^2} = 2.93 \text{ gal /yd}^2 \end{aligned}$$

*See Fundamentals of Building Construction: Materials and Methods Wiley

71. A construction company is working on a new roadway construction with which the sub base soil is to have an optimum moisture content (OMC) of 10.5%. A smooth drum roller will be used to compact the soil in 6-inch-thick lifts while the width of the road is 25ft. The soil has been tested in place and the results show moisture content of 7%. The contractor will need to add 2.34 gallons per yard of water to the stationing length of 100ft to obtain the required moisture content for compaction. What is the desired dry density of the road sub base?

- (A) 61.9 pcf
- (B) 123.8 pcf
- (C) 138.2 pcf
- (D) 115.1 pcf

The Answer is B

$$\text{Gallons/yd}^2 = 2.34 \text{ gal /yd}^2 = \frac{x \text{ gal/sta.}}{(25\text{ft} \times 100\text{ft/sta})/9 \text{ ft}^2 \text{ /yd}^2}$$

Gallons of Water = 650 gallons

$$650 \text{ gal} = \text{Desired dry density (pcf)} \times \frac{\text{Compacted cubic feet of soil}}{8.33 \text{ lb/gal}} \\ \times \frac{\text{Goal water content\%} - \text{Existing water content \%}}{100}$$

$$650 \text{ gal} = X \text{ pcf} \times \frac{25\text{ft} \times 100\text{ft(sta.)} \times \left(\frac{6\text{in}}{12\text{ft}}\right)}{8.33 \frac{\text{lb}}{\text{gal}}} \times \frac{10.5\% - 7\%}{100}$$

$$X = 123.8 \text{ pcf}$$

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72. A mold has a volume of 0.15 ft^3 and a mass of 3 lb. After the mold is filled with moist soil, it weighs 16 lbf. The mold with the soil is then oven dried at 110°C to a constant mass of 15 lb. It is known that the soil solids have a specific gravity of 2.71. Determine the porosity of the soil.

A) 0.33

B) 0.43

C) 0.53

D) 0.63

The Answer is C

Step 1:

From the given information, it is known that the moist soil sample with a volume of $V=0.15 \text{ ft}^3$ weighs $(16-3)=13 \text{ lbf}$. After drying in oven, its mass is $W_s=(15-3)=12 \text{ lb}$.

Step 2:

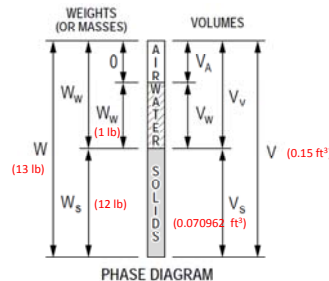
The volume of the soil solids is

$$V_s = \frac{W_s}{\gamma_s} = \frac{W_s}{G\gamma_w} = \frac{12 \text{ lb}}{2.71 \times 62.4 \text{ lb/ft}^3} = 0.070962 \text{ ft}^3$$

The volume of water and air is

$$V_v = V - V_s = 0.15 \text{ ft}^3 - 0.070962 \text{ ft}^3 = 0.079038 \text{ ft}^3$$

Plot the known information in the phase diagram, as follows.



Step 3:

$$\text{Porosity of the soil is } n = \frac{V_v}{V} = \frac{0.079038 \text{ ft}^3}{0.15 \text{ ft}^3} \cong 0.53$$

73. A mold has a volume of 0.15 ft^3 and a mass of 3 lb. After the mold is filled with moist soil, it weighs 16 lbf. The mold with the soil is then oven dried to a constant mass of 15 lb. It is known that the soil solids have a specific gravity of 2.71. What is the degree of saturation of the soil?
- A) 30.2% B) 20.3% C) 25.2% D) 31.5%

The Answer is B**Step 1:**

From the given information, it is known that the moist soil sample with a volume of $V=0.15 \text{ ft}^3$ weighs $(16-3)=13 \text{ lb}$. After drying in oven, its mass is $W_s=(15-3)=12 \text{ lb}$. Therefore, the mass of water in the moist soil sample is $W_w=(13-12)=1 \text{ lb}$.

Step 2:

The volume of the soil solids is

$$V_s = \frac{W_s}{\gamma_s} = \frac{W_s}{G\gamma_w} = \frac{12 \text{ lbf}}{2.71 \times 62.4 \text{ lbf/ft}^3} = 0.070962 \text{ ft}^3$$

The volume of water is

$$V_w = \frac{W_w}{\gamma_w} = \frac{1 \text{ lbf}}{62.4 \text{ lbf/ft}^3} = 0.016026 \text{ ft}^3$$

The volume of water and air is

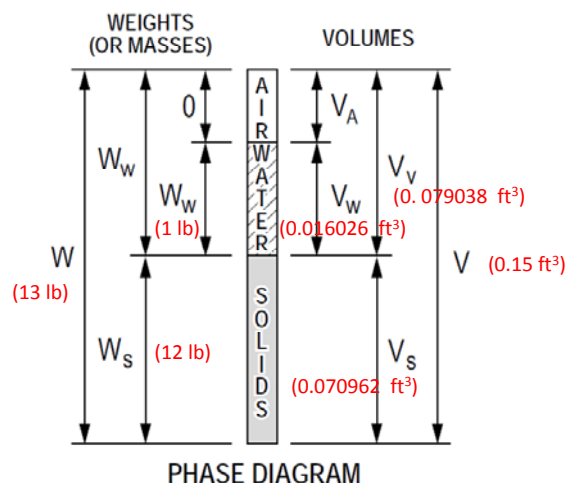
$$V_v = V - V_s = 0.15 \text{ ft}^3 - 0.070962 \text{ ft}^3 = 0.079038 \text{ ft}^3$$

Plot the known information in the phase diagram, as follows.

Step 3:

The degree of saturation of the soil is

$$S = \frac{V_w}{V_v} \times 100\% = \frac{0.016026 \text{ ft}^3}{0.079038 \text{ ft}^3} \cong 20.3\%$$



74. A mold has a volume of 0.15 ft^3 and a mass of 3 lb. After the mold is filled with moist soil, it weighs 16 lbf. The mold with the soil is then oven dried to a constant mass of 15 lb. It is known that the soil solids have a specific gravity of 2.71. What is the saturated unit weight of the soil if it is saturated by water while retaining the same total volume?
- A) 102.9 lbf/ft^3 B) 112.9 lbf/ft^3
C) 62.9 lbf/ft^3 D) 144.2 lbf/ft^3

The Answer is B**Step 1:**

From the given information, it is known that the moist soil sample with a volume of $V=0.15 \text{ ft}^3$ weighs $(16-3)=13 \text{ lbf}$. After drying in oven, its mass is $W_S=(15-3)=12 \text{ lbf}$.

Step 2:

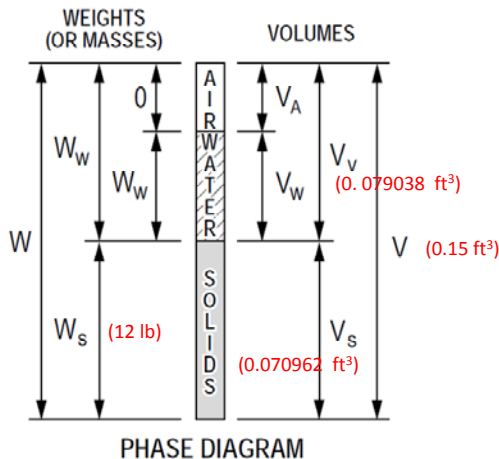
The volume of the soil solids is

$$V_S = \frac{W_S}{\gamma_S} = \frac{W_S}{G\gamma_W} = \frac{12 \text{ lbf}}{2.71 \times 62.4 \text{ lbf/ft}^3} = 0.070962 \text{ ft}^3$$

The volume of water and air is

$$V_V = V - V_S = 0.15 \text{ ft}^3 - 0.070962 \text{ ft}^3 = 0.079038 \text{ ft}^3$$

Plot the known information in the phase diagram, as follows.

**Step 3:**

When the soil is saturated by water, the air volume is occupied by water. Therefore, the weight of water in the saturated soil is

$$W_W = V_V \gamma_W = 0.079038 \text{ ft}^3 \times 62.4 \frac{\text{lbf}}{\text{ft}^3} \cong 4.932 \text{ lbf}$$

The saturated unit weight of the soil is

$$\gamma_{\text{sat}} = \frac{W_W + W_S}{V} = \frac{4.932 \text{ lbf} + 12 \text{ lbf}}{0.15 \text{ ft}^3} \cong 112.9 \text{ lbf/ft}^3$$

75. Laboratory tests on a soil revealed a maximum dry unit weight of 19.0 kN/m^3 , and a minimum dry unit weight of 17.5 kN/m^3 . A sample of the soil obtained from the field has a dry unit weight of 18.4 kN/m^3 . What is the relative density of the soil sample?

A) 38%

B) 62%

C) 68%

D) 72%

The Answer is B**Step 1:**

Based on the NCEES reference manual, the relative density (D_r) is defined by

$$D_r = \frac{(Y_{D-\text{Field}} - Y_{D-\text{min}}) Y_{D-\text{max}}}{(Y_{D-\text{max}} - Y_{D-\text{min}}) Y_{D-\text{Field}}} \times 100\%$$

Step 2:

Since the maximum and minimum dry unit weights and the field dry unit weight are all given, the relative density (D_r) of the soil sample is

$$D_r = \frac{\left(18.4 \frac{\text{kN}}{\text{m}^3} - 17.5 \text{ kN/m}^3\right) 19.0 \text{ kN/m}^3}{\left(19.0 \frac{\text{kN}}{\text{m}^3} - 17.5 \text{ kN/m}^3\right) 18.4 \text{ kN/m}^3} \times 100\% \cong 62\%$$

76. A soil has a void ratio of 0.700 and a water content of 22 percent. If the soil solids have a specific gravity of 2.70, what is the total unit weight of the soil?

- A) 102 lbf/ft³ B) 112 lbf/ft³
C) 210 lbf/ft³ D) 121 lbf/ft³

The Answer is D

Step 1:

Assume that the volume of soil solids $V_s = 1 \text{ ft}^3$. Since the void ratio $e=0.700$, the volume of water and air $V_v = eV_s = 0.7 \text{ ft}^3$.

Step 2:

The weight of the soil solids is

$$W_s = V_s \gamma_s = V_s G \gamma_w = (1 \text{ ft}^3)(2.70) \left(62.4 \frac{\text{lbf}}{\text{ft}^3} \right) = 168.48 \text{ lbf}$$

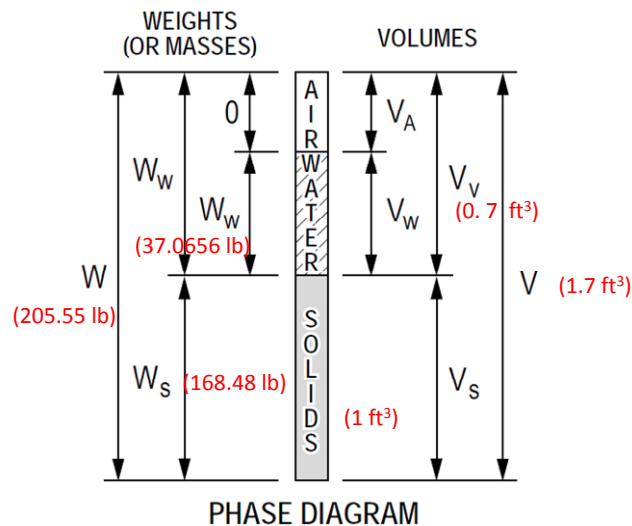
The weight of water is

$$W_w = W_s \omega = 168.48 \text{ lbf} \times 22\% = 37.0656 \text{ lbf}$$

The weight of water and soil solids is

$$W = W_s + W_w = 168.48 \text{ lbf} + 37.0656 \text{ lbf} = 205.55 \text{ lbf}$$

Plot the known information in the phase diagram, as follows.



Step 3:

The total unit weight of the soil is

$$\gamma = \frac{W}{V} = \frac{205.55 \text{ lbf}}{1.7 \text{ ft}^3} \cong 120.9 \text{ lbf/ft}^3$$

77. In a falling head permeability test on a soil sample, the following data are available:

Cross-sectional area of soil = 60 cm^2

Length of soil = 10 cm

Initial head = 120 cm

Final head = 108 cm

Duration of test = 20 minutes

Diameter of tube = 8 mm

Determine the coefficient of permeability of the soil, k .

A) 7.4 cm/sec

B) $7.36 \times 10^{-6} \text{ cm/sec}$

C) $5.8 \times 10^{-6} \text{ cm/sec}$

D) $4.60 \times 10^{-6} \text{ mm/sec}$

The Answer is C**Step 1:**

From a falling head test, coefficient of permeability, k , is calculated by

$$k = \frac{2.303aL}{At_E} \log_{10} \left(\frac{h_1}{h_2} \right)$$

where a = area of reservoir tube; L = length of flow; A = cross-sectional area of soil;

t_E = elapsed time during falling head test; h_1 = initial head; h_2 = final head.

Step 2:

From the given information, it is known that

$$a = \frac{\pi D^2}{4} = \frac{\pi (8 \text{ mm})^2}{4} = 0.503 \text{ cm}^2, L = 10 \text{ cm}, A = 60 \text{ cm}^2, t_E = 20 \text{ minutes} =$$

$$20 \text{ minutes} \times 60 \frac{\text{seconds}}{\text{minute}} = 1200 \text{ seconds}, h_1 = 120 \text{ cm}, h_2 = 108 \text{ cm}.$$

Therefore,

$$k = \frac{2.303aL}{At_E} \log_{10} \left(\frac{h_1}{h_2} \right) = \frac{2.303(0.503 \text{ cm}^2)(10 \text{ cm})}{(60 \text{ cm}^2)(1200 \text{ seconds})} \log_{10} \left(\frac{120 \text{ cm}}{108 \text{ cm}} \right)$$

$$= 7.36 \times 10^{-6} \text{ cm/sec}$$

78. A permeability test is conducted with a sample of soil that is 80 mm in diameter and 140 mm long. The head is kept constant at 240 mm. The flow is 1.8 mL in 10 minutes. What is the coefficient of permeability?

- A) 4.38×10^{-6} cm/sec B) 3.48×10^{-5} cm/sec
C) 5.8×10^{-6} cm/sec D) 3.48×10^{-6} cm/sec

The Answer is B**Step 1:**

From a constant head test, coefficient of permeability, k , is calculated by

$$k = \frac{Q}{iAt_E}$$

where Q = total quantity of water collected over time t_E ; i = hydraulic gradient = h/L ;
 A = cross-sectional area of soil; t_E = elapsed time during falling head test; h = head; L = length of soil.

Step 2:

From the given information, it is known that

$$Q = 1.8 \text{ mL}, h = 240 \text{ mm}, L = 140 \text{ mm}, A = \frac{\pi D^2}{4} = \frac{\pi (80 \text{ mm})^2}{4} = 50.3 \text{ cm}^2, t_E =$$

$$10 \text{ minutes} = 10 \text{ minutes} \times 60 \frac{\text{seconds}}{\text{minute}} = 600 \text{ seconds},$$

Therefore,

$$k = \frac{Q}{iAt_E} = \frac{(1.8 \text{ mL})}{\left(\frac{240 \text{ mm}}{140 \text{ mm}}\right) (50.3 \text{ cm}^2) (600 \text{ seconds})} = 3.48 \times 10^{-5} \text{ cm/sec}$$

79. A triaxial shear test is performed on a well-drained sand soil sample. At failure, the normal stress on the failure plane was 4000 lbf/ft^2 and the shear stress on the failure plane was 3200 lbf/ft^2 . What is the angle of internal friction?

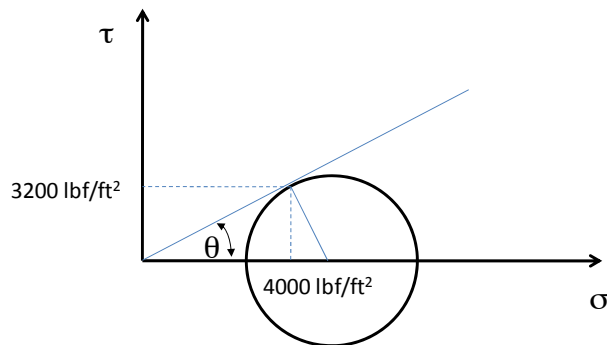
- A) 16° B) 25° C) 39° D) 62°

The Answer is C**Step 1:**

For a well-drained sand soil sample, the cohesion c is zero.

Step 2:

Use Mohr's circle to solve the problem, as shown below.



Therefore,

$$\theta = \arctan\left(\frac{3200 \frac{\text{lbf}}{\text{ft}^2}}{4000 \frac{\text{lbf}}{\text{ft}^2}}\right) = \arctan(0.8) = 38.7^\circ$$

80. A triaxial shear test is performed on an undrained soil sample with the following results:

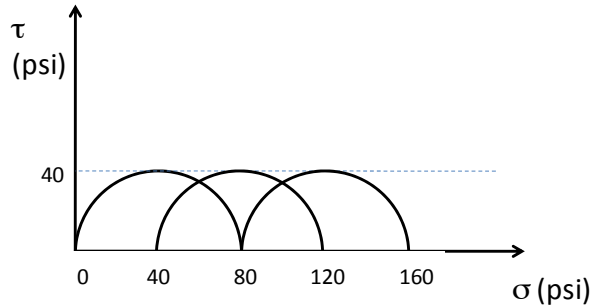
Confining Pressure (psi)	Total Axial Stress (psi)
0	80
40	120
80	160

What is the angle of internal friction?

- A) 0° B) 4° C) 16° D) 20°

The Answer is A

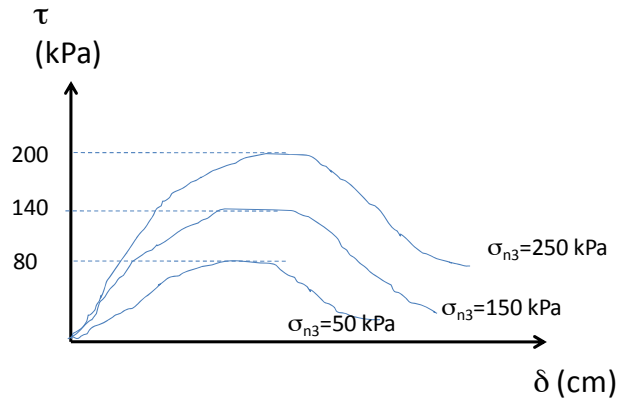
Use Mohr's circle to solve the problem, as shown below.



From the plot, it can be seen that the angle of internal friction is

$$\theta = 0^\circ$$

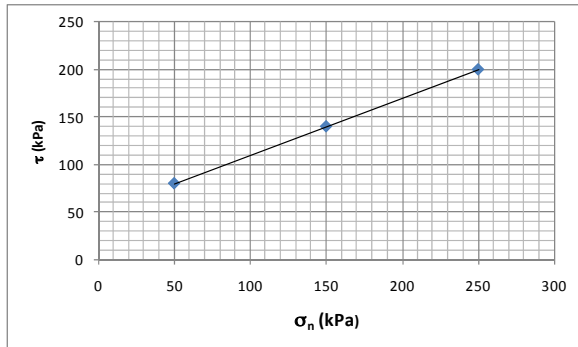
81. A soil sample was tested in a direct shear apparatus, with the results shown in the following stress-displacement curves under different normal stresses. What is the cohesion intercept of the soil?



- A) 20 kPa B) 40 kPa C) 50 kPa D) 60 kPa

The Answer is C**Step 1:**

Plot the maximum shear stress from the stress-displacement curve versus the normal stress from each individual test.

**Step 2:**

The best fit line through the three data points represents the failure envelope for the soil. Measuring from the plot, the cohesion intercept is $c = 50 \text{ kPa}$.

82. A cylindrical sample of clay soil with a diameter of 3 inches and a height of 6 inches is tested in an unconfined compression test. The sample fails under a force of 400 lbf, and at failure the sample height is reduced to 5.5 inches. What is the cohesion of the soil sample?

- A) 14.2 psi B) 28.3 psi C) 56.7 psi D) 113.4 psi

The Answer is B**Step 1:**

The normal stress at failure, σ_N , is calculated by $\sigma_N = \frac{P}{A}$, where P= normal force at failure; A=cross-sectional area of sample over which force acts.

Step 2:

From the given information, it is known that $P = 400 \text{ lbf}$, $A = \frac{\pi D^2}{4} = \frac{\pi(3 \text{ inches})^2}{4} = 7.07 \text{ in}^2$. Therefore, $\sigma_N = \frac{P}{A} = \frac{400 \text{ lbf}}{7.07 \text{ in}^2} = 56.59 \text{ psi}$.

Step 3:

The unconfined compressive strength, q_u , is equal to the normal stress at failure.

$$q_u = \sigma_N = 56.59 \text{ psi}.$$

Step 4:

In unconfined compression test, cohesion is equal to half of the unconfined compressive strength. Therefore,

$$c = \frac{1}{2} q_u = 0.5 \times 56.59 \text{ psi} = 28.3 \text{ psi}.$$

83. A sand soil sample is tested in direct shear in a 4 inch by 4 inch square shear box under a normal load of 200 lbf. it is known that the soil has an angle of internal friction $\phi=30^\circ$, a dry specific gravity of 2.6, and a coefficient of permeability $k=2$ inches/hour. What is the shear stress on the sample at failure?

- A) 7.2 psi B) 18.3 psi C) 26.7 psi D) 33.4 psi

The Answer is A**Step 1:**

The normal stress on the specimen, σ , is calculated by $\sigma = \frac{P}{A}$, where P= normal force;
A=cross-sectional area of sample over which force acts.

Step 2:

From the given information, it is known that $P = 200 \text{ lbf}$, $A = (4 \text{ inches})^2 = 16 \text{ in}^2$.

Therefore, $\sigma = \frac{P}{A} = \frac{200 \text{ lbf}}{16 \text{ in}^2} = 12.5 \text{ psi}$.

Step 3:

For sand soil, its cohesion (c) is zero. Based on Mohr-Coulomb strength equation, the shear stress on the sample at failure is $\tau_f = c + \sigma \tan \phi = 0 + (12.5 \text{ psi}) \tan 30^\circ = 7.2 \text{ psi}$.

84) An engineer specified a concrete strength of 5500 psi for a plant where the standard deviation of 17 tests results was 400 psi. Determine the required average compressive strength given the equations. The following table may be consulted.

$$f'_{cr} = f'_c + 1.34s \quad f'_{cr} = f'_c + 2.33s - 500$$

Number of tests	Modification Factor, F
15	1.16
20	1.08
25	1.03
30 or more	1.00

- (A) 6106 psi
- (B) 6321 psi
- (C) 6599 psi
- (D) 6845 psi

The Answer is A

Step1. Determine the modification factor by interpolating the values from the provided table.

$$F = 1.16 - \left(\frac{1.16 - 1.08}{20 - 15} \right) (17 - 15) = 1.13$$

Step2. Multiply standard deviation by modification factor:

$$s' = sF$$

$$s' = 400 * 1.13 = 452\text{psi}$$

Step3. Determine max f'_{cr} by taking the higher of the following two equations:

$$f'_{cr} = f'_c + 1.34s$$

$$f'_{cr} = f'_c + 2.33s - 500$$

$$f'_{cr} = 5500 + 1.34(452) = 6106\text{psi}$$

85) An engineer is using a plant where the standard deviation for concrete structures is 1000 psi. Determine the minimum number of tests required to have a modified standard deviation less than 1100psi. The following table may be consulted.

Number of tests	Modification Factor, F
15	1.16
20	1.08
25	1.03
30 or more	1.00

- (A) 16
- (B) 17
- (C) 19
- (D) 22

The Answers is C

Step1. Determine the modification factor by rearranging the following equation:

$$s' = sF$$

$$F = \frac{s'}{s}$$

$$F = \frac{1100}{1000} = 1.10$$

Step2. Iterate the number of tests. Note: The ideal modification factor is 1.10, so we know the number of tests is between 15 and 20. Also, since we want the standard deviation to be less than 1100psi, we want the number of tests that produces a modification factor as close to but not exceeding 1.10.

$$F = 1.16 - \left(\frac{1.16 - 1.08}{20 - 15} \right) (16 - 15) = 1.14$$

$$F = 1.16 - \left(\frac{1.16 - 1.08}{20 - 15} \right) (17 - 15) = 1.13$$

$$F = 1.16 - \left(\frac{1.16 - 1.08}{20 - 15} \right) (18 - 15) = 1.11$$

86) A concrete mixture requires 1150 kg/m^3 of gravel in dry condition, 600 kg/m^3 of sand in dry condition, and 165 kg/m^3 of free water. The gravel has a moisture content of 0.7% and absorption of 2.0%. The sand has a moisture content of 1.2% and absorption of 1.0%. What is the total mass of water needed for this mixture?

- (A) 175 kg/m^3
- (B) 179 kg/m^3
- (C) 182 kg/m^3
- (D) 185 kg/m^3

The Answers is B

Step1. Determine the amount of water that the gravel will absorb or release with the following equation:

$$W_{\text{gravelwater}} = W_{\text{dry}} * (\text{Moisture Content} - \text{Absorption})$$

$$W_{\text{gravelwater}} = 1150 * (0.007 - 0.020) = -15 \frac{\text{kg}}{\text{m}^3}$$

Step2. Determine the amount of water that the sand will absorb or release with the following equation:

$$W_{\text{sandwater}} = W_{\text{dry}} * (\text{Moisture Content} - \text{Absorption})$$

$$W_{\text{sandwater}} = 600 * (0.012 - 0.010) = 1 \frac{\text{kg}}{\text{m}^3}$$

Step3. Determine the amount of water needed for the concrete mixture with the following equation:

$$W_{\text{water}} = W_{\text{waterneeded}} - W_{\text{aggwater}} - W_{\text{sandwater}}$$

$$W_{\text{water}} = 165 - (-15) - 1 = 179 \frac{\text{kg}}{\text{m}^3}$$

87) A concrete mixture contains the following materials: cement, water, gravel, and sand. The weights for the cement, water, and gravel are 600 lbs/yd³, 250 lbs/yd³, and 2050 lbs/yd³ respectively. The specific gravity of the cement, gravel, and sand are 3.1, 2.6, and 2.4 respectively. Considering the mixture has 4% air content, determine the weight of sand in the concrete mixture.

- (A) 27 lbs/ft³
- (B) 30 lbs/ft³
- (C) 34 lbs/ft³
- (D) 39 lbs/ft³

The Answers is B

Step1. Determine the volume of cement in the mixture with the following equation:

$$V_{\text{cement}} = \frac{W_{\text{cement}}}{G_{\text{cement}} * \gamma_{\text{water}}}$$

$$V_{\text{cement}} = \frac{600}{3.1 * 1684.8} = 0.115 \text{yd}^3$$

Step2. Determine the volume of water in the mixture with the following equation.

$$V_{\text{water}} = \frac{W_{\text{water}}}{\gamma_{\text{water}}}$$

$$V_{\text{water}} = \frac{250}{1684.8} = 0.148 \text{yd}^3$$

Step3. Determine the volume of gravel in the mixture with the following equation:

$$V_{\text{gravel}} = \frac{W_{\text{gravel}}}{G_{\text{gravel}} * \gamma_{\text{water}}}$$

$$V_{\text{gravel}} = \frac{2050}{2.6 * 1684.8} = 0.468 \text{yd}^3$$

Step4. Determine the volume of sand in the mixture with the following formula:

Step5. Determine the weight of sand needed in the mixture by rearranging the following equation:

$$V_{\text{sand}} = \frac{W_{\text{sand}}}{G_{\text{sand}} * \gamma_{\text{water}}}$$

$$W_{\text{sand}} = V_{\text{sand}} * G_{\text{sand}} * \gamma_{\text{water}}$$

$$W_{\text{sand}} = 0.229 * 2.4 * 62.4 = 34 \frac{\text{lbs}}{\text{ft}^3}$$

88) The modulus of elasticity of concrete is given by the following equation:

$$E = 39700\sqrt{f'_{cr}}$$

The specified compressive strength for a concrete structure is 4500psi for a plant where the standard deviation is 400 psi for 18 tests. Determine the modulus of elasticity for the concrete given the equations. The following table may be consulted.

$$f'_{cr} = f'_c + 1.34s \quad f'_{cr} = f'_c + 2.33s - 500$$

Number of tests	Modification Factor, F
15	1.16
20	1.08
25	1.03
30 or more	1.00

- (A) 2.0×10^6 psi
- (B) 2.5×10^6 psi
- (C) 2.8×10^6 psi
- (D) 3.2×10^6 psi

The Answers is C

Step1. Determine the modification factor for the standard deviation by interpolation.

$$F = 1.16 - \left(\frac{1.16 - 1.08}{20 - 15} \right) (18 - 15) = 1.11$$

Step2. Determine the modified standard deviation.

$$s' = sF$$

$$s' = 400 * 1.11 = 444\text{psi}$$

Step3. Determine the required average compressive strength by choosing the higher of the two equations.

$$f'_{cr} = f'_c + 1.34s$$

$$f'_{cr} = f'_c + 2.33s - 500$$

$$f'_{cr} = 4500 + 1.34 * 444 = 5095\text{psi}$$

$$f'_{cr} = 4500 + 2.33(444) - 500 = 5035\text{psi}$$

Step4. Determine the modulus of elasticity given the equation in problem:

$$E = 39700\sqrt{f'_{cr}}$$

$$E = 39700\sqrt{5095} = 2.8 * 10^6\text{psi}$$

89) Consider a steel structure that has a strain fracture toughness of $50 \text{ MPa}\sqrt{\text{m}}$. Through testing, it has been determined that the fracture results in a stress of 350 MPa when the maximum internal crack length is 3mm. For this steel structure, what is the difference in the stress level at which the fracture will occur for a critical internal crack length of 4.5mm when compared to the crack length of 3.0mm.

- (A) 64 MPa
- (B) 80 MPa
- (C) 92 MPa
- (D) 105 MPa

The Answers is A

Step 1: Rearrange the fracture toughness equation 1 to determine the geometric factor Y, given the parameters in the problem. NOTE: we are dealing with an interior crack, so the crack length must be divided by 2.

$$K_{IC} = Y\sigma\sqrt{\pi a}$$

$$Y = \frac{K_{IC}}{\sigma\sqrt{\pi a}}$$

$$Y = \frac{50}{350\sqrt{\pi\left(\frac{3 * 10^{-3}}{2}\right)}} = 2.08$$

Step 2: Rearrange the fracture toughness equation again to determine the stress at the new crack length.

$$K_{IC} = Y\sigma\sqrt{\pi a}$$

$$\sigma = \frac{K_{IC}}{Y\sqrt{\pi a}}$$

$$\sigma = \frac{50}{2.08\sqrt{\pi\left(\frac{4.5 * 10^{-3}}{2}\right)}} = 286\text{MPa}$$

Step 3: Determine the difference between the stress levels.

$$350 - 286 = 64\text{MPa}$$

90) Tension testing is performed on a steel bar with a length of 1 inch and a cross sectional area of 0.25in^2 . The steel bar has a modulus of elasticity of 29,000 ksi. The tension testing increases the bar's length to 1.005 inches. Determine the force that was placed on the bar.

- (A) 25kips
- (B) 36kips
- (C) 43kips
- (D) 49kips

The Answers is B

Step1. Determine the strain on the steel bar using the following equation.

$$\epsilon = \frac{\Delta L}{L}$$
$$\epsilon = \frac{0.005}{1} = 0.005$$

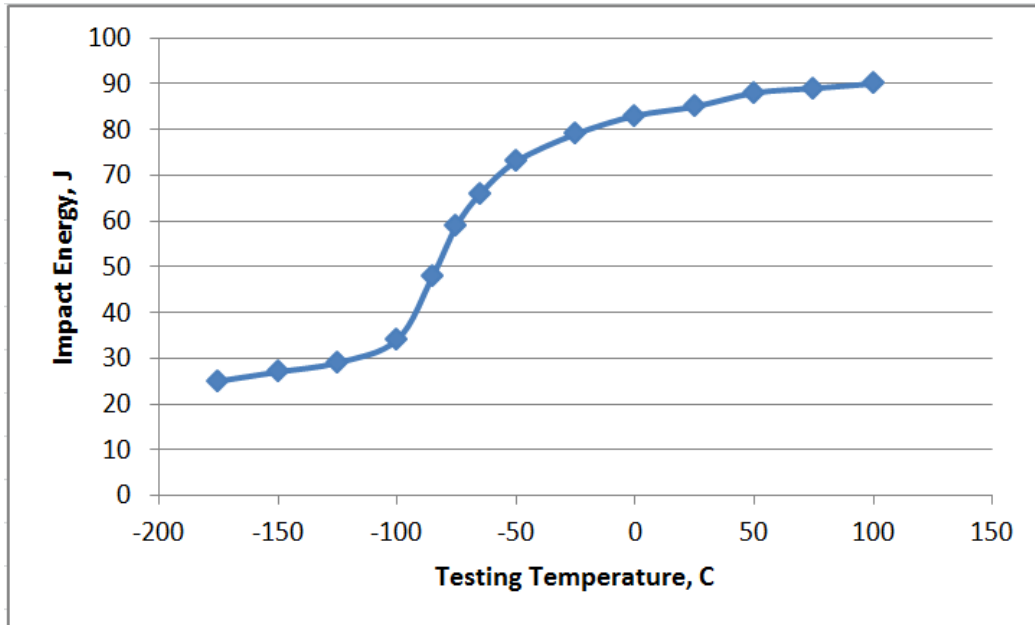
Step2. Determine the stress on the steel bar by rearranging the equation

$$E = \frac{\sigma}{\epsilon}$$
$$\sigma = E\epsilon$$
$$\sigma = 29000 * 0.005 = 145\text{ksi}$$

Step3. Determine the force on the steel bar by rearranging the equation

$$\sigma = \frac{F}{A}$$
$$F = \sigma * A$$
$$F = 145 * 0.25 = 36.3\text{kips}$$

91) A Charpy Impact Test was conducted on a steel bar that yielded the following plot. A researcher is trying to determine how they want to determine the ductility-to-brittle transition temperature. The first way they want to consider it is the temperature corresponding to the average between the maximum and minimum impact energy. The second way was the testing temperature corresponding to the median impact energy. What is the difference between these two methods?



- (A) 5°C
- (B) 20°C
- (C) 35°C
- (D) 50°C

The Answers is B

Step1. Determine the average between the maximum and minimum impact energies.

$$\text{Average} = \frac{\text{Maximum} + \text{Minimum}}{2}$$

$$\text{Average} = \frac{90 + 25}{2} = 57.5 \text{ J}$$

Step2. Determine the median impact energy. NOTE: there are 14 data points, so the median will be the average between the 7th and 8th data point.

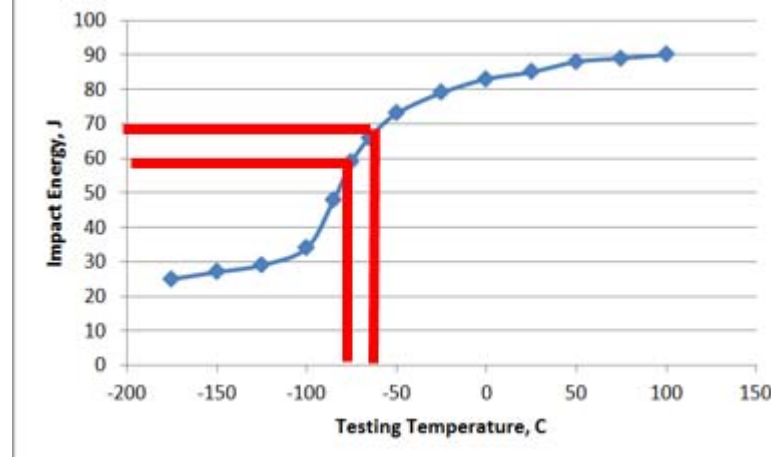
$$\text{Median} = \frac{\text{Point}_7 + \text{Point}_8}{2}$$

$$\text{Median} = \frac{73 + 66}{2} = 69.5 \text{ J}$$

Step3. Determine the ductile-to-brittle transition temperatures.

The two temperatures are approximately -60°C and -80°C

Step4. Determine the difference.



$$80 - 60 = 20^\circ\text{C}$$

92) Failure due to fatigue is considered to be at 50% of the original stiffness of the beam. A fatigue test was conducted on a steel beam while holding the strain constant at 500 microstrains. The area of the beam 0.25in^2 and the initial force on the beam is 3600lbs. Determine the stiffness at which the beam fails.

- (A) 7945 ksi
- (B) 9505 ksi
- (C) 11250 ksi
- (D) 14400 ksi

The Answer is D

Step1: Determine the stress on the steel beam using this equation

$$\sigma = \frac{F}{A}$$
$$\sigma = \frac{3600}{0.25} = 14400\text{psi}$$

Step2: Determine the initial stiffness of the beam using this equation

$$E = \frac{\sigma}{\epsilon}$$
$$E = \frac{14400}{500 * 10^{-6}} = 28,800,000\text{psi}$$

Step3: Determine the stiffness at which the beam fails.

$$\text{Failure} = 0.5E$$

$$\text{Failure} = 0.5 * 28,800,000 = 14,400\text{ksi}$$

93) Consider a steel structure that has a strain fracture toughness of $40 \text{ MPa}\sqrt{\text{m}}$. Through testing, it has been determined that the geometric factor is 1.75 when the maximum external crack length is 2mm. For this steel structure, what is the difference in the stress level at which the fracture will occur for a critical external crack length of 5mm when compared to the crack length of 2mm.

- (A) 64 MPa
- (B) 88 MPa
- (C) 106 MPa
- (D) 122 MPa

The Answers is C

Step 1: Rearrange the fracture toughness equation to determine the engineering stress, given the parameters in the problem.

$$K_{IC} = Y\sigma\sqrt{\pi a}$$

$$\sigma = \frac{K_{IC}}{Y\sqrt{\pi a}}$$

$$\sigma = \frac{40}{1.75\sqrt{\pi(2 * 10^{-3})}} = 288\text{MPa}$$

Step 2: Rearrange the fracture toughness equation again to determine the stress at the new crack length.

$$K_{IC} = Y\sigma\sqrt{\pi a}$$

$$\sigma = \frac{K_{IC}}{Y\sqrt{\pi a}}$$

$$\sigma = \frac{40}{1.75\sqrt{\pi(5 * 10^{-3})}} = 182\text{MPa}$$

Step 3: Determine the difference between the stress levels.

$$288 - 182 = 106\text{MPa}$$

94) A soil has the following Atterburg limits: shrinkage limit = 12%, plastic limit = 34%, liquid limit = 55%. What is the value of the plastic index for this soil?

- | | |
|---------|---------|
| (A) 21% | (B) 34% |
| (C) 43% | (D) 55% |

The Answer is A

Step 1: The plastic index = (liquid limit) – (plastic limit)

Step 2: Here plastic index = $55 - 34 = 21\%$

95) In the Unified Soil Classification System, what is the definition of a soil whose only known characteristic is more than 50% of soil mass passes the No. 200 sieve?

- | | |
|--------------|-----------------------------|
| (A) group GW | (B) coarse grained |
| (C) group CH | (D) inorganic silt and clay |

The Answer is B

Step 1: In the first division of soils in the USCS, if more than 50% passes the No. 200 sieve, the soil is defined as coarse-grained.

96) An organic clay soil has a liquid limit greater than 50 and falls below the “A-line” on the plasticity diagram. What is the group symbol for this soil?

- | | |
|--------|--------|
| (A) OH | (B) CH |
| (C) CL | (D) ML |

The Answers is A

Step 1: Referring to the plasticity diagram at the bottom of page 146, for an organic soil with these characteristics, the group name is OH.

97) A soil sample has less than 25% of mass passing the No. 200 sieve. According to the AASHTO classification, what is the general rating for this soil as a subgrade material?

- | | |
|---------------|-----------------------|
| (A) poor | (B) fair to poor |
| (C) not known | (D) excellent to good |

The Answer is D

Step 1: According to the chart on page 145, less than 25% passing the 200 sieve is in group A-1, which is excellent to good as a subgrade material (from the bottom of the chart).

98) A soil sample has the following characteristics: more than 35% passing No. 200 sieve, liquid limit < 40 , and plasticity index < 10 . What is the AASHTO group classification for this soil?

- | | |
|---------|---------|
| (A) A-2 | (B) A-4 |
| (C) A-5 | (D) A-6 |

The Answer is B

Step 1: Using the AASHTO classification table, these characteristics apply to group A-4.

99) A sieve analysis on a soil reveals that 82% of the soil passes No. 200 sieve (0.075 mm). The liquid limit of the soil is 24% and the plastic limit of the soil is 16%. Determine the group index (GI) of the soil according to the AASHTO system.

- A) 2 B) 3 C) 6 D) 9

The Answer is D**Step 1:**

Based on the given information, the liquid limit (LL) of the soil is 24%, the plastic limit (PL) is 16%, so the plasticity index (PI) is

$$PI = LL - PL = 24\% - 16\% = 8\%$$

The percent passing No. 200 of the soil is given as $F = 82$.

Step 2:

According to the formula Shown, the group index (GI) is calculated as

$$\begin{aligned} GI &= (F - 35)[0.2 + 0.005(LL - 40)] + 0.01(F - 15)(PI - 10) \\ &= (82 - 35)[0.2 + 0.005(16 - 40)] + 0.01(82 - 15)(8 - 10) \\ &= 47[0.2 + 0.005 \times 0] + 0.01 \times 67 \times 0 \\ &= 9.4 \cong 9 \end{aligned}$$

Note: (LL-40) and (PI-10) are both limited in the range of 0 to 20. If the computed value falls outside limiting value, then the limiting value should be used.

Part 8 : Site Development

65 Problems

- 1) In order for an employee to be able to work with hazardous materials they must:
 - a. Be supervised by a manager
 - b. Wash their hands with soap and water
 - c. Have proper training
 - d. Tape off the area with caution tape

Answer is C

Answer "A", "B" and "D" are not correct: While these all might be good practice for handling hazardous materials they are not a requirement for working with hazardous materials.

Answer "C" is correct: Before handling or working with hazardous materials employees must have the proper training on them. Having the proper training will teach them how to store, work with, and dispose of the hazard as well as the necessary emergency procedures and appropriate safety equipment to use.

- 2) Which is the most common cause of injuries on a construction site?
- a. Falls
 - b. Chemical burns
 - c. Radiation Poisoning
 - d. Drowning

Answer is A

Answer “A” is correct: Falling hazards is one of the fatal four most common hazards on site. Nearly 36% of the total construction deaths were related to the falls on site

- 3) Which of the following is not one of the “fatal four” leading causes of worker deaths?
- a. Struck by Object
 - b. Electrocutions
 - c. Chemicals
 - d. Caught-in/between

Answer is C

Answer “A”, “B” and “D” are not correct: These are fatal four hazards

Answer “C” is correct: Chemical hazards are not one of the fatal four leading causes of worker deaths.

- 4) What should you NOT wear while on a construction site?
- a. Hard hat
 - b. Jewelry
 - c. Boots having slip-resistant soles
 - d. Shirts with closely fitted sleeves

Answer is B

Answer “B” is correct: Jewelry should never be worn on a construction site as it can easily get caught and tangled in equipment. This means the wearer is at risk of being stuck or injuring a body part if the machinery begins moving.

- 5) At work an employee notices nails sticking out of the floorboards of the break room, what should they do?
- a. Remove the nails themselves
 - b. Bend the nails down
 - c. Surround the area with caution tape
 - d. Bring it to the attention of a safety monitor

Answer is D

Answer “D” is correct: While this situation seems pretty minor the best course of action is to bring the situation to the attention of the safety monitor. Just in case there is more to the situation the employee should not be handling the nails themselves or dealing with the situation by just taping off the area. Contacting the safety monitor is the best way to get the situation under control and resolved.

- 6) What is a key part in a comprehensive safety & health program?
- a. Safety Training
 - b. Prejob safety meeting
 - c. Compliance with OSHA regulations
 - d. Controlling on site hazards
 - e. All of the above

Answer is E

Answer "E" is correct: All four of these tools are an essential part in a comprehensive safety & health program. All employees must have gone through safety training with a company that is OSHA compliant where they learned how to identify and control hazards on site. Another way to do this is to have a prejob safety meeting where hazards on the job and job procedures are discussed before they are executed.

- 7) What does OSHA stand for?
- a. Occupational Safety and Health Administration
 - b. Occupational Safety and Health Act
 - c. Organizational Sanitary Health Administration
 - d. Occupational Safety and Hazards Act

Answer is A

Answer “A” is correct: OSHA stands for the Occupational Safety and Health Administration.

- 8) Which of the following does OSHA not have regulations for?
- a. Performing site safety inspections
 - b. Safety training for employees
 - c. Offering accident insurance
 - d. Protective equipment requirements

Answer is C

Answer “C” is correct: OSHA is an organization that assists in performing site safety inspections, safety training and requirements for protective equipment. However OSHA does not offer insurance for injuries that happen on site caused by safety hazards.

- 9) Research on the hazards of upcoming and new technology is mainly the responsibility of what organization?
- a. OSHA
 - b. US Safety Department
 - c. The Department of Labor
 - d. NIOSH

Answer is D

Answer “D” is correct: One of the main responsibilities of NIOSH is to do research on the hazards of new technology as well as determining new safety techniques. While NIOSH does not have regulatory capabilities the research done can lead to future regulation done by organizations like OSHA.

- 10) What is a safe practice when lifting a heavy load?
- a. Back straight, Knees straight
 - b. Knees straight, back bent
 - c. Back straight, knees bent
 - d. None of the above

Answer is C

Answer “C” is correct: The proper technique for lifting is to bend your knees and keep your back straight. In this motion the spine is kept straight and the raising and lowering towards the ground is done by bending your knees.

- 11) What are the three steps to controlling a hazard?
- a. Identify the hazard, assess the risk, eliminate/reduce the risk
 - b. Identify the hazard, assess the hazard, contact OSHA
 - c. Control the hazard, make appropriate changes, eliminate the risk
 - d. Identify the hazard, control the risks, make the hazard safe

Answer is A

Answer “A” is correct: The three steps in hazard control are to first identify the hazard in the situation, then assessing the risks in the hazard and finally eliminating or reducing it to an appropriate level. Contracting OSHA is not always necessary.

- 12) A hazard risk is defined by what two things?
- a. Probability and Magnitude of Impact
 - b. Timing and Likelihood
 - c. Harm and Safety factor
 - d. Amount of personal protection and Magnitude of Impact

Answer is C

Answer C” is correct: Risk is the probability of a hazard occurring compared to the expected impact if the hazard occurs.

- 13) What is the least effective hazard control method in the control hierarchy?
- a. Eliminate the hazard
 - b. Engineering Controls
 - c. Administrative Controls
 - d. Personal Protective Equipment

Answer C is correct

Answer "A", "B" and "D" are not correct: These controls are all higher on the hazard control hierarchy than PPE.

Answer "C" is correct: Personal Protective Equipment is the last line of defense against potential hazards. PPE must be used if the hazard is constantly present or cannot be reduced.

- 14) At which phase in the project is the ability to influence safety the highest?
- a. Conceptual Design
 - b. Detailed Engineering
 - c. Procurement
 - d. Construction

Answer is A

Answer “A” is correct: The earlier in the project the easier it is to influence safety on site. The closer to the end date the harder it is to influence safety due to high costs. Given the multiple choice questions, conceptual design is the closest to the start date.

- 15) What is not a method for hazard elimination/reduction?
- a. Safety Training
 - b. Risk Assessment
 - c. Preparing a hazard communication program
 - d. Providing PPE

Answer is B

Answer “A”, “C” and “D” are not correct: These are all concrete methods that can be implemented that can control and eliminate a hazard.

Answer “B” is correct: A risk assessment helps determine the probability and impact that a hazard will have and the risk rating it has. This information can help us determine which hazards to prioritize and control first. However a risk assessment is not a method for hazard elimination it is a tool to determine which risks must be eliminated/reduced.

- 16) If a hazard is assessed to cause permanent injury that impacts enjoyment of life and may require continued treatment, what type of magnitude category is it in?
- a. Severe
 - b. Serious
 - c. Moderate
 - d. Slight

Answer is B

Answer “B” is correct: This is the definition of a serious risk.

- 17) What does MSDS stand for?
- a. Management Survey Data Sheet
 - b. Material Standards Direct Support
 - c. Maintaining Standards for Directing Safety
 - d. Material Safety Data Sheet

Answer is D

Answer “D” is correct: MSDS stands for Material Safety Data Sheet. The purpose of this document is accompany a chemical and provide workers with procedures for handling and working with the substance in a safe manner. This includes critical information such as toxicity, health effects, boiling point storage etc.

- 18) As a part of hazard communication, the manufacturer of a chemical is required to provide
- a. MSDS
 - b. Work plan
 - c. Safety Meeting agendas
 - d. Hazardous substance training

Answer is A

Answer “A” is correct: The manufacturer is not required to provide the work plan, safety meeting agenda, or hazardous substance training. This is the responsibility of the employer. What they are required to provide are material safety data sheets (MSDS) for their chemicals which give detailed information about their product and how to handle and dispose of it etc.

- 19) Which of the following will you not find on an MSDS?
- a. Chemical components
 - b. Exposure limits
 - c. Cost of the chemical
 - d. Directions for safe handling

Answer is C

Answer “C” is correct: The MSDS holds a lot of information about the technical components of the chemical and its effect. It however does not contain the cost of the chemical.

- 20) You are on a construction crew working near a school and there is a light post that needs to be removed in order to install a new sewer system. The electrical equipment on site must be installed and maintained by whom?
- a. The inspector
 - b. The employer
 - c. An electrician
 - d. Experienced employees

Answer is C

Answer “C” is correct: An electrician or the electrical company is the qualified party to handle any electrical equipment on site. They are professionals that know the different hazards related to the equipment and how to handle it. While the inspector, employer or employees may have seen someone else do it before, it is imperative that for safety reasons only the electrician handle the light post.

- 21) What is the main reason that the minimum distance to stay away from the edge of an excavation is 2 feet?
- a. There may not be a ladder in the excavated area
 - b. The soil may collapse
 - c. The 2 foot area is where all the equipment needed for the excavation is stored here
 - d. This is where the trench box is placed for placement into the excavation

Answer is B

Answer “B” is correct: Soils are unpredictable and if anyone is standing in a 2 foot range of the excavated area the likelihood that the soil will cave in and collapse is much greater. This is dangerous for both the person on high ground as well as the people inside the excavated area.

- 22) What is not a proper technique to keep tools and equipment properly maintained?
- a. Regular Maintenance Schedule
 - b. Equipment inventory
 - c. Thorough documentation/recordkeeping
 - d. Equipment training
 - e. Frequent washing

Answer is E

Answer “E” is correct: Frequent washing will not solve the internal problems with the equipment. Washing will only make sure that the equipment does not get caked with debris. However to really keep tools and equipment properly maintained you must develop a maintenance schedule, inventory, documentation and hold equipment specific training.

- 23) Which of the following does not help determine which PPE to wear?
- a. NIOSH regulations
 - b. MSDS
 - c. Manufacturer Specifications
 - d. Previous injury reports

Answer is A

Answer “A” is correct: NIOSH is not a regulatory agency and while they can recommend PPE for different situations they do not have regulations for PPE. The MSDS, manufacturer specifications and injury reports can all help determine the proper PPE to wear.

- 24) What is not true about designing for safety?
- a. It increases workers compensations premiums
 - b. Promotes collaboration between design and build teams
 - c. Increased productivity
 - d. Decreased site hazards

Answer is B

Answer “B” is correct: When using designing for safety, there are a decrease in on site hazards thus leading to less injuries and fatalities. This means that less people are filing insurance claims thus workers compensation premiums decrease not increase.

25) According to the inspector report, after a deep excavation under a layer which contain water a water started to come out to the excavation. Immediately the contractor made a drainage system and has pumped the discharged water out of the site safely. What would be the effect of it on the adjacent buildings?

- A) Nothing, since there is not water leakage the excavation will be safe.
- B) Nothing, there is no relation to the adjacent buildings.
- C) Although the water safely pumped out, but still there would be the risk of settlement for the adjacent buildings.
- D) Since the water go out of the soil system the bearing capacity will increase and it has made more safety for the adjacent buildings.

The Answer is C

When the water goes out from the soil layer, the porosity of soil will increase. Increasing the porosity will decrease the soil bearing capacity and consequently will increase the risk of settlement for adjacent buildings and all neighbors. So, with the drainage system the contractor can make sure that the excavation will not fail, but the more pumping the water, means the more risk for settlement.

26) Once the contractor submit the drawings to the city the adjacent construction office may review drawing and the required item for submission are:

- A) Project design: General plans of the project showing relative proximity of the work, Preconstruction survey data, Design calculations, material and catalog cut information
- B) Support of Excavation, Structural monitoring and contingency plans, Updated construction bar chart, Phasing plans and Maintenance of Traffic
- C) A general plan indicating the positioning of construction equipment which may operate,
Insurance policies
- D) All of them

The Answers is D

Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 (Revision 5)

All answers are the definition of the required submittals. I would take 6 month to pull the permit for this part.

27) Which set of information are not required to submit to the city to check the effect of project on adjacent buildings?

A) A set of Civil Drawings - site plan showing all existing conditions, including building and basement level(s), parcel limit, distances from public facilities, structures, and utilities. Show demolition area if any.

B) A set of Architectural Drawings - showing new project layout, including plans, elevations and sections.

C) A set of Mechanical/HVAC & Electrical Drawings - showing piping, ducting, cable trays, cabling and the related calculation.

D) A set of Structural Drawings - showing foundation plans, elevations, sections, and column loads and Geotechnical report.

The Answers is C

Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 (Revision 5)

According to the instructions the mechanical and electrical drawings and related documents are not necessary to submit.

28) What is the zone of influence for the excavation for a building project? (If the proposed augering of soldier piles and bearing piles are within 25 feet of public facilities.)

A) 100 ft

B) 25 ft

C) 50 ft

D) 75 ft

The Answer is B

Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 (Revision 5)

Zone of influence for the excavation is considered as 25' according to the city instructions.

- 29) Find the zone of influence for the following excavation where there is no soldier piles. Neglect the brick made counterforts.



Depth of excavation = 30 ft

- A) 22.6 ft B) 30 ft C) 32 ft D) 21.2 ft

The Answers is A

[Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 \(Revision 5\)](#)

An envelope starting at a point two feet below the lowest point of the underground structure or excavation continuing upward at a forty five (45) degree angle from the horizontal at the vertical projection of the outside limits of the Public structure.

An envelope starting at a point two feet below the lowest point of Public structure continuing upwards at a forty five (45) degree angle from the horizontal, up to the horizontal projection of the outside limits of the adjacent underground structure or excavation, projected at grade level.

So, depth of excavation = 30 and according to the code 2' should be added so, $30+2=32$

Horizontal distance considering the 45 degree = $32 \cdot \cos(45) = 22.6 \text{ ft}$

- 30) For the excavation shown in the following figure, find the lateral earth pressure on each supporting structures if the spacing of them are about 10'. ($K_a=0.3$, Soil density = 150 pcf)



Spacing=10'

20' including height of foundation

- A) 11250 lb/ft B) 1125 lb/ft C) 9000 lb/ft D) 900 lb/ft

The Answers is A

Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 (Revision 5)

Lateral Earth Pressure and Groundwater Pressure. The basic horizontal earth pressures shall be computed using the active earth pressure. The resultant or total active earth pressure shall be multiplied by a stiffness factor depending upon the required stiffness. The resulting load shall be redistributed on the cofferdam in a trapezoidal pressure diagram. The stiffness factors shall be applied to both the cofferdam design and the bracing system. The stiffness factors shall be assigned as follows:

1. Use stiffness factor = 1.25 for a soldier pile and lagging or a sheet pile support system.
2. Use stiffness factor = 1.5 for a slurry wall, secant and tangent pile wall support system.

So: $P = \gamma * h * k_a * 1.25 = 150 * 20 * 0.3 * 1.25 = 1125 \text{ psf}$

Pressure * spacing of the soldiers = $1125 * 10 = 11250 \text{ lb/ft}$

- 31) Which one is not considered as the limitation of construction for the public or adjacent structures?
- A) Excavation or tunneling under Public structures is prohibited, except for access for underpinning.
 - B) Excavation within 10 feet of existing WMATA facilities is prohibited or Installation of pre-augured piles within five (5) feet of the bored tunnel liners.
 - C) Pile driving within 50 feet of WMATA structures, and tracks.
 - D) Pile driving within 25 feet of WMATA structures, and tracks.

The Answer is C

Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 (Revision 5)

All explanations except choice “C” are the definition of limitations for construction.

- 32) Excavation support systems for the portion of the adjacent construction within the public structures zone of influence will be instrumented and monitored to:
- A) Measure the movement and deflection of the sheeting/cofferdam wall, etc. by optical surveying
 - B) Monitor the horizontal and vertical movement of heel blocks where lateral support to the sheeting/cofferdam wall is provided by rakers.
 - C) Monitor movement of the ground between the Public facility and the support of excavation system by means of inclinometers, movement detection points, or other city approved instruments.
 - D) All of the answers

The Answers is D

Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 (Revision 5)

All answers are the definitions for the instrumentation, so:

33) What is shown by the contingency plan?

- A) A contingency plan will be required for all major projects adjacent to an underground tunnel
- B) Ground or structure movement exceeding the limits or threshold values.
- C) Cracking of concrete structures, Excessive opening of joints, movement and translation of joints in tunnel liners.
- D) All of above

The Answers is D

Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 (Revision 5)

All answers are the definitions for the instrumentation, so:

34) Which one as the demolishing plan adjacent to other buildings are allowed or required?

A) Demolition of structures adjacent to public facilities by blasting, or by implosion of the structure by blasting shall not be allowed.

B) Demolition adjacent to a WMATA station entrance, that requires protection from dust, shall be done during non-revenue hours if using piece-by-piece demolition. During demolition WMATA vents located adjacent to the site shall be protected with a vent cover

C) The ODC must provide the complete demolition plan for WMATA review and approval. Based on the demolition plan and the nature of adjacent WMATA structure, WMATA may require the ODC to check the structural adequacy of the WMATA structure due to the effects of the impact of the demolition.

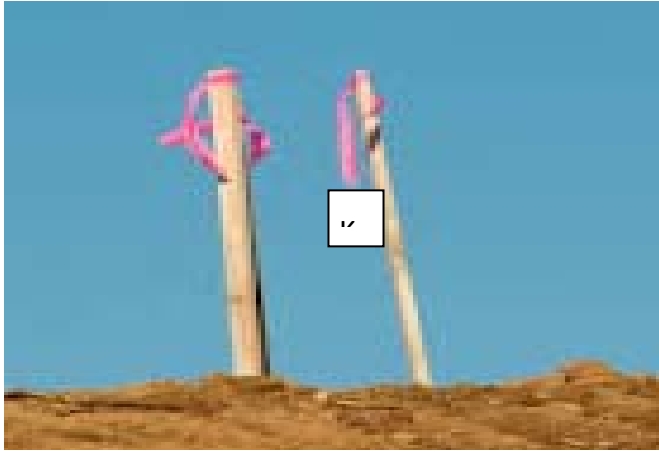
D) All above answers

The Answers is D

[Adjacent Construction Project Manual, Office of Joint Development & Adjacent Construction, September 16, 2013 \(Revision 5\)](#)

All answers are the definitions for the instrumentation, so:

- 35) The site manager put the “K” on the lath stake. The required elevation is given equal to +20.00 With that sign what would be the final elevation:

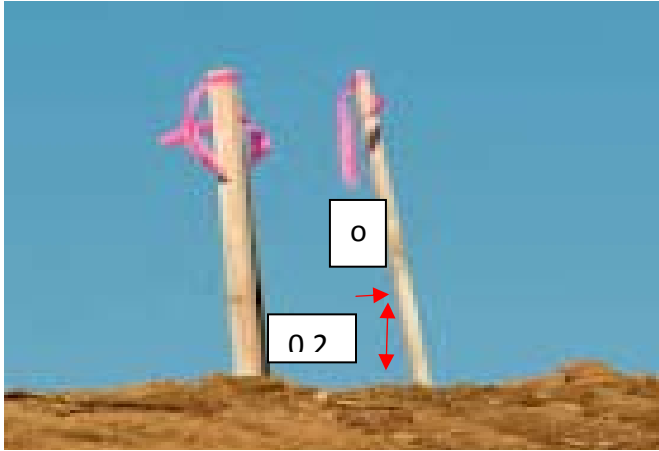


- A) +20.00
B) $> +20.00$
C) $< +20.00$
D) none of them

The Answers is B

The sign “K” means: Fill is required. So, the final elevation will be more than +20.00.

- 36) The site manager put the circle and a horizontal line on the lath stake. The required elevation is given equal to +20.00. The surveyor measures the distance between horizontal line and the ground level equal to 0.20. With that sign what would be the final elevation:



- A) +20.20
B) +18.80
C) 18.00
D) none of them

The Answer is A

The sign of circle and means: Fill is required. And the horizontal line shows the required level for filling. So, the final elevation will be more than $+20.00 + 0.20 = +20.20$.

- 37) After which step the surveyor can set the elevation lath stakes.
- A) Establish a local control
 - B) Topography map
 - C) Correct Azimuth for declination
 - D) locate the parcel boundaries according to the deed

The Answers is A

Obtaining the topography map is the first step and correction of azimuth and locate the boundaries are all required actions prior to any action. After all, the surveyor shall establish the local control to make sure about the elevations according to the federal reference points. After this local control he can start setting the lath strakes.

- 38) How can a surveyor avoid to have negative number for his station numbers?
- A) The starting (0+00) point should be within the boundary.
 - B) The starting (0+00) point should be outside of the boundary.
 - C) Station 0+00 is selected some significant distance from the project boundary.
 - D) Report the negative station numbers as positive numbers and neglect them.

The Answer is C

To avoid having the negative station number, the station 0+00 should have significant distance from project boundary and the stations inside the boundary with that distance can be started with 10+00 or 100+00.

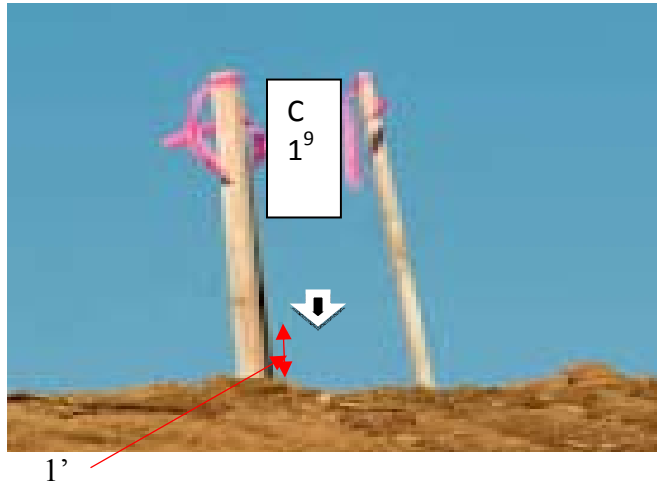
So, choice A and D are incorrect. Choice B is not wrong but choice C has given a better explanation for the problem though.

- 39) Which one should be applied if the surveyor wants to use total station?
- A) It should set up over the control point
 - B) It should be validated by shooting at least 2 offsite control points.
 - C) It should be validated by shooting at least 1 onsite control points.
 - D) All 3 above choices

The Answers is D

The total station set needs to have all those validations from a free station to a resection points and the control points.

- 40) The site manager put the mark of "C" and the following numbers on the lath stake. The ground elevation is given equal to +20.00. The surveyor measured 1' for the distance between horizontal line and the ground surface. With that sign what would be the final elevation:



- A) +21.90
B) +19.10
C) +17.10
D) +18.10

The Answers is B

The sign of “C” means: Cut is required. And the horizontal line shows the reference point for it. The number on stake shows the required cut below the horizontal line.

So, the final elevation will be equal to 1.9’ below the horizontal line. Since the distance between the ground and the horizontal is 1’, then it is necessary for the $(1.9-1)=0.9$ ’ for excavation. The final ground elevation will be: $+20.00 - 0.9 = 19.10$

41) In a highway project the following information for the station 100+25 are available:

- I) Slope: 2:1
- II) Super elevation (SE) 0.03
- III) Required excavation: 1.2'

Which information should be in the back of the lath stake and which one should be in the front side?

- A) Back side: II & III, Front side: I
- B) Back side: Station Number, Front side: I, II, and III
- C) Back side: I, II & III, Front side: Station Number I
- D) Back side: I, Front side: Station Number, II & III

The Answers is B

This is the common procedure that the station number should be in back side, and in the front side all other require information like super elevation, cut or fill should be mentioned.

42) In a highway project the following information for the station 100+25 are available. The ground level is +15.00 and the required level is +16.20. How the surveyor can show those information?

A) C +16.²⁰

B) F +16.²⁰

C) F 1²⁰

D) C 1²⁰

The Answer is C

The ground level is given equal to $+15.00$

Required level $= +16.20$

$16.20 - 15.00 = 1.20$ which means: Fill is required.

So, $F 1^{20}$ is the answer.

43) In the tunnel project (below figure) a point of intersection in the horizontal curve has made the station 150+20. What should be written on the stake?



- A) PI, +150+20
- C) CL, +150+20

- B) +150+20, TP
- D) EC, +150+20

The Answers is A

According to the common marking abbreviation the basic item is the station number which is +120+20.

For the point of intersection PI is correct. CL shows the centerline, EC end of curve, TP turning point which all of them are not related to the case.

- 44) Stakes or surveying markers are used for the following purposes:
- A) Construction stakes, alignment stakes
 - B) Offset stakes, grade stakes
 - C) Slope of stakes
 - D) All of above

The Answer is D

This is the definition of the stake. So all choices are the definitions.

45) What is the definition According to the US environment protection (EPA) which area needs permit for the land disturbing and it will called as “disturb”.

- | | |
|----------------------|-----------------------|
| A) 10 or more acres | B) 1 or more acres |
| C) 100 or more acres | D) 1000 or more acres |

The Answers is B

US environment protection (EPA) effective March 10, 2003 any activity in the area of 1 or more acres needs NPDES (National Pollutant Discharge Elimination System) permit.

46) The land disturbing according to the EPA means: Any activity that alter.....

- A) The surface of the land
- B) Clearing, grading, excavation
- C) Building construction access road
- D) All of them

The Answer is D

All three choices are the definition of the code. So for all those activities the client should pull the permit.

- 47) Is it required to pull the permit for storm water discharge for the 2 acres site?
- A) No because it is not necessary for the storm water discharge
 - B) No because the site is very small. There would not be a flood.
 - C) Yes, for the area 1 or more the permit for storm water is necessary.
 - D) Storm water is not the item of the permit.

The Answer is C

According to the EPA, any surface runoff from the disturbed areas means storm water discharge and needs to pull the permit regardless of location and disturbing activities.

- 48) Storm water management should study the following items:
- A) The pollutant should be measured in the water discharge, practice sediment removal
 - B) Erosion control, velocity dissipation devices to provide any non-erosive flow
 - C) Adequate size for the sediment basin
 - D) All of them

The Answer is D

All choices show the definition of storm water management.

49) Which one is not the natural method for the erosion control?

- A) Compost blanket
- B) Grass channels
- C) A geotextile fabric barrier to intercept sediments
- D) Channels made of Synthetic materials

The Answer is C

The geotextile fabrics are not classified as the natural erosion control solutions. All other choices are the natural solution.

50) For the following picture which shows a ditch checks for erosion control, which answer is the best description of both ditches?



Figure 1



Figure 2

- A) 1: Rock ditch for grade $<6\%$, 2: Bale ditch for grade $>6\%$
- B) 1: Bale ditch for grade $>6\%$, 2: Rock ditch for grade $<6\%$,
- C) 1: Rock ditch for grade $>6\%$, 2: Bale ditch for grade $<6\%$
- D) 1: Rock ditch for grade $>2\%$, 2: Bale ditch for grade $<2\%$

The Answer is C

Experiences show that for the grade $< 6\%$ the bale ditch check the erosion, and the rock ditch works for grade $> 6\%$.

51) When does the silt fence barrier (below picture) be used for the sediment control?



- A) It may be constructed of hay and it operates by intercepting and ponding sediment-laden runoff.
- B) It may be constructed of fence and it operates by intercepting and ponding sediment-laden runoff.
- C) It is a mechanical system spillway
- D) It is a silt fence ditch check and shall be used at 100' spacing.

The Answer is B

Choice "A" is the definition for bale slope barrier, "C" is the definition of a spillway, "D" is the ditch with silt fencing. The barrier should be used at the toe of a slope when the ditch does not exist

52) Pollution prevention plan is about:

- A) The storm water prevention plan including a listing of all planned erosion and sediment control and address to the inspection and maintenance procedures.
- B) For all projects waste management system should be designed.
- C) Requires to do all chemical tests for water and soil at site
- D) None of them

The Answer is C

Choice “A” is the definition for the pollution prevention plan.

53) Soil erosion can be caused by the following items:

A) Water & Ice

B) Wind

C) Gravity

D) All of them

The Answer is D

All choices are the cause of erosion.

- 54) Why the sediment control and study is so important for the projects?
- A) Stream channels can be filled with sediment causing flooding.
 - B) Transported sediment can cause damage by covering agriculture lands
 - C) Sediment deposited into reservoirs reduces their storage capacity.
 - D) All of above

The Answer is D

This is the definition of the effects of the sediment. So all choices are the definitions.

55) What is a characteristic of a storm water pollution prevention plan?

- A) Decrease the amount of disturbed soil
- B) Slow runoff that flows across the site
- C) Eliminate sediment from runoff that leaves the site
- D) All of the above

The Answer is D

Answer “D” is correct: A storm water pollution prevention plan (SWPPP) should have all these characteristics in order to control both soil and sediment runoff.

56) Which of the following geosynthetics may be used for soil stabilization?

(I) Geotextiles

(II) Geogrids

(III) Geonets

(IV) Geomembranes

A) I and II

B) I, II, and III

C) I, II, and IV

D) All of them

The Answer is D

Geosynthetics include a variety of manufactured products that are used in drainage, earthwork, erosion control, and soil reinforcement applications. Geosynthetics include woven and nonwoven geotextiles, geogrids, geonets, and geomembranes. Geotextiles are made of polymer tapes, monofilament fibers, or fibrillated yarns, have various strength, stiffness, and drainage properties. Geogrids are polymer grid mats, usually have high strength and stiffness and are used primarily for soil reinforcement. Geonets are similar to geogrids, but typically lighter weight and weaker, with smaller mesh openings. Geomembranes are impervious polymer sheets that are typically used to line ponds or landfills. All of the geosynthetics may be used for soil stabilization.

(A) is incorrect. (III) and (IV) are also correct.

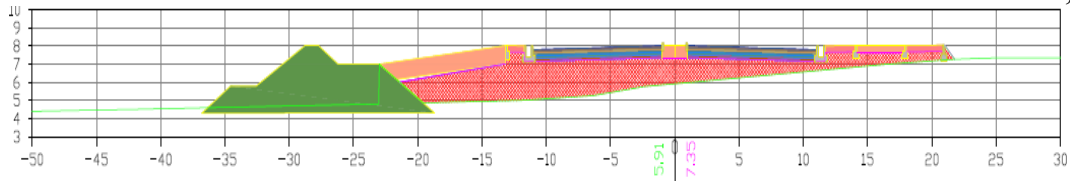
(B) is incorrect. (IV) is also correct.

(C) is incorrect. (III) is also correct.

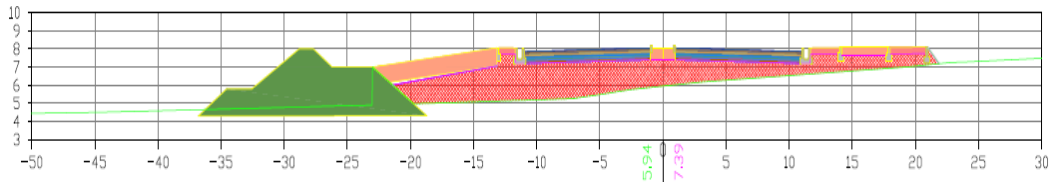
(D) is correct.

57). Refer to the figures, the net excavation (yd³) from section 78 (14+20) (100s of ft) to section 80 (15+80) is most nearly:

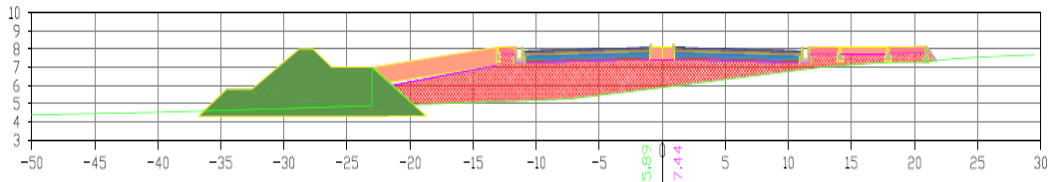
Cut= 0.00 ft² Fill=555.6 ft³ Sec. 80, 15+80



Cut= 0.00 ft² Fill=680.2 ft³ Sec. 79, 15+00



Cut= 421.3 ft² Fill=0.00 ft³ Sec. 78, 14+20



- A) 3462
C) 4000

- B) 59788
D) 2214

The Answers is D

Sec 80 & 79: No excavation, both sections shows embankment:

$$V_{\text{fill}} = \frac{(555.6+680.2)}{2} * (1580 - 1500) = 49432 \text{ ft}^3$$

Sec 79 & 78: There is excavation and embankment between these two sections:

$$V_{\text{fill}} = \frac{(680.2+0)}{2} * (1500 - 1420) = 27208 \text{ ft}^3$$

$$V_{\text{Cut}} = \frac{(421.3+0)}{2} * (1500 - 1420) = 16852 \text{ ft}^3$$

$$\text{Net excavation: } V_{\text{Net}} = \sum V_{\text{fill}} - \sum V_{\text{Cut}} = 49432 + 27208 - 16852 = 59788 \text{ ft}^3$$

$$\text{Net excavation in yd}^3 = \frac{59788}{27} = 2214.37 \text{ yd}^3$$

58) 100 yd³ of bank run soil is excavated and stockpiled before being transported and subsequently compacted. Swell and shrinkage factors for the soil are given equal to 0.30 and 0.12 respectively. The final volume of the compacted earth is most nearly.

A) 130

B) 65

C) 88

D) 112

The Answer is C

Swell is measured with respect to the banked condition. The stockpiled and transported volume will be: $V_t = (1 + \text{Swell coefficient}) * \text{bank} = (1 + 0.30) * 100 = 130 \text{ yd}^3$

Shrinkage coefficient gives the compacted volume, so the answer is:

$$V_c = (1 - \text{Shrinkage Coefficient}) * \text{bank} = (1 - 0.12) * 100 = 88 \text{ yd}^3$$

59) 1000 yd³ of bank run soil is excavated and stockpiled before being transported and subsequently compacted. Swell and shrinkage factors for the soil are given equal to 0.35 and 0.10 respectively. The truck capacity for the transportation of the soil is 5 yd³. If the contractor use 2 trucks, then find the number of trips for each truck.

A) 270
C) 900

B) 135
D) 1350

The Answers is B

Shrinkage coefficient gives the compacted volume, so the answer is:

$$V_c = (1 - \text{Shrinkage Coefficient}) * \text{bank} = (1 - 0.10) * 1000 = 900 \text{ yd}^3$$

Swell is measured with respect to the banked condition. The stockpiled and transported volume will be: $V_t = (1 + \text{Swell coefficient}) * \text{bank} = (1 + 0.35) * 1000 = 1350 \text{ yd}^3$

Each truck carries the 5 yd^3 and for two is 10 yd^3 . Therefore,

$$\text{Total Number of trips for each truck} = \frac{1350}{2 * 5} = 135$$

60) Which one is not the responsibility for the inspector in the construction site?

- A) Inspector should determine how fast the emergency personnel will arrive at site
- B) Inspector must ensure about the safety issues at site.
- C) To ensure about the hazardous material and atmosphere
- D) Likelihood of any collapse

The Answer is A

OSHA 29 CFR 1926.32(f) and 1926.651 (k)

This is not the responsibility of the inspector to determine how fast the emergency personnel will arrive.

61) For the following excavation shown below the contractor has used the nailing system to secure the project. Who should be the first person for inspection of the nailing system?



- A) Trained person
- B) Site supervisor
- C) Competent person
- D) anyone in the site with engineering degree

The Answer is A

The competent person is not necessarily have sufficient knowledge about the nailing. Choices "B" and "D" are also are not correct. For all excavations a trained person should be the first who enter.

62) For the following bridge shown in the following picture, what is the maximum distance between the net and the bridge deck that a fall can be arrested by that net?



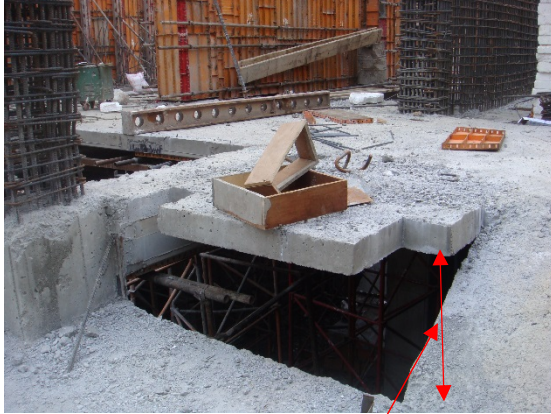
- A) 10' B) 15'
C) 30' D) 20'

The Answers is c

OSHA 29 CFR 1926.502(c)

According to the code: 30'

63) For the following picture, and height which floor has been secured properly to protect from falling according to OSHA?



Picture 1 5'



8' Picture 2

- A) 1 needs protection; 2 needs protection.
- B) 1 does not need protection; 2 needs protection.
- C) 1 needs protection; 2 does not need protection
- D) None of them needs protection.

The Answers is B

OSHA 29 CFR 1926.501(c)

Employee worker on the concrete formworks must be protected from falling if the height is more than 6'.

Picture 1 in 5' < 6' so it does not need protection and Picture 2 needs protection.

64) In the following picture what is required height for the guard rail around the scaffolding?



Height of guard rail =?

- A) 42
- B) 45
- C) 39
- D) All 3 above choices

The Answer is D

OSHA 29 CFR 1926.502(b) and 505 (g) (3) i

Height of guardrail should be: 42 +/- 3 in.

65) What is the minimum required safe load for the following pictures according to the OSHA?



Scaffolding for a building- Figure 1



Scaffolding for the bridge deck-Figure 2

A) 1: 25 psf; 2: 50 psf

C) 1: 75 psf; 2: 100 psf

B) 1: 50 psf; 2: 75 psf

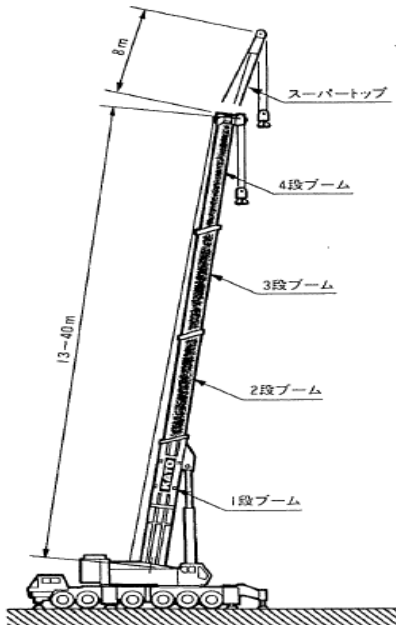
D) 1: 25 psf; 2: 50 psf

The Answers is B

OSHA 29 CFR 1926 Subpart L App. A (1)

According to the code: Light weight scaffolding is 25 psf, usual structures is 50 psf, and heavy scaffolding is 75 psf. So, for the building 50 psf, and for the bridges 75 psf would be sufficient enough.

- 66) For the mobile crane like the following picture, if the crane capacity is given equal to 360 tons and the length of boom and jib is equal to 150' and the boom stands in 80 degree position. Find the safe load if the boom stands in 45 degree?



- A) 26 ton
- B) 1 ton
- C) 0.25 tons
- D) 5 tons

The Answers is C

The load radius is the distance from vertical centerline of the load to the vertical centerline of the crane or center pin. So:

$$M = 150 * \cos(80) = 26. \text{ Ton-ft}$$

For the new boom position:

$$\text{Distance to the center pin} = 150' * \cos(45) = 106 \text{ ft}, \text{ Safe load} = 26/106 = 0.24 \text{ Tons}$$

67) Which item may cancel the lift by a crane?

- A) Wind speed and extreme air temperature
- B) inflated tires
- C) Counterweighting
- D) Supported by its outriggers

The Answer is A

When a crane supported on its outriggers it can lift the loads and the crane lifted off on the tires. If the load is within the load chart then the only item that may cancel the operation is just the wind or extreme air temperature.

68) Inspector observed 7 randomly broken individual strands in the hoist rope. What should he order?

- A) Wait for more inspection
- B) neglect it because they are just 7 strands
- C) Replace it with the new one
- D) cut the damaged part and splice it

Answers is C

OSHA 29 CFR 1926.550 (a) (7) (i)

The rope should be taken out of service when 6 or more randomly broken strands be observed.

- 69) A net should be used under a bridge like the following picture. If 6 labor and 1 engineer work on deck, find the required fall anchorage force.



- A) 5000 lbs
- B) 7000 lbs
- C) 25000 lbs
- D) 35000 lbs

The Answers is D

OSHA 29 CFR 1926.502 (d) (15)

The fall anchorage must be able to support a 5000 lbf loads per person. So, for 7 person they must support: $7 \times 5000 = 35000$ lbf

Biography :

Shahriar Jahanian received his PE in 1988 after several years of industrial experiences. Later he obtained his Ph.D from Louisiana State University in 1992. After receiving his Ph.D, he taught at Southern University Baton Rouge, Penn state, Temple University and Cal state system. While working in Academia he taught variety courses and published more than 50 papers in the areas of interest. In 2003 he started his company called EITExperts with Only 9 students to prepare them pass the EIT exam. Now he has more than 800 students across US and Canada who are prepared to pass PE and EIT in different disciplines.